

# Multinational Firms and Trade in Latin America

Stylized Facts by Sector: All Tables and Figures

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## Data

The data are the customs export records of nine Latin American countries at the firm, destination, HS6 product and year level, matched to the ownership records of Orbis and Dun & Bradstreet. The years covered are Argentina 2011–2019, Chile 2009–2022, Colombia 2010–2021, Costa Rica 2010–2019, the Dominican Republic 2012–2019, El Salvador 2006–2018, Paraguay 2012–2020, Peru 2010–2019 and Uruguay 2010–2019.

An exporter found in Orbis or Dun & Bradstreet is a multinational affiliate (MNE). It is a *domestic* MNE when its ultimate parent is located in the exporting country and a *foreign* MNE otherwise; matched exporters without a recorded parent country (7.5 percent of foreign-MNE export value) are counted as foreign. Exporters not found in either database are *local* firms. Sectors are defined on HS chapters: agriculture 01–24, mining and fuels 25–27 and 71, manufacturing 28–97 excluding 71. Each part of this document repeats the exhibits within one sector: agriculture, manufacturing, mining and fuels, and other goods (HS 98–99 and unclassified codes). Quintiles, rankings and top lists are computed within the sector.

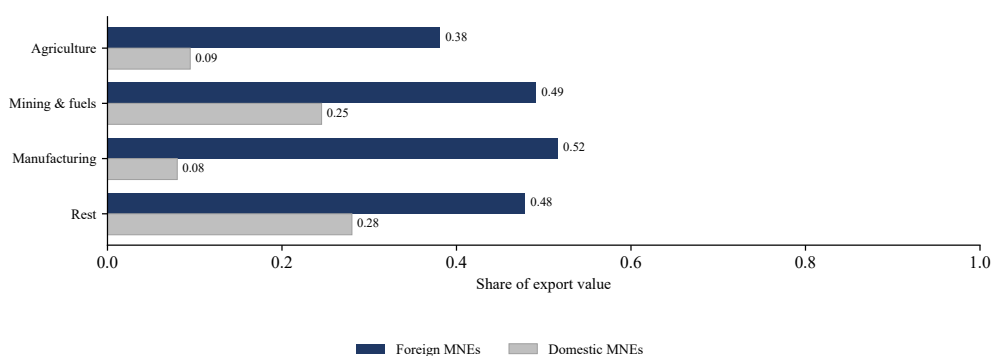
## 1 The Four Sectors

**Table 1:** Exports and Multinational Shares, by Sector

| Sector         | Total exports (US\$bn/yr) | MNE exports (US\$bn/yr) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|----------------|---------------------------|-------------------------|-------------|--------------|------------------------------------|
| Agriculture    | 78.9                      | 37.8                    | 38.1        | 9.5          | USA 0.24, CHE 0.11, NLD 0.10       |
| Mining & fuels | 72.4                      | 52.4                    | 49.2        | 24.5         | GBR 0.36, USA 0.21, CAN 0.16       |
| Manufacturing  | 72.7                      | 43.9                    | 51.6        | 8.0          | USA 0.25, GBR 0.14, DEU 0.09       |
| Rest           | 0.5                       | 0.4                     | 47.9        | 28.0         | USA 0.39, GBR 0.12, ESP 0.11       |
| All            | 224.5                     | 134.5                   | 46.3        | 14.1         |                                    |

*Notes:* Foreign and domestic: percent of the sector’s export value. Other goods: HS 98–99 and unclassified codes. Leading parents: the three largest parent countries and their share of the row’s foreign-MNE exports. Values are annual averages in US\$ billion.

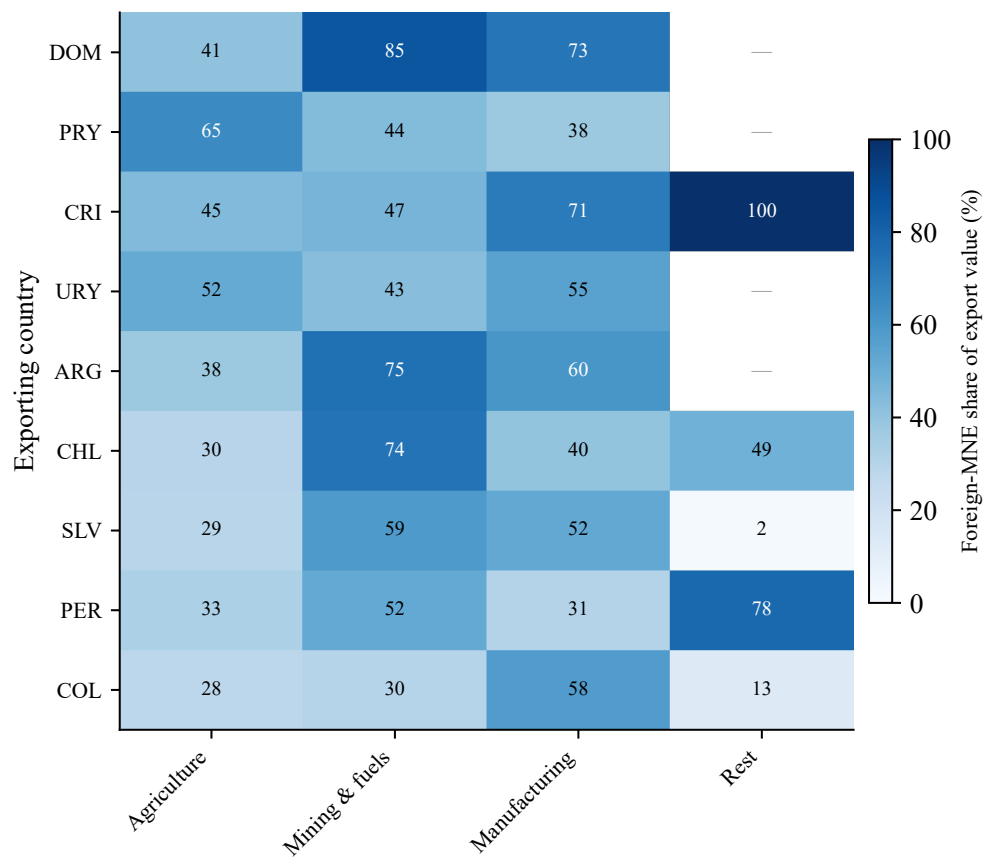
*Source:* Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 1:** Multinational Share of Export Value, by Sector

*Notes:* Shares of export value, all available years pooled. Agriculture: HS 01–24. Mining and fuels: HS 25–27 and 71. Manufacturing: HS 28–97 excluding 71.

*Source:* Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 2:** Foreign-Multinational Share of Export Value, by Exporting Country and Sector

*Notes:* Percent of each country's export value in each sector shipped by foreign MNEs.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 2:** Exports by Exporting Country and Sector

| Origin                                                                   | Agriculture | Mining & fuels | Manufacturing | Rest  | All   |
|--------------------------------------------------------------------------|-------------|----------------|---------------|-------|-------|
| <i>Panel A: composition of the origin's exports, % (rows sum to 100)</i> |             |                |               |       |       |
| ARG                                                                      | 57.4        | 9.9            | 32.7          | 0.0   | 100.0 |
| CHL                                                                      | 21.9        | 29.7           | 46.9          | 1.4   | 100.0 |
| COL                                                                      | 16.3        | 63.4           | 20.3          | 0.0   | 100.0 |
| CRI                                                                      | 41.5        | 1.0            | 57.5          | 0.0   | 100.0 |
| DOM                                                                      | 23.1        | 22.8           | 54.1          | 0.0   | 100.0 |
| PER                                                                      | 19.4        | 61.0           | 19.6          | 0.0   | 100.0 |
| PRY                                                                      | 85.2        | 1.1            | 13.7          | 0.0   | 100.0 |
| SLV                                                                      | 22.1        | 3.0            | 74.9          | 0.0   | 100.0 |
| URY                                                                      | 63.0        | 2.4            | 34.6          | 0.0   | 100.0 |
| All origins                                                              | 32.9        | 33.9           | 32.9          | 0.3   | 100.0 |
| <i>Panel B: foreign-MNE share of export value, %</i>                     |             |                |               |       |       |
| ARG                                                                      | 37.8        | 75.1           | 60.4          | –     | 48.9  |
| CHL                                                                      | 29.6        | 73.9           | 39.9          | 48.6  | 47.9  |
| COL                                                                      | 28.0        | 30.1           | 57.8          | 12.7  | 35.3  |
| CRI                                                                      | 44.5        | 46.9           | 70.9          | 100.0 | 59.7  |
| DOM                                                                      | 40.7        | 85.3           | 73.2          | –     | 68.4  |
| PER                                                                      | 33.0        | 52.2           | 30.6          | 78.0  | 44.2  |
| PRY                                                                      | 65.0        | 44.1           | 37.9          | –     | 61.1  |
| SLV                                                                      | 28.9        | 58.8           | 52.5          | 1.5   | 47.4  |
| URY                                                                      | 51.8        | 42.9           | 55.3          | –     | 52.8  |
| All origins                                                              | 38.1        | 49.2           | 51.6          | 47.9  | 46.3  |

*Notes:* Panel A: composition of each country's exports by sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs.

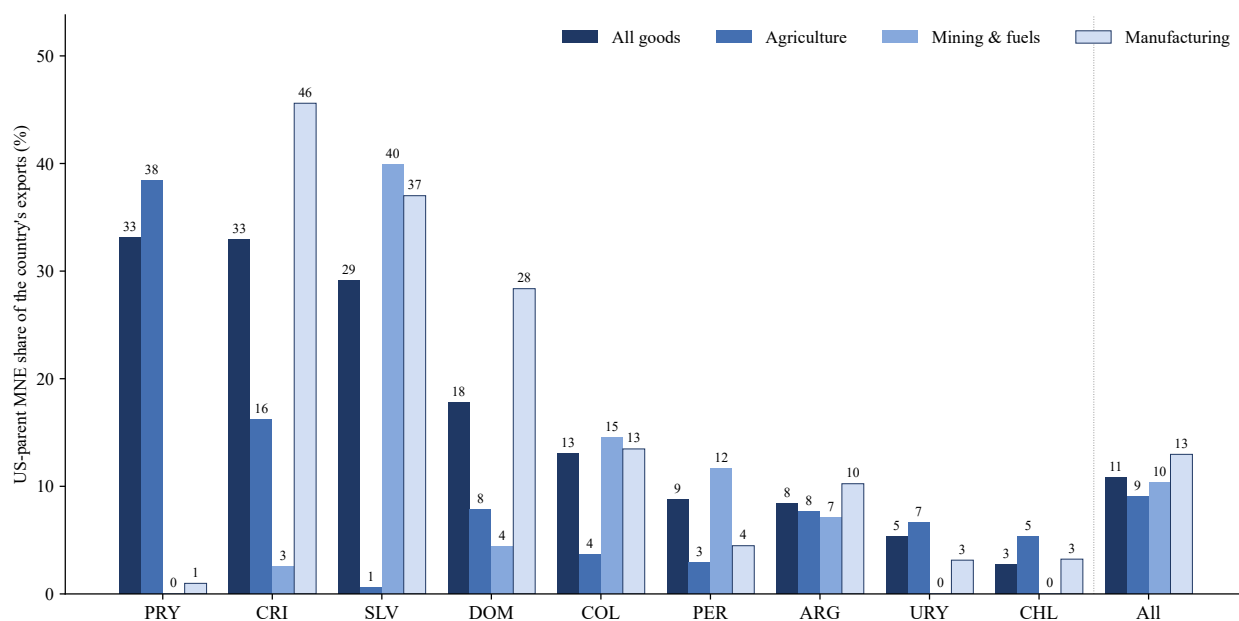
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 3:** Industry of Matched Exporting Affiliates

| NAICS sector of the matched exporting affiliate      | % of matched export value |
|------------------------------------------------------|---------------------------|
| goods-producing or unknown                           | 82.4                      |
| information, finance, professional, holdings (51–56) | 7.7                       |
| wholesale / retail trade (42–45)                     | 7.4                       |
| other services (61–92)                               | 1.8                       |
| transport, logistics (48–49)                         | 0.6                       |

*Notes:* Percent of matched export value, by NAICS sector of the exporting affiliate. Services are not observed in customs records; the table shows how much of goods exports is shipped by affiliates classified in service industries.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 3:** Exports by US-Parent Multinationals, by Exporting Country and Sector

*Notes:* Percent of each country's exports in each sector shipped by US-parent MNEs, to all destinations.

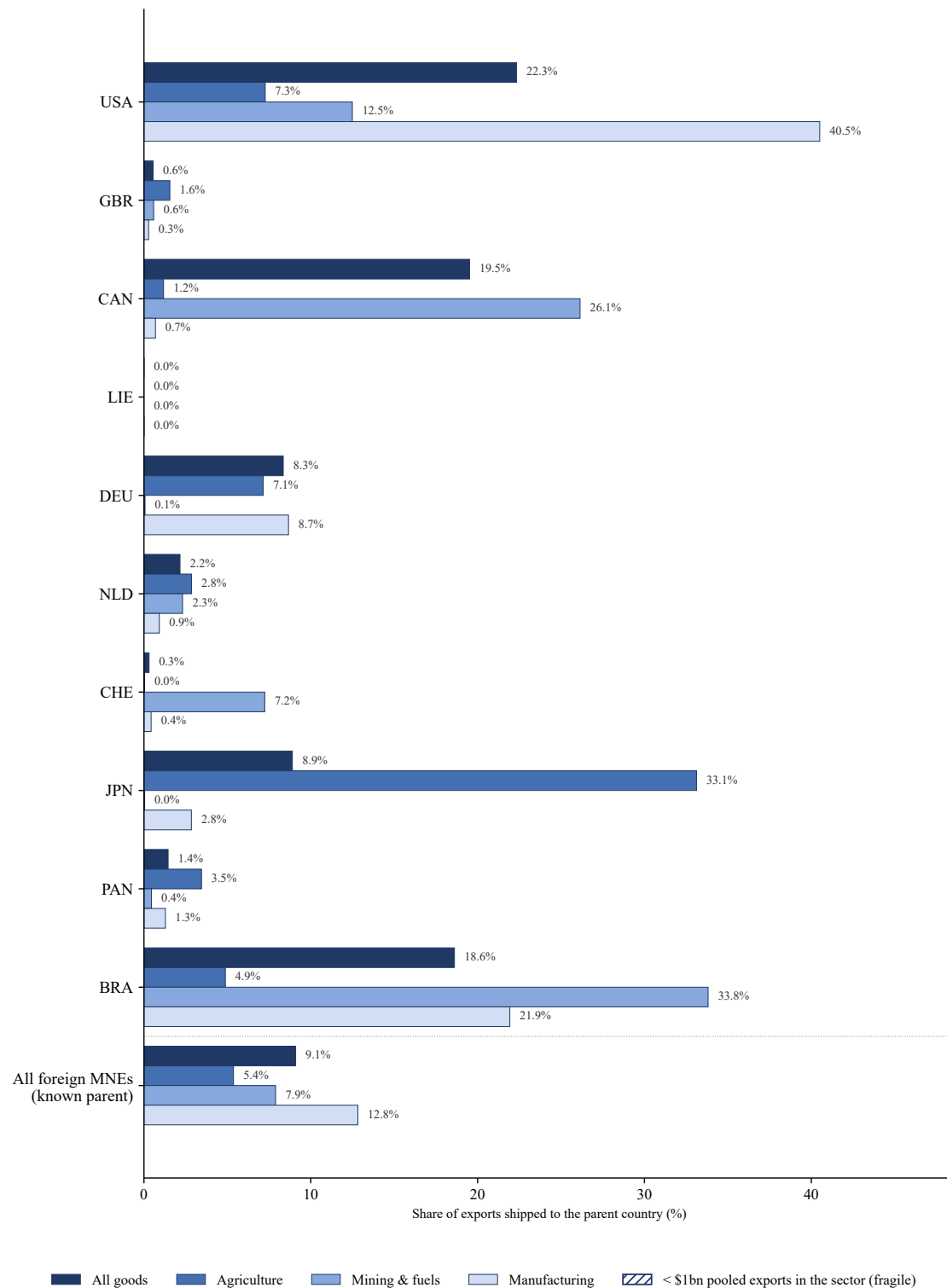
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 4:** Exports by US-Parent Multinationals, by Exporting Country and Sector

| Origin | US-parent MNE share of the origin's exports in the sector (%) |             |                |               | US-parent MNE exports (US\$bn/yr) |             |                |               |
|--------|---------------------------------------------------------------|-------------|----------------|---------------|-----------------------------------|-------------|----------------|---------------|
|        | All goods                                                     | Agriculture | Mining & fuels | Manufacturing | All goods                         | Agriculture | Mining & fuels | Manufacturing |
| PRY    | 33.1                                                          | 38.5        | 0.0            | 1.0           | 2.2                               | 2.2         | 0.0            | 0.0           |
| CRI    | 33.0                                                          | 16.2        | 2.6            | 45.6          | 3.2                               | 0.6         | 0.0            | 2.5           |
| SLV    | 29.1                                                          | 0.6         | 39.9           | 37.0          | 1.4                               | 0.0         | 0.1            | 1.3           |
| DOM    | 17.8                                                          | 7.9         | 4.4            | 28.4          | 1.5                               | 0.2         | 0.1            | 1.2           |
| COL    | 13.1                                                          | 3.7         | 14.6           | 13.5          | 5.1                               | 0.2         | 3.9            | 1.0           |
| PER    | 8.8                                                           | 2.9         | 11.7           | 4.5           | 3.2                               | 0.2         | 2.8            | 0.3           |
| ARG    | 8.4                                                           | 7.7         | 7.1            | 10.2          | 5.4                               | 2.7         | 0.5            | 2.2           |
| URY    | 5.3                                                           | 6.6         | 0.0            | 3.1           | 0.3                               | 0.3         | 0.0            | 0.1           |
| CHL    | 2.8                                                           | 5.4         | 0.1            | 3.2           | 0.9                               | 0.3         | 0.0            | 0.5           |
| All    | 10.8                                                          | 9.1         | 10.3           | 13.0          | 23.1                              | 6.7         | 7.3            | 9.1           |

*Notes:* Left panel: percent of each country's exports in the sector shipped by US-parent MNEs. Right panel: exports by US-parent MNEs. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 4: Share of Foreign-Multinational Exports Shipped to the Parent Country, by Sector**

*Notes:* Percent of each parent country's exports shipped to that same country, by sector. Hatched bars: the parent country's exports in the sector are below US\$1 billion over the period.

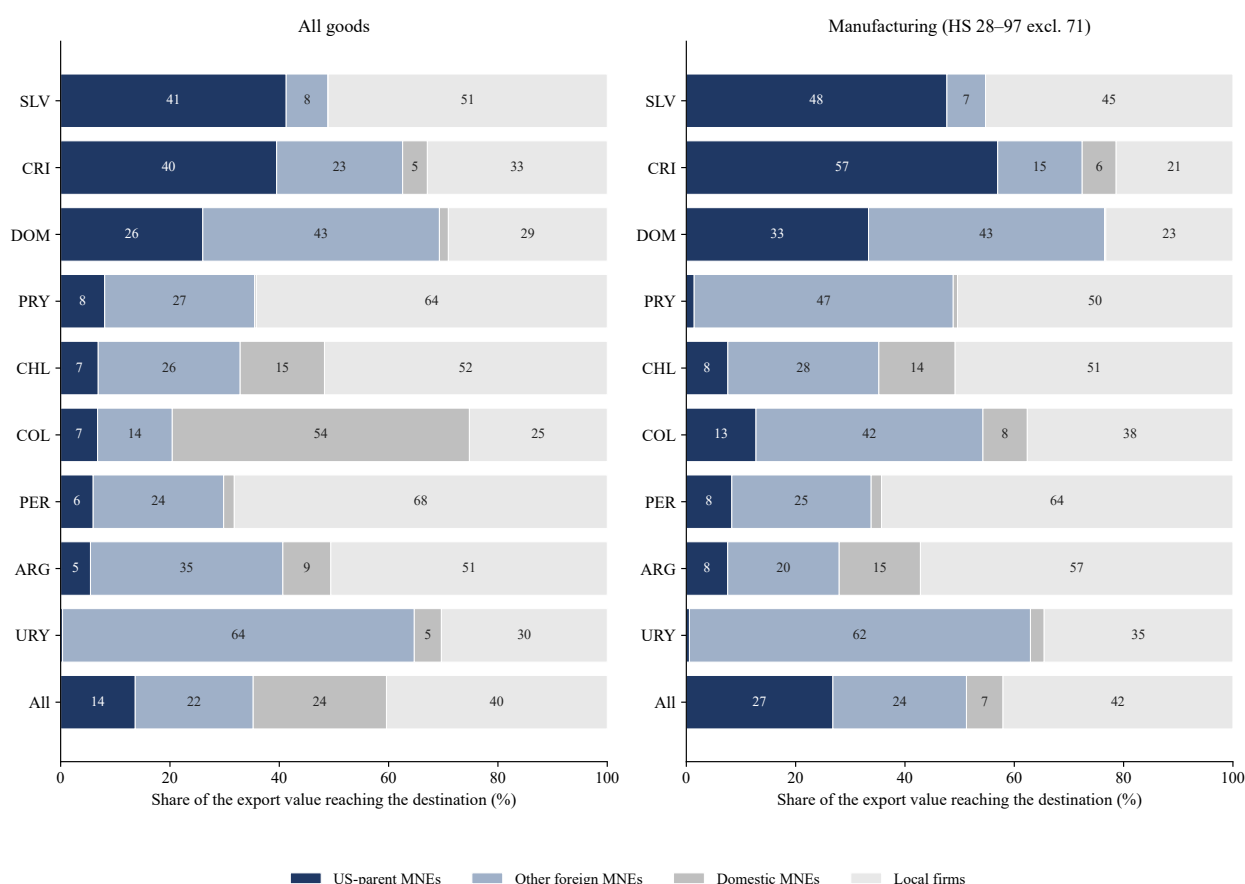
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 5:** Share of Foreign-Multinational Exports Shipped to the Parent Country, by Sector

| Parent                        | % of the parent's exports shipped to the parent's country |             |                |               | Parent's exports (US\$bn/yr) |             |                |               |
|-------------------------------|-----------------------------------------------------------|-------------|----------------|---------------|------------------------------|-------------|----------------|---------------|
|                               | All goods                                                 | Agriculture | Mining & fuels | Manufacturing | All goods                    | Agriculture | Mining & fuels | Manufacturing |
| USA                           | 22.3                                                      | 7.3         | 12.5           | 40.5          | 23.1                         | 6.7         | 7.3            | 9.1           |
| GBR                           | 0.6                                                       | 1.6         | 0.6            | 0.3           | 16.7                         | 0.7         | 11.9           | 4.2           |
| CAN                           | 19.5                                                      | 1.2         | 26.1           | 0.7           | 7.4                          | 0.4         | 5.7            | 1.3           |
| LIE                           | 0.0                                                       | 0.0         | 0.0            | 0.0           | 4.9                          | 2.2         | 2.0            | 0.7           |
| DEU                           | 8.3                                                       | 7.1         | 0.1            | 8.7           | 4.6                          | 0.8         | 0.0            | 3.7           |
| NLD                           | 2.2                                                       | 2.8         | 2.3            | 0.9           | 4.4                          | 2.7         | 0.1            | 1.5           |
| CHE                           | 0.3                                                       | 0.0         | 7.2            | 0.4           | 3.9                          | 3.1         | 0.1            | 0.7           |
| JPN                           | 8.9                                                       | 33.1        | 0.0            | 2.8           | 3.5                          | 0.6         | 0.0            | 2.9           |
| PAN                           | 1.4                                                       | 3.5         | 0.4            | 1.3           | 3.0                          | 0.6         | 1.0            | 1.3           |
| BRA                           | 18.6                                                      | 4.9         | 33.8           | 21.9          | 3.0                          | 1.1         | 0.7            | 1.3           |
| All known-parent foreign MNEs | 9.1                                                       | 5.4         | 7.9            | 12.8          | 97.6                         | 27.2        | 34.5           | 35.7          |

*Notes:* Left panel: percent of each parent country's exports shipped to that same country. Right panel: the parent country's exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Figure 5:** Exports to the United States, by Type of Exporter and Exporting Country

*Notes:* Percent of each country's exports to the United States, by type of exporter. Left: all goods. Right: manufacturing.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 6:** Exports to the United States, by Type of Exporter and Sector

| Sector         | Exports to the USA (US\$bn/yr) | US-parent MNEs (%) | Other foreign MNEs (%) | Domestic MNEs (%) | Local firms (%) |
|----------------|--------------------------------|--------------------|------------------------|-------------------|-----------------|
| All goods      | 40.0                           | 13.6               | 21.6                   | 24.4              | 40.4            |
| Agriculture    | 10.7                           | 4.8                | 28.0                   | 9.2               | 58.0            |
| Mining & fuels | 14.6                           | 6.6                | 14.8                   | 52.0              | 26.6            |
| Manufacturing  | 14.5                           | 26.8               | 24.4                   | 6.7               | 42.1            |
| Rest           | 0.1                            | 40.0               | 38.4                   | 4.5               | 17.1            |

*Notes:* Percent of each sector's exports to the United States, by type of exporter. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 7: Multinational Presence and Exports by Sector: Intensive Margin**

| Sample                                                                                                 | All goods             | Agriculture           | Mining & fuels        | Manufacturing         |
|--------------------------------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| <i>Panel A. Dependent variable: ln exports, all firms</i>                                              |                       |                       |                       |                       |
| Fixed effects: origin $\times$ year, destination $\times$ year, product                                |                       |                       |                       |                       |
| ln(# foreign MNEs)                                                                                     | 1.2129***<br>(0.0285) | 1.2954***<br>(0.0275) | 1.1301***<br>(0.1799) | 1.1644***<br>(0.0358) |
| ln(# domestic MNEs)                                                                                    | 0.4382***<br>(0.0259) | 0.5233***<br>(0.0266) | 0.3172<br>(0.2062)    | 0.3833***<br>(0.0319) |
| Observations                                                                                           | 113,422               | 25,437                | 1,417                 | 85,453                |
| Fixed effects: origin $\times$ destination $\times$ product, origin $\times$ destination $\times$ year |                       |                       |                       |                       |
| ln(# foreign MNEs)                                                                                     | 0.4506***<br>(0.0124) | 0.4625***<br>(0.0188) | 0.2479<br>(0.1778)    | 0.4435***<br>(0.0148) |
| ln(# domestic MNEs)                                                                                    | 0.2214***<br>(0.0151) | 0.2831***<br>(0.0234) | 0.3216*<br>(0.1639)   | 0.1904***<br>(0.0175) |
| Observations                                                                                           | 98,223                | 22,002                | 1,027                 | 73,318                |
| <i>Panel B. Dependent variable: ln exports, non-MNE firms</i>                                          |                       |                       |                       |                       |
| Fixed effects: origin $\times$ year, destination $\times$ year, product                                |                       |                       |                       |                       |
| ln(# foreign MNEs)                                                                                     | 1.1331***<br>(0.0375) | 1.2191***<br>(0.0363) | 0.9420***<br>(0.2641) | 1.0546***<br>(0.0493) |
| ln(# domestic MNEs)                                                                                    | 0.4707***<br>(0.0341) | 0.5676***<br>(0.0339) | -0.6594<br>(0.4830)   | 0.4194***<br>(0.0419) |
| Observations                                                                                           | 93,010                | 22,503                | 995                   | 68,534                |
| Fixed effects: origin $\times$ destination $\times$ product, origin $\times$ destination $\times$ year |                       |                       |                       |                       |
| ln(# foreign MNEs)                                                                                     | 0.2413***<br>(0.0217) | 0.3602***<br>(0.0292) | -0.1512<br>(0.3320)   | 0.1865***<br>(0.0250) |
| ln(# domestic MNEs)                                                                                    | 0.1250***<br>(0.0218) | 0.2461***<br>(0.0341) | -0.7273<br>(0.6536)   | 0.0742***<br>(0.0238) |
| Observations                                                                                           | 81,563                | 19,535                | 726                   | 59,772                |

Notes: Observations are origin–destination–HS6–year cells. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Regressors: ln number of foreign and of domestic MNEs; cells with at least one of each. Blank: too few observations. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 8: Distance and Firm Exports, by Sector**

| Dep. var.: ln firm exports                                     | All goods              | Agriculture            | Mining & fuels        | Manufacturing          |
|----------------------------------------------------------------|------------------------|------------------------|-----------------------|------------------------|
| Origin, destination, year and product FE                       |                        |                        |                       |                        |
| ln distance                                                    | -0.1645***<br>(0.0454) | -0.2135***<br>(0.0748) | -0.0926<br>(0.0819)   | -0.1510***<br>(0.0454) |
| ln distance $\times$ foreign MNE                               | 0.0487***<br>(0.0041)  | 0.0366***<br>(0.0034)  | 0.0566***<br>(0.0086) | 0.0515***<br>(0.0048)  |
| ln distance $\times$ domestic MNE                              | 0.0318***<br>(0.0042)  | 0.0187***<br>(0.0046)  | 0.0706***<br>(0.0180) | 0.0388***<br>(0.0052)  |
| Observations                                                   | 5,081,547              | 1,045,009              | 79,938                | 3,917,960              |
| Origin $\times$ year, destination $\times$ year and product FE |                        |                        |                       |                        |
| ln distance                                                    | -0.1853***<br>(0.0460) | -0.2285***<br>(0.0711) | -0.1001<br>(0.0828)   | -0.1742***<br>(0.0459) |
| ln distance $\times$ foreign MNE                               | 0.0483***<br>(0.0040)  | 0.0358***<br>(0.0034)  | 0.0557***<br>(0.0087) | 0.0512***<br>(0.0048)  |
| ln distance $\times$ domestic MNE                              | 0.0318***<br>(0.0042)  | 0.0182***<br>(0.0046)  | 0.0673***<br>(0.0183) | 0.0390***<br>(0.0052)  |
| Observations                                                   | 5,081,360              | 1,044,787              | 79,612                | 3,917,765              |

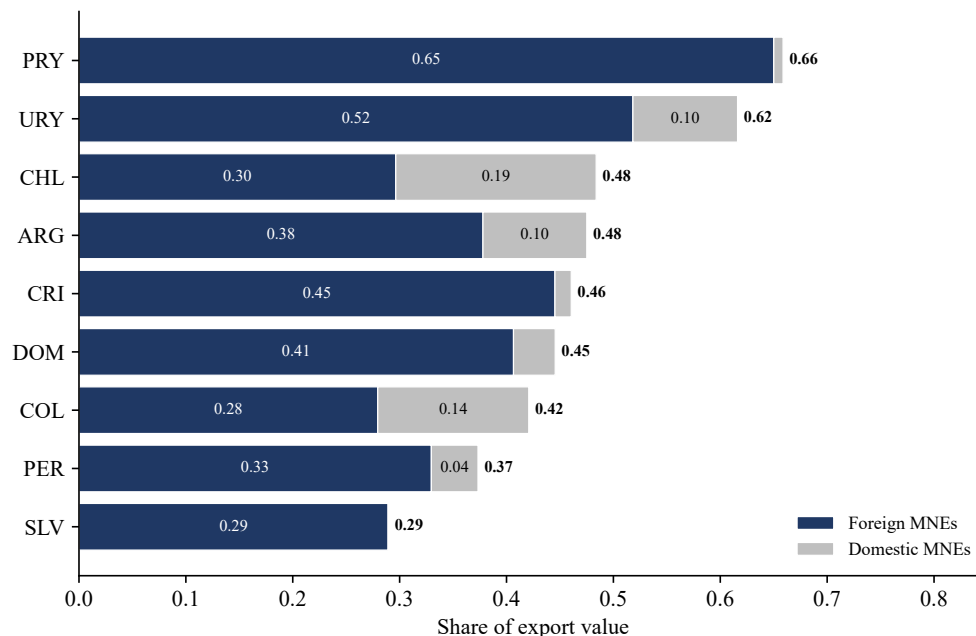
Notes: Observations are firm–destination–HS6–year cells. Dependent variable: ln exports. Non-MNE firms are the omitted group. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## Part I

# Agriculture

## 2 Fact 1: Multinationals Account for a Large Share of Exports



**Figure 6:** Agriculture: Multinational Share of Export Value, by Exporting Country

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. The figure at the end of each bar is the total MNE share.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

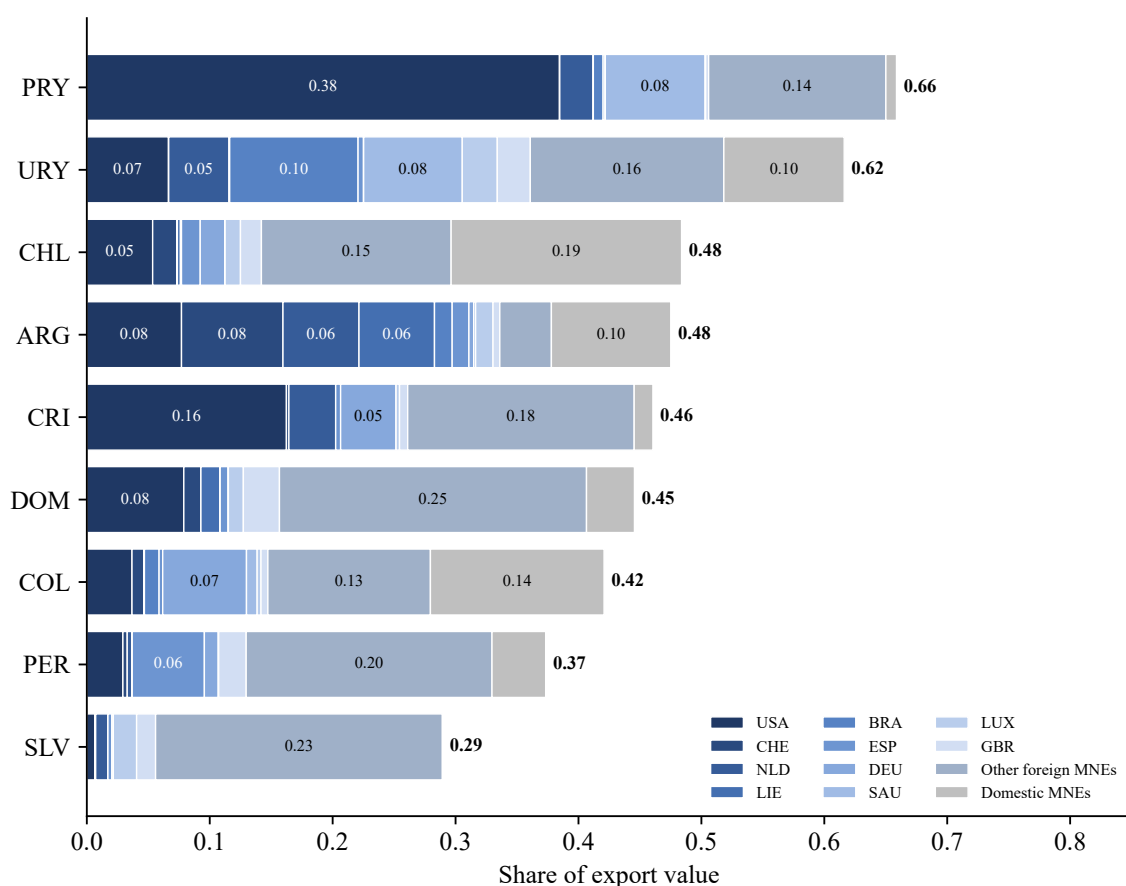
**Table 9:** Agriculture: Multinational Share of Export Value, by Exporting Country

| Origin | Foreign | Domestic | All MNEs | Value (US\$bn/yr) |
|--------|---------|----------|----------|-------------------|
| PRY    | 0.650   | 0.009    | 0.659    | 6.2               |
| URY    | 0.518   | 0.098    | 0.616    | 4.4               |
| CHL    | 0.296   | 0.187    | 0.484    | 7.5               |
| ARG    | 0.378   | 0.097    | 0.475    | 38.1              |
| CRI    | 0.445   | 0.015    | 0.461    | 4.3               |
| DOM    | 0.407   | 0.039    | 0.446    | 2.0               |
| COL    | 0.280   | 0.141    | 0.421    | 7.2               |
| PER    | 0.330   | 0.044    | 0.373    | 8.1               |
| SLV    | 0.289   | 0.000    | 0.289    | 1.1               |
| All    | 0.381   | 0.095    | 0.476    | 78.9              |

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 2.1 By Parent Country



**Figure 7:** Agriculture: Multinational Share of Export Value, by Exporting Country and Parent Country

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. Foreign MNEs split by country of the ultimate parent; the ten largest parent countries are shown separately.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 10:** Agriculture: Multinational Share of Export Value, by Exporting Country and Parent Country

| Origin | USA   | CHE   | NLD   | LIE   | BRA   | ESP   | DEU   | SAU   | LUX   | GBR   | Other | Domestic | All MNEs | Value (US\$bn/yr) |
|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|----------|-------------------|
| PRY    | 0.385 | 0.000 | 0.027 | 0.000 | 0.008 | 0.002 | 0.000 | 0.081 | 0.000 | 0.002 | 0.144 | 0.009    | 0.659    | 6.2               |
| URY    | 0.066 | 0.000 | 0.049 | 0.001 | 0.104 | 0.004 | 0.000 | 0.080 | 0.028 | 0.027 | 0.157 | 0.098    | 0.616    | 4.4               |
| CHL    | 0.054 | 0.020 | 0.000 | 0.002 | 0.001 | 0.015 | 0.020 | 0.000 | 0.012 | 0.017 | 0.154 | 0.187    | 0.484    | 7.5               |
| ARG    | 0.077 | 0.083 | 0.062 | 0.061 | 0.014 | 0.014 | 0.004 | 0.002 | 0.014 | 0.006 | 0.042 | 0.097    | 0.475    | 38.1              |
| CRI    | 0.162 | 0.002 | 0.038 | 0.000 | 0.000 | 0.004 | 0.045 | 0.000 | 0.003 | 0.007 | 0.184 | 0.015    | 0.461    | 4.3               |
| DOM    | 0.079 | 0.014 | 0.000 | 0.016 | 0.000 | 0.007 | 0.000 | 0.000 | 0.012 | 0.029 | 0.250 | 0.039    | 0.446    | 2.0               |
| COL    | 0.037 | 0.010 | 0.000 | 0.000 | 0.012 | 0.003 | 0.068 | 0.009 | 0.003 | 0.006 | 0.132 | 0.141    | 0.421    | 7.2               |
| PER    | 0.029 | 0.004 | 0.004 | 0.000 | 0.000 | 0.059 | 0.011 | 0.000 | 0.000 | 0.022 | 0.200 | 0.044    | 0.373    | 8.1               |
| SLV    | 0.006 | 0.001 | 0.010 | 0.000 | 0.000 | 0.003 | 0.001 | 0.000 | 0.019 | 0.015 | 0.233 | 0.000    | 0.289    | 1.1               |

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. Other: all remaining foreign parents. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



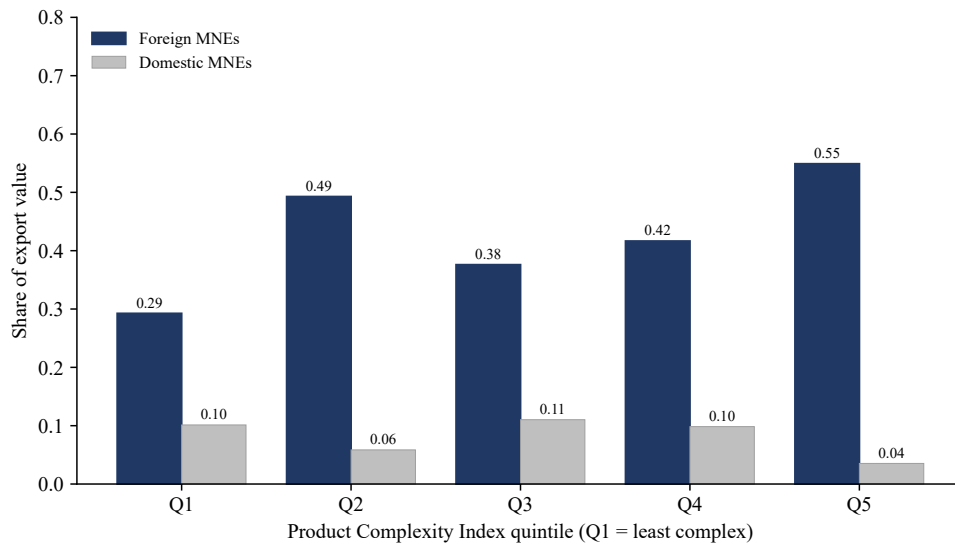
**Figure 8:** Agriculture: Export Value by Exporting Country and Parent Country

*Notes:* Percent of each country's agricultural export value, by country of the ultimate parent of the exporter.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

### 3 Fact 2: Foreign Multinationals Specialize in Complex Products, Domestic Multinationals in Primary Goods

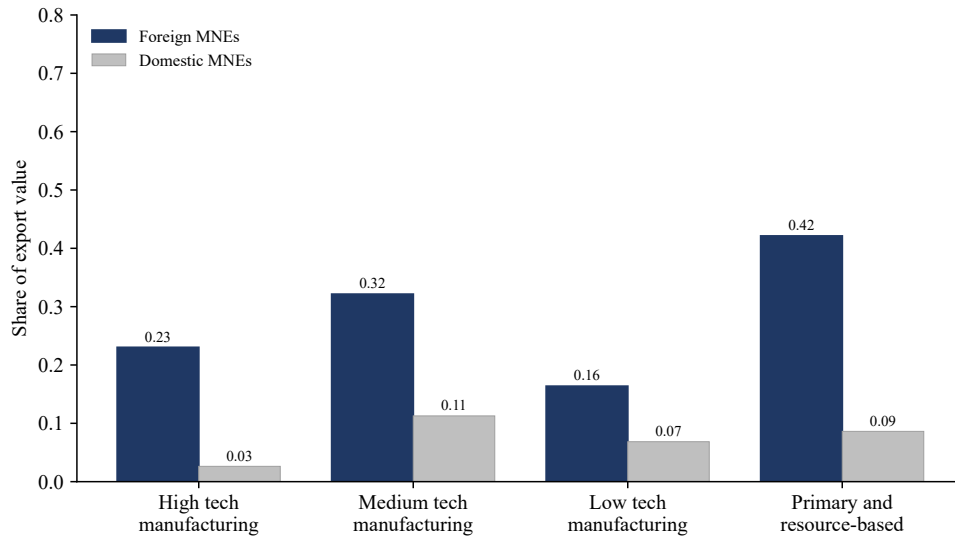
#### 3.1 Product Complexity and Technology



**Figure 9:** Agriculture: Multinational Share of Export Value, by Product Complexity Quintile

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. Quintiles of the Hausmann–Hidalgo Product Complexity Index across HS6 products (Q1 = least complex).

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

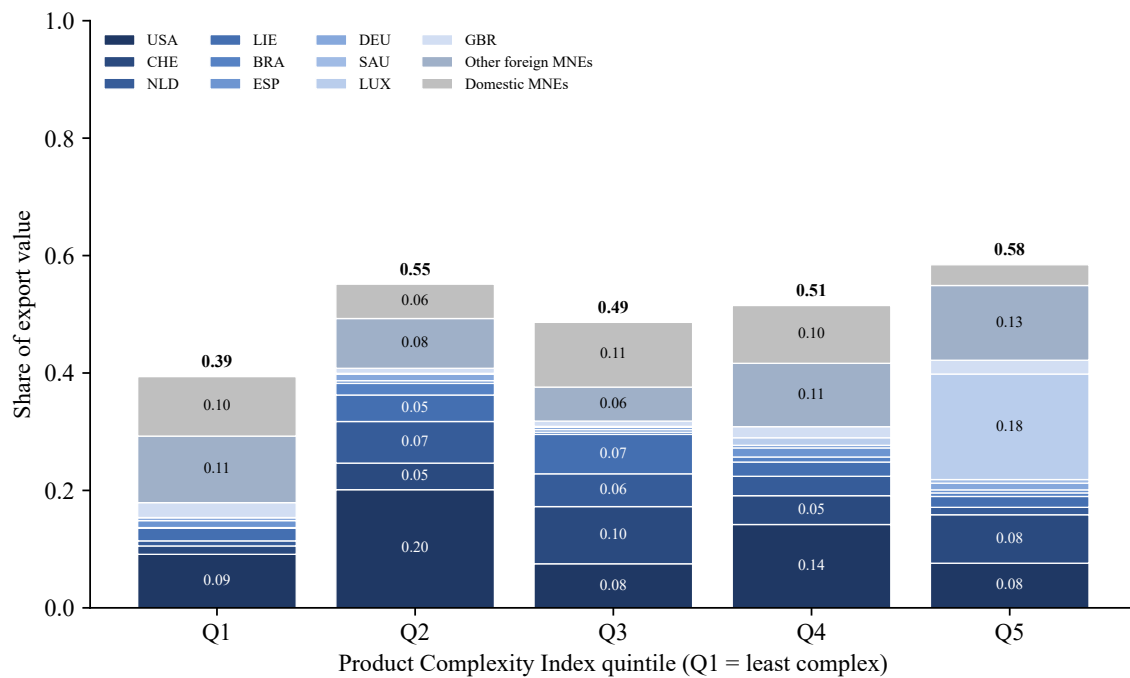


**Figure 10:** Agriculture: Multinational Share of Export Value, by Technology Class

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. Lall (2000) technology classes: primary and resource-based products, and low-, medium- and high-technology manufactures.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

### 3.2 By Parent Country



**Figure 11:** Agriculture: Multinational Share of Export Value, by Product Complexity Quintile and Parent Country

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. Foreign MNEs split by country of the ultimate parent. Quintiles of the Hausmann–Hidalgo Product Complexity Index across HS6 products (Q1 = least complex).

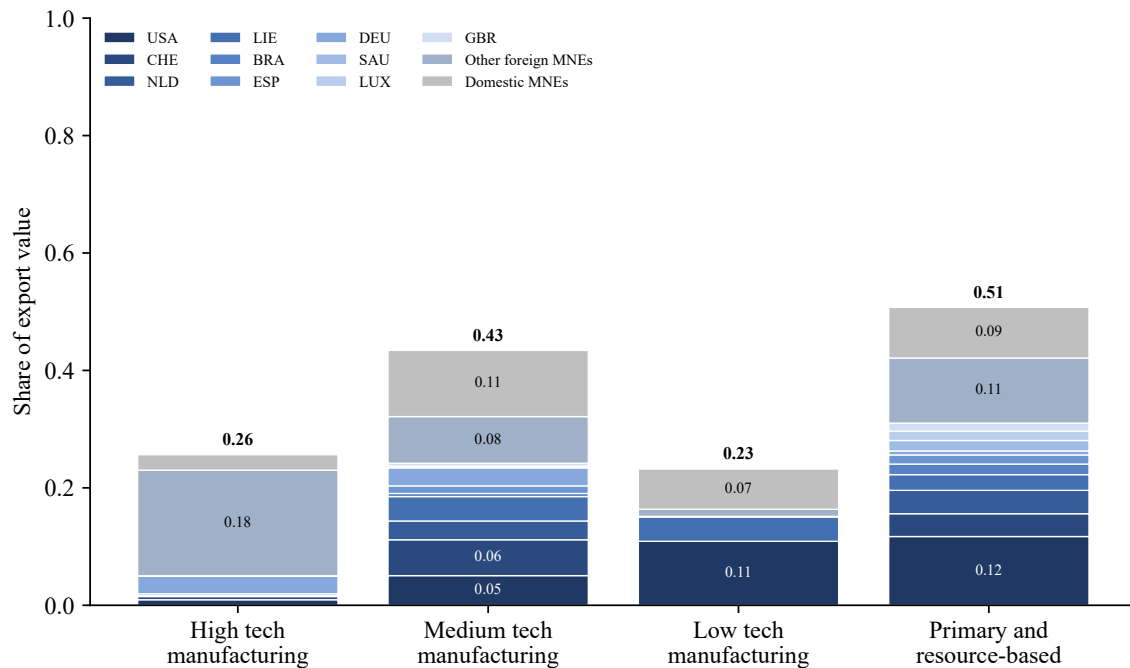
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 11: Agriculture: Multinational Share of Export Value, by Product Complexity Quintile and Parent Country**

| PCI quintile | USA   | CHE   | NLD   | LIE   | BRA   | ESP   | DEU   | SAU   | LUX   | GBR   | Other | Domestic | All MNEs | Value (US\$bn/yr) |
|--------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|----------|-------------------|
| Q1           | 0.091 | 0.014 | 0.009 | 0.022 | 0.001 | 0.012 | 0.005 | 0.000 | 0.000 | 0.025 | 0.113 | 0.101    | 0.394    | 4.5               |
| Q2           | 0.201 | 0.045 | 0.071 | 0.045 | 0.020 | 0.005 | 0.011 | 0.001 | 0.001 | 0.009 | 0.085 | 0.059    | 0.551    | 15.8              |
| Q3           | 0.075 | 0.097 | 0.056 | 0.067 | 0.004 | 0.004 | 0.004 | 0.001 | 0.000 | 0.009 | 0.058 | 0.110    | 0.486    | 17.8              |
| Q4           | 0.142 | 0.049 | 0.033 | 0.024 | 0.009 | 0.015 | 0.004 | 0.000 | 0.013 | 0.019 | 0.108 | 0.098    | 0.515    | 8.3               |
| Q5           | 0.076 | 0.083 | 0.013 | 0.019 | 0.006 | 0.005 | 0.011 | 0.006 | 0.180 | 0.024 | 0.127 | 0.035    | 0.584    | 2.4               |

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. Other: all remaining foreign parents. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Figure 12: Agriculture: Multinational Share of Export Value, by Technology Class and Parent Country**

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. Foreign MNEs split by country of the ultimate parent. Lall (2000) technology classes: primary and resource-based products, and low-, medium- and high-technology manufactures.

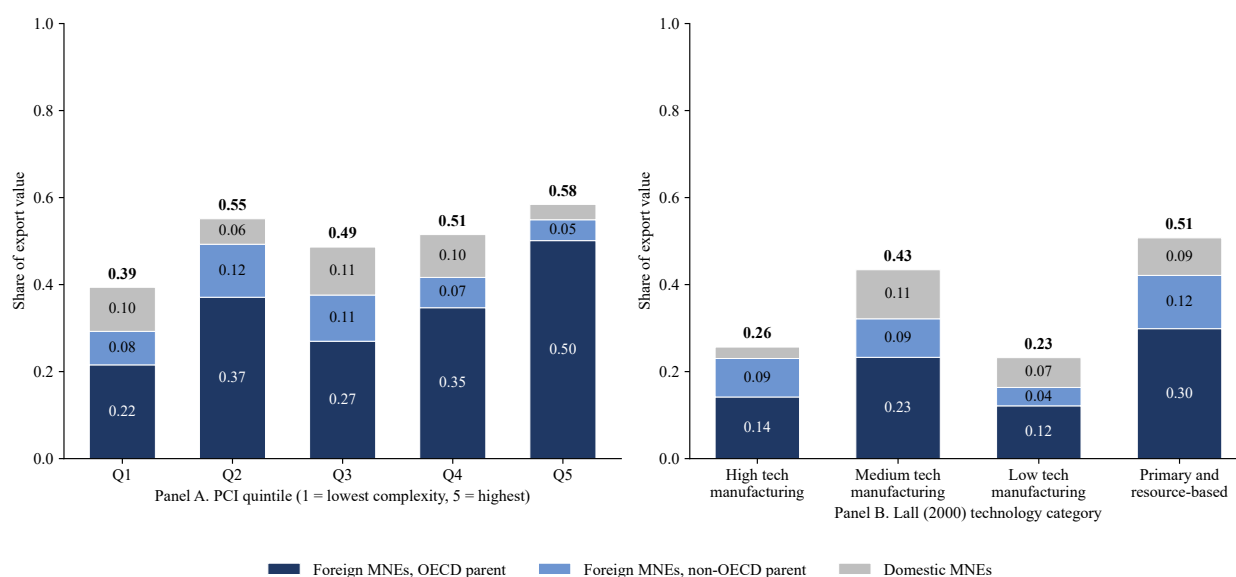
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 12: Agriculture: Multinational Share of Export Value, by Technology Class and Parent Country**

| Technology category        | USA   | CHE   | NLD   | LIE   | BRA   | ESP   | DEU   | SAU   | LUX   | GBR   | Other | Domestic | All MNEs | Value (US\$bn/yr) |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|----------|-------------------|
| High tech manufacturing    | 0.009 | 0.005 | 0.001 | 0.000 | 0.001 | 0.003 | 0.030 | 0.000 | 0.000 | 0.001 | 0.180 | 0.026    | 0.256    | 0.7               |
| Medium tech manufacturing  | 0.050 | 0.061 | 0.032 | 0.042 | 0.005 | 0.013 | 0.031 | 0.001 | 0.002 | 0.005 | 0.079 | 0.113    | 0.434    | 23.0              |
| Low tech manufacturing     | 0.109 | 0.000 | 0.000 | 0.042 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.012 | 0.068    | 0.232    | 0.7               |
| Primary and resource-based | 0.117 | 0.039 | 0.040 | 0.027 | 0.018 | 0.015 | 0.007 | 0.018 | 0.016 | 0.014 | 0.111 | 0.086    | 0.507    | 48.5              |

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. Other: all remaining foreign parents. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

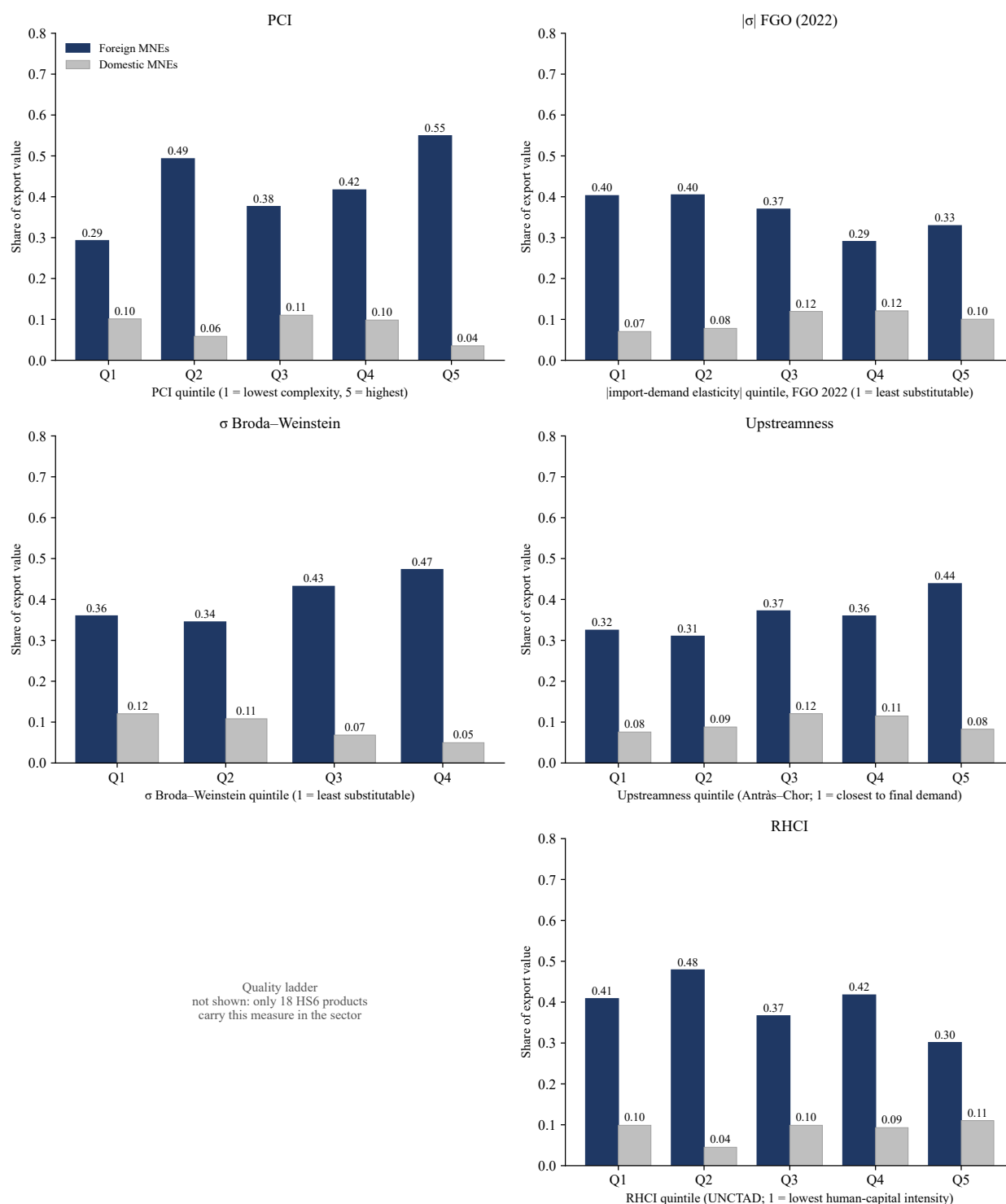


**Figure 13: Agriculture: Multinational Share of Export Value, by Product Sophistication and Location of the Parent**

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. Foreign MNEs are split by whether the ultimate parent is in an OECD member country. Panel A: Quintiles of the Hausmann–Hidalgo Product Complexity Index across HS6 products (Q1 = least complex). Panel B: Lall (2000) technology classes: primary and resource-based products, and low-, medium- and high-technology manufactures.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

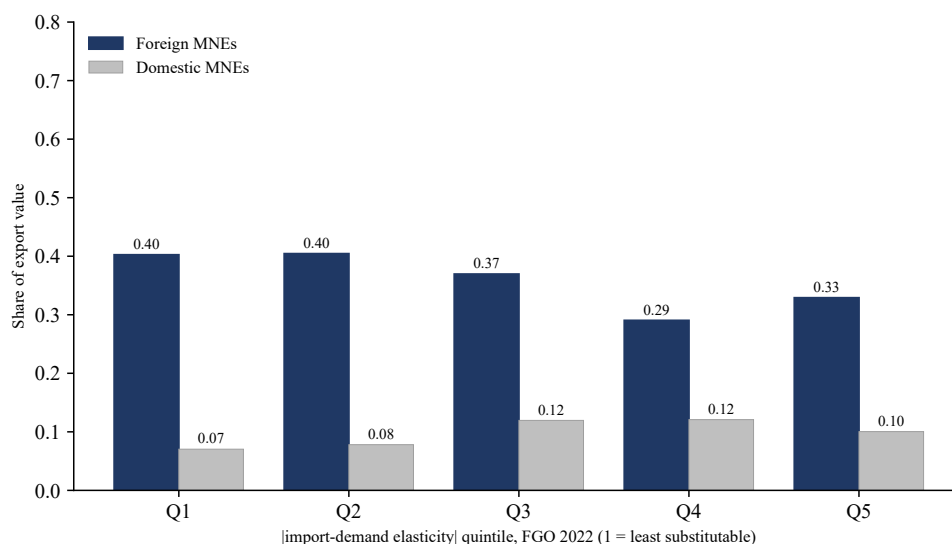
### 3.3 Other Measures of Product Sophistication



**Figure 14:** Agriculture: Multinational Share of Export Value, by Quintile of Six Sophistication Measures

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. Quintiles computed over HS6 products. The six measures are the Product Complexity Index, the absolute import-demand elasticity of Fontagné, Guimbard and Orefice (2022), the elasticity of substitution of Broda and Weinstein, upstreamness, the quality ladder and the revealed human-capital intensity of UNCTAD.

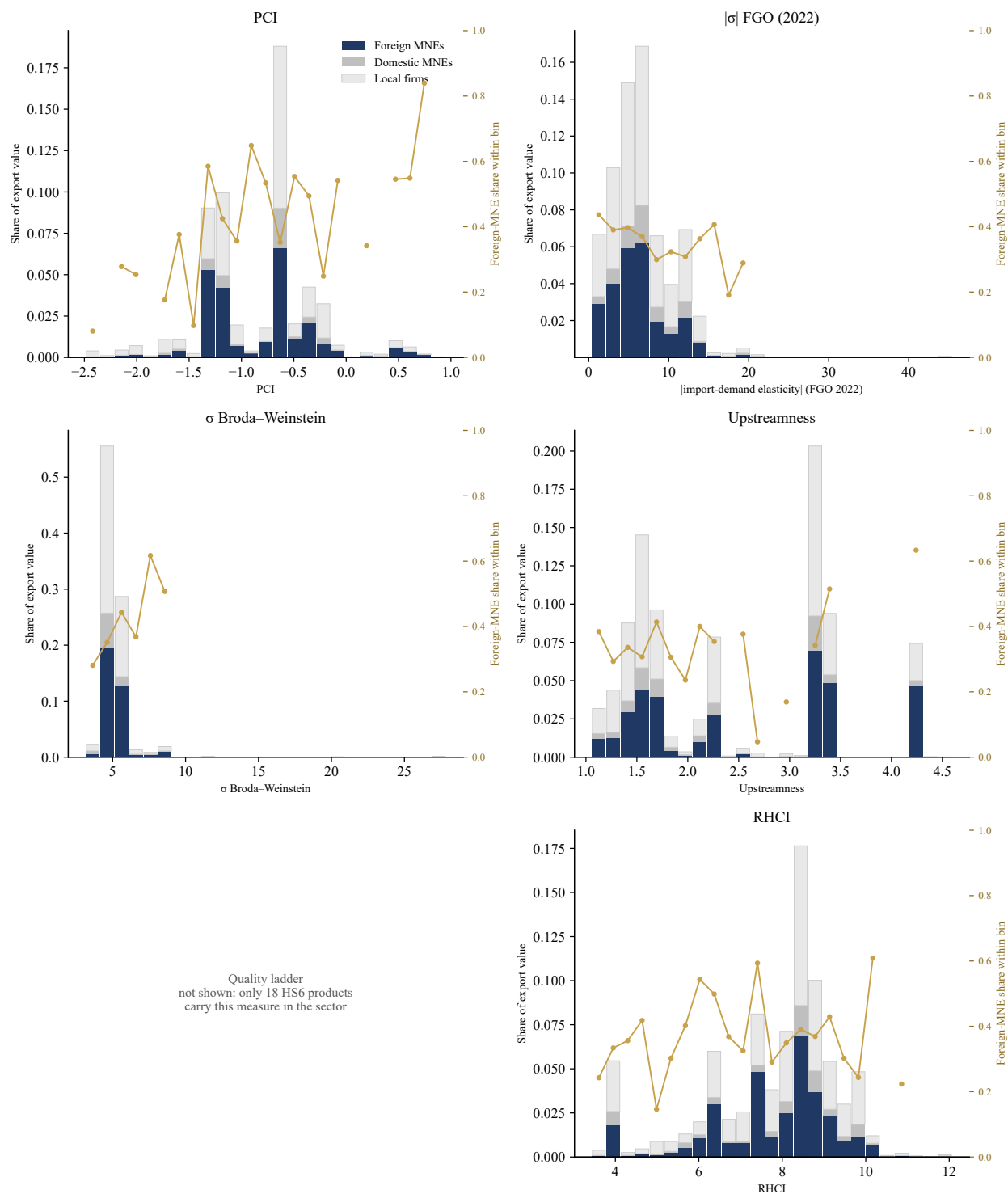
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 15:** Agriculture: Multinational Share of Export Value, by Quintile of the Import-Demand Elasticity

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. Quintiles of the absolute import-demand elasticity of Fontagné, Guimbard and Orefice (2022) across HS6 products; higher quintiles are more substitutable products.

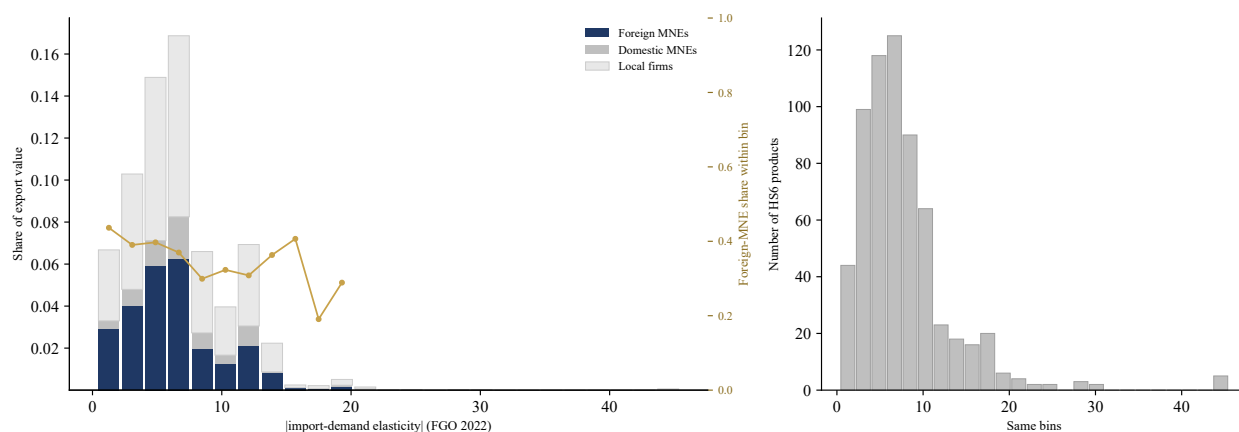
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 16:** Agriculture: Distribution of Export Value across Six Sophistication Measures

*Notes:* Bars: agricultural export value (HS 01–24) in each bin, by type of exporter. Line (right axis): foreign-MNE share of each bin. The six measures are the Product Complexity Index, the absolute import-demand elasticity of Fontagné, Guimbard and Orefice (2022), the elasticity of substitution of Broda and Weinstein, upstreamness, the quality ladder and the revealed human-capital intensity of UNCTAD.

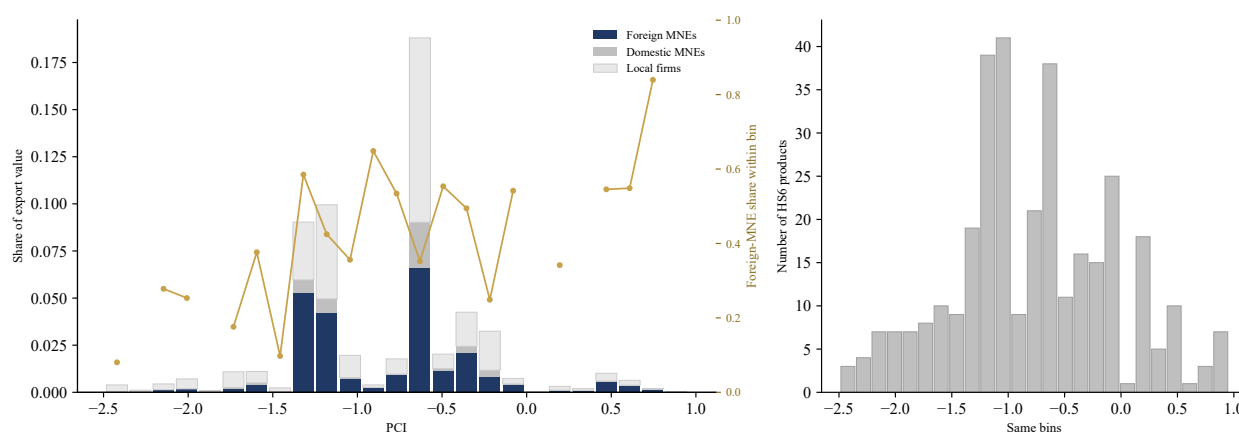
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 17:** Agriculture: Distribution of Export Value across the Import-Demand Elasticity

*Notes:* Left: export value in each bin, by type of exporter, and the foreign-MNE share of each bin. Right: number of HS6 products in each bin.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 18:** Agriculture: Distribution of Export Value across the Product Complexity Index

*Notes:* Left: export value in each bin, by type of exporter, and the foreign-MNE share of each bin. Right: number of HS6 products in each bin.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 13:** Agriculture: Multinational Share of Export Value, by Quintile of Six Sophistication Measures

| Measure                  | Foreign MNE share |      |      |      |      | Domestic MNE share |      |      |      |      |
|--------------------------|-------------------|------|------|------|------|--------------------|------|------|------|------|
|                          | Q1                | Q2   | Q3   | Q4   | Q5   | Q1                 | Q2   | Q3   | Q4   | Q5   |
| PCI                      | 0.29              | 0.49 | 0.38 | 0.42 | 0.55 | 0.10               | 0.06 | 0.11 | 0.10 | 0.04 |
| $ \sigma $ FGO (2022)    | 0.40              | 0.40 | 0.37 | 0.29 | 0.33 | 0.07               | 0.08 | 0.12 | 0.12 | 0.10 |
| $\sigma$ Broda–Weinstein | 0.36              | 0.34 | 0.43 | 0.47 | –    | 0.12               | 0.11 | 0.07 | 0.05 | –    |
| Upstreamness             | 0.32              | 0.31 | 0.37 | 0.36 | 0.44 | 0.08               | 0.09 | 0.12 | 0.11 | 0.08 |
| Quality ladder           | –                 | –    | –    | –    | –    | –                  | –    | –    | –    | –    |
| RHCI                     | 0.41              | 0.48 | 0.37 | 0.42 | 0.30 | 0.10               | 0.04 | 0.10 | 0.09 | 0.11 |

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. Quintiles computed over HS6 products. –: fewer than 30 HS6 products carry the measure.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 14:** Agriculture: Correlations between Sophistication Measures

|                          | PCI   | $ \sigma $ FGO (2022) | $\sigma$ Broda–Weinstein | Upstreamness | Quality ladder | RHCI  |
|--------------------------|-------|-----------------------|--------------------------|--------------|----------------|-------|
| PCI                      | 1.00  | -0.12                 | -0.16                    | -0.21        | –              | 0.72  |
| $ \sigma $ FGO (2022)    | -0.12 | 1.00                  | 0.01                     | 0.13         | –              | 0.03  |
| $\sigma$ Broda–Weinstein | -0.16 | 0.01                  | 1.00                     | 0.08         | –              | -0.09 |
| Upstreamness             | -0.21 | 0.13                  | 0.08                     | 1.00         | –              | 0.01  |
| Quality ladder           | –     | –                     | –                        | –            | –              | –     |
| RHCI                     | 0.72  | 0.03                  | -0.09                    | 0.01         | –              | 1.00  |

Notes: Correlations across HS6 products, weighted by export value.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

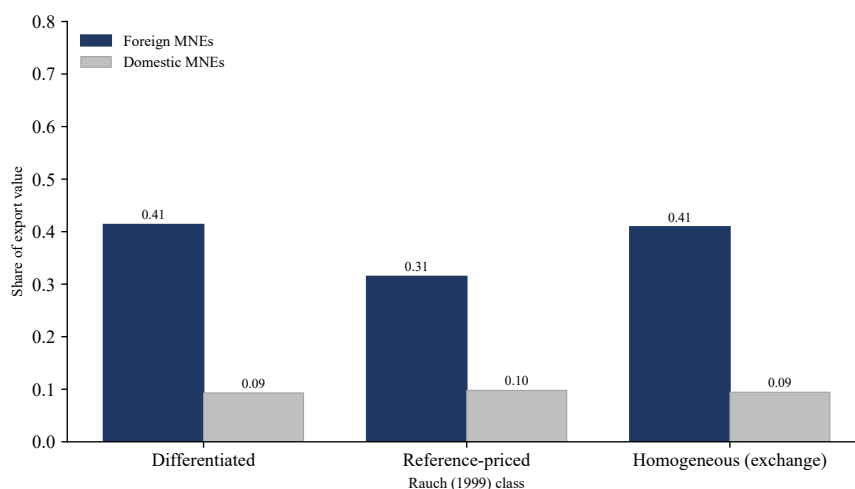
### 3.4 Rauch Classes and End Use

**Table 15:** Agriculture: Multinational Share of Export Value, by Rauch Product Class

| Rauch (1999) class     | Foreign | Domestic | All MNEs | HS6 products | Value (US\$bn/yr) |
|------------------------|---------|----------|----------|--------------|-------------------|
| Differentiated         | 0.413   | 0.093    | 0.506    | 167          | 7.8               |
| Reference-priced       | 0.315   | 0.098    | 0.413    | 357          | 20.1              |
| Homogeneous (exchange) | 0.409   | 0.094    | 0.503    | 164          | 43.7              |

Notes: Shares of agricultural export value (HS 01–24), all available years pooled. Rauch (1999) conservative classification: homogeneous goods are traded on organized exchanges, reference-priced goods have a price quoted in trade publications, and differentiated goods have neither. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Figure 19:** Agriculture: Multinational Share of Export Value, by Rauch Product Class

Notes: Shares of agricultural export value (HS 01–24), all available years pooled. Rauch (1999) conservative classification: homogeneous goods are traded on organized exchanges, reference-priced goods have a price quoted in trade publications, and differentiated goods have neither.

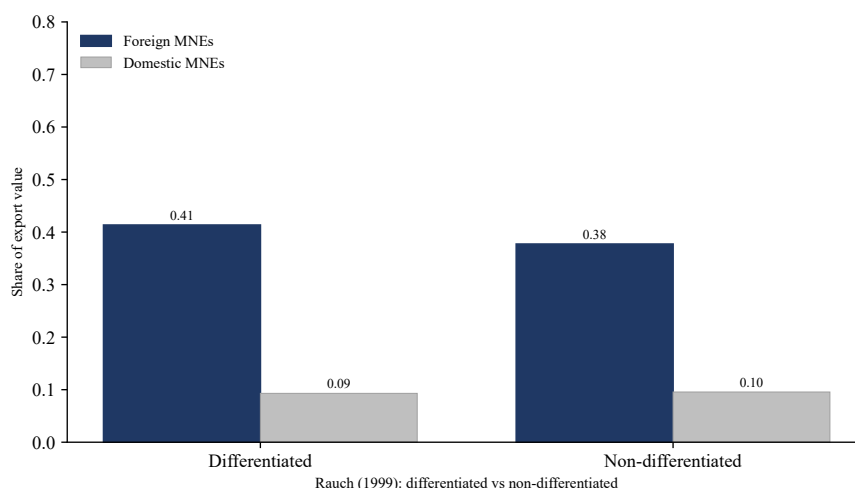
Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 16:** Agriculture: Multinational Share of Export Value: Differentiated and Other Products

| Rauch (1999): differentiated vs non-differentiated | Foreign | Domestic | All MNEs | HS6 products | Value (US\$bn/yr) |
|----------------------------------------------------|---------|----------|----------|--------------|-------------------|
| Differentiated                                     | 0.413   | 0.093    | 0.506    | 167          | 7.8               |
| Non-differentiated                                 | 0.377   | 0.096    | 0.473    | 521          | 63.8              |

Notes: Shares of agricultural export value (HS 01–24), all available years pooled. Rauch (1999) conservative classification; other products are reference-priced and homogeneous goods. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 20:** Agriculture: Multinational Share of Export Value: Differentiated and Other Products

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. Rauch (1999) conservative classification; other products are reference-priced and homogeneous goods.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 17:** Agriculture: Multinational Share of Export Value, by End Use

| BEC end use  | Foreign | Domestic | All MNEs | HS6 products | Value (US\$bn/yr) |
|--------------|---------|----------|----------|--------------|-------------------|
| Intermediate | 0.411   | 0.090    | 0.501    | 243          | 42.5              |
| Consumption  | 0.351   | 0.099    | 0.450    | 456          | 30.4              |
| Capital      | 0.168   | 0.020    | 0.187    | 3            | 0.0               |

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. End use from BEC Rev. 4; HS6 products without a BEC correspondence are excluded. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

### 3.5 Regressions

**Table 18:** Agriculture: Product Characteristics and Multinational Shares

|                                                            | (1)                    | (2)                    | (3)                    |
|------------------------------------------------------------|------------------------|------------------------|------------------------|
| <i>Panel A. Dependent variable: share of all MNEs</i>      |                        |                        |                        |
| Complexity (PCI)                                           | 0.0838***<br>(0.0067)  | 0.0831***<br>(0.0065)  | 0.0763***<br>(0.0067)  |
| $ \sigma $ FGO (per s.d.)                                  | -0.0225***<br>(0.0038) | -0.0231***<br>(0.0036) | -0.0268***<br>(0.0036) |
| Upstreamness                                               | 0.0129**<br>(0.0055)   | 0.0112**<br>(0.0052)   | 0.0086<br>(0.0053)     |
| <i>Panel B. Dependent variable: share of foreign MNEs</i>  |                        |                        |                        |
| Complexity (PCI)                                           | 0.0757***<br>(0.0070)  | 0.0771***<br>(0.0069)  | 0.0737***<br>(0.0071)  |
| $ \sigma $ FGO (per s.d.)                                  | -0.0499***<br>(0.0037) | -0.0499***<br>(0.0035) | -0.0539***<br>(0.0035) |
| Upstreamness                                               | 0.0241***<br>(0.0055)  | 0.0230***<br>(0.0053)  | 0.0141***<br>(0.0054)  |
| <i>Panel C. Dependent variable: share of domestic MNEs</i> |                        |                        |                        |
| Complexity (PCI)                                           | 0.0081**<br>(0.0039)   | 0.0061<br>(0.0038)     | 0.0026<br>(0.0038)     |
| $ \sigma $ FGO (per s.d.)                                  | 0.0273***<br>(0.0020)  | 0.0268***<br>(0.0018)  | 0.0271***<br>(0.0018)  |
| Upstreamness                                               | -0.0112***<br>(0.0027) | -0.0118***<br>(0.0026) | -0.0055**<br>(0.0023)  |
| Origin FE                                                  | ✓                      |                        |                        |
| Destination FE                                             | ✓                      |                        |                        |
| Year FE                                                    | ✓                      |                        |                        |
| Origin $\times$ year FE                                    |                        | ✓                      |                        |
| Destination $\times$ year FE                               |                        | ✓                      |                        |
| Origin $\times$ dest. $\times$ year FE                     |                        |                        | ✓                      |
| Observations                                               | 150,776                | 150,391                | 148,224                |

*Notes:* Observations are origin–destination–HS6–year cells. Agriculture products only. Dependent variable: share of the cell's export value shipped by each type of MNE. The import-demand elasticity is standardized (mean 0, standard deviation 1). Regressions weighted by the cell's export value. Robust standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 19: Agriculture: Product Sophistication and Multinational Shares**

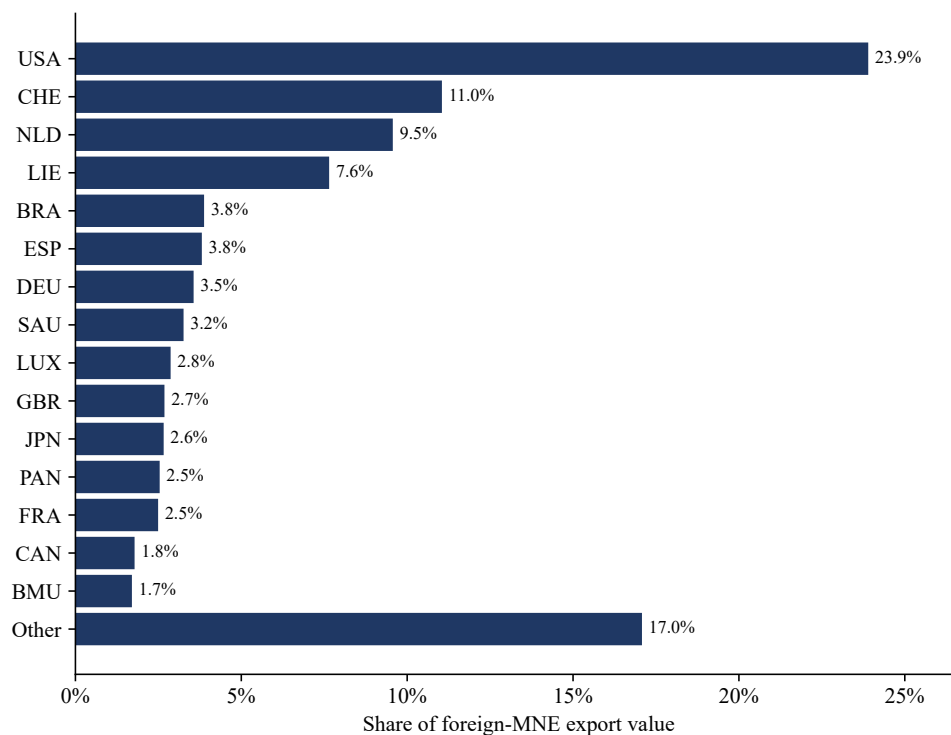
|                                                            | PCI                   |                       | Lall                   |                        | Rauch                  |                        |
|------------------------------------------------------------|-----------------------|-----------------------|------------------------|------------------------|------------------------|------------------------|
|                                                            | (1)                   | (2)                   | (3)                    | (4)                    | (5)                    | (6)                    |
| <i>Panel A. Dependent variable: share of all MNEs</i>      |                       |                       |                        |                        |                        |                        |
| Complexity (PCI)                                           | 0.0733***<br>(0.0061) | 0.0614***<br>(0.0064) |                        |                        |                        |                        |
| Lall: high tech (vs primary and resource-based)            |                       |                       | -0.2016***<br>(0.0104) | -0.2103***<br>(0.0102) |                        |                        |
| Lall: medium tech                                          |                       |                       | -0.0395***<br>(0.0060) | -0.0465***<br>(0.0066) |                        |                        |
| Lall: low tech                                             |                       |                       | -0.2467***<br>(0.0278) | -0.2272***<br>(0.0268) |                        |                        |
| Rauch: differentiated (vs homogeneous)                     |                       |                       |                        |                        | 0.0435***<br>(0.0088)  | 0.0488***<br>(0.0098)  |
| Rauch: reference-priced                                    |                       |                       |                        |                        | -0.0460***<br>(0.0068) | -0.0393***<br>(0.0066) |
| Observations                                               | 158,616               | 156,478               | 274,948                | 272,985                | 271,105                | 269,190                |
| <i>Panel B. Dependent variable: share of foreign MNEs</i>  |                       |                       |                        |                        |                        |                        |
| Complexity (PCI)                                           | 0.0562***<br>(0.0067) | 0.0490***<br>(0.0072) |                        |                        |                        |                        |
| Lall: high tech (vs primary and resource-based)            |                       |                       | -0.1480***<br>(0.0091) | -0.1576***<br>(0.0094) |                        |                        |
| Lall: medium tech                                          |                       |                       | -0.0623***<br>(0.0067) | -0.0678***<br>(0.0073) |                        |                        |
| Lall: low tech                                             |                       |                       | -0.2301***<br>(0.0153) | -0.2262***<br>(0.0139) |                        |                        |
| Rauch: differentiated (vs homogeneous)                     |                       |                       |                        |                        | 0.0554***<br>(0.0097)  | 0.0560***<br>(0.0107)  |
| Rauch: reference-priced                                    |                       |                       |                        |                        | -0.0439***<br>(0.0069) | -0.0383***<br>(0.0066) |
| Observations                                               | 158,616               | 156,478               | 274,948                | 272,985                | 271,105                | 269,190                |
| <i>Panel C. Dependent variable: share of domestic MNEs</i> |                       |                       |                        |                        |                        |                        |
| Complexity (PCI)                                           | 0.0172***<br>(0.0031) | 0.0123***<br>(0.0033) |                        |                        |                        |                        |
| Lall: high tech (vs primary and resource-based)            |                       |                       | -0.0536***<br>(0.0051) | -0.0527***<br>(0.0053) |                        |                        |
| Lall: medium tech                                          |                       |                       | 0.0229***<br>(0.0032)  | 0.0212***<br>(0.0033)  |                        |                        |
| Lall: low tech                                             |                       |                       | -0.0165<br>(0.0247)    | -0.0010<br>(0.0273)    |                        |                        |
| Rauch: differentiated (vs homogeneous)                     |                       |                       |                        |                        | -0.0120***<br>(0.0044) | -0.0071*<br>(0.0043)   |
| Rauch: reference-priced                                    |                       |                       |                        |                        | -0.0021<br>(0.0038)    | -0.0010<br>(0.0040)    |
| Observations                                               | 158,616               | 156,478               | 274,948                | 272,985                | 271,105                | 269,190                |
| Origin × year and destination × year FE                    | ✓                     |                       | ✓                      |                        | ✓                      |                        |
| Origin × dest. × year FE                                   |                       | ✓                     |                        | ✓                      |                        | ✓                      |

*Notes:* Observations are origin–destination–HS6–year cells. Agriculture products only. Dependent variable: share of the cell's export value shipped by each type of MNE. Omitted categories: primary and resource-based products (Lall) and homogeneous products (Rauch). Robust standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 4 Fact 3: Multinationals from a Small Set of Countries Dominate Exports

### 4.1 Parent Countries



**Figure 21:** Agriculture: Foreign-Multinational Exports, by Parent Country

*Notes:* Percent of foreign-MNE agricultural export value, by country of the ultimate parent.

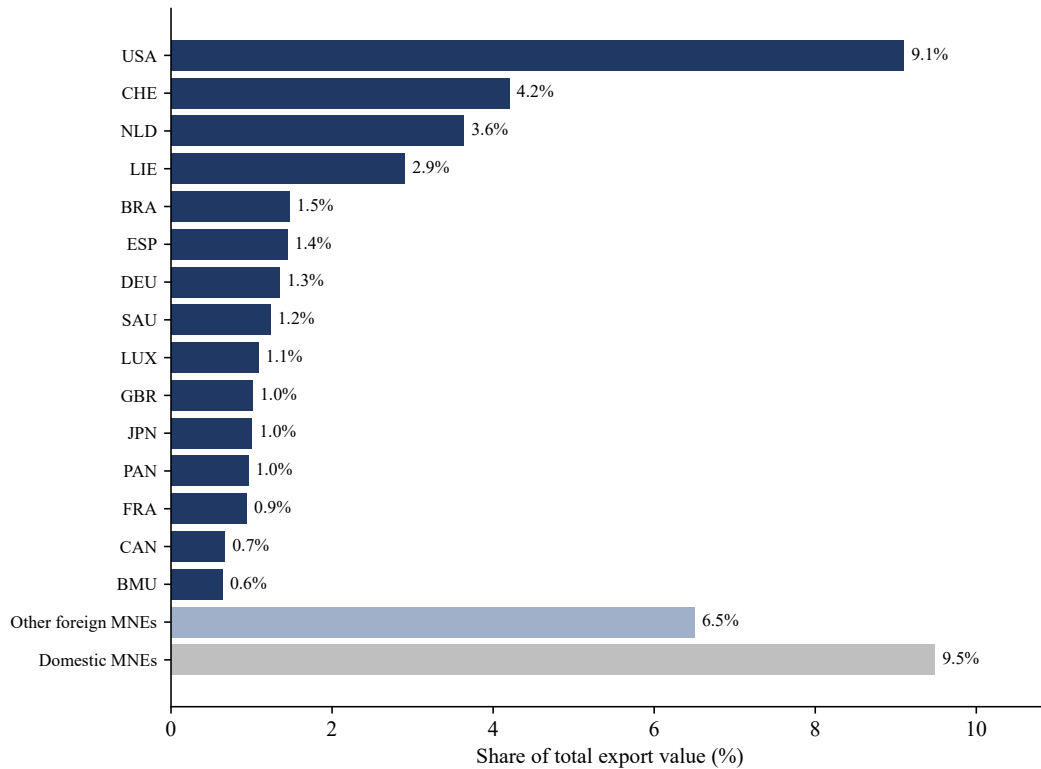
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 20:** Agriculture: Foreign-Multinational Exports, by Parent Country

| Parent country        | Share (%) | Value (US\$bn/yr) |
|-----------------------|-----------|-------------------|
| USA                   | 23.9      | 6.7               |
| CHE                   | 11.0      | 3.1               |
| NLD                   | 9.5       | 2.7               |
| LIE                   | 7.6       | 2.2               |
| BRA                   | 3.8       | 1.1               |
| ESP                   | 3.8       | 1.0               |
| DEU                   | 3.5       | 0.8               |
| SAU                   | 3.2       | 0.9               |
| LUX                   | 2.8       | 0.8               |
| GBR                   | 2.7       | 0.7               |
| JPN                   | 2.6       | 0.6               |
| PAN                   | 2.5       | 0.6               |
| FRA                   | 2.5       | 0.7               |
| CAN                   | 1.8       | 0.4               |
| BMU                   | 1.7       | 0.5               |
| Other (115 countries) | 17.0      | 4.4               |
| All foreign MNEs      | 100.0     | 30.7              |

*Notes:* Percent of foreign-MNE agricultural export value, by country of the ultimate parent. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 22:** Agriculture: Share of Total Exports, by Parent Country

*Notes:* Percent of total agricultural export value shipped by foreign MNEs of each parent country, other foreign MNEs and domestic MNEs; the remainder is local firms.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 21:** Agriculture: Share of Total Exports, by Parent Country

| Exporter type / parent country | Share of total exports (%) |
|--------------------------------|----------------------------|
| USA                            | 9.1                        |
| CHE                            | 4.2                        |
| NLD                            | 3.6                        |
| LIE                            | 2.9                        |
| BRA                            | 1.5                        |
| ESP                            | 1.4                        |
| DEU                            | 1.3                        |
| SAU                            | 1.2                        |
| LUX                            | 1.1                        |
| GBR                            | 1.0                        |
| JPN                            | 1.0                        |
| PAN                            | 1.0                        |
| FRA                            | 0.9                        |
| CAN                            | 0.7                        |
| BMU                            | 0.6                        |
| Other foreign MNEs             | 6.5                        |
| Domestic MNEs                  | 9.5                        |
| All MNEs                       | 47.6                       |
| Local firms                    | 52.4                       |

*Notes:* Percent of total agricultural export value, by type of exporter and parent country.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 4.2 Parent Region and Destination Region

**Table 22:** Agriculture: Multinational Exports by Parent Region and Destination Region: Export Value

| Parent region / Destination region | Latin America | North America | Europe | Asia | Africa | Oceania | Rest of World | Total |
|------------------------------------|---------------|---------------|--------|------|--------|---------|---------------|-------|
| Latin America                      | 1.0           | 0.6           | 0.7    | 1.0  | 0.1    | 0.0     | 0.0           | 3.4   |
| North America                      | 3.4           | 0.5           | 0.7    | 1.8  | 0.6    | 0.0     | 0.0           | 7.1   |
| Europe                             | 3.8           | 1.2           | 3.0    | 4.5  | 1.2    | 0.1     | 0.0           | 13.8  |
| Asia                               | 0.4           | 0.8           | 0.7    | 0.8  | 0.1    | 0.0     | 0.0           | 2.8   |
| Africa                             | 0.0           | 0.0           | 0.0    | 0.0  | 0.0    | 0.0     | 0.0           | 0.1   |
| Oceania                            | 0.0           | 0.0           | 0.0    | 0.0  | 0.0    | 0.0     | 0.0           | 0.0   |
| Rest of World                      | 0.0           | 0.0           | 0.0    | 0.0  | 0.0    | 0.0     | 0.0           | 0.0   |
| Domestic MNEs                      | 1.5           | 1.0           | 1.4    | 2.5  | 0.6    | 0.1     | 0.0           | 7.1   |
| Local firms                        | 9.5           | 6.7           | 11.0   | 10.8 | 2.5    | 0.5     | 0.1           | 41.1  |
| Total                              | 19.7          | 10.9          | 17.5   | 21.5 | 5.0    | 0.6     | 0.1           | 75.4  |

*Notes:* Values are annual averages in US\$ billion. Rows: region of the ultimate parent; columns: destination region.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 23:** Agriculture: Multinational Exports by Parent Region and Destination Region: Row Percentages

| Parent region / Destination region | Latin America | North America | Europe | Asia | Africa | Oceania | Rest of World | Total |
|------------------------------------|---------------|---------------|--------|------|--------|---------|---------------|-------|
| Latin America                      | 30.0          | 17.1          | 20.5   | 29.3 | 2.8    | 0.3     | 0.1           | 100.0 |
| North America                      | 47.9          | 8.3           | 10.2   | 25.6 | 7.6    | 0.1     | 0.4           | 100.0 |
| Europe                             | 27.3          | 10.1          | 22.0   | 31.6 | 8.2    | 0.6     | 0.2           | 100.0 |
| Asia                               | 14.2          | 29.2          | 24.7   | 29.1 | 2.4    | 0.3     | 0.2           | 100.0 |
| Africa                             | 29.5          | 15.3          | 30.8   | 13.1 | 9.2    | 0.3     | 1.8           | 100.0 |
| Oceania                            | 59.2          | 21.5          | 8.5    | 6.1  | 0.7    | 4.1     | 0.0           | 100.0 |
| Rest of World                      | 17.8          | 36.0          | 28.7   | 16.8 | 0.0    | 0.7     | 0.0           | 100.0 |
| Domestic MNEs                      | 20.5          | 16.0          | 20.4   | 34.9 | 7.0    | 1.1     | 0.1           | 100.0 |
| Local firms                        | 22.8          | 17.9          | 26.8   | 25.6 | 5.6    | 1.1     | 0.2           | 100.0 |

*Notes:* Percent of each group's exports, by destination region (rows add to 100). Domestic MNEs and local firms are shown for comparison.

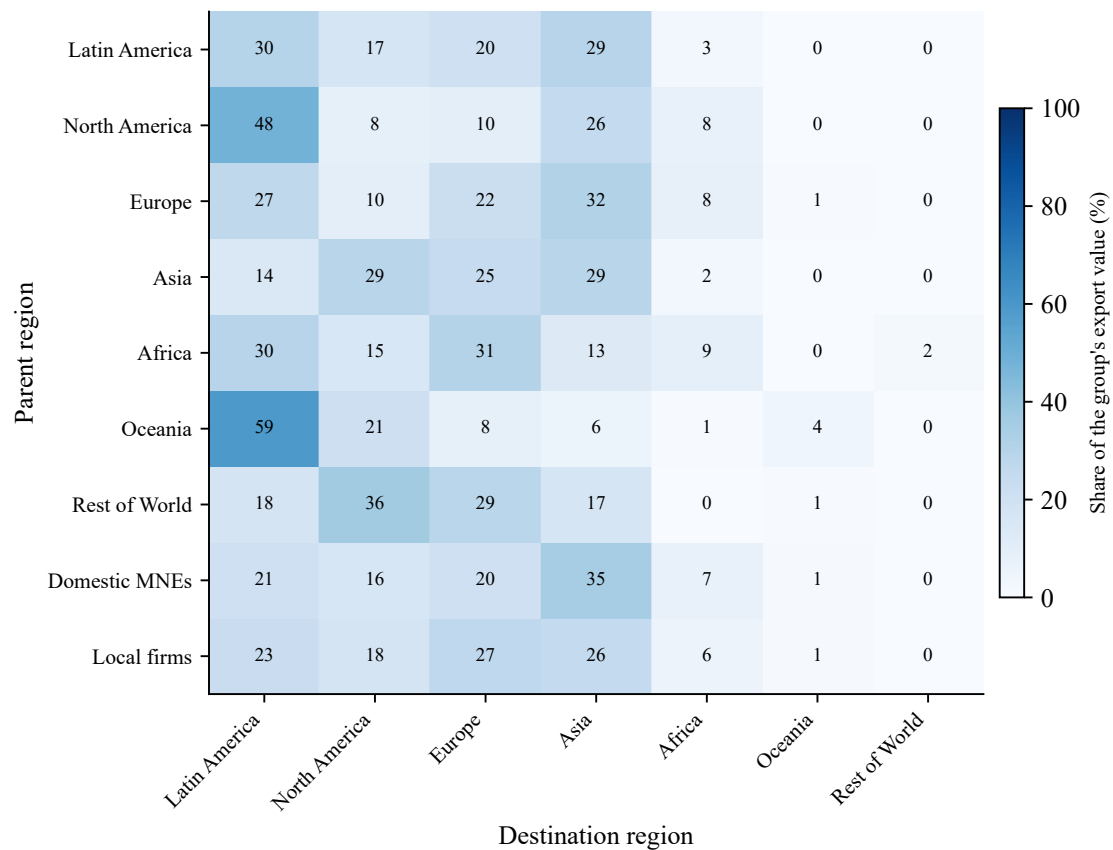
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 24:** Agriculture: Multinational Exports by Parent Region and Destination Region: Column Percentages

| Parent region / Destination region | Latin America | North America | Europe | Asia  | Africa | Oceania | Rest of World |
|------------------------------------|---------------|---------------|--------|-------|--------|---------|---------------|
| Latin America                      | 5.3           | 4.9           | 3.9    | 4.7   | 2.0    | 1.5     | 3.4           |
| North America                      | 16.8          | 4.7           | 3.9    | 8.3   | 11.1   | 0.8     | 19.1          |
| Europe                             | 18.7          | 11.2          | 16.5   | 20.0  | 23.5   | 13.4    | 17.8          |
| Asia                               | 2.1           | 6.9           | 4.0    | 3.9   | 1.5    | 1.2     | 3.5           |
| Africa                             | 0.2           | 0.1           | 0.2    | 0.1   | 0.2    | 0.0     | 1.5           |
| Oceania                            | 0.1           | 0.1           | 0.0    | 0.0   | 0.0    | 0.2     | 0.0           |
| Rest of World                      | 0.0           | 0.0           | 0.0    | 0.0   | 0.0    | 0.0     | 0.0           |
| Domestic MNEs                      | 8.0           | 10.0          | 8.6    | 12.5  | 11.4   | 12.8    | 7.2           |
| Local firms                        | 48.9          | 62.0          | 62.8   | 50.5  | 50.2   | 70.1    | 47.5          |
| Total                              | 100.0         | 100.0         | 100.0  | 100.0 | 100.0  | 100.0   | 100.0         |

*Notes:* Percent of the exports reaching each destination region, by parent region (columns add to 100).

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 23:** Agriculture: Multinational Exports by Parent Region and Destination Region

*Notes:* Percent of each group's exports, by destination region (rows add to 100).

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 25:** Agriculture: Foreign-Multinational Exports by Exporting Country and Destination Region

| Origin | Destination region | Latin America | North America | Europe | Asia | Africa | Oceania | Rest of World | Total |
|--------|--------------------|---------------|---------------|--------|------|--------|---------|---------------|-------|
| ARG    |                    | 24.3          | 3.2           | 17.2   | 42.4 | 12.6   | 0.3     | 0.0           | 14.4  |
| CHL    |                    | 26.6          | 22.8          | 20.3   | 27.8 | 1.1    | 1.3     | 0.0           | 2.2   |
| COL    |                    | 18.0          | 41.2          | 27.8   | 9.5  | 0.6    | 1.0     | 2.0           | 2.0   |
| CRI    |                    | 31.5          | 36.6          | 29.7   | 1.7  | 0.3    | 0.2     | 0.0           | 1.9   |
| DOM    |                    | 25.5          | 58.0          | 13.9   | 2.1  | 0.2    | 0.0     | 0.4           | 0.8   |
| PER    |                    | 15.7          | 24.8          | 32.1   | 25.9 | 0.9    | 0.6     | 0.0           | 2.7   |
| PRY    |                    | 79.7          | 0.5           | 11.7   | 6.9  | 1.2    | 0.0     | 0.0           | 4.0   |
| SLV    |                    | 58.1          | 12.7          | 26.5   | 1.8  | 0.4    | 0.5     | 0.0           | 0.3   |
| URY    |                    | 26.0          | 8.8           | 21.2   | 37.5 | 3.9    | 0.1     | 2.4           | 2.3   |

*Notes:* Percent of each country's foreign-MNE agricultural exports, by destination region (rows add to 100). Last column: foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 26: Agriculture: Local-Firm Exports by Exporting Country and Destination Region**

| Origin | Destination region | Latin America | North America | Europe | Asia | Africa | Oceania | Rest of World | Total |
|--------|--------------------|---------------|---------------|--------|------|--------|---------|---------------|-------|
| ARG    |                    | 21.6          | 5.0           | 23.8   | 36.9 | 11.0   | 1.7     | 0.0           | 20.0  |
| CHL    |                    | 17.9          | 27.4          | 24.9   | 28.3 | 1.1    | 0.4     | 0.0           | 3.9   |
| COL    |                    | 16.1          | 42.5          | 31.7   | 7.3  | 1.2    | 0.7     | 0.5           | 4.1   |
| CRI    |                    | 27.3          | 33.2          | 35.3   | 3.3  | 0.3    | 0.6     | 0.0           | 2.3   |
| DOM    |                    | 25.6          | 38.8          | 33.3   | 1.7  | 0.1    | 0.0     | 0.5           | 1.1   |
| PER    |                    | 13.7          | 25.8          | 35.5   | 23.4 | 0.5    | 1.0     | 0.0           | 5.0   |
| PRY    |                    | 62.9          | 3.1           | 21.7   | 10.8 | 1.4    | 0.0     | 0.0           | 2.1   |
| SLV    |                    | 41.9          | 34.7          | 13.4   | 8.8  | 0.2    | 0.8     | 0.2           | 0.8   |
| URY    |                    | 33.8          | 4.9           | 23.8   | 27.8 | 7.9    | 0.1     | 1.8           | 1.7   |

*Notes:* Percent of each country's agricultural exports by local firms, by destination region (rows add to 100). Last column: local-firm exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

### 4.3 Parent Country and Destination Country

**Table 27: Agriculture: Foreign-Multinational Exports by Parent Country and Destination: Export Value**

| Parent / Destination | USA | CHN | BRA | ARG | ESP | CHL | JPN | NLD | URY | ITA | Other destinations | Total |
|----------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|--------------------|-------|
| USA                  | 0.4 | 1.1 | 0.2 | 1.2 | 0.1 | 0.3 | 0.1 | 0.1 | 0.4 | 0.0 | 2.7                | 6.7   |
| CHE                  | 0.0 | 0.2 | 0.2 | 0.0 | 0.1 | 0.1 | 0.0 | 0.0 | 0.0 | 0.1 | 2.3                | 3.1   |
| NLD                  | 0.1 | 0.4 | 0.1 | 0.1 | 0.0 | 0.1 | 0.0 | 0.1 | 0.1 | 0.2 | 1.6                | 2.7   |
| LIE                  | 0.0 | 0.3 | 0.1 | 0.0 | 0.1 | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 1.5                | 2.2   |
| BRA                  | 0.1 | 0.3 | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.4                | 1.1   |
| ESP                  | 0.1 | 0.1 | 0.0 | 0.0 | 0.3 | 0.0 | 0.0 | 0.1 | 0.0 | 0.0 | 0.3                | 1.0   |
| DEU                  | 0.3 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.4                | 0.8   |
| SAU                  | 0.1 | 0.1 | 0.1 | 0.0 | 0.0 | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.5                | 0.9   |
| LUX                  | 0.0 | 0.0 | 0.3 | 0.0 | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.2                | 0.8   |
| GBR                  | 0.1 | 0.1 | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.3                | 0.7   |
| Other parents        | 1.7 | 0.6 | 0.6 | 0.6 | 0.3 | 0.1 | 0.3 | 0.3 | 0.1 | 0.1 | 2.5                | 7.2   |
| Total                | 3.0 | 3.2 | 1.9 | 1.9 | 1.0 | 0.9 | 0.6 | 0.7 | 0.7 | 0.6 | 12.7               | 27.2  |

*Notes:* Values are annual averages in US\$ billion. Ten largest parent countries and ten largest destinations by foreign-MNE export value.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 28: Agriculture: Foreign-Multinational Exports by Parent Country and Destination: Row Percentages**

| Parent / Destination | USA  | CHN  | BRA  | ARG  | ESP  | CHL  | JPN | NLD | URY | ITA | Other destinations | Total |
|----------------------|------|------|------|------|------|------|-----|-----|-----|-----|--------------------|-------|
| USA                  | 7.3  | 15.5 | 3.7  | 16.7 | 1.2  | 4.1  | 1.8 | 1.7 | 5.5 | 0.7 | 41.8               | 100.0 |
| CHE                  | 1.7  | 6.7  | 7.2  | 0.2  | 2.6  | 3.4  | 0.4 | 1.2 | 0.7 | 3.7 | 72.3               | 100.0 |
| NLD                  | 3.3  | 14.4 | 5.1  | 3.9  | 1.4  | 3.6  | 0.6 | 2.8 | 2.4 | 5.6 | 57.0               | 100.0 |
| LIE                  | 1.3  | 14.4 | 6.1  | 0.0  | 2.9  | 2.6  | 0.4 | 0.9 | 0.2 | 2.2 | 69.1               | 100.0 |
| BRA                  | 11.4 | 31.2 | 4.9  | 1.9  | 1.2  | 1.7  | 1.1 | 4.3 | 0.9 | 2.2 | 39.2               | 100.0 |
| ESP                  | 15.5 | 8.3  | 3.0  | 0.4  | 26.1 | 2.5  | 3.9 | 5.8 | 0.2 | 2.9 | 31.4               | 100.0 |
| DEU                  | 37.3 | 0.6  | 1.8  | 0.4  | 3.5  | 0.9  | 5.0 | 4.3 | 0.7 | 1.8 | 43.6               | 100.0 |
| SAU                  | 6.4  | 12.3 | 6.9  | 0.1  | 2.3  | 15.8 | 0.1 | 2.6 | 0.3 | 2.1 | 51.1               | 100.0 |
| LUX                  | 6.5  | 2.5  | 40.8 | 0.0  | 8.1  | 2.0  | 2.1 | 0.3 | 2.8 | 4.5 | 30.4               | 100.0 |
| GBR                  | 8.9  | 16.9 | 8.5  | 0.6  | 3.5  | 2.7  | 2.2 | 2.9 | 2.1 | 0.8 | 50.9               | 100.0 |
| Other parents        | 23.1 | 7.9  | 7.9  | 7.5  | 4.0  | 1.5  | 5.5 | 3.5 | 1.8 | 2.0 | 35.4               | 100.0 |

*Notes:* Percent of each parent country's exports, by destination (rows add to 100). Ten largest parent countries and destinations.

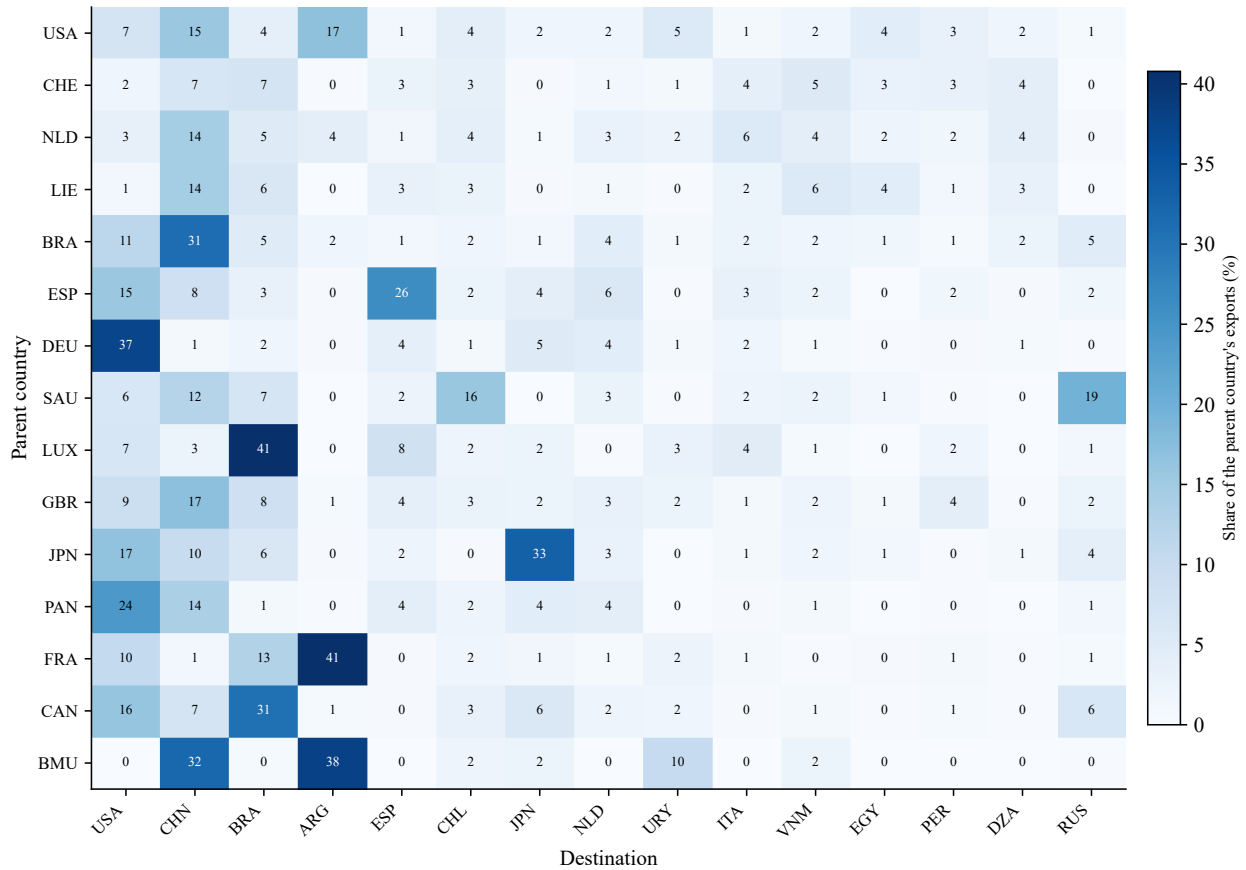
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 29:** Agriculture: Foreign-Multinational Exports by Parent Country and Destination: Column Percentages

| Parent / Destination | USA   | CHN   | BRA   | ARG   | ESP   | CHL   | JPN   | NLD   | URY   | ITA   | Other destinations |
|----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------------------|
| USA                  | 14.7  | 32.0  | 12.9  | 60.5  | 8.2   | 31.2  | 16.7  | 15.5  | 55.8  | 7.1   | 21.4               |
| CHE                  | 1.6   | 6.3   | 11.6  | 0.3   | 7.8   | 12.0  | 1.7   | 5.3   | 3.3   | 17.6  | 17.1               |
| NLD                  | 2.7   | 11.9  | 7.1   | 5.6   | 3.6   | 10.8  | 2.3   | 10.6  | 9.7   | 23.1  | 11.6               |
| LIE                  | 0.8   | 9.5   | 6.8   | 0.0   | 6.1   | 6.2   | 1.1   | 2.7   | 0.6   | 7.3   | 11.3               |
| BRA                  | 3.7   | 10.4  | 2.8   | 1.1   | 1.3   | 2.1   | 1.6   | 6.5   | 1.5   | 3.7   | 3.2                |
| ESP                  | 5.0   | 2.7   | 1.7   | 0.2   | 27.4  | 3.0   | 5.6   | 8.6   | 0.3   | 4.8   | 2.5                |
| DEU                  | 11.2  | 0.2   | 0.9   | 0.2   | 3.5   | 1.0   | 6.8   | 6.0   | 1.0   | 2.7   | 3.3                |
| SAU                  | 1.7   | 3.4   | 3.3   | 0.0   | 2.1   | 16.1  | 0.2   | 3.3   | 0.4   | 2.9   | 3.5                |
| LUX                  | 1.6   | 0.6   | 17.0  | 0.0   | 6.4   | 1.8   | 2.3   | 0.3   | 3.4   | 5.5   | 1.8                |
| GBR                  | 2.0   | 3.9   | 3.3   | 0.2   | 2.6   | 2.2   | 2.3   | 3.1   | 2.3   | 1.0   | 2.9                |
| Other parents        | 55.1  | 19.1  | 32.6  | 31.7  | 31.0  | 13.5  | 59.4  | 38.1  | 21.6  | 24.4  | 21.3               |
| Total                | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0              |

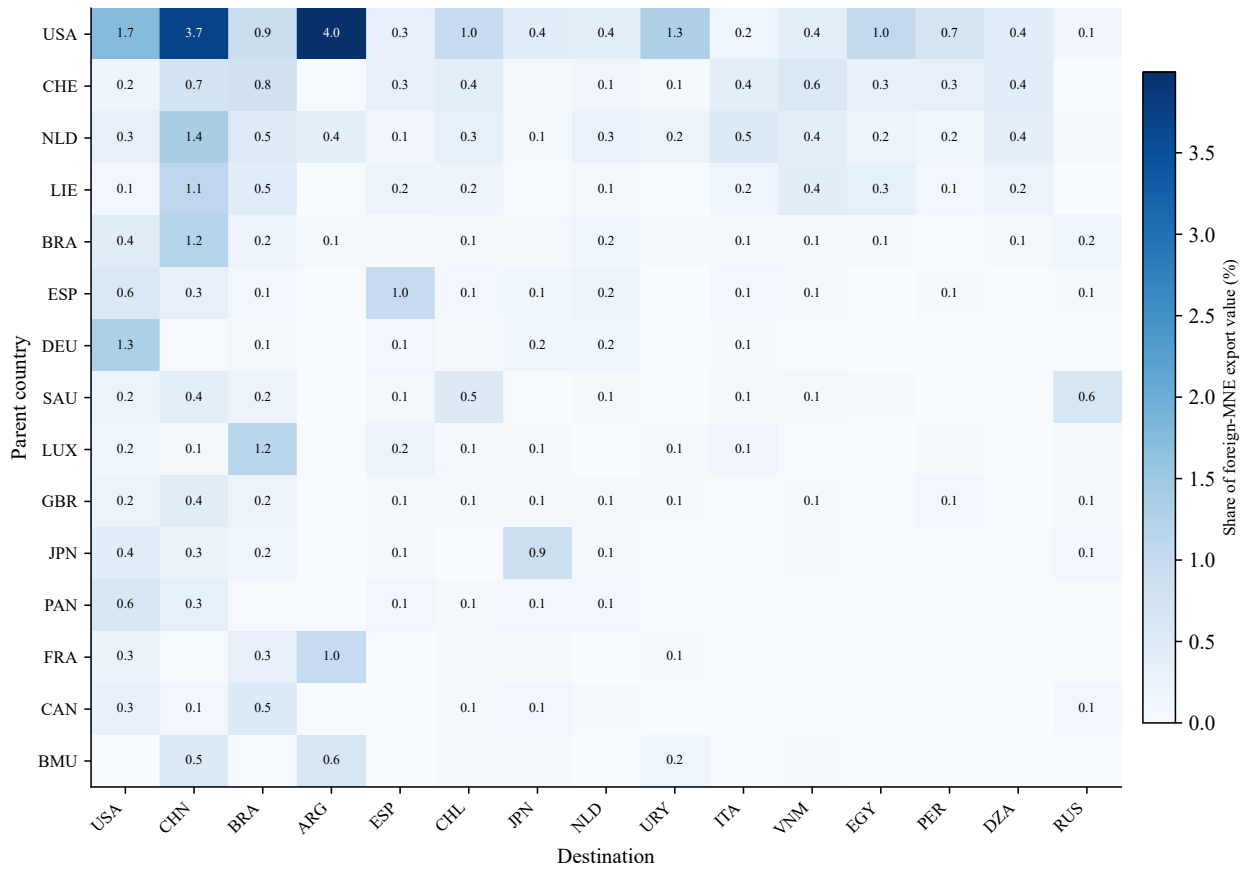
*Notes:* Percent of the foreign-MNE exports reaching each destination, by parent country (columns add to 100). Ten largest parent countries and destinations.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Figure 24:** Agriculture: Destination of Foreign-Multinational Exports: Fifteen Largest Parent Countries and Destinations

*Notes:* Percent of each parent country's exports going to each destination (rows add to 100 including other destinations).

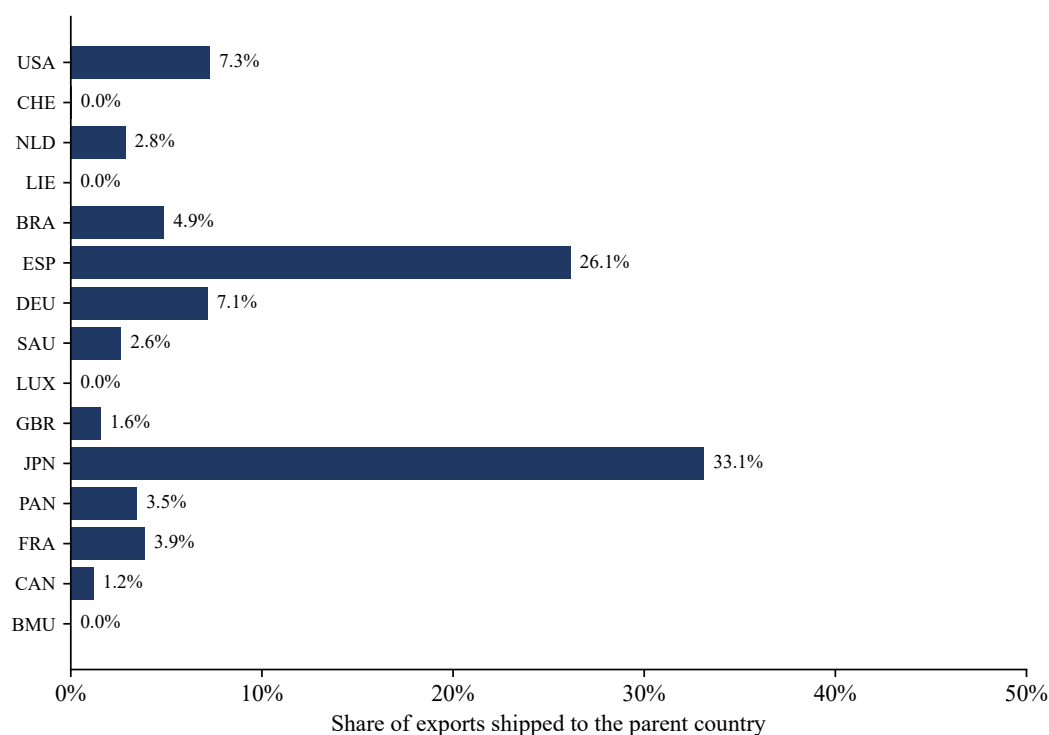
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 25: Agriculture: Foreign-Multinational Exports by Parent Country and Destination: Share of the Total**  
*Notes:* Percent of all foreign-MNE export value in each parent country and destination pair; fifteen largest parent countries and destinations.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 4.4 Exports to the Parent Country



**Figure 26:** Agriculture: Share of Foreign-Multinational Exports Shipped to the Parent Country

*Notes:* Percent of each parent country's exports shipped to that same country.

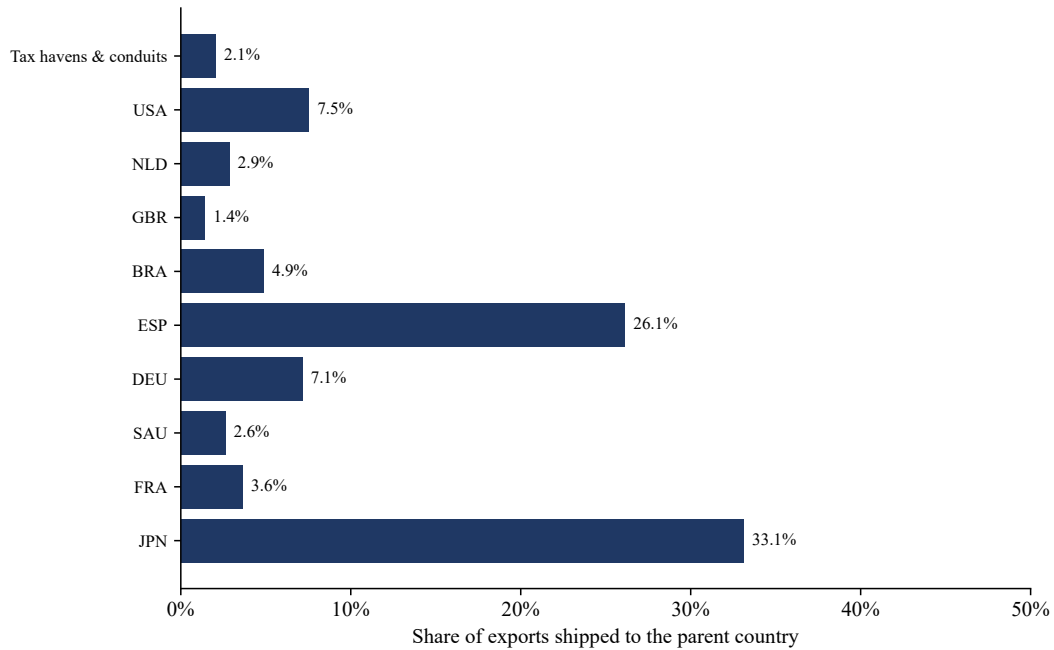
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 30:** Agriculture: Foreign-Multinational Exports to the Parent Countries, by Parent Country

| Parent / Destination | USA  | CHE | NLD | BRA  | ESP  | DEU | SAU | LUX | GBR | Other destinations | Total |
|----------------------|------|-----|-----|------|------|-----|-----|-----|-----|--------------------|-------|
| USA                  | 7.3  | 0.0 | 1.7 | 3.7  | 1.2  | 0.6 | 1.0 | 0.0 | 0.5 | 84.1               | 100.0 |
| CHE                  | 1.7  | 0.0 | 1.2 | 7.2  | 2.6  | 0.3 | 3.8 | 0.0 | 0.9 | 82.3               | 100.0 |
| NLD                  | 3.3  | 0.0 | 2.8 | 5.1  | 1.4  | 0.6 | 1.0 | 0.0 | 0.8 | 85.0               | 100.0 |
| LIE                  | 1.3  | 0.0 | 0.9 | 6.1  | 2.9  | 0.1 | 3.0 | 0.0 | 0.5 | 85.3               | 100.0 |
| BRA                  | 11.4 | 0.6 | 4.3 | 4.9  | 1.2  | 5.1 | 0.9 | 0.0 | 1.3 | 70.3               | 100.0 |
| ESP                  | 15.5 | 0.0 | 5.8 | 3.0  | 26.1 | 1.6 | 0.3 | 0.0 | 1.9 | 45.7               | 100.0 |
| DEU                  | 37.3 | 0.1 | 4.3 | 1.8  | 3.5  | 7.1 | 0.2 | 0.0 | 1.2 | 44.4               | 100.0 |
| SAU                  | 6.4  | 0.7 | 2.6 | 6.9  | 2.3  | 1.2 | 2.6 | 0.0 | 1.5 | 75.7               | 100.0 |
| LUX                  | 6.5  | 0.1 | 0.3 | 40.8 | 8.1  | 0.7 | 0.0 | 0.0 | 0.4 | 43.1               | 100.0 |
| GBR                  | 8.9  | 0.2 | 2.9 | 8.5  | 3.5  | 2.1 | 0.0 | 0.0 | 1.6 | 72.3               | 100.0 |
| Other parents        | 23.1 | 0.3 | 3.5 | 7.9  | 4.0  | 2.0 | 0.3 | 0.0 | 1.9 | 56.9               | 100.0 |

*Notes:* Percent of each parent country's exports, by destination (rows add to 100). Destinations are the ten largest parent countries; the diagonal is the share shipped to the parent's own country.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 27:** Agriculture: Share of Foreign-Multinational Exports Shipped to the Parent Jurisdiction, Consolidated Parents

*Notes:* Percent of each parent's exports shipped to its own jurisdiction. Dependencies are assigned to their sovereign state and tax havens and conduit jurisdictions are pooled.

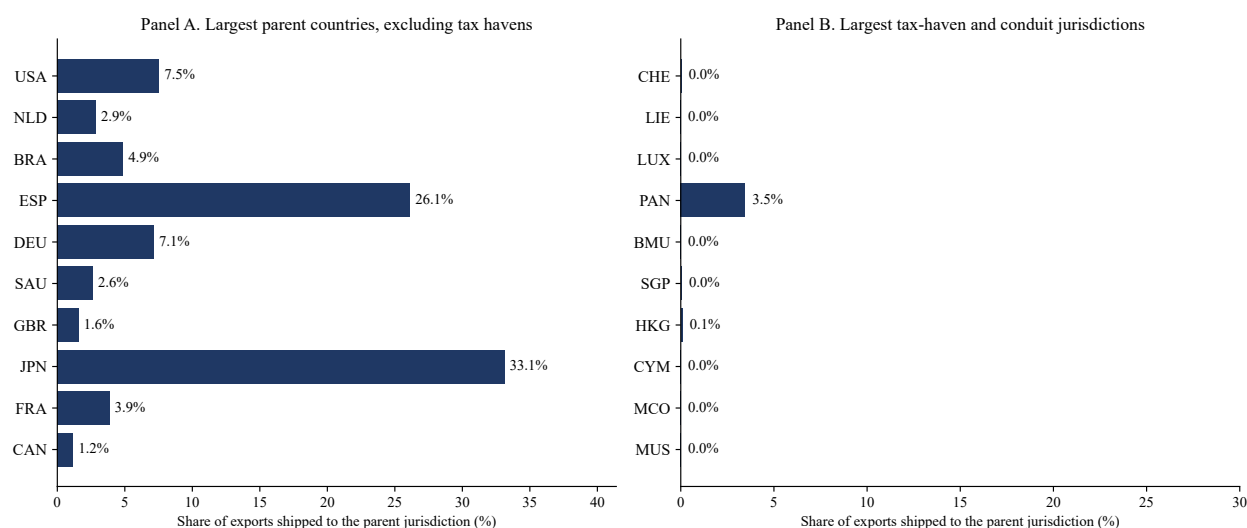
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 31:** Agriculture: Share of Foreign-Multinational Exports Shipped to the Parent Jurisdiction, Consolidated Parents

| Consolidated parent   | Home share (%) | Exports (US\$bn/yr) |
|-----------------------|----------------|---------------------|
| Tax havens & conduits | 2.1            | 7.1                 |
| USA                   | 7.5            | 6.7                 |
| NLD                   | 2.9            | 2.8                 |
| GBR                   | 1.4            | 1.3                 |
| BRA                   | 4.9            | 1.1                 |
| ESP                   | 26.1           | 1.0                 |
| DEU                   | 7.1            | 0.8                 |
| SAU                   | 2.6            | 0.9                 |
| FRA                   | 3.6            | 0.7                 |
| JPN                   | 33.1           | 0.6                 |

*Notes:* Home share: percent of the parent's exports shipped to its own jurisdiction. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 28:** Agriculture: Share of Foreign-Multinational Exports Shipped to the Parent Jurisdiction

*Notes:* Percent of each parent jurisdiction's exports shipped to that same jurisdiction. Panel A: largest parent countries, excluding tax havens; dependencies are assigned to their sovereign state. Panel B: largest tax-haven and conduit jurisdictions.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

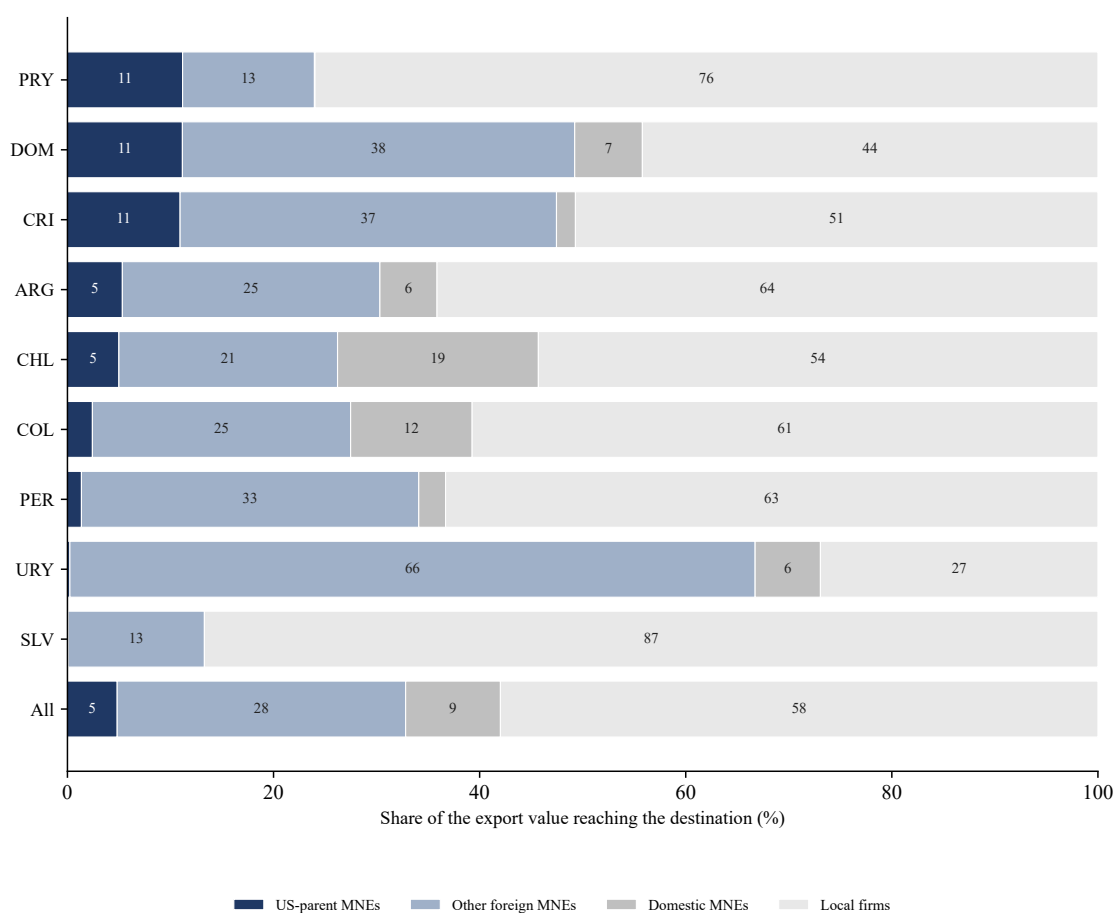
**Table 32:** Agriculture: Share of Foreign-Multinational Exports Shipped to the Parent Jurisdiction

| Panel   | Parent | Home share (%) | Exports (US\$bn/yr) |
|---------|--------|----------------|---------------------|
| Panel A | USA    | 7.5            | 6.7                 |
| Panel A | NLD    | 2.9            | 2.7                 |
| Panel A | BRA    | 4.9            | 1.1                 |
| Panel A | ESP    | 26.1           | 1.0                 |
| Panel A | DEU    | 7.1            | 0.8                 |
| Panel A | SAU    | 2.6            | 0.9                 |
| Panel A | GBR    | 1.6            | 0.7                 |
| Panel A | JPN    | 33.1           | 0.6                 |
| Panel A | FRA    | 3.9            | 0.7                 |
| Panel A | CAN    | 1.2            | 0.4                 |
| Panel B | CHE    | 0.0            | 3.1                 |
| Panel B | LIE    | 0.0            | 2.2                 |
| Panel B | LUX    | 0.0            | 0.8                 |
| Panel B | PAN    | 3.5            | 0.6                 |
| Panel B | BMU    | 0.0            | 0.5                 |
| Panel B | SGP    | 0.0            | 0.2                 |
| Panel B | HKG    | 0.1            | 0.2                 |
| Panel B | CYM    | 0.0            | 0.1                 |
| Panel B | MCO    | 0.0            | 0.1                 |
| Panel B | MUS    | 0.0            | 0.0                 |

*Notes:* Home share: percent of the parent's exports shipped to its own jurisdiction. Panel A: largest parent countries, excluding tax havens. Panel B: largest tax-haven and conduit jurisdictions. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 4.5 Destination Markets



**Figure 29:** Agriculture: Exports to the United States, by Type of Exporter and Exporting Country

*Notes:* Percent of each country's agricultural exports to the United States, by type of exporter.

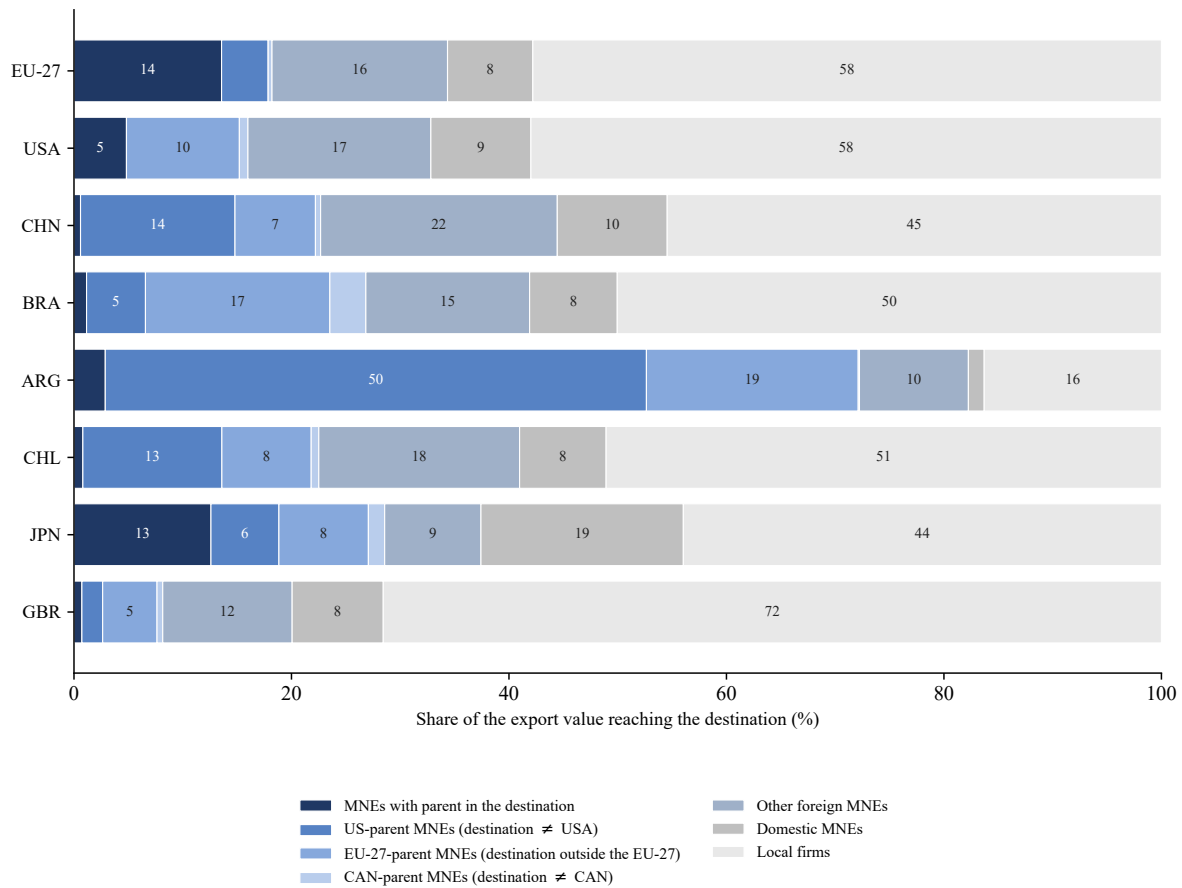
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 33:** Agriculture: Exports to the United States, by Type of Exporter and Exporting Country

| Origin | Exports to the USA (US\$bn/yr) | US-parent MNEs (%) | Other foreign MNEs (%) | Domestic MNEs (%) | Local firms (%) |
|--------|--------------------------------|--------------------|------------------------|-------------------|-----------------|
| PRY    | 0.1                            | 11.2               | 12.8                   | 0.1               | 76.0            |
| DOM    | 0.9                            | 11.2               | 38.1                   | 6.6               | 44.2            |
| CRI    | 1.5                            | 10.9               | 36.5                   | 1.8               | 50.7            |
| ARG    | 1.4                            | 5.3                | 25.0                   | 5.5               | 64.1            |
| CHL    | 1.8                            | 5.0                | 21.2                   | 19.5              | 54.3            |
| COL    | 2.7                            | 2.4                | 25.1                   | 11.8              | 60.7            |
| PER    | 1.8                            | 1.4                | 32.7                   | 2.6               | 63.3            |
| URY    | 0.3                            | 0.2                | 66.5                   | 6.3               | 26.9            |
| SLV    | 0.3                            | 0.1                | 13.2                   | 0.0               | 86.7            |
| All    | 10.7                           | 4.8                | 28.0                   | 9.2               | 58.0            |

*Notes:* Percent of each country's agricultural exports to the United States, by type of exporter. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 30: Agriculture: Ownership of Exports to the Main Destination Markets**

*Notes:* Percent of the export value reaching each destination, by type of exporter. The first segment is exports by MNEs whose parent is located in that destination. EU-27 member states are pooled.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 34: Agriculture: Ownership of Exports to the Main Destination Markets**

| Destination | Exports (US\$bn/yr) | MNEs with parent in the destination (%) | US-parent MNEs (%) | EU-27-parent MNEs (%) | CAN-parent MNEs (%) | Other foreign MNEs (%) | Domestic MNEs (%) | Local firms (%) |
|-------------|---------------------|-----------------------------------------|--------------------|-----------------------|---------------------|------------------------|-------------------|-----------------|
| EU-27       | 13.8                | 13.6                                    | 4.3                | —                     | 0.3                 | 16.1                   | 7.8               | 57.8            |
| USA         | 10.7                | 4.8                                     | —                  | 10.4                  | 0.8                 | 16.8                   | 9.2               | 58.0            |
| CHN         | 7.5                 | 0.6                                     | 14.2               | 7.4                   | 0.5                 | 21.8                   | 10.1              | 45.4            |
| BRA         | 5.1                 | 1.2                                     | 5.4                | 17.0                  | 3.3                 | 15.1                   | 8.1               | 50.0            |
| ARG         | 2.5                 | 2.9                                     | 49.8               | 19.5                  | 0.1                 | 10.0                   | 1.5               | 16.3            |
| CHL         | 2.4                 | 0.8                                     | 12.8               | 8.2                   | 0.7                 | 18.5                   | 8.0               | 51.1            |
| JPN         | 1.9                 | 12.6                                    | 6.2                | 8.2                   | 1.5                 | 8.9                    | 18.6              | 44.0            |
| GBR         | 1.9                 | 0.7                                     | 1.9                | 5.0                   | 0.5                 | 11.9                   | 8.4               | 71.6            |

*Notes:* Percent of the export value reaching each destination, by type of exporter. EU-27 member states are pooled. Values are annual averages in US\$ billion.

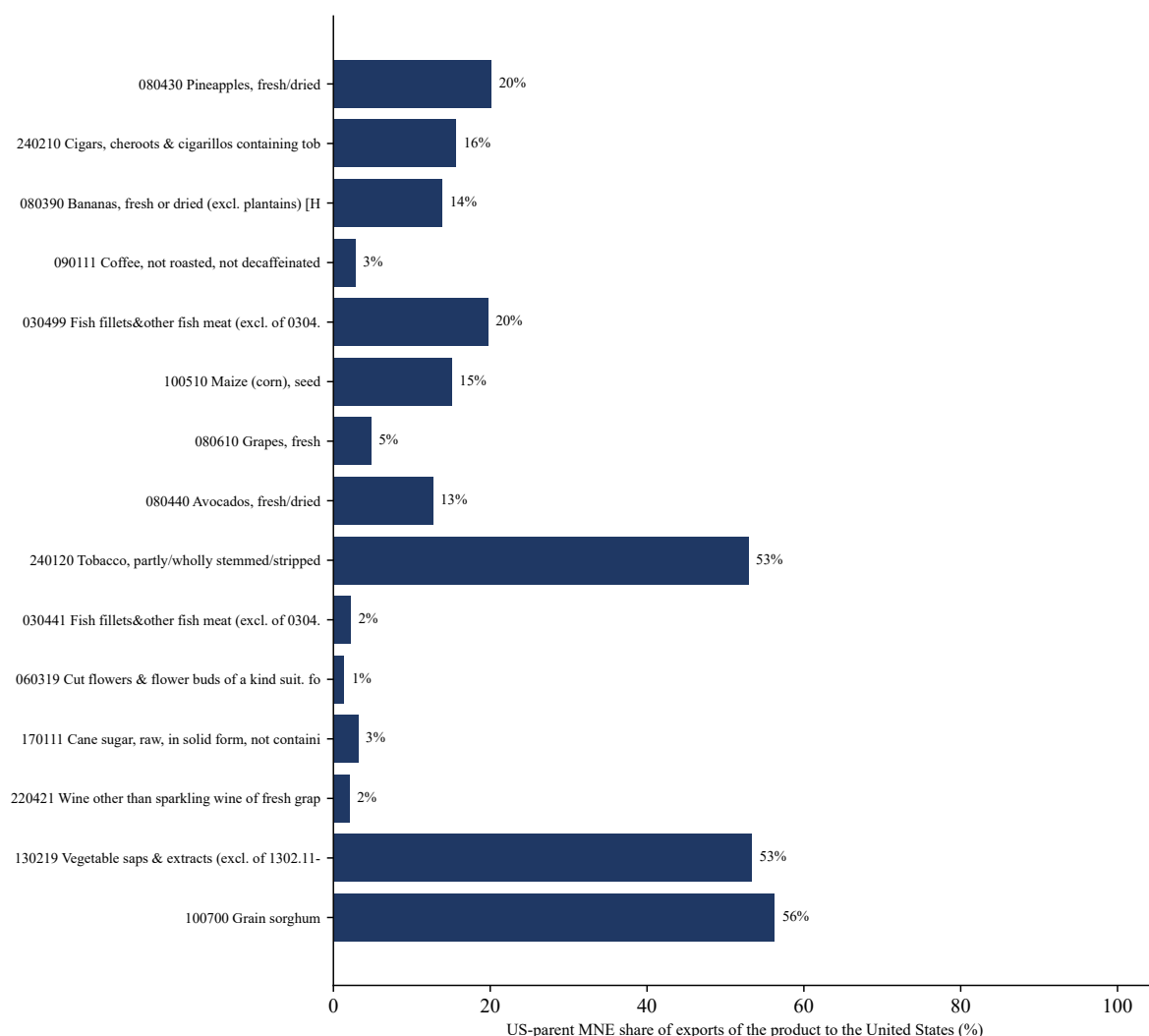
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 35: Agriculture: Main Products Shipped by US-Parent Multinationals to the United States**

| HS6       | Description                                      | US-parent exports to the USA (US\$bn/yr) | % of all US-parent exports to the USA | % of US-parent exports of the product going to the USA | All exports of the product to the USA (US\$bn/yr) | US-parent share of exports of the product to the USA (%) | Main origins of the US-parent shipments (% of the line) |
|-----------|--------------------------------------------------|------------------------------------------|---------------------------------------|--------------------------------------------------------|---------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------|
| 080430    | Pineapples, fresh/dried                          | 0.08                                     | 18.1                                  | 66                                                     | 0.42                                              | 20                                                       | CRI 100%                                                |
| 240210    | Cigars, cheroots & cigarillos containing toba... | 0.09                                     | 15.4                                  | 100                                                    | 0.57                                              | 16                                                       | DOM 100%                                                |
| 080390    | Bananas, fresh or dried (excl. plantains) [HS... | 0.06                                     | 13.3                                  | 66                                                     | 0.43                                              | 14                                                       | CRI 100%                                                |
| 090111    | Coffee, not roasted, not decaffeinated           | 0.03                                     | 8.6                                   | 52                                                     | 1.42                                              | 3                                                        | COL 100%, PER 0%, SLV 0%                                |
| 030499    | Fish fillets&other fish meat (excl. of 0304.1... | 0.02                                     | 5.5                                   | 60                                                     | 0.10                                              | 20                                                       | CHL 100%, PER 0%                                        |
| 100510    | Maize (corn), seed                               | 0.01                                     | 3.7                                   | 61                                                     | 0.11                                              | 15                                                       | CHL 85%, ARG 15%, CRI 0%                                |
| 080610    | Grapes, fresh                                    | 0.01                                     | 3.3                                   | 39                                                     | 0.39                                              | 5                                                        | CHL 100%                                                |
| 080440    | Avocados, fresh/dried                            | 0.01                                     | 3.0                                   | 48                                                     | 0.14                                              | 13                                                       | PER 99%, CHL 0%, COL 0%                                 |
| 240120    | Tobacco, partly/wholly stemmed/stripped          | 0.01                                     | 2.8                                   | 12                                                     | 0.03                                              | 53                                                       | ARG 76%, DOM 24%, PRY 1%                                |
| 030441    | Fish fillets&other fish meat (excl. of 0304.1... | 0.01                                     | 2.5                                   | 99                                                     | 0.38                                              | 2                                                        | CHL 100%                                                |
| 060319    | Cut flowers & flower buds of a kind suit. for... | 0.01                                     | 1.5                                   | 98                                                     | 0.61                                              | 1                                                        | COL 100%                                                |
| 170111    | Cane sugar, raw, in solid form, not containin... | 0.01                                     | 1.4                                   | 43                                                     | 0.22                                              | 3                                                        | ARG 100%, DOM 0%, CRI 0%                                |
| 220421    | Wine other than sparkling wine of fresh grape... | 0.01                                     | 1.3                                   | 59                                                     | 0.34                                              | 2                                                        | ARG 92%, CHL 8%                                         |
| 130219    | Vegetable saps & extracts (excl. of 1302.11-1... | 0.01                                     | 1.2                                   | 85                                                     | 0.01                                              | 53                                                       | DOM 100%                                                |
| 100700    | Grain sorghum                                    | 0.01                                     | 1.1                                   | 4                                                      | 0.01                                              | 56                                                       | ARG 100%                                                |
| All lines |                                                  | 0.4                                      | 100.0                                 | 7                                                      | 10.7                                              | 5                                                        |                                                         |

*Notes:* Fifteen HS6 products with the largest exports by US-parent MNEs to the United States. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Figure 31: Agriculture: US-Parent Share of Exports to the United States, Main Products Shipped by US-Parent Multinationals**

*Notes:* Percent of the nine countries' exports of each product to the United States shipped by US-parent MNEs; the fifteen products with the largest US-parent exports to the United States.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 36:** Agriculture: Largest Export Products to the United States, by Type of Exporter

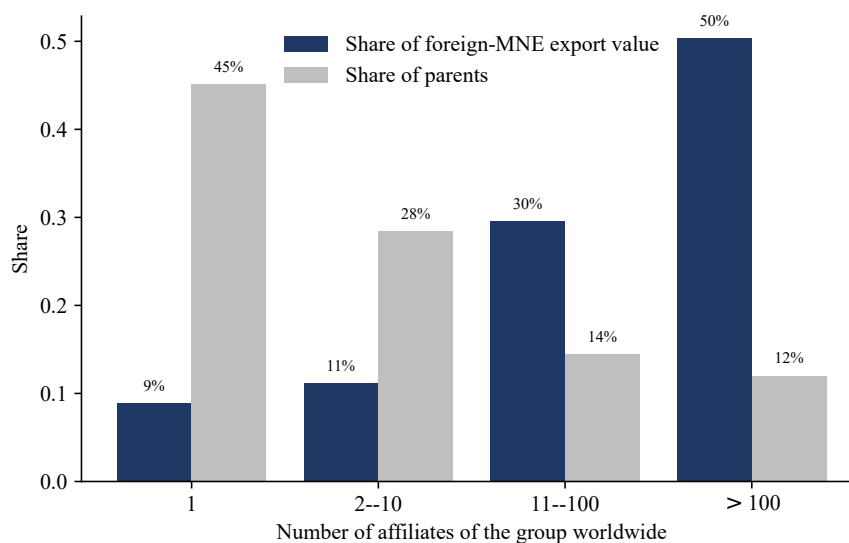
| HS6    | Description                                      | Exports to the<br>USA<br>(US\$bn/yr) | US-parent<br>MNEs (%) | Other foreign<br>MNEs (%) | Domestic<br>MNEs (%) | Local firms<br>(%) |
|--------|--------------------------------------------------|--------------------------------------|-----------------------|---------------------------|----------------------|--------------------|
| 090111 | Coffee, not roasted, not decaffeinated           | 1.42                                 | 3                     | 36                        | 8                    | 53                 |
| 060319 | Cut flowers & flower buds of a kind suit. for... | 0.61                                 | 1                     | 19                        | 19                   | 60                 |
| 030441 | Fish fillets&other fish meat (excl. of 0304.1... | 0.38                                 | 2                     | 18                        | 24                   | 56                 |
| 080610 | Grapes, fresh                                    | 0.39                                 | 5                     | 27                        | 10                   | 58                 |
| 240210 | Cigars, cheroots & cigarillos containing toba... | 0.57                                 | 16                    | 47                        | 0                    | 37                 |
| 080390 | Bananas, fresh or dried (excl. plantains) [HS... | 0.43                                 | 14                    | 27                        | 1                    | 58                 |
| 080430 | Pineapples, fresh/dried                          | 0.42                                 | 20                    | 32                        | 0                    | 48                 |
| 220421 | Wine other than sparkling wine of fresh grape... | 0.34                                 | 2                     | 21                        | 7                    | 70                 |
| 081040 | Cranberries, bilberries & other fruits of the... | 0.28                                 | 1                     | 33                        | 11                   | 54                 |
| 060311 | Fresh roses                                      | 0.25                                 | 2                     | 19                        | 8                    | 70                 |
| 070920 | Asparagus, fresh/chilled                         | 0.23                                 | 1                     | 31                        | 0                    | 68                 |
| 170111 | Cane sugar, raw, in solid form, not containin... | 0.22                                 | 3                     | 10                        | 22                   | 64                 |
| 080300 | Bananas, incl. plantains, fresh/dried            | 0.14                                 | 4                     | 22                        | 4                    | 71                 |
| 080440 | Avocados, fresh/dried                            | 0.14                                 | 13                    | 29                        | 3                    | 55                 |
| 030499 | Fish fillets&other fish meat (excl. of 0304.1... | 0.10                                 | 20                    | 15                        | 19                   | 47                 |

*Notes:* Fifteen largest HS6 products exported to the United States; percent of each product's exports to the United States by type of exporter. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 5 Fact 4: A Small Set of Large Multinational Groups Accounts for Most Exports

### 5.1 Size of the Group's Global Network

**Figure 32:** Agriculture: Foreign-Multinational Exports and Number of Parents, by Size of the Group's Global Network

*Notes:* Shares of foreign-MNE agricultural export value and of parents, by the number of affiliates the group has worldwide.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

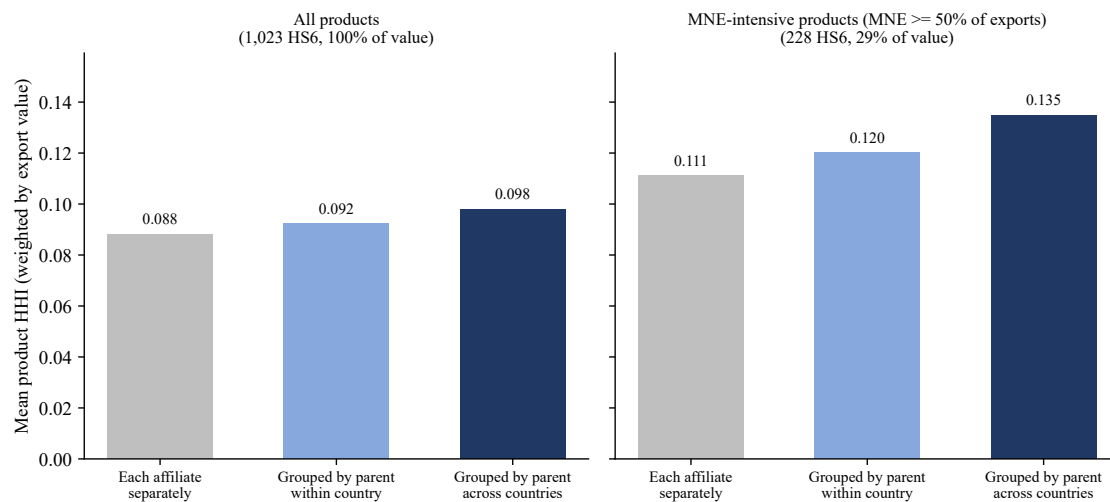
**Table 37:** Agriculture: Foreign-Multinational Exports and Number of Parents, by Size of the Group's Global Network

| Affiliates worldwide | Parents | Share of parents (%) | Share of foreign-MNE export value (%) |
|----------------------|---------|----------------------|---------------------------------------|
| 1                    | 1,436   | 45.1                 | 8.9                                   |
| 2–10                 | 904     | 28.4                 | 11.2                                  |
| 11–100               | 459     | 14.4                 | 29.6                                  |
| >100                 | 383     | 12.0                 | 50.3                                  |
| All parents          | 3,182   | 100                  | 100                                   |

Notes: Number and share of parents, and share of foreign-MNE agricultural export value, by the number of affiliates the group has worldwide.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 5.2 Concentration within Products

**Figure 33:** Agriculture: Export Concentration within Products: Firms and Groups

Notes: Mean Herfindahl index of exporters within each HS6 product, weighted by export value, counting each affiliate separately or grouping affiliates by parent within or across countries. Right panel: products where MNEs account for at least half of exports.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 38:** Agriculture: Export Concentration within Products: Firms and Groups

| Sample                                                                      | Measure                               | Firm level | Groups within country | Groups across countries |
|-----------------------------------------------------------------------------|---------------------------------------|------------|-----------------------|-------------------------|
| <i>All products: 1,023 HS6, 100% of value</i>                               |                                       |            |                       |                         |
|                                                                             | Mean HHI                              | 0.088      | 0.092                 | 0.098                   |
|                                                                             | Mean top-exporter share               | 17.7%      | 18.1%                 | 19.2%                   |
|                                                                             | Effective number of exporters (1/HHI) | 18.5       | 17.3                  | 16.4                    |
| <i>MNE-intensive products (MNE ≥ 50% of exports): 228 HS6, 29% of value</i> |                                       |            |                       |                         |
|                                                                             | Mean HHI                              | 0.111      | 0.120                 | 0.135                   |
|                                                                             | Mean top-exporter share               | 20.7%      | 21.6%                 | 24.4%                   |
|                                                                             | Effective number of exporters (1/HHI) | 15.8       | 14.1                  | 12.8                    |

Notes: Means across HS6 products, weighted by export value. Firm level: each affiliate counted separately. Groups: affiliates grouped by ultimate parent within each exporting country or across countries.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

### 5.3 Groups and Affiliates

**Table 39:** Agriculture: Multinational Affiliates and Groups in Origin–Destination–Product–Year Cells

| Origin-destination-product-year cells with $\geq 1$ MNE   |         |
|-----------------------------------------------------------|---------|
| Cells                                                     | 154,256 |
| MNE affiliates per cell (mean)                            | 2.17    |
| MNE groups per cell (mean)                                | 2.13    |
| Affiliates per group, pooled                              | 1.02    |
| Affiliates per group, mean weighted by export value       | 1.06    |
| Cells where one group exports through $> 1$ affiliate (%) | 2.6     |

*Notes:* Cells with at least one MNE. Group: ultimate parent in Orbis or Dun & Bradstreet (parent name when the identifier is missing).

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 40:** Agriculture: Multinational Presence and Exports: Affiliates and Groups

|                                                               | Origin $\times$ year, destination $\times$ year, product FE |                       |                       | Origin $\times$ dest. $\times$ product, origin $\times$ dest. $\times$ year FE |                       |                       |
|---------------------------------------------------------------|-------------------------------------------------------------|-----------------------|-----------------------|--------------------------------------------------------------------------------|-----------------------|-----------------------|
|                                                               | (1)                                                         | (2)                   | (3)                   | (4)                                                                            | (5)                   | (6)                   |
| <i>Panel A. Dependent variable: ln exports, all firms</i>     |                                                             |                       |                       |                                                                                |                       |                       |
| ln(# MNE firms)                                               | 2.1861***<br>(0.0279)                                       |                       |                       | 0.8509***<br>(0.0190)                                                          |                       |                       |
| ln(# MNE groups)                                              |                                                             | 2.2049***<br>(0.0283) | 2.1827***<br>(0.0285) |                                                                                | 0.8525***<br>(0.0191) | 0.8528***<br>(0.0191) |
| ln(affiliates per group)                                      |                                                             |                       | 2.3343***<br>(0.2073) |                                                                                |                       | 0.7524***<br>(0.0777) |
| <i>Panel B. Dependent variable: ln exports, non-MNE firms</i> |                                                             |                       |                       |                                                                                |                       |                       |
| ln(# MNE firms)                                               | 1.5628***<br>(0.0332)                                       |                       |                       | 0.4048***<br>(0.0245)                                                          |                       |                       |
| ln(# MNE groups)                                              |                                                             | 1.5822***<br>(0.0335) | 1.5743***<br>(0.0338) |                                                                                | 0.4073***<br>(0.0241) | 0.4073***<br>(0.0241) |
| ln(affiliates per group)                                      |                                                             |                       | 0.9591***<br>(0.2648) |                                                                                |                       | 0.2200*<br>(0.1333)   |
| Product FE                                                    | ✓                                                           | ✓                     | ✓                     |                                                                                |                       |                       |
| Origin $\times$ year FE                                       | ✓                                                           | ✓                     | ✓                     |                                                                                |                       |                       |
| Destination $\times$ year FE                                  | ✓                                                           | ✓                     | ✓                     |                                                                                |                       |                       |
| Origin $\times$ dest. $\times$ product FE                     |                                                             |                       |                       | ✓                                                                              | ✓                     | ✓                     |
| Origin $\times$ dest. $\times$ year FE                        |                                                             |                       |                       | ✓                                                                              | ✓                     | ✓                     |
| Observations (Panel A)                                        | 153,847                                                     | 153,847               | 153,847               | 137,324                                                                        | 137,324               | 137,324               |
| Observations (Panel B)                                        | 90,655                                                      | 90,655                | 90,655                | 81,670                                                                         | 81,670                | 81,670                |

*Notes:* Observations are origin–destination–HS6–year cells. Agriculture products only. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Regressors: ln number of MNE affiliates or ln number of MNE groups (distinct ultimate parents). Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 6 Fact 5: Multinational Presence Is Associated with Higher Exports

**Table 41:** Agriculture: Multinational Presence and Exports

|                                                               | Intensive: ln(# MNE firms) |                       |                       | Extensive: any MNE present |                       |                       |
|---------------------------------------------------------------|----------------------------|-----------------------|-----------------------|----------------------------|-----------------------|-----------------------|
|                                                               | (1)                        | (2)                   | (3)                   | (1)                        | (2)                   | (3)                   |
| <i>Panel A. Dependent variable: ln exports, all firms</i>     |                            |                       |                       |                            |                       |                       |
| ln(# MNE firms)                                               | 2.1821***<br>(0.0283)      | 2.1861***<br>(0.0279) | 0.8509***<br>(0.0190) |                            |                       |                       |
| Any MNE present                                               |                            |                       |                       | 1.7857***<br>(0.0611)      | 1.7863***<br>(0.0604) | 0.6860***<br>(0.0232) |
| <i>Panel B. Dependent variable: ln exports, non-MNE firms</i> |                            |                       |                       |                            |                       |                       |
| ln(# MNE firms)                                               | 1.5538***<br>(0.0332)      | 1.5628***<br>(0.0332) | 0.4048***<br>(0.0245) |                            |                       |                       |
| Any MNE present                                               |                            |                       |                       | 1.9365***<br>(0.0567)      | 1.9313***<br>(0.0554) | 0.2812***<br>(0.0181) |
| Origin FE                                                     | ✓                          |                       |                       | ✓                          |                       |                       |
| Destination FE                                                | ✓                          |                       |                       | ✓                          |                       |                       |
| Product FE                                                    | ✓                          | ✓                     |                       | ✓                          | ✓                     |                       |
| Year FE                                                       | ✓                          |                       |                       | ✓                          |                       |                       |
| Origin × year FE                                              |                            | ✓                     |                       |                            | ✓                     |                       |
| Destination × year FE                                         |                            | ✓                     |                       |                            | ✓                     |                       |
| Origin × dest. × product FE                                   |                            |                       | ✓                     |                            |                       | ✓                     |
| Origin × dest. × year FE                                      |                            |                       | ✓                     |                            |                       | ✓                     |
| Observations (Panel A)                                        | 154,199                    | 153,847               | 137,324               | 296,092                    | 295,788               | 268,332               |
| Observations (Panel B)                                        | 91,020                     | 90,655                | 81,670                | 232,907                    | 232,617               | 208,479               |

*Notes:* Observations are origin–destination–HS6–year cells. Agriculture products only. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Intensive margin: ln number of MNEs, cells with at least one MNE. Extensive margin: indicator for at least one MNE. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 42:** Agriculture: Multinational Exports by Exporting Country and Destination Type

| Origin | MNE value (US\$bn/yr) | of which domestic (%) | Foreign-MNE value (US\$bn/yr) | % of foreign-MNE value, by destination |                                         |
|--------|-----------------------|-----------------------|-------------------------------|----------------------------------------|-----------------------------------------|
|        |                       |                       |                               | parent country                         | other (of which: group affiliate there) |
| ARG    | 18.1                  | 20.4                  | 14.4                          | 2.8                                    | 97.2 (30.6)                             |
| CHL    | 3.6                   | 38.7                  | 2.2                           | 12.1                                   | 87.9 (29.7)                             |
| COL    | 3.0                   | 33.6                  | 2.0                           | 4.4                                    | 95.6 (33.7)                             |
| CRI    | 2.0                   | 3.3                   | 1.9                           | 10.0                                   | 90.0 (45.9)                             |
| DOM    | 0.9                   | 8.8                   | 0.8                           | 13.7                                   | 86.3 (9.7)                              |
| PER    | 3.0                   | 11.7                  | 2.7                           | 4.7                                    | 95.3 (15.5)                             |
| PRY    | 4.1                   | 1.3                   | 4.0                           | 1.9                                    | 98.1 (67.8)                             |
| SLV    | 0.3                   | 0.0                   | 0.3                           | 9.5                                    | 90.5 (58.9)                             |
| URY    | 2.7                   | 15.9                  | 2.3                           | 2.8                                    | 97.2 (42.8)                             |
| All    | 37.8                  | 19.9                  | 30.7                          | 4.7                                    | 95.3 (35.8)                             |

*Notes:* MNE and foreign-MNE agricultural export value by country, and the split of foreign-MNE value by destination. Parent country: exports to the country of the ultimate parent. Other: all remaining destinations; in parentheses, destinations where the group has another affiliate. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 43:** Agriculture: Multinational Presence and Exports: Intensive Margin, by Type of Multinational

|                                                               | All MNEs              |                       | Foreign / domestic    |                       | Foreign by destination, domestic |                       |
|---------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------------------------|-----------------------|
|                                                               | (1)                   | (2)                   | (3)                   | (4)                   | (5)                              | (6)                   |
| <i>Panel A. Dependent variable: ln exports, all firms</i>     |                       |                       |                       |                       |                                  |                       |
| ln(# MNE firms)                                               | 2.1861***<br>(0.0279) | 0.8509***<br>(0.0190) |                       |                       |                                  |                       |
| ln(# foreign MNEs)                                            |                       |                       | 1.2954***<br>(0.0275) | 0.4625***<br>(0.0188) |                                  |                       |
| ln(# foreign MNEs, parent-country destination)                |                       |                       |                       |                       | 0.4928***<br>(0.1145)            | 0.2334***<br>(0.0804) |
| ln(# foreign MNEs, other destinations)                        |                       |                       |                       |                       | 0.9569***<br>(0.0754)            | 0.3804***<br>(0.0360) |
| ln(# domestic MNEs)                                           |                       |                       | 0.5233***<br>(0.0266) | 0.2831***<br>(0.0234) | 0.4096***<br>(0.0753)            | 0.1935***<br>(0.0421) |
| <i>Panel B. Dependent variable: ln exports, non-MNE firms</i> |                       |                       |                       |                       |                                  |                       |
| ln(# MNE firms)                                               | 1.5628***<br>(0.0332) | 0.4048***<br>(0.0245) |                       |                       |                                  |                       |
| ln(# foreign MNEs)                                            |                       |                       | 1.2191***<br>(0.0363) | 0.3602***<br>(0.0292) |                                  |                       |
| ln(# foreign MNEs, parent-country destination)                |                       |                       |                       |                       | 0.3864***<br>(0.1357)            | 0.2698**<br>(0.1255)  |
| ln(# foreign MNEs, other destinations)                        |                       |                       |                       |                       | 1.1003***<br>(0.1197)            | 0.4395***<br>(0.0693) |
| ln(# domestic MNEs)                                           |                       |                       | 0.5676***<br>(0.0339) | 0.2461***<br>(0.0341) | 0.4367***<br>(0.1236)            | 0.2032**<br>(0.0759)  |
| Product FE                                                    | ✓                     |                       | ✓                     |                       | ✓                                |                       |
| Origin × year FE                                              | ✓                     |                       | ✓                     |                       | ✓                                |                       |
| Destination × year FE                                         | ✓                     |                       | ✓                     |                       | ✓                                |                       |
| Origin × dest. × product FE                                   |                       | ✓                     |                       | ✓                     |                                  | ✓                     |
| Origin × dest. × year FE                                      |                       | ✓                     |                       | ✓                     |                                  | ✓                     |
| Observations (Panel A)                                        | 153,847               | 137,324               | 25,437                | 22,002                | 1,952                            | 1,604                 |
| Observations (Panel B)                                        | 90,655                | 81,670                | 22,503                | 19,535                | 1,903                            | 1,567                 |

*Notes:* Observations are origin–destination–HS6–year cells. Agriculture products only. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Regressors: ln number of MNEs of each type. Columns (5)–(6) split foreign MNEs by whether the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 44:** Agriculture: Multinational Presence and Exports: Extensive Margin, by Type of Multinational

|                                                               | All MNEs              |                       | Foreign / domestic    |                       | Foreign by destination, domestic |                       |
|---------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------------------------|-----------------------|
|                                                               | (1)                   | (2)                   | (3)                   | (4)                   | (5)                              | (6)                   |
| <i>Panel A. Dependent variable: ln exports, all firms</i>     |                       |                       |                       |                       |                                  |                       |
| Any MNE present                                               | 1.7863***<br>(0.0604) | 0.6860***<br>(0.0232) |                       |                       |                                  |                       |
| Any foreign MNE                                               |                       |                       | 1.7972***<br>(0.0558) | 0.6929***<br>(0.0213) |                                  |                       |
| Any foreign MNE, parent-country destination                   |                       |                       |                       |                       | 1.6277***<br>(0.1122)            | 0.5045***<br>(0.0603) |
| Any foreign MNE, other destination                            |                       |                       |                       |                       | 1.7886***<br>(0.0563)            | 0.6904***<br>(0.0217) |
| Any domestic MNE                                              |                       |                       | 1.6133***<br>(0.0577) | 0.5683***<br>(0.0237) | 1.5621***<br>(0.0556)            | 0.5650***<br>(0.0238) |
| <i>Panel B. Dependent variable: ln exports, non-MNE firms</i> |                       |                       |                       |                       |                                  |                       |
| Any MNE present                                               | 1.9313***<br>(0.0554) | 0.2812***<br>(0.0181) |                       |                       |                                  |                       |
| Any foreign MNE                                               |                       |                       | 1.7649***<br>(0.0546) | 0.2854***<br>(0.0179) |                                  |                       |
| Any foreign MNE, parent-country destination                   |                       |                       |                       |                       | 1.4262***<br>(0.1036)            | 0.2467***<br>(0.0420) |
| Any foreign MNE, other destination                            |                       |                       |                       |                       | 1.7354***<br>(0.0537)            | 0.2888***<br>(0.0179) |
| Any domestic MNE                                              |                       |                       | 1.3082***<br>(0.0539) | 0.2610***<br>(0.0201) | 1.2541***<br>(0.0504)            | 0.2590***<br>(0.0201) |
| Product FE                                                    | ✓                     |                       | ✓                     |                       | ✓                                |                       |
| Origin × year FE                                              | ✓                     |                       | ✓                     |                       | ✓                                |                       |
| Destination × year FE                                         | ✓                     |                       | ✓                     |                       | ✓                                |                       |
| Origin × dest. × product FE                                   |                       | ✓                     |                       | ✓                     |                                  | ✓                     |
| Origin × dest. × year FE                                      |                       | ✓                     |                       | ✓                     |                                  | ✓                     |
| Observations (Panel A)                                        | 295,788               | 268,332               | 295,788               | 268,332               | 295,788                          | 268,332               |
| Observations (Panel B)                                        | 232,617               | 208,479               | 232,617               | 208,479               | 232,617                          | 208,479               |

*Notes:* Observations are origin–destination–HS6–year cells. Agriculture products only. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Regressors: indicators for at least one MNE of each type. Columns (5)–(6) split foreign MNEs by whether the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 7 Fact 6: Distance Is a Weaker Barrier for Multinationals

**Table 45:** Agriculture: Distance and Firm Exports: Multinationals by Presence at the Destination

| Dep. var.: ln exports                         | (1)                    | (2)                    | (3)                    | (4)                    |
|-----------------------------------------------|------------------------|------------------------|------------------------|------------------------|
| ln distance                                   | -0.2118***<br>(0.0748) | -0.2269***<br>(0.0712) | -0.2078***<br>(0.0745) | -0.2224***<br>(0.0708) |
| ln distance × MNE                             | 0.0326***<br>(0.0031)  | 0.0318***<br>(0.0031)  |                        |                        |
| ln distance × MNE, present at destination     |                        |                        | 0.0890***<br>(0.0070)  | 0.0875***<br>(0.0069)  |
| ln distance × MNE, not present at destination |                        |                        | 0.0228***<br>(0.0032)  | 0.0221***<br>(0.0032)  |
| Origin FE                                     | ✓                      |                        | ✓                      |                        |
| Destination FE                                | ✓                      |                        | ✓                      |                        |
| Year FE                                       | ✓                      |                        | ✓                      |                        |
| Origin × year FE                              |                        | ✓                      |                        | ✓                      |
| Destination × year FE                         |                        | ✓                      |                        | ✓                      |
| Product FE                                    | ✓                      | ✓                      | ✓                      | ✓                      |
| Observations                                  | 1,045,009              | 1,044,787              | 1,045,009              | 1,044,787              |

*Notes:* Observations are firm–destination–HS6–year cells. Agriculture products only. Dependent variable: ln exports. Non-MNE firms are the omitted group. Present: the group has an affiliate in the destination or the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

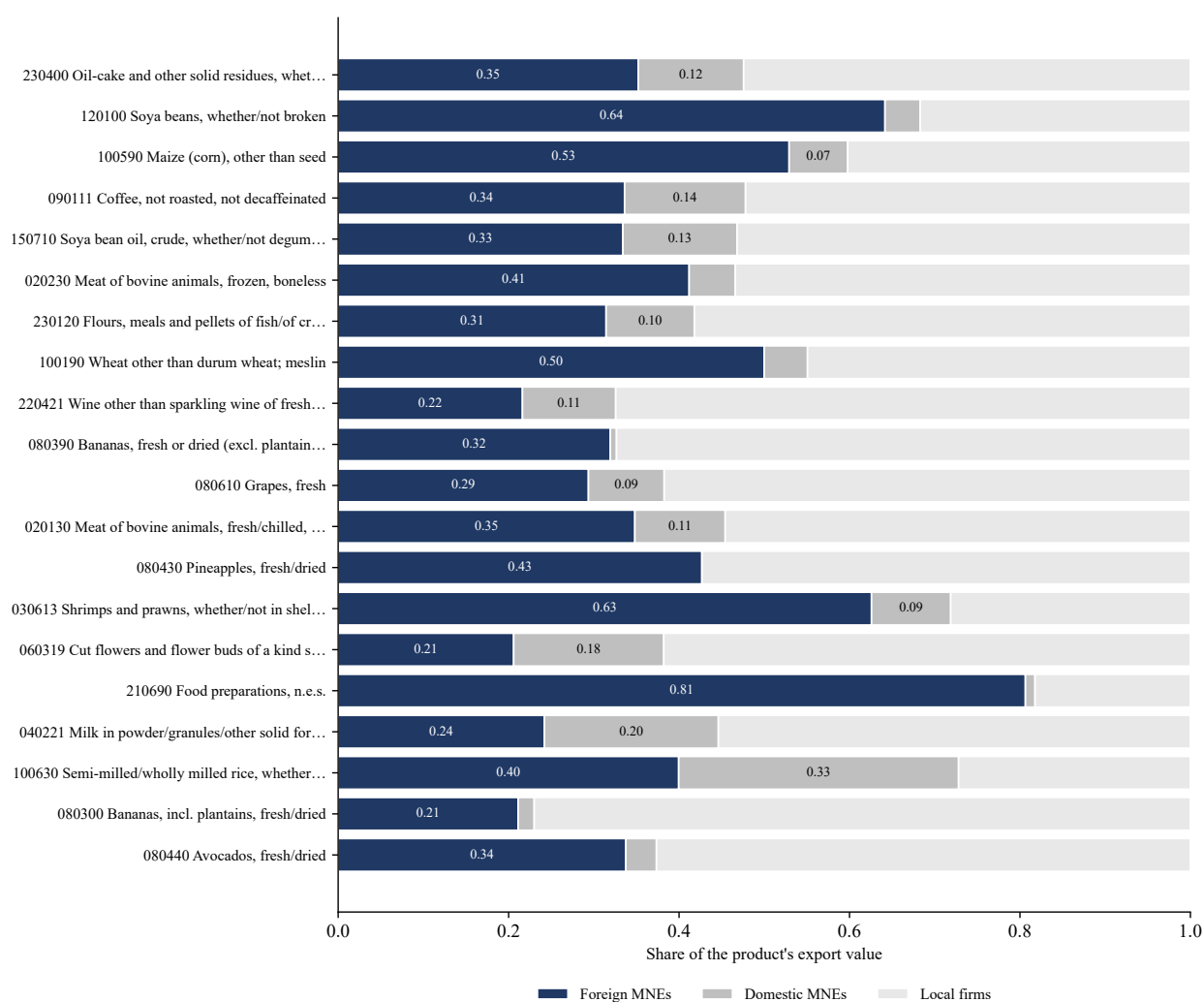
**Table 46:** Agriculture: Distance and Firm Exports: Foreign and Domestic Multinationals

| Dep. var.: ln exports                                        | (1)                    | (2)                    | (3)                    | (4)                    |
|--------------------------------------------------------------|------------------------|------------------------|------------------------|------------------------|
| ln distance                                                  | -0.2135***<br>(0.0748) | -0.2285***<br>(0.0711) | -0.2130***<br>(0.0748) | -0.2280***<br>(0.0711) |
| ln distance $\times$ foreign MNE                             | 0.0366***<br>(0.0034)  | 0.0358***<br>(0.0034)  |                        |                        |
| ln distance $\times$ foreign MNE, parent-country destination |                        |                        | 0.0610***<br>(0.0165)  | 0.0590***<br>(0.0165)  |
| ln distance $\times$ foreign MNE, other destination          |                        |                        | 0.0359***<br>(0.0035)  | 0.0351***<br>(0.0034)  |
| ln distance $\times$ domestic MNE                            | 0.0187***<br>(0.0046)  | 0.0182***<br>(0.0046)  | 0.0187***<br>(0.0046)  | 0.0182***<br>(0.0046)  |
| Origin FE                                                    | ✓                      |                        | ✓                      |                        |
| Destination FE                                               | ✓                      |                        | ✓                      |                        |
| Year FE                                                      | ✓                      |                        | ✓                      |                        |
| Origin $\times$ year FE                                      |                        | ✓                      |                        | ✓                      |
| Destination $\times$ year FE                                 |                        | ✓                      |                        | ✓                      |
| Product FE                                                   | ✓                      | ✓                      | ✓                      | ✓                      |
| Observations                                                 | 1,045,009              | 1,044,787              | 1,045,009              | 1,044,787              |

*Notes:* Observations are firm–destination–HS6–year cells. Agriculture products only. Dependent variable: ln exports. Non-MNE firms are the omitted group. Columns (3)–(4) split foreign MNEs by whether the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

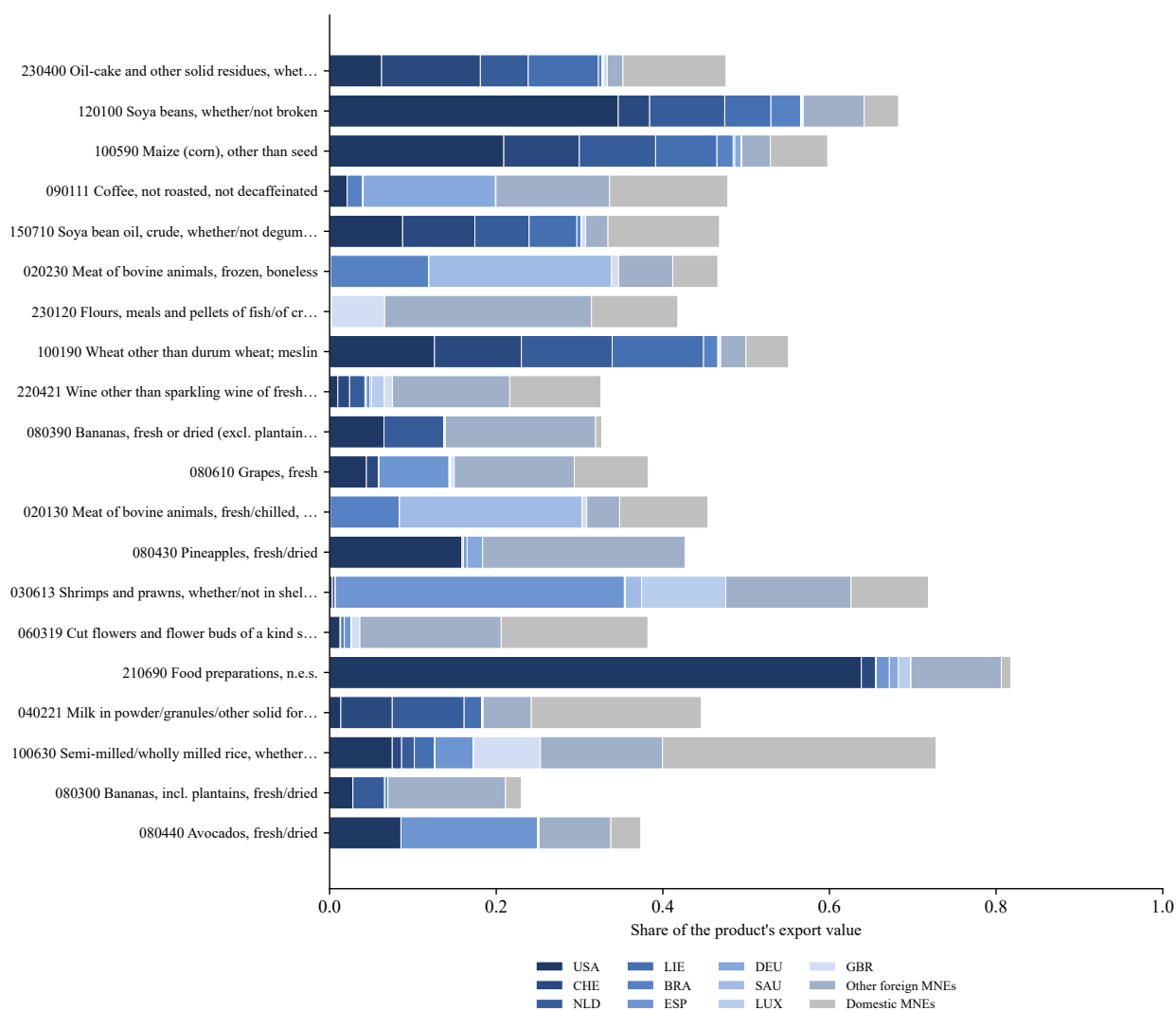
## 8 Products



**Figure 34: Agriculture: Ownership of the Twenty Largest Export Products**

*Notes:* Shares of each product's export value. Twenty largest HS6 products by export value.

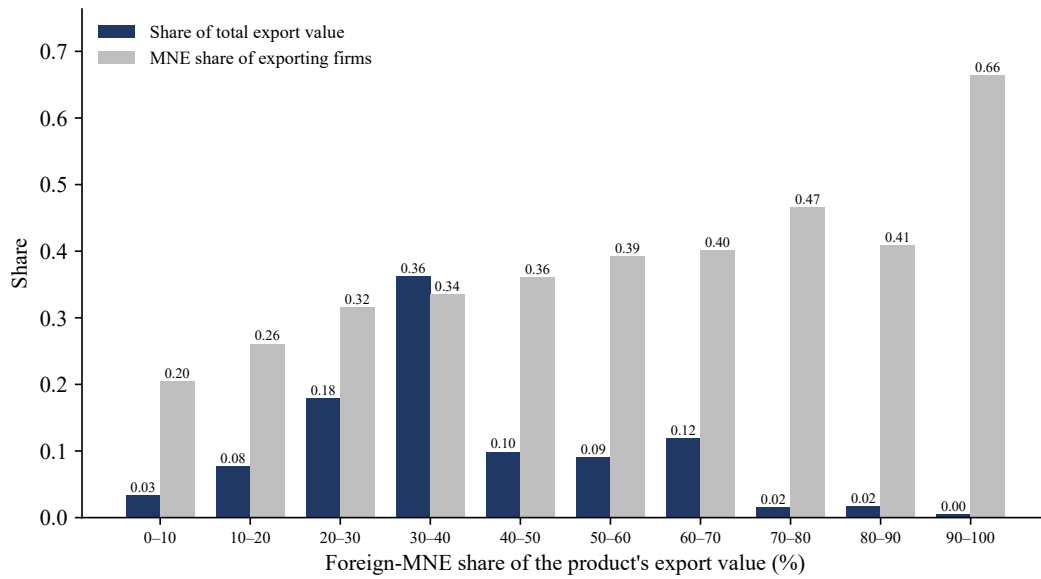
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 35:** Agriculture: Ownership of the Twenty Largest Export Products, by Parent Country

*Notes:* Shares of each product's export value. Foreign-MNE share split by country of the ultimate parent; the unfilled part of each bar is local firms.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 36:** Agriculture: Distribution of Export Value by Product-Level Foreign-Multinational Share

*Notes:* Share of agricultural export value and MNE share of exporting firms, by the foreign-MNE share of each HS6 product's export value.

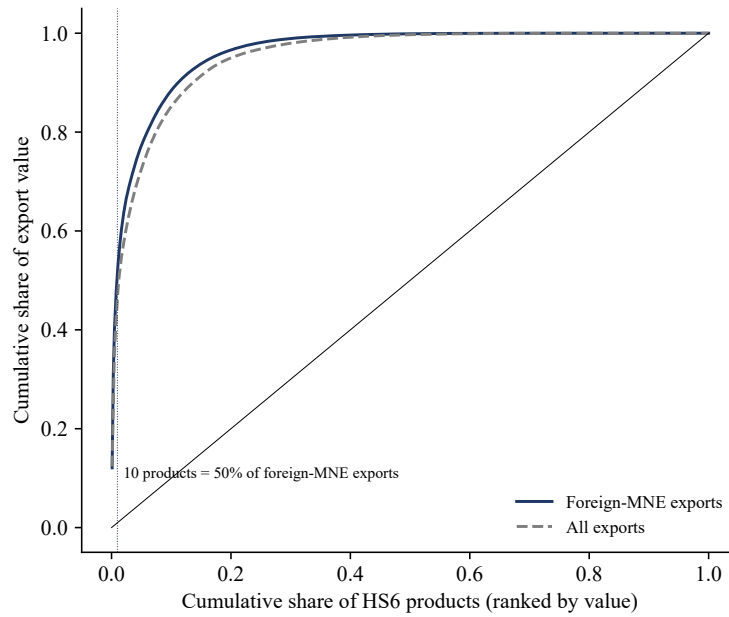
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 47:** Agriculture: Distribution of Export Value by Product-Level Foreign-Multinational Share

| Foreign share bin (%) | Share of value | MNE share of exporting firms | HS6 products |
|-----------------------|----------------|------------------------------|--------------|
| 0-10                  | 0.034          | 0.205                        | 349          |
| 10-20                 | 0.077          | 0.261                        | 159          |
| 20-30                 | 0.179          | 0.316                        | 134          |
| 30-40                 | 0.363          | 0.335                        | 103          |
| 40-50                 | 0.099          | 0.361                        | 60           |
| 50-60                 | 0.091          | 0.393                        | 53           |
| 60-70                 | 0.119          | 0.402                        | 49           |
| 70-80                 | 0.016          | 0.466                        | 37           |
| 80-90                 | 0.018          | 0.409                        | 30           |
| 90-100                | 0.005          | 0.664                        | 49           |

*Notes:* Share of agricultural export value and MNE share of exporting firms, by the foreign-MNE share of each HS6 product's export value. MNE share of exporting firms: MNE firm-destination-product-year observations over all observations in the bin.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 37:** Agriculture: Concentration of Exports across Products

*Notes:* Cumulative share of export value against the cumulative share of HS6 products, ranked by value, for foreign-MNE exports and for all exports.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 48:** Agriculture: Exports and Multinational Shares, by HS Section

| HS section                                 | US\$bn/yr | Foreign (%) | Domestic (%) | Local (%) | Leading parent |
|--------------------------------------------|-----------|-------------|--------------|-----------|----------------|
| II Vegetable products                      | 33.5      | 42          | 7            | 51        | USA (0.33)     |
| IV Prepared foodstuffs, beverages, tobacco | 25.9      | 37          | 10           | 53        | USA (0.21)     |
| I Live animals; animal products            | 12.5      | 34          | 14           | 51        | SAU (0.19)     |
| III Fats and oils                          | 6.9       | 31          | 10           | 59        | CHE (0.26)     |
| All sections                               | 78.9      | 38          | 9            | 52        |                |

*Notes:* Foreign, domestic and local: percent of each section's export value. Leading parent: largest parent country and its share of the section's foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 49: Agriculture: Thirty Largest Export Products**

| HS6    | Description                                                                                        | US\$bn/yr | Foreign (%) | Domestic (%) | Local (%) | Leading parent |
|--------|----------------------------------------------------------------------------------------------------|-----------|-------------|--------------|-----------|----------------|
| 230400 | Oil-cake and other solid residues, whether/not ground/in pellets, from extraction of soyabean ...  | 10.8      | 35          | 12           | 52        | CHE (0.34)     |
| 120100 | Soya beans, whether/not broken                                                                     | 6.2       | 64          | 4            | 32        | USA (0.54)     |
| 100590 | Maize (corn), other than seed                                                                      | 4.8       | 53          | 7            | 40        | USA (0.40)     |
| 090111 | Coffee, not roasted, not decaffeinated                                                             | 3.7       | 34          | 14           | 52        | DEU (0.47)     |
| 150710 | Soya bean oil, crude, whether/not degummed, not chemically modified                                | 4.3       | 33          | 13           | 53        | USA (0.26)     |
| 020230 | Meat of bovine animals, frozen, boneless                                                           | 2.1       | 41          | 5            | 53        | SAU (0.53)     |
| 230120 | Flours, meals and pellets of fish/of crustaceans, molluscs/other aquatic invertebrates             | 1.7       | 31          | 10           | 58        | BMU (0.38)     |
| 100190 | Wheat other than durum wheat; meslin                                                               | 1.9       | 50          | 5            | 45        | USA (0.25)     |
| 220421 | Wine other than sparkling wine of fresh grapes, incl. fortified; grape must with fermentation ...  | 1.4       | 22          | 11           | 67        | FRA (0.25)     |
| 080390 | Bananas, fresh or dried (excl. plantains) [HS2012 code]                                            | 1.4       | 32          | 1            | 67        | JOR (0.38)     |
| 080610 | Grapes, fresh                                                                                      | 1.1       | 29          | 9            | 62        | ESP (0.29)     |
| 020130 | Meat of bovine animals, fresh/chilled, boneless                                                    | 1.4       | 35          | 11           | 55        | SAU (0.63)     |
| 080430 | Pineapples, fresh/dried                                                                            | 0.9       | 43          | 0            | 57        | JOR (0.47)     |
| 030613 | Shrimps and prawns, whether/not in shell, frozen                                                   | 0.9       | 63          | 9            | 28        | ESP (0.55)     |
| 060319 | Cut flowers and flower buds of a kind suit. for bouquets/ornamental purposes(excl. of 0603.11-...  | 0.7       | 21          | 18           | 62        | ARG (0.30)     |
| 210690 | Food preparations, n.e.s.                                                                          | 0.7       | 81          | 1            | 18        | USA (0.79)     |
| 040221 | Milk in powder/granules/other solid form, unsweetened, fat content by weight >1.5%                 | 0.8       | 24          | 20           | 55        | NLD (0.36)     |
| 100630 | Semi-milled/wholly milled rice, whether/not polished/glazed                                        | 0.6       | 40          | 33           | 27        | CHN (0.25)     |
| 080300 | Bananas, incl. plantains, fresh/dried                                                              | 0.6       | 21          | 2            | 77        | JOR (0.34)     |
| 080440 | Avocados, fresh/dried                                                                              | 0.5       | 34          | 4            | 63        | ESP (0.49)     |
| 081040 | Cranberries, bilberries and other fruits of the genus Vaccinium, fresh                             | 0.5       | 37          | 11           | 52        | ESP (0.39)     |
| 030441 | Fish filletsandother fish meat (excl. of 0304.11-0304.12, whether/not minced), fresh/chilled [...] | 0.4       | 20          | 24           | 56        | JPN (0.50)     |
| 080810 | Apples, fresh                                                                                      | 0.4       | 25          | 14           | 61        | USA (0.39)     |
| 100300 | Barley                                                                                             | 0.6       | 55          | 4            | 41        | USA (0.37)     |
| 240210 | Cigars, cheroots and cigarillos containing tobacco                                                 | 0.6       | 62          | 0            | 38        | HKG (0.40)     |
| 151211 | Sunflower seed/safflower oil, crude                                                                | 0.6       | 19          | 2            | 79        | NLD (0.69)     |
| 230990 | Preparations of a kind used in animal feeding other than dog/cat food put up for RS                | 0.5       | 57          | 3            | 40        | USA (0.44)     |
| 170199 | Cane/beet sugar and chemically pure sucrose, in solid form, not containing added flavouring/co...  | 0.4       | 19          | 0            | 81        | USA (0.95)     |
| 150420 | Fats and oils and their fractions, of fish, other than liver oils, whether/not refined but not...  | 0.4       | 23          | 12           | 65        | BMU (0.37)     |
| 110710 | Malt, not roasted                                                                                  | 0.5       | 71          | 0            | 29        | LUX (0.86)     |

*Notes:* Thirty largest HS6 products by export value. Foreign, domestic and local: percent of the product's export value. Leading parent: largest parent country and its share of the product's foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 50:** Agriculture: Thirty Largest Export Products of Foreign Multinationals

| HS6    | Description                                                                                       | US\$bn/yr | Foreign (%) | Domestic (%) | Local (%) | Leading parent |
|--------|---------------------------------------------------------------------------------------------------|-----------|-------------|--------------|-----------|----------------|
| 120100 | Soya beans, whether/not broken                                                                    | 6.2       | 64          | 4            | 32        | USA (0.54)     |
| 230400 | Oil-cake and other solid residues, whether/not ground/in pellets, from extraction of soyabean ... | 10.8      | 35          | 12           | 52        | CHE (0.34)     |
| 100590 | Maize (corn), other than seed                                                                     | 4.8       | 53          | 7            | 40        | USA (0.40)     |
| 090111 | Coffee, not roasted, not decaffeinated                                                            | 3.7       | 34          | 14           | 52        | DEU (0.47)     |
| 150710 | Soya bean oil, crude, whether/not degummed, not chemically modified                               | 4.3       | 33          | 13           | 53        | USA (0.26)     |
| 100190 | Wheat other than durum wheat; meslin                                                              | 1.9       | 50          | 5            | 45        | USA (0.25)     |
| 020230 | Meat of bovine animals, frozen, boneless                                                          | 2.1       | 41          | 5            | 53        | SAU (0.53)     |
| 210690 | Food preparations, n.e.s.                                                                         | 0.7       | 81          | 1            | 18        | USA (0.79)     |
| 230120 | Flours, meals and pellets of fish/of crustaceans, molluscs/other aquatic invertebrates            | 1.7       | 31          | 10           | 58        | BMU (0.38)     |
| 030613 | Shrimps and prawns, whether/not in shell, frozen                                                  | 0.9       | 63          | 9            | 28        | ESP (0.55)     |
| 080390 | Bananas, fresh or dried (excl. plantains) [HS2012 code]                                           | 1.4       | 32          | 1            | 67        | JOR (0.38)     |
| 020130 | Meat of bovine animals, fresh/chilled, boneless                                                   | 1.4       | 35          | 11           | 55        | SAU (0.63)     |
| 080610 | Grapes, fresh                                                                                     | 1.1       | 29          | 9            | 62        | ESP (0.29)     |
| 080430 | Pineapples, fresh/dried                                                                           | 0.9       | 43          | 0            | 57        | JOR (0.47)     |
| 220421 | Wine other than sparkling wine of fresh grapes, incl. fortified; grape must with fermentation ... | 1.4       | 22          | 11           | 67        | FRA (0.25)     |
| 240210 | Cigars, cheroots and cigarillos containing tobacco                                                | 0.6       | 62          | 0            | 38        | HKG (0.40)     |
| 110710 | Malt, not roasted                                                                                 | 0.5       | 71          | 0            | 29        | LUX (0.86)     |
| 230990 | Preparations of a kind used in animal feeding other than dog/cat food put up for RS               | 0.5       | 57          | 3            | 40        | USA (0.44)     |
| 100300 | Barley                                                                                            | 0.6       | 55          | 4            | 41        | USA (0.37)     |
| 100630 | Semi-milled/wholly milled rice, whether/not polished/glazed                                       | 0.6       | 40          | 33           | 27        | CHN (0.25)     |
| 120190 | Soya beans, whether/not broken [HS 2007 line 120100]                                              | 0.4       | 69          | 0            | 31        | USA (0.50)     |
| 081040 | Cranberries, bilberries and other fruits of the genus Vaccinium, fresh                            | 0.5       | 37          | 11           | 52        | ESP (0.39)     |
| 202300 | HS 202300 (no description available)                                                              | 0.3       | 68          | 7            | 25        | BRA (0.37)     |
| 080440 | Avocados, fresh/dried                                                                             | 0.5       | 34          | 4            | 63        | ESP (0.49)     |
| 040221 | Milk in powder/granules/other solid form, unsweetened, fat content by weight >1.5%                | 0.8       | 24          | 20           | 55        | NLD (0.36)     |
| 060319 | Cut flowers and flower buds of a kind suit. for bouquets/ornamental purposes(excl. of 0603.11-... | 0.7       | 21          | 18           | 62        | ARG (0.30)     |
| 100700 | Grain sorghum                                                                                     | 0.2       | 74          | 1            | 25        | USA (0.78)     |
| 070920 | Asparagus, fresh/chilled                                                                          | 0.4       | 41          | 0            | 59        | PAN (0.53)     |
| 100199 | Wheat other than durum wheat; meslin [HS 2007 line 100190]                                        | 0.3       | 62          | 2            | 36        | NLD (0.32)     |
| 240120 | Tobacco, partly/wholly stemmed/stripped                                                           | 0.4       | 44          | 0            | 56        | USA (0.74)     |

*Notes:* Thirty largest HS6 products by foreign-MNE export value. Foreign, domestic and local: percent of the product's export value. Leading parent: largest parent country and its share of the product's foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 51: Agriculture: Products with the Highest Foreign-Multinational Share**

| HS6    | Description                                                                                        | US\$bn/yr | Foreign (%) | Domestic (%) | Local (%) | Leading parent |
|--------|----------------------------------------------------------------------------------------------------|-----------|-------------|--------------|-----------|----------------|
| 190110 | Preparations for infant use, put up for RS                                                         | 0.1       | 92          | 3            | 5         | CHE (0.63)     |
| 200410 | Potatoes, prepared/preserved othw. than by vine-gar/acetic acid, frozen, other than products of... | 0.2       | 85          | 0            | 14        | CAN (0.98)     |
| 040291 | Milk and cream, concentrated (excl. in powder), unsweetened                                        | 0.1       | 85          | 0            | 15        | ROU (0.99)     |
| 230910 | Dog/cat food, put up for RS                                                                        | 0.2       | 81          | 5            | 13        | USA (0.60)     |
| 210690 | Food preparations, n.e.s.                                                                          | 0.7       | 81          | 1            | 18        | USA (0.79)     |
| 200949 | Pineapple juice (excl. of 2009.41), unfermented and not containing added spirit, whether/not c...  | 0.1       | 79          | 0            | 20        | PAN (0.69)     |
| 220300 | Beer made from malt                                                                                | 0.1       | 78          | 0            | 22        | LUX (0.80)     |
| 200941 | Pineapple juice, of a Brix value not >20, unfermented and not containing added spirit, whether...  | 0.1       | 75          | 0            | 25        | PAN (0.51)     |
| 100700 | Grain sorghum                                                                                      | 0.2       | 74          | 1            | 25        | USA (0.78)     |
| 220290 | Non-alcoholic beverages other than waters of 2202.10 (not incl. fruit/vegetable juices of 20.0...  | 0.1       | 73          | 3            | 24        | PAN (0.71)     |
| 110710 | Malt, not roasted                                                                                  | 0.5       | 71          | 0            | 29        | LUX (0.86)     |
| 200560 | Asparagus, preserved/preserved othw. than by vinegar/acetic acid, not frozen, other than produ...  | 0.1       | 71          | 0            | 29        | ESP (0.47)     |
| 120190 | Soya beans, whether/not broken [HS 2007 line 120100]                                               | 0.4       | 69          | 0            | 31        | USA (0.50)     |
| 200290 | Tomatoes, prepared/preserved othw. than by vine-gar/acetic acid, other than whole/in pieces        | 0.1       | 68          | 0            | 32        | DEU (0.48)     |
| 202300 | HS 202300 (no description available)                                                               | 0.3       | 68          | 7            | 25        | BRA (0.37)     |
| 120991 | Vegetable seeds, of a kind used for sowing                                                         | 0.1       | 67          | 1            | 32        | DEU (0.54)     |
| 160414 | Tunas, skipjack and bonito (Sarda spp.), prepared/preserved, whole/in pieces (excl. minced)        | 0.1       | 67          | 0            | 33        | ITA (0.97)     |
| 220840 | Rum and tafia                                                                                      | 0.1       | 66          | 2            | 32        | GBR (0.76)     |
| 120100 | Soya beans, whether/not broken                                                                     | 6.2       | 64          | 4            | 32        | USA (0.54)     |
| 200190 | Vegetables, fruit, nuts and other edible parts of plants (excl. cucumbers and gherkins), prepa...  | 0.1       | 64          | 2            | 34        | PAN (0.51)     |
| 151790 | Edible mixtures/preparations of animal/vegetable fats/oils/fractions of different fats/oils of...  | 0.2       | 64          | 5            | 31        | SWE (0.49)     |
| 210390 | Sauces and preparations therefor, n.e.s.; mixed condiments and mixed seasonings, n.e.s.            | 0.2       | 63          | 1            | 35        | USA (0.36)     |
| 030613 | Shrimps and prawns, whether/not in shell, frozen                                                   | 0.9       | 63          | 9            | 28        | ESP (0.55)     |
| 240210 | Cigars, cheroots and cigarillos containing tobacco                                                 | 0.6       | 62          | 0            | 38        | HKG (0.40)     |
| 100199 | Wheat other than durum wheat; meslin [HS 2007 line 100190]                                         | 0.3       | 62          | 2            | 36        | NLD (0.32)     |
| 190190 | Malt extract; food preparations of flour/groats/meal/starch/malt extact,not containing cocoa/c...  | 0.1       | 62          | 5            | 34        | CHE (0.50)     |
| 200979 | Apple juice (excl. of 2009.71), unfermented and not containing added spirit, whether/not conta...  | 0.1       | 62          | 1            | 37        | DEU (0.75)     |
| 201300 | HS 201300 (no description available)                                                               | 0.1       | 60          | 13           | 27        | BRA (0.44)     |
| 100510 | Maize (corn), seed                                                                                 | 0.2       | 60          | 3            | 37        | DEU (0.57)     |
| 120600 | Sunflower seeds, whether/not broken                                                                | 0.1       | 58          | 4            | 38        | USA (0.31)     |

*Notes:* HS6 products with exports of at least US\$500 million over the period, ranked by foreign-MNE share. Foreign, domestic and local: percent of the product's export value. Leading parent: largest parent country and its share of the product's foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 52:** Agriculture: Products with the Highest Domestic-Multinational Share

| HS6    | Description                                                                                                         | US\$bn/yr | Foreign (%) | Domestic (%) | Local (%) | Leading parent |
|--------|---------------------------------------------------------------------------------------------------------------------|-----------|-------------|--------------|-----------|----------------|
| 020727 | Cuts and edible offal of turkey, frozen                                                                             | 0.04      | 1           | 82           | 17        | JPN (0.45)     |
| 040610 | Fresh (unripened/uncured) cheese, incl. whey cheese, and curd                                                       | 0.11      | 12          | 66           | 22        | MUS (0.36)     |
| 020329 | Meat of swine (excl. carcasses/half-carcasses/hams/shoulders and cuts thereof ), froxen                             | 0.19      | 29          | 54           | 17        | JPN (1.00)     |
| 210111 | Extracts, essences and concentrates of coffee                                                                       | 0.24      | 23          | 48           | 29        | CHE (0.70)     |
| 030319 | Pacific salmon ( <i>Oncorhynchus gorbuscha/keta/tshawytscha/kisutch/masou/rhodurus</i> ), frozen (excl...           | 0.05      | 8           | 44           | 47        | CAN (0.39)     |
| 180400 | Cocoa butter, fat and oil                                                                                           | 0.07      | 1           | 38           | 61        | ARG (0.58)     |
| 170490 | Sugar confectionery other than chewing gum (incl. white chocolate), not containing cocoa                            | 0.32      | 9           | 37           | 54        | USA (0.67)     |
| 100630 | Semi-milled/wholly milled rice, whether/not polished/glazed                                                         | 0.63      | 40          | 33           | 27        | CHN (0.25)     |
| 100640 | Broken rice                                                                                                         | 0.07      | 39          | 32           | 28        | USA (0.24)     |
| 020714 | Cuts and edible offal of species <i>Gallus domesticus</i> , frozen                                                  | 0.33      | 20          | 31           | 49        | MCO (0.46)     |
| 040510 | Butter                                                                                                              | 0.11      | 12          | 28           | 60        | PER (0.47)     |
| 030312 | Pacific salmon ( <i>Oncorhynchus gorbuscha/keta/tshawytscha/kisutch/masou/rhodurus</i> ), frozen (excl...           | 0.23      | 23          | 27           | 50        | JPN (0.78)     |
| 030321 | Trout ( <i>Salmo trutta</i> , <i>Oncorhynchus mykiss/clarki/aguabonita/gilae/apache/chrysogaster</i> ), frozen ...  | 0.05      | 17          | 27           | 56        | JPN (0.64)     |
| 090300 | MatT                                                                                                                | 0.09      | 22          | 27           | 51        | CAN (0.25)     |
| 040690 | Cheese (excl. of 0406.10-0406.40)                                                                                   | 0.28      | 25          | 25           | 50        | ALB (0.29)     |
| 040410 | Whey and modified whey, whether/not concentrated/sweetened                                                          | 0.12      | 30          | 25           | 45        | DNK (0.73)     |
| 100620 | Husked (brown) rice                                                                                                 | 0.13      | 34          | 24           | 41        | USA (0.35)     |
| 030441 | Fish filletsandother fish meat (excl. of 0304.11-0304.12, whether/not minced), fresh/chilled [...]                  | 0.40      | 20          | 24           | 56        | JPN (0.50)     |
| 030419 | Fish filletsandother fish meat (excl. of 0304.11-0304.12, whether/not minced), fresh/chilled                        | 0.05      | 14          | 23           | 63        | CAN (0.40)     |
| 090240 | Tea, black (fermented) and partly fermented tea, whether/not flavoured, in immediate packings ...                   | 0.10      | 13          | 23           | 64        | TUR (0.32)     |
| 030481 | Fish filletsandother fish meat (excl. of 0304.21-0304.92, whether/not minced), frozen fillets ...                   | 0.25      | 15          | 22           | 63        | JPN (0.38)     |
| 200570 | Olives, preserved/preserved othw. than by vinegar/acetic acid, not frozen, other than products...                   | 0.11      | 24          | 21           | 55        | ESP (0.25)     |
| 030314 | Trout ( <i>Salmo trutta</i> , <i>Oncorhynchus mykiss/clarki/aguabonita/gilae/apache/chrysogaster</i> ), frozen ...  | 0.07      | 54          | 21           | 26        | JPN (0.66)     |
| 030313 | Atlantic salmon ( <i>Salmo salar</i> ) and Danube salmon ( <i>Hucho hucho</i> ), frozen (excl. fillets/other fis... | 0.20      | 33          | 21           | 46        | JPN (0.42)     |
| 030214 | Pacific salmon/Atlantic salmon/Danube salmon [see list of conventions for species incld.], fre...                   | 0.25      | 21          | 21           | 58        | JPN (0.42)     |
| 080940 | Plums and sloes, fresh                                                                                              | 0.07      | 17          | 20           | 62        | CAN (0.37)     |
| 040221 | Milk in powder/granules/other solid form, unsweetened, fat content by weight >1.5%                                  | 0.81      | 24          | 20           | 55        | NLD (0.36)     |
| 160553 | Molluscs and other aquatic invertebrates, prepared/preserved [HS 2007 line ...]                                     | 0.08      | 46          | 20           | 34        | ESP (0.67)     |
| 080930 | Peaches, incl. nectarines, fresh                                                                                    | 0.06      | 13          | 19           | 67        | CAN (0.41)     |
| 081050 | Kiwifruit, fresh                                                                                                    | 0.10      | 20          | 19           | 61        | USA (0.49)     |

*Notes:* HS6 products with exports of at least US\$500 million over the period, ranked by domestic-MNE share. Foreign, domestic and local: percent of the product's export value. Leading parent: largest parent country and its share of the product's foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 9 Product Groups within the Sector

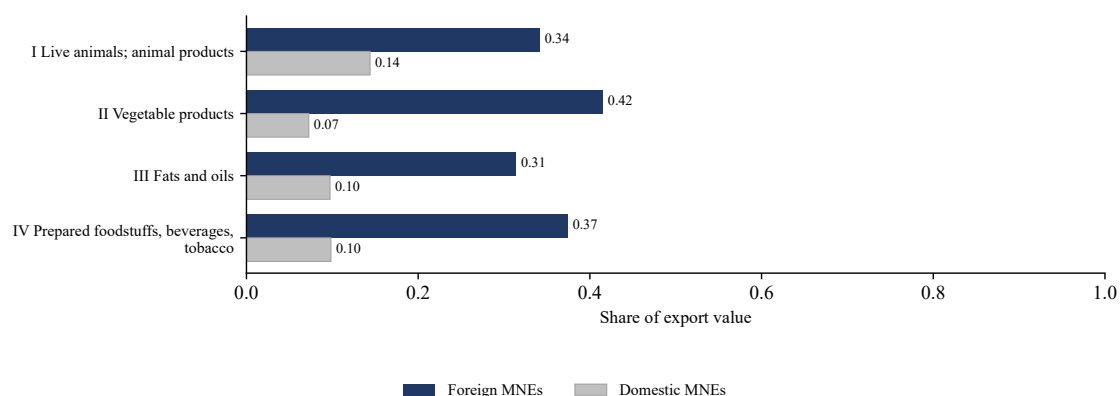
### 9.1 By HS Section

**Table 53:** Agriculture: Exports and Multinational Shares, by HS Section

| HS section                                 | Total exports (US\$bn/yr) | MNE exports (US\$bn/yr) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|--------------------------------------------|---------------------------|-------------------------|-------------|--------------|------------------------------------|
| I Live animals; animal products            | 12.5                      | 6.0                     | 34.1        | 14.4         | SAU 0.19, JPN 0.14, ESP 0.10       |
| II Vegetable products                      | 33.5                      | 16.6                    | 41.5        | 7.3          | USA 0.33, NLD 0.11, LIE 0.07       |
| III Fats and oils                          | 6.9                       | 2.9                     | 31.4        | 9.7          | CHE 0.26, NLD 0.21, USA 0.19       |
| IV Prepared foodstuffs, beverages, tobacco | 25.9                      | 12.3                    | 37.4        | 9.8          | USA 0.21, CHE 0.17, LIE 0.10       |
| All                                        | 78.9                      | 37.8                    | 38.1        | 9.5          |                                    |

*Notes:* HS sections. Foreign and domestic: percent of the group's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

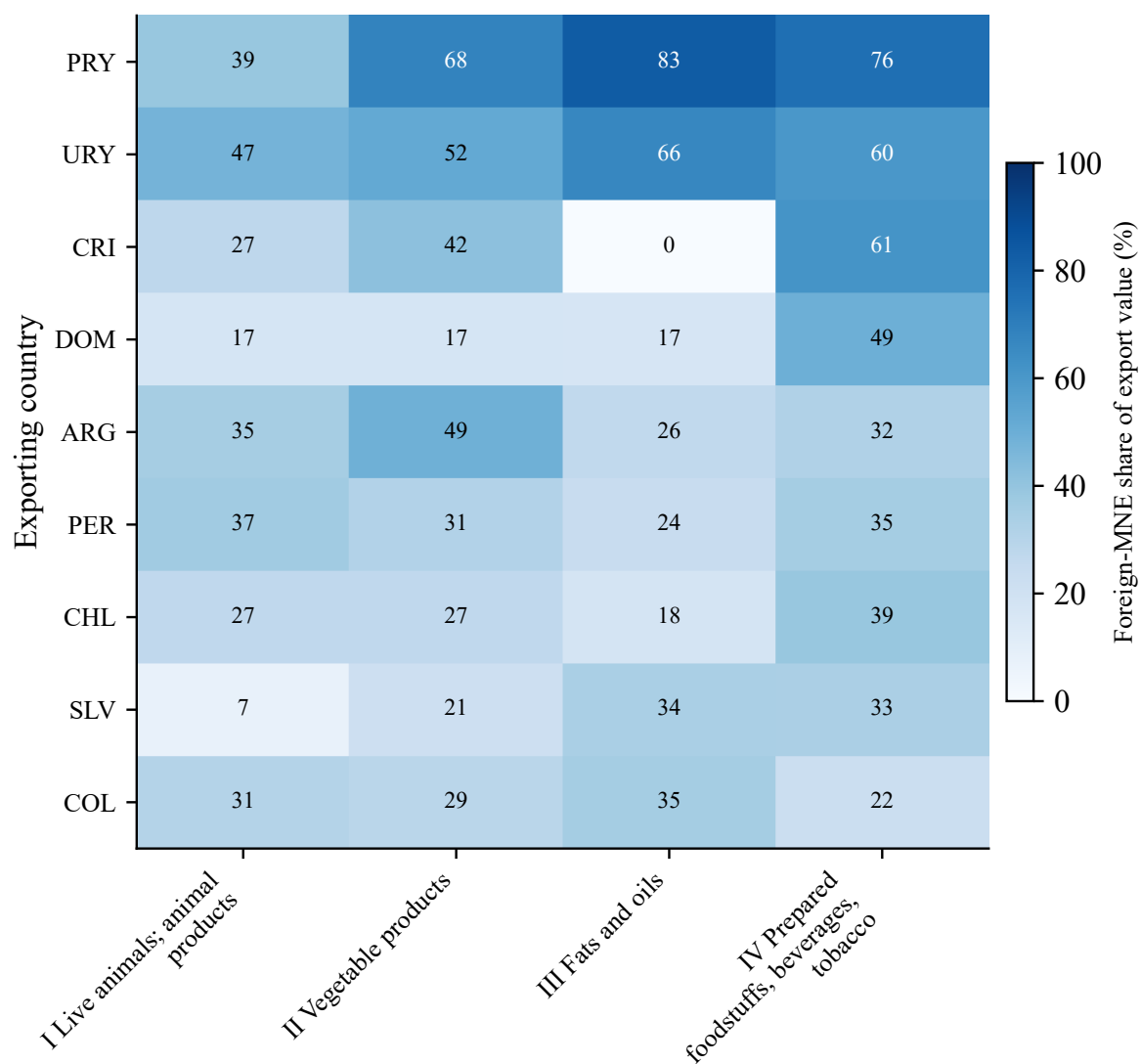
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 38:** Agriculture: Multinational Share of Export Value, by HS Section

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. HS sections.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 39:** Agriculture: Foreign-Multinational Share of Export Value, by Exporting Country and HS Section

*Notes:* Percent of each group's export value in each country shipped by foreign MNEs. HS sections.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 54:** Agriculture: Exports by Exporting Country and HS Section

| Origin                                                                   | I Live animals; animal products | II Vegetable products | III Fats and oils | IV Prepared foodstuffs, beverages, tobacco | All   |
|--------------------------------------------------------------------------|---------------------------------|-----------------------|-------------------|--------------------------------------------|-------|
| <i>Panel A: composition of the origin's exports, % (rows sum to 100)</i> |                                 |                       |                   |                                            |       |
| ARG                                                                      | 13.1                            | 36.4                  | 13.2              | 37.3                                       | 100.0 |
| CHL                                                                      | 36.9                            | 36.5                  | 1.3               | 25.4                                       | 100.0 |
| COL                                                                      | 5.3                             | 69.1                  | 6.0               | 19.6                                       | 100.0 |
| CRI                                                                      | 7.4                             | 59.3                  | 4.3               | 29.0                                       | 100.0 |
| DOM                                                                      | 1.9                             | 23.2                  | 1.5               | 73.4                                       | 100.0 |
| PER                                                                      | 11.3                            | 45.5                  | 5.1               | 38.1                                       | 100.0 |
| PRY                                                                      | 19.6                            | 51.8                  | 9.4               | 19.2                                       | 100.0 |
| SLV                                                                      | 4.6                             | 27.2                  | 1.9               | 66.3                                       | 100.0 |
| URY                                                                      | 42.3                            | 38.7                  | 2.8               | 16.2                                       | 100.0 |
| All origins                                                              | 16.6                            | 42.9                  | 8.2               | 32.2                                       | 100.0 |
| <i>Panel B: foreign-MNE share of export value, %</i>                     |                                 |                       |                   |                                            |       |
| ARG                                                                      | 34.8                            | 49.0                  | 26.4              | 31.9                                       | 37.8  |
| CHL                                                                      | 26.6                            | 26.8                  | 17.7              | 38.8                                       | 29.6  |
| COL                                                                      | 30.7                            | 28.7                  | 35.4              | 22.2                                       | 28.0  |
| CRI                                                                      | 27.4                            | 41.6                  | 0.3               | 61.5                                       | 44.5  |
| DOM                                                                      | 16.8                            | 17.0                  | 16.7              | 49.3                                       | 40.7  |
| PER                                                                      | 36.7                            | 31.0                  | 23.7              | 35.5                                       | 33.0  |
| PRY                                                                      | 38.9                            | 67.6                  | 83.1              | 75.7                                       | 65.0  |
| SLV                                                                      | 7.4                             | 21.1                  | 33.9              | 33.5                                       | 28.9  |
| URY                                                                      | 47.4                            | 52.2                  | 66.4              | 60.1                                       | 51.8  |
| All origins                                                              | 34.1                            | 41.5                  | 31.4              | 37.4                                       | 38.1  |

*Notes:* Panel A: composition of each country's exports in the sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs. HS sections.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 55:** Agriculture: Exports by HS Section and Parent Country

| HS section                                 | USA  | CHE | NLD | LIE | BRA | ESP | DEU | SAU | LUX | GBR | Other | Domestic |
|--------------------------------------------|------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-------|----------|
| I Live animals; animal products            | 1.0  | 0.7 | 0.8 | 0.1 | 2.9 | 2.9 | 0.0 | 5.5 | 1.1 | 0.9 | 18.2  | 14.4     |
| II Vegetable products                      | 11.8 | 2.5 | 4.0 | 2.7 | 1.2 | 1.0 | 2.3 | 0.0 | 1.3 | 0.5 | 14.2  | 7.3      |
| III Fats and oils                          | 5.4  | 7.4 | 6.0 | 3.5 | 0.3 | 0.1 | 0.0 | 0.1 | 0.1 | 0.9 | 7.6   | 9.7      |
| IV Prepared foodstuffs, beverages, tobacco | 7.2  | 5.8 | 2.6 | 3.4 | 0.8 | 1.1 | 0.6 | 0.5 | 0.7 | 1.4 | 13.4  | 9.8      |

*Notes:* Percent of each group's export value, by country of the ultimate parent of foreign MNEs; domestic MNEs shown for comparison. HS sections.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

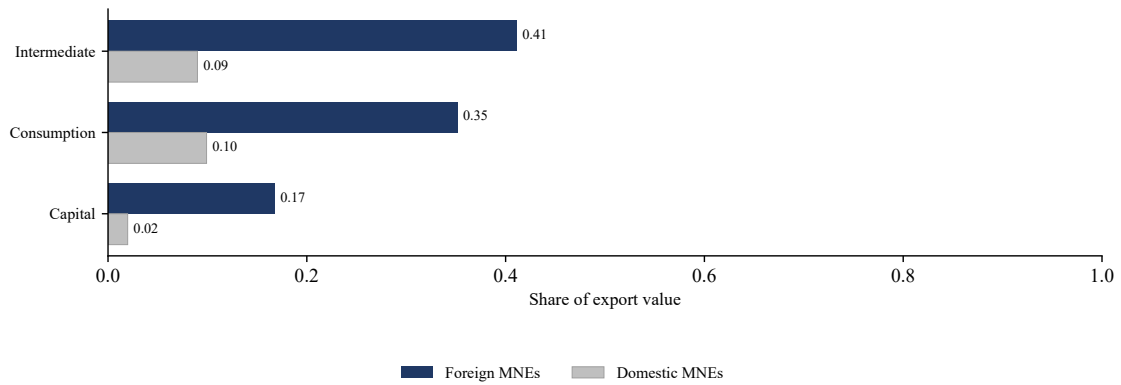
## 9.2 By End Use

**Table 56:** Agriculture: Exports and Multinational Shares, by End Use

| BEC end use  | Total exports (US\$bn/yr) | MNE exports (US\$bn/yr) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|--------------|---------------------------|-------------------------|-------------|--------------|------------------------------------|
| Intermediate | 42.5                      | 21.4                    | 41.1        | 9.0          | USA 0.30, CHE 0.16, NLD 0.13       |
| Consumption  | 30.4                      | 13.7                    | 35.1        | 9.9          | USA 0.16, ESP 0.09, SAU 0.08       |
| Capital      | 0.0                       | 0.0                     | 16.8        | 2.0          | GBR 0.35, SAU 0.33, NLD 0.28       |
| All          | 72.9                      | 35.0                    | 38.5        | 9.4          |                                    |

*Notes:* End use from BEC Rev. 4. Foreign and domestic: percent of the group's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

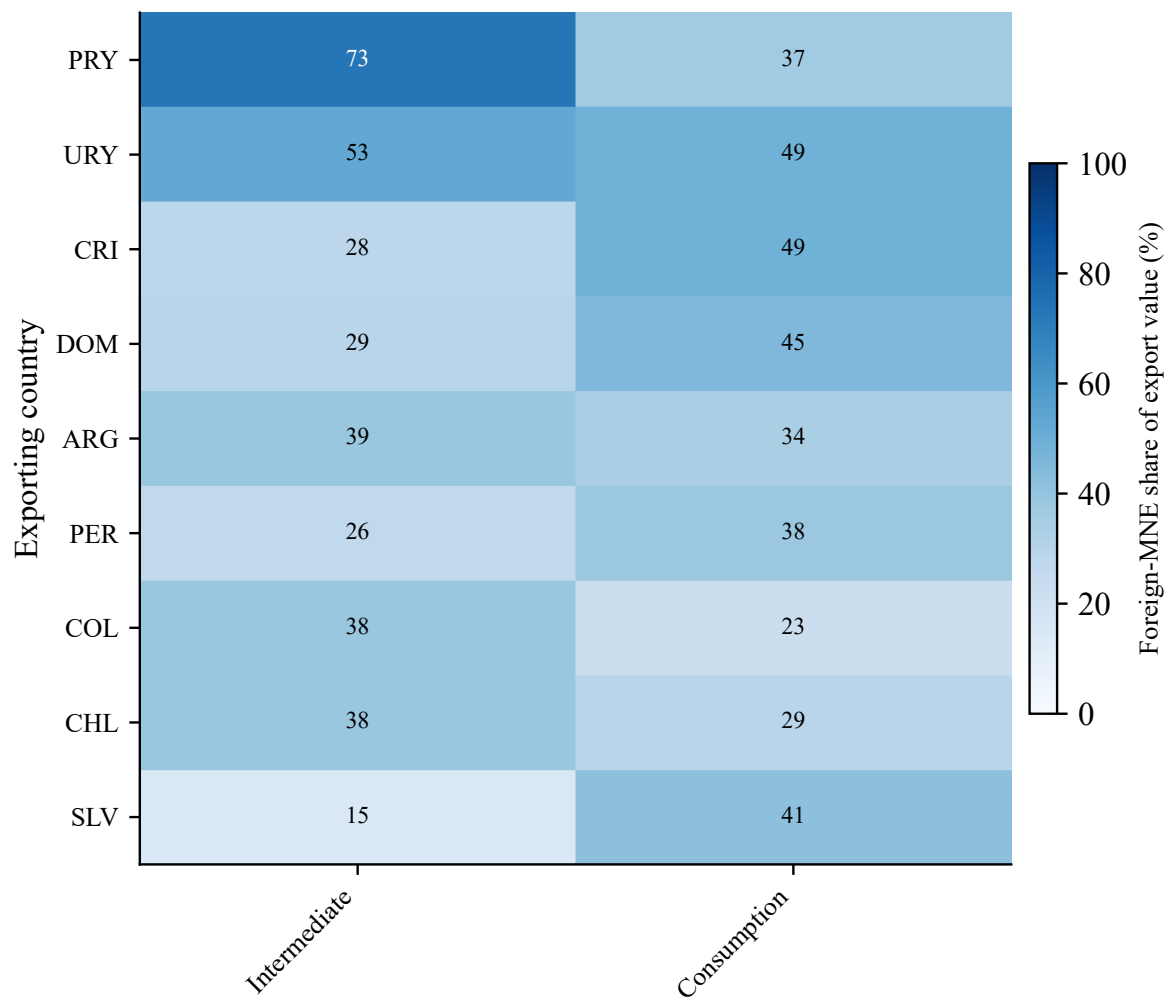
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 40: Agriculture: Multinational Share of Export Value, by End Use**

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. End use from BEC Rev. 4.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 41: Agriculture: Foreign-Multinational Share of Export Value, by Exporting Country and End Use**

*Notes:* Percent of each group's export value in each country shipped by foreign MNEs. End use from BEC Rev. 4.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 57:** Agriculture: Exports by Exporting Country and End Use

| Origin                                                                   | Intermediate | Consumption | Capital | All   |
|--------------------------------------------------------------------------|--------------|-------------|---------|-------|
| <i>Panel A: composition of the origin's exports, % (rows sum to 100)</i> |              |             |         |       |
| ARG                                                                      | 73.1         | 26.9        | 0.0     | 100.0 |
| CHL                                                                      | 12.6         | 87.3        | 0.0     | 100.0 |
| COL                                                                      | 49.6         | 50.4        | 0.0     | 100.0 |
| CRI                                                                      | 20.9         | 79.0        | 0.0     | 100.0 |
| DOM                                                                      | 27.9         | 72.1        | 0.0     | 100.0 |
| PER                                                                      | 44.1         | 55.9        | 0.0     | 100.0 |
| PRY                                                                      | 78.3         | 21.7        | 0.0     | 100.0 |
| SLV                                                                      | 44.2         | 55.8        | 0.0     | 100.0 |
| URY                                                                      | 40.1         | 59.8        | 0.2     | 100.0 |
| All origins                                                              | 56.0         | 44.0        | 0.0     | 100.0 |
| <i>Panel B: foreign-MNE share of export value, %</i>                     |              |             |         |       |
| ARG                                                                      | 38.5         | 33.7        | 5.4     | 37.2  |
| CHL                                                                      | 38.4         | 29.1        | 95.3    | 30.3  |
| COL                                                                      | 38.4         | 22.5        | 4.4     | 30.4  |
| CRI                                                                      | 28.0         | 48.7        | 85.6    | 44.4  |
| DOM                                                                      | 29.3         | 45.0        | 0.1     | 40.7  |
| PER                                                                      | 26.2         | 38.3        | 4.9     | 33.0  |
| PRY                                                                      | 72.9         | 36.5        | 32.6    | 65.0  |
| SLV                                                                      | 14.7         | 41.5        | 5.4     | 29.6  |
| URY                                                                      | 52.7         | 48.6        | 16.4    | 50.2  |
| All origins                                                              | 41.1         | 35.1        | 16.8    | 38.5  |

Notes: Panel A: composition of each country's exports in the sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs. End use from BEC Rev. 4.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 58:** Agriculture: Exports by End Use and Parent Country

| BEC end use  | USA  | CHE | NLD | LIE | BRA | ESP | DEU | SAU | LUX | GBR | Other | Domestic |
|--------------|------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-------|----------|
| Intermediate | 11.3 | 6.0 | 5.0 | 4.8 | 1.2 | 0.2 | 1.9 | 0.1 | 0.9 | 0.7 | 9.0   | 9.0      |
| Consumption  | 4.4  | 1.4 | 0.9 | 0.1 | 1.2 | 2.6 | 0.5 | 2.2 | 1.2 | 1.2 | 19.5  | 9.9      |
| Capital      | 0.2  | 0.0 | 3.8 | 0.0 | 0.0 | 0.0 | 0.0 | 4.5 | 0.0 | 4.9 | 3.4   | 2.0      |

Notes: Percent of each group's export value, by country of the ultimate parent of foreign MNEs; domestic MNEs shown for comparison. End use from BEC Rev. 4.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

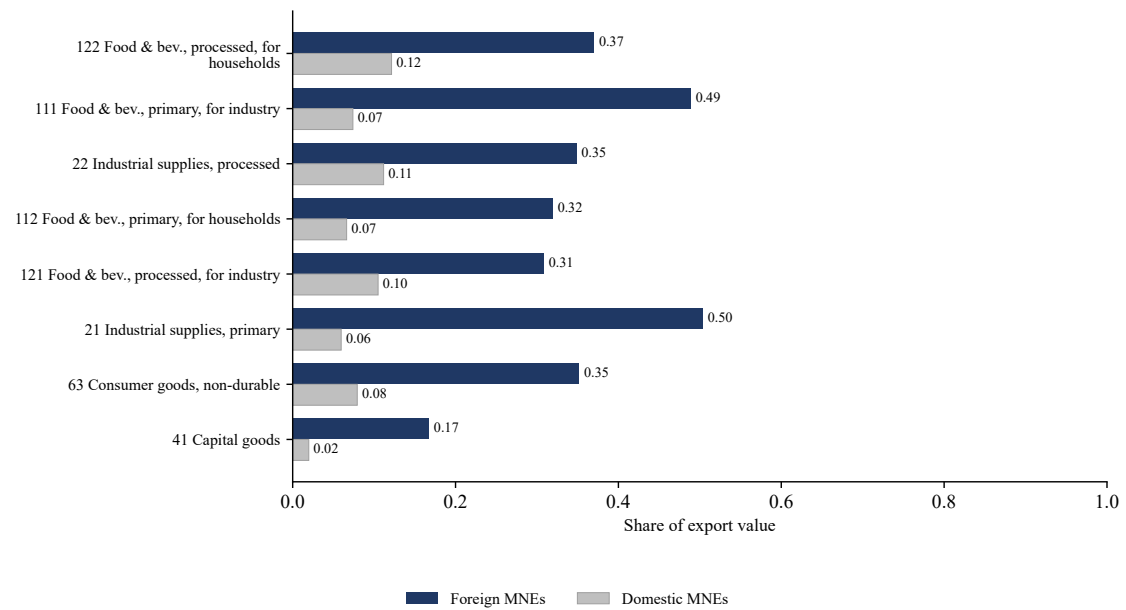
### 9.3 By BEC Category

**Table 59:** Agriculture: Exports and Multinational Shares, by BEC Category

| BEC Rev. 4 category                        | Total exports (US\$bn/yr) | MNE exports (US\$bn/yr) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|--------------------------------------------|---------------------------|-------------------------|-------------|--------------|------------------------------------|
| 122 Food & bev., processed, for households | 17.7                      | 8.6                     | 37.0        | 12.1         | USA 0.16, SAU 0.12, BRA 0.07       |
| 111 Food & bev., primary, for industry     | 14.0                      | 7.9                     | 48.9        | 7.4          | USA 0.40, NLD 0.11, DEU 0.10       |
| 22 Industrial supplies, processed          | 13.8                      | 6.4                     | 34.9        | 11.2         | CHE 0.28, LIE 0.19, USA 0.17       |
| 112 Food & bev., primary, for households   | 10.3                      | 4.0                     | 31.9        | 6.6          | ESP 0.22, USA 0.13, JOR 0.09       |
| 121 Food & bev., processed, for industry   | 8.4                       | 3.5                     | 30.9        | 10.5         | CHE 0.20, USA 0.18, NLD 0.16       |
| 21 Industrial supplies, primary            | 6.3                       | 3.6                     | 50.4        | 6.0          | USA 0.37, NLD 0.15, CHE 0.14       |
| 63 Consumer goods, non-durable             | 2.4                       | 1.1                     | 35.2        | 7.9          | USA 0.25, HKG 0.17, GBR 0.09       |
| 41 Capital goods                           | 0.0                       | 0.0                     | 16.8        | 2.0          | GBR 0.35, SAU 0.33, NLD 0.28       |
| All                                        | 72.9                      | 35.0                    | 38.5        | 9.4          |                                    |

Notes: Categories of BEC Rev. 4. Foreign and domestic: percent of the group's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

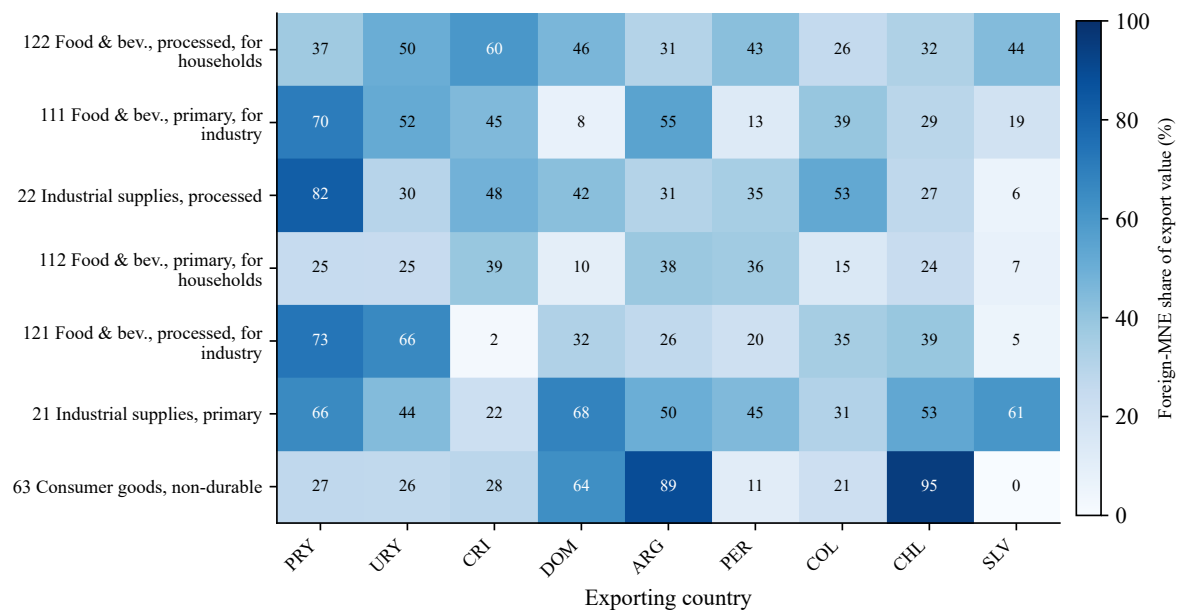
Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 42: Agriculture: Multinational Share of Export Value, by BEC Category**

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. Categories of BEC Rev. 4.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 43: Agriculture: Foreign-Multinational Share of Export Value, by Exporting Country and BEC Category**

*Notes:* Percent of each group's export value in each country shipped by foreign MNEs. Categories of BEC Rev. 4.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 60:** Agriculture: Exports by Exporting Country and BEC Category

| Sub-sector                                                                  | ARG   | CHL   | COL   | CRI   | DOM   | PER   | PRY   | SLV   | URY   | All   |
|-----------------------------------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| <i>Panel A: composition of the origin's exports, % (columns sum to 100)</i> |       |       |       |       |       |       |       |       |       |       |
| 122 Food & bev., processed, for households                                  | 18.5  | 46.2  | 22.5  | 37.7  | 21.2  | 18.2  | 20.8  | 51.8  | 55.8  | 25.4  |
| 111 Food & bev., primary, for industry                                      | 16.1  | 0.5   | 39.2  | 9.7   | 11.1  | 13.7  | 42.0  | 22.2  | 28.2  | 19.0  |
| 22 Industrial supplies, processed                                           | 28.2  | 5.4   | 0.6   | 1.4   | 2.0   | 21.5  | 17.3  | 6.9   | 2.2   | 17.9  |
| 112 Food & bev., primary, for households                                    | 8.0   | 40.3  | 5.2   | 39.8  | 18.0  | 37.6  | 0.3   | 3.4   | 3.2   | 15.1  |
| 121 Food & bev., processed, for industry                                    | 15.3  | 2.3   | 8.1   | 6.6   | 9.9   | 6.8   | 10.7  | 13.3  | 7.5   | 11.0  |
| 21 Industrial supplies, primary                                             | 13.4  | 4.4   | 1.6   | 3.2   | 4.9   | 2.1   | 8.3   | 1.9   | 2.2   | 8.2   |
| 63 Consumer goods, non-durable                                              | 0.4   | 0.8   | 22.8  | 1.6   | 33.0  | 0.2   | 0.6   | 0.5   | 0.8   | 3.5   |
| 41 Capital goods                                                            | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.2   | 0.0   |
| All sub-sectors                                                             | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 |
| <i>Panel B: foreign-MNE share of export value, %</i>                        |       |       |       |       |       |       |       |       |       |       |
| 122 Food & bev., processed, for households                                  | 30.8  | 32.3  | 25.6  | 60.1  | 46.2  | 42.6  | 37.0  | 44.1  | 50.3  | 37.0  |
| 111 Food & bev., primary, for industry                                      | 54.8  | 28.7  | 39.2  | 44.7  | 7.7   | 12.9  | 70.4  | 19.1  | 51.7  | 48.9  |
| 22 Industrial supplies, processed                                           | 30.5  | 27.1  | 52.6  | 48.1  | 42.4  | 34.7  | 82.1  | 6.0   | 29.5  | 34.9  |
| 112 Food & bev., primary, for households                                    | 37.9  | 24.0  | 14.7  | 38.8  | 9.7   | 36.3  | 24.6  | 7.1   | 24.5  | 31.9  |
| 121 Food & bev., processed, for industry                                    | 26.3  | 38.9  | 35.0  | 2.2   | 31.8  | 20.5  | 73.4  | 5.3   | 65.7  | 30.9  |
| 21 Industrial supplies, primary                                             | 49.8  | 53.0  | 31.0  | 21.6  | 67.6  | 44.8  | 65.6  | 60.5  | 44.0  | 50.4  |
| 63 Consumer goods, non-durable                                              | 88.8  | 94.6  | 21.2  | 28.3  | 63.6  | 11.2  | 26.8  | 0.2   | 26.4  | 35.2  |
| 41 Capital goods                                                            | 5.4   | 95.3  | 4.4   | 85.6  | 0.1   | 4.9   | 32.6  | 5.4   | 16.4  | 16.8  |
| All sub-sectors                                                             | 37.2  | 30.3  | 30.4  | 44.4  | 40.7  | 33.0  | 65.0  | 29.6  | 50.2  | 38.5  |

*Notes:* Panel A: composition of each country's exports in the sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs. Categories of BEC Rev. 4.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 61:** Agriculture: Exports by BEC Category and Parent Country

| BEC Rev. 4 category                        | USA  | CHE | NLD | LIE | BRA | ESP | DEU | SAU | LUX | GBR | Other | Domestic |
|--------------------------------------------|------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-------|----------|
| 122 Food & bev., processed, for households | 4.9  | 2.1 | 1.2 | 0.2 | 2.1 | 1.3 | 0.5 | 3.8 | 1.1 | 1.4 | 18.3  | 12.1     |
| 111 Food & bev., primary, for industry     | 17.7 | 3.0 | 5.1 | 3.9 | 2.2 | 0.1 | 4.3 | 0.0 | 0.5 | 0.2 | 11.8  | 7.4      |
| 22 Industrial supplies, processed          | 5.7  | 9.2 | 4.5 | 6.5 | 0.3 | 0.5 | 0.2 | 0.1 | 0.0 | 1.2 | 6.5   | 11.2     |
| 112 Food & bev., primary, for households   | 3.1  | 0.2 | 0.4 | 0.0 | 0.0 | 5.2 | 0.5 | 0.1 | 1.6 | 0.4 | 20.5  | 6.6      |
| 121 Food & bev., processed, for industry   | 4.9  | 5.5 | 4.5 | 3.3 | 0.3 | 0.0 | 0.2 | 0.0 | 3.4 | 1.1 | 7.6   | 10.5     |
| 21 Industrial supplies, primary            | 17.3 | 6.4 | 6.7 | 5.1 | 1.7 | 0.2 | 2.5 | 0.5 | 0.1 | 0.3 | 9.7   | 6.0      |
| 63 Consumer goods, non-durable             | 7.1  | 1.9 | 0.0 | 0.0 | 0.1 | 0.2 | 0.0 | 0.0 | 0.0 | 2.7 | 23.1  | 7.9      |
| 41 Capital goods                           | 0.2  | 0.0 | 3.8 | 0.0 | 0.0 | 0.0 | 0.0 | 4.5 | 0.0 | 4.9 | 3.4   | 2.0      |

*Notes:* Percent of each group's export value, by country of the ultimate parent of foreign MNEs; domestic MNEs shown for comparison. Categories of BEC Rev. 4.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

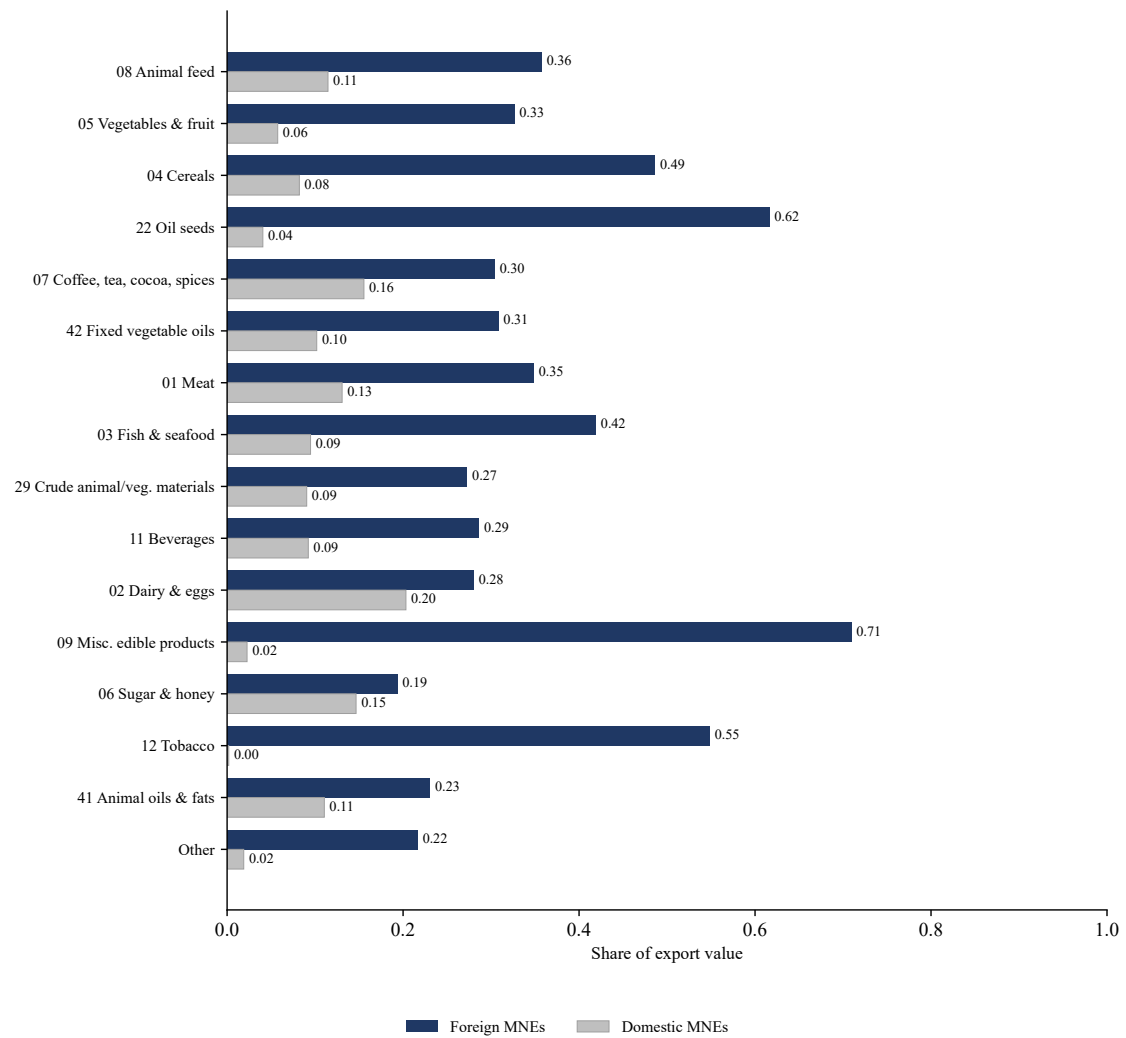
## 9.4 By SITC Division

**Table 62:** Agriculture: Exports and Multinational Shares, by SITC Division

| SITC Rev. 3 division           | Total exports (US\$bn/yr) | MNE exports (US\$bn/yr) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|--------------------------------|---------------------------|-------------------------|-------------|--------------|------------------------------------|
| 08 Animal feed                 | 13.8                      | 6.6                     | 35.7        | 11.5         | CHE 0.28, LIE 0.19, USA 0.18       |
| 05 Vegetables & fruit          | 10.6                      | 4.0                     | 32.6        | 5.8          | ESP 0.14, USA 0.13, PAN 0.11       |
| 04 Cereals                     | 10.3                      | 5.9                     | 48.6        | 8.2          | USA 0.33, CHE 0.14, NLD 0.14       |
| 22 Oil seeds                   | 6.7                       | 4.4                     | 61.6        | 4.1          | USA 0.54, NLD 0.14, LIE 0.08       |
| 07 Coffee, tea, cocoa, spices  | 5.1                       | 2.3                     | 30.4        | 15.6         | DEU 0.41, ITA 0.14, SGP 0.12       |
| 42 Fixed vegetable oils        | 6.0                       | 2.5                     | 30.8        | 10.2         | CHE 0.27, NLD 0.24, USA 0.23       |
| 01 Meat                        | 5.2                       | 2.4                     | 34.8        | 13.1         | SAU 0.46, BRA 0.26, JPN 0.08       |
| 03 Fish & seafood              | 3.4                       | 1.8                     | 41.9        | 9.5          | ESP 0.32, LUX 0.11, ITA 0.10       |
| 29 Crude animal/veg. materials | 2.2                       | 0.8                     | 27.3        | 9.0          | DEU 0.10, ARG 0.09, BGR 0.08       |
| 11 Beverages                   | 2.2                       | 0.8                     | 28.6        | 9.2          | LUX 0.17, PAN 0.15, GBR 0.12       |
| 02 Dairy & eggs                | 1.9                       | 1.0                     | 28.0        | 20.3         | NLD 0.20, CHE 0.19, ROU 0.19       |
| 09 Misc. edible products       | 1.7                       | 1.2                     | 70.9        | 2.3          | USA 0.46, CHE 0.17, FRA 0.09       |
| 06 Sugar & honey               | 1.4                       | 0.5                     | 19.3        | 14.7         | USA 0.50, KOR 0.18, LIE 0.10       |
| 12 Tobacco                     | 1.2                       | 0.7                     | 54.8        | 0.2          | USA 0.36, HKG 0.25, GBR 0.12       |
| 41 Animal oils & fats          | 0.5                       | 0.2                     | 23.0        | 11.0         | BMU 0.31, PAN 0.23, GBR 0.16       |
| Other                          | 0.6                       | 0.1                     | 21.6        | 1.9          | CHE 0.21, USA 0.20, ROU 0.10       |
| All                            | 72.9                      | 35.0                    | 38.5        | 9.4          |                                    |

*Notes:* SITC Rev. 3 divisions; HS6 products without an SITC correspondence are excluded. Foreign and domestic: percent of the group's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

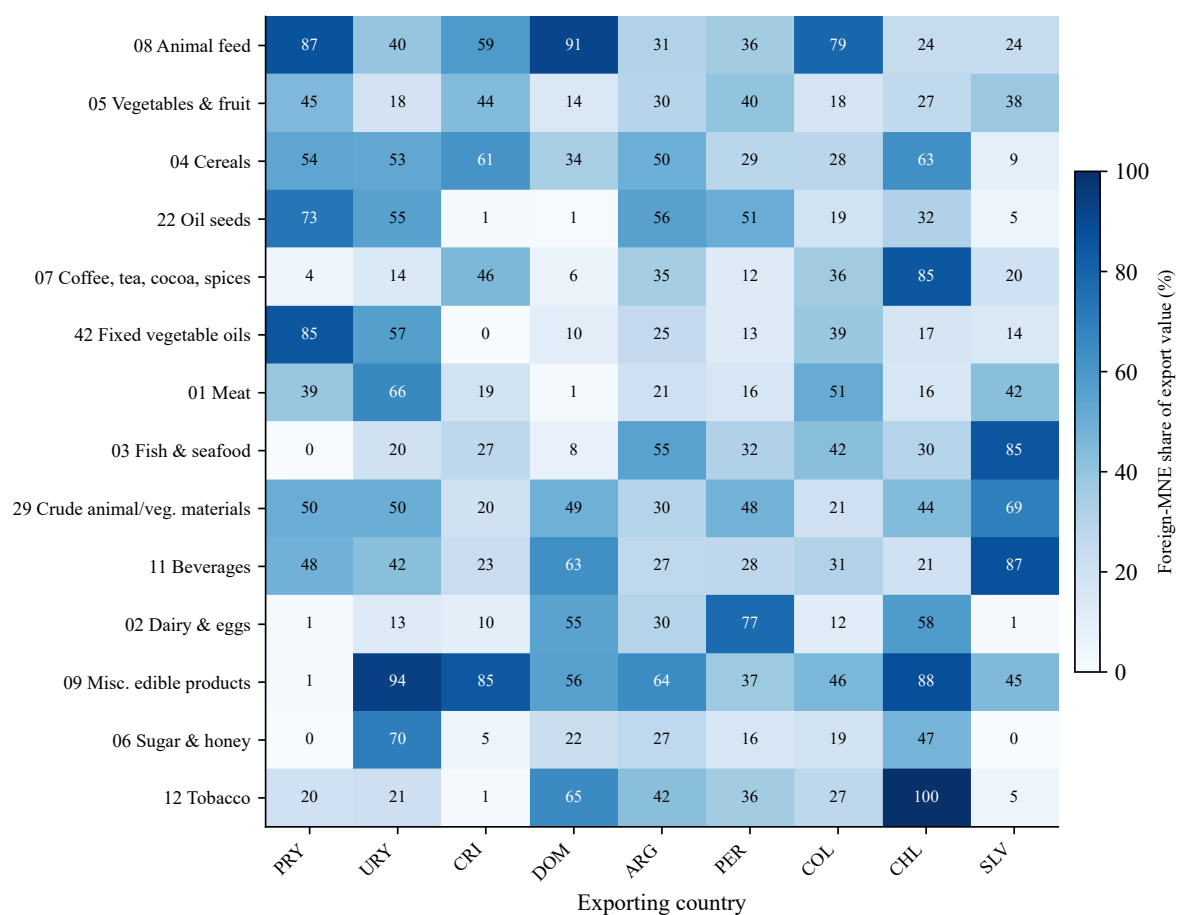
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 44: Agriculture: Multinational Share of Export Value, by SITC Division**

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. SITC Rev. 3 divisions; HS6 products without an SITC correspondence are excluded.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 45:** Agriculture: Foreign-Multinational Share of Export Value, by Exporting Country and SITC Division

*Notes:* Percent of each group's export value in each country shipped by foreign MNEs. SITC Rev. 3 divisions; HS6 products without an SITC correspondence are excluded.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 63:** Agriculture: Exports by Exporting Country and SITC Division

| Sub-sector                                                                  | ARG   | CHL   | COL   | CRI   | DOM   | PER   | PRY   | SLV   | URY   | All   |
|-----------------------------------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| <i>Panel A: composition of the origin's exports, % (columns sum to 100)</i> |       |       |       |       |       |       |       |       |       |       |
| 08 Animal feed                                                              | 29.1  | 4.4   | 0.6   | 1.5   | 0.7   | 20.5  | 16.5  | 1.7   | 1.0   | 17.8  |
| 05 Vegetables & fruit                                                       | 7.6   | 43.7  | 5.9   | 48.1  | 20.8  | 36.4  | 0.3   | 7.2   | 2.0   | 15.6  |
| 04 Cereals                                                                  | 20.9  | 2.8   | 2.5   | 3.6   | 5.6   | 3.3   | 12.5  | 19.6  | 20.5  | 13.4  |
| 22 Oil seeds                                                                | 9.5   | 0.4   | 0.0   | 0.1   | 0.0   | 0.3   | 38.6  | 0.1   | 20.6  | 8.6   |
| 07 Coffee, tea, cocoa, spices                                               | 0.9   | 0.7   | 45.4  | 9.9   | 11.1  | 14.5  | 0.1   | 21.8  | 0.3   | 7.9   |
| 42 Fixed vegetable oils                                                     | 12.8  | 0.5   | 6.2   | 5.2   | 1.2   | 0.5   | 9.1   | 0.2   | 0.1   | 7.7   |
| 01 Meat                                                                     | 5.9   | 9.2   | 1.4   | 2.8   | 0.7   | 0.1   | 18.3  | 1.3   | 30.2  | 7.0   |
| 03 Fish & seafood                                                           | 3.9   | 10.4  | 1.6   | 1.8   | 0.6   | 13.1  | 0.0   | 9.6   | 3.0   | 4.9   |
| 29 Crude animal/veg. materials                                              | 0.4   | 3.8   | 23.4  | 3.7   | 1.0   | 2.1   | 1.1   | 1.6   | 1.5   | 3.6   |
| 11 Beverages                                                                | 2.5   | 16.4  | 0.4   | 1.7   | 6.8   | 0.5   | 0.0   | 9.2   | 0.5   | 3.4   |
| 02 Dairy & eggs                                                             | 2.8   | 1.4   | 0.3   | 3.9   | 0.6   | 1.5   | 0.2   | 2.4   | 12.7  | 2.6   |
| 09 Misc. edible products                                                    | 1.3   | 3.5   | 2.2   | 15.1  | 6.4   | 0.9   | 0.2   | 5.3   | 2.4   | 2.4   |
| 06 Sugar & honey                                                            | 1.1   | 0.5   | 8.8   | 1.1   | 6.7   | 0.8   | 1.1   | 12.1  | 0.6   | 2.1   |
| 12 Tobacco                                                                  | 0.9   | 0.8   | 0.6   | 0.3   | 37.1  | 0.1   | 0.8   | 0.0   | 1.1   | 1.5   |
| 41 Animal oils & fats                                                       | 0.0   | 1.0   | 0.0   | 0.0   | 0.0   | 4.4   | 0.3   | 0.0   | 1.3   | 0.7   |
| Other                                                                       | 0.5   | 0.5   | 0.7   | 0.9   | 0.7   | 1.1   | 0.7   | 7.7   | 2.2   | 0.9   |
| All sub-sectors                                                             | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 |
| <i>Panel B: foreign-MNE share of export value, %</i>                        |       |       |       |       |       |       |       |       |       |       |
| 08 Animal feed                                                              | 30.9  | 24.2  | 79.1  | 58.9  | 91.5  | 35.7  | 86.6  | 24.1  | 39.6  | 35.7  |
| 05 Vegetables & fruit                                                       | 29.6  | 26.5  | 17.9  | 44.0  | 14.3  | 40.0  | 45.0  | 38.0  | 18.1  | 32.6  |
| 04 Cereals                                                                  | 49.9  | 63.0  | 28.2  | 61.0  | 34.2  | 28.6  | 53.7  | 8.8   | 52.8  | 48.6  |
| 22 Oil seeds                                                                | 55.8  | 31.8  | 18.8  | 1.2   | 0.8   | 50.5  | 73.1  | 4.8   | 55.5  | 61.6  |
| 07 Coffee, tea, cocoa, spices                                               | 35.4  | 85.1  | 36.2  | 45.6  | 6.3   | 12.4  | 4.0   | 20.2  | 13.7  | 30.4  |
| 42 Fixed vegetable oils                                                     | 25.2  | 17.1  | 38.6  | 0.0   | 10.2  | 12.8  | 85.2  | 14.4  | 56.6  | 30.8  |
| 01 Meat                                                                     | 20.8  | 16.3  | 50.8  | 19.3  | 1.4   | 16.3  | 38.6  | 42.4  | 66.0  | 34.8  |
| 03 Fish & seafood                                                           | 55.4  | 30.4  | 42.4  | 27.4  | 7.5   | 32.0  | 0.0   | 85.4  | 19.6  | 41.9  |
| 29 Crude animal/veg. materials                                              | 29.7  | 43.9  | 20.6  | 19.7  | 49.2  | 47.8  | 50.5  | 69.3  | 50.2  | 27.3  |
| 11 Beverages                                                                | 27.3  | 20.7  | 31.0  | 22.9  | 63.0  | 27.7  | 48.3  | 87.3  | 42.4  | 28.6  |
| 02 Dairy & eggs                                                             | 29.8  | 58.4  | 11.7  | 9.6   | 54.9  | 77.3  | 0.8   | 0.8   | 13.1  | 28.0  |
| 09 Misc. edible products                                                    | 64.5  | 88.2  | 45.8  | 84.6  | 55.7  | 37.1  | 1.1   | 44.7  | 93.6  | 70.9  |
| 06 Sugar & honey                                                            | 26.7  | 47.4  | 19.2  | 4.9   | 22.1  | 16.2  | 0.1   | 0.1   | 69.8  | 19.3  |
| 12 Tobacco                                                                  | 42.3  | 99.8  | 27.4  | 1.0   | 65.0  | 35.6  | 20.0  | 5.0   | 21.5  | 54.8  |
| 41 Animal oils & fats                                                       | 26.3  | 13.3  | 41.6  | 36.8  | 89.4  | 24.9  | 28.7  | 49.8  | 20.5  | 23.0  |
| Other                                                                       | 36.1  | 45.3  | 16.5  | 26.3  | 5.1   | 7.5   | 1.6   | 0.1   | 34.5  | 21.6  |
| All sub-sectors                                                             | 37.2  | 30.3  | 30.4  | 44.4  | 40.7  | 33.0  | 65.0  | 29.6  | 50.2  | 38.5  |

Notes: Panel A: composition of each country's exports in the sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs. SITC Rev. 3 divisions; HS6 products without an SITC correspondence are excluded.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 64:** Agriculture: Exports by SITC Division and Parent Country

| SITC Rev. 3 division           | USA  | CHE  | NLD | LIE | BRA | ESP  | DEU  | SAU  | LUX | GBR | Other | Domestic |
|--------------------------------|------|------|-----|-----|-----|------|------|------|-----|-----|-------|----------|
| 08 Animal feed                 | 6.3  | 9.8  | 4.7 | 6.5 | 0.3 | 0.5  | 0.2  | 0.1  | 0.0 | 1.2 | 6.2   | 11.5     |
| 05 Vegetables & fruit          | 3.2  | 0.2  | 0.7 | 0.0 | 0.0 | 3.3  | 1.2  | 0.1  | 0.8 | 0.4 | 22.7  | 5.8      |
| 04 Cereals                     | 15.0 | 6.2  | 6.2 | 5.8 | 1.3 | 0.5  | 1.0  | 0.0  | 3.5 | 1.0 | 8.2   | 8.2      |
| 22 Oil seeds                   | 30.7 | 3.2  | 7.8 | 4.8 | 3.1 | 0.1  | 0.1  | 0.0  | 0.0 | 0.3 | 11.5  | 4.1      |
| 07 Coffee, tea, cocoa, spices  | 1.5  | 0.8  | 0.0 | 0.0 | 1.2 | 0.1  | 10.2 | 0.0  | 1.1 | 0.0 | 15.6  | 15.6     |
| 42 Fixed vegetable oils        | 6.3  | 7.3  | 6.7 | 4.0 | 0.3 | 0.1  | 0.0  | 0.0  | 0.1 | 0.4 | 5.6   | 10.2     |
| 01 Meat                        | 0.2  | 0.0  | 0.0 | 0.0 | 7.7 | 0.1  | 0.0  | 13.4 | 0.0 | 0.9 | 12.5  | 13.1     |
| 03 Fish & seafood              | 1.7  | 0.0  | 0.1 | 0.0 | 0.0 | 10.1 | 0.0  | 0.4  | 3.4 | 0.7 | 25.4  | 9.5      |
| 29 Crude animal/veg. materials | 0.9  | 0.1  | 0.7 | 0.7 | 1.0 | 0.3  | 1.8  | 0.7  | 0.2 | 0.5 | 20.4  | 9.0      |
| 11 Beverages                   | 0.7  | 0.6  | 0.8 | 0.1 | 0.0 | 1.2  | 0.1  | 0.0  | 3.6 | 2.6 | 19.0  | 9.2      |
| 02 Dairy & eggs                | 0.8  | 5.2  | 5.3 | 0.9 | 0.1 | 0.1  | 0.0  | 0.0  | 0.0 | 0.1 | 15.6  | 20.3     |
| 09 Misc. edible products       | 31.8 | 12.0 | 1.0 | 0.0 | 0.1 | 0.7  | 0.5  | 0.0  | 0.7 | 4.5 | 19.7  | 2.3      |
| 06 Sugar & honey               | 7.4  | 0.0  | 0.6 | 1.5 | 0.0 | 0.1  | 0.7  | 0.0  | 0.0 | 0.1 | 8.8   | 14.7     |
| 12 Tobacco                     | 19.3 | 0.0  | 0.0 | 0.0 | 0.0 | 0.0  | 0.0  | 0.0  | 0.0 | 6.4 | 29.2  | 0.2      |
| 41 Animal oils & fats          | 0.0  | 0.0  | 1.7 | 0.0 | 0.4 | 0.0  | 0.0  | 1.4  | 0.0 | 3.3 | 16.2  | 11.0     |
| Other                          | 3.2  | 3.4  | 0.3 | 0.8 | 0.1 | 0.6  | 1.6  | 0.3  | 0.5 | 1.3 | 9.7   | 1.9      |

*Notes:* Percent of each group's export value, by country of the ultimate parent of foreign MNEs; domestic MNEs shown for comparison. SITC Rev. 3 divisions; HS6 products without an SITC correspondence are excluded.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

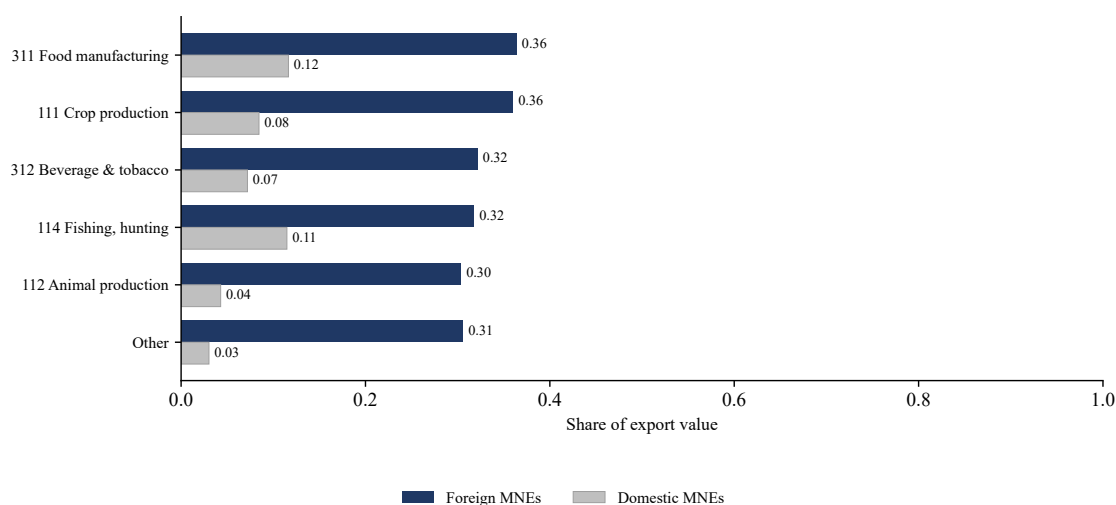
## 9.5 By NAICS Industry

**Table 65:** Agriculture: Exports and Multinational Shares, by NAICS Industry

| NAICS 3-digit (product-based) | Total exports (US\$bn/yr) | MNE exports (US\$bn/yr) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|-------------------------------|---------------------------|-------------------------|-------------|--------------|------------------------------------|
| 311 Food manufacturing        | 35.1                      | 16.7                    | 36.4        | 11.6         | CHE 0.19, USA 0.18, NLD 0.10       |
| 111 Crop production           | 17.7                      | 7.9                     | 36.0        | 8.4          | USA 0.25, DEU 0.14, NLD 0.07       |
| 312 Beverage & tobacco        | 3.0                       | 1.2                     | 32.1        | 7.2          | HKG 0.15, USA 0.13, GBR 0.13       |
| 114 Fishing, hunting          | 2.5                       | 1.1                     | 31.8        | 11.5         | BMU 0.31, PAN 0.21, GBR 0.18       |
| 112 Animal production         | 0.4                       | 0.1                     | 30.3        | 4.3          | KOR 0.52, ROU 0.14, NLD 0.11       |
| Other                         | 0.3                       | 0.1                     | 30.6        | 3.0          | USA 0.20, ITA 0.15, GBR 0.14       |
| All                           | 59.0                      | 27.2                    | 35.8        | 10.3         |                                    |

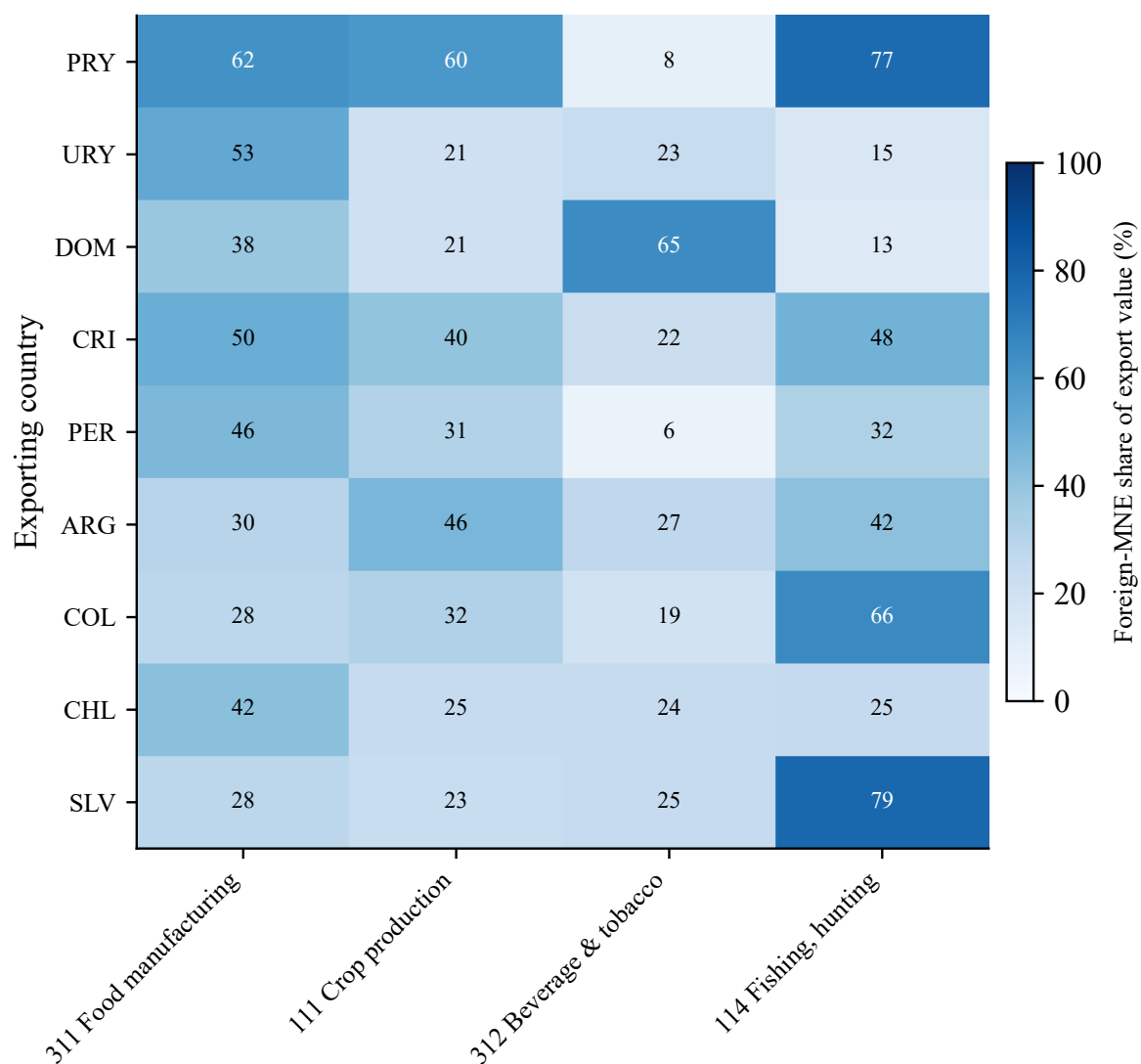
*Notes:* NAICS 3-digit industries assigned to HS6 products. Foreign and domestic: percent of the group's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Figure 46:** Agriculture: Multinational Share of Export Value, by NAICS Industry

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. NAICS 3-digit industries assigned to HS6 products.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 47:** Agriculture: Foreign-Multinational Share of Export Value, by Exporting Country and NAICS Industry

*Notes:* Percent of each group's export value in each country shipped by foreign MNEs. NAICS 3-digit industries assigned to HS6 products.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 66: Agriculture: Exports by Exporting Country and NAICS Industry**

| Origin                                                                   | 311 Food manufacturing | 111 Crop production | 312 Beverage & tobacco | 114 Fishing, hunting | 112 Animal production | Other | All   |
|--------------------------------------------------------------------------|------------------------|---------------------|------------------------|----------------------|-----------------------|-------|-------|
| <i>Panel A: composition of the origin's exports, % (rows sum to 100)</i> |                        |                     |                        |                      |                       |       |       |
| ARG                                                                      | 74.8                   | 20.7                | 3.2                    | 0.2                  | 0.7                   | 0.4   | 100.0 |
| CHL                                                                      | 34.3                   | 36.8                | 18.3                   | 9.7                  | 0.3                   | 0.7   | 100.0 |
| COL                                                                      | 31.3                   | 66.6                | 0.4                    | 1.1                  | 0.3                   | 0.4   | 100.0 |
| CRI                                                                      | 49.8                   | 46.8                | 2.3                    | 0.4                  | 0.2                   | 0.5   | 100.0 |
| DOM                                                                      | 25.6                   | 25.3                | 47.8                   | 0.2                  | 0.6                   | 0.6   | 100.0 |
| PER                                                                      | 23.1                   | 46.7                | 1.4                    | 28.0                 | 0.1                   | 0.6   | 100.0 |
| PRY                                                                      | 82.9                   | 15.4                | 1.5                    | 0.1                  | 0.0                   | 0.0   | 100.0 |
| SLV                                                                      | 60.1                   | 26.8                | 10.4                   | 0.7                  | 1.7                   | 0.2   | 100.0 |
| URY                                                                      | 90.3                   | 3.6                 | 1.5                    | 0.4                  | 3.7                   | 0.4   | 100.0 |
| All origins                                                              | 57.4                   | 31.6                | 5.3                    | 4.6                  | 0.6                   | 0.4   | 100.0 |
| <i>Panel B: foreign-MNE share of export value, %</i>                     |                        |                     |                        |                      |                       |       |       |
| ARG                                                                      | 29.5                   | 46.1                | 26.8                   | 41.8                 | 33.3                  | 29.0  | 32.9  |
| CHL                                                                      | 42.0                   | 24.8                | 24.2                   | 25.1                 | 17.5                  | 8.4   | 30.5  |
| COL                                                                      | 28.0                   | 31.9                | 19.4                   | 65.6                 | 2.4                   | 12.0  | 30.9  |
| CRI                                                                      | 50.1                   | 39.9                | 22.2                   | 48.0                 | 17.9                  | 34.3  | 44.6  |
| DOM                                                                      | 38.5                   | 20.6                | 64.5                   | 12.8                 | 1.4                   | 88.8  | 46.4  |
| PER                                                                      | 45.6                   | 31.0                | 6.5                    | 32.1                 | 6.3                   | 57.9  | 34.5  |
| PRY                                                                      | 62.3                   | 60.0                | 8.2                    | 77.1                 | 0.4                   | 0.5   | 61.1  |
| SLV                                                                      | 27.7                   | 23.3                | 24.7                   | 78.8                 | 0.0                   | 17.6  | 26.1  |
| URY                                                                      | 52.5                   | 21.1                | 23.5                   | 15.1                 | 43.3                  | 35.3  | 50.4  |
| All origins                                                              | 36.4                   | 36.0                | 32.1                   | 31.8                 | 30.3                  | 30.6  | 35.8  |

*Notes:* Panel A: composition of each country's exports in the sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs. NAICS 3-digit industries assigned to HS6 products.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 67: Agriculture: Exports by NAICS Industry and Parent Country**

| NAICS 3-digit (product-based) | USA | CHE | NLD | LIE | BRA | ESP | DEU | SAU | LUX | GBR | Other | Domestic |
|-------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-------|----------|
| 311 Food manufacturing        | 6.0 | 6.0 | 3.4 | 3.3 | 1.4 | 0.7 | 0.4 | 2.1 | 1.1 | 0.9 | 10.9  | 11.6     |
| 111 Crop production           | 7.5 | 2.1 | 2.2 | 1.7 | 0.8 | 1.4 | 4.1 | 0.1 | 0.2 | 0.3 | 15.7  | 8.4      |
| 312 Beverage & tobacco        | 3.3 | 0.6 | 0.6 | 0.1 | 0.0 | 0.9 | 0.1 | 0.0 | 2.7 | 3.3 | 20.6  | 7.2      |
| 114 Fishing, hunting          | 1.6 | 0.0 | 0.3 | 0.0 | 0.0 | 0.2 | 0.1 | 0.0 | 0.2 | 4.9 | 24.4  | 11.5     |
| 112 Animal production         | 0.1 | 0.0 | 2.3 | 0.0 | 0.0 | 0.3 | 0.0 | 0.3 | 0.0 | 0.6 | 26.6  | 4.3      |
| Other                         | 4.2 | 1.2 | 1.0 | 1.9 | 0.1 | 0.5 | 0.3 | 0.0 | 0.0 | 2.9 | 18.6  | 3.0      |

*Notes:* Percent of each group's export value, by country of the ultimate parent of foreign MNEs; domestic MNEs shown for comparison. NAICS 3-digit industries assigned to HS6 products.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

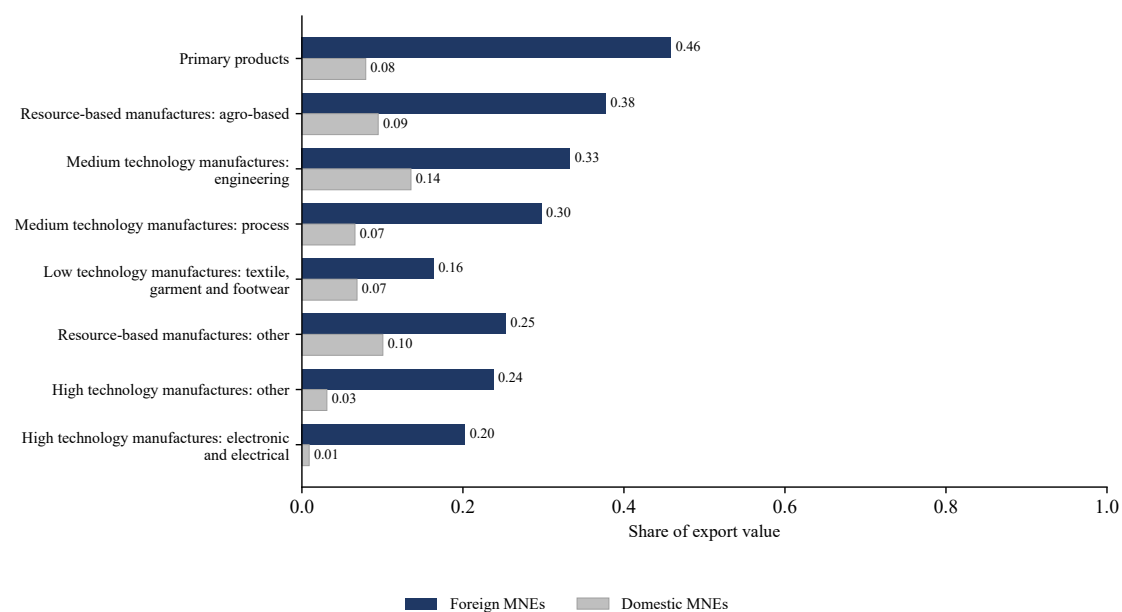
## 9.6 By Technology Class

**Table 68: Agriculture: Exports and Multinational Shares, by Technology Class**

| Lall (2000) category                                       | Total exports (US\$bn/yr) | MNE exports (US\$bn/yr) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|------------------------------------------------------------|---------------------------|-------------------------|-------------|--------------|------------------------------------|
| Primary products                                           | 27.3                      | 14.9                    | 45.9        | 7.9          | USA 0.34, NLD 0.11, CHE 0.09       |
| Resource-based manufactures: agro-based                    | 20.6                      | 9.6                     | 37.7        | 9.5          | USA 0.18, SAU 0.10, CHE 0.09       |
| Medium technology manufactures: engineering                | 16.1                      | 7.5                     | 33.3        | 13.5         | CHE 0.25, LIE 0.17, USA 0.14       |
| Medium technology manufactures: process                    | 6.9                       | 2.5                     | 29.8        | 6.6          | USA 0.19, ESP 0.15, JOR 0.13       |
| Low technology manufactures: textile, garment and footwear | 0.7                       | 0.2                     | 16.4        | 6.8          | USA 0.67, LIE 0.26, COL 0.06       |
| Resource-based manufactures: other                         | 0.6                       | 0.2                     | 25.3        | 10.1         | BMU 0.24, PAN 0.18, USA 0.13       |
| High technology manufactures: other                        | 0.5                       | 0.1                     | 23.8        | 3.1          | ITA 0.17, DEU 0.16, SGP 0.15       |
| High technology manufactures: electronic and electrical    | 0.1                       | 0.0                     | 20.3        | 0.9          | KOR 0.33, SLV 0.19, CHE 0.10       |
| All                                                        | 72.9                      | 35.0                    | 38.5        | 9.4          |                                    |

*Notes:* Lall (2000) technology classes. Foreign and domestic: percent of the group's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

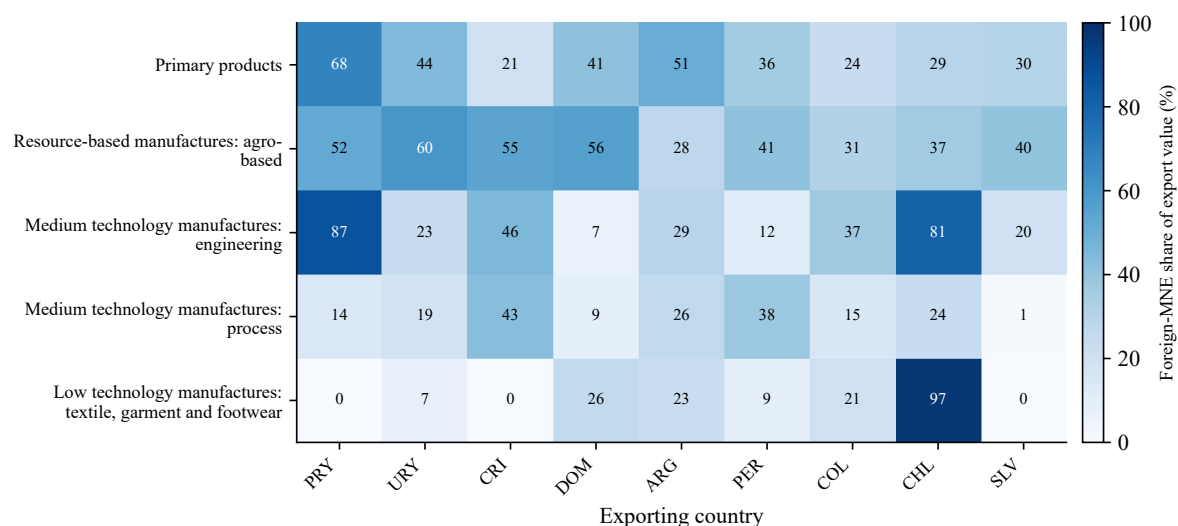
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 48:** Agriculture: Multinational Share of Export Value, by Technology Class

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. Lall (2000) technology classes.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 49:** Agriculture: Foreign-Multinational Share of Export Value, by Exporting Country and Technology Class

*Notes:* Percent of each group's export value in each country shipped by foreign MNEs. Lall (2000) technology classes.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 69: Agriculture: Exports by Exporting Country and Technology Class**

| Sub-sector                                                                  | ARG   | CHL   | COL   | CRI   | DOM   | PER   | PRY   | SLV   | URY   | All   |
|-----------------------------------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| <i>Panel A: composition of the origin's exports, % (columns sum to 100)</i> |       |       |       |       |       |       |       |       |       |       |
| Primary products                                                            | 39.4  | 28.3  | 26.9  | 13.3  | 11.4  | 42.4  | 52.6  | 9.2   | 55.0  | 36.8  |
| Resource-based manufactures: agro-based                                     | 27.3  | 34.6  | 18.7  | 38.0  | 57.6  | 15.1  | 28.6  | 56.0  | 41.4  | 28.3  |
| Medium technology manufactures: engineering                                 | 28.4  | 0.6   | 43.8  | 9.6   | 10.6  | 12.9  | 16.3  | 21.4  | 0.4   | 21.6  |
| Medium technology manufactures: process                                     | 3.4   | 35.2  | 4.7   | 38.2  | 14.6  | 22.4  | 0.6   | 6.3   | 1.9   | 10.5  |
| Low technology manufactures: textile, garment and footwear                  | 0.3   | 0.0   | 4.8   | 0.6   | 5.2   | 0.7   | 1.1   | 6.5   | 0.0   | 1.0   |
| Resource-based manufactures: other                                          | 0.1   | 1.2   | 0.1   | 0.0   | 0.0   | 4.5   | 0.8   | 0.1   | 1.3   | 0.8   |
| High technology manufactures: other                                         | 1.1   | 0.1   | 0.9   | 0.2   | 0.6   | 0.5   | 0.0   | 0.4   | 0.0   | 0.7   |
| High technology manufactures: electronic and electrical                     | 0.0   | 0.0   | 0.1   | 0.1   | 0.0   | 1.5   | 0.0   | 0.1   | 0.0   | 0.2   |
| All sub-sectors                                                             | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 |
| <i>Panel B: foreign-MNE share of export value, %</i>                        |       |       |       |       |       |       |       |       |       |       |
| Primary products                                                            | 50.8  | 29.3  | 23.8  | 20.5  | 41.2  | 35.9  | 67.8  | 29.8  | 44.4  | 45.9  |
| Resource-based manufactures: agro-based                                     | 27.7  | 36.6  | 31.0  | 54.6  | 56.4  | 41.0  | 52.3  | 39.9  | 60.4  | 37.7  |
| Medium technology manufactures: engineering                                 | 29.5  | 80.8  | 37.1  | 45.7  | 6.8   | 11.6  | 86.8  | 20.2  | 23.5  | 33.3  |
| Medium technology manufactures: process                                     | 26.4  | 24.2  | 15.4  | 42.9  | 9.3   | 37.9  | 14.4  | 0.9   | 19.4  | 29.8  |
| Low technology manufactures: textile, garment and footwear                  | 22.8  | 96.9  | 20.9  | 0.0   | 26.0  | 9.0   | 0.0   | 0.0   | 7.4   | 16.4  |
| Resource-based manufactures: other                                          | 29.9  | 22.7  | 91.7  | 41.7  | 75.4  | 25.2  | 18.9  | 36.1  | 21.2  | 25.3  |
| High technology manufactures: other                                         | 26.3  | 37.3  | 14.4  | 52.2  | 4.1   | 14.0  | 25.4  | 18.7  | 34.9  | 23.8  |
| High technology manufactures: electronic and electrical                     | 22.6  | 27.0  | 17.2  | 38.7  | 26.1  | 19.7  | 13.9  | 41.6  | 16.7  | 20.3  |
| All sub-sectors                                                             | 37.2  | 30.3  | 30.4  | 44.4  | 40.7  | 33.0  | 65.0  | 29.6  | 50.2  | 38.5  |

*Notes:* Panel A: composition of each country's exports in the sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs. Lall (2000) technology classes.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 70: Agriculture: Exports by Technology Class and Parent Country**

| Lall (2000) category                                       | USA  | CHE | NLD | LIE | BRA | ESP | DEU | SAU | LUX | GBR | Other | Domestic |
|------------------------------------------------------------|------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-------|----------|
| Primary products                                           | 13.8 | 3.7 | 4.5 | 3.3 | 1.4 | 1.8 | 0.6 | 0.3 | 0.9 | 1.0 | 14.5  | 7.9      |
| Resource-based manufactures: agro-based                    | 5.9  | 3.1 | 2.2 | 1.2 | 1.8 | 0.9 | 0.5 | 3.2 | 2.1 | 1.4 | 15.5  | 9.5      |
| Medium technology manufactures: engineering                | 4.5  | 7.9 | 3.8 | 5.4 | 0.7 | 0.1 | 3.8 | 0.1 | 0.0 | 0.3 | 6.8   | 13.5     |
| Medium technology manufactures: process                    | 4.3  | 0.2 | 0.7 | 0.1 | 0.0 | 3.3 | 0.5 | 0.0 | 0.7 | 0.6 | 19.3  | 6.6      |
| Low technology manufactures: textile, garment and footwear | 7.2  | 0.0 | 0.0 | 2.8 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 6.4   | 6.8      |
| Resource-based manufactures: other                         | 3.0  | 0.0 | 1.6 | 0.0 | 0.4 | 0.0 | 1.6 | 1.2 | 0.0 | 2.9 | 14.6  | 10.1     |
| High technology manufactures: other                        | 0.6  | 0.1 | 0.0 | 0.0 | 0.1 | 0.2 | 2.3 | 0.0 | 0.0 | 0.0 | 20.5  | 3.1      |
| High technology manufactures: electronic and electrical    | 0.5  | 1.2 | 0.0 | 0.0 | 0.0 | 0.2 | 0.1 | 0.0 | 0.0 | 0.0 | 18.1  | 0.9      |

*Notes:* Percent of each group's export value, by country of the ultimate parent of foreign MNEs; domestic MNEs shown for comparison. Lall (2000) technology classes.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

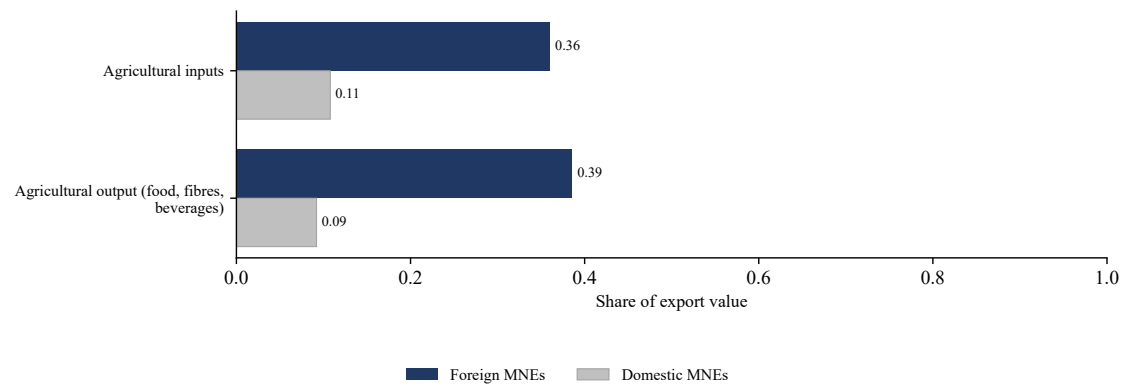
## 9.7 By Inputs and Other Products

**Table 71: Agriculture: Exports and Multinational Shares, by Inputs and Other Products**

| Inputs vs output                              | Total exports (US\$bn/yr) | MNE exports (US\$bn/yr) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|-----------------------------------------------|---------------------------|-------------------------|-------------|--------------|------------------------------------|
| Agricultural inputs                           | 14.9                      | 7.0                     | 36.0        | 10.8         | CHE 0.26, LIE 0.18, USA 0.18       |
| Agricultural output (food, fibres, beverages) | 64.0                      | 30.8                    | 38.6        | 9.2          | USA 0.25, NLD 0.09, CHE 0.08       |
| All                                           | 78.9                      | 37.8                    | 38.1        | 9.5          |                                    |

*Notes:* Inputs: seeds, animal feed and live animals. Foreign and domestic: percent of the group's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

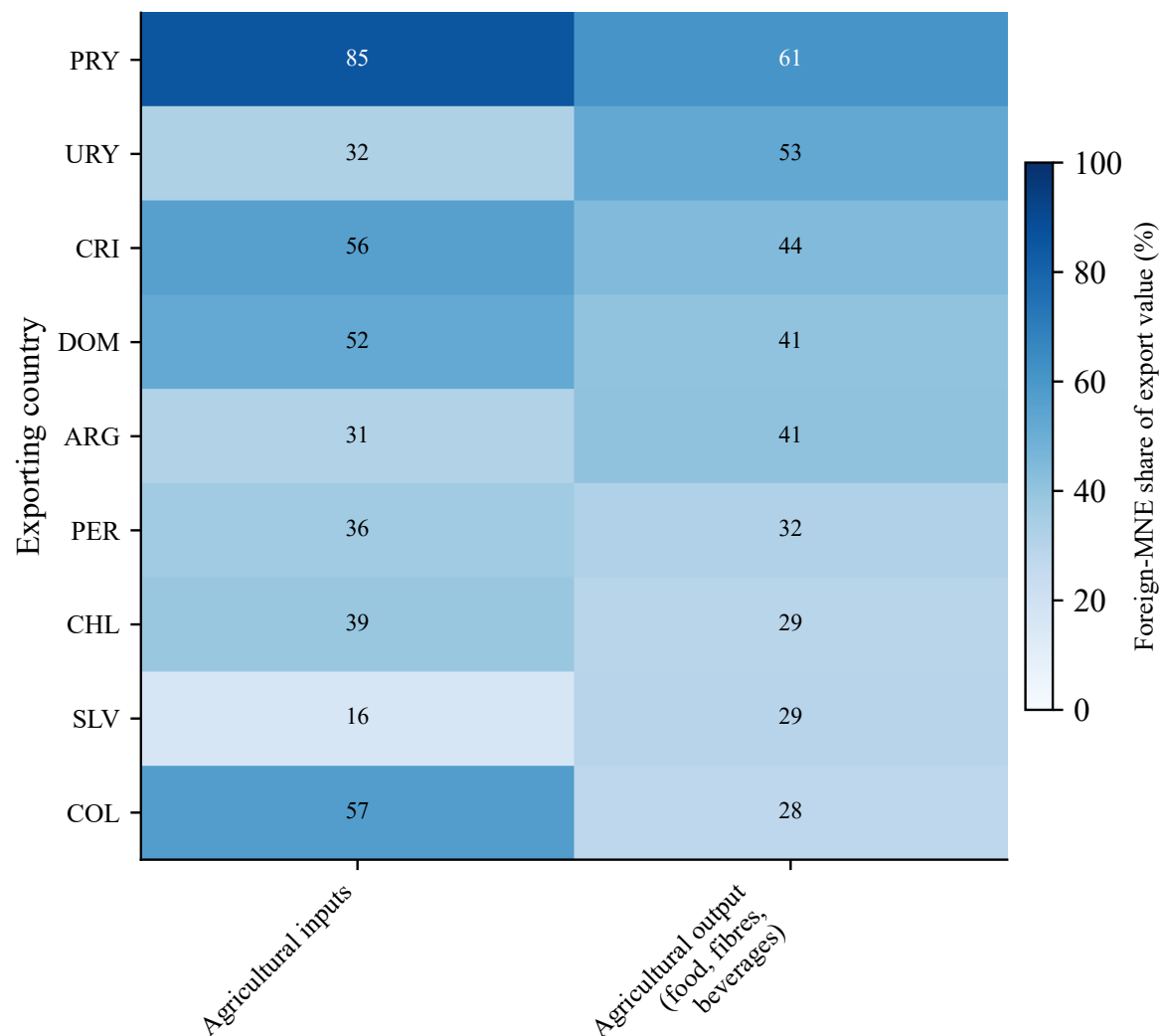
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 50:** Agriculture: Multinational Share of Export Value, by Inputs and Other Products

*Notes:* Shares of agricultural export value (HS 01–24), all available years pooled. Inputs: seeds, animal feed and live animals.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 51:** Agriculture: Foreign-Multinational Share of Export Value, by Exporting Country and Inputs and Other Products

*Notes:* Percent of each group's export value in each country shipped by foreign MNEs. Inputs: seeds, animal feed and live animals.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 72:** Agriculture: Exports by Exporting Country and Inputs and Other Products

| Origin                                                                   | Agricultural inputs | Agricultural output (food, fibres, beverages) | All   |
|--------------------------------------------------------------------------|---------------------|-----------------------------------------------|-------|
| <i>Panel A: composition of the origin's exports, % (rows sum to 100)</i> |                     |                                               |       |
| ARG                                                                      | 29.9                | 70.1                                          | 100.0 |
| CHL                                                                      | 5.0                 | 95.0                                          | 100.0 |
| COL                                                                      | 0.9                 | 99.1                                          | 100.0 |
| CRI                                                                      | 1.7                 | 98.3                                          | 100.0 |
| DOM                                                                      | 1.3                 | 98.7                                          | 100.0 |
| PER                                                                      | 21.7                | 78.3                                          | 100.0 |
| PRY                                                                      | 17.0                | 83.0                                          | 100.0 |
| SLV                                                                      | 2.8                 | 97.2                                          | 100.0 |
| URY                                                                      | 3.4                 | 96.6                                          | 100.0 |
| All origins                                                              | 17.5                | 82.5                                          | 100.0 |
| <i>Panel B: foreign-MNE share of export value, %</i>                     |                     |                                               |       |
| ARG                                                                      | 31.1                | 40.7                                          | 37.8  |
| CHL                                                                      | 38.7                | 29.2                                          | 29.6  |
| COL                                                                      | 57.1                | 27.7                                          | 28.0  |
| CRI                                                                      | 56.2                | 44.3                                          | 44.5  |
| DOM                                                                      | 52.0                | 40.5                                          | 40.7  |
| PER                                                                      | 36.4                | 32.0                                          | 33.0  |
| PRY                                                                      | 85.4                | 60.8                                          | 65.0  |
| SLV                                                                      | 15.7                | 29.3                                          | 28.9  |
| URY                                                                      | 32.3                | 52.5                                          | 51.8  |
| All origins                                                              | 36.0                | 38.6                                          | 38.1  |

*Notes:* Panel A: composition of each country's exports in the sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs. Inputs: seeds, animal feed and live animals.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 73:** Agriculture: Exports by Inputs and Other Products and Parent Country

| Inputs vs output                              | USA | CHE | NLD | LIE | BRA | ESP | DEU | SAU | LUX | GBR | Other | Domestic |
|-----------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-------|----------|
| Agricultural inputs                           | 6.0 | 9.0 | 4.4 | 6.1 | 0.3 | 0.4 | 1.0 | 0.1 | 0.0 | 1.1 | 7.6   | 10.8     |
| Agricultural output (food, fibres, beverages) | 8.4 | 2.6 | 2.9 | 1.8 | 1.5 | 1.4 | 1.2 | 1.3 | 1.2 | 0.8 | 15.4  | 9.2      |

*Notes:* Percent of each group's export value, by country of the ultimate parent of foreign MNEs; domestic MNEs shown for comparison. Inputs: seeds, animal feed and live animals.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 74:** Agriculture: Largest Input Products

| HS6    | Description                                                                                                                            | \$bn/yr | Foreign (%) | Domestic (%) |
|--------|----------------------------------------------------------------------------------------------------------------------------------------|---------|-------------|--------------|
| 230400 | Oil-cake & other solid residues, whether/not ground/in pellets, from extraction of soyabean oil                                        | 10.84   | 35          | 12           |
| 230120 | Flours, meals & pellets of fish/of crustaceans, molluscs/other aquatic invertebrates                                                   | 1.70    | 31          | 10           |
| 230990 | Preparations of a kind used in animal feeding other than dog/cat food put up for RS                                                    | 0.51    | 57          | 3            |
| 071333 | Kidney beans, incl. white pea beans ( <i>Phaseolus vulgaris</i> ), dried, shelled, whether/not skinned/split                           | 0.30    | 28          | 4            |
| 230250 | Bran, sharps & other residues, whether/not in the form of pellets, derived from the sifting/milling/other working of leguminous plants | 0.28    | 30          | 16           |
| 100510 | Maize (corn), seed                                                                                                                     | 0.19    | 60          | 3            |
| 230910 | Dog/cat food, put up for RS                                                                                                            | 0.17    | 81          | 5            |
| 120991 | Vegetable seeds, of a kind used for sowing                                                                                             | 0.12    | 67          | 1            |
| 230630 | Oil-cake & other solid residues, whether/not ground/in pellets, from extraction of sunflower seeds                                     | 0.12    | 22          | 2            |
| 010290 | Live bovine animals other than pure-bred breeding animals                                                                              | 0.08    | 31          | 0            |
| 071320 | Chickpeas ( <i>garbanzos</i> ), dried, shelled, whether/not skinned/split                                                              | 0.08    | 23          | 2            |
| 230800 | Vegetable materials/waste/residues/by-products, whether/not in pellets, of a kind used in animal feeding, n.e.s.                       | 0.08    | 19          | 0            |
| 100610 | Rice in the husk (paddy/rough)                                                                                                         | 0.06    | 45          | 5            |
| 230110 | Flours, meals & pellets of meat/meat offal; greaves                                                                                    | 0.05    | 31          | 0            |
| 051199 | Animal products not elsewhere specified/incl. (excl. of 0511.10); dead animals of Ch. 1, unfit for human consumption                   | 0.04    | 39          | 5            |
| 071310 | Peas ( <i>Pisum sativum</i> ), dried, shelled, whether/not skinned/split                                                               | 0.03    | 22          | 1            |
| 071339 | Beans ( <i>Vigna</i> spp., <i>Phaseolus</i> spp. (excl. of 0713.31-0713.33)), dried, shelled, whether/not skinned/split                | 0.03    | 11          | 2            |
| 230230 | Bran, sharps & other residues, whether/not in the form of pellets, derived from the sifting/milling/other working of wheat             | 0.03    | 16          | 1            |
| 010511 | Live fowls of species <i>Gallus domesticus</i> , weighing not >185g                                                                    | 0.02    | 3           | 1            |
| 120930 | Seeds of herbaceous plants cultivated principally for their flowers, of a kind used for sowing                                         | 0.02    | 76          | 2            |

*Notes:* HS6 input products with the largest export value. Foreign and domestic: percent of the product's export value. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 9.8 HS Section and End Use

**Table 75:** Agriculture: Exports by HS Section and End Use

| HS section / BEC end use                   | Intermediate | Consumption | Capital | Total |
|--------------------------------------------|--------------|-------------|---------|-------|
| I Live animals; animal products            | 0.8          | 9.6         | 0.0     | 10.4  |
| II Vegetable products                      | 20.0         | 10.6        | 0.0     | 30.6  |
| III Fats and oils                          | 6.3          | 0.6         | 0.0     | 6.9   |
| IV Prepared foodstuffs, beverages, tobacco | 15.4         | 9.6         | 0.0     | 25.0  |
| Total                                      | 42.5         | 30.4        | 0.0     | 72.9  |

*Notes:* Export value by HS section and BEC Rev. 4 end use. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 76:** Agriculture: Foreign-Multinational Share of Exports by HS Section and End Use

| HS section / BEC end use                   | Intermediate | Consumption | Capital |
|--------------------------------------------|--------------|-------------|---------|
| I Live animals; animal products            | 33.1         | 35.8        | 16.8    |
| II Vegetable products                      | 50.2         | 28.3        | –       |
| III Fats and oils                          | 29.8         | 47.0        | –       |
| IV Prepared foodstuffs, beverages, tobacco | 33.9         | 41.6        | –       |

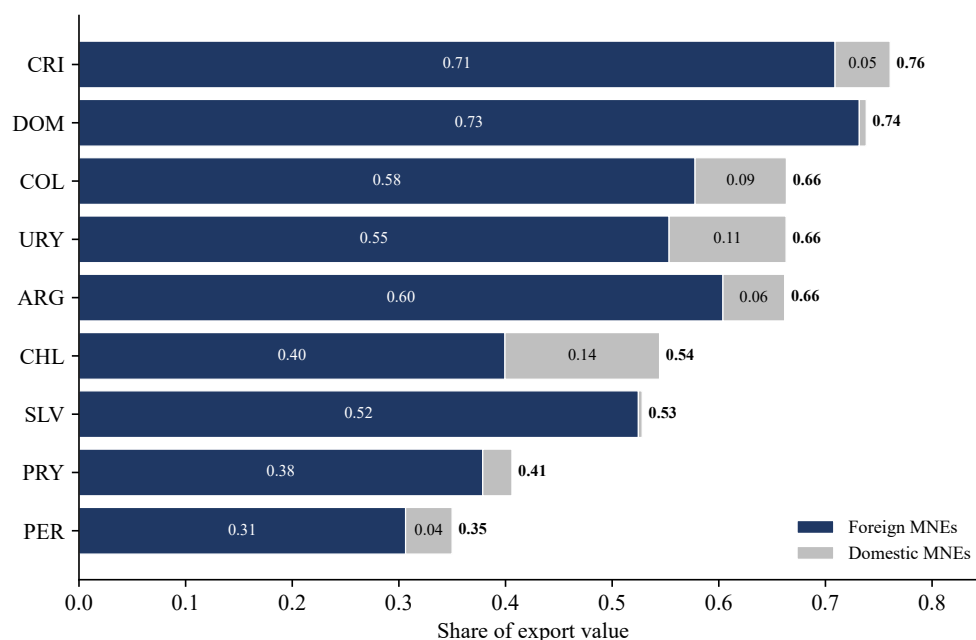
*Notes:* Percent of each cell's export value shipped by foreign MNEs.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## Part II

# Manufacturing

## 10 Fact 1: Multinationals Account for a Large Share of Exports



**Figure 52:** Manufacturing: Multinational Share of Export Value, by Exporting Country

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. The figure at the end of each bar is the total MNE share.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

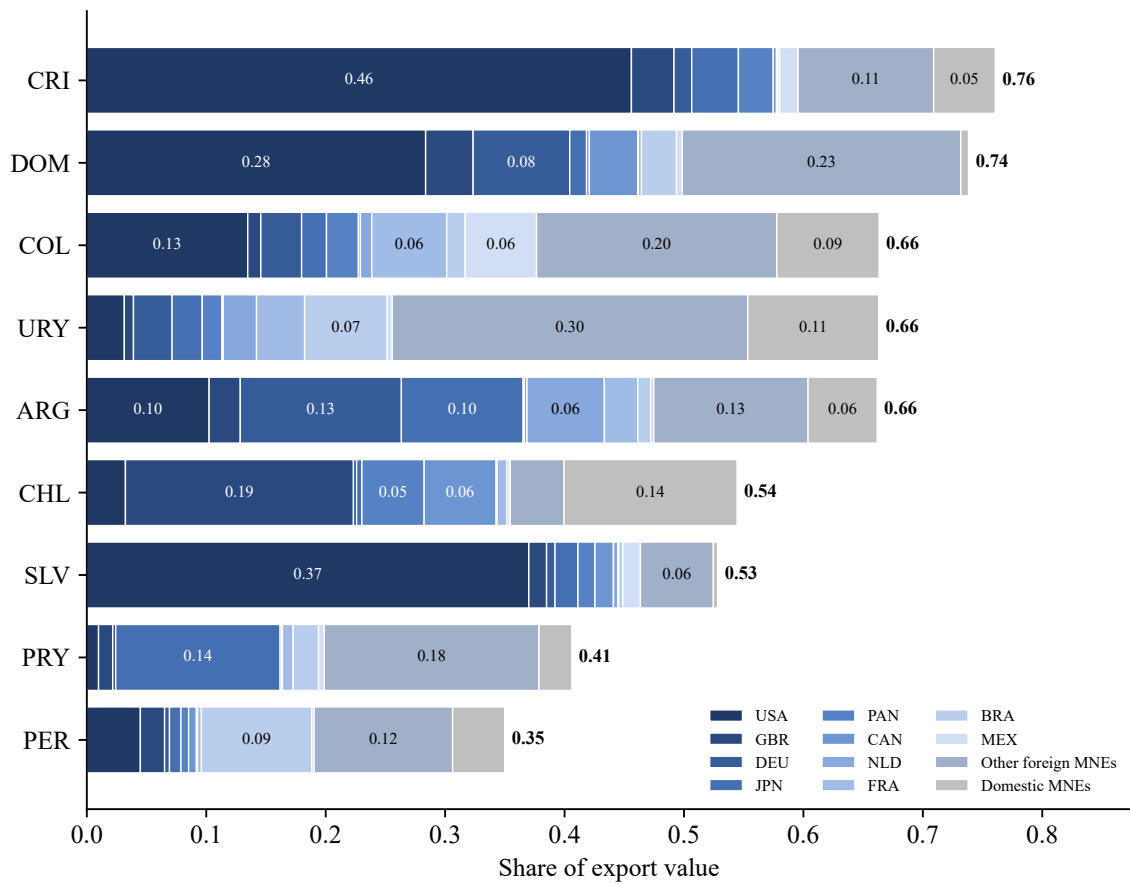
**Table 77:** Manufacturing: Multinational Share of Export Value, by Exporting Country

| Origin | Foreign | Domestic | All MNEs | Value (US\$bn/yr) |
|--------|---------|----------|----------|-------------------|
| CRI    | 0.709   | 0.052    | 0.761    | 6.0               |
| DOM    | 0.732   | 0.006    | 0.738    | 4.7               |
| COL    | 0.578   | 0.086    | 0.663    | 8.9               |
| URY    | 0.553   | 0.110    | 0.663    | 2.4               |
| ARG    | 0.604   | 0.058    | 0.662    | 21.7              |
| CHL    | 0.399   | 0.145    | 0.544    | 16.0              |
| SLV    | 0.525   | 0.004    | 0.528    | 3.8               |
| PRY    | 0.379   | 0.027    | 0.406    | 1.0               |
| PER    | 0.306   | 0.044    | 0.350    | 8.1               |
| All    | 0.516   | 0.080    | 0.596    | 72.7              |

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 10.1 By Parent Country



**Figure 53:** Manufacturing: Multinational Share of Export Value, by Exporting Country and Parent Country

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. Foreign MNEs split by country of the ultimate parent; the ten largest parent countries are shown separately.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 78:** Manufacturing: Multinational Share of Export Value, by Exporting Country and Parent Country

| Origin | USA   | GBR   | DEU   | JPN   | PAN   | CAN   | NLD   | FRA   | BRA   | MEX   | Other | Domestic | All MNEs | Value (US\$bn/yr) |
|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|----------|-------------------|
| CRI    | 0.456 | 0.036 | 0.015 | 0.039 | 0.029 | 0.003 | 0.001 | 0.001 | 0.000 | 0.015 | 0.113 | 0.052    | 0.761    | 6.0               |
| DOM    | 0.284 | 0.040 | 0.081 | 0.014 | 0.002 | 0.041 | 0.000 | 0.003 | 0.029 | 0.005 | 0.233 | 0.006    | 0.738    | 4.7               |
| COL    | 0.135 | 0.011 | 0.034 | 0.021 | 0.027 | 0.002 | 0.010 | 0.063 | 0.016 | 0.060 | 0.201 | 0.086    | 0.663    | 8.9               |
| URY    | 0.031 | 0.008 | 0.032 | 0.025 | 0.016 | 0.001 | 0.028 | 0.040 | 0.069 | 0.004 | 0.298 | 0.110    | 0.663    | 2.4               |
| ARG    | 0.102 | 0.026 | 0.135 | 0.102 | 0.001 | 0.002 | 0.065 | 0.028 | 0.011 | 0.003 | 0.129 | 0.058    | 0.662    | 21.7              |
| CHL    | 0.032 | 0.191 | 0.003 | 0.004 | 0.052 | 0.060 | 0.001 | 0.008 | 0.002 | 0.001 | 0.045 | 0.145    | 0.544    | 16.0              |
| SLV    | 0.370 | 0.015 | 0.007 | 0.019 | 0.014 | 0.015 | 0.004 | 0.001 | 0.003 | 0.015 | 0.061 | 0.004    | 0.528    | 3.8               |
| PRY    | 0.010 | 0.012 | 0.003 | 0.138 | 0.002 | 0.000 | 0.000 | 0.009 | 0.021 | 0.005 | 0.180 | 0.027    | 0.406    | 1.0               |
| PER    | 0.045 | 0.020 | 0.004 | 0.010 | 0.006 | 0.006 | 0.001 | 0.003 | 0.092 | 0.002 | 0.116 | 0.044    | 0.350    | 8.1               |

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. Other: all remaining foreign parents. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



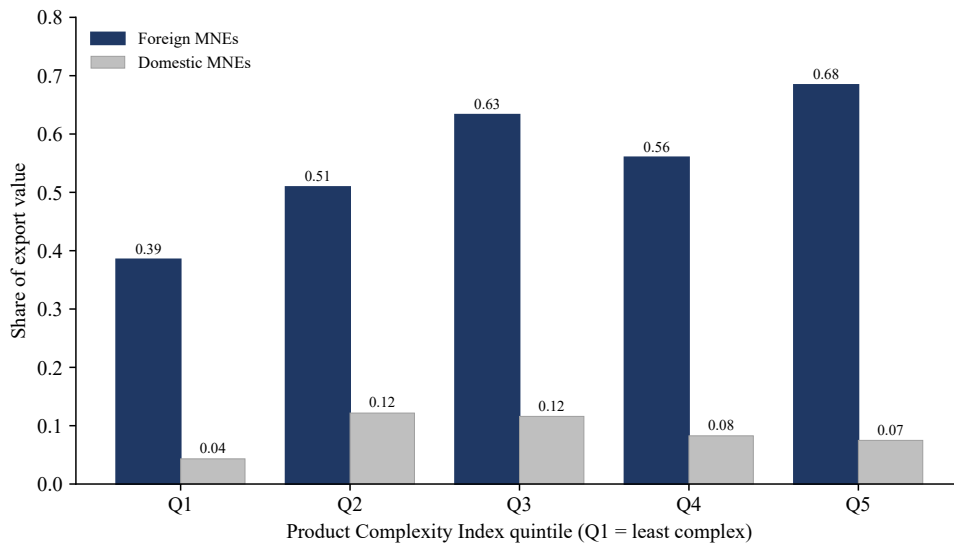
**Figure 54:** Manufacturing: Export Value by Exporting Country and Parent Country

*Notes:* Percent of each country's manufacturing export value, by country of the ultimate parent of the exporter.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 11 Fact 2: Foreign Multinationals Specialize in Complex Products, Domestic Multinationals in Primary Goods

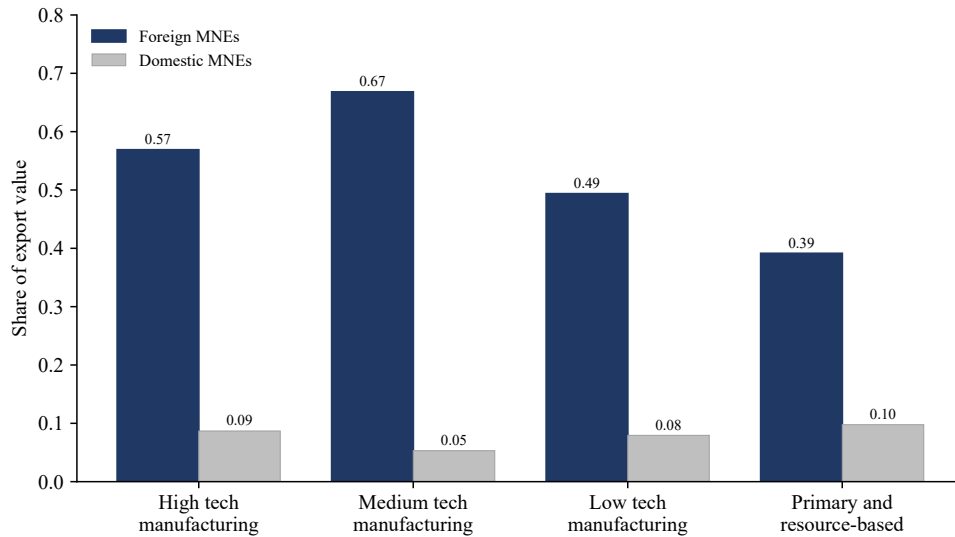
### 11.1 Product Complexity and Technology



**Figure 55:** Manufacturing: Multinational Share of Export Value, by Product Complexity Quintile

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. Quintiles of the Hausmann–Hidalgo Product Complexity Index across HS6 products (Q1 = least complex).

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

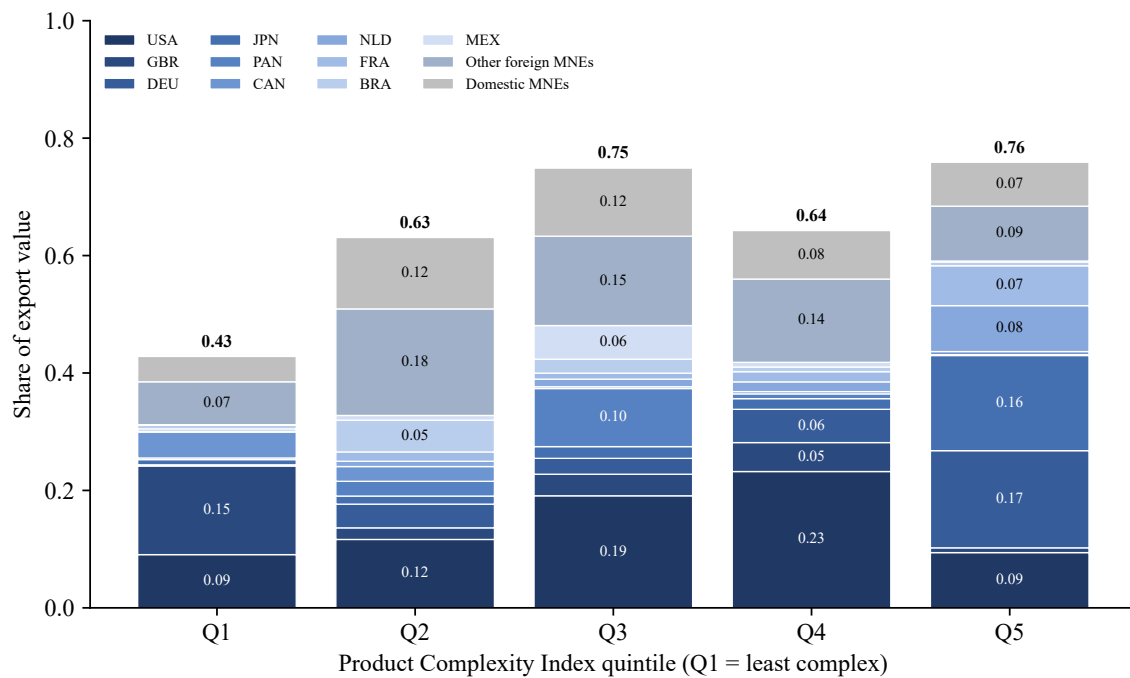


**Figure 56: Manufacturing: Multinational Share of Export Value, by Technology Class**

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. Lall (2000) technology classes: primary and resource-based products, and low-, medium- and high-technology manufactures.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 11.2 By Parent Country



**Figure 57: Manufacturing: Multinational Share of Export Value, by Product Complexity Quintile and Parent Country**

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. Foreign MNEs split by country of the ultimate parent. Quintiles of the Hausmann–Hidalgo Product Complexity Index across HS6 products (Q1 = least complex).

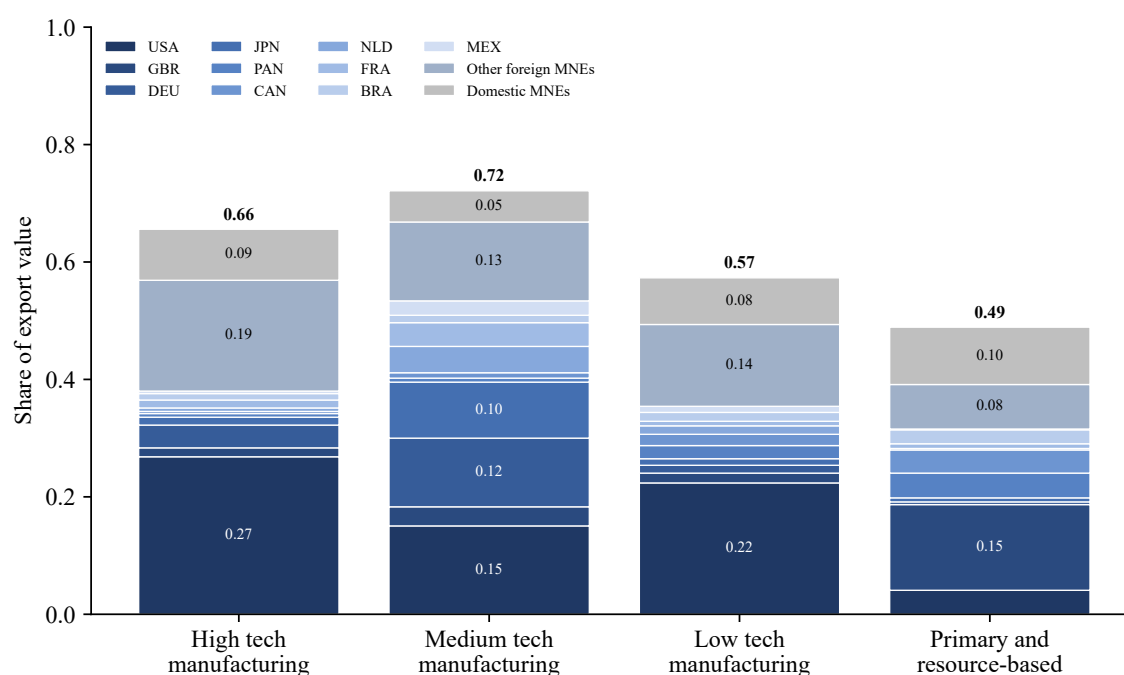
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 79: Manufacturing: Multinational Share of Export Value, by Product Complexity Quintile and Parent Country**

| PCI quintile | USA   | GBR   | DEU   | JPN   | PAN   | CAN   | NLD   | FRA   | BRA   | MEX   | Other | Domestic | All MNEs | Value (US\$bn/yr) |
|--------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|----------|-------------------|
| Q1           | 0.091 | 0.151 | 0.002 | 0.009 | 0.003 | 0.044 | 0.002 | 0.004 | 0.007 | 0.001 | 0.073 | 0.043    | 0.428    | 24.3              |
| Q2           | 0.116 | 0.020 | 0.040 | 0.014 | 0.025 | 0.025 | 0.009 | 0.016 | 0.054 | 0.008 | 0.181 | 0.122    | 0.631    | 12.7              |
| Q3           | 0.191 | 0.037 | 0.027 | 0.020 | 0.099 | 0.003 | 0.013 | 0.010 | 0.024 | 0.057 | 0.152 | 0.116    | 0.749    | 10.4              |
| Q4           | 0.232 | 0.049 | 0.057 | 0.018 | 0.008 | 0.004 | 0.017 | 0.017 | 0.008 | 0.008 | 0.142 | 0.083    | 0.643    | 11.0              |
| Q5           | 0.094 | 0.008 | 0.165 | 0.162 | 0.002 | 0.005 | 0.078 | 0.068 | 0.006 | 0.002 | 0.093 | 0.075    | 0.759    | 13.2              |

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. Other: all remaining foreign parents. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Figure 58: Manufacturing: Multinational Share of Export Value, by Technology Class and Parent Country**

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. Foreign MNEs split by country of the ultimate parent. Lall (2000) technology classes: primary and resource-based products, and low-, medium- and high-technology manufactures.

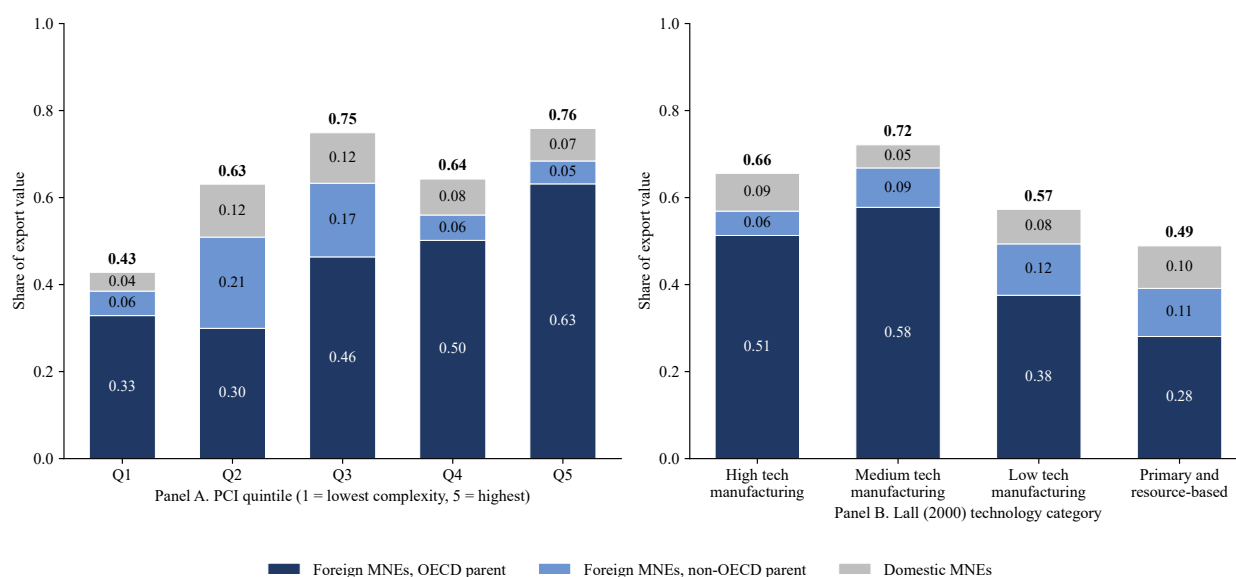
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 80: Manufacturing: Multinational Share of Export Value, by Technology Class and Parent Country**

| Technology category        | USA   | GBR   | DEU   | JPN   | PAN   | CAN   | NLD   | FRA   | BRA   | MEX   | Other | Domestic | All MNEs | Value (US\$bn/yr) |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|----------|-------------------|
| High tech manufacturing    | 0.268 | 0.015 | 0.039 | 0.013 | 0.006 | 0.004 | 0.006 | 0.014 | 0.011 | 0.004 | 0.189 | 0.087    | 0.656    | 5.4               |
| Medium tech manufacturing  | 0.151 | 0.033 | 0.117 | 0.095 | 0.007 | 0.009 | 0.045 | 0.040 | 0.013 | 0.025 | 0.134 | 0.053    | 0.721    | 26.5              |
| Low tech manufacturing     | 0.224 | 0.017 | 0.013 | 0.011 | 0.022 | 0.019 | 0.014 | 0.008 | 0.015 | 0.010 | 0.139 | 0.079    | 0.573    | 14.7              |
| Primary and resource-based | 0.041 | 0.146 | 0.005 | 0.007 | 0.042 | 0.039 | 0.003 | 0.008 | 0.024 | 0.001 | 0.076 | 0.098    | 0.489    | 25.1              |

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. Other: all remaining foreign parents. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

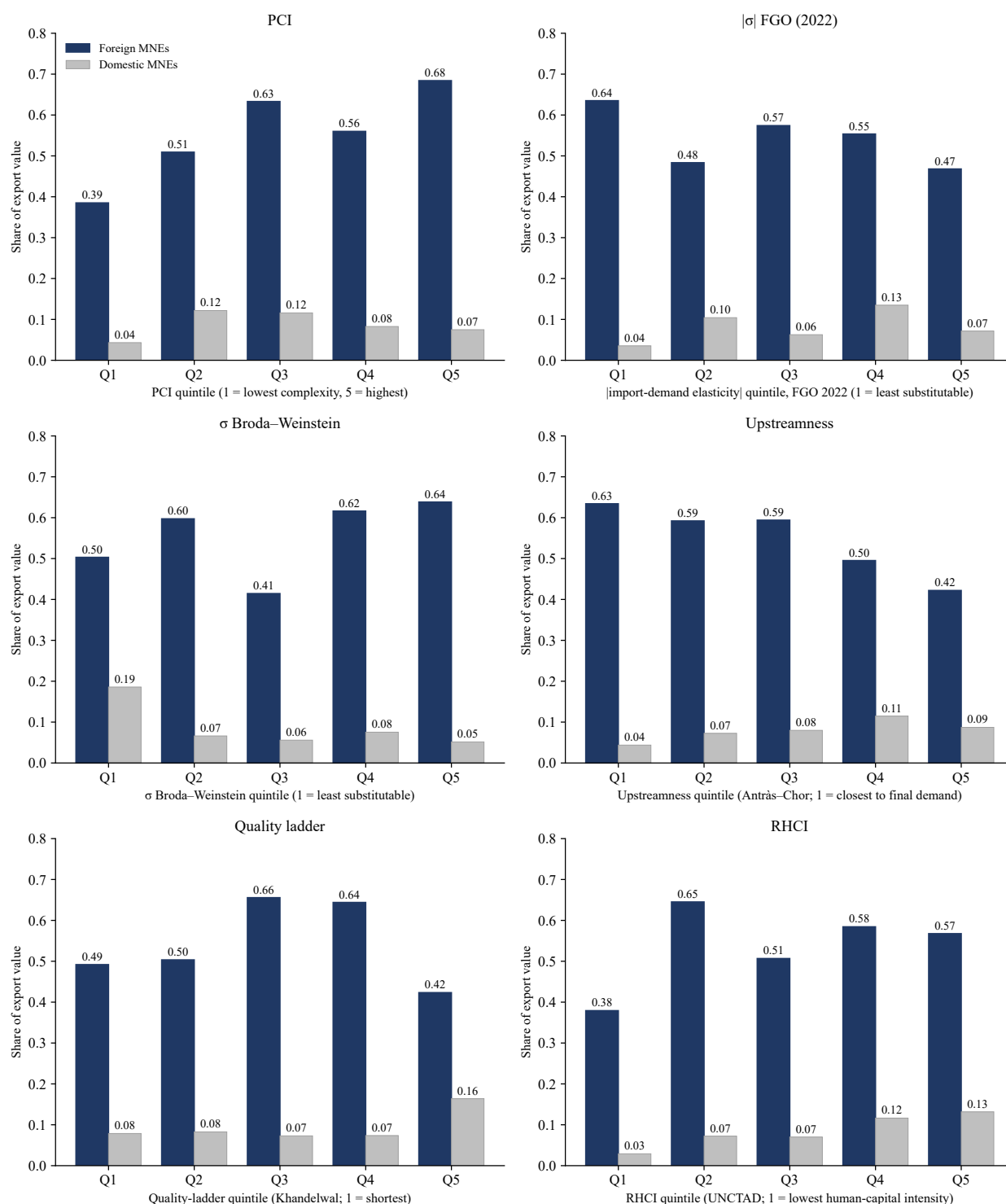


**Figure 59: Manufacturing: Multinational Share of Export Value, by Product Sophistication and Location of the Parent**

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. Foreign MNEs are split by whether the ultimate parent is in an OECD member country. Panel A: Quintiles of the Hausmann–Hidalgo Product Complexity Index across HS6 products (Q1 = least complex). Panel B: Lall (2000) technology classes: primary and resource-based products, and low-, medium- and high-technology manufactures.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

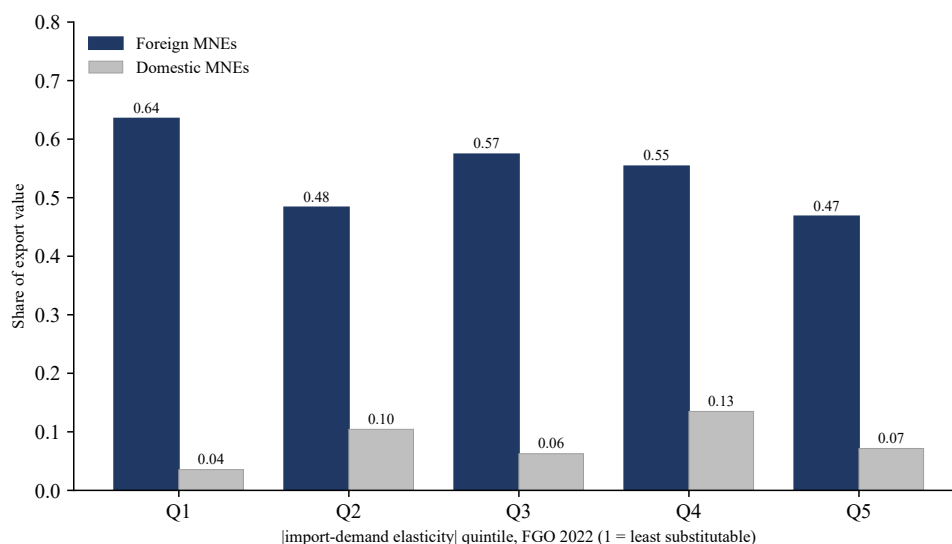
## 11.3 Other Measures of Product Sophistication



**Figure 60:** Manufacturing: Multinational Share of Export Value, by Quintile of Six Sophistication Measures

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. Quintiles computed over HS6 products. The six measures are the Product Complexity Index, the absolute import-demand elasticity of Fontagné, Guimbard and Orefice (2022), the elasticity of substitution of Broda and Weinstein, upstreamness, the quality ladder and the revealed human-capital intensity of UNCTAD.

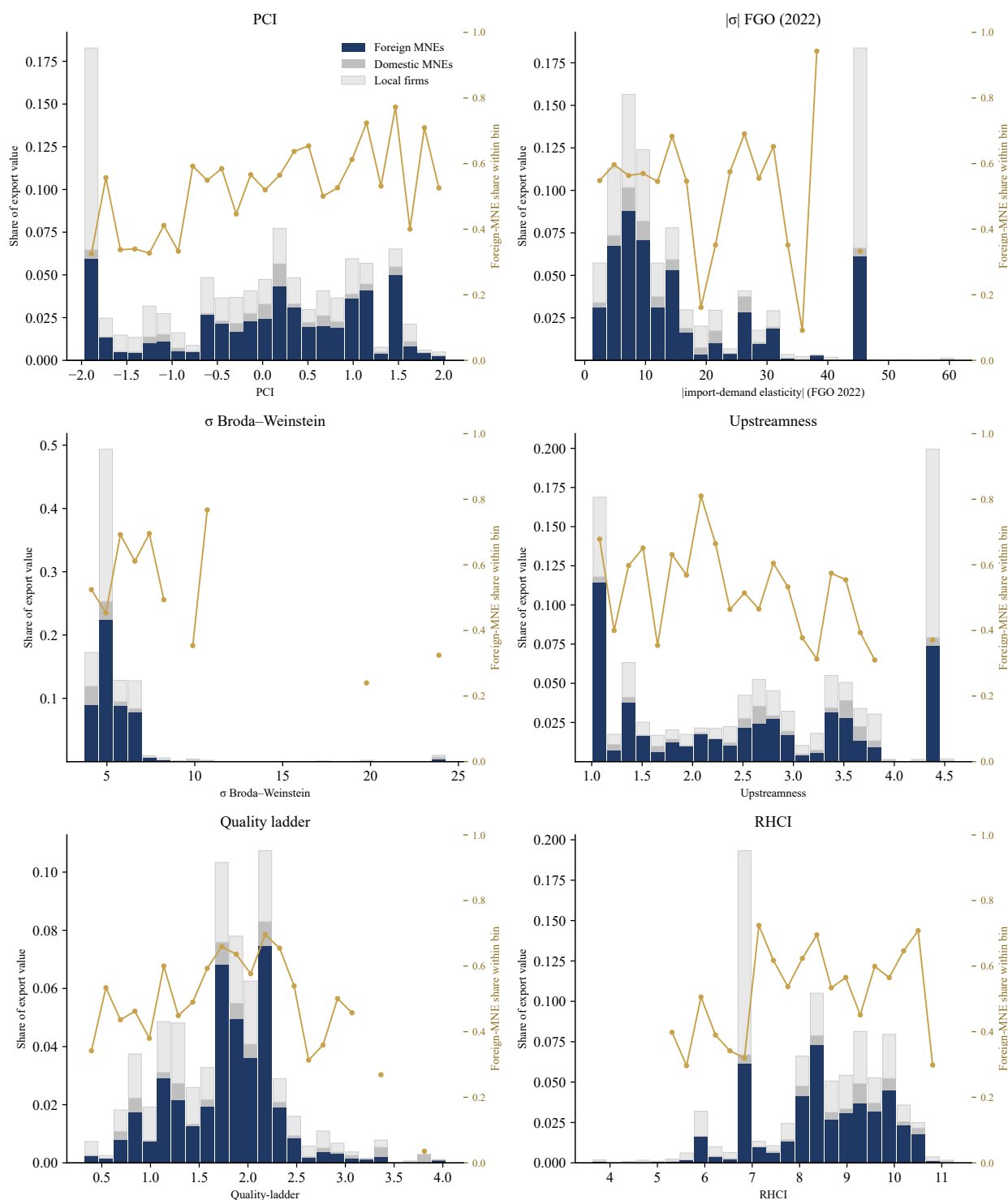
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 61:** Manufacturing: Multinational Share of Export Value, by Quintile of the Import-Demand Elasticity

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. Quintiles of the absolute import-demand elasticity of Fontagné, Guimbard and Orefice (2022) across HS6 products; higher quintiles are more substitutable products.

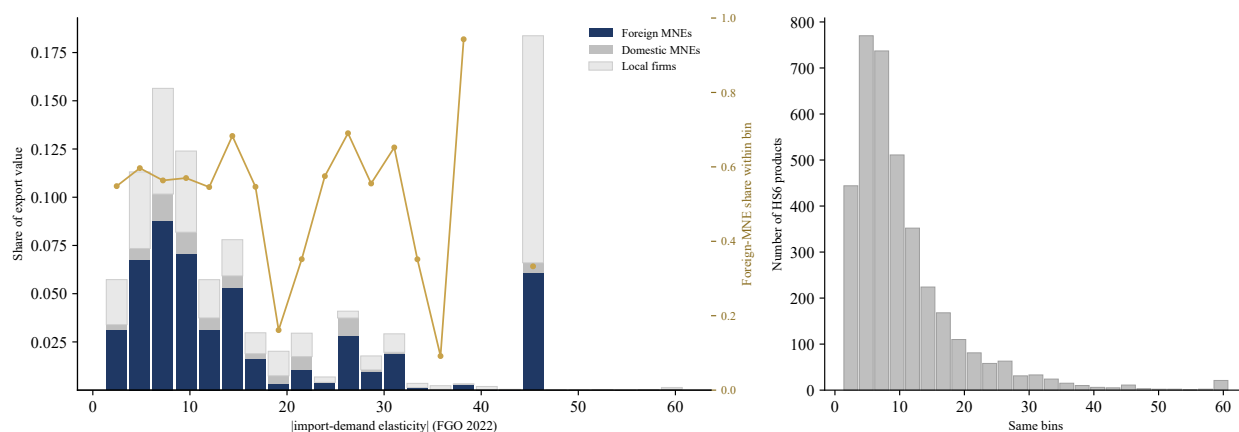
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 62: Manufacturing: Distribution of Export Value across Six Sophistication Measures**

*Notes:* Bars: manufacturing export value (HS 28–97 excluding 71) in each bin, by type of exporter. Line (right axis): foreign-MNE share of each bin. The six measures are the Product Complexity Index, the absolute import-demand elasticity of Fontagné, Guimbard and Oreife (2022), the elasticity of substitution of Broda and Weinstein, upstreamness, the quality ladder and the revealed human-capital intensity of UNCTAD.

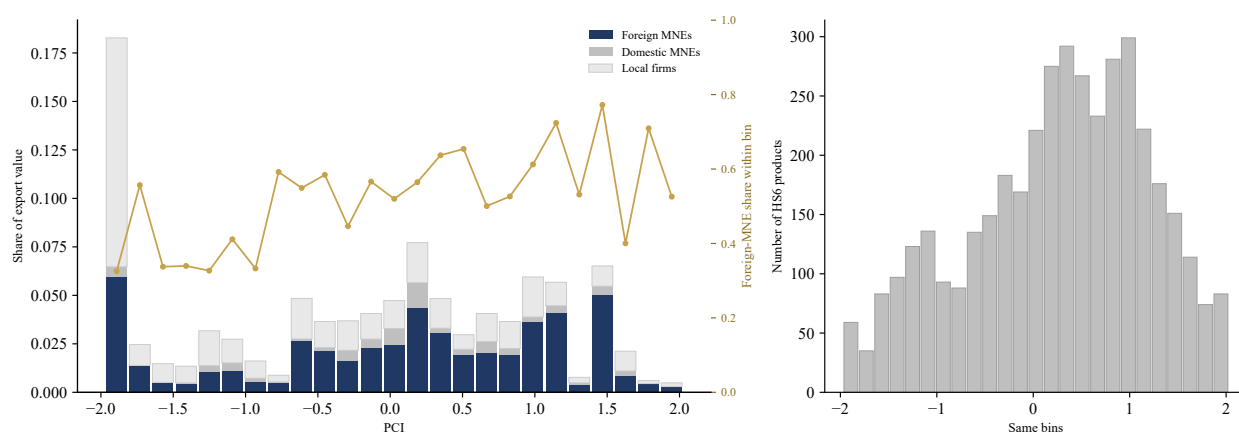
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 63:** Manufacturing: Distribution of Export Value across the Import-Demand Elasticity

*Notes:* Left: export value in each bin, by type of exporter, and the foreign-MNE share of each bin. Right: number of HS6 products in each bin.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 64:** Manufacturing: Distribution of Export Value across the Product Complexity Index

*Notes:* Left: export value in each bin, by type of exporter, and the foreign-MNE share of each bin. Right: number of HS6 products in each bin.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 81:** Manufacturing: Multinational Share of Export Value, by Quintile of Six Sophistication Measures

| Measure                  | Foreign MNE share |      |      |      |      | Domestic MNE share |      |      |      |      |
|--------------------------|-------------------|------|------|------|------|--------------------|------|------|------|------|
|                          | Q1                | Q2   | Q3   | Q4   | Q5   | Q1                 | Q2   | Q3   | Q4   | Q5   |
| PCI                      | 0.39              | 0.51 | 0.63 | 0.56 | 0.68 | 0.04               | 0.12 | 0.12 | 0.08 | 0.07 |
| $ \sigma $ FGO (2022)    | 0.64              | 0.48 | 0.57 | 0.55 | 0.47 | 0.04               | 0.10 | 0.06 | 0.13 | 0.07 |
| $\sigma$ Broda–Weinstein | 0.50              | 0.60 | 0.41 | 0.62 | 0.64 | 0.19               | 0.07 | 0.06 | 0.08 | 0.05 |
| Upstreamness             | 0.63              | 0.59 | 0.59 | 0.50 | 0.42 | 0.04               | 0.07 | 0.08 | 0.11 | 0.09 |
| Quality ladder           | 0.49              | 0.50 | 0.66 | 0.64 | 0.42 | 0.08               | 0.08 | 0.07 | 0.07 | 0.16 |
| RHCI                     | 0.38              | 0.65 | 0.51 | 0.58 | 0.57 | 0.03               | 0.07 | 0.07 | 0.12 | 0.13 |

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. Quintiles computed over HS6 products. —: fewer than 30 HS6 products carry the measure.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 82:** Manufacturing: Correlations between Sophistication Measures

|                          | PCI   | $ \sigma $ FGO (2022) | $\sigma$ Broda–Weinstein | Upstreamness | Quality ladder | RHCI  |
|--------------------------|-------|-----------------------|--------------------------|--------------|----------------|-------|
| PCI                      | 1.00  | -0.61                 | 0.06                     | -0.52        | 0.29           | 0.79  |
| $ \sigma $ FGO (2022)    | -0.61 | 1.00                  | -0.05                    | 0.74         | 0.12           | -0.43 |
| $\sigma$ Broda–Weinstein | 0.06  | -0.05                 | 1.00                     | -0.15        | -0.12          | 0.08  |
| Upstreamness             | -0.52 | 0.74                  | -0.15                    | 1.00         | 0.13           | -0.32 |
| Quality ladder           | 0.29  | 0.12                  | -0.12                    | 0.13         | 1.00           | 0.36  |
| RHCI                     | 0.79  | -0.43                 | 0.08                     | -0.32        | 0.36           | 1.00  |

Notes: Correlations across HS6 products, weighted by export value.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

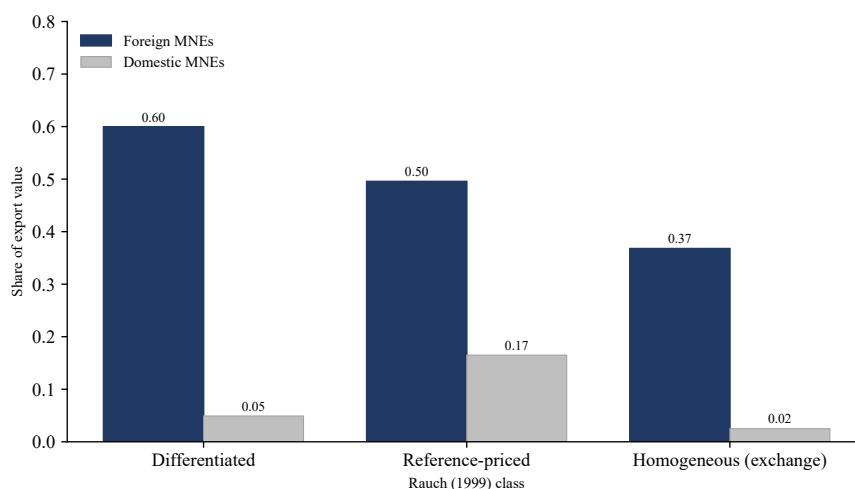
## 11.4 Rauch Classes and End Use

**Table 83:** Manufacturing: Multinational Share of Export Value, by Rauch Product Class

| Rauch (1999) class     | Foreign | Domestic | All MNEs | HS6 products | Value (US\$bn/yr) |
|------------------------|---------|----------|----------|--------------|-------------------|
| Differentiated         | 0.599   | 0.049    | 0.649    | 2,505        | 36.2              |
| Reference-priced       | 0.495   | 0.165    | 0.661    | 1,180        | 18.6              |
| Homogeneous (exchange) | 0.368   | 0.025    | 0.393    | 128          | 14.4              |

Notes: Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. Rauch (1999) conservative classification: homogeneous goods are traded on organized exchanges, reference-priced goods have a price quoted in trade publications, and differentiated goods have neither. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Figure 65:** Manufacturing: Multinational Share of Export Value, by Rauch Product Class

Notes: Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. Rauch (1999) conservative classification: homogeneous goods are traded on organized exchanges, reference-priced goods have a price quoted in trade publications, and differentiated goods have neither.

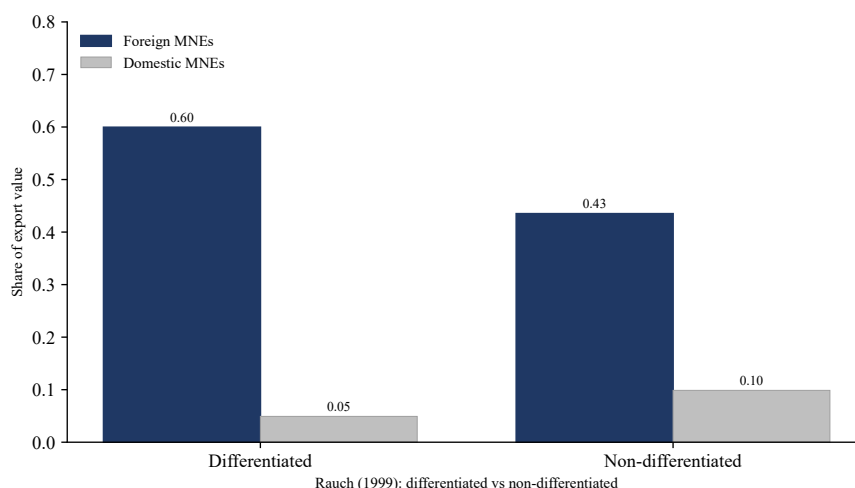
Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 84:** Manufacturing: Multinational Share of Export Value: Differentiated and Other Products

| Rauch (1999): differentiated vs non-differentiated | Foreign | Domestic | All MNEs | HS6 products | Value (US\$bn/yr) |
|----------------------------------------------------|---------|----------|----------|--------------|-------------------|
| Differentiated                                     | 0.599   | 0.049    | 0.649    | 2,505        | 36.2              |
| Non-differentiated                                 | 0.435   | 0.099    | 0.534    | 1,308        | 33.0              |

Notes: Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. Rauch (1999) conservative classification; other products are reference-priced and homogeneous goods. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 66:** Manufacturing: Multinational Share of Export Value: Differentiated and Other Products

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. Rauch (1999) conservative classification; other products are reference-priced and homogeneous goods.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 85:** Manufacturing: Multinational Share of Export Value, by End Use

| BEC end use  | Foreign | Domestic | All MNEs | HS6 products | Value (US\$bn/yr) |
|--------------|---------|----------|----------|--------------|-------------------|
| Intermediate | 0.469   | 0.091    | 0.560    | 2,668        | 47.2              |
| Consumption  | 0.606   | 0.049    | 0.655    | 734          | 15.2              |
| Capital      | 0.648   | 0.056    | 0.704    | 654          | 9.3               |

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. End use from BEC Rev. 4; HS6 products without a BEC correspondence are excluded. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 11.5 Regressions

**Table 86:** Manufacturing: Product Characteristics and Multinational Shares

|                                                            | (1)                    | (2)                    | (3)                    |
|------------------------------------------------------------|------------------------|------------------------|------------------------|
| <i>Panel A. Dependent variable: share of all MNEs</i>      |                        |                        |                        |
| Complexity (PCI)                                           | 0.0873***<br>(0.0038)  | 0.0859***<br>(0.0037)  | 0.0825***<br>(0.0038)  |
| $ \sigma $ FGO (per s.d.)                                  | -0.0153***<br>(0.0022) | -0.0169***<br>(0.0021) | -0.0200***<br>(0.0021) |
| Upstreamness                                               | -0.0330***<br>(0.0043) | -0.0328***<br>(0.0041) | -0.0216***<br>(0.0042) |
| <i>Panel B. Dependent variable: share of foreign MNEs</i>  |                        |                        |                        |
| Complexity (PCI)                                           | 0.0607***<br>(0.0043)  | 0.0604***<br>(0.0041)  | 0.0602***<br>(0.0041)  |
| $ \sigma $ FGO (per s.d.)                                  | 0.0092***<br>(0.0026)  | 0.0065***<br>(0.0024)  | 0.0033<br>(0.0024)     |
| Upstreamness                                               | -0.0434***<br>(0.0045) | -0.0435***<br>(0.0043) | -0.0267***<br>(0.0041) |
| <i>Panel C. Dependent variable: share of domestic MNEs</i> |                        |                        |                        |
| Complexity (PCI)                                           | 0.0266***<br>(0.0031)  | 0.0255***<br>(0.0028)  | 0.0222***<br>(0.0029)  |
| $ \sigma $ FGO (per s.d.)                                  | -0.0245***<br>(0.0020) | -0.0235***<br>(0.0018) | -0.0233***<br>(0.0021) |
| Upstreamness                                               | 0.0104***<br>(0.0023)  | 0.0107***<br>(0.0024)  | 0.0052*<br>(0.0030)    |
| Origin FE                                                  | ✓                      |                        |                        |
| Destination FE                                             | ✓                      |                        |                        |
| Year FE                                                    | ✓                      |                        |                        |
| Origin $\times$ year FE                                    |                        | ✓                      |                        |
| Destination $\times$ year FE                               |                        | ✓                      |                        |
| Origin $\times$ dest. $\times$ year FE                     |                        |                        | ✓                      |
| Observations                                               | 1,388,296              | 1,388,073              | 1,386,580              |

*Notes:* Observations are origin–destination–HS6–year cells. Manufacturing products only. Dependent variable: share of the cell's export value shipped by each type of MNE. The import-demand elasticity is standardized (mean 0, standard deviation 1). Regressions weighted by the cell's export value. Robust standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 87: Manufacturing: Product Sophistication and Multinational Shares**

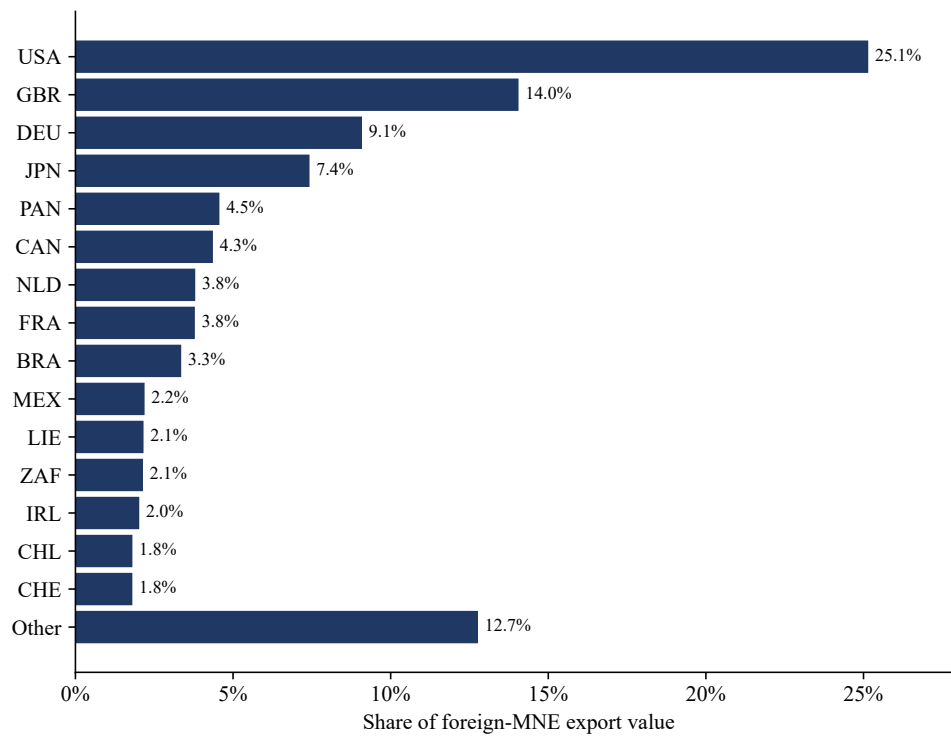
|                                                            | PCI                   |                       | Lall                  |                        | Rauch                 |                       |
|------------------------------------------------------------|-----------------------|-----------------------|-----------------------|------------------------|-----------------------|-----------------------|
|                                                            | (1)                   | (2)                   | (3)                   | (4)                    | (5)                   | (6)                   |
| <i>Panel A. Dependent variable: share of all MNEs</i>      |                       |                       |                       |                        |                       |                       |
| Complexity (PCI)                                           | 0.1107***<br>(0.0041) | 0.1062***<br>(0.0040) |                       |                        |                       |                       |
| Lall: high tech (vs primary and resource-based)            |                       |                       | 0.0969***<br>(0.0142) | 0.0729***<br>(0.0152)  |                       |                       |
| Lall: medium tech                                          |                       |                       | 0.1720***<br>(0.0099) | 0.1443***<br>(0.0090)  |                       |                       |
| Lall: low tech                                             |                       |                       | 0.0799***<br>(0.0091) | 0.0562***<br>(0.0085)  |                       |                       |
| Rauch: differentiated (vs homogeneous)                     |                       |                       |                       |                        | 0.2448***<br>(0.0136) | 0.2232***<br>(0.0131) |
| Rauch: reference-priced                                    |                       |                       |                       |                        | 0.2642***<br>(0.0133) | 0.2662***<br>(0.0135) |
| Observations                                               | 1,480,362             | 1,478,870             | 1,481,180             | 1,479,691              | 1,370,899             | 1,369,380             |
| <i>Panel B. Dependent variable: share of foreign MNEs</i>  |                       |                       |                       |                        |                       |                       |
| Complexity (PCI)                                           | 0.0696***<br>(0.0046) | 0.0676***<br>(0.0044) |                       |                        |                       |                       |
| Lall: high tech (vs primary and resource-based)            |                       |                       | 0.0603***<br>(0.0145) | 0.0389***<br>(0.0134)  |                       |                       |
| Lall: medium tech                                          |                       |                       | 0.1797***<br>(0.0097) | 0.1558***<br>(0.0087)  |                       |                       |
| Lall: low tech                                             |                       |                       | 0.0336***<br>(0.0093) | 0.0162*<br>(0.0087)    |                       |                       |
| Rauch: differentiated (vs homogeneous)                     |                       |                       |                       |                        | 0.0980***<br>(0.0162) | 0.0738***<br>(0.0160) |
| Rauch: reference-priced                                    |                       |                       |                       |                        | 0.0518***<br>(0.0162) | 0.0494***<br>(0.0168) |
| Observations                                               | 1,480,362             | 1,478,870             | 1,481,180             | 1,479,691              | 1,370,899             | 1,369,380             |
| <i>Panel C. Dependent variable: share of domestic MNEs</i> |                       |                       |                       |                        |                       |                       |
| Complexity (PCI)                                           | 0.0411***<br>(0.0036) | 0.0386***<br>(0.0039) |                       |                        |                       |                       |
| Lall: high tech (vs primary and resource-based)            |                       |                       | 0.0366***<br>(0.0138) | 0.0340***<br>(0.0128)  |                       |                       |
| Lall: medium tech                                          |                       |                       | -0.0078*<br>(0.0043)  | -0.0115***<br>(0.0045) |                       |                       |
| Lall: low tech                                             |                       |                       | 0.0463***<br>(0.0057) | 0.0400***<br>(0.0054)  |                       |                       |
| Rauch: differentiated (vs homogeneous)                     |                       |                       |                       |                        | 0.1468***<br>(0.0109) | 0.1494***<br>(0.0128) |
| Rauch: reference-priced                                    |                       |                       |                       |                        | 0.2125***<br>(0.0131) | 0.2168***<br>(0.0153) |
| Observations                                               | 1,480,362             | 1,478,870             | 1,481,180             | 1,479,691              | 1,370,899             | 1,369,380             |
| Origin × year and destination × year FE                    | ✓                     |                       | ✓                     |                        | ✓                     |                       |
| Origin × dest. × year FE                                   |                       | ✓                     |                       | ✓                      |                       | ✓                     |

*Notes:* Observations are origin–destination–HS6–year cells. Manufacturing products only. Dependent variable: share of the cell's export value shipped by each type of MNE. Omitted categories: primary and resource-based products (Lall) and homogeneous products (Rauch). Robust standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 12 Fact 3: Multinationals from a Small Set of Countries Dominate Exports

### 12.1 Parent Countries



**Figure 67:** Manufacturing: Foreign-Multinational Exports, by Parent Country

*Notes:* Percent of foreign-MNE manufacturing export value, by country of the ultimate parent.

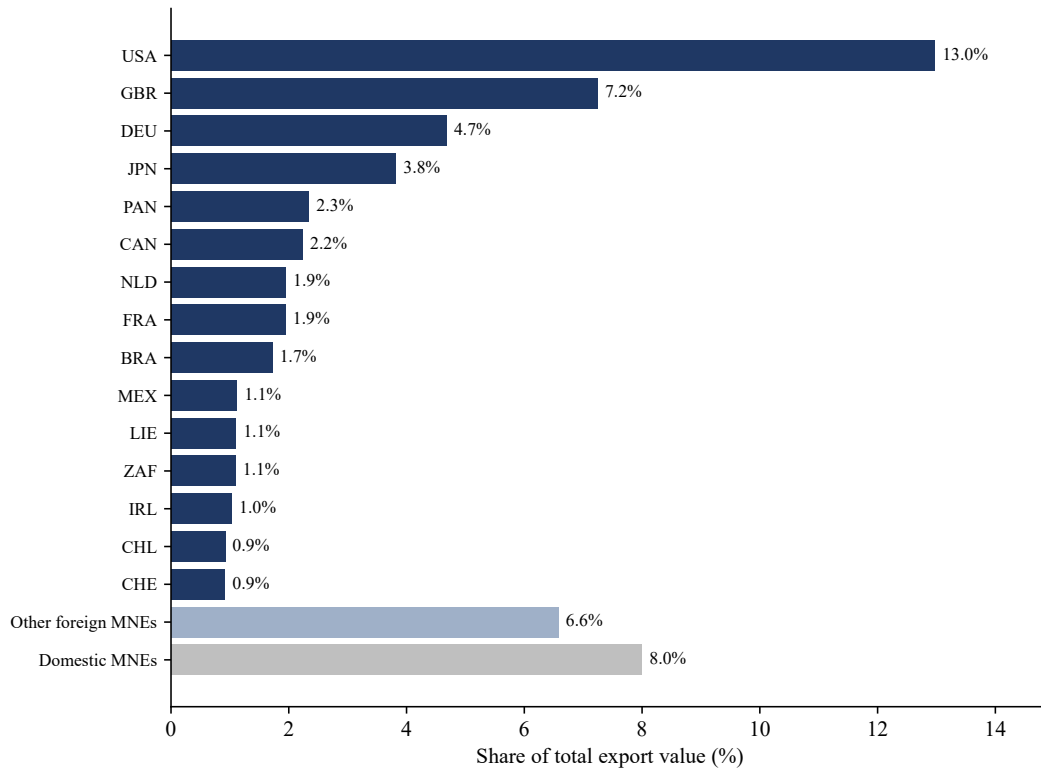
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 88:** Manufacturing: Foreign-Multinational Exports, by Parent Country

| Parent country        | Share (%) | Value (US\$bn/yr) |
|-----------------------|-----------|-------------------|
| USA                   | 25.1      | 9.1               |
| GBR                   | 14.0      | 4.2               |
| DEU                   | 9.1       | 3.7               |
| JPN                   | 7.4       | 2.9               |
| PAN                   | 4.5       | 1.3               |
| CAN                   | 4.3       | 1.3               |
| NLD                   | 3.8       | 1.5               |
| FRA                   | 3.8       | 1.3               |
| BRA                   | 3.3       | 1.3               |
| MEX                   | 2.2       | 0.7               |
| LIE                   | 2.1       | 0.7               |
| ZAF                   | 2.1       | 0.7               |
| IRL                   | 2.0       | 0.9               |
| CHL                   | 1.8       | 0.7               |
| CHE                   | 1.8       | 0.7               |
| Other (143 countries) | 12.7      | 4.7               |
| All foreign MNEs      | 100.0     | 38.5              |

*Notes:* Percent of foreign-MNE manufacturing export value, by country of the ultimate parent. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 68:** Manufacturing: Share of Total Exports, by Parent Country

*Notes:* Percent of total manufacturing export value shipped by foreign MNEs of each parent country, other foreign MNEs and domestic MNEs; the remainder is local firms.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 89:** Manufacturing: Share of Total Exports, by Parent Country

| Exporter type / parent country | Share of total exports (%) |
|--------------------------------|----------------------------|
| USA                            | 13.0                       |
| GBR                            | 7.2                        |
| DEU                            | 4.7                        |
| JPN                            | 3.8                        |
| PAN                            | 2.3                        |
| CAN                            | 2.2                        |
| NLD                            | 1.9                        |
| FRA                            | 1.9                        |
| BRA                            | 1.7                        |
| MEX                            | 1.1                        |
| LIE                            | 1.1                        |
| ZAF                            | 1.1                        |
| IRL                            | 1.0                        |
| CHL                            | 0.9                        |
| CHE                            | 0.9                        |
| Other foreign MNEs             | 6.6                        |
| Domestic MNEs                  | 8.0                        |
| All MNEs                       | 59.6                       |
| Local firms                    | 40.4                       |

*Notes:* Percent of total manufacturing export value, by type of exporter and parent country.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 12.2 Parent Region and Destination Region

**Table 90:** Manufacturing: Multinational Exports by Parent Region and Destination Region: Export Value

| Parent region / Destination region | Latin America | North America | Europe | Asia | Africa | Oceania | Rest of World | Total |
|------------------------------------|---------------|---------------|--------|------|--------|---------|---------------|-------|
| Latin America                      | 3.1           | 0.8           | 0.5    | 1.2  | 0.1    | 0.0     | 0.0           | 5.6   |
| North America                      | 4.5           | 3.8           | 0.7    | 1.3  | 0.0    | 0.0     | 0.0           | 10.4  |
| Europe                             | 7.8           | 2.3           | 1.9    | 2.3  | 0.2    | 0.4     | 0.0           | 14.8  |
| Asia                               | 3.3           | 0.4           | 0.1    | 0.2  | 0.0    | 0.0     | 0.0           | 4.0   |
| Africa                             | 0.1           | 0.1           | 0.2    | 0.4  | 0.0    | 0.0     | 0.0           | 0.7   |
| Oceania                            | 0.0           | 0.0           | 0.0    | 0.1  | 0.0    | 0.0     | 0.0           | 0.1   |
| Rest of World                      | 0.0           | 0.0           | 0.0    | 0.0  | 0.0    | 0.0     | 0.0           | 0.0   |
| Domestic MNEs                      | 2.4           | 1.0           | 0.8    | 1.1  | 0.0    | 0.1     | 0.0           | 5.4   |
| Local firms                        | 11.9          | 6.1           | 3.7    | 6.6  | 0.3    | 0.1     | 0.1           | 28.8  |
| Total                              | 33.2          | 14.5          | 7.8    | 13.1 | 0.6    | 0.6     | 0.1           | 69.8  |

*Notes:* Values are annual averages in US\$ billion. Rows: region of the ultimate parent; columns: destination region.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 91:** Manufacturing: Multinational Exports by Parent Region and Destination Region: Row Percentages

| Parent region / Destination region | Latin America | North America | Europe | Asia | Africa | Oceania | Rest of World | Total |
|------------------------------------|---------------|---------------|--------|------|--------|---------|---------------|-------|
| Latin America                      | 53.0          | 14.1          | 8.2    | 23.2 | 0.9    | 0.2     | 0.4           | 100.0 |
| North America                      | 42.5          | 36.5          | 7.3    | 13.4 | 0.2    | 0.2     | 0.0           | 100.0 |
| Europe                             | 49.5          | 15.1          | 12.2   | 19.5 | 1.1    | 2.6     | 0.0           | 100.0 |
| Asia                               | 81.3          | 10.0          | 1.8    | 6.1  | 0.2    | 0.1     | 0.4           | 100.0 |
| Africa                             | 11.2          | 7.1           | 22.5   | 57.5 | 1.7    | 0.0     | 0.0           | 100.0 |
| Oceania                            | 33.7          | 11.5          | 14.9   | 39.3 | 0.2    | 0.3     | 0.0           | 100.0 |
| Rest of World                      | 71.1          | 0.7           | 4.0    | 0.7  | 0.0    | 23.5    | 0.1           | 100.0 |
| Domestic MNEs                      | 42.1          | 17.7          | 14.1   | 23.9 | 0.8    | 1.4     | 0.0           | 100.0 |
| Local firms                        | 39.2          | 21.2          | 12.9   | 25.2 | 0.9    | 0.3     | 0.2           | 100.0 |

*Notes:* Percent of each group's exports, by destination region (rows add to 100). Domestic MNEs and local firms are shown for comparison.

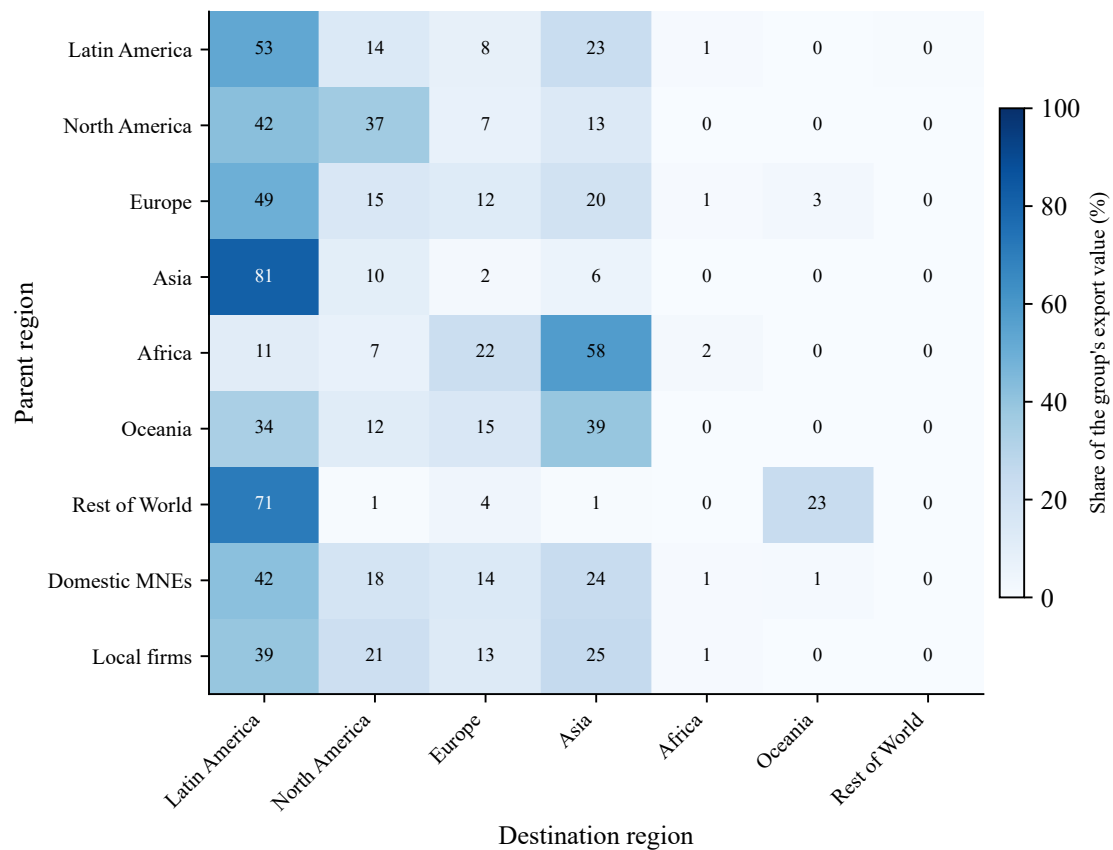
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 92:** Manufacturing: Multinational Exports by Parent Region and Destination Region: Column Percentages

| Parent region / Destination region | Latin America | North America | Europe | Asia  | Africa | Oceania | Rest of World |
|------------------------------------|---------------|---------------|--------|-------|--------|---------|---------------|
| Latin America                      | 9.6           | 5.5           | 5.9    | 8.8   | 8.7    | 1.8     | 21.4          |
| North America                      | 13.8          | 26.0          | 9.5    | 9.1   | 3.0    | 2.8     | 1.8           |
| Europe                             | 22.7          | 15.2          | 22.4   | 18.8  | 27.5   | 63.4    | 5.7           |
| Asia                               | 9.1           | 2.4           | 0.8    | 1.4   | 1.4    | 0.8     | 11.2          |
| Africa                             | 0.3           | 0.4           | 2.2    | 3.0   | 2.4    | 0.0     | 0.0           |
| Oceania                            | 0.1           | 0.1           | 0.3    | 0.4   | 0.1    | 0.1     | 0.0           |
| Rest of World                      | 0.0           | 0.0           | 0.0    | 0.0   | 0.0    | 0.2     | 0.0           |
| Domestic MNEs                      | 7.8           | 7.2           | 10.4   | 9.2   | 8.3    | 13.7    | 2.4           |
| Local firms                        | 36.6          | 43.2          | 48.4   | 49.3  | 48.7   | 17.3    | 57.5          |
| Total                              | 100.0         | 100.0         | 100.0  | 100.0 | 100.0  | 100.0   | 100.0         |

*Notes:* Percent of the exports reaching each destination region, by parent region (columns add to 100).

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 69:** Manufacturing: Multinational Exports by Parent Region and Destination Region

Notes: Percent of each group's exports, by destination region (rows add to 100).

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 93:** Manufacturing: Foreign-Multinational Exports by Exporting Country and Destination Region

| Origin | Destination region | Latin America | North America | Europe | Asia | Africa | Oceania | Rest of World | Total |
|--------|--------------------|---------------|---------------|--------|------|--------|---------|---------------|-------|
| ARG    |                    | 78.8          | 4.5           | 9.7    | 3.9  | 1.5    | 1.7     | 0.0           | 13.1  |
| CHL    |                    | 19.1          | 16.4          | 11.3   | 50.5 | 0.2    | 2.5     | 0.0           | 6.4   |
| COL    |                    | 67.7          | 13.7          | 5.2    | 11.9 | 0.5    | 0.1     | 0.8           | 5.2   |
| CRI    |                    | 33.2          | 42.8          | 10.2   | 13.6 | 0.1    | 0.1     | 0.0           | 4.3   |
| DOM    |                    | 26.8          | 60.7          | 6.3    | 5.4  | 0.2    | 0.3     | 0.1           | 3.4   |
| PER    |                    | 43.5          | 22.1          | 18.0   | 14.9 | 1.2    | 0.3     | 0.0           | 2.5   |
| PRY    |                    | 79.5          | 7.2           | 9.4    | 3.9  | 0.1    | 0.0     | 0.0           | 0.4   |
| SLV    |                    | 43.0          | 55.7          | 0.8    | 0.4  | 0.1    | 0.1     | 0.0           | 2.0   |
| URY    |                    | 62.9          | 6.2           | 16.0   | 12.9 | 1.5    | 0.1     | 0.4           | 1.3   |

Notes: Percent of each country's foreign-MNE manufacturing exports, by destination region (rows add to 100). Last column: foreign-MNE exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 94:** Manufacturing: Local-Firm Exports by Exporting Country and Destination Region

| Origin | Destination region | Latin America | North America | Europe | Asia | Africa | Oceania | Rest of World | Total |
|--------|--------------------|---------------|---------------|--------|------|--------|---------|---------------|-------|
| ARG    |                    | 56.3          | 16.3          | 13.4   | 11.2 | 2.5    | 0.3     | 0.0           | 7.3   |
| CHL    |                    | 14.5          | 16.9          | 16.6   | 51.0 | 0.4    | 0.6     | 0.0           | 7.3   |
| COL    |                    | 66.1          | 16.2          | 5.8    | 9.5  | 0.5    | 0.1     | 1.8           | 3.0   |
| CRI    |                    | 40.4          | 37.5          | 12.6   | 8.9  | 0.2    | 0.4     | 0.0           | 1.4   |
| DOM    |                    | 36.4          | 51.8          | 3.1    | 8.4  | 0.1    | 0.1     | 0.2           | 1.2   |
| PER    |                    | 40.4          | 19.7          | 16.0   | 23.1 | 0.3    | 0.4     | 0.0           | 5.3   |
| PRY    |                    | 69.5          | 4.7           | 15.0   | 10.6 | 0.1    | 0.0     | 0.0           | 0.6   |
| SLV    |                    | 46.4          | 51.1          | 0.9    | 1.5  | 0.1    | 0.0     | 0.0           | 1.8   |
| URY    |                    | 43.5          | 5.8           | 19.4   | 24.1 | 6.7    | 0.1     | 0.4           | 0.8   |

*Notes:* Percent of each country's manufacturing exports by local firms, by destination region (rows add to 100). Last column: local-firm exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 12.3 Parent Country and Destination Country

**Table 95:** Manufacturing: Foreign-Multinational Exports by Parent Country and Destination: Export Value

| Parent / Destination | BRA | USA | CHN | ECU | PER | CHL | MEX | NLD | KOR | VEN | Other destinations | Total |
|----------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|--------------------|-------|
| USA                  | 1.5 | 3.6 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.3 | 0.0 | 0.1 | 2.4                | 9.1   |
| GBR                  | 0.4 | 0.6 | 1.3 | 0.0 | 0.0 | 0.1 | 0.1 | 0.1 | 0.3 | 0.0 | 1.2                | 4.2   |
| DEU                  | 1.5 | 0.3 | 0.0 | 0.1 | 0.1 | 0.1 | 0.2 | 0.0 | 0.0 | 0.0 | 1.3                | 3.7   |
| JPN                  | 1.7 | 0.2 | 0.0 | 0.1 | 0.1 | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.7                | 2.9   |
| PAN                  | 0.0 | 0.1 | 0.5 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.1 | 0.0 | 0.5                | 1.3   |
| CAN                  | 0.1 | 0.1 | 0.4 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.1 | 0.0 | 0.5                | 1.3   |
| NLD                  | 1.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.3                | 1.5   |
| FRA                  | 0.5 | 0.0 | 0.0 | 0.1 | 0.1 | 0.1 | 0.2 | 0.0 | 0.0 | 0.0 | 0.3                | 1.3   |
| BRA                  | 0.3 | 0.2 | 0.1 | 0.1 | 0.0 | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.6                | 1.3   |
| MEX                  | 0.2 | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.3                | 0.7   |
| Other parents        | 1.0 | 1.7 | 0.7 | 0.2 | 0.3 | 0.4 | 0.3 | 0.2 | 0.1 | 0.2 | 3.1                | 8.3   |
| Total                | 8.3 | 7.0 | 3.2 | 0.9 | 0.9 | 1.1 | 1.0 | 0.8 | 0.6 | 0.6 | 11.3               | 35.7  |

*Notes:* Values are annual averages in US\$ billion. Ten largest parent countries and ten largest destinations by foreign-MNE export value.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 96:** Manufacturing: Foreign-Multinational Exports by Parent Country and Destination: Row Percentages

| Parent / Destination | BRA  | USA  | CHN  | ECU | PER | CHL | MEX  | NLD | KOR | VEN | Other destinations | Total |
|----------------------|------|------|------|-----|-----|-----|------|-----|-----|-----|--------------------|-------|
| USA                  | 15.0 | 40.5 | 2.5  | 3.1 | 2.6 | 2.6 | 2.0  | 3.6 | 0.4 | 1.4 | 26.3               | 100.0 |
| GBR                  | 7.6  | 14.6 | 33.4 | 0.6 | 0.9 | 1.3 | 1.6  | 2.7 | 7.4 | 0.8 | 29.2               | 100.0 |
| DEU                  | 40.5 | 8.8  | 1.1  | 2.2 | 2.2 | 3.3 | 4.5  | 0.1 | 0.2 | 1.2 | 36.0               | 100.0 |
| JPN                  | 54.8 | 6.8  | 0.6  | 5.6 | 3.5 | 1.7 | 1.1  | 0.4 | 0.1 | 1.1 | 24.2               | 100.0 |
| PAN                  | 0.7  | 4.8  | 41.3 | 3.0 | 2.8 | 1.2 | 0.9  | 2.5 | 5.6 | 2.6 | 34.5               | 100.0 |
| CAN                  | 10.0 | 9.8  | 31.0 | 0.5 | 0.9 | 0.5 | 0.7  | 3.6 | 4.8 | 0.6 | 37.6               | 100.0 |
| NLD                  | 67.2 | 2.5  | 0.0  | 1.3 | 1.5 | 2.5 | 2.4  | 0.9 | 0.0 | 2.6 | 19.0               | 100.0 |
| FRA                  | 32.4 | 3.0  | 2.4  | 7.6 | 5.1 | 5.6 | 14.6 | 0.1 | 0.8 | 2.1 | 26.3               | 100.0 |
| BRA                  | 21.9 | 11.9 | 4.0  | 4.6 | 3.2 | 4.5 | 0.8  | 0.3 | 0.8 | 1.4 | 46.6               | 100.0 |
| MEX                  | 33.7 | 7.4  | 0.2  | 5.6 | 4.9 | 3.6 | 2.0  | 0.1 | 0.1 | 3.7 | 38.8               | 100.0 |
| Other parents        | 11.4 | 20.0 | 9.3  | 3.0 | 4.1 | 4.0 | 3.4  | 2.7 | 1.8 | 2.8 | 37.4               | 100.0 |

*Notes:* Percent of each parent country's exports, by destination (rows add to 100). Ten largest parent countries and destinations.

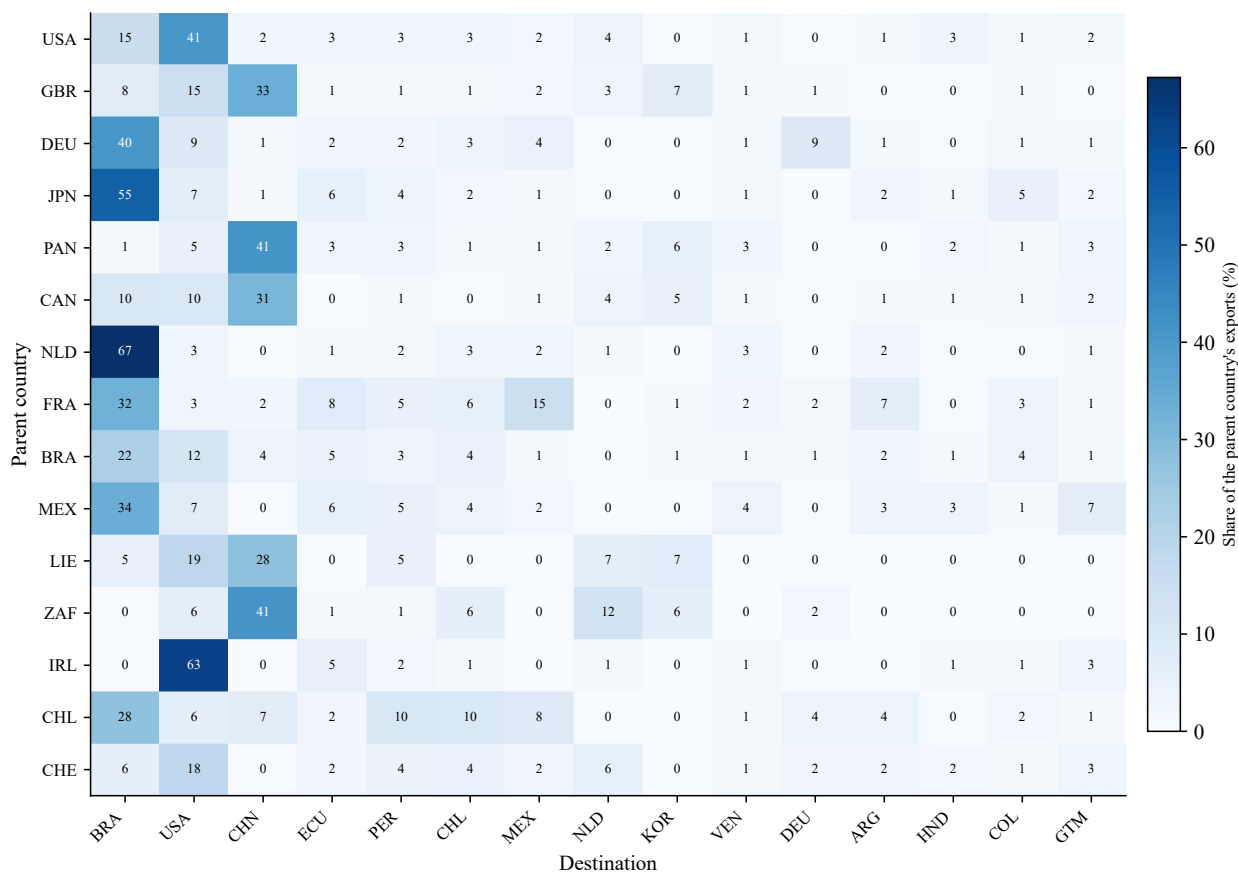
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 97:** Manufacturing: Foreign-Multinational Exports by Parent Country and Destination: Column Percentages

| Parent / Destination | BRA   | USA   | CHN   | ECU   | PER   | CHL   | MEX   | NLD   | KOR   | VEN   | Other destinations |
|----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------------------|
| USA                  | 18.1  | 52.3  | 5.7   | 26.7  | 23.3  | 23.2  | 18.1  | 40.2  | 5.0   | 20.5  | 21.0               |
| GBR                  | 5.1   | 10.5  | 42.6  | 2.9   | 4.5   | 6.4   | 8.1   | 16.9  | 49.9  | 6.3   | 13.0               |
| DEU                  | 17.6  | 4.1   | 0.9   | 6.8   | 7.3   | 10.7  | 14.8  | 0.5   | 0.7   | 6.1   | 10.4               |
| JPN                  | 19.5  | 2.6   | 0.4   | 14.1  | 9.4   | 4.6   | 3.1   | 1.3   | 0.3   | 4.6   | 5.7                |
| PAN                  | 0.2   | 1.1   | 17.0  | 4.6   | 4.5   | 2.1   | 1.5   | 5.0   | 12.3  | 6.9   | 5.0                |
| CAN                  | 2.1   | 2.2   | 12.2  | 0.7   | 1.3   | 0.8   | 1.0   | 7.0   | 10.0  | 1.6   | 5.2                |
| NLD                  | 12.2  | 0.5   | 0.0   | 1.6   | 2.0   | 3.5   | 3.3   | 1.5   | 0.0   | 5.6   | 2.3                |
| FRA                  | 5.9   | 0.6   | 0.8   | 9.8   | 6.8   | 7.6   | 19.9  | 0.1   | 1.4   | 4.6   | 3.1                |
| BRA                  | 3.5   | 2.0   | 1.2   | 5.2   | 3.8   | 5.4   | 1.0   | 0.4   | 1.3   | 2.6   | 4.9                |
| MEX                  | 3.5   | 0.8   | 0.0   | 4.2   | 3.8   | 2.8   | 1.5   | 0.1   | 0.1   | 4.6   | 2.7                |
| Other parents        | 12.4  | 23.2  | 19.1  | 23.4  | 33.2  | 33.0  | 27.7  | 26.8  | 19.1  | 36.5  | 26.7               |
| Total                | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0              |

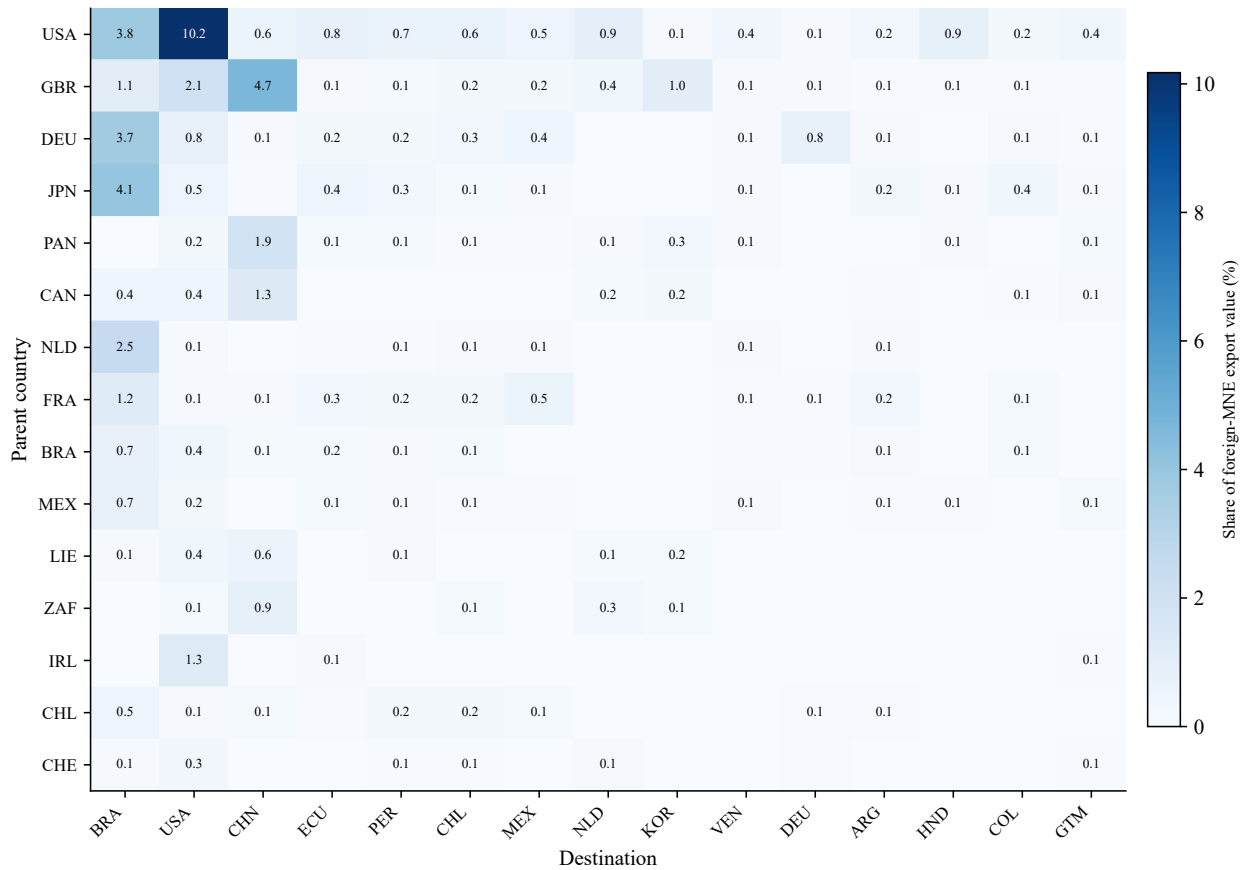
Notes: Percent of the foreign-MNE exports reaching each destination, by parent country (columns add to 100). Ten largest parent countries and destinations.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Figure 70:** Manufacturing: Destination of Foreign-Multinational Exports: Fifteen Largest Parent Countries and Destinations

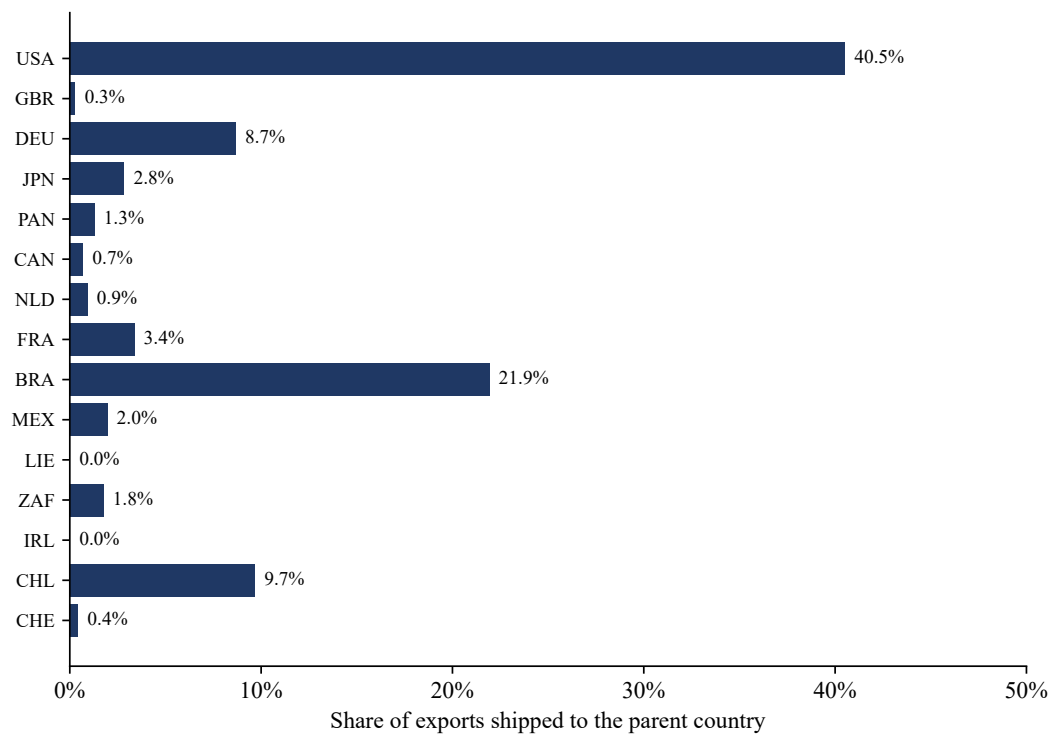
Notes: Percent of each parent country's exports going to each destination (rows add to 100 including other destinations).

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 71: Manufacturing: Foreign-Multinational Exports by Parent Country and Destination: Share of the Total**  
*Notes:* Percent of all foreign-MNE export value in each parent country and destination pair; fifteen largest parent countries and destinations.  
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 12.4 Exports to the Parent Country



**Figure 72:** Manufacturing: Share of Foreign-Multinational Exports Shipped to the Parent Country

*Notes:* Percent of each parent country's exports shipped to that same country.

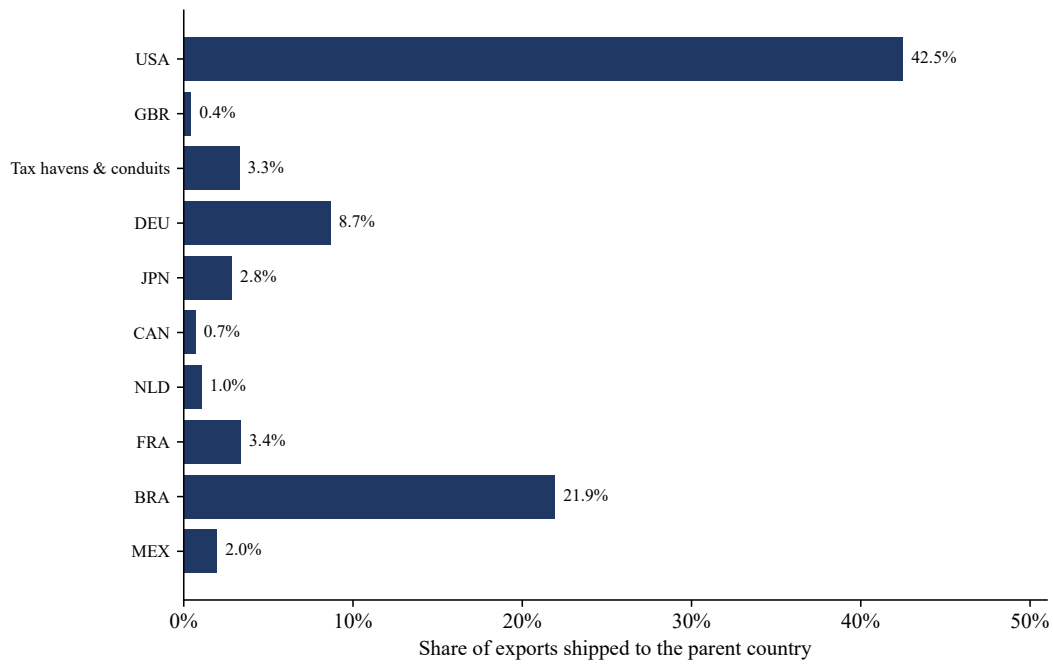
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 98:** Manufacturing: Foreign-Multinational Exports to the Parent Countries, by Parent Country

| Parent / Destination | USA  | GBR | DEU | JPN | PAN | CAN | NLD | FRA | BRA  | MEX  | Other destinations | Total |
|----------------------|------|-----|-----|-----|-----|-----|-----|-----|------|------|--------------------|-------|
| USA                  | 40.5 | 0.2 | 0.4 | 0.7 | 1.2 | 0.5 | 3.6 | 0.1 | 15.0 | 2.0  | 35.8               | 100.0 |
| GBR                  | 14.6 | 0.3 | 0.8 | 0.5 | 0.3 | 5.3 | 2.7 | 0.7 | 7.6  | 1.6  | 65.7               | 100.0 |
| DEU                  | 8.8  | 0.0 | 8.7 | 0.0 | 1.0 | 0.1 | 0.1 | 0.0 | 40.5 | 4.5  | 36.2               | 100.0 |
| JPN                  | 6.8  | 0.1 | 0.1 | 2.8 | 1.2 | 0.3 | 0.4 | 0.1 | 54.8 | 1.1  | 32.2               | 100.0 |
| PAN                  | 4.8  | 0.1 | 0.3 | 0.9 | 1.3 | 0.4 | 2.5 | 1.4 | 0.7  | 0.9  | 86.5               | 100.0 |
| CAN                  | 9.8  | 0.2 | 0.4 | 0.2 | 0.9 | 0.7 | 3.6 | 0.2 | 10.0 | 0.7  | 73.3               | 100.0 |
| NLD                  | 2.5  | 0.0 | 0.0 | 0.0 | 0.8 | 0.2 | 0.9 | 0.3 | 67.2 | 2.4  | 25.6               | 100.0 |
| FRA                  | 3.0  | 0.4 | 1.6 | 0.1 | 1.8 | 0.1 | 0.1 | 3.4 | 32.4 | 14.6 | 42.6               | 100.0 |
| BRA                  | 11.9 | 0.1 | 1.3 | 2.5 | 1.5 | 0.1 | 0.3 | 0.0 | 21.9 | 0.8  | 59.6               | 100.0 |
| MEX                  | 7.4  | 0.1 | 0.3 | 0.0 | 1.9 | 0.0 | 0.1 | 0.0 | 33.7 | 2.0  | 54.5               | 100.0 |
| Other parents        | 20.0 | 0.6 | 2.5 | 1.0 | 2.1 | 0.6 | 2.7 | 0.3 | 11.4 | 3.4  | 55.3               | 100.0 |

*Notes:* Percent of each parent country's exports, by destination (rows add to 100). Destinations are the ten largest parent countries; the diagonal is the share shipped to the parent's own country.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 73:** Manufacturing: Share of Foreign-Multinational Exports Shipped to the Parent Jurisdiction, Consolidated Parents

*Notes:* Percent of each parent's exports shipped to its own jurisdiction. Dependencies are assigned to their sovereign state and tax havens and conduit jurisdictions are pooled.

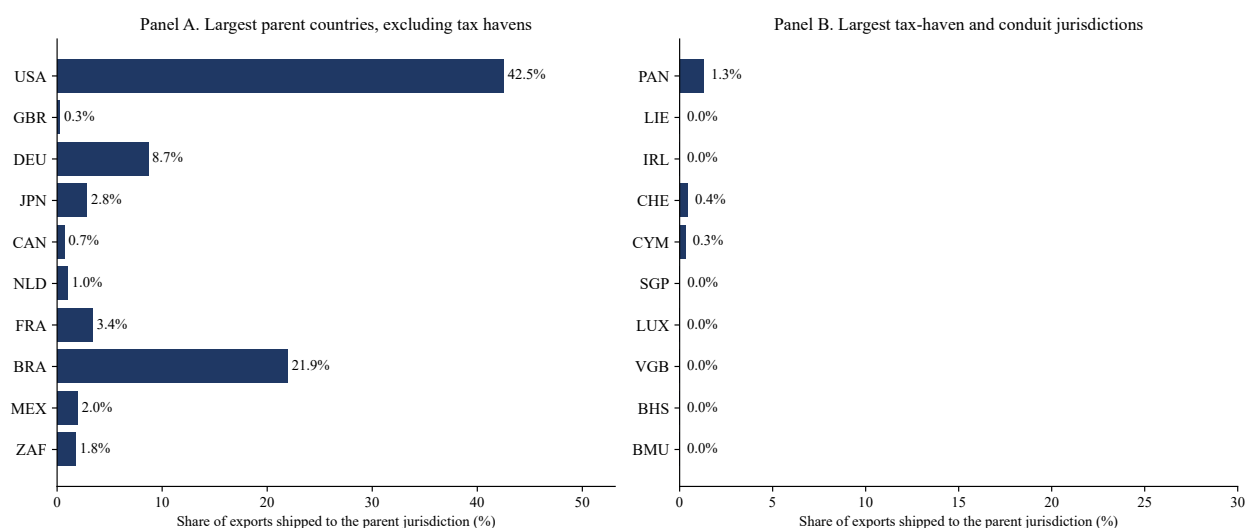
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 99:** Manufacturing: Share of Foreign-Multinational Exports Shipped to the Parent Jurisdiction, Consolidated Parents

| Consolidated parent   | Home share (%) | Exports (US\$bn/yr) |
|-----------------------|----------------|---------------------|
| USA                   | 42.5           | 9.1                 |
| GBR                   | 0.4            | 4.9                 |
| Tax havens & conduits | 3.3            | 4.3                 |
| DEU                   | 8.7            | 3.7                 |
| JPN                   | 2.8            | 2.9                 |
| CAN                   | 0.7            | 1.3                 |
| NLD                   | 1.0            | 1.6                 |
| FRA                   | 3.4            | 1.3                 |
| BRA                   | 21.9           | 1.3                 |
| MEX                   | 2.0            | 0.7                 |

*Notes:* Home share: percent of the parent's exports shipped to its own jurisdiction. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 74:** Manufacturing: Share of Foreign-Multinational Exports Shipped to the Parent Jurisdiction

*Notes:* Percent of each parent jurisdiction's exports shipped to that same jurisdiction. Panel A: largest parent countries, excluding tax havens; dependencies are assigned to their sovereign state. Panel B: largest tax-haven and conduit jurisdictions.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

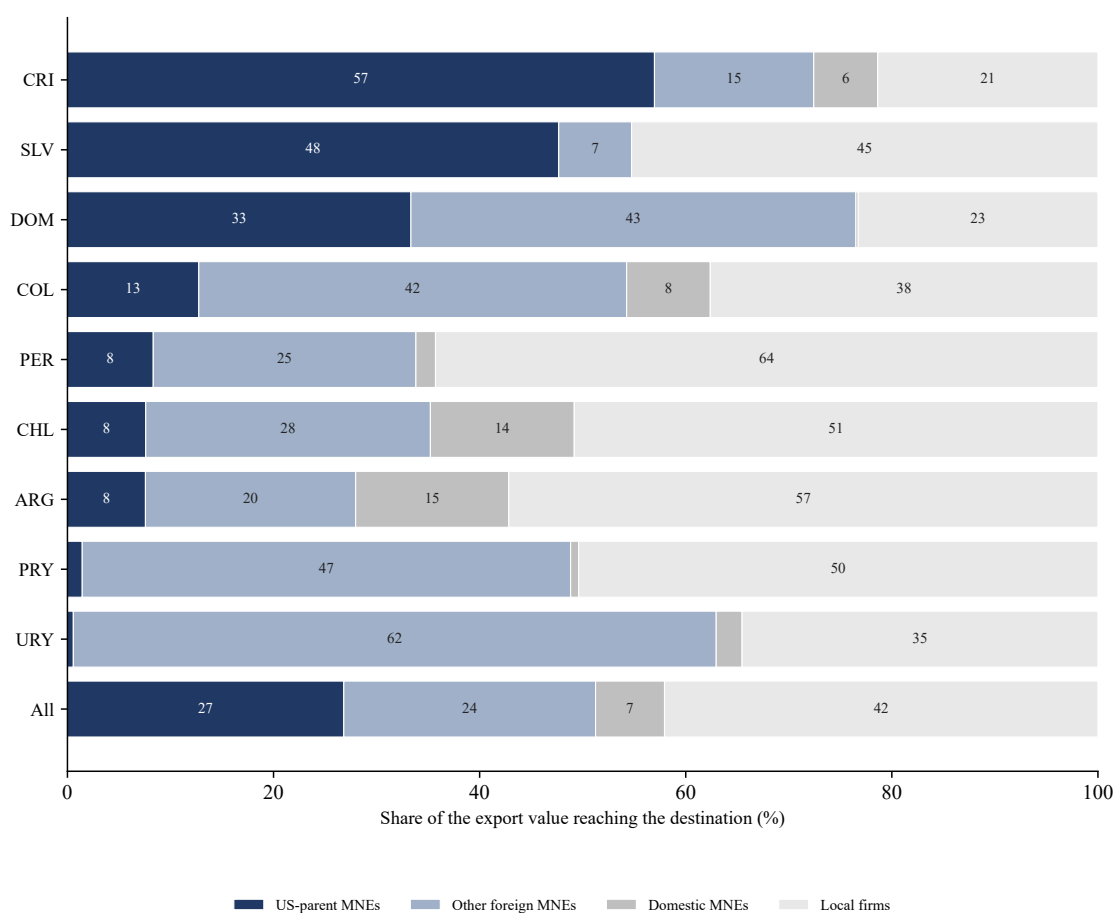
**Table 100:** Manufacturing: Share of Foreign-Multinational Exports Shipped to the Parent Jurisdiction

| Panel   | Parent | Home share (%) | Exports (US\$bn/yr) |
|---------|--------|----------------|---------------------|
| Panel A | USA    | 42.5           | 9.1                 |
| Panel A | GBR    | 0.3            | 4.2                 |
| Panel A | DEU    | 8.7            | 3.7                 |
| Panel A | JPN    | 2.8            | 2.9                 |
| Panel A | CAN    | 0.7            | 1.3                 |
| Panel A | NLD    | 1.0            | 1.5                 |
| Panel A | FRA    | 3.4            | 1.3                 |
| Panel A | BRA    | 21.9           | 1.3                 |
| Panel A | MEX    | 2.0            | 0.7                 |
| Panel A | ZAF    | 1.8            | 0.7                 |
| Panel B | PAN    | 1.3            | 1.3                 |
| Panel B | LIE    | 0.0            | 0.7                 |
| Panel B | IRL    | 0.0            | 0.9                 |
| Panel B | CHE    | 0.4            | 0.7                 |
| Panel B | CYM    | 0.3            | 0.3                 |
| Panel B | SGP    | 0.0            | 0.3                 |
| Panel B | LUX    | 0.0            | 0.2                 |
| Panel B | VGB    | 0.0            | 0.2                 |
| Panel B | BHS    | 0.0            | 0.2                 |
| Panel B | BMU    | 0.0            | 0.2                 |

*Notes:* Home share: percent of the parent's exports shipped to its own jurisdiction. Panel A: largest parent countries, excluding tax havens. Panel B: largest tax-haven and conduit jurisdictions. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 12.5 Destination Markets



**Figure 75:** Manufacturing: Exports to the United States, by Type of Exporter and Exporting Country

*Notes:* Percent of each country's manufacturing exports to the United States, by type of exporter.

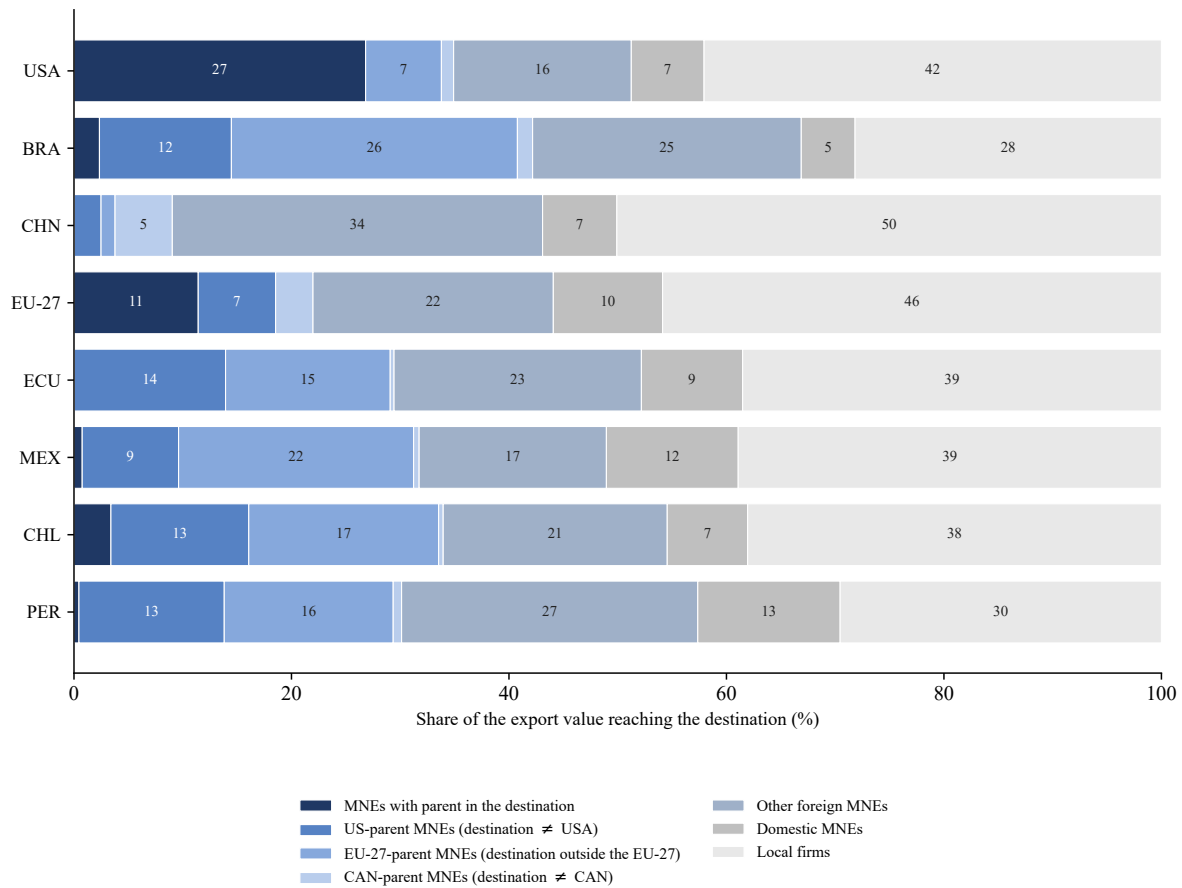
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 101:** Manufacturing: Exports to the United States, by Type of Exporter and Exporting Country

| Origin | Exports to the USA (US\$bn/yr) | US-parent MNEs (%) | Other foreign MNEs (%) | Domestic MNEs (%) | Local firms (%) |
|--------|--------------------------------|--------------------|------------------------|-------------------|-----------------|
| CRI    | 2.5                            | 57.0               | 15.5                   | 6.2               | 21.4            |
| SLV    | 2.0                            | 47.7               | 7.1                    | 0.0               | 45.3            |
| DOM    | 2.7                            | 33.3               | 43.2                   | 0.2               | 23.3            |
| COL    | 1.3                            | 12.7               | 41.5                   | 8.1               | 37.6            |
| PER    | 1.6                            | 8.3                | 25.5                   | 1.9               | 64.3            |
| CHL    | 2.4                            | 7.6                | 27.6                   | 14.0              | 50.8            |
| ARG    | 2.0                            | 7.6                | 20.4                   | 14.9              | 57.2            |
| PRY    | 0.1                            | 1.4                | 47.4                   | 0.8               | 50.4            |
| URY    | 0.1                            | 0.6                | 62.4                   | 2.5               | 34.5            |
| All    | 14.5                           | 26.8               | 24.4                   | 6.7               | 42.1            |

*Notes:* Percent of each country's manufacturing exports to the United States, by type of exporter. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 76: Manufacturing: Ownership of Exports to the Main Destination Markets**

*Notes:* Percent of the export value reaching each destination, by type of exporter. The first segment is exports by MNEs whose parent is located in that destination. EU-27 member states are pooled.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 102: Manufacturing: Ownership of Exports to the Main Destination Markets**

| Destination | Exports (US\$bn/yr) | MNEs with parent in the destination (%) | US-parent MNEs (%) | EU-27-parent MNEs (%) | CAN-parent MNEs (%) | Other foreign MNEs (%) | Domestic MNEs (%) | Local firms (%) |
|-------------|---------------------|-----------------------------------------|--------------------|-----------------------|---------------------|------------------------|-------------------|-----------------|
| USA         | 14.5                | 26.8                                    | –                  | 7.0                   | 1.1                 | 16.3                   | 6.7               | 42.1            |
| BRA         | 12.4                | 2.3                                     | 12.1               | 26.3                  | 1.4                 | 24.7                   | 4.9               | 28.2            |
| CHN         | 7.7                 | 0.0                                     | 2.5                | 1.3                   | 5.3                 | 34.0                   | 6.8               | 50.1            |
| EU-27       | 7.3                 | 11.4                                    | 7.1                | –                     | 3.4                 | 22.1                   | 10.0              | 45.9            |
| ECU         | 2.2                 | 0.0                                     | 13.9               | 15.1                  | 0.4                 | 22.7                   | 9.3               | 38.5            |
| MEX         | 2.2                 | 0.8                                     | 8.9                | 21.6                  | 0.5                 | 17.2                   | 12.1              | 38.9            |
| CHL         | 2.2                 | 3.4                                     | 12.7               | 17.5                  | 0.4                 | 20.6                   | 7.4               | 38.1            |
| PER         | 1.8                 | 0.5                                     | 13.4               | 15.5                  | 0.8                 | 27.2                   | 13.1              | 29.5            |

*Notes:* Percent of the export value reaching each destination, by type of exporter. EU-27 member states are pooled. Values are annual averages in US\$ billion.

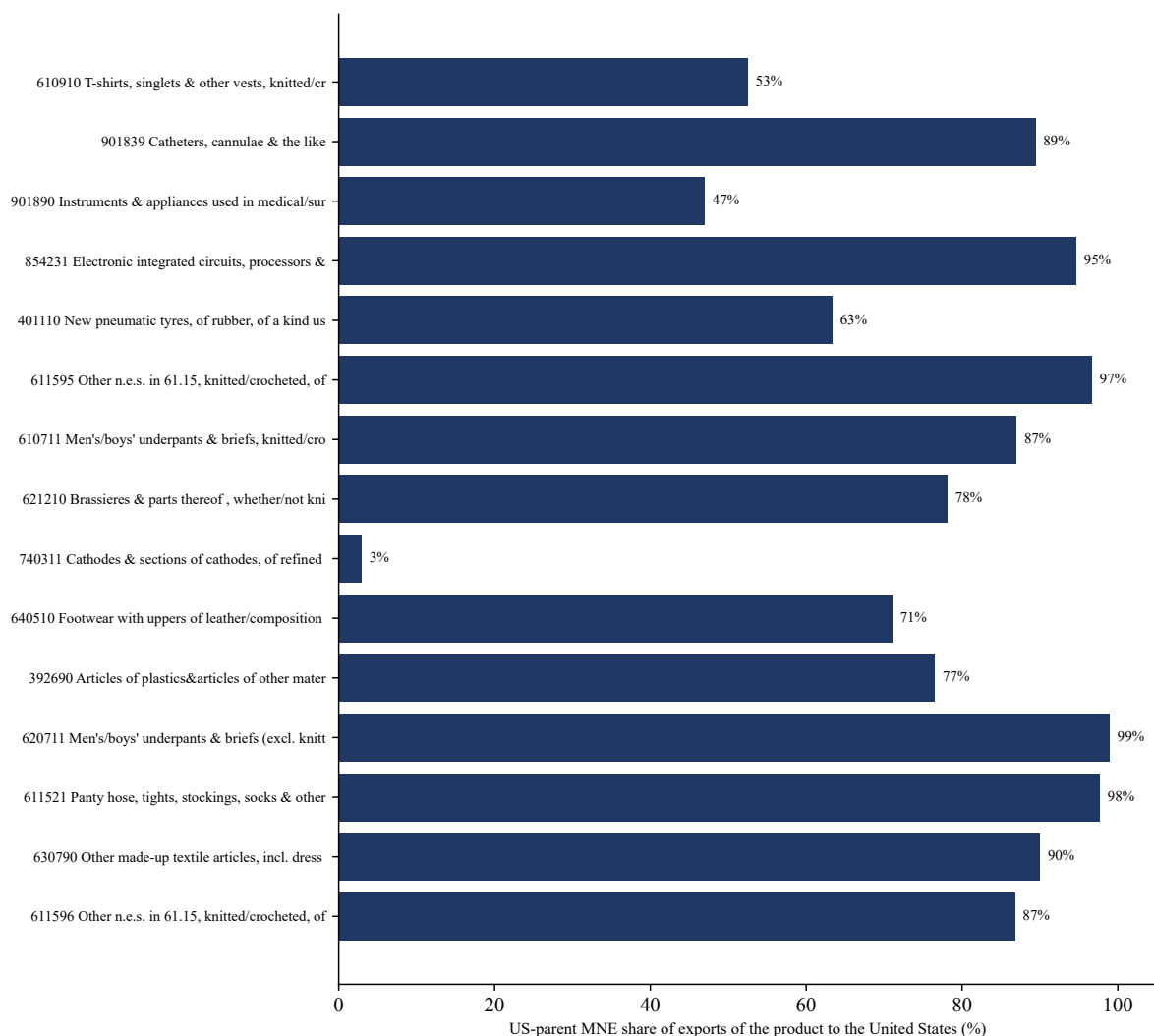
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 103: Manufacturing: Main Products Shipped by US-Parent Multinationals to the United States**

| HS6       | Description                                      | US-parent exports to the USA (US\$bn/yr) | % of all US-parent exports to the USA | % of US-parent exports of the product going to the USA | All exports of the product to the USA (US\$bn/yr) | US-parent share of exports of the product to the USA (%) | Main origins of the US-parent shipments (% of the line) |
|-----------|--------------------------------------------------|------------------------------------------|---------------------------------------|--------------------------------------------------------|---------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------|
| 610910    | T-shirts, singlets & other vests, knitted/cro... | 0.49                                     | 15.4                                  | 84                                                     | 1.01                                              | 53                                                       | SLV 85%, DOM 10%, PER 5%                                |
| 901839    | Catheters, cannulae & the like                   | 0.52                                     | 13.4                                  | 73                                                     | 0.63                                              | 89                                                       | CRI 92%, DOM 8%, COL 0%                                 |
| 901890    | Instruments & appliances used in medical/surg... | 0.39                                     | 9.5                                   | 69                                                     | 0.94                                              | 47                                                       | CRI 71%, DOM 29%, COL 0%                                |
| 854231    | Electronic integrated circuits, processors & ... | 0.21                                     | 5.4                                   | 28                                                     | 0.22                                              | 95                                                       | CRI 100%, COL 0%, ARG 0%                                |
| 401110    | New pneumatic tyres, of rubber, of a kind use... | 0.13                                     | 4.8                                   | 93                                                     | 0.24                                              | 63                                                       | CHL 100%, DOM 0%, COL 0%                                |
| 611595    | Other n.e.s. in 61.15, knitted/crocheted, of ... | 0.10                                     | 3.3                                   | 100                                                    | 0.11                                              | 97                                                       | SLV 100%, DOM 0%                                        |
| 610711    | Men's/boys' underpants & briefs, knitted/croc... | 0.09                                     | 3.0                                   | 87                                                     | 0.10                                              | 87                                                       | SLV 100%, DOM 0%, CRI 0%                                |
| 621210    | Brassieres & parts thereof, whether/not knit...  | 0.09                                     | 2.5                                   | 98                                                     | 0.12                                              | 78                                                       | SLV 74%, DOM 26%, CRI 0%                                |
| 740311    | Cathodes & sections of cathodes, of refined c... | 0.06                                     | 1.6                                   | 24                                                     | 1.56                                              | 3                                                        | PER 64%, CHL 36%                                        |
| 640510    | Footwear with uppers of leather/composition l... | 0.08                                     | 1.6                                   | 70                                                     | 0.11                                              | 71                                                       | DOM 100%                                                |
| 392690    | Articles of plastics&articles of other materi... | 0.06                                     | 1.5                                   | 79                                                     | 0.08                                              | 77                                                       | CRI 66%, DOM 33%, SLV 0%                                |
| 620711    | Men's/boys' underpants & briefs (excl. knitt...  | 0.04                                     | 1.2                                   | 99                                                     | 0.05                                              | 99                                                       | CRI 73%, SLV 27%, DOM 0%                                |
| 611521    | Panty hose, tights, stockings, socks & other ... | 0.03                                     | 1.1                                   | 77                                                     | 0.03                                              | 98                                                       | SLV 100%, DOM 0%                                        |
| 630790    | Other made-up textile articles, incl. dress p... | 0.05                                     | 1.1                                   | 97                                                     | 0.06                                              | 90                                                       | DOM 100%, COL 0%, CRI 0%                                |
| 611596    | Other n.e.s. in 61.15, knitted/crocheted, of ... | 0.03                                     | 1.1                                   | 100                                                    | 0.04                                              | 87                                                       | SLV 100%, CRI 0%, PER 0%                                |
| All lines |                                                  | 3.6                                      | 100.0                                 | 41                                                     | 14.5                                              | 27                                                       |                                                         |

*Notes:* Fifteen HS6 products with the largest exports by US-parent MNEs to the United States. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Figure 77: Manufacturing: US-Parent Share of Exports to the United States, Main Products Shipped by US-Parent Multinationals**

*Notes:* Percent of the nine countries' exports of each product to the United States shipped by US-parent MNEs; the fifteen products with the largest US-parent exports to the United States.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 104:** Manufacturing: Largest Export Products to the United States, by Type of Exporter

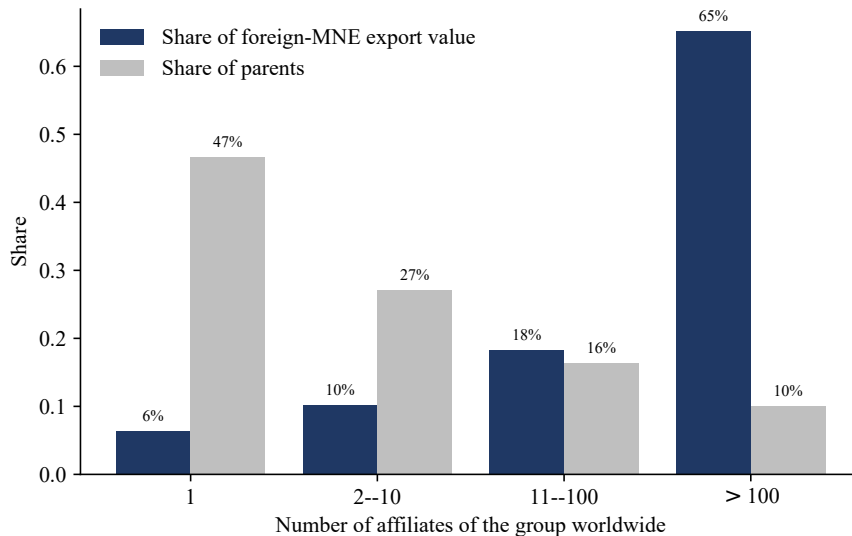
| HS6    | Description                                      | Exports to the<br>USA<br>(US\$bn/yr) | US-parent<br>MNEs (%) | Other foreign<br>MNEs (%) | Domestic<br>MNEs (%) | Local firms<br>(%) |
|--------|--------------------------------------------------|--------------------------------------|-----------------------|---------------------------|----------------------|--------------------|
| 740311 | Cathodes & sections of cathodes, of refined c... | 1.56                                 | 3                     | 33                        | 4                    | 60                 |
| 610910 | T-shirts, singlets & other vests, knitted/cro... | 1.01                                 | 53                    | 7                         | 0                    | 40                 |
| 901890 | Instruments & appliances used in medical/surg... | 0.94                                 | 47                    | 44                        | 0                    | 9                  |
| 901839 | Catheters, cannulae & the like                   | 0.63                                 | 89                    | 2                         | 0                    | 9                  |
| 401110 | New pneumatic tyres, of rubber, of a kind use... | 0.24                                 | 63                    | 16                        | 0                    | 20                 |
| 853620 | Automatic circuit breakers, for a voltage not... | 0.35                                 | 9                     | 91                        | 0                    | 0                  |
| 382490 | Chemical products&preparations of the chemica... | 0.30                                 | 0                     | 36                        | 14                   | 50                 |
| 854231 | Electronic integrated circuits, processors & ... | 0.22                                 | 95                    | 4                         | 0                    | 2                  |
| 880240 | Aeroplanes & other aircraft, of an unladen we... | 0.21                                 | 0                     | 6                         | 83                   | 11                 |
| 902139 | Artificial parts of the body other than teeth... | 0.19                                 | 5                     | 1                         | 70                   | 25                 |
| 800110 | Tin, not alloyed, unwrought                      | 0.19                                 | 0                     | 0                         | 0                    | 100                |
| 611020 | Jerseys, pullovers, cardigans, waist-coats & ... | 0.15                                 | 23                    | 7                         | 0                    | 70                 |
| 610990 | T-shirts, singlets & other vests, knitted/cro... | 0.14                                 | 25                    | 12                        | 1                    | 62                 |
| 853221 | Fixed electrical capacitors, other than those... | 0.12                                 | 0                     | 0                         | 0                    | 100                |
| 611030 | Jerseys, pullovers, cardigans, waist-coats & ... | 0.13                                 | 11                    | 21                        | 0                    | 68                 |

*Notes:* Fifteen largest HS6 products exported to the United States; percent of each product's exports to the United States by type of exporter. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 13 Fact 4: A Small Set of Large Multinational Groups Accounts for Most Exports

### 13.1 Size of the Group's Global Network



**Figure 78:** Manufacturing: Foreign-Multinational Exports and Number of Parents, by Size of the Group's Global Network

*Notes:* Shares of foreign-MNE manufacturing export value and of parents, by the number of affiliates the group has worldwide.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

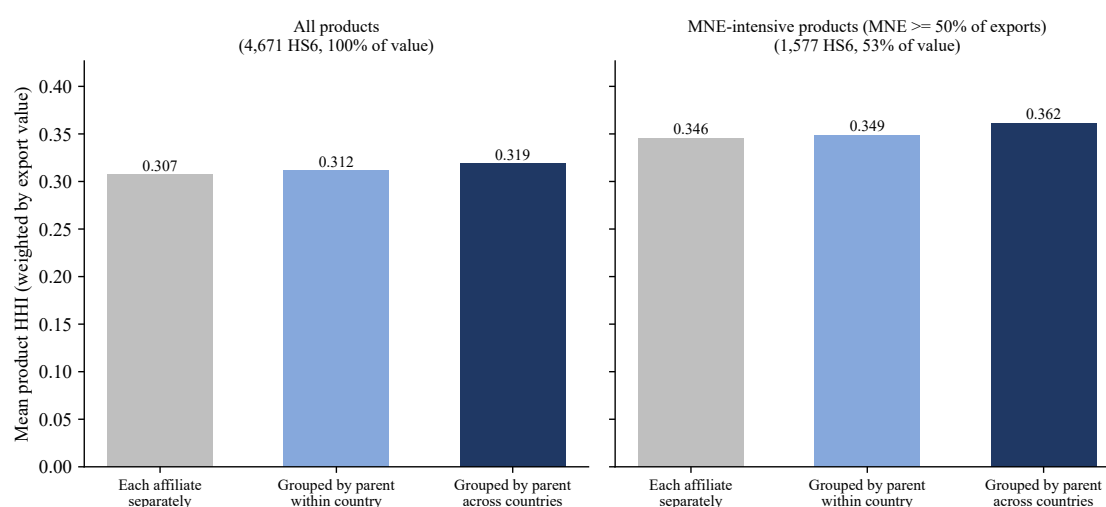
**Table 105:** Manufacturing: Foreign-Multinational Exports and Number of Parents, by Size of the Group's Global Network

| Affiliates worldwide | Parents | Share of parents (%) | Share of foreign-MNE export value (%) |
|----------------------|---------|----------------------|---------------------------------------|
| 1                    | 4,801   | 46.6                 | 6.4                                   |
| 2–10                 | 2,787   | 27.0                 | 10.2                                  |
| 11–100               | 1,679   | 16.3                 | 18.3                                  |
| >100                 | 1,037   | 10.1                 | 65.2                                  |
| All parents          | 10,304  | 100                  | 100                                   |

*Notes:* Number and share of parents, and share of foreign-MNE manufacturing export value, by the number of affiliates the group has worldwide.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 13.2 Concentration within Products



**Figure 79:** Manufacturing: Export Concentration within Products: Firms and Groups

*Notes:* Mean Herfindahl index of exporters within each HS6 product, weighted by export value, counting each affiliate separately or grouping affiliates by parent within or across countries. Right panel: products where MNEs account for at least half of exports.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 106:** Manufacturing: Export Concentration within Products: Firms and Groups

| Sample                                                                        | Measure                               | Firm level | Groups within country | Groups across countries |
|-------------------------------------------------------------------------------|---------------------------------------|------------|-----------------------|-------------------------|
| <i>All products: 4,671 HS6, 100% of value</i>                                 |                                       |            |                       |                         |
|                                                                               | Mean HHI                              | 0.307      | 0.312                 | 0.319                   |
|                                                                               | Mean top-exporter share               | 44.3%      | 44.4%                 | 45.3%                   |
|                                                                               | Effective number of exporters (1/HHI) | 7.5        | 7.2                   | 6.9                     |
| <i>MNE-intensive products (MNE ≥ 50% of exports): 1,577 HS6, 53% of value</i> |                                       |            |                       |                         |
|                                                                               | Mean HHI                              | 0.346      | 0.349                 | 0.362                   |
|                                                                               | Mean top-exporter share               | 46.8%      | 47.1%                 | 48.7%                   |
|                                                                               | Effective number of exporters (1/HHI) | 6.8        | 6.6                   | 6.0                     |

*Notes:* Means across HS6 products, weighted by export value. Firm level: each affiliate counted separately. Groups: affiliates grouped by ultimate parent within each exporting country or across countries.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

### 13.3 Groups and Affiliates

**Table 107:** Manufacturing: Multinational Affiliates and Groups in Origin–Destination–Product–Year Cells

| Origin-destination-product-year cells with $\geq 1$ MNE   |         |
|-----------------------------------------------------------|---------|
| Cells                                                     | 834,508 |
| MNE affiliates per cell (mean)                            | 1.86    |
| MNE groups per cell (mean)                                | 1.85    |
| Affiliates per group, pooled                              | 1.00    |
| Affiliates per group, mean weighted by export value       | 1.08    |
| Cells where one group exports through $> 1$ affiliate (%) | 0.7     |

*Notes:* Cells with at least one MNE. Group: ultimate parent in Orbis or Dun & Bradstreet (parent name when the identifier is missing).

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 108:** Manufacturing: Multinational Presence and Exports: Affiliates and Groups

|                                                               | Origin $\times$ year, destination $\times$ year, product FE |                       |                       | Origin $\times$ dest. $\times$ product, origin $\times$ dest. $\times$ year FE |                       |                       |
|---------------------------------------------------------------|-------------------------------------------------------------|-----------------------|-----------------------|--------------------------------------------------------------------------------|-----------------------|-----------------------|
|                                                               | (1)                                                         | (2)                   | (3)                   | (4)                                                                            | (5)                   | (6)                   |
| <i>Panel A. Dependent variable: ln exports, all firms</i>     |                                                             |                       |                       |                                                                                |                       |                       |
| ln(# MNE firms)                                               | 2.0010***<br>(0.0414)                                       |                       |                       | 0.9522***<br>(0.0238)                                                          |                       |                       |
| ln(# MNE groups)                                              |                                                             | 2.0024***<br>(0.0413) | 1.9951***<br>(0.0409) |                                                                                | 0.9521***<br>(0.0240) | 0.9526***<br>(0.0240) |
| ln(affiliates per group)                                      |                                                             |                       | 2.9370***<br>(0.4047) |                                                                                |                       | 0.8982***<br>(0.0892) |
| <i>Panel B. Dependent variable: ln exports, non-MNE firms</i> |                                                             |                       |                       |                                                                                |                       |                       |
| ln(# MNE firms)                                               | 1.2489***<br>(0.0367)                                       |                       |                       | 0.1925***<br>(0.0202)                                                          |                       |                       |
| ln(# MNE groups)                                              |                                                             | 1.2533***<br>(0.0364) | 1.2511***<br>(0.0361) |                                                                                | 0.1935***<br>(0.0201) | 0.1935***<br>(0.0201) |
| ln(affiliates per group)                                      |                                                             |                       | 0.8359<br>(0.5918)    |                                                                                |                       | -0.0193<br>(0.1346)   |
| Product FE                                                    | ✓                                                           | ✓                     | ✓                     |                                                                                |                       |                       |
| Origin $\times$ year FE                                       | ✓                                                           | ✓                     | ✓                     |                                                                                |                       |                       |
| Destination $\times$ year FE                                  | ✓                                                           | ✓                     | ✓                     |                                                                                |                       |                       |
| Origin $\times$ dest. $\times$ product FE                     |                                                             |                       |                       | ✓                                                                              | ✓                     | ✓                     |
| Origin $\times$ dest. $\times$ year FE                        |                                                             |                       |                       | ✓                                                                              | ✓                     | ✓                     |
| Observations (Panel A)                                        | 834,033                                                     | 834,033               | 834,033               | 726,525                                                                        | 726,525               | 726,525               |
| Observations (Panel B)                                        | 375,761                                                     | 375,761               | 375,761               | 338,556                                                                        | 338,556               | 338,556               |

*Notes:* Observations are origin–destination–HS6–year cells. Manufacturing products only. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Regressors: ln number of MNE affiliates or ln number of MNE groups (distinct ultimate parents). Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 14 Fact 5: Multinational Presence Is Associated with Higher Exports

**Table 109:** Manufacturing: Multinational Presence and Exports

|                                                               | Intensive: ln(# MNE firms) |                       |                       | Extensive: any MNE present |                       |                       |
|---------------------------------------------------------------|----------------------------|-----------------------|-----------------------|----------------------------|-----------------------|-----------------------|
|                                                               | (1)                        | (2)                   | (3)                   | (1)                        | (2)                   | (3)                   |
| <i>Panel A. Dependent variable: ln exports, all firms</i>     |                            |                       |                       |                            |                       |                       |
| ln(# MNE firms)                                               | 2.0181***<br>(0.0409)      | 2.0010***<br>(0.0414) | 0.9522***<br>(0.0238) |                            |                       |                       |
| Any MNE present                                               |                            |                       |                       | 1.3806***<br>(0.0397)      | 1.3800***<br>(0.0391) | 0.6850***<br>(0.0182) |
| <i>Panel B. Dependent variable: ln exports, non-MNE firms</i> |                            |                       |                       |                            |                       |                       |
| ln(# MNE firms)                                               | 1.2657***<br>(0.0367)      | 1.2489***<br>(0.0367) | 0.1925***<br>(0.0202) |                            |                       |                       |
| Any MNE present                                               |                            |                       |                       | 1.2400***<br>(0.0379)      | 1.2208***<br>(0.0372) | 0.0959***<br>(0.0109) |
| Origin FE                                                     | ✓                          |                       |                       | ✓                          |                       |                       |
| Destination FE                                                | ✓                          |                       |                       | ✓                          |                       |                       |
| Product FE                                                    | ✓                          | ✓                     |                       | ✓                          | ✓                     |                       |
| Year FE                                                       | ✓                          |                       |                       | ✓                          |                       |                       |
| Origin × year FE                                              |                            | ✓                     |                       |                            | ✓                     |                       |
| Destination × year FE                                         |                            | ✓                     |                       |                            | ✓                     |                       |
| Origin × dest. × product FE                                   |                            |                       | ✓                     |                            |                       | ✓                     |
| Origin × dest. × year FE                                      |                            |                       | ✓                     |                            |                       | ✓                     |
| Observations (Panel A)                                        | 834,285                    | 834,033               | 726,525               | 1,496,595                  | 1,496,381             | 1,329,815             |
| Observations (Panel B)                                        | 376,034                    | 375,761               | 338,556               | 1,038,432                  | 1,038,206             | 904,114               |

*Notes:* Observations are origin–destination–HS6–year cells. Manufacturing products only. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Intensive margin: ln number of MNEs, cells with at least one MNE. Extensive margin: indicator for at least one MNE. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 110:** Manufacturing: Multinational Exports by Exporting Country and Destination Type

| Origin | MNE value (US\$bn/yr) | of which domestic (%) | Foreign-MNE value (US\$bn/yr) | % of foreign-MNE value, by destination |                                         |
|--------|-----------------------|-----------------------|-------------------------------|----------------------------------------|-----------------------------------------|
|        |                       |                       |                               | parent country                         | other (of which: group affiliate there) |
| ARG    | 14.4                  | 8.7                   | 13.1                          | 5.5                                    | 94.5 (55.3)                             |
| CHL    | 8.7                   | 26.6                  | 6.4                           | 4.1                                    | 95.9 (57.8)                             |
| COL    | 5.9                   | 12.9                  | 5.2                           | 4.0                                    | 96.0 (46.2)                             |
| CRI    | 4.6                   | 6.8                   | 4.3                           | 32.7                                   | 67.3 (28.0)                             |
| DOM    | 3.4                   | 0.9                   | 3.4                           | 25.3                                   | 74.7 (30.7)                             |
| PER    | 2.8                   | 12.5                  | 2.5                           | 5.0                                    | 95.0 (19.7)                             |
| PRY    | 0.4                   | 6.7                   | 0.4                           | 4.3                                    | 95.7 (45.2)                             |
| SLV    | 2.0                   | 0.7                   | 2.0                           | 47.4                                   | 52.6 (23.4)                             |
| URY    | 1.6                   | 16.5                  | 1.3                           | 9.6                                    | 90.4 (30.8)                             |
| All    | 43.9                  | 13.4                  | 38.5                          | 11.9                                   | 88.1 (44.9)                             |

*Notes:* MNE and foreign-MNE manufacturing export value by country, and the split of foreign-MNE value by destination. Parent country: exports to the country of the ultimate parent. Other: all remaining destinations; in parentheses, destinations where the group has another affiliate. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 111:** Manufacturing: Multinational Presence and Exports: Intensive Margin, by Type of Multinational

|                                                               | All MNEs              |                       | Foreign / domestic    |                       | Foreign by destination, domestic |                       |
|---------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------------------------|-----------------------|
|                                                               | (1)                   | (2)                   | (3)                   | (4)                   | (5)                              | (6)                   |
| <i>Panel A. Dependent variable: ln exports, all firms</i>     |                       |                       |                       |                       |                                  |                       |
| ln(# MNE firms)                                               | 2.0010***<br>(0.0414) | 0.9522***<br>(0.0238) |                       |                       |                                  |                       |
| ln(# foreign MNEs)                                            |                       |                       | 1.1644***<br>(0.0358) | 0.4435***<br>(0.0148) |                                  |                       |
| ln(# foreign MNEs, parent-country destination)                |                       |                       |                       |                       | 0.4294***<br>(0.1086)            | 0.1248**<br>(0.0474)  |
| ln(# foreign MNEs, other destinations)                        |                       |                       |                       |                       | 0.9347***<br>(0.0710)            | 0.3008***<br>(0.0431) |
| ln(# domestic MNEs)                                           |                       |                       | 0.3833***<br>(0.0319) | 0.1904***<br>(0.0175) | 0.3269***<br>(0.0729)            | 0.1636***<br>(0.0401) |
| <i>Panel B. Dependent variable: ln exports, non-MNE firms</i> |                       |                       |                       |                       |                                  |                       |
| ln(# MNE firms)                                               | 1.2489***<br>(0.0367) | 0.1925***<br>(0.0202) |                       |                       |                                  |                       |
| ln(# foreign MNEs)                                            |                       |                       | 1.0546***<br>(0.0493) | 0.1865***<br>(0.0250) |                                  |                       |
| ln(# foreign MNEs, parent-country destination)                |                       |                       |                       |                       | 0.2820**<br>(0.1140)             | 0.1364**<br>(0.0611)  |
| ln(# foreign MNEs, other destinations)                        |                       |                       |                       |                       | 0.9954***<br>(0.0680)            | 0.1594**<br>(0.0641)  |
| ln(# domestic MNEs)                                           |                       |                       | 0.4194***<br>(0.0419) | 0.0742***<br>(0.0238) | 0.3667***<br>(0.0885)            | 0.1106**<br>(0.0440)  |
| Product FE                                                    | ✓                     |                       | ✓                     |                       | ✓                                |                       |
| Origin × year FE                                              | ✓                     |                       | ✓                     |                       | ✓                                |                       |
| Destination × year FE                                         | ✓                     |                       | ✓                     |                       | ✓                                |                       |
| Origin × dest. × product FE                                   |                       | ✓                     |                       | ✓                     |                                  | ✓                     |
| Origin × dest. × year FE                                      |                       | ✓                     |                       | ✓                     |                                  | ✓                     |
| Observations (Panel A)                                        | 834,033               | 726,525               | 85,453                | 73,318                | 6,936                            | 5,563                 |
| Observations (Panel B)                                        | 375,761               | 338,556               | 68,534                | 59,772                | 6,579                            | 5,352                 |

*Notes:* Observations are origin–destination–HS6–year cells. Manufacturing products only. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Regressors: ln number of MNEs of each type. Columns (5)–(6) split foreign MNEs by whether the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 112: Manufacturing: Multinational Presence and Exports: Extensive Margin, by Type of Multinational**

|                                                               | All MNEs              |                       | Foreign / domestic    |                       | Foreign by destination, domestic |                       |
|---------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------------------------|-----------------------|
|                                                               | (1)                   | (2)                   | (3)                   | (4)                   | (5)                              | (6)                   |
| <i>Panel A. Dependent variable: ln exports, all firms</i>     |                       |                       |                       |                       |                                  |                       |
| Any MNE present                                               | 1.3800***<br>(0.0391) | 0.6850***<br>(0.0182) |                       |                       |                                  |                       |
| Any foreign MNE                                               |                       |                       | 1.4231***<br>(0.0344) | 0.7021***<br>(0.0178) |                                  |                       |
| Any foreign MNE, parent-country destination                   |                       |                       |                       |                       | 1.3645***<br>(0.1114)            | 0.5991***<br>(0.0488) |
| Any foreign MNE, other destination                            |                       |                       |                       |                       | 1.4329***<br>(0.0324)            | 0.7010***<br>(0.0164) |
| Any domestic MNE                                              |                       |                       | 1.3878***<br>(0.0466) | 0.4684***<br>(0.0186) | 1.3589***<br>(0.0440)            | 0.4682***<br>(0.0186) |
| <i>Panel B. Dependent variable: ln exports, non-MNE firms</i> |                       |                       |                       |                       |                                  |                       |
| Any MNE present                                               | 1.2208***<br>(0.0372) | 0.0959***<br>(0.0109) |                       |                       |                                  |                       |
| Any foreign MNE                                               |                       |                       | 1.1402***<br>(0.0344) | 0.0997***<br>(0.0114) |                                  |                       |
| Any foreign MNE, parent-country destination                   |                       |                       |                       |                       | 0.9721***<br>(0.1138)            | 0.1045***<br>(0.0202) |
| Any foreign MNE, other destination                            |                       |                       |                       |                       | 1.1200***<br>(0.0329)            | 0.1030***<br>(0.0115) |
| Any domestic MNE                                              |                       |                       | 1.0491***<br>(0.0490) | 0.0984***<br>(0.0167) | 1.0171***<br>(0.0467)            | 0.0983***<br>(0.0167) |
| Product FE                                                    | ✓                     |                       | ✓                     |                       | ✓                                |                       |
| Origin × year FE                                              | ✓                     |                       | ✓                     |                       | ✓                                |                       |
| Destination × year FE                                         | ✓                     |                       | ✓                     |                       | ✓                                |                       |
| Origin × dest. × product FE                                   |                       | ✓                     |                       | ✓                     |                                  | ✓                     |
| Origin × dest. × year FE                                      |                       | ✓                     |                       | ✓                     |                                  | ✓                     |
| Observations (Panel A)                                        | 1,496,381             | 1,329,815             | 1,496,381             | 1,329,815             | 1,496,381                        | 1,329,815             |
| Observations (Panel B)                                        | 1,038,206             | 904,114               | 1,038,206             | 904,114               | 1,038,206                        | 904,114               |

*Notes:* Observations are origin–destination–HS6–year cells. Manufacturing products only. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Regressors: indicators for at least one MNE of each type. Columns (5)–(6) split foreign MNEs by whether the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 15 Fact 6: Distance Is a Weaker Barrier for Multinationals

**Table 113: Manufacturing: Distance and Firm Exports: Multinationals by Presence at the Destination**

| Dep. var.: ln exports                         | (1)                    | (2)                    | (3)                    | (4)                    |
|-----------------------------------------------|------------------------|------------------------|------------------------|------------------------|
| ln distance                                   | -0.1509***<br>(0.0454) | -0.1741***<br>(0.0460) | -0.1499***<br>(0.0453) | -0.1730***<br>(0.0458) |
| ln distance × MNE                             | 0.0499***<br>(0.0045)  | 0.0496***<br>(0.0045)  |                        |                        |
| ln distance × MNE, present at destination     |                        |                        | 0.0659***<br>(0.0081)  | 0.0653***<br>(0.0081)  |
| ln distance × MNE, not present at destination |                        |                        | 0.0444***<br>(0.0037)  | 0.0443***<br>(0.0037)  |
| Origin FE                                     | ✓                      |                        | ✓                      |                        |
| Destination FE                                | ✓                      |                        | ✓                      |                        |
| Year FE                                       | ✓                      |                        | ✓                      |                        |
| Origin × year FE                              |                        | ✓                      |                        | ✓                      |
| Destination × year FE                         |                        | ✓                      |                        | ✓                      |
| Product FE                                    | ✓                      | ✓                      | ✓                      | ✓                      |
| Observations                                  | 3,917,960              | 3,917,765              | 3,917,960              | 3,917,765              |

*Notes:* Observations are firm–destination–HS6–year cells. Manufacturing products only. Dependent variable: ln exports. Non-MNE firms are the omitted group. Present: the group has an affiliate in the destination or the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

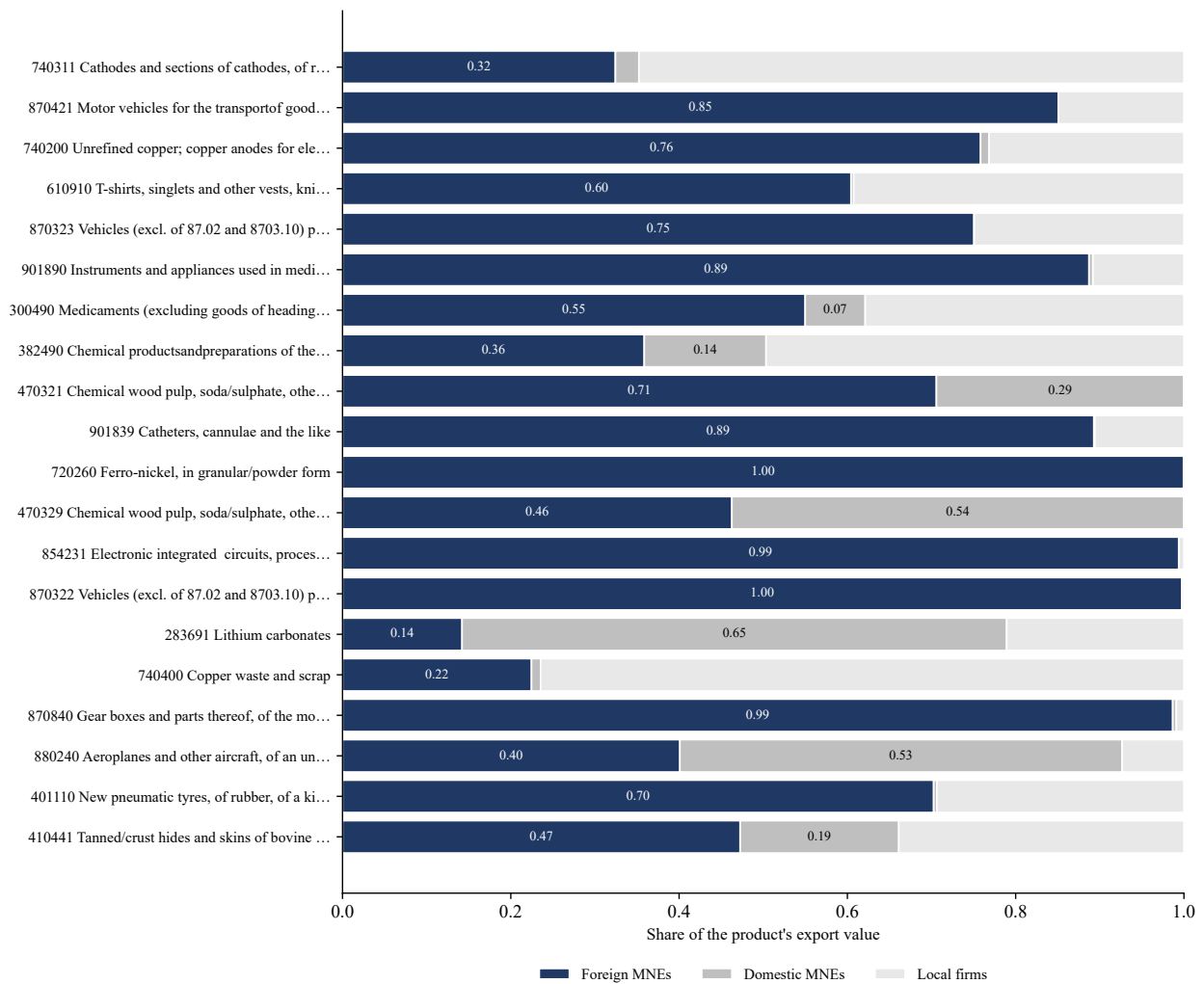
**Table 114:** Manufacturing: Distance and Firm Exports: Foreign and Domestic Multinationals

| Dep. var.: ln exports                                        | (1)                    | (2)                    | (3)                    | (4)                    |
|--------------------------------------------------------------|------------------------|------------------------|------------------------|------------------------|
| ln distance                                                  | -0.1510***<br>(0.0454) | -0.1742***<br>(0.0459) | -0.1507***<br>(0.0455) | -0.1738***<br>(0.0460) |
| ln distance $\times$ foreign MNE                             | 0.0515***<br>(0.0048)  | 0.0512***<br>(0.0048)  |                        |                        |
| ln distance $\times$ foreign MNE, parent-country destination |                        |                        | 0.0690***<br>(0.0189)  | 0.0687***<br>(0.0193)  |
| ln distance $\times$ foreign MNE, other destination          |                        |                        | 0.0502***<br>(0.0041)  | 0.0498***<br>(0.0040)  |
| ln distance $\times$ domestic MNE                            | 0.0388***<br>(0.0052)  | 0.0390***<br>(0.0052)  | 0.0386***<br>(0.0051)  | 0.0388***<br>(0.0051)  |
| Origin FE                                                    | ✓                      |                        | ✓                      |                        |
| Destination FE                                               | ✓                      |                        | ✓                      |                        |
| Year FE                                                      | ✓                      |                        | ✓                      |                        |
| Origin $\times$ year FE                                      |                        | ✓                      |                        | ✓                      |
| Destination $\times$ year FE                                 |                        | ✓                      |                        | ✓                      |
| Product FE                                                   | ✓                      | ✓                      | ✓                      | ✓                      |
| Observations                                                 | 3,917,960              | 3,917,765              | 3,917,960              | 3,917,765              |

*Notes:* Observations are firm–destination–HS6–year cells. Manufacturing products only. Dependent variable: ln exports. Non-MNE firms are the omitted group. Columns (3)–(4) split foreign MNEs by whether the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

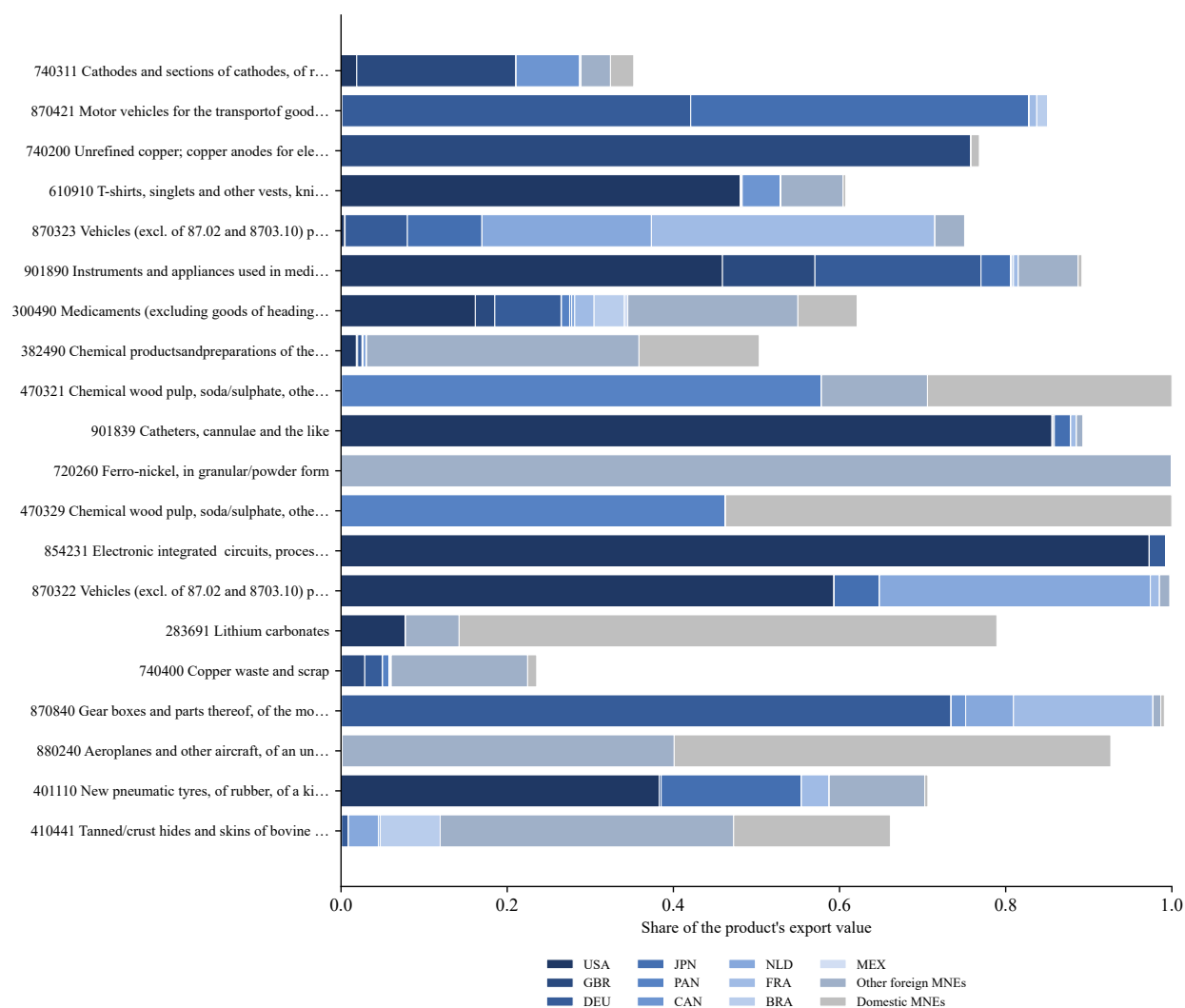
## 16 Products



**Figure 80:** Manufacturing: Ownership of the Twenty Largest Export Products

*Notes:* Shares of each product's export value. Twenty largest HS6 products by export value.

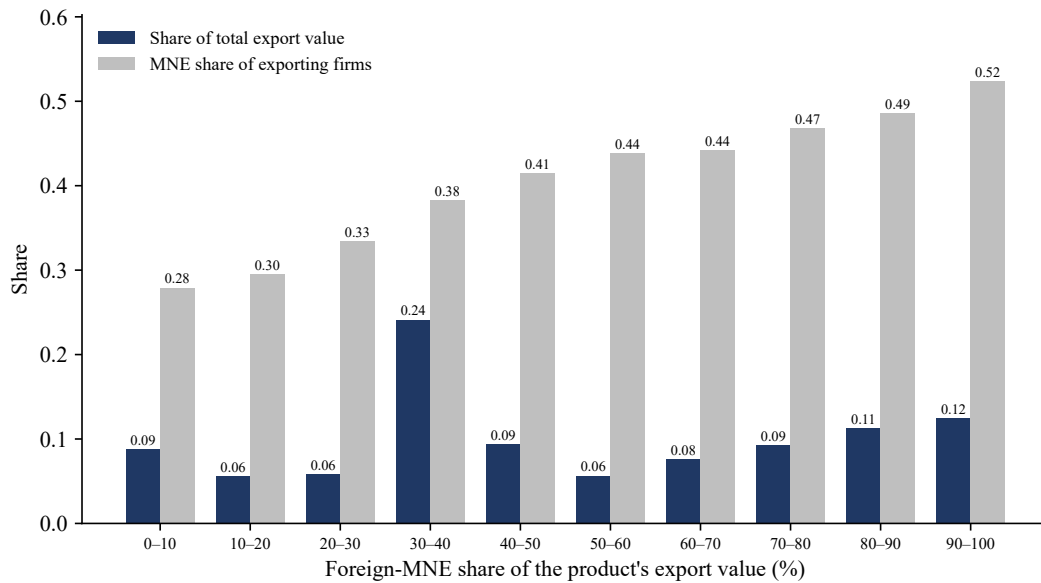
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 81:** Manufacturing: Ownership of the Twenty Largest Export Products, by Parent Country

*Notes:* Shares of each product's export value. Foreign-MNE share split by country of the ultimate parent; the unfilled part of each bar is local firms.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 82:** Manufacturing: Distribution of Export Value by Product-Level Foreign-Multinational Share

*Notes:* Share of manufacturing export value and MNE share of exporting firms, by the foreign-MNE share of each HS6 product's export value.

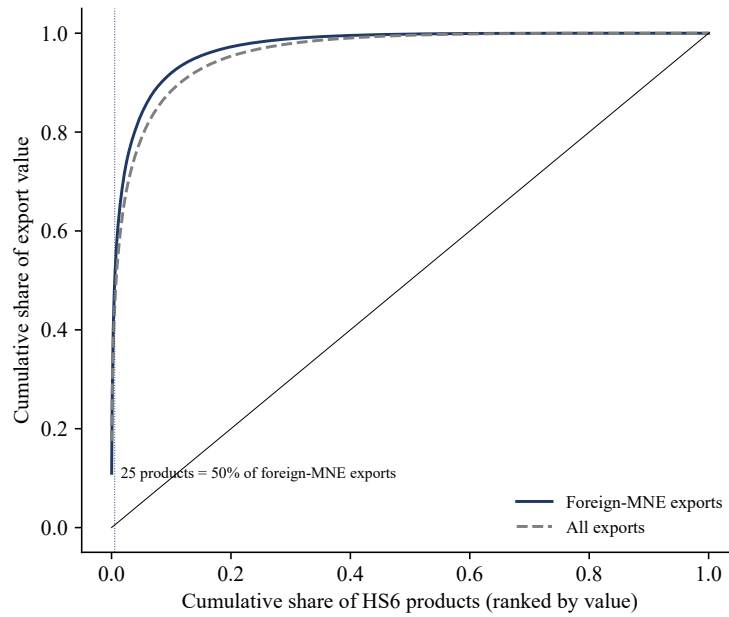
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 115:** Manufacturing: Distribution of Export Value by Product-Level Foreign-Multinational Share

| Foreign share bin (%) | Share of value | MNE share of exporting firms | HS6 products |
|-----------------------|----------------|------------------------------|--------------|
| 0-10                  | 0.088          | 0.279                        | 1,005        |
| 10-20                 | 0.056          | 0.296                        | 537          |
| 20-30                 | 0.058          | 0.334                        | 489          |
| 30-40                 | 0.241          | 0.383                        | 472          |
| 40-50                 | 0.094          | 0.415                        | 451          |
| 50-60                 | 0.057          | 0.438                        | 368          |
| 60-70                 | 0.076          | 0.442                        | 323          |
| 70-80                 | 0.093          | 0.468                        | 267          |
| 80-90                 | 0.113          | 0.486                        | 275          |
| 90-100                | 0.125          | 0.524                        | 484          |

*Notes:* Share of manufacturing export value and MNE share of exporting firms, by the foreign-MNE share of each HS6 product's export value. MNE share of exporting firms: MNE firm-destination-product-year observations over all observations in the bin.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 83:** Manufacturing: Concentration of Exports across Products

*Notes:* Cumulative share of export value against the cumulative share of HS6 products, ranked by value, for foreign-MNE exports and for all exports.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 116:** Manufacturing: Exports and Multinational Shares, by HS Section

| HS section                             | US\$bn/yr | Foreign (%) | Domestic (%) | Local (%) | Leading parent |
|----------------------------------------|-----------|-------------|--------------|-----------|----------------|
| XV Base metals                         | 19.6      | 41          | 3            | 56        | GBR (0.44)     |
| VI Chemicals                           | 11.6      | 49          | 14           | 37        | USA (0.25)     |
| XVII Transport equipment               | 9.7       | 78          | 4            | 18        | DEU (0.30)     |
| XI Textiles and apparel                | 6.9       | 49          | 3            | 48        | USA (0.62)     |
| XVI Machinery and electrical equipment | 6.2       | 63          | 4            | 33        | USA (0.41)     |
| VII Plastics and rubber                | 5.8       | 51          | 8            | 42        | USA (0.34)     |
| X Pulp, paper                          | 3.4       | 54          | 28           | 18        | PAN (0.54)     |
| XVIII Optical, precision instruments   | 3.4       | 72          | 8            | 19        | USA (0.66)     |
| IX Wood and articles                   | 1.9       | 27          | 34           | 39        | BHS (0.24)     |
| VIII Hides, skins, leather             | 1.6       | 40          | 12           | 49        | BGR (0.19)     |
| XIII Stone, ceramics, glass            | 0.9       | 49          | 4            | 46        | USA (0.32)     |
| XX Miscellaneous manufactures          | 0.9       | 53          | 7            | 40        | USA (0.48)     |
| XII Footwear, headgear                 | 0.5       | 48          | 1            | 50        | USA (0.66)     |
| XXI Works of art                       | 0.1       | 58          | 1            | 41        | BGR (0.94)     |
| XIX Arms                               | 0.0       | 1           | 0            | 99        | CHE (0.82)     |
| All sections                           | 72.7      | 52          | 8            | 40        |                |

*Notes:* Foreign, domestic and local: percent of each section's export value. Leading parent: largest parent country and its share of the section's foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 117: Manufacturing: Thirty Largest Export Products**

| HS6    | Description                                                                                         | US\$bn/yr | Foreign (%) | Domestic (%) | Local (%) | Leading parent |
|--------|-----------------------------------------------------------------------------------------------------|-----------|-------------|--------------|-----------|----------------|
| 740311 | Cathodes and sections of cathodes, of refined copper, unwrought                                     | 10.4      | 32          | 3            | 65        | GBR (0.59)     |
| 870421 | Motor vehicles for the transport of goods (excl. of 8704.10), with C-I internal combustion pist...  | 3.2       | 85          | 0            | 15        | DEU (0.49)     |
| 740200 | Unrefined copper; copper anodes for electrolytic refining                                           | 1.3       | 76          | 1            | 23        | GBR (1.00)     |
| 610910 | T-shirts, singlets and other vests, knitted/crocheted, of cotton                                    | 1.3       | 60          | 0            | 39        | USA (0.79)     |
| 870323 | Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s...  | 1.3       | 75          | 0            | 25        | FRA (0.45)     |
| 901890 | Instruments and appliances used in medical/surgical/veterinary sciences, incl. other electro-m...   | 1.4       | 89          | 0            | 11        | USA (0.52)     |
| 300490 | Medicaments (excluding goods of heading 30.02/30.05/30.06/3004.10-3004.50) consisting of mixed...   | 1.0       | 55          | 7            | 38        | USA (0.29)     |
| 382490 | Chemical products and preparations of the chemical/allied industries (including those consisting... | 1.2       | 36          | 14           | 50        | LIE (0.61)     |
| 470321 | Chemical wood pulp, soda/sulphate, other than dissolving grades, semi-bleached/bleached, conif...   | 0.7       | 71          | 29           | 0         | PAN (0.82)     |
| 901839 | Catheters, cannulae and the like                                                                    | 0.9       | 89          | 0            | 11        | USA (0.96)     |
| 720260 | Ferro-nickel, in granular/powder form                                                               | 0.8       | 100         | 0            | 0         | ZAF (0.85)     |
| 470329 | Chemical wood pulp, soda/sulphate, other than dissolving grades, semi-bleached/bleached, non-c...   | 0.6       | 46          | 54           | 0         | PAN (1.00)     |
| 854231 | Electronic integrated circuits, processors and controllers, whether/not combined with memorie...    | 0.8       | 99          | 0            | 1         | USA (0.98)     |
| 870322 | Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s...  | 0.8       | 100         | 0            | 0         | USA (0.59)     |
| 283691 | Lithium carbonates                                                                                  | 0.6       | 14          | 65           | 21        | USA (0.54)     |
| 740400 | Copper waste and scrap                                                                              | 0.5       | 22          | 1            | 76        | BGR (0.39)     |
| 870840 | Gear boxes and parts thereof, of the motor vehicles of headings 87.01 to 87.05.                     | 0.6       | 99          | 0            | 1         | DEU (0.74)     |
| 880240 | Aeroplanes and other aircraft, of an unladen weight >15000kg                                        | 0.6       | 40          | 53           | 7         | CHL (0.99)     |
| 401110 | New pneumatic tyres, of rubber, of a kind used on motor cars (incl. station wagons and racing ...   | 0.5       | 70          | 0            | 29        | USA (0.55)     |
| 410441 | Tanned/crust hides and skins of bovine (incl. buffalo)/equine animals, without hair on, in the...   | 0.5       | 47          | 19           | 34        | BGR (0.33)     |
| 730429 | Other casing, tubing, of a kind used in drilling for oil/gas (excl. of 7304.22/23/24), other ...    | 0.5       | 4           | 0            | 96        | CHE (0.70)     |
| 392330 | Carboys, bottles, flasks and similar articles, of plastics                                          | 0.5       | 46          | 26           | 28        | CAN (0.54)     |
| 870321 | Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s...  | 0.5       | 100         | 0            | 0         | NLD (0.49)     |
| 740811 | Copper wire, of refined copper of which the maximum cross-sectional dim. exceeds 6mm                | 0.4       | 62          | 0            | 38        | USA (0.50)     |
| 800110 | Tin, not alloyed, unwrought                                                                         | 0.4       | 0           | 0            | 100       | FRA (0.49)     |
| 390410 | Poly(vinyl chloride), not mixed with any other substance, in primary forms                          | 0.4       | 100         | 0            | 0         | MEX (0.83)     |
| 390210 | Polypropylene, in primary forms                                                                     | 0.4       | 16          | 0            | 84        | ESP (0.55)     |
| 380892 | Fungicides, put up in forms/packings for retail sale/as preparations/articles                       | 0.4       | 82          | 2            | 16        | USA (0.42)     |
| 280120 | Iodine                                                                                              | 0.3       | 8           | 17           | 76        | CAN (1.00)     |
| 902139 | Artificial parts of the body other than teeth, dental fittings and joints                           | 0.4       | 10          | 60           | 30        | VGB (0.46)     |

*Notes:* Thirty largest HS6 products by export value. Foreign, domestic and local: percent of the product's export value. Leading parent: largest parent country and its share of the product's foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 118: Manufacturing: Thirty Largest Export Products of Foreign Multinationals**

| HS6    | Description                                                                                         | US\$bn/yr | Foreign (%) | Domestic (%) | Local (%) | Leading parent |
|--------|-----------------------------------------------------------------------------------------------------|-----------|-------------|--------------|-----------|----------------|
| 740311 | Cathodes and sections of cathodes, of refined copper, unwrought                                     | 10.4      | 32          | 3            | 65        | GBR (0.59)     |
| 870421 | Motor vehicles for the transport of goods (excl. of 8704.10), with C-I internal combustion pist...  | 3.2       | 85          | 0            | 15        | DEU (0.49)     |
| 740200 | Unrefined copper; copper anodes for electrolytic refining                                           | 1.3       | 76          | 1            | 23        | GBR (1.00)     |
| 901890 | Instruments and appliances used in medical/surgical/veterinary sciences, incl. other electro-m...   | 1.4       | 89          | 0            | 11        | USA (0.52)     |
| 870323 | Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s...  | 1.3       | 75          | 0            | 25        | FRA (0.45)     |
| 610910 | T-shirts, singlets and other vests, knitted/crocheted, of cotton                                    | 1.3       | 60          | 0            | 39        | USA (0.79)     |
| 720260 | Ferro-nickel, in granular/powder form                                                               | 0.8       | 100         | 0            | 0         | ZAF (0.85)     |
| 901839 | Catheters, cannulae and the like                                                                    | 0.9       | 89          | 0            | 11        | USA (0.96)     |
| 854231 | Electronic integrated circuits, processors and controllers, whether/not combined with memorie...    | 0.8       | 99          | 0            | 1         | USA (0.98)     |
| 870322 | Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s...  | 0.8       | 100         | 0            | 0         | USA (0.59)     |
| 470321 | Chemical wood pulp, soda/sulphate, other than dissolving grades, semi-bleached/bleached, conif...   | 0.7       | 71          | 29           | 0         | PAN (0.82)     |
| 300490 | Medicaments (excluding goods of heading 30.02/30.05/30.06/3004.10-3004.50) consisting of mixed...   | 1.0       | 55          | 7            | 38        | USA (0.29)     |
| 870840 | Gear boxes and parts thereof, of the motor vehicles of headings 87.01 to 87.05.                     | 0.6       | 99          | 0            | 1         | DEU (0.74)     |
| 870321 | Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s...  | 0.5       | 100         | 0            | 0         | NLD (0.49)     |
| 390410 | Poly(vinyl chloride), not mixed with any other substance, in primary forms                          | 0.4       | 100         | 0            | 0         | MEX (0.83)     |
| 382490 | Chemical products and preparations of the chemical/allied industries (including those consisting... | 1.2       | 36          | 14           | 50        | LIE (0.61)     |
| 470329 | Chemical wood pulp, soda/sulphate, other than dissolving grades, semi-bleached/bleached, non-c...   | 0.6       | 46          | 54           | 0         | PAN (1.00)     |
| 853620 | Automatic circuit breakers, for a voltage not >1000V                                                | 0.4       | 99          | 0            | 1         | IRL (0.89)     |
| 401110 | New pneumatic tyres, of rubber, of a kind used on motor cars (incl. station wagons and racing ...   | 0.5       | 70          | 0            | 29        | USA (0.55)     |
| 870333 | Vehicles principally designed for the transport of persons (excl. of 87.02 and 8703.10-8703.24)...  | 0.4       | 100         | 0            | 0         | JPN (1.00)     |
| 380892 | Fungicides, put up in forms/packings for retail sale/as preparations/articles                       | 0.4       | 82          | 2            | 16        | USA (0.42)     |
| 790111 | Zinc, not alloyed, unwrought, containing by weight 99.99%/more of zinc                              | 0.3       | 97          | 0            | 3         | BRA (0.90)     |
| 740811 | Copper wire, of refined copper of which the maximum cross-sectional dim. exceeds 6mm                | 0.4       | 62          | 0            | 38        | USA (0.50)     |
| 380891 | Insecticides, put up in forms/packings for retail sale/as preparations/articles                     | 0.3       | 77          | 4            | 18        | DEU (0.52)     |
| 410441 | Tanned/crust hides and skins of bovine (incl. buffalo)/equine animals, without hair on, in the...   | 0.5       | 47          | 19           | 34        | BGR (0.33)     |
| 961900 | Sanitary towels and tampons, napkins and napkin liners for babies and similar sanitary article...   | 0.2       | 97          | 1            | 1         | USA (0.79)     |
| 790112 | Zinc, not alloyed, unwrought, containing by weight <99.99% of zinc                                  | 0.2       | 100         | 0            | 0         | BRA (1.00)     |
| 392330 | Carboys, bottles, flasks and similar articles, of plastics                                          | 0.5       | 46          | 26           | 28        | CAN (0.54)     |
| 854430 | Ignition wiring sets and other wiring sets of a kind used in vehicles/aircraft/ships                | 0.2       | 98          | 0            | 2         | JPN (0.81)     |
| 880240 | Aeroplanes and other aircraft, of an unladen weight >15000kg                                        | 0.6       | 40          | 53           | 7         | CHL (0.99)     |

*Notes:* Thirty largest HS6 products by foreign-MNE export value. Foreign, domestic and local: percent of the product's export value. Leading parent: largest parent country and its share of the product's foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 119: Manufacturing: Products with the Highest Foreign-Multinational Share**

| HS6    | Description                                                                                        | US\$bn/yr | Foreign (%) | Domestic (%) | Local (%) | Leading parent |
|--------|----------------------------------------------------------------------------------------------------|-----------|-------------|--------------|-----------|----------------|
| 720260 | Ferro-nickel, in granular/powder form                                                              | 0.76      | 100         | 0            | 0         | ZAF (0.85)     |
| 790500 | Zinc plates, sheets, strip and foil                                                                | 0.05      | 100         | 0            | 0         | GBR (0.94)     |
| 790112 | Zinc, not alloyed, unwrought, containing by weight <99.99% of zinc                                 | 0.22      | 100         | 0            | 0         | BRA (1.00)     |
| 870321 | Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s... | 0.53      | 100         | 0            | 0         | NLD (0.49)     |
| 390410 | Poly(vinyl chloride), not mixed with any other substance, in primary forms                         | 0.38      | 100         | 0            | 0         | MEX (0.83)     |
| 870322 | Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s... | 0.82      | 100         | 0            | 0         | USA (0.59)     |
| 870333 | Vehicles principally designed for the transport of persons (excl. of 87.02 and 8703.10-8703.24)... | 0.39      | 100         | 0            | 0         | JPN (1.00)     |
| 280300 | Carbon (carbon blacks and other forms of carbon, n.e.s.)                                           | 0.08      | 100         | 0            | 0         | USA (0.99)     |
| 470311 | Chemical wood pulp, soda/sulphate, other than dissolving grades, unbleached, coniferous            | 0.14      | 100         | 0            | 0         | PAN (1.00)     |
| 300691 | Appliances identifiable for ostomy use                                                             | 0.19      | 100         | 0            | 0         | GBR (0.80)     |
| 853620 | Automatic circuit breakers, for a voltage not >1000V                                               | 0.45      | 99          | 0            | 1         | IRL (0.89)     |
| 854231 | Electronic integrated circuits, processors and controllers, whether/not combined with memorie...   | 0.76      | 99          | 0            | 1         | USA (0.98)     |
| 381512 | Supported catalysts, with precious metal/precious metal comps. as the active substance             | 0.08      | 99          | 0            | 1         | GBR (0.80)     |
| 870840 | Gear boxes and parts thereof, of the motor vehicles of headings 87.01 to 87.05.                    | 0.63      | 99          | 0            | 1         | DEU (0.74)     |
| 620711 | Men's/boys' underpants and briefs (excl. knitted/crocheted), of cotton                             | 0.05      | 98          | 0            | 2         | USA (1.00)     |
| 854430 | Ignition wiring sets and other wiring sets of a kind used in vehicles/aircraft/ships               | 0.23      | 98          | 0            | 2         | JPN (0.81)     |
| 740710 | Bars, rods and profiles, of refined copper                                                         | 0.06      | 98          | 0            | 1         | MEX (0.73)     |
| 390120 | Polyethylene having a sp.gr. of 0.94/more, in primary forms                                        | 0.13      | 98          | 0            | 2         | USA (0.98)     |
| 961900 | Sanitary towels and tampons, napkins and napkin liners for babies and similar sanitary article...  | 0.22      | 97          | 1            | 1         | USA (0.79)     |
| 350300 | Gelatin, incl. gelatin in rectangular (incl. square) sheets, whether/not surface-worked /colou...  | 0.08      | 97          | 0            | 3         | BEL (0.43)     |
| 790111 | Zinc, not alloyed, unwrought, containing by weight 99.99%/more of zinc                             | 0.35      | 97          | 0            | 3         | BRA (0.90)     |
| 521213 | Woven fabrics of cotton (excl. of 52.08-52.11), dyed, weighing not >200 g/m2                       | 0.13      | 96          | 0            | 4         | USA (1.00)     |
| 390190 | Polymers of ethylene, in primary forms (excl. of 3901.10-3901.30)                                  | 0.13      | 96          | 0            | 4         | USA (1.00)     |
| 840820 | Compression-ignition internal combustion piston engines (diesel/semi-diesel engines) of a kind...  | 0.19      | 96          | 0            | 4         | USA (0.54)     |
| 761010 | Doors, windows and their frames and thresholds for doors, of aluminium                             | 0.13      | 96          | 0            | 4         | CYM (0.99)     |
| 870422 | Motor vehicles for the transport of goods (excl. of 8704.10), with C-I internal combustion pist... | 0.08      | 96          | 0            | 4         | JPN (0.88)     |
| 281700 | Zinc oxide; zinc peroxide                                                                          | 0.07      | 95          | 0            | 5         | BRA (0.80)     |
| 300610 | Sterile surgical catgut, similar sterile suture materials (including sterile absorbable surgic...  | 0.14      | 95          | 3            | 2         | IRL (0.98)     |
| 611595 | Other n.e.s. in 61.15, knitted/crocheted, of cotton.                                               | 0.12      | 94          | 1            | 5         | USA (0.97)     |
| 611521 | Panty hose, tights, stockings, socks and other hosiery, and footwear without applied soles, of...  | 0.05      | 94          | 1            | 5         | USA (0.99)     |

*Notes:* HS6 products with exports of at least US\$500 million over the period, ranked by foreign-MNE share. Foreign, domestic and local: percent of the product's export value. Leading parent: largest parent country and its share of the product's foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 120: Manufacturing: Products with the Highest Domestic-Multinational Share**

| HS6    | Description                                                                                       | US\$bn/yr | Foreign (%) | Domestic (%) | Local (%) | Leading parent |
|--------|---------------------------------------------------------------------------------------------------|-----------|-------------|--------------|-----------|----------------|
| 282520 | Lithium oxide and hydroxide                                                                       | 0.04      | 0           | 100          | 0         | –              |
| 481092 | Paper and paperboard(excl. of 4810.13-4810.39), coated on one/both sides with kaolin (China cl... | 0.16      | 1           | 99           | 0         | IRL (0.84)     |
| 310420 | Potassium chloride                                                                                | 0.14      | 3           | 95           | 2         | DZA (0.58)     |
| 441193 | Fibreboard of wood/other ligneous materials, whether/not bonded with resins/other organic subs... | 0.04      | 1           | 94           | 5         | ESP (0.94)     |
| 350110 | Casein                                                                                            | 0.06      | 0           | 84           | 16        | CHL (0.80)     |
| 730640 | Tubes, pipes and hollow profiles (excl. of 7306.10-7306.30), welded, of circular cross-section... | 0.06      | 15          | 82           | 3         | DEU (0.60)     |
| 390390 | Polymers of styrene, in primary forms (excl. of 3903.11-3903.30)                                  | 0.05      | 15          | 73           | 12        | NLD (0.78)     |
| 440711 | Wood sawn/chipped length wise, sliced/peeled, whether/not planed, sanded/end-jointed, of a thi... | 0.18      | 10          | 71           | 19        | BHS (0.67)     |
| 441239 | Plywood, consisting solely of sheets of wood (other than bamboo,excl. of 4412.31 and4412.32), ... | 0.21      | 28          | 71           | 1         | VGB (0.46)     |
| 390319 | Polystyrene other than expansible, in primary forms                                               | 0.05      | 22          | 67           | 11        | PAN (0.99)     |
| 283691 | Lithium carbonates                                                                                | 0.55      | 14          | 65           | 21        | USA (0.54)     |
| 540219 | High tenacity yarn other than textured yarn/sewing thread, of nylon/other polyamides (excl. of... | 0.05      | 36          | 64           | 0         | GBR (0.80)     |
| 481141 | Gummed/adhesive paper,paperboard,coated, impregnated, covered, surface-coloured, surface-decor... | 0.07      | 26          | 62           | 12        | GBR (0.56)     |
| 902139 | Artificial parts of the body other than teeth, dental fittings and joints                         | 0.38      | 10          | 60           | 30        | VGB (0.46)     |
| 722830 | Bars and rods of other alloy steel (excl. of 72.27, 7228.10 and 7228.20), not further worked t... | 0.06      | 25          | 56           | 19        | SGP (0.59)     |
| 470329 | Chemical wood pulp, soda/sulphate, other than dissolving grades, semi-bleached/bleached, non-c... | 0.56      | 46          | 54           | 0         | PAN (1.00)     |
| 491110 | Trade advertising material, commercial catalogue-sandthe like                                     | 0.06      | 16          | 53           | 31        | USA (0.45)     |
| 880240 | Aeroplanes and other aircraft, of an unladen weight >15000kg                                      | 0.60      | 40          | 53           | 7         | CHL (0.99)     |
| 410449 | Tanned/crust hides and skins of bovine (incl. buffalo)/equine animals, without hair on, in the... | 0.08      | 37          | 52           | 12        | URY (0.71)     |
| 330300 | Perfumes and toilet waters                                                                        | 0.13      | 34          | 46           | 20        | PAN (0.38)     |
| 391990 | Self-adhesive plates, sheets, film, foil, tape, strip and other flat shapes, of plastics, whet... | 0.07      | 41          | 44           | 15        | GBR (0.57)     |
| 440910 | Wood (including strips and friezes for parquet flooring, not assembled) continuously shaped (t... | 0.14      | 16          | 43           | 41        | USA (0.47)     |
| 330420 | Eye make-up preparations                                                                          | 0.06      | 18          | 40           | 42        | PAN (0.31)     |
| 392112 | Plates, sheets, film, foil and strip, cellular, of polymers of vinyl chloride                     | 0.09      | 59          | 39           | 2         | COL (0.66)     |
| 721420 | Bars and rods of iron/non-alloy steel (excl. of 72.13), containing indentations/ribs/grooves/o... | 0.23      | 56          | 38           | 7         | SGP (0.39)     |
| 902190 | Appliances which are worn/carried/implanted in the body, to compensate for a defect/disability... | 0.09      | 63          | 31           | 6         | CHE (0.40)     |
| 847490 | Parts of the machinery of 84.74                                                                   | 0.05      | 40          | 31           | 28        | FIN (0.24)     |
| 441114 | Medium density of fibreboard of wood/other ligneous materials, whether/not bonded with resins/... | 0.13      | 68          | 30           | 2         | BHS (0.60)     |
| 470321 | Chemical wood pulp, soda/sulphate, other than dissolving grades, semi-bleached/bleached, confi... | 0.72      | 71          | 29           | 0         | PAN (0.82)     |
| 870850 | Drive-axes with differential, whether/not provided with other transmission components, and no...  | 0.14      | 67          | 29           | 4         | DEU (0.54)     |

*Notes:* HS6 products with exports of at least US\$500 million over the period, ranked by domestic-MNE share. Foreign, domestic and local: percent of the product's export value. Leading parent: largest parent country and its share of the product's foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 17 Product Groups within the Sector

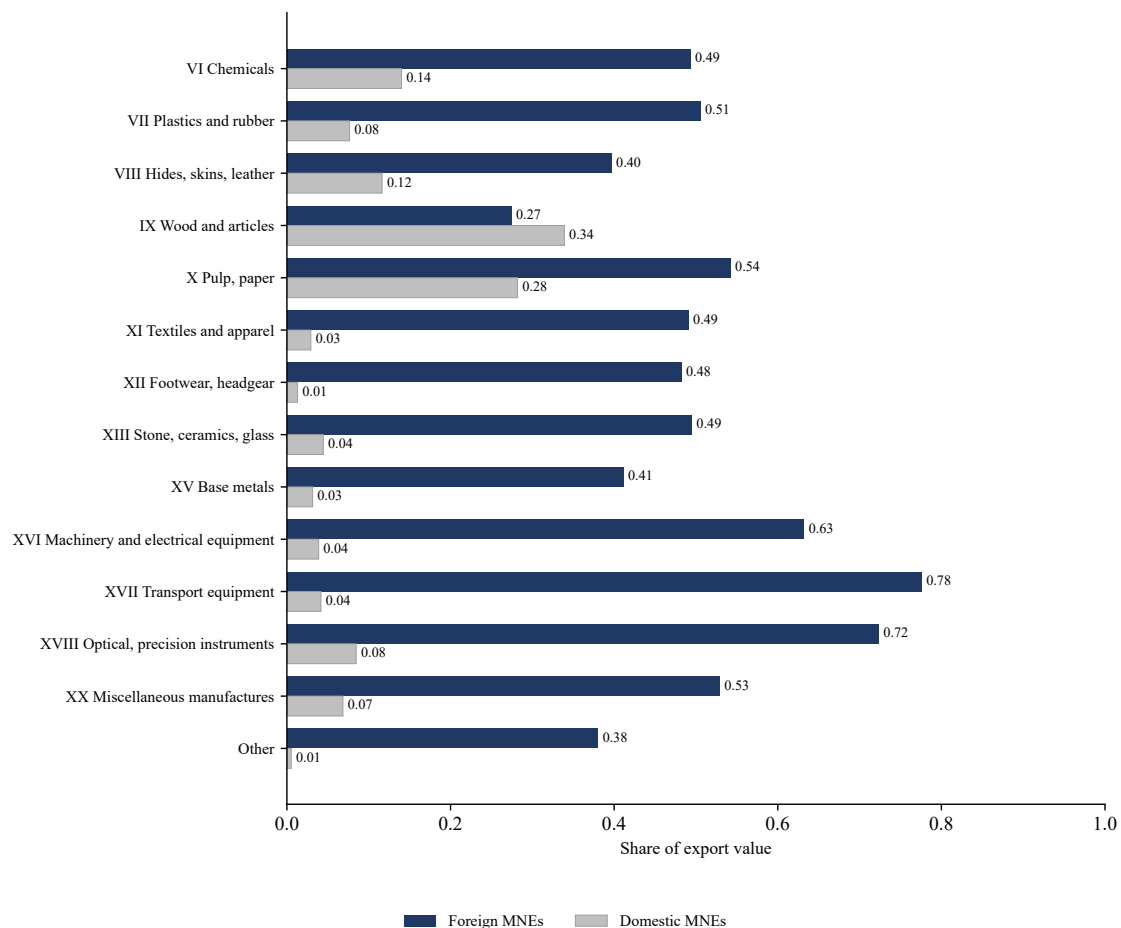
### 17.1 By HS Section

**Table 121:** Manufacturing: Exports and Multinational Shares, by HS Section

| HS section                             | Total exports (US\$bn/yr) | MNE exports (US\$bn/yr) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|----------------------------------------|---------------------------|-------------------------|-------------|--------------|------------------------------------|
| VI Chemicals                           | 11.6                      | 7.4                     | 49.3        | 14.0         | USA 0.25, DEU 0.13, GBR 0.12       |
| VII Plastics and rubber                | 5.8                       | 3.4                     | 50.5        | 7.7          | USA 0.34, MEX 0.17, JPN 0.06       |
| VIII Hides, skins, leather             | 1.6                       | 0.8                     | 39.7        | 11.6         | BGR 0.19, URY 0.17, SGP 0.16       |
| IX Wood and articles                   | 1.9                       | 1.1                     | 27.4        | 33.9         | BHS 0.24, JPN 0.12, VGB 0.11       |
| X Pulp, paper                          | 3.4                       | 2.7                     | 54.2        | 28.2         | PAN 0.54, USA 0.19, CHL 0.08       |
| XI Textiles and apparel                | 6.9                       | 3.6                     | 49.1        | 2.9          | USA 0.62, CAN 0.04, FRA 0.03       |
| XII Footwear, headgear                 | 0.5                       | 0.3                     | 48.2        | 1.3          | USA 0.66, BRA 0.13, JPN 0.08       |
| XIII Stone, ceramics, glass            | 0.9                       | 0.5                     | 49.5        | 4.5          | USA 0.32, PAN 0.20, CYM 0.11       |
| XV Base metals                         | 19.6                      | 8.5                     | 41.1        | 3.1          | GBR 0.44, CAN 0.12, ZAF 0.08       |
| XVI Machinery and electrical equipment | 6.2                       | 4.2                     | 63.2        | 3.9          | USA 0.41, IRL 0.10, JPN 0.08       |
| XVII Transport equipment               | 9.7                       | 7.9                     | 77.6        | 4.2          | DEU 0.30, JPN 0.28, NLD 0.12       |
| XVIII Optical, precision instruments   | 3.4                       | 2.8                     | 72.3        | 8.5          | USA 0.66, DEU 0.12, GBR 0.08       |
| XX Miscellaneous manufactures          | 0.9                       | 0.6                     | 52.9        | 6.9          | USA 0.48, SWE 0.10, DEU 0.10       |
| Other                                  | 0.1                       | 0.0                     | 38.0        | 0.5          | BGR 0.92, USA 0.03, CHE 0.01       |
| All                                    | 72.7                      | 43.9                    | 51.6        | 8.0          |                                    |

*Notes:* HS sections. Foreign and domestic: percent of the group's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

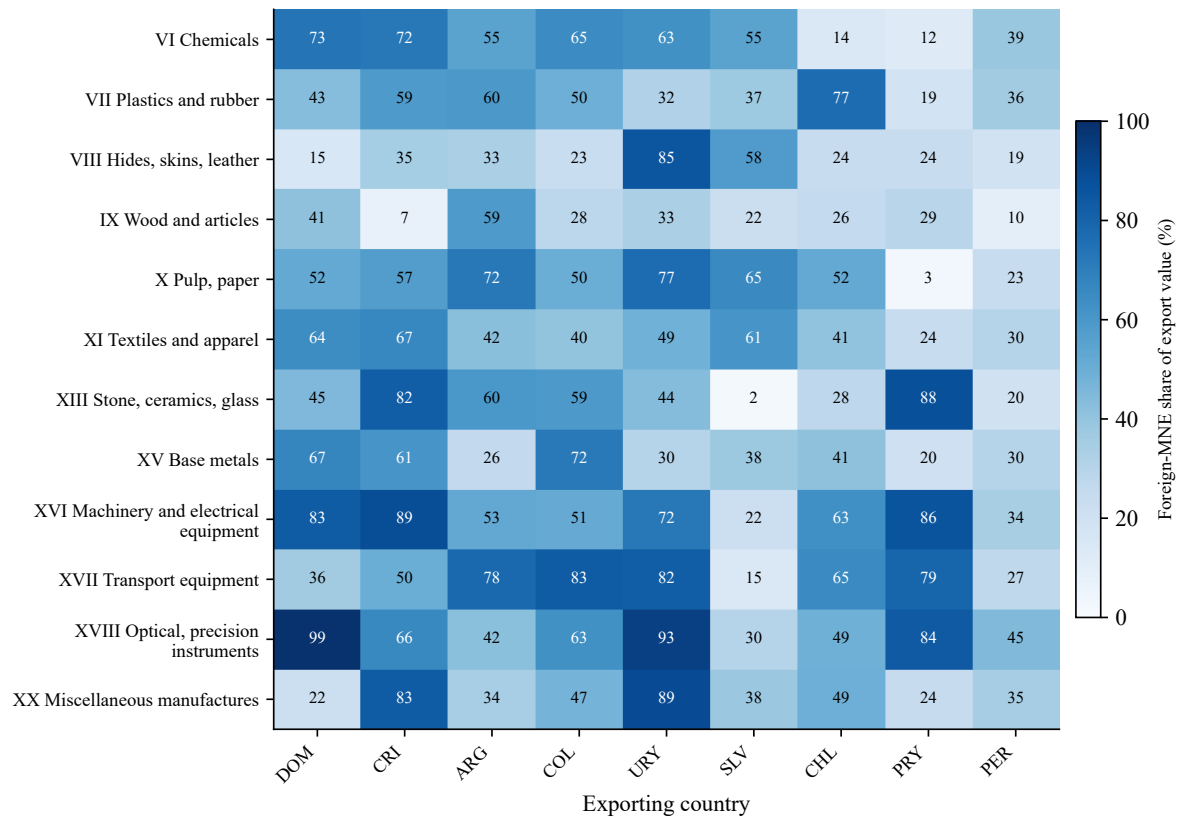
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 84:** Manufacturing: Multinational Share of Export Value, by HS Section

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. HS sections.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 85: Manufacturing: Foreign-Multinational Share of Export Value, by Exporting Country and HS Section**

*Notes:* Percent of each group's export value in each country shipped by foreign MNEs. HS sections.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 122: Manufacturing: Exports by Exporting Country and HS Section**

| Sub-sector                                                                  | ARG   | CHL   | COL   | CRI   | DOM   | PER   | PRY   | SLV   | URY   | All   |
|-----------------------------------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| <i>Panel A: composition of the origin's exports, % (columns sum to 100)</i> |       |       |       |       |       |       |       |       |       |       |
| VI Chemicals                                                                | 23.2  | 10.7  | 22.7  | 10.4  | 13.2  | 10.2  | 12.4  | 7.0   | 16.6  | 15.4  |
| VII Plastics and rubber                                                     | 6.8   | 2.2   | 17.9  | 10.0  | 7.0   | 7.0   | 10.3  | 7.6   | 22.3  | 7.8   |
| VIII Hides, skins, leather                                                  | 4.0   | 0.1   | 2.2   | 0.4   | 0.5   | 0.5   | 13.1  | 0.3   | 11.1  | 1.9   |
| IX Wood and articles                                                        | 0.8   | 7.0   | 0.5   | 1.2   | 0.2   | 1.8   | 7.5   | 0.2   | 11.7  | 3.0   |
| X Pulp, paper                                                               | 2.4   | 10.0  | 4.7   | 2.4   | 1.8   | 2.7   | 2.7   | 7.4   | 3.2   | 5.2   |
| XI Textiles and apparel                                                     | 2.6   | 0.3   | 9.6   | 2.6   | 21.0  | 20.0  | 17.0  | 58.2  | 12.7  | 9.5   |
| XII Footwear, headgear                                                      | 0.1   | 0.0   | 0.5   | 0.0   | 7.3   | 0.4   | 1.9   | 1.1   | 0.1   | 0.6   |
| XIII Stone, ceramics, glass                                                 | 0.7   | 0.2   | 3.9   | 2.0   | 0.6   | 2.6   | 1.6   | 1.0   | 0.4   | 1.3   |
| XV Base metals                                                              | 10.2  | 65.8  | 17.1  | 6.6   | 10.0  | 48.3  | 7.9   | 7.2   | 4.6   | 30.2  |
| XVI Machinery and electrical equipment                                      | 8.2   | 2.3   | 9.9   | 24.8  | 16.6  | 4.3   | 18.5  | 7.5   | 4.0   | 8.0   |
| XVII Transport equipment                                                    | 39.6  | 1.0   | 6.4   | 0.6   | 0.4   | 0.9   | 3.2   | 0.2   | 8.2   | 11.4  |
| XVIII Optical, precision instruments                                        | 0.8   | 0.1   | 1.0   | 35.9  | 20.3  | 0.2   | 1.6   | 0.1   | 1.5   | 4.1   |
| XX Miscellaneous manufactures                                               | 0.5   | 0.2   | 3.4   | 3.1   | 1.1   | 0.9   | 2.2   | 2.1   | 3.5   | 1.3   |
| Other                                                                       | 0.3   | 0.0   | 0.2   | 0.0   | 0.0   | 0.1   | 0.0   | 0.0   | 0.1   | 0.1   |
| All sub-sectors                                                             | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 |
| <i>Panel B: foreign-MNE share of export value, %</i>                        |       |       |       |       |       |       |       |       |       |       |
| VI Chemicals                                                                | 55.0  | 14.3  | 64.6  | 72.2  | 73.4  | 38.8  | 12.2  | 55.1  | 62.7  | 49.3  |
| VII Plastics and rubber                                                     | 59.7  | 76.7  | 49.5  | 59.0  | 43.3  | 36.2  | 18.5  | 37.2  | 32.0  | 50.5  |
| VIII Hides, skins, leather                                                  | 33.2  | 23.6  | 23.1  | 35.3  | 15.3  | 18.8  | 23.6  | 57.9  | 85.5  | 39.7  |
| IX Wood and articles                                                        | 58.9  | 25.7  | 27.8  | 7.3   | 41.1  | 9.7   | 29.1  | 21.9  | 33.4  | 27.4  |
| X Pulp, paper                                                               | 72.0  | 52.2  | 50.1  | 57.2  | 52.3  | 23.5  | 2.9   | 65.4  | 76.8  | 54.2  |
| XI Textiles and apparel                                                     | 42.5  | 40.6  | 40.1  | 66.6  | 64.3  | 30.0  | 24.1  | 60.8  | 48.6  | 49.1  |
| XII Footwear, headgear                                                      | 41.4  | 52.4  | 22.2  | 18.4  | 63.3  | 15.4  | 83.2  | 7.7   | 39.8  | 48.2  |
| XIII Stone, ceramics, glass                                                 | 59.8  | 27.6  | 59.4  | 82.3  | 45.0  | 19.7  | 87.8  | 2.2   | 43.6  | 49.5  |
| XV Base metals                                                              | 26.1  | 41.4  | 71.9  | 60.8  | 66.9  | 30.0  | 20.4  | 37.5  | 30.1  | 41.1  |
| XVI Machinery and electrical equipment                                      | 52.5  | 63.2  | 51.3  | 88.7  | 83.1  | 34.0  | 86.0  | 21.7  | 72.2  | 63.2  |
| XVII Transport equipment                                                    | 78.1  | 65.2  | 82.9  | 50.3  | 36.2  | 27.0  | 79.4  | 14.6  | 82.0  | 77.6  |
| XVIII Optical, precision instruments                                        | 42.1  | 49.2  | 62.9  | 65.7  | 99.2  | 44.9  | 83.8  | 30.3  | 93.5  | 72.3  |
| XX Miscellaneous manufactures                                               | 34.4  | 49.2  | 47.3  | 83.0  | 21.6  | 34.8  | 24.5  | 38.0  | 89.3  | 52.9  |
| Other                                                                       | 52.7  | 1.3   | 7.2   | 12.5  | 10.5  | 4.5   | 1.1   | 33.7  | 5.1   | 38.0  |
| All sub-sectors                                                             | 60.4  | 39.9  | 57.8  | 70.9  | 73.2  | 30.6  | 37.9  | 52.5  | 55.3  | 51.6  |

*Notes:* Panel A: composition of each country's exports in the sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs. HS sections.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 123: Manufacturing: Exports by HS Section and Parent Country**

| HS section                             | USA  | GBR  | DEU  | JPN  | PAN  | CAN | NLD | FRA | BRA | MEX | Other | Domestic |
|----------------------------------------|------|------|------|------|------|-----|-----|-----|-----|-----|-------|----------|
| VI Chemicals                           | 11.0 | 5.5  | 5.8  | 0.7  | 1.0  | 1.4 | 0.8 | 0.7 | 1.7 | 0.1 | 20.4  | 14.0     |
| VII Plastics and rubber                | 15.7 | 1.8  | 0.7  | 2.7  | 1.8  | 2.3 | 1.8 | 1.3 | 1.9 | 8.0 | 12.6  | 7.7      |
| VIII Hides, skins, leather             | 1.0  | 0.1  | 1.9  | 0.0  | 0.5  | 0.0 | 1.5 | 0.2 | 2.8 | 0.2 | 31.6  | 11.6     |
| IX Wood and articles                   | 2.4  | 0.5  | 0.1  | 2.7  | 0.1  | 1.2 | 0.0 | 0.3 | 0.1 | 0.1 | 19.8  | 33.9     |
| X Pulp, paper                          | 10.3 | 0.4  | 0.1  | 0.1  | 28.5 | 0.0 | 0.0 | 0.0 | 0.0 | 0.3 | 14.4  | 28.2     |
| XI Textiles and apparel                | 25.8 | 1.2  | 0.5  | 0.7  | 0.9  | 1.6 | 0.0 | 1.4 | 0.4 | 0.1 | 16.6  | 2.9      |
| XII Footwear, headgear                 | 27.8 | 0.1  | 0.9  | 3.5  | 0.0  | 0.0 | 0.0 | 0.6 | 5.3 | 0.0 | 9.9   | 1.3      |
| XIII Stone, ceramics, glass            | 13.0 | 2.1  | 1.9  | 0.2  | 8.0  | 0.0 | 2.6 | 2.4 | 0.5 | 2.3 | 16.6  | 4.5      |
| XV Base metals                         | 2.5  | 17.1 | 0.1  | 0.2  | 0.5  | 4.5 | 0.8 | 0.5 | 2.9 | 0.4 | 11.8  | 3.1      |
| XVI Machinery and electrical equipment | 24.5 | 1.7  | 2.3  | 4.5  | 0.3  | 1.2 | 1.6 | 2.9 | 0.8 | 2.4 | 21.1  | 3.9      |
| XVII Transport equipment               | 7.6  | 0.1  | 23.2 | 21.8 | 0.0  | 0.2 | 9.5 | 8.6 | 0.9 | 0.0 | 5.7   | 4.2      |
| XVIII Optical, precision instruments   | 43.8 | 5.1  | 7.7  | 2.8  | 0.1  | 0.2 | 0.3 | 0.7 | 0.1 | 0.0 | 11.6  | 8.5      |
| XX Miscellaneous manufactures          | 21.9 | 1.4  | 4.3  | 1.5  | 0.5  | 0.0 | 0.7 | 2.3 | 0.1 | 0.9 | 19.2  | 6.9      |
| Other                                  | 0.6  | 0.1  | 0.0  | 0.0  | 0.0  | 0.0 | 0.1 | 0.0 | 0.0 | 0.0 | 37.3  | 0.5      |

*Notes:* Percent of each group's export value, by country of the ultimate parent of foreign MNEs; domestic MNEs shown for comparison. HS sections.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

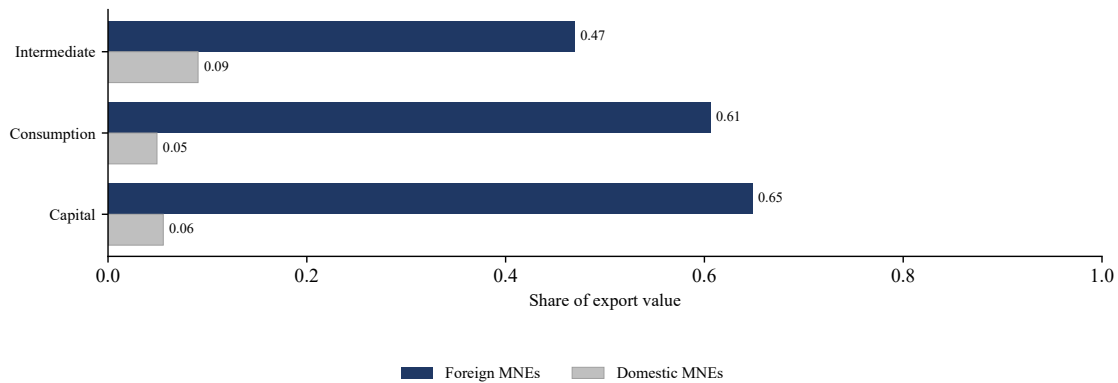
## 17.2 By End Use

**Table 124:** Manufacturing: Exports and Multinational Shares, by End Use

| BEC end use  | Total exports (US\$bn/yr) | MNE exports (US\$bn/yr) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|--------------|---------------------------|-------------------------|-------------|--------------|------------------------------------|
| Intermediate | 47.2                      | 26.6                    | 46.9        | 9.1          | GBR 0.20, USA 0.20, PAN 0.07       |
| Consumption  | 15.2                      | 10.0                    | 60.6        | 4.9          | USA 0.43, NLD 0.09, DEU 0.08       |
| Capital      | 9.3                       | 6.7                     | 64.8        | 5.6          | DEU 0.31, JPN 0.28, USA 0.18       |
| All          | 71.7                      | 43.3                    | 51.7        | 7.8          |                                    |

*Notes:* End use from BEC Rev. 4. Foreign and domestic: percent of the group's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

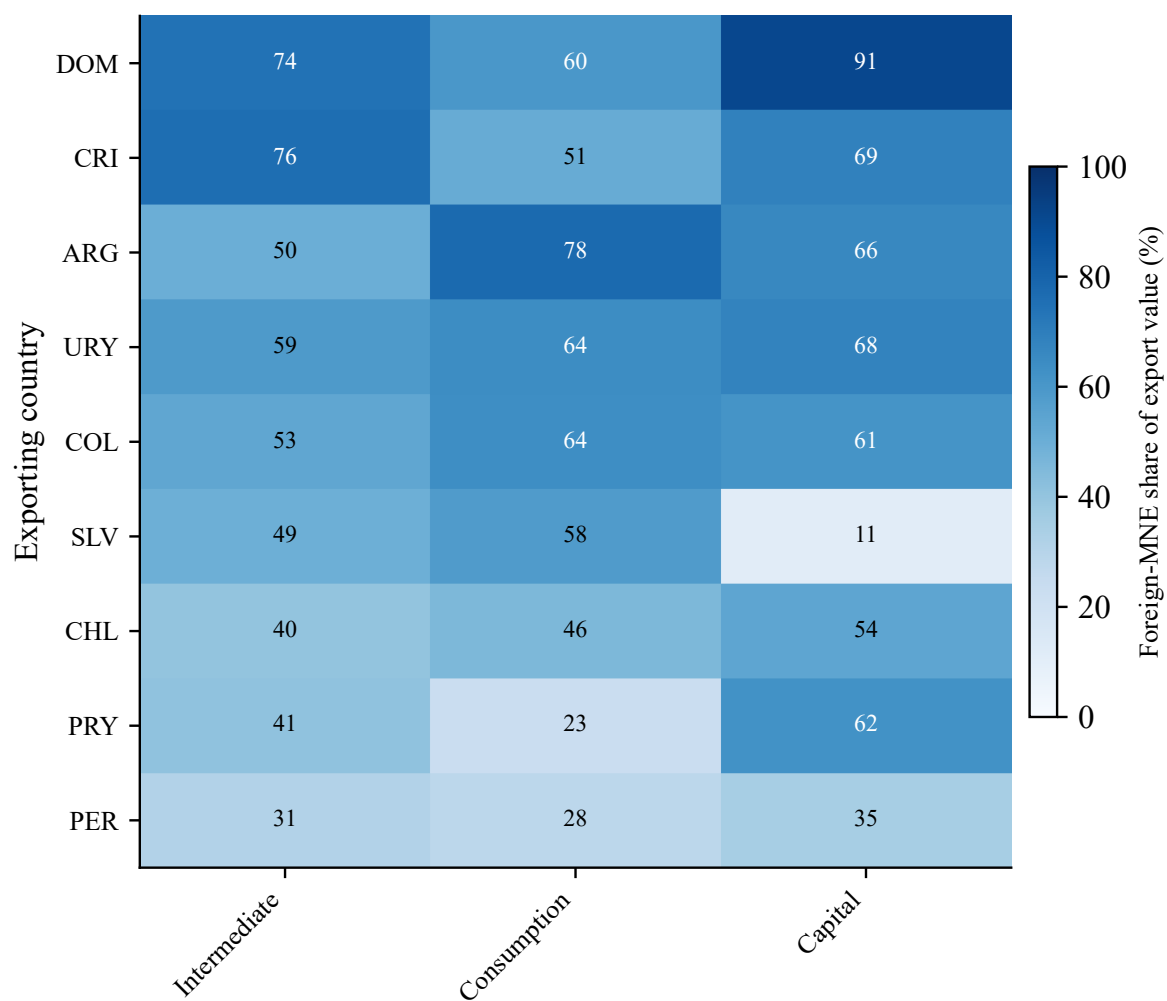
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 86:** Manufacturing: Multinational Share of Export Value, by End Use

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. End use from BEC Rev. 4.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 87:** Manufacturing: Foreign-Multinational Share of Export Value, by Exporting Country and End Use  
*Notes:* Percent of each group's export value in each country shipped by foreign MNEs. End use from BEC Rev. 4.  
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 125: Manufacturing: Exports by Exporting Country and End Use**

| Origin                                                                   | Intermediate | Consumption | Capital | All   |
|--------------------------------------------------------------------------|--------------|-------------|---------|-------|
| <i>Panel A: composition of the origin's exports, % (rows sum to 100)</i> |              |             |         |       |
| ARG                                                                      | 51.1         | 22.3        | 26.7    | 100.0 |
| CHL                                                                      | 96.6         | 1.5         | 1.9     | 100.0 |
| COL                                                                      | 60.5         | 32.5        | 7.0     | 100.0 |
| CRI                                                                      | 66.5         | 19.0        | 14.6    | 100.0 |
| DOM                                                                      | 42.9         | 34.3        | 22.8    | 100.0 |
| PER                                                                      | 76.1         | 20.6        | 3.3     | 100.0 |
| PRY                                                                      | 70.4         | 23.6        | 5.9     | 100.0 |
| SLV                                                                      | 29.3         | 64.0        | 6.7     | 100.0 |
| URY                                                                      | 78.3         | 15.4        | 6.4     | 100.0 |
| All origins                                                              | 68.3         | 20.3        | 11.4    | 100.0 |
| <i>Panel B: foreign-MNE share of export value, %</i>                     |              |             |         |       |
| ARG                                                                      | 49.9         | 77.7        | 66.3    | 60.4  |
| CHL                                                                      | 39.9         | 45.7        | 53.6    | 40.3  |
| COL                                                                      | 53.5         | 63.9        | 61.4    | 57.4  |
| CRI                                                                      | 76.0         | 51.0        | 68.6    | 70.2  |
| DOM                                                                      | 74.3         | 60.1        | 90.6    | 73.2  |
| PER                                                                      | 31.2         | 27.8        | 34.5    | 30.6  |
| PRY                                                                      | 40.8         | 23.0        | 62.2    | 37.9  |
| SLV                                                                      | 49.4         | 58.2        | 11.0    | 52.4  |
| URY                                                                      | 58.7         | 64.1        | 67.9    | 60.2  |
| All origins                                                              | 46.9         | 60.6        | 64.8    | 51.7  |

Notes: Panel A: composition of each country's exports in the sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs. End use from BEC Rev. 4.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 126: Manufacturing: Exports by End Use and Parent Country**

| BEC end use  | USA  | GBR | DEU  | JPN  | PAN | CAN | NLD | FRA | BRA | MEX | Other | Domestic |
|--------------|------|-----|------|------|-----|-----|-----|-----|-----|-----|-------|----------|
| Intermediate | 8.5  | 8.6 | 1.9  | 1.2  | 2.9 | 2.8 | 1.1 | 1.2 | 2.0 | 1.2 | 15.5  | 9.1      |
| Consumption  | 23.9 | 3.3 | 4.6  | 4.1  | 0.8 | 0.5 | 4.8 | 4.0 | 0.8 | 1.0 | 12.8  | 4.9      |
| Capital      | 10.7 | 1.9 | 19.1 | 16.9 | 0.2 | 0.5 | 0.7 | 1.5 | 0.9 | 0.3 | 12.0  | 5.6      |

Notes: Percent of each group's export value, by country of the ultimate parent of foreign MNEs; domestic MNEs shown for comparison. End use from BEC Rev. 4.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

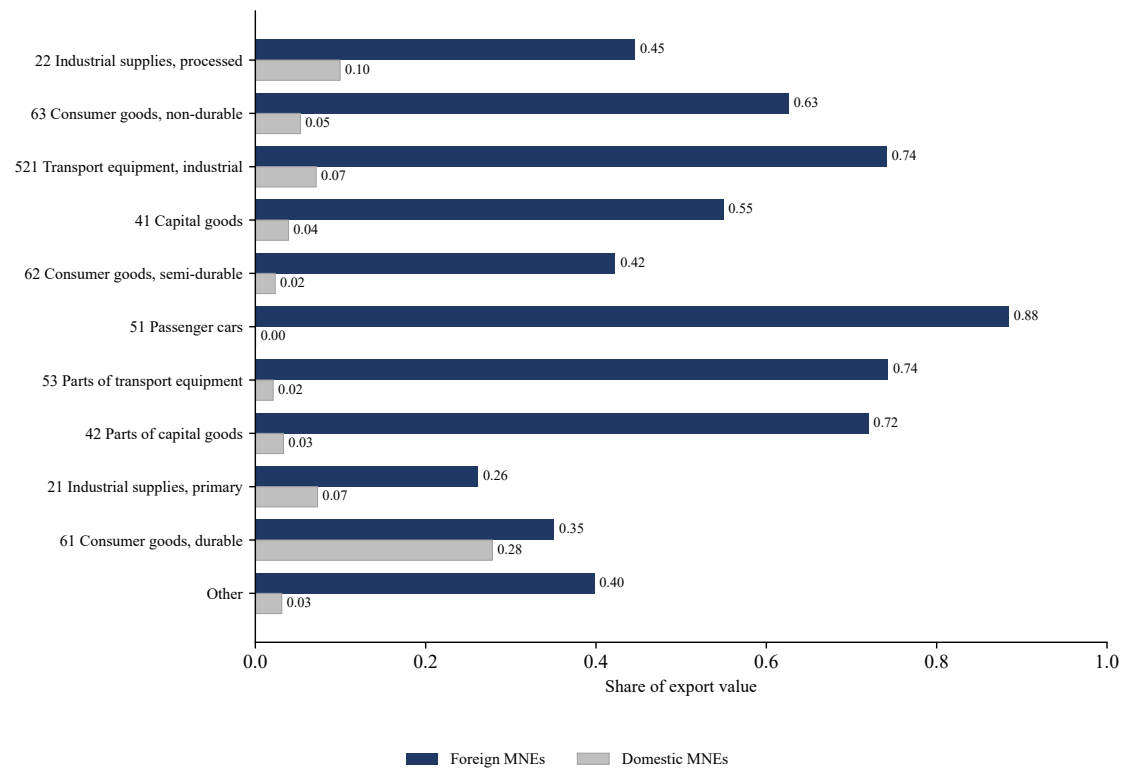
## 17.3 By BEC Category

**Table 127: Manufacturing: Exports and Multinational Shares, by BEC Category**

| BEC Rev. 4 category                 | Total exports (US\$bn/yr) | MNE exports (US\$bn/yr) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|-------------------------------------|---------------------------|-------------------------|-------------|--------------|------------------------------------|
| 22 Industrial supplies, processed   | 39.4                      | 21.5                    | 44.6        | 10.0         | GBR 0.24, USA 0.16, PAN 0.08       |
| 63 Consumer goods, non-durable      | 7.4                       | 5.0                     | 62.6        | 5.3          | USA 0.54, DEU 0.13, GBR 0.08       |
| 521 Transport equipment, industrial | 4.9                       | 4.0                     | 74.1        | 7.1          | DEU 0.42, JPN 0.42, CHL 0.07       |
| 41 Capital goods                    | 4.4                       | 2.6                     | 55.0        | 3.9          | USA 0.45, DEU 0.14, GBR 0.08       |
| 62 Consumer goods, semi-durable     | 3.8                       | 1.7                     | 42.3        | 2.3          | USA 0.49, GBR 0.10, CHE 0.05       |
| 51 Passenger cars                   | 3.1                       | 2.7                     | 88.5        | 0.0          | NLD 0.29, USA 0.23, FRA 0.21       |
| 53 Parts of transport equipment     | 2.9                       | 2.2                     | 74.3        | 2.1          | DEU 0.31, USA 0.22, JPN 0.17       |
| 42 Parts of capital goods           | 2.9                       | 2.2                     | 72.0        | 3.3          | USA 0.56, IRL 0.19, GBR 0.05       |
| 21 Industrial supplies, primary     | 1.8                       | 0.6                     | 26.1        | 7.3          | JPN 0.20, BGR 0.11, GBR 0.09       |
| 61 Consumer goods, durable          | 0.9                       | 0.6                     | 35.1        | 27.8         | MEX 0.43, USA 0.10, JPN 0.09       |
| Other                               | 0.2                       | 0.1                     | 39.8        | 3.1          | JPN 0.32, ESP 0.23, DEU 0.09       |
| All                                 | 71.7                      | 43.3                    | 51.7        | 7.8          |                                    |

Notes: Categories of BEC Rev. 4. Foreign and domestic: percent of the group's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

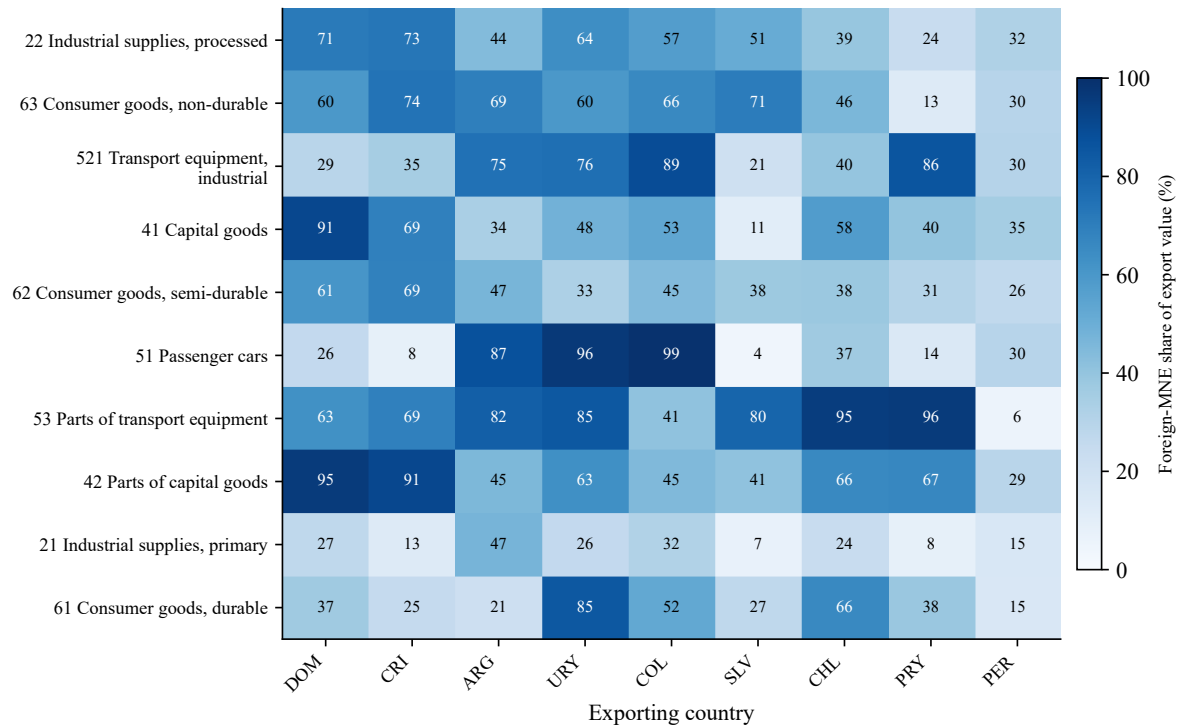
Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 88: Manufacturing: Multinational Share of Export Value, by BEC Category**

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. Categories of BEC Rev. 4.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 89: Manufacturing: Foreign-Multinational Share of Export Value, by Exporting Country and BEC Category**

*Notes:* Percent of each group's export value in each country shipped by foreign MNEs. Categories of BEC Rev. 4.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 128: Manufacturing: Exports by Exporting Country and BEC Category**

| Sub-sector                                                                  | ARG   | CHL   | COL   | CRI   | DOM   | PER   | PRY   | SLV   | URY   | All   |
|-----------------------------------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| <i>Panel A: composition of the origin's exports, % (columns sum to 100)</i> |       |       |       |       |       |       |       |       |       |       |
| 22 Industrial supplies, processed                                           | 39.2  | 91.5  | 48.9  | 41.1  | 27.6  | 71.5  | 42.0  | 26.1  | 60.3  | 58.2  |
| 63 Consumer goods, non-durable                                              | 8.4   | 1.0   | 19.5  | 8.1   | 13.4  | 10.4  | 10.3  | 40.1  | 9.2   | 10.2  |
| 521 Transport equipment, industrial                                         | 21.0  | 0.4   | 1.6   | 0.1   | 0.1   | 0.6   | 2.9   | 0.1   | 4.6   | 5.8   |
| 41 Capital goods                                                            | 5.6   | 1.5   | 5.4   | 14.4  | 22.7  | 2.7   | 3.1   | 6.7   | 1.8   | 5.5   |
| 62 Consumer goods, semi-durable                                             | 0.9   | 0.2   | 7.1   | 2.3   | 20.5  | 9.9   | 12.2  | 23.1  | 2.2   | 5.1   |
| 51 Passenger cars                                                           | 12.6  | 0.1   | 3.5   | 0.0   | 0.0   | 0.0   | 0.1   | 0.0   | 2.6   | 3.7   |
| 53 Parts of transport equipment                                             | 7.7   | 1.5   | 3.8   | 3.3   | 0.7   | 1.4   | 15.5  | 1.0   | 4.3   | 3.7   |
| 42 Parts of capital goods                                                   | 2.8   | 0.9   | 3.0   | 19.8  | 11.5  | 1.7   | 2.4   | 0.6   | 0.9   | 3.7   |
| 21 Industrial supplies, primary                                             | 1.0   | 2.7   | 4.5   | 2.2   | 3.2   | 1.4   | 7.1   | 1.6   | 12.7  | 2.6   |
| 61 Consumer goods, durable                                                  | 0.3   | 0.1   | 2.3   | 8.6   | 0.3   | 0.3   | 1.1   | 0.7   | 1.3   | 1.2   |
| Other                                                                       | 0.3   | 0.0   | 0.4   | 0.1   | 0.1   | 0.1   | 3.5   | 0.1   | 0.1   | 0.2   |
| All sub-sectors                                                             | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 |
| <i>Panel B: foreign-MNE share of export value, %</i>                        |       |       |       |       |       |       |       |       |       |       |
| 22 Industrial supplies, processed                                           | 44.1  | 39.2  | 57.2  | 72.9  | 71.4  | 32.1  | 24.1  | 51.0  | 63.8  | 44.6  |
| 63 Consumer goods, non-durable                                              | 69.2  | 45.8  | 65.9  | 73.8  | 59.7  | 29.7  | 12.5  | 70.6  | 59.7  | 62.6  |
| 521 Transport equipment, industrial                                         | 74.9  | 39.6  | 88.8  | 35.0  | 29.2  | 30.4  | 85.8  | 20.6  | 75.6  | 74.1  |
| 41 Capital goods                                                            | 34.0  | 57.8  | 53.5  | 69.0  | 90.9  | 35.4  | 40.0  | 10.9  | 48.1  | 55.0  |
| 62 Consumer goods, semi-durable                                             | 46.5  | 38.1  | 44.5  | 68.5  | 60.8  | 26.2  | 30.5  | 37.7  | 32.7  | 42.3  |
| 51 Passenger cars                                                           | 87.2  | 37.0  | 98.9  | 8.3   | 26.0  | 29.7  | 14.4  | 4.1   | 96.2  | 88.5  |
| 53 Parts of transport equipment                                             | 81.9  | 94.9  | 40.7  | 69.1  | 62.7  | 5.7   | 95.6  | 79.6  | 84.5  | 74.3  |
| 42 Parts of capital goods                                                   | 45.4  | 66.0  | 44.6  | 90.8  | 95.4  | 28.9  | 66.5  | 41.0  | 63.3  | 72.0  |
| 21 Industrial supplies, primary                                             | 47.1  | 23.9  | 31.6  | 13.1  | 27.1  | 15.4  | 8.2   | 7.5   | 25.9  | 26.1  |
| 61 Consumer goods, durable                                                  | 21.3  | 65.8  | 52.4  | 25.2  | 37.1  | 15.2  | 38.4  | 27.1  | 84.5  | 35.1  |
| Other                                                                       | 37.9  | 17.7  | 42.9  | 47.7  | 15.0  | 59.2  | 46.2  | 8.8   | 58.4  | 39.8  |
| All sub-sectors                                                             | 60.4  | 40.3  | 57.4  | 70.2  | 73.2  | 30.6  | 37.9  | 52.4  | 60.2  | 51.7  |

Notes: Panel A: composition of each country's exports in the sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs. Categories of BEC Rev. 4.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 129: Manufacturing: Exports by BEC Category and Parent Country**

| BEC Rev. 4 category                 | USA  | GBR | DEU  | JPN  | PAN | CAN | NLD  | FRA  | BRA | MEX  | Other | Domestic |
|-------------------------------------|------|-----|------|------|-----|-----|------|------|-----|------|-------|----------|
| 22 Industrial supplies, processed   | 6.5  | 9.8 | 0.6  | 0.4  | 3.4 | 3.2 | 1.0  | 0.9  | 2.2 | 1.4  | 15.2  | 10.0     |
| 63 Consumer goods, non-durable      | 31.4 | 4.7 | 7.3  | 1.0  | 1.1 | 0.9 | 0.1  | 0.5  | 1.2 | 0.3  | 14.3  | 5.3      |
| 521 Transport equipment, industrial | 0.3  | 0.1 | 31.0 | 30.6 | 0.0 | 0.1 | 0.5  | 1.9  | 1.4 | 0.0  | 8.2   | 7.1      |
| 41 Capital goods                    | 21.8 | 3.9 | 6.6  | 2.4  | 0.4 | 0.9 | 0.9  | 1.0  | 0.5 | 0.6  | 16.0  | 3.9      |
| 62 Consumer goods, semi-durable     | 16.4 | 3.4 | 1.3  | 0.5  | 1.0 | 0.0 | 0.2  | 0.6  | 0.8 | 0.2  | 17.8  | 2.3      |
| 51 Passenger cars                   | 20.5 | 0.0 | 3.4  | 17.9 | 0.0 | 0.0 | 25.7 | 19.0 | 0.0 | 0.0  | 2.0   | 0.0      |
| 53 Parts of transport equipment     | 15.9 | 0.2 | 22.8 | 12.3 | 0.1 | 1.1 | 4.6  | 6.4  | 1.3 | 0.1  | 9.7   | 2.1      |
| 42 Parts of capital goods           | 38.4 | 3.4 | 1.6  | 1.0  | 0.2 | 0.9 | 1.0  | 2.1  | 0.3 | 0.2  | 22.9  | 3.3      |
| 21 Industrial supplies, primary     | 0.9  | 1.5 | 0.6  | 3.2  | 0.4 | 0.0 | 0.0  | 0.5  | 0.6 | 0.1  | 18.3  | 7.3      |
| 61 Consumer goods, durable          | 3.2  | 0.2 | 0.1  | 2.8  | 1.0 | 0.0 | 0.1  | 1.8  | 0.1 | 13.7 | 12.0  | 27.8     |
| Other                               | 1.3  | 0.2 | 2.5  | 8.8  | 0.0 | 0.4 | 0.0  | 0.2  | 0.9 | 0.0  | 25.5  | 3.1      |

Notes: Percent of each group's export value, by country of the ultimate parent of foreign MNEs; domestic MNEs shown for comparison. Categories of BEC Rev. 4.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

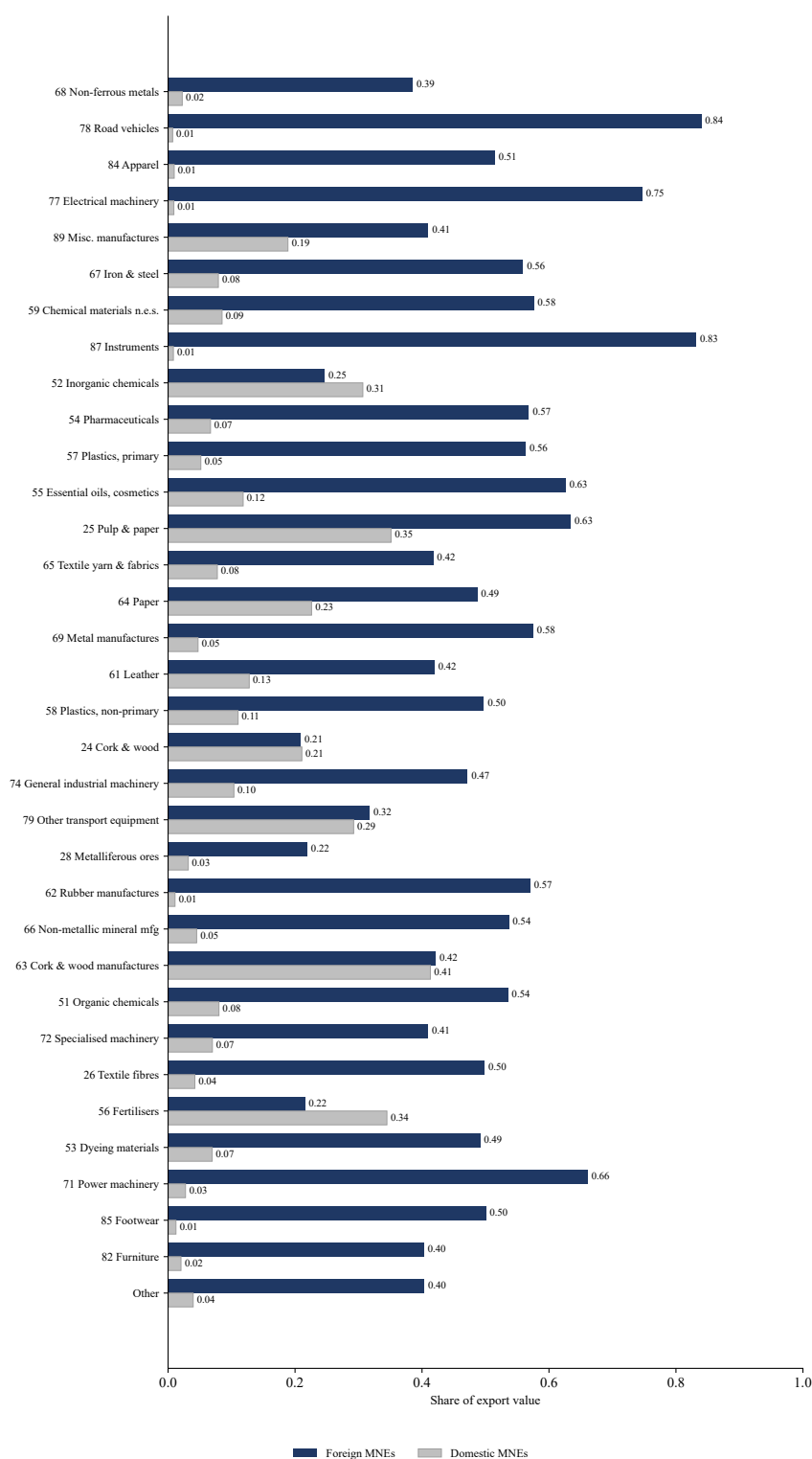
## 17.4 By SITC Division

**Table 130:** Manufacturing: Exports and Multinational Shares, by SITC Division

| SITC Rev. 3 division            | Total exports (US\$bn/yr) | MNE exports (US\$bn/yr) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|---------------------------------|---------------------------|-------------------------|-------------|--------------|------------------------------------|
| 68 Non-ferrous metals           | 14.5                      | 5.7                     | 38.6        | 2.2          | GBR 0.59, CAN 0.16, BRA 0.09       |
| 78 Road vehicles                | 8.5                       | 7.2                     | 84.0        | 0.7          | DEU 0.32, JPN 0.30, NLD 0.13       |
| 84 Apparel                      | 4.5                       | 2.3                     | 51.5        | 1.0          | USA 0.73, CHE 0.04, CAN 0.03       |
| 77 Electrical machinery         | 3.2                       | 2.5                     | 74.7        | 0.9          | USA 0.46, IRL 0.15, JPN 0.11       |
| 89 Misc. manufactures           | 2.9                       | 1.8                     | 41.0        | 18.9         | USA 0.29, GBR 0.15, CAN 0.12       |
| 67 Iron & steel                 | 2.8                       | 1.7                     | 55.9        | 7.9          | ZAF 0.47, SGP 0.10, NLD 0.10       |
| 59 Chemical materials n.e.s.    | 3.0                       | 1.9                     | 57.6        | 8.5          | USA 0.29, DEU 0.23, LIE 0.14       |
| 87 Instruments                  | 2.7                       | 2.3                     | 83.2        | 0.8          | USA 0.69, DEU 0.12, GBR 0.07       |
| 52 Inorganic chemicals          | 1.9                       | 1.1                     | 24.7        | 30.7         | USA 0.36, AUS 0.12, BRA 0.08       |
| 54 Pharmaceuticals              | 2.2                       | 1.4                     | 56.8        | 6.7          | USA 0.21, DEU 0.13, CHE 0.12       |
| 57 Plastics, primary            | 1.9                       | 1.2                     | 56.4        | 5.2          | MEX 0.38, USA 0.30, NLD 0.07       |
| 55 Essential oils, cosmetics    | 2.0                       | 1.4                     | 62.6        | 11.8         | GBR 0.32, USA 0.25, BRA 0.09       |
| 25 Pulp & paper                 | 1.5                       | 1.5                     | 63.5        | 35.1         | PAN 0.89, CHL 0.10, IRL 0.00       |
| 65 Textile yarn & fabrics       | 1.7                       | 0.9                     | 41.8        | 7.8          | USA 0.51, CAN 0.09, GBR 0.05       |
| 64 Paper                        | 1.6                       | 1.1                     | 48.7        | 22.6         | USA 0.46, IRL 0.14, ZAF 0.08       |
| 69 Metal manufactures           | 1.3                       | 0.8                     | 57.5        | 4.7          | USA 0.26, CYM 0.22, JPN 0.06       |
| 61 Leather                      | 1.4                       | 0.8                     | 42.0        | 12.8         | BGR 0.19, URY 0.18, SGP 0.17       |
| 58 Plastics, non-primary        | 1.2                       | 0.7                     | 49.8        | 11.0         | USA 0.24, GBR 0.13, PAN 0.08       |
| 24 Cork & wood                  | 1.0                       | 0.4                     | 20.9        | 21.1         | JPN 0.34, CHL 0.16, USA 0.10       |
| 74 General industrial machinery | 1.2                       | 0.7                     | 47.2        | 10.4         | USA 0.24, GBR 0.08, ITA 0.07       |
| 79 Other transport equipment    | 1.2                       | 0.7                     | 31.8        | 29.2         | CHL 0.66, BHS 0.11, ESP 0.03       |
| 28 Metalliferous ores           | 0.9                       | 0.2                     | 21.9        | 3.2          | BGR 0.29, TUR 0.12, BRA 0.11       |
| 62 Rubber manufactures          | 0.9                       | 0.5                     | 57.1        | 1.1          | USA 0.55, JPN 0.23, CHN 0.11       |
| 66 Non-metallic mineral mfg     | 0.9                       | 0.5                     | 53.7        | 4.5          | USA 0.31, PAN 0.19, CYM 0.11       |
| 63 Cork & wood manufactures     | 0.7                       | 0.6                     | 42.2        | 41.3         | BHS 0.31, VGB 0.16, USA 0.11       |
| 51 Organic chemicals            | 0.8                       | 0.5                     | 53.6        | 8.0          | CAN 0.29, USA 0.23, GBR 0.12       |
| 72 Specialised machinery        | 0.7                       | 0.3                     | 41.0        | 7.0          | USA 0.19, CAN 0.10, GBR 0.08       |
| 26 Textile fibres               | 0.8                       | 0.4                     | 49.9        | 4.2          | FRA 0.28, GBR 0.16, CYM 0.14       |
| 56 Fertilisers                  | 0.6                       | 0.3                     | 21.6        | 34.5         | VEN 0.39, PAN 0.17, CHE 0.10       |
| 53 Dyeing materials             | 0.6                       | 0.3                     | 49.2        | 6.9          | USA 0.31, LUX 0.13, NLD 0.10       |
| 71 Power machinery              | 0.6                       | 0.4                     | 66.1        | 2.7          | USA 0.31, DEU 0.14, ITA 0.12       |
| 85 Footwear                     | 0.5                       | 0.3                     | 50.1        | 1.2          | USA 0.67, BRA 0.13, JPN 0.08       |
| 82 Furniture                    | 0.4                       | 0.2                     | 40.3        | 2.0          | CHL 0.21, DEU 0.19, FRA 0.18       |
| Other                           | 1.2                       | 0.5                     | 40.4        | 4.0          | USA 0.48, GBR 0.09, MEX 0.04       |
| All                             | 71.7                      | 43.3                    | 51.7        | 7.8          |                                    |

*Notes:* SITC Rev. 3 divisions; HS6 products without an SITC correspondence are excluded. Foreign and domestic: percent of the group's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

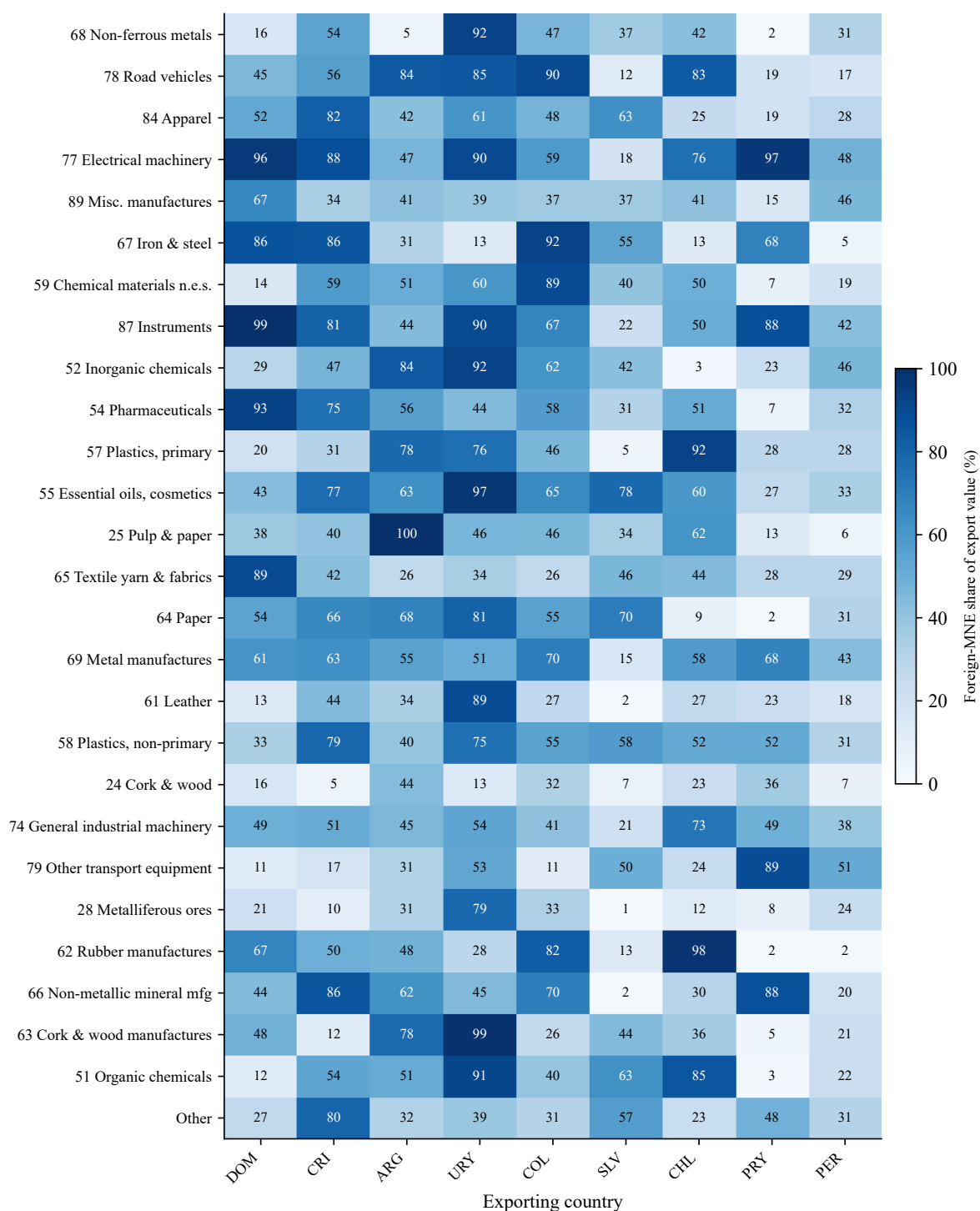
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 90: Manufacturing: Multinational Share of Export Value, by SITC Division**

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. SITC Rev. 3 divisions; HS6 products without an SITC correspondence are excluded.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 91: Manufacturing: Foreign-Multinational Share of Export Value, by Exporting Country and SITC Division**

*Notes:* Percent of each group's export value in each country shipped by foreign MNEs. SITC Rev.3 divisions; HS6 products without an SITC correspondence are excluded.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 131: Manufacturing: Exports by Exporting Country and SITC Division**

| Sub-sector                                                                  | ARG   | CHL   | COL   | CRI   | DOM   | PER   | PRY   | SLV   | URY   | All   |
|-----------------------------------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| <i>Panel A: composition of the origin's exports, % (columns sum to 100)</i> |       |       |       |       |       |       |       |       |       |       |
| 68 Non-ferrous metals                                                       | 3.6   | 63.0  | 0.7   | 1.2   | 0.5   | 43.4  | 0.6   | 1.0   | 0.4   | 23.7  |
| 78 Road vehicles                                                            | 35.2  | 0.7   | 6.0   | 0.5   | 0.3   | 0.6   | 0.4   | 0.2   | 8.4   | 10.2  |
| 84 Apparel                                                                  | 0.3   | 0.1   | 6.0   | 1.7   | 13.8  | 14.2  | 5.2   | 50.2  | 1.5   | 6.4   |
| 77 Electrical machinery                                                     | 1.0   | 0.4   | 5.4   | 22.5  | 13.8  | 0.6   | 15.5  | 6.0   | 2.8   | 4.2   |
| 89 Misc. manufactures                                                       | 1.8   | 0.5   | 4.9   | 12.3  | 10.2  | 3.6   | 8.7   | 7.7   | 8.7   | 3.9   |
| 67 Iron & steel                                                             | 5.0   | 1.1   | 8.7   | 2.4   | 6.3   | 1.9   | 0.5   | 3.9   | 2.7   | 3.7   |
| 59 Chemical materials n.e.s.                                                | 9.3   | 0.5   | 5.5   | 2.4   | 0.3   | 0.9   | 3.7   | 0.3   | 4.6   | 3.7   |
| 87 Instruments                                                              | 0.6   | 0.1   | 0.8   | 26.9  | 19.0  | 0.2   | 1.4   | 0.1   | 0.4   | 3.3   |
| 52 Inorganic chemicals                                                      | 1.6   | 6.6   | 1.3   | 0.3   | 0.2   | 3.9   | 0.4   | 0.1   | 0.7   | 2.9   |
| 54 Pharmaceuticals                                                          | 4.4   | 0.5   | 4.7   | 4.0   | 4.1   | 0.6   | 4.8   | 3.2   | 5.3   | 2.8   |
| 57 Plastics, primary                                                        | 3.3   | 0.3   | 10.9  | 0.4   | 0.9   | 0.8   | 1.5   | 0.5   | 2.0   | 2.6   |
| 55 Essential oils, cosmetics                                                | 3.9   | 0.2   | 7.2   | 1.1   | 1.8   | 2.0   | 2.4   | 1.6   | 3.3   | 2.6   |
| 25 Pulp & paper                                                             | 0.6   | 8.3   | 0.0   | 0.2   | 0.3   | 0.0   | 0.1   | 0.2   | 0.2   | 2.6   |
| 65 Textile yarn & fabrics                                                   | 0.8   | 0.2   | 3.9   | 1.0   | 7.5   | 4.4   | 10.8  | 7.8   | 1.7   | 2.3   |
| 64 Paper                                                                    | 1.5   | 1.8   | 3.8   | 1.8   | 1.2   | 1.9   | 2.6   | 6.3   | 3.2   | 2.3   |
| 69 Metal manufactures                                                       | 1.7   | 1.3   | 4.0   | 2.0   | 1.2   | 2.1   | 1.2   | 1.4   | 0.3   | 1.9   |
| 61 Leather                                                                  | 3.9   | 0.1   | 1.5   | 0.1   | 0.2   | 0.2   | 12.7  | 0.1   | 11.7  | 1.7   |
| 58 Plastics, non-primary                                                    | 1.2   | 0.6   | 3.5   | 2.3   | 1.6   | 3.1   | 1.0   | 1.4   | 2.9   | 1.7   |
| 24 Cork & wood                                                              | 0.4   | 3.1   | 0.3   | 0.7   | 0.0   | 1.4   | 5.8   | 0.1   | 10.1  | 1.6   |
| 74 General industrial machinery                                             | 3.5   | 0.6   | 2.0   | 0.8   | 0.7   | 1.0   | 0.6   | 0.4   | 0.8   | 1.6   |
| 79 Other transport equipment                                                | 4.6   | 0.3   | 0.5   | 0.1   | 0.1   | 0.3   | 2.8   | 0.0   | 0.9   | 1.4   |
| 28 Metalliferous ores                                                       | 0.0   | 1.3   | 3.9   | 1.2   | 1.9   | 0.8   | 5.6   | 0.8   | 0.6   | 1.3   |
| 62 Rubber manufactures                                                      | 1.3   | 1.0   | 1.0   | 4.2   | 0.2   | 1.0   | 0.3   | 0.0   | 4.0   | 1.3   |
| 66 Non-metallic mineral mfg                                                 | 0.6   | 0.2   | 3.4   | 1.9   | 0.5   | 2.5   | 1.6   | 1.0   | 0.4   | 1.2   |
| 63 Cork & wood manufactures                                                 | 0.4   | 2.8   | 0.2   | 0.4   | 0.1   | 0.3   | 1.7   | 0.1   | 3.1   | 1.1   |
| 51 Organic chemicals                                                        | 2.2   | 0.7   | 1.6   | 0.1   | 0.5   | 0.3   | 0.5   | 0.1   | 0.6   | 1.0   |
| 72 Specialised machinery                                                    | 1.2   | 0.7   | 1.2   | 0.6   | 0.5   | 1.7   | 1.1   | 0.5   | 0.5   | 1.0   |
| 26 Textile fibres                                                           | 1.5   | 0.1   | 0.1   | 0.0   | 0.5   | 1.6   | 1.2   | 0.2   | 10.6  | 0.9   |
| 56 Fertilisers                                                              | 0.3   | 1.8   | 1.0   | 0.6   | 0.6   | 0.3   | 0.1   | 0.5   | 0.9   | 0.9   |
| 53 Dyeing materials                                                         | 0.7   | 0.1   | 1.4   | 1.3   | 0.4   | 2.0   | 0.4   | 1.0   | 1.8   | 0.8   |
| 71 Power machinery                                                          | 2.1   | 0.2   | 0.7   | 0.2   | 0.5   | 0.6   | 0.3   | 0.1   | 0.2   | 0.8   |
| 85 Footwear                                                                 | 0.1   | 0.0   | 0.4   | 0.0   | 7.2   | 0.3   | 1.9   | 1.1   | 0.1   | 0.6   |
| 82 Furniture                                                                | 0.2   | 0.0   | 1.2   | 0.6   | 0.3   | 0.3   | 1.1   | 1.1   | 3.8   | 0.5   |
| Other                                                                       | 1.4   | 0.8   | 2.3   | 4.0   | 2.6   | 1.0   | 1.4   | 0.7   | 1.1   | 1.5   |
| All sub-sectors                                                             | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 |
| <i>Panel B: foreign-MNE share of export value, %</i>                        |       |       |       |       |       |       |       |       |       |       |
| 68 Non-ferrous metals                                                       | 5.1   | 42.1  | 46.9  | 54.5  | 15.6  | 30.6  | 1.8   | 36.6  | 92.2  | 38.6  |
| 78 Road vehicles                                                            | 84.3  | 83.4  | 89.8  | 56.5  | 45.5  | 16.9  | 18.9  | 12.3  | 85.1  | 84.0  |
| 84 Apparel                                                                  | 41.9  | 25.0  | 47.6  | 81.7  | 52.5  | 27.7  | 19.0  | 63.1  | 60.6  | 51.5  |
| 77 Electrical machinery                                                     | 46.6  | 75.8  | 58.7  | 87.9  | 96.0  | 47.8  | 96.7  | 18.1  | 90.1  | 74.7  |
| 89 Misc. manufactures                                                       | 41.2  | 41.4  | 36.9  | 33.8  | 66.7  | 45.9  | 15.2  | 36.8  | 38.7  | 41.0  |
| 67 Iron & steel                                                             | 31.4  | 12.7  | 92.3  | 85.8  | 86.4  | 5.3   | 68.5  | 54.6  | 12.8  | 55.9  |
| 59 Chemical materials n.e.s.                                                | 50.7  | 49.6  | 89.2  | 58.5  | 14.5  | 18.8  | 7.2   | 40.1  | 60.2  | 57.6  |
| 87 Instruments                                                              | 43.8  | 50.2  | 67.4  | 80.6  | 99.5  | 42.2  | 88.2  | 21.6  | 89.8  | 83.2  |
| 52 Inorganic chemicals                                                      | 84.3  | 3.3   | 61.6  | 47.3  | 29.1  | 45.7  | 23.2  | 41.9  | 92.3  | 24.7  |
| 54 Pharmaceuticals                                                          | 56.2  | 51.0  | 57.7  | 75.1  | 92.6  | 31.9  | 6.5   | 30.8  | 44.3  | 56.8  |
| 57 Plastics, primary                                                        | 77.9  | 92.4  | 46.0  | 30.5  | 19.6  | 28.2  | 28.0  | 4.9   | 75.7  | 56.4  |
| 55 Essential oils, cosmetics                                                | 63.2  | 60.1  | 65.3  | 76.8  | 43.0  | 33.2  | 26.6  | 77.8  | 97.2  | 62.6  |
| 25 Pulp & paper                                                             | 100.0 | 61.7  | 46.0  | 40.5  | 37.6  | 6.1   | 13.1  | 34.3  | 46.1  | 63.5  |
| 65 Textile yarn & fabrics                                                   | 26.4  | 44.2  | 26.3  | 42.3  | 89.3  | 28.5  | 28.1  | 46.0  | 34.1  | 41.8  |
| 64 Paper                                                                    | 68.3  | 8.5   | 55.3  | 66.1  | 54.4  | 30.8  | 1.8   | 70.4  | 80.5  | 48.7  |
| 69 Metal manufactures                                                       | 55.2  | 57.9  | 69.9  | 63.2  | 61.0  | 43.1  | 67.8  | 15.5  | 50.6  | 57.5  |
| 61 Leather                                                                  | 33.7  | 26.5  | 26.7  | 44.2  | 13.0  | 17.5  | 22.9  | 1.9   | 89.0  | 42.0  |
| 58 Plastics, non-primary                                                    | 39.7  | 51.6  | 54.9  | 79.1  | 33.3  | 30.7  | 51.7  | 57.5  | 75.1  | 49.8  |
| 24 Cork & wood                                                              | 43.6  | 22.7  | 32.3  | 5.0   | 16.0  | 7.1   | 36.4  | 6.9   | 13.4  | 20.9  |
| 74 General industrial machinery                                             | 45.4  | 73.0  | 41.1  | 50.8  | 49.3  | 38.3  | 49.2  | 20.9  | 53.6  | 47.2  |
| 79 Other transport equipment                                                | 31.2  | 24.3  | 11.2  | 17.2  | 11.4  | 51.4  | 88.9  | 50.0  | 53.5  | 31.8  |
| 28 Metalliferous ores                                                       | 31.3  | 12.5  | 32.7  | 10.3  | 21.2  | 23.8  | 7.6   | 1.1   | 78.7  | 21.9  |
| 62 Rubber manufactures                                                      | 47.8  | 97.8  | 82.4  | 49.5  | 67.5  | 1.7   | 1.6   | 13.0  | 28.0  | 57.1  |
| 66 Non-metallic mineral mfg                                                 | 62.3  | 29.5  | 69.7  | 85.9  | 44.1  | 19.9  | 87.8  | 2.1   | 44.9  | 53.7  |
| 63 Cork & wood manufactures                                                 | 77.6  | 35.6  | 25.9  | 12.4  | 48.2  | 21.5  | 4.5   | 44.3  | 98.9  | 42.2  |
| 51 Organic chemicals                                                        | 51.0  | 85.2  | 39.5  | 53.7  | 11.7  | 22.5  | 2.8   | 62.8  | 91.2  | 53.6  |
| 72 Specialised machinery                                                    | 35.4  | 58.4  | 37.5  | 43.5  | 57.4  | 32.1  | 12.2  | 37.4  | 51.2  | 41.0  |
| 26 Textile fibres                                                           | 50.0  | 52.6  | 69.5  | 71.5  | 52.6  | 53.5  | 9.8   | 12.4  | 49.9  | 49.9  |
| 56 Fertilisers                                                              | 25.5  | 1.8   | 54.0  | 82.3  | 0.4   | 72.6  | 33.2  | 93.5  | 1.5   | 21.6  |
| 53 Dyeing materials                                                         | 38.2  | 63.3  | 28.8  | 76.4  | 46.5  | 40.8  | 37.3  | 84.0  | 96.3  | 49.2  |
| 71 Power machinery                                                          | 78.9  | 72.3  | 38.0  | 12.7  | 71.8  | 16.4  | 56.1  | 26.8  | 7.6   | 66.1  |
| 85 Footwear                                                                 | 44.0  | 55.3  | 24.2  | 15.9  | 64.4  | 15.6  | 84.5  | 6.4   | 42.7  | 50.1  |
| 82 Furniture                                                                | 30.7  | 24.3  | 19.5  | 61.7  | 28.0  | 10.9  | 4.3   | 30.7  | 90.5  | 40.3  |
| Other                                                                       | 31.5  | 23.3  | 30.8  | 80.1  | 26.6  | 30.6  | 48.2  | 57.1  | 39.0  | 40.4  |
| All sub-sectors                                                             | 60.4  | 40.3  | 57.4  | 70.2  | 73.2  | 30.6  | 37.9  | 52.4  | 60.2  | 51.7  |

*Notes:* Panel A: composition of each country's exports in the sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs. SITC Rev. 3 divisions; HS6 products without an SITC correspondence are excluded.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 132: Manufacturing: Exports by SITC Division and Parent Country**

| SITC Rev. 3 division            | USA  | GBR  | DEU  | JPN  | PAN  | CAN  | NLD  | FRA  | BRA | MEX  | Other | Domestic |
|---------------------------------|------|------|------|------|------|------|------|------|-----|------|-------|----------|
| 68 Non-ferrous metals           | 2.2  | 21.9 | 0.0  | 0.0  | 0.0  | 5.7  | 0.0  | 0.3  | 3.2 | 0.1  | 5.1   | 2.2      |
| 78 Road vehicles                | 8.6  | 0.1  | 26.4 | 24.7 | 0.0  | 0.3  | 10.8 | 9.8  | 0.9 | 0.0  | 2.4   | 0.7      |
| 84 Apparel                      | 32.4 | 0.2  | 0.5  | 0.4  | 0.2  | 1.3  | 0.0  | 0.3  | 0.1 | 0.1  | 16.0  | 1.0      |
| 77 Electrical machinery         | 33.4 | 1.0  | 1.5  | 7.8  | 0.2  | 0.1  | 0.7  | 4.4  | 0.1 | 3.7  | 21.8  | 0.9      |
| 89 Misc. manufactures           | 10.5 | 5.5  | 2.3  | 2.0  | 1.7  | 4.3  | 0.3  | 0.9  | 1.1 | 1.0  | 11.4  | 18.9     |
| 67 Iron & steel                 | 0.4  | 0.2  | 0.2  | 0.0  | 3.2  | 0.0  | 5.1  | 0.1  | 2.5 | 2.1  | 42.0  | 7.9      |
| 59 Chemical materials n.e.s.    | 16.2 | 3.3  | 13.1 | 0.7  | 0.6  | 0.3  | 0.7  | 0.5  | 0.7 | 0.1  | 21.4  | 8.5      |
| 87 Instruments                  | 52.9 | 5.2  | 9.2  | 2.4  | 0.1  | 0.2  | 0.3  | 0.6  | 0.0 | 0.0  | 12.2  | 0.8      |
| 52 Inorganic chemicals          | 7.2  | 0.1  | 1.4  | 0.0  | 0.1  | 1.6  | 0.8  | 0.1  | 1.6 | 0.0  | 11.6  | 30.7     |
| 54 Pharmaceuticals              | 10.1 | 1.0  | 6.4  | 0.0  | 1.0  | 0.2  | 0.2  | 2.3  | 2.0 | 0.2  | 33.5  | 6.7      |
| 57 Plastics, primary            | 16.5 | 0.1  | 0.9  | 0.9  | 0.8  | 0.3  | 3.7  | 0.7  | 3.6 | 20.7 | 8.3   | 5.2      |
| 55 Essential oils, cosmetics    | 14.0 | 18.3 | 4.1  | 2.1  | 2.9  | 0.1  | 0.3  | 0.6  | 5.0 | 0.4  | 15.0  | 11.8     |
| 25 Pulp & paper                 | 0.1  | 0.0  | 0.0  | 0.0  | 56.5 | 0.0  | 0.0  | 0.0  | 0.0 | 0.0  | 6.8   | 35.1     |
| 65 Textile yarn & fabrics       | 17.6 | 1.7  | 0.5  | 1.6  | 1.0  | 3.0  | 0.1  | 0.3  | 1.1 | 0.3  | 14.6  | 7.8      |
| 64 Paper                        | 21.3 | 0.7  | 0.1  | 0.0  | 2.5  | 0.0  | 0.0  | 0.0  | 0.1 | 0.6  | 23.4  | 22.6     |
| 69 Metal manufactures           | 12.4 | 1.1  | 1.0  | 2.8  | 1.1  | 0.6  | 2.6  | 2.1  | 0.5 | 1.5  | 31.8  | 4.7      |
| 61 Leather                      | 1.0  | 0.0  | 2.1  | 0.0  | 0.0  | 0.0  | 1.7  | 0.1  | 3.1 | 0.2  | 33.7  | 12.8     |
| 58 Plastics, non-primary        | 10.2 | 5.6  | 0.2  | 0.7  | 3.4  | 0.4  | 1.9  | 0.9  | 0.8 | 2.8  | 22.9  | 11.0     |
| 24 Cork & wood                  | 1.5  | 0.8  | 0.0  | 5.3  | 0.0  | 0.1  | 0.0  | 0.2  | 0.0 | 0.2  | 12.8  | 21.1     |
| 74 General industrial machinery | 9.9  | 3.4  | 2.6  | 1.7  | 0.4  | 1.9  | 1.9  | 1.5  | 1.4 | 0.7  | 21.6  | 10.4     |
| 79 Other transport equipment    | 0.3  | 0.1  | 0.1  | 0.9  | 0.0  | 0.1  | 0.0  | 0.1  | 0.6 | 0.0  | 29.6  | 29.2     |
| 28 Metalliferous ores           | 0.3  | 0.9  | 0.9  | 0.0  | 0.3  | 0.0  | 0.0  | 0.3  | 1.3 | 0.1  | 17.7  | 3.2      |
| 62 Rubber manufactures          | 29.6 | 0.3  | 1.3  | 12.5 | 0.0  | 0.1  | 0.3  | 3.0  | 0.1 | 0.0  | 10.0  | 1.1      |
| 66 Non-metallic mineral mfg     | 13.9 | 2.5  | 2.0  | 0.2  | 8.7  | 0.0  | 3.0  | 2.6  | 0.6 | 2.4  | 17.7  | 4.5      |
| 63 Cork & wood manufactures     | 4.5  | 0.1  | 0.3  | 0.0  | 0.2  | 2.9  | 0.0  | 0.5  | 0.3 | 0.1  | 33.1  | 41.3     |
| 51 Organic chemicals            | 11.5 | 5.9  | 2.4  | 0.4  | 0.6  | 14.6 | 1.1  | 0.2  | 0.3 | 0.1  | 16.4  | 8.0      |
| 72 Specialised machinery        | 5.9  | 2.6  | 1.7  | 0.7  | 0.4  | 2.9  | 0.6  | 1.4  | 0.3 | 0.4  | 24.1  | 7.0      |
| 26 Textile fibres               | 0.9  | 6.7  | 0.2  | 0.0  | 4.7  | 0.0  | 0.0  | 11.6 | 0.0 | 0.0  | 25.7  | 4.2      |
| 56 Fertilisers                  | 0.2  | 0.1  | 0.0  | 0.0  | 3.0  | 0.1  | 0.4  | 0.0  | 0.0 | 0.0  | 17.8  | 34.5     |
| 53 Dyeing materials             | 12.2 | 0.2  | 2.3  | 1.9  | 0.1  | 0.1  | 4.0  | 0.2  | 1.0 | 0.1  | 27.1  | 6.9      |
| 71 Power machinery              | 20.1 | 1.7  | 9.0  | 0.4  | 0.5  | 4.2  | 6.9  | 1.7  | 4.2 | 0.3  | 17.1  | 2.7      |
| 85 Footwear                     | 29.3 | 0.1  | 1.0  | 3.7  | 0.0  | 0.0  | 0.0  | 0.6  | 5.6 | 0.0  | 9.8   | 1.2      |
| 82 Furniture                    | 2.5  | 0.3  | 5.5  | 0.6  | 0.1  | 0.0  | 1.1  | 5.3  | 0.1 | 2.2  | 22.6  | 2.0      |
| Other                           | 16.7 | 3.2  | 1.1  | 0.7  | 0.7  | 0.2  | 1.1  | 0.6  | 0.3 | 1.4  | 14.4  | 4.0      |

*Notes:* Percent of each group's export value, by country of the ultimate parent of foreign MNEs; domestic MNEs shown for comparison. SITC Rev. 3 divisions; HS6 products without an SITC correspondence are excluded.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

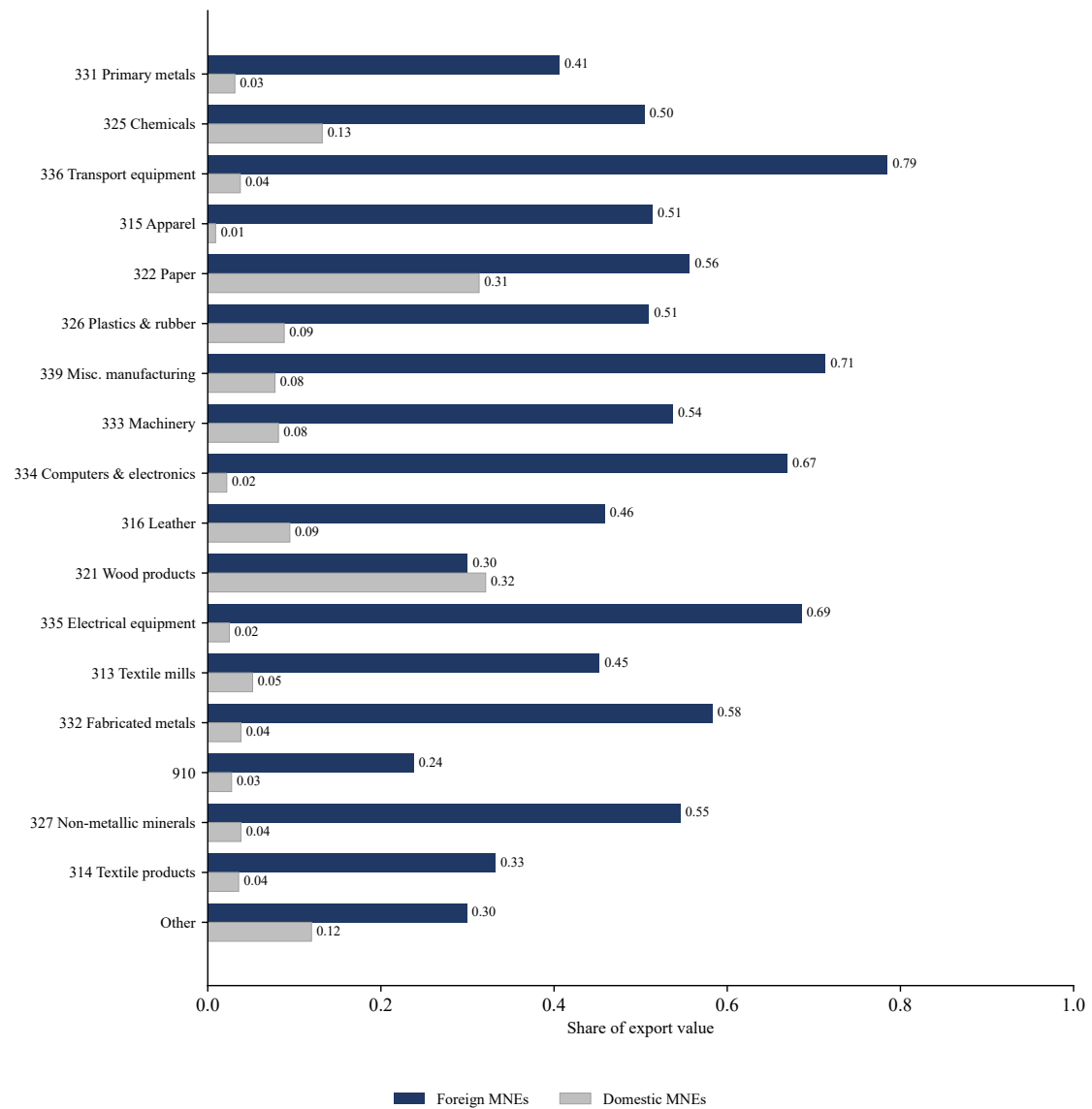
## 17.5 By NAICS Industry

**Table 133:** Manufacturing: Exports and Multinational Shares, by NAICS Industry

| NAICS 3-digit (product-based) | Total exports (US\$bn/yr) | MNE exports (US\$bn/yr) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|-------------------------------|---------------------------|-------------------------|-------------|--------------|------------------------------------|
| 331 Primary metals            | 17.3                      | 7.5                     | 40.6        | 3.1          | GBR 0.48, CAN 0.13, ZAF 0.09       |
| 325 Chemicals                 | 11.5                      | 7.4                     | 50.4        | 13.2         | USA 0.30, DEU 0.12, GBR 0.10       |
| 336 Transport equipment       | 9.7                       | 8.0                     | 78.5        | 3.8          | JPN 0.30, DEU 0.29, NLD 0.12       |
| 315 Apparel                   | 4.4                       | 2.3                     | 51.4        | 0.9          | USA 0.73, CHE 0.04, CAN 0.03       |
| 322 Paper                     | 2.9                       | 2.5                     | 55.7        | 31.3         | PAN 0.59, USA 0.14, CHL 0.09       |
| 326 Plastics & rubber         | 3.4                       | 2.0                     | 50.9        | 8.8          | USA 0.36, JPN 0.08, CAN 0.08       |
| 339 Misc. manufacturing       | 3.8                       | 3.0                     | 71.3        | 7.8          | USA 0.59, DEU 0.13, GBR 0.11       |
| 333 Machinery                 | 2.4                       | 1.5                     | 53.7        | 8.2          | USA 0.28, DEU 0.14, NLD 0.07       |
| 334 Computers & electronics   | 2.2                       | 1.5                     | 66.9        | 2.2          | USA 0.79, DEU 0.03, JPN 0.03       |
| 316 Leather                   | 2.0                       | 1.1                     | 45.8        | 9.5          | USA 0.19, BGR 0.13, CHL 0.12       |
| 321 Wood products             | 1.5                       | 0.9                     | 30.0        | 32.1         | BHS 0.25, JPN 0.13, USA 0.11       |
| 335 Electrical equipment      | 1.9                       | 1.4                     | 68.6        | 2.5          | IRL 0.29, USA 0.16, ITA 0.11       |
| 313 Textile mills             | 1.6                       | 0.9                     | 45.2        | 5.2          | USA 0.36, FRA 0.12, GBR 0.09       |
| 332 Fabricated metals         | 1.5                       | 0.9                     | 58.3        | 3.8          | USA 0.25, CYM 0.20, GBR 0.07       |
| 910                           | 1.0                       | 0.3                     | 23.8        | 2.8          | BGR 0.25, TUR 0.10, BRA 0.10       |
| 327 Non-metallic minerals     | 0.8                       | 0.5                     | 54.6        | 3.8          | USA 0.32, PAN 0.20, CYM 0.11       |
| 314 Textile products          | 0.4                       | 0.2                     | 33.2        | 3.6          | USA 0.36, PAN 0.16, GBR 0.12       |
| Other                         | 1.2                       | 0.5                     | 29.9        | 12.0         | USA 0.20, DNK 0.09, AUS 0.08       |
| All                           | 69.7                      | 42.2                    | 52.0        | 7.8          |                                    |

*Notes:* NAICS 3-digit industries assigned to HS6 products. Foreign and domestic: percent of the group's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

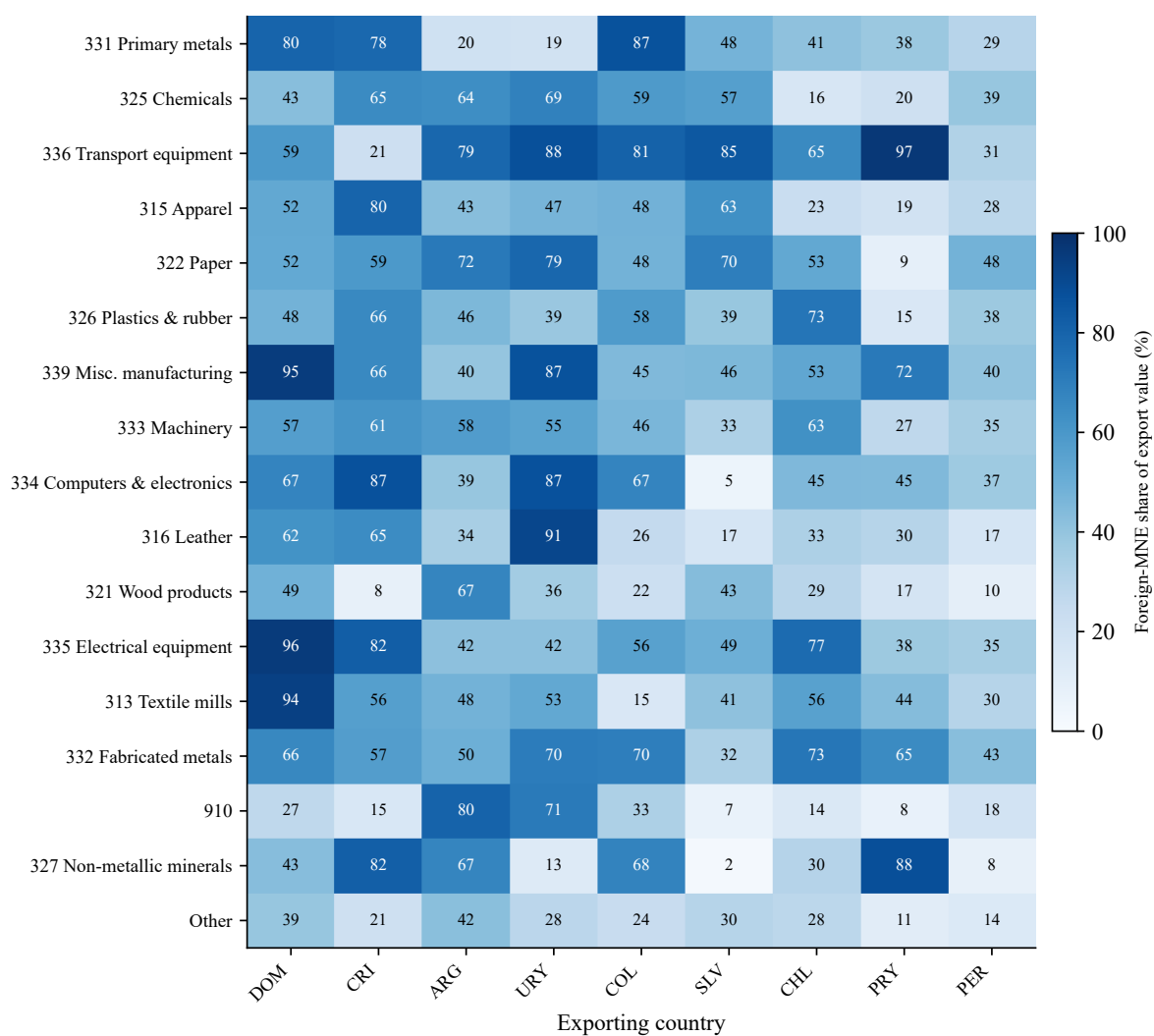
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 92: Manufacturing: Multinational Share of Export Value, by NAICS Industry**

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. NAICS 3-digit industries assigned to HS6 products.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 93:** Manufacturing: Foreign-Multinational Share of Export Value, by Exporting Country and NAICS Industry

*Notes:* Percent of each group's export value in each country shipped by foreign MNEs. NAICS 3-digit industries assigned to HS6 products.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 134: Manufacturing: Exports by Exporting Country and NAICS Industry**

| Sub-sector                                                                  | ARG   | CHL   | COL   | CRI   | DOM   | PER   | PRY   | SLV   | URY   | All   |
|-----------------------------------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| <i>Panel A: composition of the origin's exports, % (columns sum to 100)</i> |       |       |       |       |       |       |       |       |       |       |
| 331 Primary metals                                                          | 9.1   | 64.5  | 9.6   | 3.7   | 6.8   | 46.8  | 1.1   | 4.6   | 3.0   | 28.2  |
| 325 Chemicals                                                               | 21.5  | 11.0  | 33.6  | 9.2   | 6.2   | 11.0  | 16.6  | 8.0   | 18.1  | 16.3  |
| 336 Transport equipment                                                     | 41.9  | 1.0   | 6.5   | 0.4   | 0.5   | 1.0   | 18.3  | 0.9   | 10.5  | 11.9  |
| 315 Apparel                                                                 | 0.3   | 0.1   | 6.0   | 1.4   | 13.7  | 14.5  | 5.2   | 50.4  | 1.1   | 6.5   |
| 322 Paper                                                                   | 2.0   | 10.1  | 3.9   | 1.7   | 1.5   | 1.2   | 2.9   | 6.5   | 2.9   | 4.8   |
| 326 Plastics & rubber                                                       | 3.3   | 1.7   | 6.8   | 8.6   | 5.9   | 6.3   | 8.8   | 7.1   | 14.2  | 4.8   |
| 339 Misc. manufacturing                                                     | 0.7   | 0.1   | 1.8   | 35.9  | 26.9  | 0.9   | 2.1   | 0.9   | 0.9   | 4.7   |
| 333 Machinery                                                               | 6.5   | 1.4   | 4.0   | 2.4   | 1.1   | 2.8   | 2.2   | 1.0   | 2.4   | 3.2   |
| 334 Computers & electronics                                                 | 1.1   | 0.5   | 1.7   | 20.6  | 5.6   | 0.6   | 1.2   | 5.1   | 1.7   | 2.9   |
| 316 Leather                                                                 | 4.4   | 0.1   | 2.3   | 0.3   | 7.4   | 0.5   | 15.6  | 1.3   | 15.4  | 2.5   |
| 321 Wood products                                                           | 0.6   | 5.9   | 0.3   | 0.6   | 0.1   | 1.8   | 3.2   | 0.1   | 10.7  | 2.5   |
| 335 Electrical equipment                                                    | 1.4   | 0.5   | 5.4   | 6.9   | 11.5  | 1.1   | 0.8   | 0.7   | 0.6   | 2.5   |
| 313 Textile mills                                                           | 1.3   | 0.2   | 2.4   | 0.6   | 6.4   | 4.7   | 3.3   | 5.6   | 9.2   | 2.2   |
| 332 Fabricated metals                                                       | 2.7   | 1.1   | 4.1   | 2.2   | 1.3   | 2.0   | 1.3   | 1.8   | 0.5   | 2.1   |
| 910                                                                         | 0.1   | 1.2   | 4.0   | 1.5   | 3.0   | 0.9   | 5.8   | 1.4   | 0.9   | 1.4   |
| 327 Non-metallic minerals                                                   | 0.5   | 0.2   | 3.5   | 2.1   | 0.6   | 1.5   | 1.7   | 1.0   | 0.3   | 1.1   |
| 314 Textile products                                                        | 0.2   | 0.0   | 1.3   | 0.4   | 0.4   | 0.8   | 7.9   | 1.7   | 1.3   | 0.6   |
| Other                                                                       | 2.3   | 0.2   | 3.0   | 1.4   | 0.7   | 1.6   | 2.0   | 2.0   | 6.4   | 1.7   |
| All sub-sectors                                                             | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 |
| <i>Panel B: foreign-MNE share of export value, %</i>                        |       |       |       |       |       |       |       |       |       |       |
| 331 Primary metals                                                          | 20.3  | 41.3  | 87.1  | 78.4  | 80.1  | 29.5  | 37.9  | 47.7  | 18.9  | 40.6  |
| 325 Chemicals                                                               | 63.9  | 16.1  | 58.9  | 65.1  | 42.7  | 39.4  | 20.3  | 56.6  | 68.7  | 50.4  |
| 336 Transport equipment                                                     | 78.7  | 65.4  | 80.6  | 21.2  | 59.4  | 31.4  | 96.8  | 84.7  | 87.7  | 78.5  |
| 315 Apparel                                                                 | 42.7  | 23.1  | 48.1  | 79.7  | 52.5  | 27.8  | 19.1  | 63.1  | 47.2  | 51.4  |
| 322 Paper                                                                   | 71.7  | 52.6  | 47.8  | 59.2  | 52.1  | 48.0  | 8.7   | 69.8  | 78.6  | 55.7  |
| 326 Plastics & rubber                                                       | 45.5  | 73.3  | 58.4  | 65.9  | 47.8  | 37.6  | 15.0  | 38.9  | 38.6  | 50.9  |
| 339 Misc. manufacturing                                                     | 39.7  | 53.4  | 44.7  | 66.0  | 95.1  | 40.4  | 71.7  | 45.6  | 87.0  | 71.3  |
| 333 Machinery                                                               | 57.7  | 62.8  | 46.4  | 61.2  | 56.9  | 35.1  | 27.2  | 32.7  | 55.1  | 53.7  |
| 334 Computers & electronics                                                 | 39.2  | 45.3  | 67.3  | 87.2  | 67.2  | 37.4  | 44.6  | 5.2   | 87.3  | 66.9  |
| 316 Leather                                                                 | 34.2  | 32.7  | 25.5  | 64.6  | 61.8  | 16.9  | 29.7  | 16.9  | 91.2  | 45.8  |
| 321 Wood products                                                           | 67.0  | 29.0  | 21.7  | 7.8   | 48.7  | 9.6   | 17.4  | 43.4  | 35.6  | 30.0  |
| 335 Electrical equipment                                                    | 41.6  | 77.1  | 55.6  | 82.3  | 95.6  | 34.8  | 37.6  | 49.3  | 41.6  | 68.6  |
| 313 Textile mills                                                           | 47.7  | 55.5  | 15.1  | 56.4  | 93.5  | 29.8  | 43.7  | 41.1  | 52.6  | 45.2  |
| 332 Fabricated metals                                                       | 49.5  | 73.0  | 69.8  | 57.4  | 66.2  | 42.9  | 65.1  | 32.5  | 70.4  | 58.3  |
| 910                                                                         | 80.3  | 14.4  | 32.7  | 15.2  | 27.3  | 18.4  | 7.7   | 6.7   | 71.3  | 23.8  |
| 327 Non-metallic minerals                                                   | 67.0  | 30.2  | 67.6  | 81.7  | 43.1  | 8.0   | 88.2  | 2.1   | 13.0  | 54.6  |
| 314 Textile products                                                        | 31.1  | 46.1  | 30.6  | 37.3  | 13.5  | 46.3  | 21.3  | 40.3  | 30.3  | 33.2  |
| Other                                                                       | 42.5  | 27.9  | 23.6  | 20.5  | 39.1  | 13.8  | 11.3  | 30.0  | 28.2  | 29.9  |
| All sub-sectors                                                             | 62.1  | 40.3  | 57.5  | 69.8  | 73.2  | 30.8  | 38.4  | 52.5  | 60.2  | 52.0  |

Notes: Panel A: composition of each country's exports in the sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs. NAICS 3-digit industries assigned to HS6 products.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 135:** Manufacturing: Exports by NAICS Industry and Parent Country

| NAICS 3-digit (product-based) | USA  | GBR  | DEU  | JPN  | PAN  | CAN | NLD | FRA | BRA | MEX | Other | Domestic |
|-------------------------------|------|------|------|------|------|-----|-----|-----|-----|-----|-------|----------|
| 331 Primary metals            | 1.9  | 18.8 | 0.1  | 0.0  | 0.4  | 5.0 | 0.7 | 0.4 | 3.1 | 0.3 | 9.9   | 3.1      |
| 325 Chemicals                 | 13.4 | 4.4  | 5.3  | 0.8  | 1.1  | 1.4 | 1.3 | 0.8 | 2.3 | 3.4 | 16.3  | 13.2     |
| 336 Transport equipment       | 7.3  | 0.2  | 22.4 | 23.3 | 0.0  | 0.2 | 9.4 | 8.6 | 0.8 | 0.0 | 6.4   | 3.8      |
| 315 Apparel                   | 32.4 | 0.2  | 0.5  | 0.4  | 0.2  | 1.3 | 0.0 | 0.3 | 0.1 | 0.1 | 15.9  | 0.9      |
| 322 Paper                     | 7.9  | 0.9  | 0.0  | 0.1  | 32.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.2 | 14.4  | 31.3     |
| 326 Plastics & rubber         | 16.4 | 2.5  | 0.7  | 3.7  | 2.6  | 3.7 | 1.0 | 1.7 | 1.2 | 1.8 | 15.7  | 8.8      |
| 339 Misc. manufacturing       | 39.1 | 7.3  | 8.3  | 2.2  | 0.2  | 0.2 | 0.1 | 0.4 | 0.2 | 0.0 | 13.4  | 7.8      |
| 333 Machinery                 | 13.1 | 2.8  | 6.6  | 2.3  | 0.4  | 2.4 | 3.4 | 1.5 | 1.9 | 0.5 | 18.8  | 8.2      |
| 334 Computers & electronics   | 51.5 | 1.0  | 2.2  | 1.8  | 0.2  | 0.1 | 0.6 | 1.0 | 0.1 | 0.7 | 7.6   | 2.2      |
| 316 Leather                   | 7.7  | 0.0  | 2.8  | 0.8  | 0.4  | 0.0 | 1.2 | 1.3 | 3.5 | 0.6 | 27.5  | 9.5      |
| 321 Wood products             | 3.0  | 0.5  | 0.1  | 3.4  | 0.1  | 1.4 | 0.0 | 0.3 | 0.1 | 0.1 | 20.8  | 32.1     |
| 335 Electrical equipment      | 10.0 | 1.5  | 0.8  | 3.8  | 0.6  | 0.5 | 1.0 | 6.5 | 0.1 | 7.0 | 36.6  | 2.5      |
| 313 Textile mills             | 15.0 | 3.6  | 0.6  | 1.5  | 2.3  | 3.3 | 0.0 | 4.8 | 1.2 | 0.3 | 12.5  | 5.2      |
| 332 Fabricated metals         | 11.1 | 2.9  | 0.9  | 2.6  | 0.6  | 1.3 | 2.8 | 1.3 | 0.4 | 0.6 | 33.7  | 3.8      |
| 910                           | 0.8  | 0.8  | 1.1  | 0.0  | 0.5  | 0.0 | 0.0 | 0.1 | 1.2 | 0.1 | 19.1  | 2.8      |
| 327 Non-metallic minerals     | 15.6 | 2.8  | 2.3  | 0.3  | 9.7  | 0.0 | 2.5 | 2.8 | 0.3 | 2.3 | 16.0  | 3.8      |
| 314 Textile products          | 7.1  | 2.3  | 0.2  | 0.2  | 3.2  | 0.0 | 0.9 | 1.8 | 0.1 | 0.1 | 17.3  | 3.6      |
| Other                         | 4.0  | 1.3  | 1.1  | 0.5  | 0.1  | 0.1 | 0.7 | 0.3 | 0.1 | 0.5 | 21.2  | 12.0     |

*Notes:* Percent of each group's export value, by country of the ultimate parent of foreign MNEs; domestic MNEs shown for comparison. NAICS 3-digit industries assigned to HS6 products.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

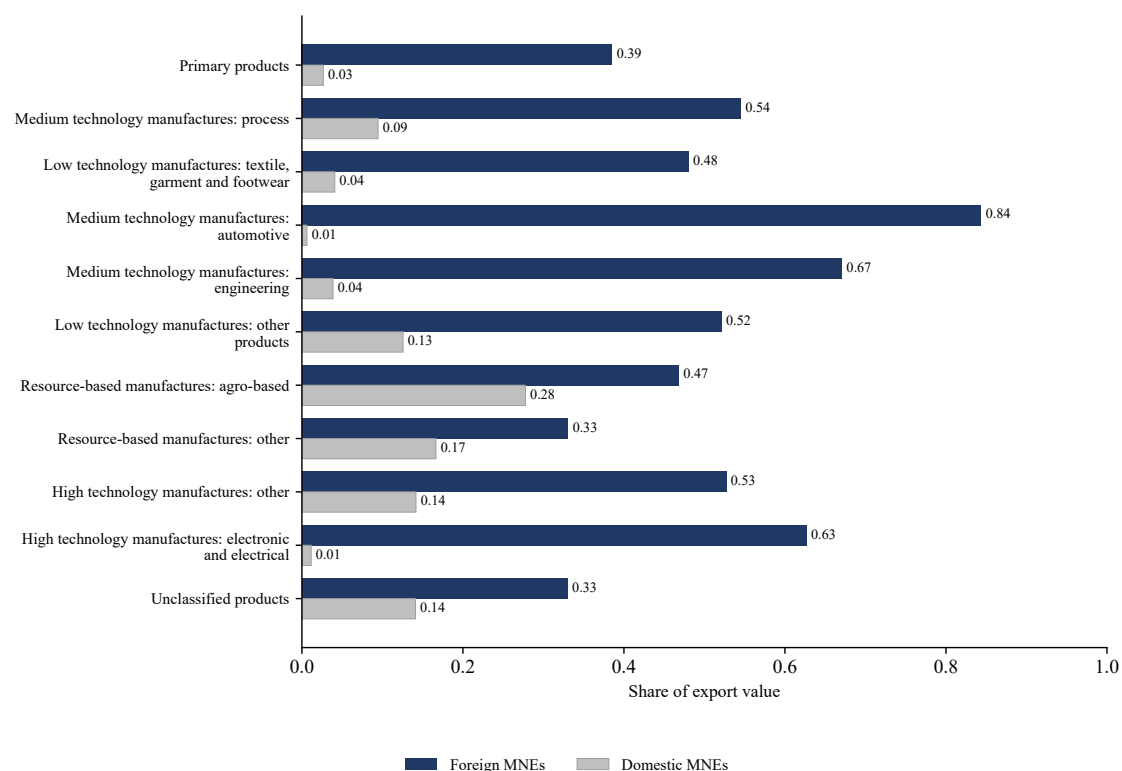
## 17.6 By Technology Class

**Table 136:** Manufacturing: Exports and Multinational Shares, by Technology Class

| Lall (2000) category                                       | Total exports (US\$bn/yr) | MNE exports (US\$bn/yr) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|------------------------------------------------------------|---------------------------|-------------------------|-------------|--------------|------------------------------------|
| Primary products                                           | 15.6                      | 6.3                     | 38.5        | 2.7          | GBR 0.57, CAN 0.15, BRA 0.08       |
| Medium technology manufactures: process                    | 10.8                      | 6.8                     | 54.5        | 9.5          | USA 0.22, ZAF 0.13, GBR 0.10       |
| Low technology manufactures: textile, garment and footwear | 8.1                       | 4.2                     | 48.0        | 4.1          | USA 0.59, URY 0.04, CAN 0.03       |
| Medium technology manufactures: automotive                 | 8.5                       | 7.2                     | 84.3        | 0.6          | DEU 0.32, JPN 0.30, NLD 0.13       |
| Medium technology manufactures: engineering                | 7.3                       | 5.2                     | 67.1        | 3.9          | USA 0.42, IRL 0.10, DEU 0.08       |
| Low technology manufactures: other products                | 6.3                       | 4.1                     | 52.1        | 12.6         | USA 0.29, PAN 0.09, GBR 0.06       |
| Resource-based manufactures: agro-based                    | 4.8                       | 3.4                     | 46.8        | 27.8         | PAN 0.44, USA 0.16, CHL 0.07       |
| Resource-based manufactures: other                         | 4.7                       | 2.4                     | 33.0        | 16.6         | USA 0.25, CHE 0.09, DEU 0.08       |
| High technology manufactures: other                        | 3.2                       | 2.2                     | 52.7        | 14.2         | USA 0.20, CHL 0.15, DEU 0.10       |
| High technology manufactures: electronic and electrical    | 2.2                       | 1.4                     | 62.7        | 1.1          | USA 0.76, JPN 0.05, DEU 0.03       |
| Unclassified products                                      | 0.4                       | 0.2                     | 33.0        | 14.1         | USA 0.41, BGR 0.20, ESP 0.07       |
| All                                                        | 71.7                      | 43.3                    | 51.7        | 7.8          |                                    |

*Notes:* Lall (2000) technology classes. Foreign and domestic: percent of the group's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

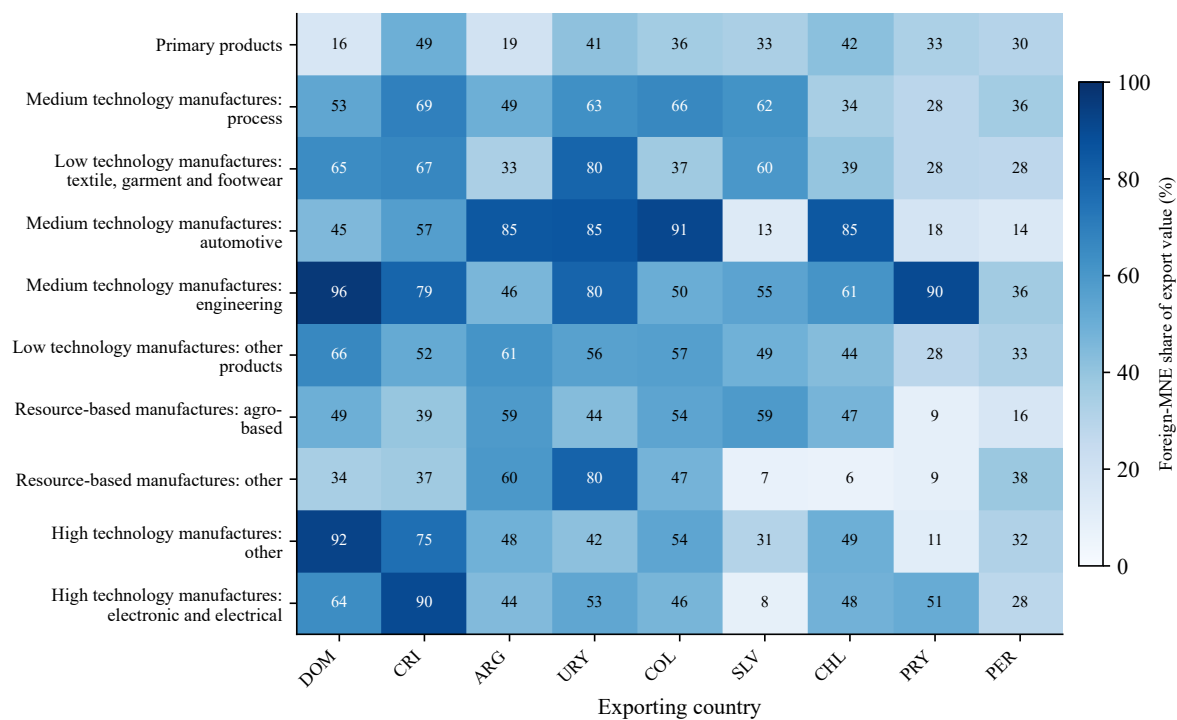
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 94: Manufacturing: Multinational Share of Export Value, by Technology Class**

*Notes:* Shares of manufacturing export value (HS 28–97 excluding 71), all available years pooled. Lall (2000) technology classes.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 95: Manufacturing: Foreign-Multinational Share of Export Value, by Exporting Country and Technology Class**

*Notes:* Percent of each group's export value in each country shipped by foreign MNEs. Lall (2000) technology classes.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 137: Manufacturing: Exports by Exporting Country and Technology Class**

| Sub-sector                                                                  | ARG   | CHL   | COL   | CRI   | DOM   | PER   | PRY   | SLV   | URY   | All   |
|-----------------------------------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| <i>Panel A: composition of the origin's exports, % (columns sum to 100)</i> |       |       |       |       |       |       |       |       |       |       |
| Primary products                                                            | 5.3   | 64.4  | 1.2   | 1.4   | 0.7   | 44.7  | 5.2   | 1.1   | 15.8  | 25.2  |
| Medium technology manufactures: process                                     | 21.0  | 4.9   | 37.4  | 8.2   | 8.9   | 8.2   | 9.2   | 6.5   | 16.3  | 14.5  |
| Low technology manufactures: textile, garment and footwear                  | 5.1   | 0.3   | 11.9  | 2.9   | 28.8  | 19.0  | 29.8  | 58.5  | 15.0  | 11.0  |
| Medium technology manufactures: automotive                                  | 35.0  | 0.7   | 5.9   | 0.4   | 0.3   | 0.6   | 0.4   | 0.2   | 8.3   | 10.1  |
| Medium technology manufactures: engineering                                 | 9.2   | 2.1   | 8.1   | 36.0  | 30.5  | 3.7   | 21.2  | 2.2   | 4.7   | 9.1   |
| Low technology manufactures: other products                                 | 5.2   | 2.4   | 13.0  | 19.2  | 15.0  | 8.8   | 12.5  | 18.2  | 14.4  | 8.5   |
| Resource-based manufactures: agro-based                                     | 3.9   | 15.6  | 3.5   | 6.0   | 1.4   | 3.4   | 6.8   | 1.5   | 14.7  | 7.4   |
| Resource-based manufactures: other                                          | 6.0   | 8.3   | 9.3   | 2.6   | 3.6   | 8.9   | 8.4   | 2.2   | 4.0   | 6.8   |
| High technology manufactures: other                                         | 7.8   | 0.6   | 5.5   | 5.1   | 5.1   | 0.8   | 5.3   | 3.3   | 5.7   | 4.0   |
| High technology manufactures: electronic and electrical                     | 0.9   | 0.6   | 3.4   | 17.7  | 5.3   | 1.1   | 1.2   | 5.4   | 0.8   | 2.9   |
| Unclassified products                                                       | 0.6   | 0.1   | 1.0   | 0.4   | 0.3   | 0.8   | 0.1   | 1.0   | 0.3   | 0.5   |
| All sub-sectors                                                             | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 |
| <i>Panel B: foreign-MNE share of export value, %</i>                        |       |       |       |       |       |       |       |       |       |       |
| Primary products                                                            | 19.2  | 42.0  | 35.9  | 49.4  | 16.3  | 30.4  | 33.1  | 33.4  | 41.4  | 38.5  |
| Medium technology manufactures: process                                     | 49.4  | 34.3  | 66.1  | 68.8  | 53.3  | 35.8  | 28.4  | 61.7  | 63.2  | 54.5  |
| Low technology manufactures: textile, garment and footwear                  | 33.3  | 39.5  | 37.1  | 66.6  | 64.6  | 27.5  | 28.3  | 60.1  | 79.7  | 48.0  |
| Medium technology manufactures: automotive                                  | 84.5  | 84.7  | 90.7  | 56.7  | 45.3  | 14.2  | 17.6  | 13.1  | 85.5  | 84.3  |
| Medium technology manufactures: engineering                                 | 46.4  | 61.1  | 50.3  | 79.3  | 96.3  | 36.2  | 89.6  | 54.6  | 79.7  | 67.1  |
| Low technology manufactures: other products                                 | 61.4  | 43.7  | 56.6  | 51.8  | 66.4  | 32.6  | 28.1  | 48.5  | 55.6  | 52.1  |
| Resource-based manufactures: agro-based                                     | 58.9  | 47.2  | 54.2  | 39.1  | 49.4  | 16.3  | 8.8   | 58.6  | 43.7  | 46.8  |
| Resource-based manufactures: other                                          | 59.6  | 6.3   | 47.4  | 36.7  | 34.3  | 38.1  | 8.7   | 7.1   | 80.2  | 33.0  |
| High technology manufactures: other                                         | 48.5  | 49.5  | 53.9  | 75.4  | 92.3  | 32.0  | 10.8  | 30.7  | 41.5  | 52.7  |
| High technology manufactures: electronic and electrical                     | 44.2  | 48.2  | 46.5  | 89.7  | 64.1  | 28.1  | 51.0  | 8.4   | 53.5  | 62.7  |
| Unclassified products                                                       | 47.3  | 39.9  | 29.8  | 27.1  | 56.1  | 7.7   | 23.9  | 38.1  | 34.3  | 33.0  |
| All sub-sectors                                                             | 60.4  | 40.3  | 57.4  | 70.2  | 73.2  | 30.6  | 37.9  | 52.4  | 60.2  | 51.7  |

*Notes:* Panel A: composition of each country's exports in the sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs. Lall (2000) technology classes.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 138: Manufacturing: Exports by Technology Class and Parent Country**

| Lall (2000) category                                       | USA  | GBR  | DEU  | JPN  | PAN  | CAN | NLD  | FRA | BRA | MEX | Other | Domestic |
|------------------------------------------------------------|------|------|------|------|------|-----|------|-----|-----|-----|-------|----------|
| Primary products                                           | 2.1  | 20.9 | 0.0  | 0.3  | 0.2  | 5.4 | 0.0  | 0.8 | 3.0 | 0.0 | 5.8   | 2.7      |
| Medium technology manufactures: process                    | 11.1 | 5.2  | 4.1  | 1.1  | 1.4  | 1.2 | 1.5  | 0.4 | 1.8 | 4.2 | 22.6  | 9.5      |
| Low technology manufactures: textile, garment and footwear | 24.3 | 0.5  | 0.8  | 0.6  | 0.4  | 1.4 | 0.3  | 0.3 | 1.1 | 0.1 | 18.2  | 4.1      |
| Medium technology manufactures: automotive                 | 8.6  | 0.0  | 26.6 | 24.9 | 0.0  | 0.3 | 10.9 | 9.9 | 0.8 | 0.0 | 2.2   | 0.6      |
| Medium technology manufactures: engineering                | 25.6 | 3.1  | 4.9  | 4.1  | 0.2  | 1.0 | 1.2  | 2.6 | 0.7 | 1.9 | 21.7  | 3.9      |
| Low technology manufactures: other products                | 13.7 | 2.8  | 1.6  | 1.5  | 4.0  | 2.1 | 2.6  | 1.2 | 1.7 | 2.0 | 18.9  | 12.6     |
| Resource-based manufactures: agro-based                    | 7.1  | 0.3  | 0.3  | 2.2  | 19.8 | 0.5 | 0.1  | 0.6 | 0.1 | 0.1 | 15.8  | 27.8     |
| Resource-based manufactures: other                         | 6.7  | 1.0  | 2.1  | 0.1  | 0.6  | 0.7 | 1.3  | 0.8 | 1.5 | 0.5 | 17.7  | 16.6     |
| High technology manufactures: other                        | 9.3  | 1.8  | 4.7  | 0.1  | 0.8  | 0.2 | 0.3  | 1.7 | 1.5 | 0.2 | 32.2  | 14.2     |
| High technology manufactures: electronic and electrical    | 45.6 | 0.8  | 2.0  | 2.8  | 0.3  | 0.5 | 0.9  | 0.6 | 0.4 | 0.7 | 8.0   | 1.1      |
| Unclassified products                                      | 9.9  | 0.7  | 1.3  | 1.3  | 0.2  | 0.3 | 0.1  | 0.1 | 0.2 | 0.4 | 18.6  | 14.1     |

*Notes:* Percent of each group's export value, by country of the ultimate parent of foreign MNEs; domestic MNEs shown for comparison. Lall (2000) technology classes.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 17.7 HS Section and End Use

**Table 139:** Manufacturing: Exports by HS Section and End Use

| HS section / BEC end use               | Intermediate | Consumption | Capital | Total |
|----------------------------------------|--------------|-------------|---------|-------|
| VI Chemicals                           | 7.1          | 4.3         | 0.0     | 11.4  |
| VII Plastics and rubber                | 5.2          | 0.4         | 0.0     | 5.6   |
| VIII Hides, skins, leather             | 1.5          | 0.1         | 0.0     | 1.6   |
| IX Wood and articles                   | 1.7          | 0.0         | 0.0     | 1.7   |
| X Pulp, paper                          | 2.7          | 0.7         | 0.0     | 3.4   |
| XI Textiles and apparel                | 2.1          | 4.7         | 0.0     | 6.9   |
| XII Footwear, headgear                 | 0.0          | 0.5         | 0.0     | 0.5   |
| XIII Stone, ceramics, glass            | 0.9          | 0.1         | 0.0     | 0.9   |
| XV Base metals                         | 19.2         | 0.1         | 0.2     | 19.5  |
| XVI Machinery and electrical equipment | 4.0          | 0.2         | 2.0     | 6.2   |
| XVII Transport equipment               | 1.4          | 3.1         | 5.2     | 9.7   |
| XVIII Optical, precision instruments   | 1.1          | 0.6         | 1.8     | 3.4   |
| XIX Arms                               | 0.0          | 0.0         | 0.0     | 0.0   |
| XX Miscellaneous manufactures          | 0.3          | 0.3         | 0.1     | 0.7   |
| XXI Works of art                       | 0.0          | 0.1         | 0.0     | 0.1   |
| Total                                  | 47.2         | 15.2        | 9.3     | 71.7  |

*Notes:* Export value by HS section and BEC Rev. 4 end use. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 140:** Manufacturing: Foreign-Multinational Share of Exports by HS Section and End Use

| HS section / BEC end use               | Intermediate | Consumption | Capital |
|----------------------------------------|--------------|-------------|---------|
| VI Chemicals                           | 40.0         | 64.8        | –       |
| VII Plastics and rubber                | 52.5         | 41.5        | –       |
| VIII Hides, skins, leather             | 40.4         | 30.4        | –       |
| IX Wood and articles                   | 29.7         | 29.8        | –       |
| X Pulp, paper                          | 53.7         | 56.7        | –       |
| XI Textiles and apparel                | 45.2         | 50.6        | –       |
| XII Footwear, headgear                 | 6.9          | 48.9        | –       |
| XIII Stone, ceramics, glass            | 46.6         | 85.6        | –       |
| XV Base metals                         | 41.0         | 42.4        | 53.3    |
| XVI Machinery and electrical equipment | 73.8         | 74.1        | 42.5    |
| XVII Transport equipment               | 80.2         | 88.1        | 70.5    |
| XVIII Optical, precision instruments   | 82.5         | 27.4        | 80.8    |
| XIX Arms                               | 1.5          | 1.2         | 13.0    |
| XX Miscellaneous manufactures          | 49.8         | 30.1        | 36.0    |
| XXI Works of art                       | –            | 57.9        | –       |

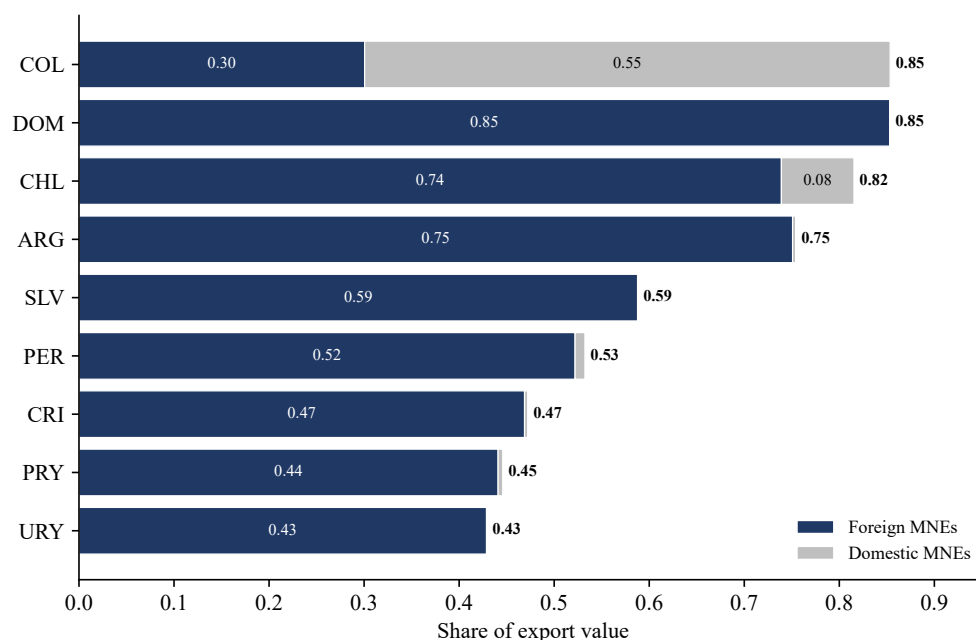
*Notes:* Percent of each cell's export value shipped by foreign MNEs.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## Part III

# Mining and Fuels

## 18 Fact 1: Multinationals Account for a Large Share of Exports



**Figure 96:** Mining and Fuels: Multinational Share of Export Value, by Exporting Country

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. The figure at the end of each bar is the total MNE share.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

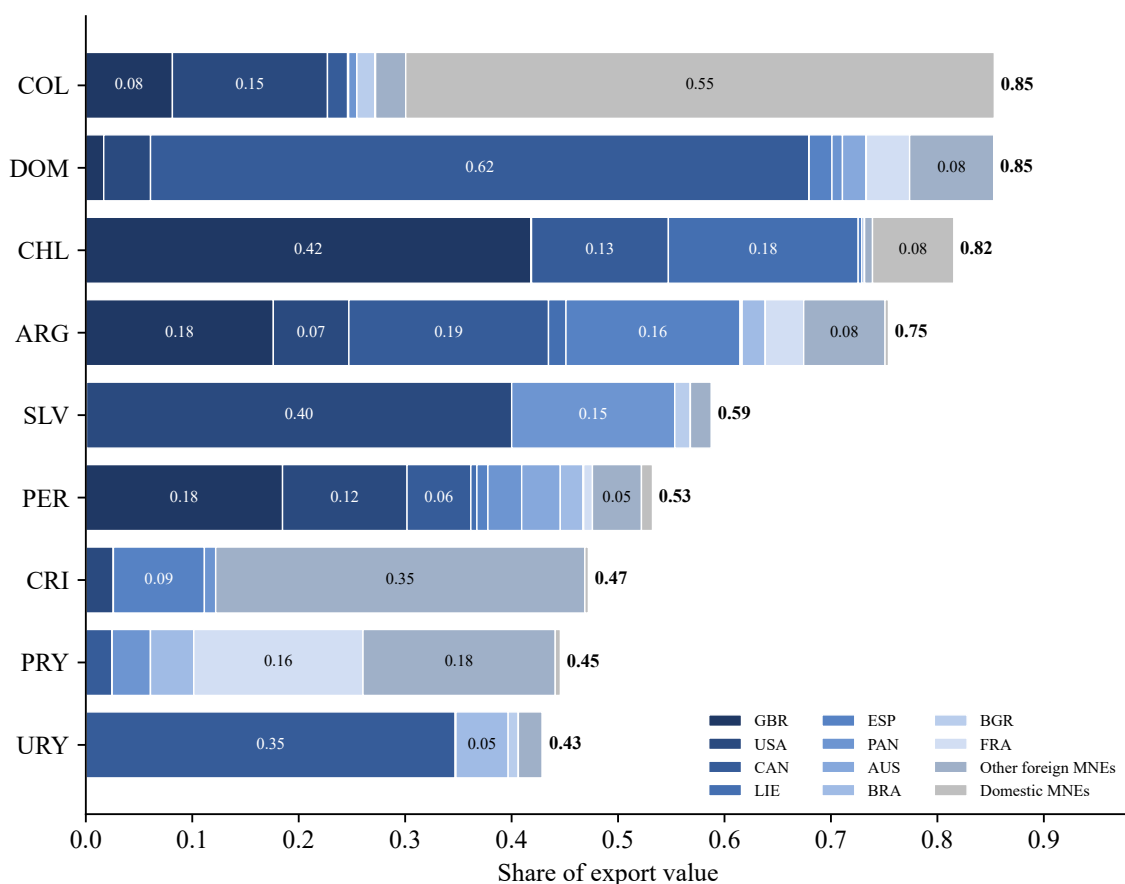
**Table 141:** Mining and Fuels: Multinational Share of Export Value, by Exporting Country

| Origin | Foreign | Domestic | All MNEs | Value (US\$bn/yr) |
|--------|---------|----------|----------|-------------------|
| COL    | 0.301   | 0.553    | 0.853    | 27.9              |
| DOM    | 0.853   | 0.000    | 0.853    | 2.0               |
| CHL    | 0.739   | 0.076    | 0.815    | 10.2              |
| ARG    | 0.751   | 0.003    | 0.754    | 6.6               |
| SLV    | 0.588   | 0.000    | 0.588    | 0.2               |
| PER    | 0.522   | 0.010    | 0.532    | 25.3              |
| CRI    | 0.469   | 0.003    | 0.472    | 0.1               |
| PRY    | 0.441   | 0.005    | 0.446    | 0.1               |
| URY    | 0.429   | 0.000    | 0.429    | 0.2               |
| All    | 0.492   | 0.245    | 0.737    | 72.4              |

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 18.1 By Parent Country



**Figure 97:** Mining and Fuels: Multinational Share of Export Value, by Exporting Country and Parent Country

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. Foreign MNEs split by country of the ultimate parent; the ten largest parent countries are shown separately.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 142:** Mining and Fuels: Multinational Share of Export Value, by Exporting Country and Parent Country

| Origin | GBR   | USA   | CAN   | LIE   | ESP   | PAN   | AUS   | BRA   | BGR   | FRA   | Other | Domestic | All MNEs | Value (US\$bn/yr) |
|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|----------|-------------------|
| COL    | 0.081 | 0.146 | 0.019 | 0.000 | 0.001 | 0.008 | 0.000 | 0.000 | 0.017 | 0.001 | 0.028 | 0.553    | 0.853    | 27.9              |
| DOM    | 0.017 | 0.044 | 0.619 | 0.000 | 0.021 | 0.010 | 0.022 | 0.000 | 0.000 | 0.041 | 0.079 | 0.000    | 0.853    | 2.0               |
| CHL    | 0.418 | 0.001 | 0.129 | 0.178 | 0.003 | 0.000 | 0.003 | 0.000 | 0.000 | 0.000 | 0.007 | 0.076    | 0.815    | 10.2              |
| ARG    | 0.176 | 0.071 | 0.188 | 0.016 | 0.163 | 0.001 | 0.001 | 0.022 | 0.000 | 0.036 | 0.076 | 0.003    | 0.754    | 6.6               |
| SLV    | 0.001 | 0.399 | 0.000 | 0.000 | 0.000 | 0.153 | 0.000 | 0.000 | 0.015 | 0.000 | 0.020 | 0.000    | 0.588    | 0.2               |
| PER    | 0.185 | 0.117 | 0.060 | 0.006 | 0.010 | 0.032 | 0.036 | 0.021 | 0.001 | 0.008 | 0.046 | 0.010    | 0.532    | 25.3              |
| CRI    | 0.000 | 0.026 | 0.000 | 0.000 | 0.085 | 0.011 | 0.000 | 0.000 | 0.000 | 0.000 | 0.347 | 0.003    | 0.472    | 0.1               |
| PRY    | 0.000 | 0.000 | 0.025 | 0.000 | 0.000 | 0.036 | 0.000 | 0.041 | 0.000 | 0.159 | 0.181 | 0.005    | 0.446    | 0.1               |
| URY    | 0.000 | 0.000 | 0.347 | 0.000 | 0.001 | 0.000 | 0.000 | 0.049 | 0.010 | 0.000 | 0.022 | 0.000    | 0.429    | 0.2               |

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. Other: all remaining foreign parents. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



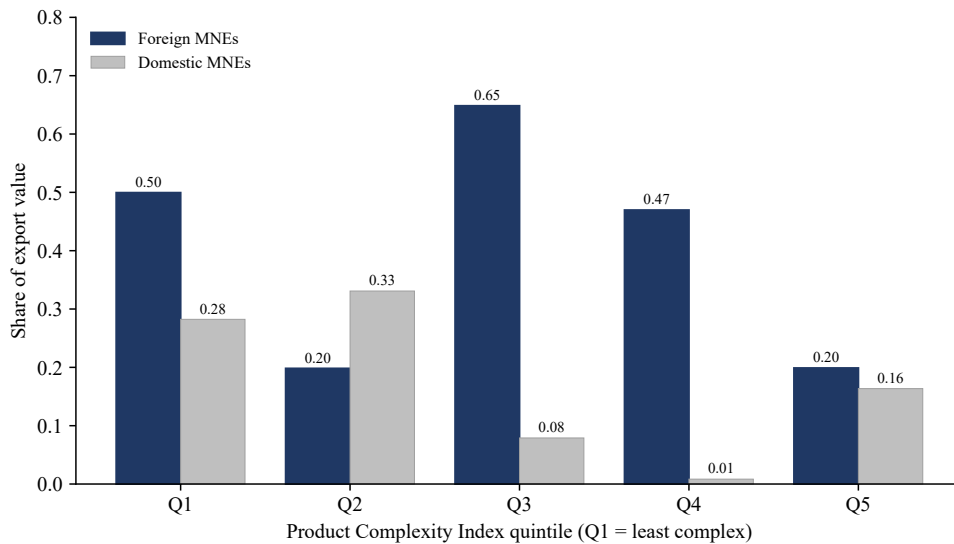
**Figure 98:** Mining and Fuels: Export Value by Exporting Country and Parent Country

*Notes:* Percent of each country's mining and fuels export value, by country of the ultimate parent of the exporter.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 19 Fact 2: Foreign Multinationals Specialize in Complex Products, Domestic Multinationals in Primary Goods

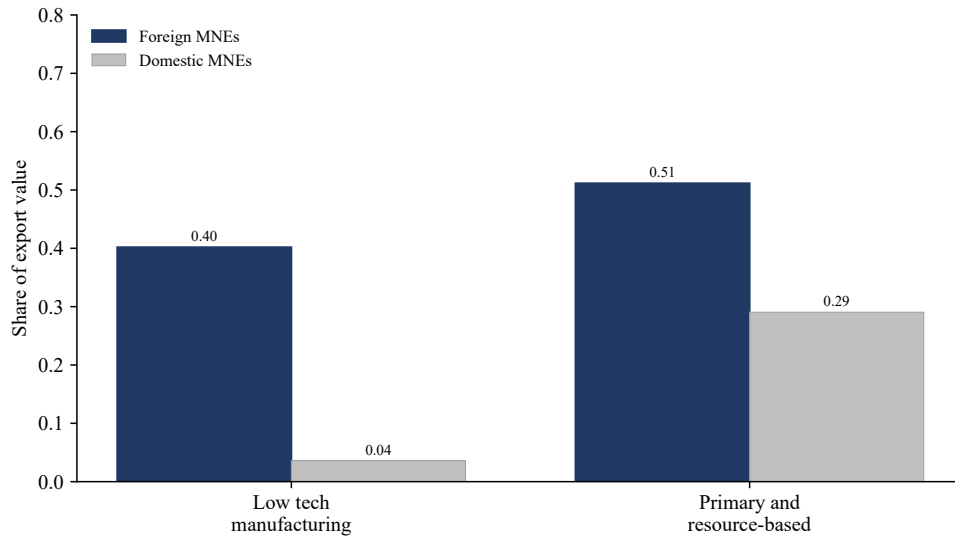
### 19.1 Product Complexity and Technology



**Figure 99:** Mining and Fuels: Multinational Share of Export Value, by Product Complexity Quintile

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. Quintiles of the Hausmann–Hidalgo Product Complexity Index across HS6 products (Q1 = least complex).

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

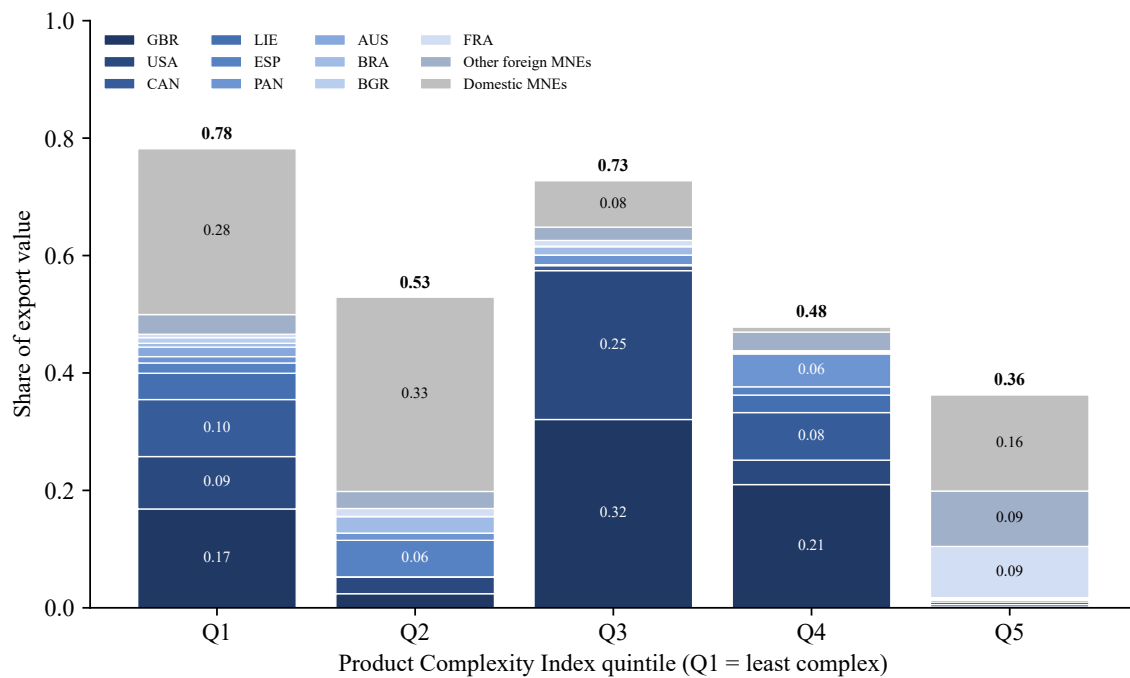


**Figure 100:** Mining and Fuels: Multinational Share of Export Value, by Technology Class

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. Lall (2000) technology classes: primary and resource-based products, and low-, medium- and high-technology manufactures.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 19.2 By Parent Country



**Figure 101:** Mining and Fuels: Multinational Share of Export Value, by Product Complexity Quintile and Parent Country

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. Foreign MNEs split by country of the ultimate parent. Quintiles of the Hausmann–Hidalgo Product Complexity Index across HS6 products (Q1 = least complex).

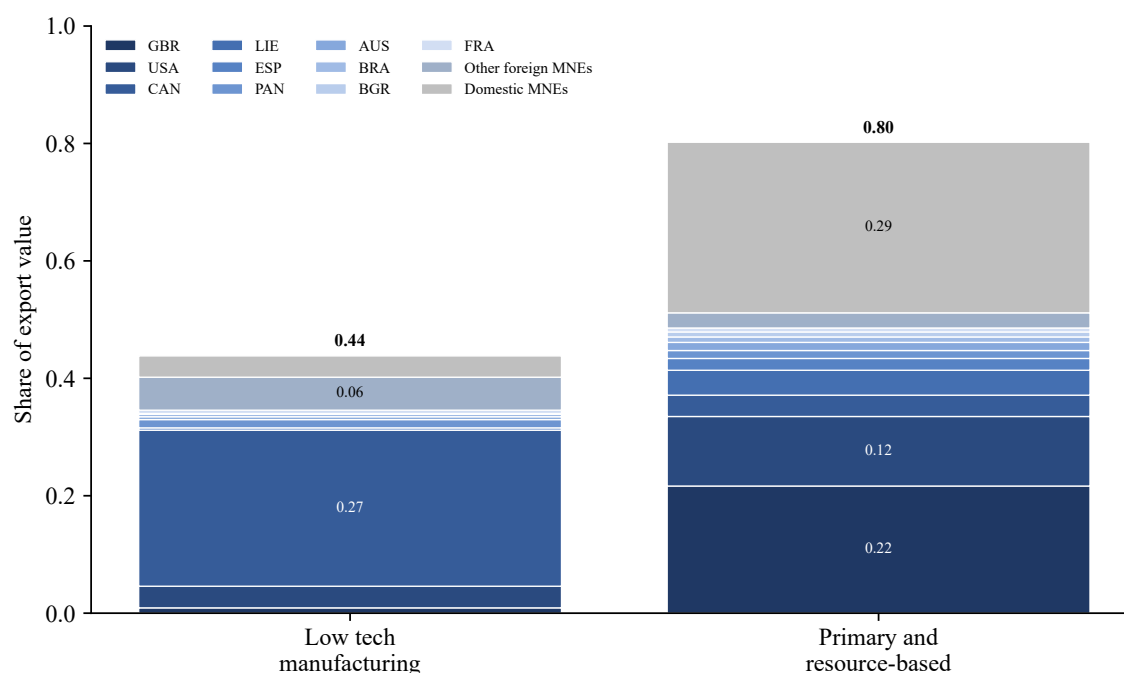
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 143:** Mining and Fuels: Multinational Share of Export Value, by Product Complexity Quintile and Parent Country

| PCI quintile | GBR   | USA   | CAN   | LIE   | ESP   | PAN   | AUS   | BRA   | BGR   | FRA   | Other | Domestic | All MNEs | Value (US\$bn/yr) |
|--------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|----------|-------------------|
| Q1           | 0.168 | 0.089 | 0.097 | 0.045 | 0.017 | 0.011 | 0.016 | 0.006 | 0.010 | 0.006 | 0.033 | 0.282    | 0.782    | 52.4              |
| Q2           | 0.024 | 0.028 | 0.000 | 0.000 | 0.062 | 0.012 | 0.000 | 0.028 | 0.000 | 0.014 | 0.029 | 0.331    | 0.529    | 6.1               |
| Q3           | 0.321 | 0.254 | 0.008 | 0.001 | 0.001 | 0.016 | 0.000 | 0.014 | 0.001 | 0.010 | 0.023 | 0.079    | 0.727    | 9.0               |
| Q4           | 0.210 | 0.042 | 0.081 | 0.030 | 0.014 | 0.056 | 0.002 | 0.002 | 0.000 | 0.001 | 0.032 | 0.008    | 0.478    | 4.0               |
| Q5           | 0.001 | 0.004 | 0.004 | 0.003 | 0.001 | 0.002 | 0.001 | 0.000 | 0.000 | 0.087 | 0.094 | 0.164    | 0.362    | 0.1               |

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. Other: all remaining foreign parents. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 102:** Mining and Fuels: Multinational Share of Export Value, by Technology Class and Parent Country

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. Foreign MNEs split by country of the ultimate parent. Lall (2000) technology classes: primary and resource-based products, and low-, medium- and high-technology manufactures.

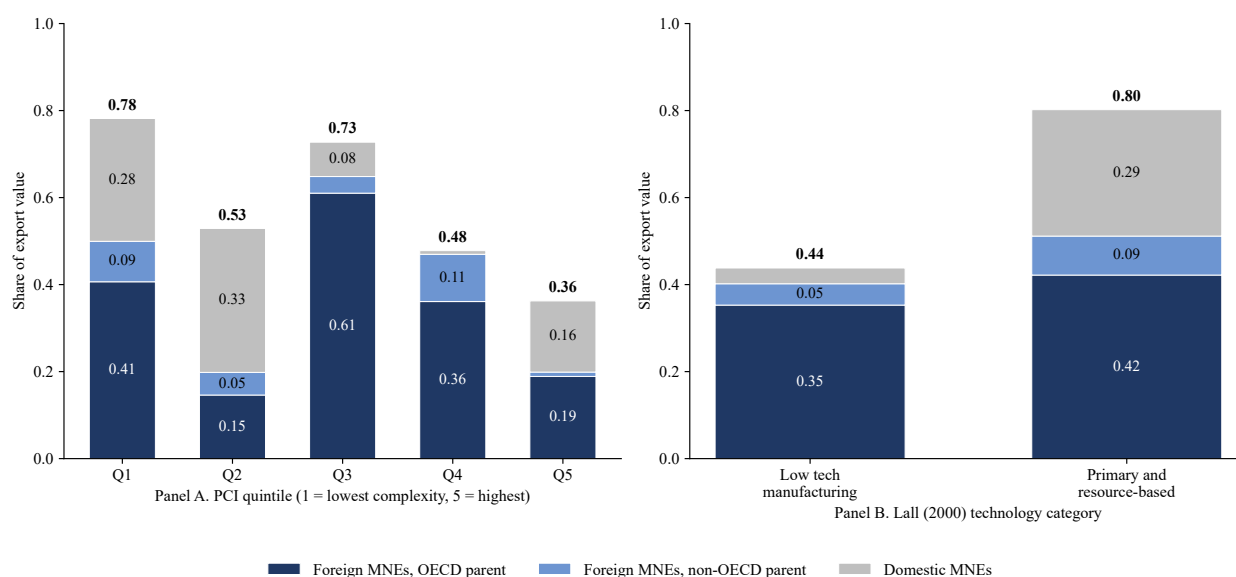
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 144:** Mining and Fuels: Multinational Share of Export Value, by Technology Class and Parent Country

| Technology category        | GBR   | USA   | CAN   | LIE   | ESP   | PAN   | AUS   | BRA   | BGR   | FRA   | Other | Domestic | All MNEs | Value (US\$bn/yr) |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|----------|-------------------|
| Low tech manufacturing     | 0.009 | 0.037 | 0.266 | 0.000 | 0.004 | 0.014 | 0.005 | 0.004 | 0.001 | 0.006 | 0.056 | 0.036    | 0.438    | 14.4              |
| Primary and resource-based | 0.217 | 0.118 | 0.036 | 0.042 | 0.020 | 0.014 | 0.014 | 0.009 | 0.009 | 0.007 | 0.026 | 0.291    | 0.802    | 57.3              |

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. Other: all remaining foreign parents. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

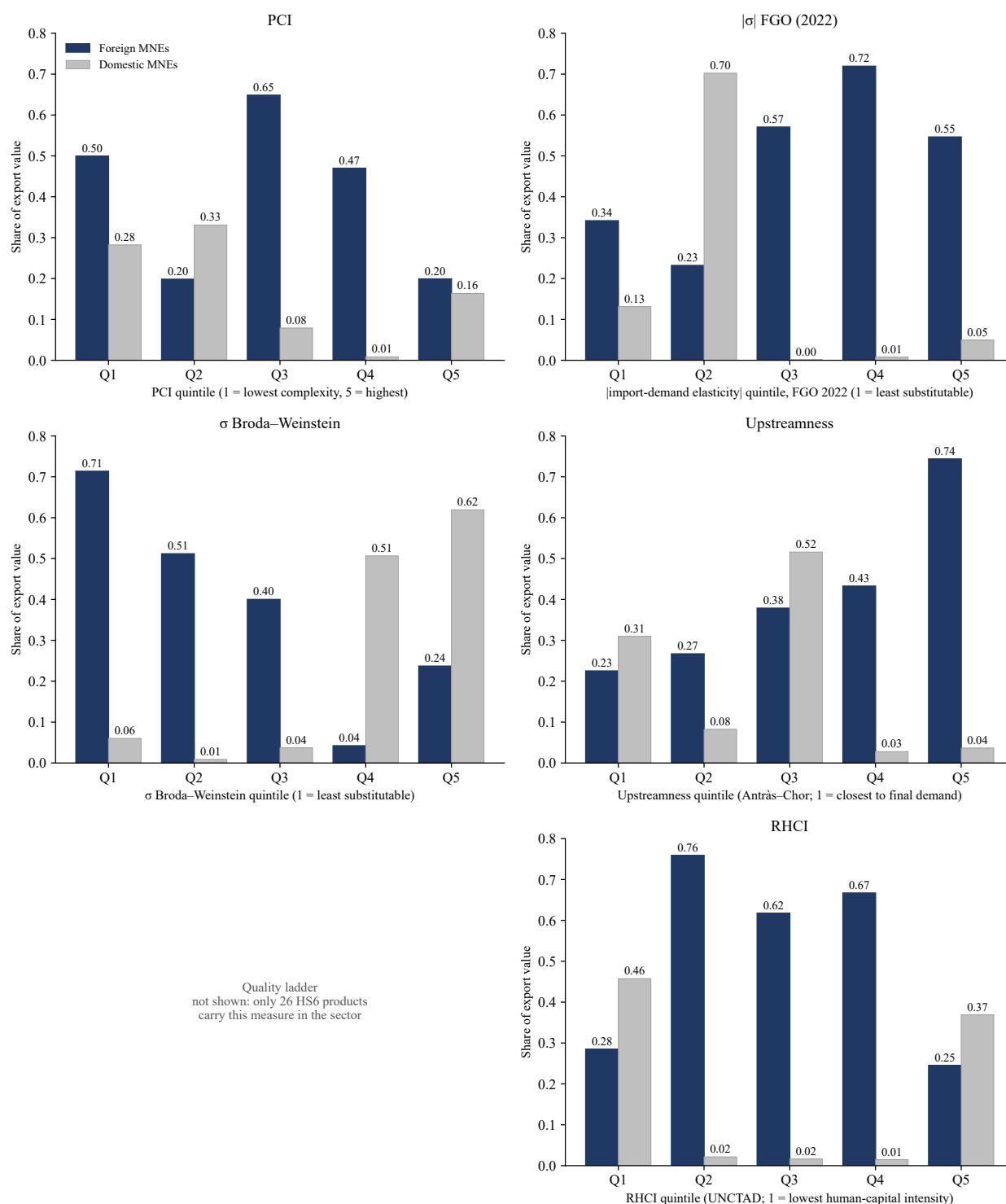


**Figure 103: Mining and Fuels: Multinational Share of Export Value, by Product Sophistication and Location of the Parent**

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. Foreign MNEs are split by whether the ultimate parent is in an OECD member country. Panel A: Quintiles of the Hausmann–Hidalgo Product Complexity Index across HS6 products (Q1 = least complex). Panel B: Lall (2000) technology classes: primary and resource-based products, and low-, medium- and high-technology manufactures.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

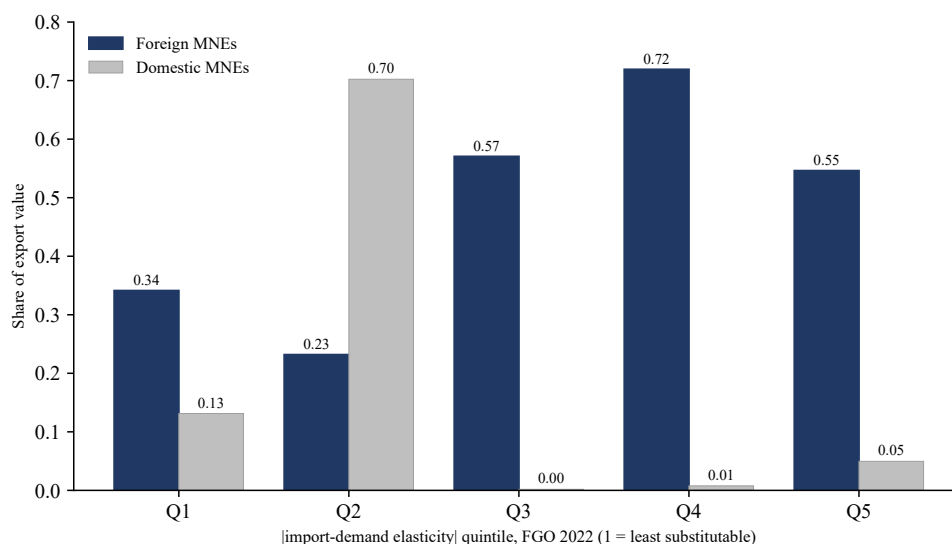
## 19.3 Other Measures of Product Sophistication



**Figure 104: Mining and Fuels: Multinational Share of Export Value, by Quintile of Six Sophistication Measures**

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. Quintiles computed over HS6 products. The six measures are the Product Complexity Index, the absolute import-demand elasticity of Fontagné, Guimbard and Orefice (2022), the elasticity of substitution of Broda and Weinstein, upstreamness, the quality ladder and the revealed human-capital intensity of UNCTAD.

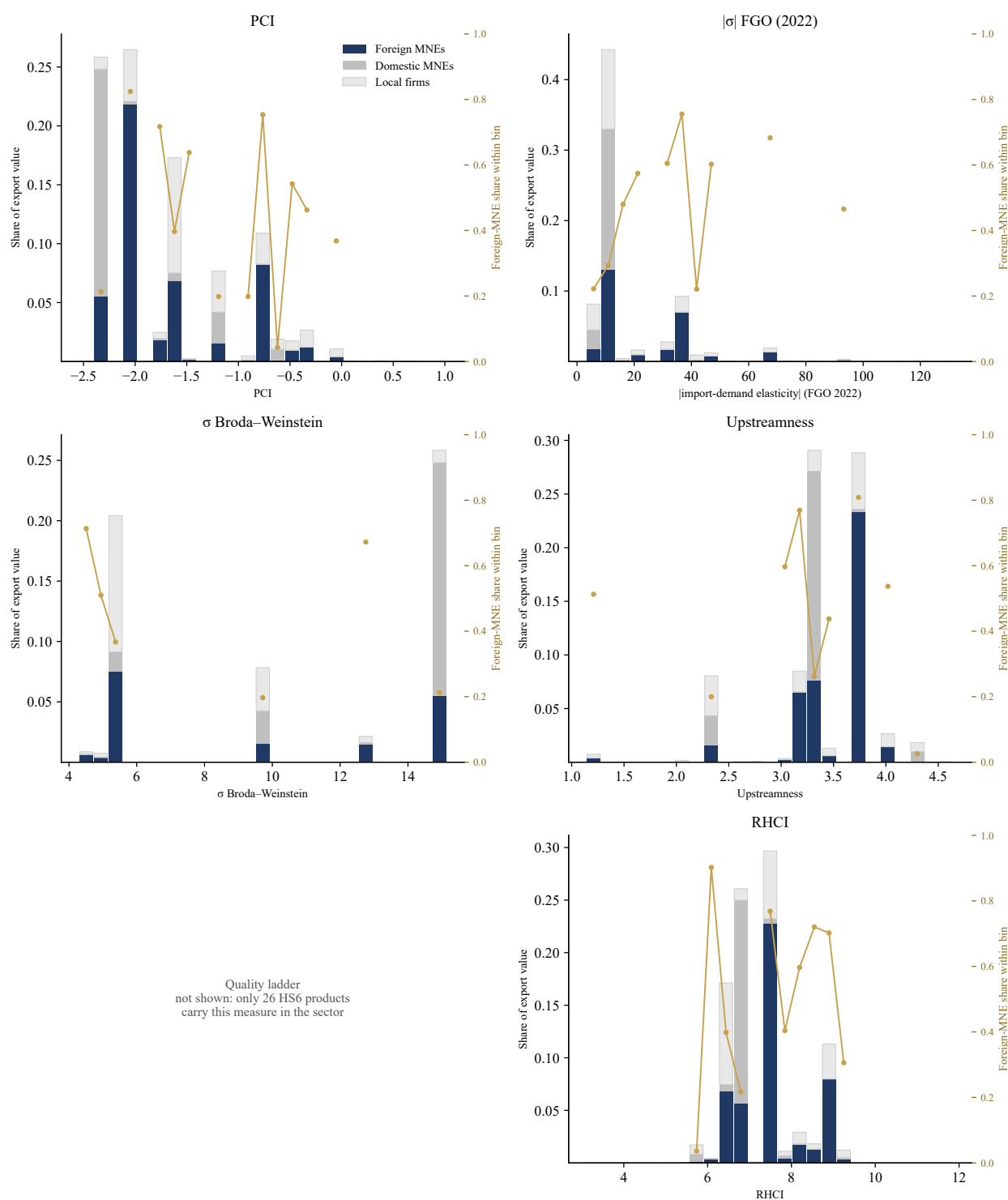
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 105:** Mining and Fuels: Multinational Share of Export Value, by Quintile of the Import-Demand Elasticity

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. Quintiles of the absolute import-demand elasticity of Fontagné, Guimbard and Orefice (2022) across HS6 products; higher quintiles are more substitutable products.

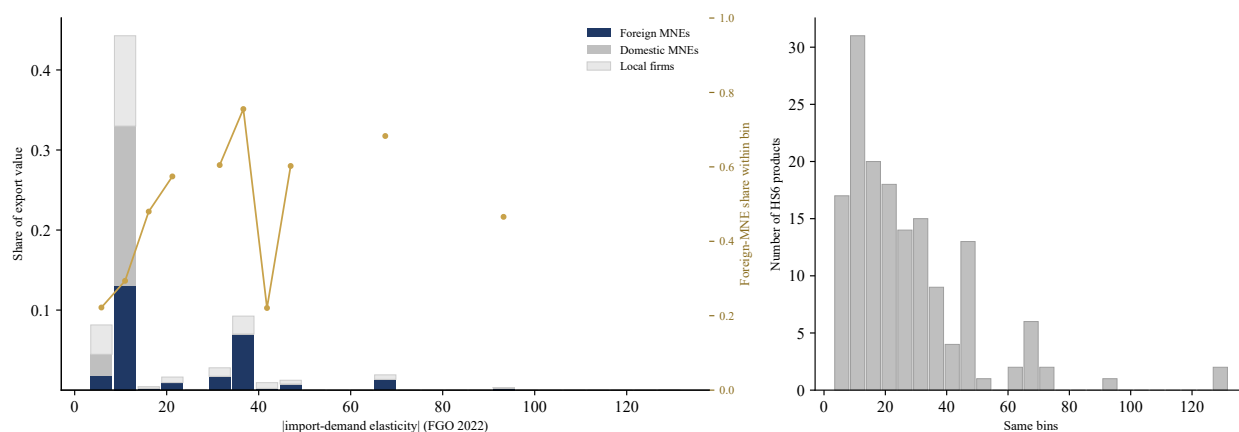
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 106:** Mining and Fuels: Distribution of Export Value across Six Sophistication Measures

*Notes:* Bars: mining and fuels export value (HS 25–27 and 71) in each bin, by type of exporter. Line (right axis): foreign-MNE share of each bin. The six measures are the Product Complexity Index, the absolute import-demand elasticity of Fontagné, Guimbard and Orefice (2022), the elasticity of substitution of Broda and Weinstein, upstreamness, the quality ladder and the revealed human-capital intensity of UNCTAD.

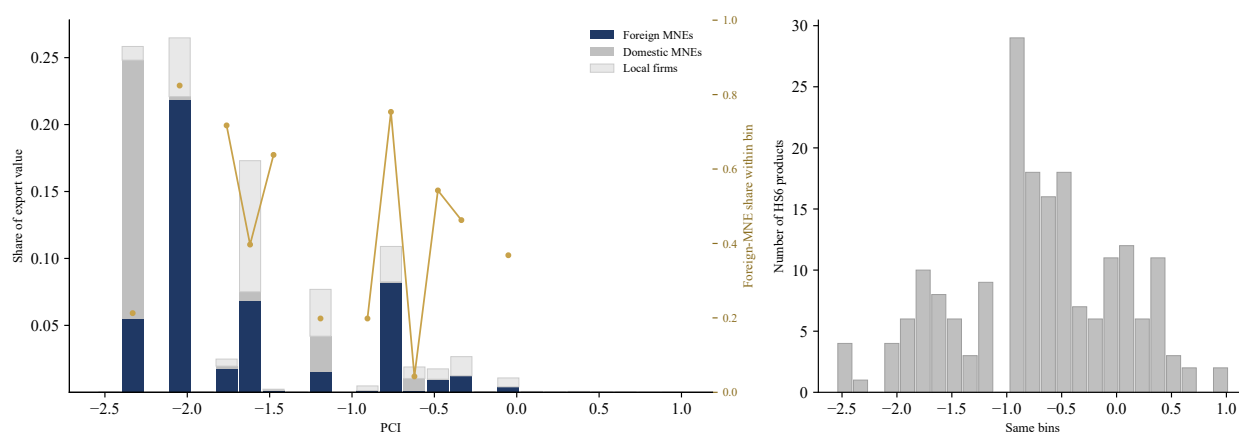
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 107:** Mining and Fuels: Distribution of Export Value across the Import-Demand Elasticity

*Notes:* Left: export value in each bin, by type of exporter, and the foreign-MNE share of each bin. Right: number of HS6 products in each bin.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 108:** Mining and Fuels: Distribution of Export Value across the Product Complexity Index

*Notes:* Left: export value in each bin, by type of exporter, and the foreign-MNE share of each bin. Right: number of HS6 products in each bin.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 145:** Mining and Fuels: Multinational Share of Export Value, by Quintile of Six Sophistication Measures

| Measure                  | Foreign MNE share |      |      |      |      | Domestic MNE share |      |      |      |      |
|--------------------------|-------------------|------|------|------|------|--------------------|------|------|------|------|
|                          | Q1                | Q2   | Q3   | Q4   | Q5   | Q1                 | Q2   | Q3   | Q4   | Q5   |
| PCI                      | 0.50              | 0.20 | 0.65 | 0.47 | 0.20 | 0.28               | 0.33 | 0.08 | 0.01 | 0.16 |
| $ \sigma $ FGO (2022)    | 0.34              | 0.23 | 0.57 | 0.72 | 0.55 | 0.13               | 0.70 | 0.00 | 0.01 | 0.05 |
| $\sigma$ Broda–Weinstein | 0.71              | 0.51 | 0.40 | 0.04 | 0.24 | 0.06               | 0.01 | 0.04 | 0.51 | 0.62 |
| Upstreamness             | 0.23              | 0.27 | 0.38 | 0.43 | 0.74 | 0.31               | 0.08 | 0.52 | 0.03 | 0.04 |
| Quality ladder           | —                 | —    | —    | —    | —    | —                  | —    | —    | —    | —    |
| RHCI                     | 0.28              | 0.76 | 0.62 | 0.67 | 0.25 | 0.46               | 0.02 | 0.02 | 0.01 | 0.37 |

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. Quintiles computed over HS6 products. —: fewer than 30 HS6 products carry the measure.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 146:** Mining and Fuels: Correlations between Sophistication Measures

|                          | PCI   | $ \sigma $ FGO (2022) | $\sigma$ Broda–Weinstein | Upstreamness | Quality ladder | RHCI  |
|--------------------------|-------|-----------------------|--------------------------|--------------|----------------|-------|
| PCI                      | 1.00  | 0.34                  | -0.84                    | -0.19        | –              | 0.59  |
| $ \sigma $ FGO (2022)    | 0.34  | 1.00                  | 0.12                     | 0.34         | –              | 0.73  |
| $\sigma$ Broda–Weinstein | -0.84 | 0.12                  | 1.00                     | 0.37         | –              | -0.07 |
| Upstreamness             | -0.19 | 0.34                  | 0.37                     | 1.00         | –              | -0.05 |
| Quality ladder           | –     | –                     | –                        | –            | –              | –     |
| RHCI                     | 0.59  | 0.73                  | -0.07                    | -0.05        | –              | 1.00  |

Notes: Correlations across HS6 products, weighted by export value.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

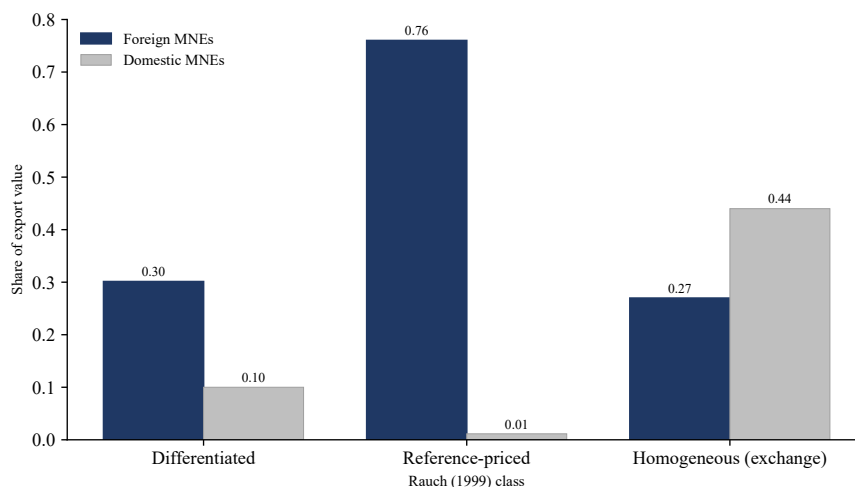
## 19.4 Rauch Classes and End Use

**Table 147:** Mining and Fuels: Multinational Share of Export Value, by Rauch Product Class

| Rauch (1999) class     | Foreign | Domestic | All MNEs | HS6 products | Value (US\$bn/yr) |
|------------------------|---------|----------|----------|--------------|-------------------|
| Differentiated         | 0.301   | 0.100    | 0.401    | 40           | 0.3               |
| Reference-priced       | 0.760   | 0.011    | 0.771    | 99           | 25.1              |
| Homogeneous (exchange) | 0.270   | 0.440    | 0.710    | 27           | 39.5              |

Notes: Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. Rauch (1999) conservative classification: homogeneous goods are traded on organized exchanges, reference-priced goods have a price quoted in trade publications, and differentiated goods have neither. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Figure 109:** Mining and Fuels: Multinational Share of Export Value, by Rauch Product Class

Notes: Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. Rauch (1999) conservative classification: homogeneous goods are traded on organized exchanges, reference-priced goods have a price quoted in trade publications, and differentiated goods have neither.

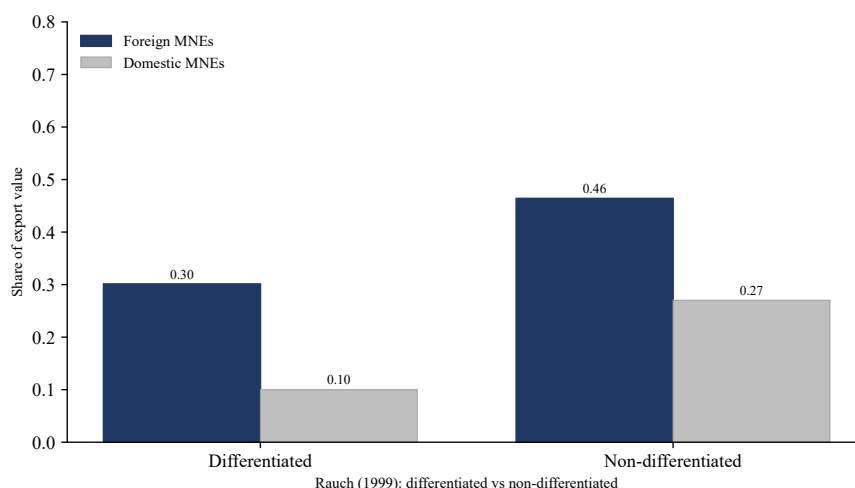
Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 148:** Mining and Fuels: Multinational Share of Export Value: Differentiated and Other Products

| Rauch (1999): differentiated vs non-differentiated | Foreign | Domestic | All MNEs | HS6 products | Value (US\$bn/yr) |
|----------------------------------------------------|---------|----------|----------|--------------|-------------------|
| Differentiated                                     | 0.301   | 0.100    | 0.401    | 40           | 0.3               |
| Non-differentiated                                 | 0.464   | 0.270    | 0.734    | 126          | 64.6              |

Notes: Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. Rauch (1999) conservative classification; other products are reference-priced and homogeneous goods. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 110:** Mining and Fuels: Multinational Share of Export Value: Differentiated and Other Products

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. Rauch (1999) conservative classification; other products are reference-priced and homogeneous goods.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 149:** Mining and Fuels: Multinational Share of Export Value, by End Use

| BEC end use  | Foreign | Domestic | All MNEs | HS6 products | Value (US\$bn/yr) |
|--------------|---------|----------|----------|--------------|-------------------|
| Intermediate | 0.516   | 0.237    | 0.753    | 179          | 65.4              |
| Consumption  | 0.617   | 0.004    | 0.621    | 11           | 0.5               |
| Capital      | 0.214   | 0.000    | 0.214    | 2            | 0.0               |

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. End use from BEC Rev. 4; HS6 products without a BEC correspondence are excluded. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 19.5 Regressions

**Table 150:** Mining and Fuels: Product Characteristics and Multinational Shares

|                                                            | (1)                    | (2)                    | (3)                    |
|------------------------------------------------------------|------------------------|------------------------|------------------------|
| <i>Panel A. Dependent variable: share of all MNEs</i>      |                        |                        |                        |
| Complexity (PCI)                                           | -0.1048***<br>(0.0132) | -0.0986***<br>(0.0120) | -0.0859***<br>(0.0128) |
| $ \sigma $ FGO (per s.d.)                                  | 0.0156*<br>(0.0085)    | 0.0064<br>(0.0087)     | -0.0016<br>(0.0077)    |
| Upstreamness                                               | 0.1419***<br>(0.0201)  | 0.1522***<br>(0.0194)  | 0.1346***<br>(0.0245)  |
| <i>Panel B. Dependent variable: share of foreign MNEs</i>  |                        |                        |                        |
| Complexity (PCI)                                           | 0.1307***<br>(0.0134)  | 0.1349***<br>(0.0128)  | 0.1174***<br>(0.0136)  |
| $ \sigma $ FGO (per s.d.)                                  | 0.1160***<br>(0.0103)  | 0.1105***<br>(0.0098)  | 0.1216***<br>(0.0109)  |
| Upstreamness                                               | 0.0877***<br>(0.0175)  | 0.0805***<br>(0.0173)  | 0.0664***<br>(0.0188)  |
| <i>Panel C. Dependent variable: share of domestic MNEs</i> |                        |                        |                        |
| Complexity (PCI)                                           | -0.2355***<br>(0.0124) | -0.2335***<br>(0.0124) | -0.2033***<br>(0.0149) |
| $ \sigma $ FGO (per s.d.)                                  | -0.1003***<br>(0.0076) | -0.1042***<br>(0.0079) | -0.1232***<br>(0.0088) |
| Upstreamness                                               | 0.0542***<br>(0.0183)  | 0.0717***<br>(0.0183)  | 0.0682***<br>(0.0225)  |
| Origin FE                                                  | ✓                      |                        |                        |
| Destination FE                                             | ✓                      |                        |                        |
| Year FE                                                    | ✓                      |                        |                        |
| Origin $\times$ year FE                                    |                        | ✓                      |                        |
| Destination $\times$ year FE                               |                        | ✓                      |                        |
| Origin $\times$ dest. $\times$ year FE                     |                        |                        | ✓                      |
| Observations                                               | 23,906                 | 23,513                 | 22,358                 |

*Notes:* Observations are origin–destination–HS6–year cells. Mining and Fuels products only. Dependent variable: share of the cell’s export value shipped by each type of MNE. The import-demand elasticity is standardized (mean 0, standard deviation 1). Regressions weighted by the cell’s export value. Robust standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 151:** Mining and Fuels: Product Sophistication and Multinational Shares

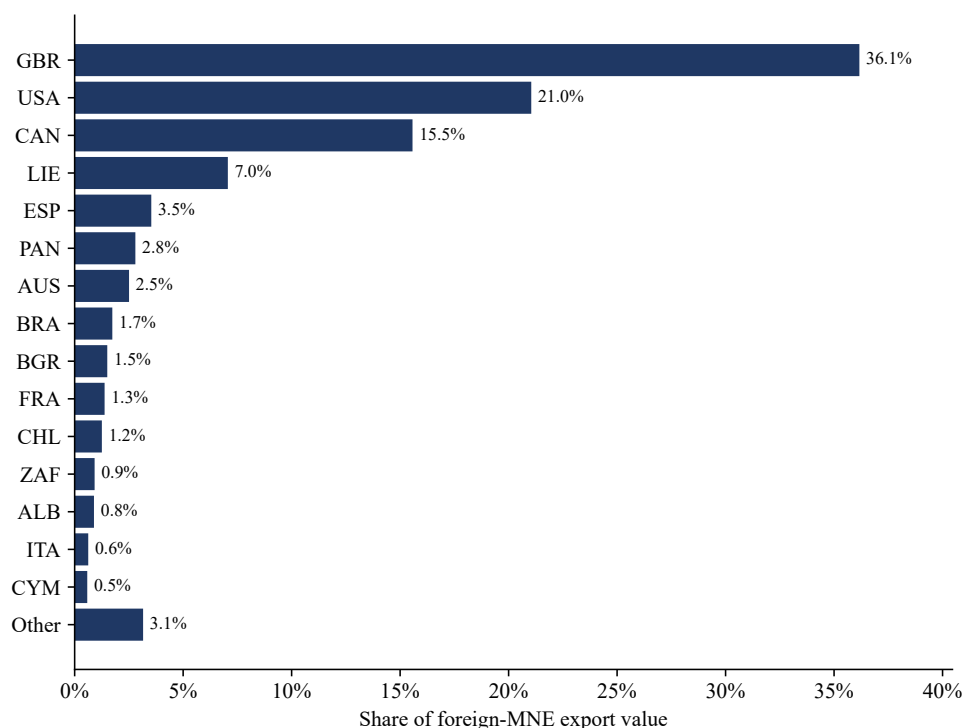
|                                                            | PCI                    |                        | Lall                   |                        | Rauch                  |                        |
|------------------------------------------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                                                            | (1)                    | (2)                    | (3)                    | (4)                    | (5)                    | (6)                    |
| <i>Panel A. Dependent variable: share of all MNEs</i>      |                        |                        |                        |                        |                        |                        |
| Complexity (PCI)                                           | -0.1846***<br>(0.0141) | -0.1722***<br>(0.0166) |                        |                        |                        |                        |
| Lall: high tech (vs primary and resource-based)            |                        |                        |                        |                        |                        |                        |
| Lall: medium tech                                          |                        |                        |                        |                        |                        |                        |
| Lall: low tech                                             |                        |                        | -0.3546***<br>(0.0396) | -0.3899***<br>(0.0501) |                        |                        |
| Rauch: differentiated (vs homogeneous)                     |                        |                        |                        |                        | -0.1843***<br>(0.0457) | -0.1742***<br>(0.0557) |
| Rauch: reference-priced                                    |                        |                        |                        |                        | 0.1987***<br>(0.0259)  | 0.2580***<br>(0.0351)  |
| Observations                                               | 32,084                 | 30,880                 | 32,087                 | 30,883                 | 24,732                 | 23,549                 |
| <i>Panel B. Dependent variable: share of foreign MNEs</i>  |                        |                        |                        |                        |                        |                        |
| Complexity (PCI)                                           | -0.0374**<br>(0.0169)  | -0.0647***<br>(0.0203) |                        |                        |                        |                        |
| Lall: high tech (vs primary and resource-based)            |                        |                        |                        |                        |                        |                        |
| Lall: medium tech                                          |                        |                        |                        |                        |                        |                        |
| Lall: low tech                                             |                        |                        | -0.1503***<br>(0.0248) | -0.1296***<br>(0.0250) |                        |                        |
| Rauch: differentiated (vs homogeneous)                     |                        |                        |                        |                        | -0.0035<br>(0.0348)    | -0.0015<br>(0.0435)    |
| Rauch: reference-priced                                    |                        |                        |                        |                        | 0.4411***<br>(0.0211)  | 0.5010***<br>(0.0254)  |
| Observations                                               | 32,084                 | 30,880                 | 32,087                 | 30,883                 | 24,732                 | 23,549                 |
| <i>Panel C. Dependent variable: share of domestic MNEs</i> |                        |                        |                        |                        |                        |                        |
| Complexity (PCI)                                           | -0.1472***<br>(0.0173) | -0.1075***<br>(0.0210) |                        |                        |                        |                        |
| Lall: high tech (vs primary and resource-based)            |                        |                        |                        |                        |                        |                        |
| Lall: medium tech                                          |                        |                        |                        |                        |                        |                        |
| Lall: low tech                                             |                        |                        | -0.2044***<br>(0.0457) | -0.2602***<br>(0.0570) |                        |                        |
| Rauch: differentiated (vs homogeneous)                     |                        |                        |                        |                        | -0.1808***<br>(0.0339) | -0.1726***<br>(0.0410) |
| Rauch: reference-priced                                    |                        |                        |                        |                        | -0.2423***<br>(0.0237) | -0.2430***<br>(0.0335) |
| Observations                                               | 32,084                 | 30,880                 | 32,087                 | 30,883                 | 24,732                 | 23,549                 |
| Origin × year and destination × year FE                    | ✓                      |                        | ✓                      |                        | ✓                      |                        |
| Origin × dest. × year FE                                   |                        | ✓                      |                        | ✓                      |                        | ✓                      |

*Notes:* Observations are origin–destination–HS6–year cells. Mining and Fuels products only. Dependent variable: share of the cell's export value shipped by each type of MNE. Omitted categories: primary and resource-based products (Lall) and homogeneous products (Rauch). Robust standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 20 Fact 3: Multinationals from a Small Set of Countries Dominate Exports

### 20.1 Parent Countries



**Figure 111:** Mining and Fuels: Foreign-Multinational Exports, by Parent Country

*Notes:* Percent of foreign-MNE mining and fuels export value, by country of the ultimate parent.

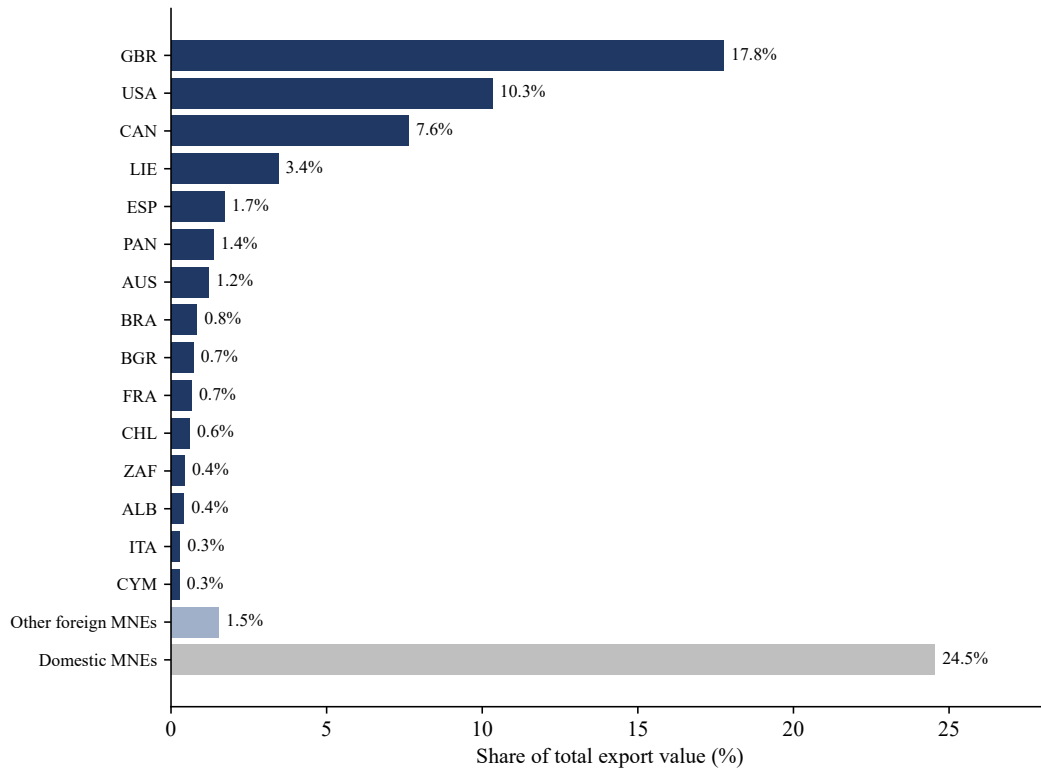
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 152:** Mining and Fuels: Foreign-Multinational Exports, by Parent Country

| Parent country       | Share (%) | Value (US\$bn/yr) |
|----------------------|-----------|-------------------|
| GBR                  | 36.1      | 11.9              |
| USA                  | 21.0      | 7.3               |
| CAN                  | 15.5      | 5.7               |
| LIE                  | 7.0       | 2.0               |
| ESP                  | 3.5       | 1.4               |
| PAN                  | 2.8       | 1.0               |
| AUS                  | 2.5       | 0.9               |
| BRA                  | 1.7       | 0.7               |
| BGR                  | 1.5       | 0.5               |
| FRA                  | 1.3       | 0.5               |
| CHL                  | 1.2       | 0.4               |
| ZAF                  | 0.9       | 0.3               |
| ALB                  | 0.8       | 0.3               |
| ITA                  | 0.6       | 0.2               |
| CYM                  | 0.5       | 0.2               |
| Other (87 countries) | 3.1       | 1.2               |
| All foreign MNEs     | 100.0     | 36.0              |

*Notes:* Percent of foreign-MNE mining and fuels export value, by country of the ultimate parent. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 112:** Mining and Fuels: Share of Total Exports, by Parent Country

*Notes:* Percent of total mining and fuels export value shipped by foreign MNEs of each parent country, other foreign MNEs and domestic MNEs; the remainder is local firms.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 153:** Mining and Fuels: Share of Total Exports, by Parent Country

| Exporter type / parent country | Share of total exports (%) |
|--------------------------------|----------------------------|
| GBR                            | 17.8                       |
| USA                            | 10.3                       |
| CAN                            | 7.6                        |
| LIE                            | 3.4                        |
| ESP                            | 1.7                        |
| PAN                            | 1.4                        |
| AUS                            | 1.2                        |
| BRA                            | 0.8                        |
| BGR                            | 0.7                        |
| FRA                            | 0.7                        |
| CHL                            | 0.6                        |
| ZAF                            | 0.4                        |
| ALB                            | 0.4                        |
| ITA                            | 0.3                        |
| CYM                            | 0.3                        |
| Other foreign MNEs             | 1.5                        |
| Domestic MNEs                  | 24.5                       |
| All MNEs                       | 73.7                       |
| Local firms                    | 26.3                       |

*Notes:* Percent of total mining and fuels export value, by type of exporter and parent country.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 20.2 Parent Region and Destination Region

**Table 154:** Mining and Fuels: Multinational Exports by Parent Region and Destination Region: Export Value

| Parent region / Destination region | Latin America | North America | Europe | Asia | Africa | Oceania | Rest of World | Total |
|------------------------------------|---------------|---------------|--------|------|--------|---------|---------------|-------|
| Latin America                      | 0.6           | 0.4           | 0.6    | 0.9  | 0.0    | 0.1     | 0.0           | 2.6   |
| North America                      | 1.7           | 2.9           | 4.5    | 3.9  | 0.0    | 0.0     | 0.0           | 13.0  |
| Europe                             | 2.2           | 2.0           | 3.5    | 9.5  | 0.2    | 0.0     | 0.0           | 17.3  |
| Asia                               | 0.0           | 0.1           | 0.0    | 0.1  | 0.0    | 0.0     | 0.0           | 0.2   |
| Africa                             | 0.0           | 0.0           | 0.2    | 0.2  | 0.0    | 0.0     | 0.0           | 0.4   |
| Oceania                            | 0.2           | 0.0           | 0.1    | 0.7  | 0.0    | 0.0     | 0.0           | 0.9   |
| Rest of World                      | 0.0           | 0.0           | 0.0    | 0.0  | 0.0    | 0.0     | 0.0           | 0.0   |
| Domestic MNEs                      | 3.7           | 7.4           | 1.1    | 4.1  | 0.2    | 0.0     | 0.1           | 16.5  |
| Local firms                        | 3.5           | 5.9           | 4.9    | 5.3  | 0.1    | 0.0     | 0.2           | 19.9  |
| Total                              | 11.9          | 18.7          | 14.8   | 24.6 | 0.5    | 0.1     | 0.3           | 71.0  |

*Notes:* Values are annual averages in US\$ billion. Rows: region of the ultimate parent; columns: destination region.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 155:** Mining and Fuels: Multinational Exports by Parent Region and Destination Region: Row Percentages

| Parent region / Destination region | Latin America | North America | Europe | Asia | Africa | Oceania | Rest of World | Total |
|------------------------------------|---------------|---------------|--------|------|--------|---------|---------------|-------|
| Latin America                      | 24.6          | 16.6          | 22.3   | 33.8 | 0.0    | 2.4     | 0.2           | 100.0 |
| North America                      | 14.3          | 20.7          | 35.0   | 29.8 | 0.2    | 0.0     | 0.0           | 100.0 |
| Europe                             | 11.5          | 9.9           | 19.4   | 58.0 | 0.9    | 0.1     | 0.2           | 100.0 |
| Asia                               | 5.8           | 32.7          | 15.1   | 45.6 | 0.1    | 0.3     | 0.4           | 100.0 |
| Africa                             | 0.7           | 3.7           | 35.6   | 58.6 | 1.4    | 0.0     | 0.1           | 100.0 |
| Oceania                            | 16.6          | 4.2           | 7.4    | 71.7 | 0.0    | 0.0     | 0.0           | 100.0 |
| Rest of World                      | 100.0         | 0.0           | 0.0    | 0.0  | 0.0    | 0.0     | 0.0           | 100.0 |
| Domestic MNEs                      | 22.1          | 44.4          | 6.8    | 25.3 | 1.0    | 0.0     | 0.4           | 100.0 |
| Local firms                        | 17.5          | 29.2          | 24.8   | 26.9 | 0.5    | 0.1     | 1.1           | 100.0 |

*Notes:* Percent of each group's exports, by destination region (rows add to 100). Domestic MNEs and local firms are shown for comparison.

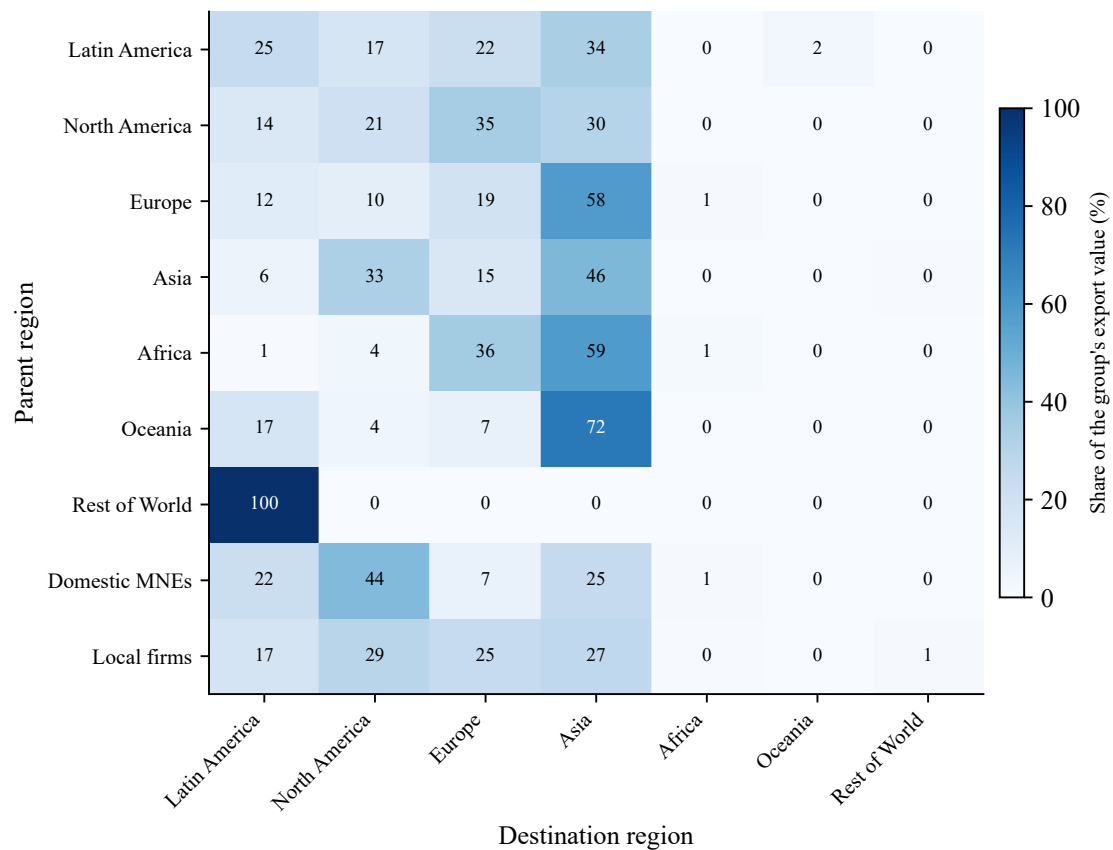
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 156:** Mining and Fuels: Multinational Exports by Parent Region and Destination Region: Column Percentages

| Parent region / Destination region | Latin America | North America | Europe | Asia  | Africa | Oceania | Rest of World |
|------------------------------------|---------------|---------------|--------|-------|--------|---------|---------------|
| Latin America                      | 5.0           | 2.2           | 3.7    | 3.2   | 0.1    | 51.7    | 1.9           |
| North America                      | 15.0          | 14.1          | 30.1   | 14.6  | 5.6    | 2.4     | 0.7           |
| Europe                             | 17.4          | 9.7           | 23.8   | 40.7  | 35.1   | 21.0    | 9.2           |
| Asia                               | 0.1           | 0.4           | 0.2    | 0.4   | 0.0    | 0.6     | 0.3           |
| Africa                             | 0.0           | 0.1           | 0.9    | 0.9   | 1.1    | 0.0     | 0.2           |
| Oceania                            | 1.2           | 0.2           | 0.4    | 2.4   | 0.0    | 0.1     | 0.0           |
| Rest of World                      | 0.0           | 0.0           | 0.0    | 0.0   | 0.0    | 0.0     | 0.0           |
| Domestic MNEs                      | 33.2          | 43.0          | 8.3    | 17.7  | 39.1   | 2.2     | 22.3          |
| Local firms                        | 28.1          | 30.3          | 32.5   | 20.1  | 19.0   | 22.0    | 65.4          |
| Total                              | 100.0         | 100.0         | 100.0  | 100.0 | 100.0  | 100.0   | 100.0         |

*Notes:* Percent of the exports reaching each destination region, by parent region (columns add to 100).

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 113:** Mining and Fuels: Multinational Exports by Parent Region and Destination Region

*Notes:* Percent of each group's exports, by destination region (rows add to 100).

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 157:** Mining and Fuels: Foreign-Multinational Exports by Exporting Country and Destination Region

| Origin | Destination region | Latin America | North America | Europe | Asia | Africa | Oceania | Rest of World | Total |
|--------|--------------------|---------------|---------------|--------|------|--------|---------|---------------|-------|
| ARG    |                    | 18.8          | 37.6          | 25.9   | 17.2 | 0.4    | 0.1     | 0.1           | 5.0   |
| CHL    |                    | 3.2           | 3.2           | 12.9   | 80.5 | 0.1    | 0.1     | 0.0           | 7.5   |
| COL    |                    | 29.4          | 18.2          | 37.2   | 13.8 | 0.5    | 0.2     | 0.6           | 8.4   |
| CRI    |                    | 76.5          | 23.1          | 0.3    | 0.1  | 0.0    | 0.0     | 0.0           | 0.1   |
| DOM    |                    | 4.4           | 56.9          | 13.5   | 25.1 | 0.0    | 0.0     | 0.0           | 1.7   |
| PER    |                    | 10.0          | 8.8           | 25.3   | 54.4 | 1.0    | 0.5     | 0.0           | 13.2  |
| PRY    |                    | 46.0          | 21.2          | 32.7   | 0.0  | 0.0    | 0.0     | 0.0           | 0.0   |
| SLV    |                    | 92.4          | 6.0           | 1.5    | 0.1  | 0.0    | 0.0     | 0.0           | 0.1   |
| URY    |                    | 12.5          | 1.8           | 80.1   | 5.1  | 0.1    | 0.0     | 0.3           | 0.1   |

*Notes:* Percent of each country's foreign-MNE mining and fuels exports, by destination region (rows add to 100). Last column: foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 158:** Mining and Fuels: Local-Firm Exports by Exporting Country and Destination Region

| Origin | Destination region | Latin America | North America | Europe | Asia | Africa | Oceania | Rest of World | Total |
|--------|--------------------|---------------|---------------|--------|------|--------|---------|---------------|-------|
| ARG    |                    | 53.6          | 12.4          | 28.5   | 3.3  | 1.9    | 0.4     | 0.0           | 1.6   |
| CHL    |                    | 12.3          | 8.5           | 25.3   | 53.3 | 0.4    | 0.2     | 0.0           | 1.9   |
| COL    |                    | 23.5          | 35.2          | 28.3   | 9.0  | 0.5    | 0.0     | 3.5           | 4.1   |
| CRI    |                    | 19.3          | 74.7          | 4.6    | 1.4  | 0.0    | 0.0     | 0.0           | 0.1   |
| DOM    |                    | 36.7          | 56.6          | 2.1    | 0.0  | 4.5    | 0.0     | 0.0           | 0.3   |
| PER    |                    | 10.3          | 32.7          | 23.8   | 32.4 | 0.2    | 0.1     | 0.5           | 11.8  |
| PRY    |                    | 85.2          | 13.0          | 1.0    | 0.7  | 0.0    | 0.0     | 0.0           | 0.0   |
| SLV    |                    | 41.3          | 58.1          | 0.1    | 0.5  | 0.0    | 0.0     | 0.0           | 0.1   |
| URY    |                    | 83.9          | 5.4           | 1.1    | 9.5  | 0.0    | 0.1     | 0.0           | 0.1   |

*Notes:* Percent of each country's mining and fuels exports by local firms, by destination region (rows add to 100). Last column: local-firm exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 20.3 Parent Country and Destination Country

**Table 159:** Mining and Fuels: Foreign-Multinational Exports by Parent Country and Destination: Export Value

| Parent / Destination | CHN | JPN | USA | CAN | CHE | KOR | ESP | IND | DEU | CHL | Other destinations | Total |
|----------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|--------------------|-------|
| GBR                  | 4.5 | 1.1 | 0.2 | 0.3 | 0.1 | 0.8 | 0.5 | 0.6 | 0.7 | 0.5 | 2.5                | 11.9  |
| USA                  | 1.1 | 0.7 | 0.9 | 0.0 | 0.5 | 0.3 | 0.8 | 0.1 | 0.1 | 0.2 | 2.6                | 7.3   |
| CAN                  | 0.4 | 0.3 | 0.3 | 1.7 | 1.1 | 0.1 | 0.2 | 0.4 | 0.2 | 0.0 | 1.0                | 5.7   |
| LIE                  | 0.6 | 0.7 | 0.0 | 0.0 | 0.0 | 0.2 | 0.1 | 0.1 | 0.0 | 0.1 | 0.2                | 2.0   |
| ESP                  | 0.4 | 0.0 | 0.6 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.1 | 0.3                | 1.4   |
| PAN                  | 0.4 | 0.0 | 0.1 | 0.0 | 0.0 | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.3                | 1.0   |
| AUS                  | 0.3 | 0.1 | 0.0 | 0.0 | 0.0 | 0.1 | 0.0 | 0.1 | 0.0 | 0.0 | 0.2                | 0.9   |
| BRA                  | 0.0 | 0.0 | 0.1 | 0.0 | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.4                | 0.7   |
| BGR                  | 0.0 | 0.0 | 0.2 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.2                | 0.5   |
| FRA                  | 0.1 | 0.0 | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.1 | 0.3                | 0.5   |
| Other parents        | 0.3 | 0.1 | 0.5 | 0.2 | 0.2 | 0.1 | 0.1 | 0.1 | 0.1 | 0.1 | 0.8                | 2.6   |
| Total                | 8.2 | 3.1 | 3.1 | 2.3 | 2.0 | 1.6 | 1.7 | 1.5 | 1.2 | 1.2 | 8.7                | 34.5  |

*Notes:* Values are annual averages in US\$ billion. Ten largest parent countries and ten largest destinations by foreign-MNE export value.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 160:** Mining and Fuels: Foreign-Multinational Exports by Parent Country and Destination: Row Percentages

| Parent / Destination | CHN  | JPN  | USA  | CAN  | CHE  | KOR  | ESP  | IND | DEU | CHL  | Other destinations | Total |
|----------------------|------|------|------|------|------|------|------|-----|-----|------|--------------------|-------|
| GBR                  | 38.8 | 9.8  | 1.9  | 2.6  | 0.9  | 6.4  | 3.8  | 5.6 | 4.8 | 4.2  | 21.1               | 100.0 |
| USA                  | 15.0 | 9.0  | 12.5 | 0.6  | 5.1  | 4.6  | 10.7 | 1.7 | 0.8 | 2.4  | 37.8               | 100.0 |
| CAN                  | 8.2  | 6.7  | 4.9  | 26.1 | 17.5 | 1.1  | 3.4  | 6.5 | 4.7 | 0.3  | 20.5               | 100.0 |
| LIE                  | 31.9 | 38.0 | 1.1  | 0.0  | 0.4  | 8.2  | 2.7  | 4.5 | 1.7 | 3.2  | 8.2                | 100.0 |
| ESP                  | 26.5 | 0.0  | 37.6 | 0.3  | 0.3  | 0.4  | 3.4  | 3.1 | 0.0 | 8.4  | 20.0               | 100.0 |
| PAN                  | 38.3 | 1.5  | 8.0  | 0.9  | 0.5  | 10.8 | 1.8  | 3.5 | 3.3 | 0.2  | 31.1               | 100.0 |
| AUS                  | 35.3 | 14.3 | 2.1  | 2.1  | 3.8  | 11.4 | 2.1  | 6.7 | 0.8 | 2.3  | 19.3               | 100.0 |
| BRA                  | 1.2  | 0.7  | 22.4 | 0.1  | 8.6  | 1.0  | 0.2  | 7.3 | 0.0 | 2.5  | 56.1               | 100.0 |
| BGR                  | 5.5  | 0.0  | 50.6 | 0.0  | 0.0  | 0.0  | 5.6  | 1.2 | 0.2 | 0.0  | 36.8               | 100.0 |
| FRA                  | 13.9 | 1.0  | 14.0 | 0.4  | 0.9  | 2.7  | 1.0  | 0.9 | 0.6 | 10.5 | 54.1               | 100.0 |
| Other parents        | 10.2 | 4.9  | 20.7 | 7.4  | 7.6  | 3.0  | 2.7  | 3.6 | 4.8 | 4.0  | 31.2               | 100.0 |

*Notes:* Percent of each parent country's exports, by destination (rows add to 100). Ten largest parent countries and destinations.

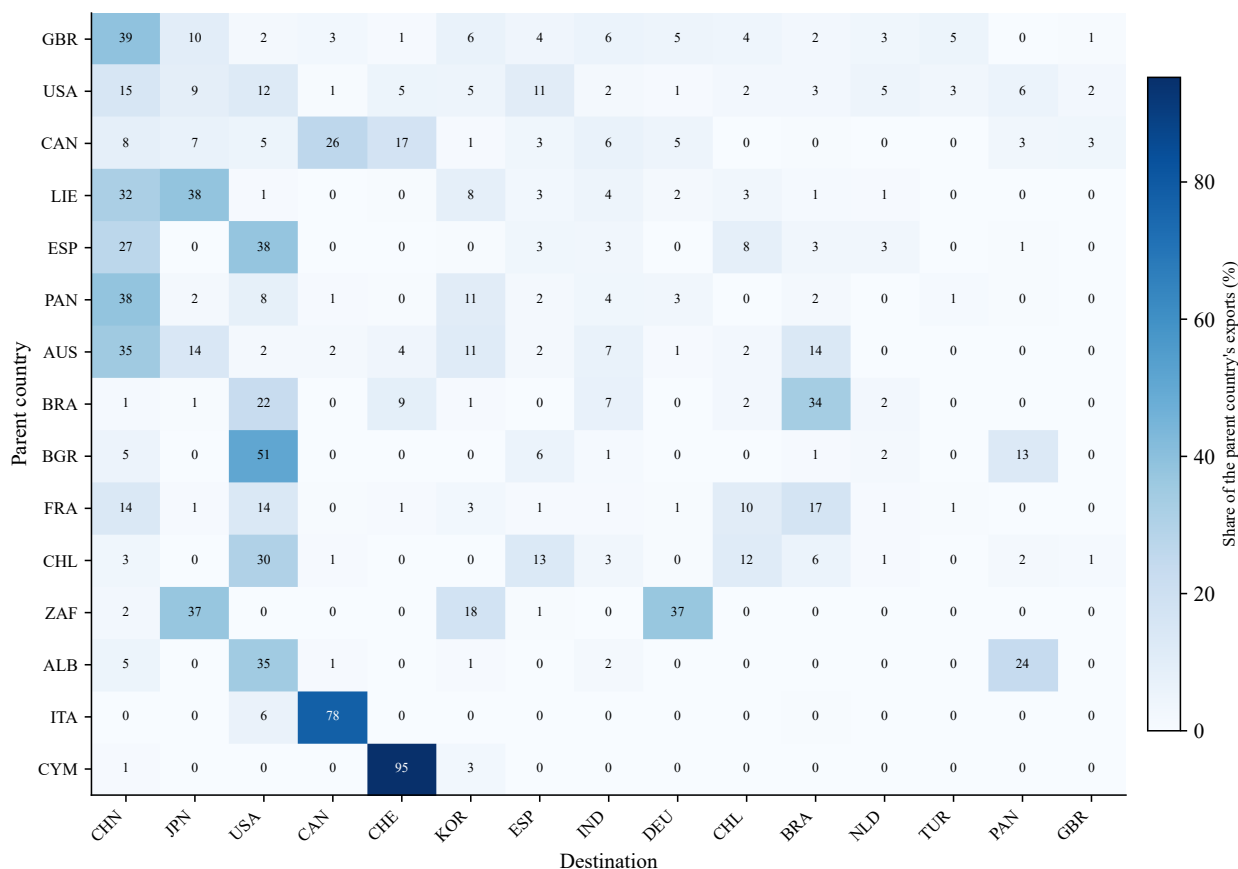
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 161:** Mining and Fuels: Foreign-Multinational Exports by Parent Country and Destination: Column Percentages

| Parent / Destination | CHN   | JPN   | USA   | CAN   | CHE   | KOR   | ESP   | IND   | DEU   | CHL   | Other destinations |
|----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------------------|
| GBR                  | 57.1  | 35.9  | 8.0   | 16.6  | 6.7   | 47.4  | 28.2  | 45.3  | 53.8  | 48.9  | 29.6               |
| USA                  | 12.8  | 19.0  | 30.8  | 2.1   | 21.5  | 19.7  | 46.3  | 8.0   | 5.2   | 15.9  | 30.8               |
| CAN                  | 5.2   | 10.5  | 9.0   | 70.6  | 54.8  | 3.5   | 11.0  | 22.4  | 22.6  | 1.7   | 12.3               |
| LIE                  | 9.1   | 26.9  | 0.9   | 0.0   | 0.6   | 11.8  | 3.9   | 7.0   | 3.7   | 7.3   | 2.2                |
| ESP                  | 3.8   | 0.0   | 15.4  | 0.2   | 0.2   | 0.3   | 2.4   | 2.4   | 0.0   | 9.4   | 2.7                |
| PAN                  | 4.3   | 0.4   | 2.6   | 0.4   | 0.3   | 6.1   | 1.0   | 2.2   | 2.9   | 0.2   | 3.3                |
| AUS                  | 3.5   | 3.6   | 0.6   | 0.9   | 1.9   | 5.7   | 1.0   | 3.6   | 0.6   | 1.8   | 1.8                |
| BRA                  | 0.1   | 0.1   | 4.4   | 0.0   | 2.9   | 0.3   | 0.1   | 2.7   | 0.0   | 1.3   | 3.7                |
| BGR                  | 0.3   | 0.0   | 8.7   | 0.0   | 0.0   | 0.0   | 1.7   | 0.4   | 0.1   | 0.0   | 2.1                |
| FRA                  | 0.8   | 0.1   | 2.2   | 0.1   | 0.2   | 0.7   | 0.3   | 0.3   | 0.2   | 4.5   | 2.8                |
| Other parents        | 3.0   | 3.5   | 17.4  | 9.1   | 10.9  | 4.4   | 4.0   | 5.8   | 10.8  | 9.1   | 8.7                |
| Total                | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0              |

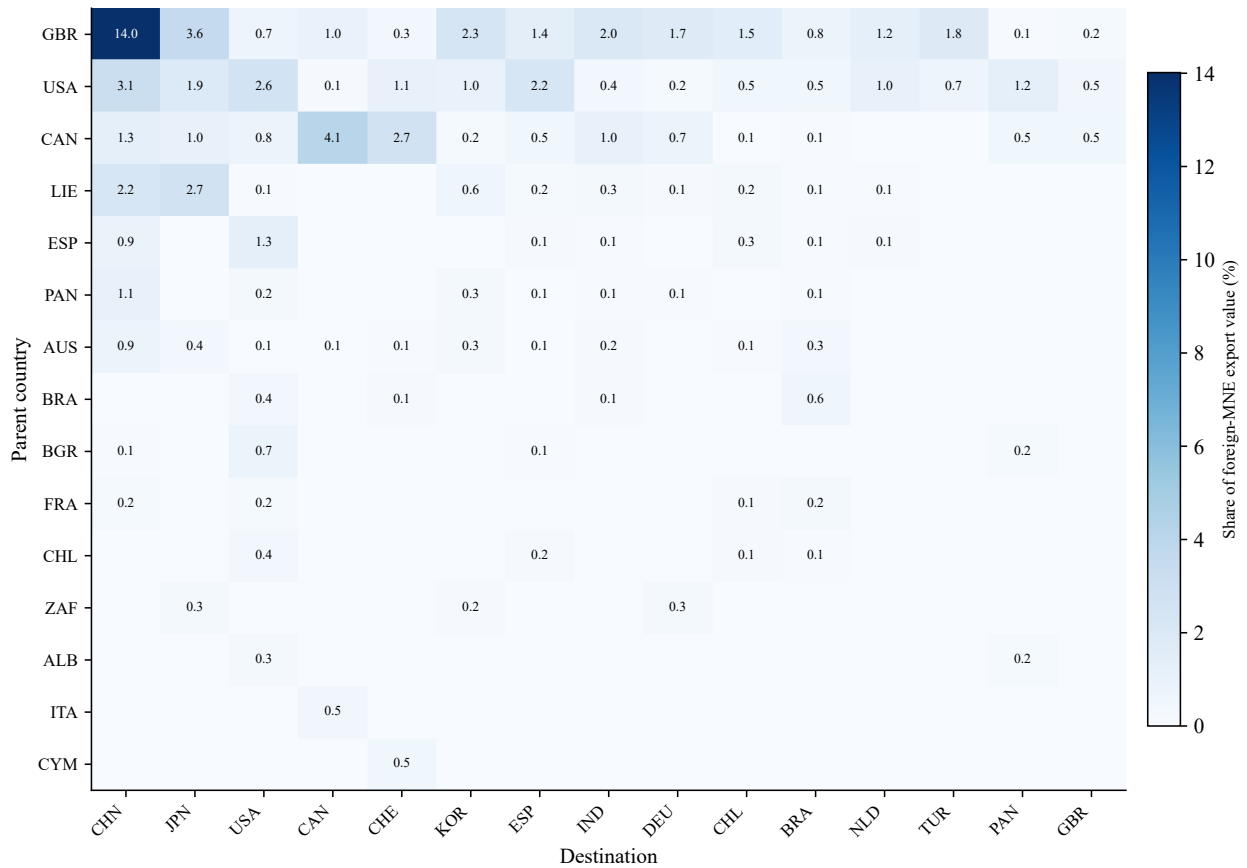
Notes: Percent of the foreign-MNE exports reaching each destination, by parent country (columns add to 100). Ten largest parent countries and destinations.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Figure 114:** Mining and Fuels: Destination of Foreign-Multinational Exports: Fifteen Largest Parent Countries and Destinations

Notes: Percent of each parent country's exports going to each destination (rows add to 100 including other destinations).

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

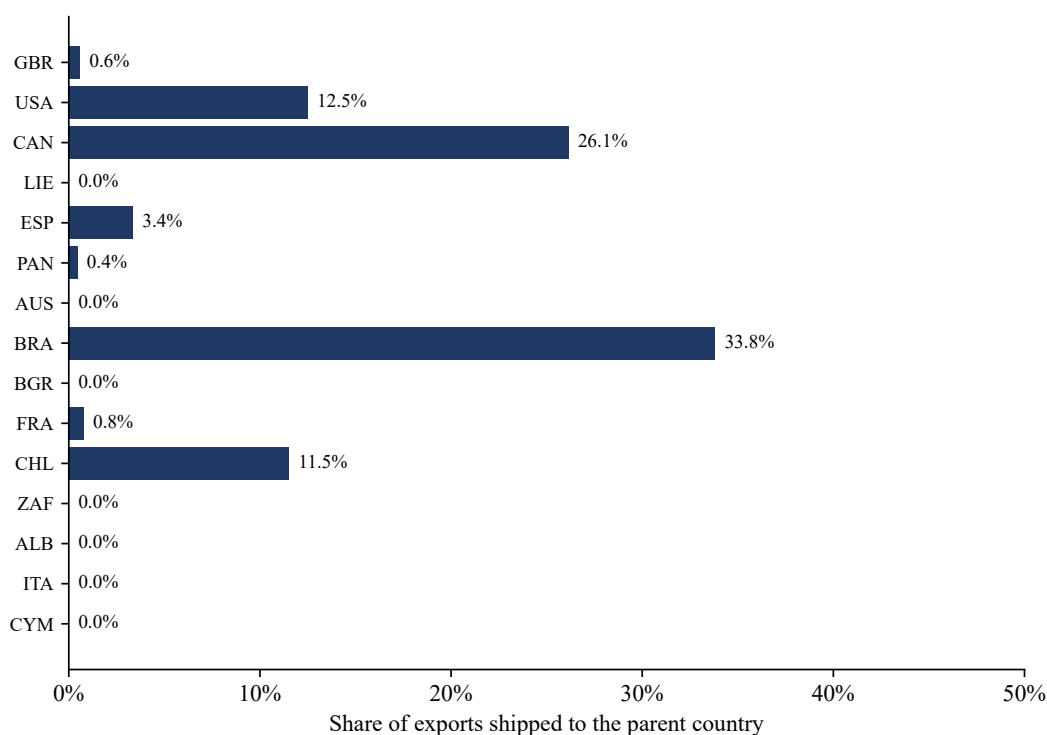


**Figure 115:** Mining and Fuels: Foreign-Multinational Exports by Parent Country and Destination: Share of the Total

*Notes:* Percent of all foreign-MNE export value in each parent country and destination pair; fifteen largest parent countries and destinations.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 20.4 Exports to the Parent Country



**Figure 116:** Mining and Fuels: Share of Foreign-Multinational Exports Shipped to the Parent Country

*Notes:* Percent of each parent country's exports shipped to that same country.

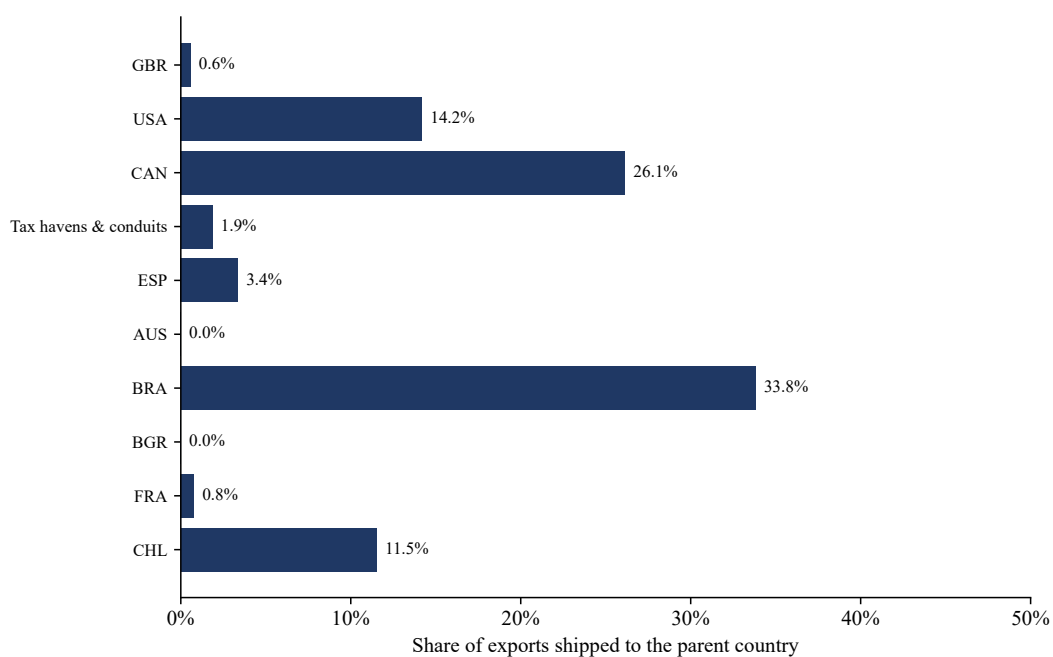
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 162:** Mining and Fuels: Foreign-Multinational Exports to the Parent Countries, by Parent Country

| Parent / Destination | GBR | USA  | CAN  | ESP  | PAN  | AUS | BRA  | BGR | FRA | Other destinations | Total |
|----------------------|-----|------|------|------|------|-----|------|-----|-----|--------------------|-------|
| GBR                  | 0.6 | 1.9  | 2.6  | 3.8  | 0.2  | 0.2 | 2.3  | 1.4 | 0.1 | 87.0               | 100.0 |
| USA                  | 2.3 | 12.5 | 0.6  | 10.7 | 5.8  | 0.0 | 2.5  | 0.3 | 1.1 | 64.2               | 100.0 |
| CAN                  | 3.4 | 4.9  | 26.1 | 3.4  | 3.3  | 0.0 | 0.5  | 0.7 | 0.0 | 57.6               | 100.0 |
| LIE                  | 0.0 | 1.1  | 0.0  | 2.7  | 0.0  | 0.0 | 1.4  | 2.3 | 0.0 | 92.4               | 100.0 |
| ESP                  | 0.3 | 37.6 | 0.3  | 3.4  | 0.7  | 0.0 | 3.0  | 0.0 | 0.7 | 54.1               | 100.0 |
| PAN                  | 0.2 | 8.0  | 0.9  | 1.8  | 0.4  | 5.6 | 2.2  | 4.8 | 0.1 | 75.9               | 100.0 |
| AUS                  | 0.0 | 2.1  | 2.1  | 2.1  | 0.0  | 0.0 | 13.7 | 0.0 | 0.0 | 80.0               | 100.0 |
| BRA                  | 0.0 | 22.4 | 0.1  | 0.2  | 0.0  | 0.8 | 33.8 | 0.0 | 0.0 | 42.8               | 100.0 |
| BGR                  | 0.0 | 50.6 | 0.0  | 5.6  | 12.8 | 0.0 | 1.1  | 0.0 | 0.0 | 29.9               | 100.0 |
| FRA                  | 0.0 | 14.0 | 0.4  | 1.0  | 0.0  | 0.5 | 16.6 | 0.0 | 0.8 | 66.6               | 100.0 |
| Other parents        | 0.9 | 20.7 | 7.4  | 2.7  | 3.7  | 0.0 | 2.1  | 0.4 | 0.6 | 61.4               | 100.0 |

*Notes:* Percent of each parent country's exports, by destination (rows add to 100). Destinations are the ten largest parent countries; the diagonal is the share shipped to the parent's own country.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 117:** Mining and Fuels: Share of Foreign-Multinational Exports Shipped to the Parent Jurisdiction, Consolidated Parents

*Notes:* Percent of each parent's exports shipped to its own jurisdiction. Dependencies are assigned to their sovereign state and tax havens and conduit jurisdictions are pooled.

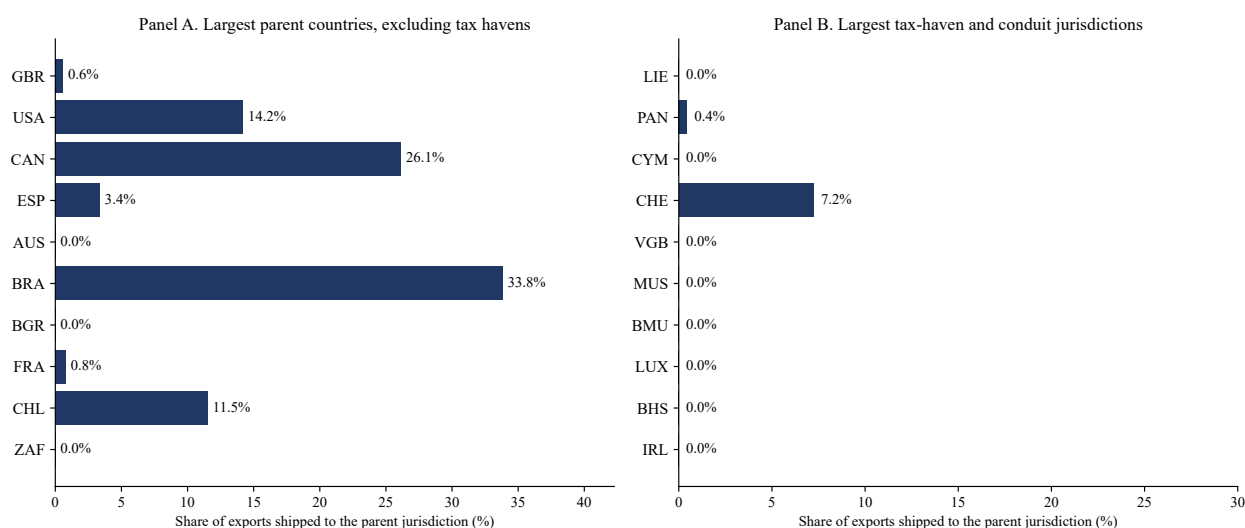
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 163:** Mining and Fuels: Share of Foreign-Multinational Exports Shipped to the Parent Jurisdiction, Consolidated Parents

| Consolidated parent   | Home share (%) | Exports (US\$bn/yr) |
|-----------------------|----------------|---------------------|
| GBR                   | 0.6            | 12.2                |
| USA                   | 14.2           | 7.3                 |
| CAN                   | 26.1           | 5.7                 |
| Tax havens & conduits | 1.9            | 3.3                 |
| ESP                   | 3.4            | 1.4                 |
| AUS                   | 0.0            | 0.9                 |
| BRA                   | 33.8           | 0.7                 |
| BGR                   | 0.0            | 0.5                 |
| FRA                   | 0.8            | 0.5                 |
| CHL                   | 11.5           | 0.4                 |

*Notes:* Home share: percent of the parent's exports shipped to its own jurisdiction. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 118: Mining and Fuels: Share of Foreign-Multinational Exports Shipped to the Parent Jurisdiction**

*Notes:* Percent of each parent jurisdiction's exports shipped to that same jurisdiction. Panel A: largest parent countries, excluding tax havens; dependencies are assigned to their sovereign state. Panel B: largest tax-haven and conduit jurisdictions.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

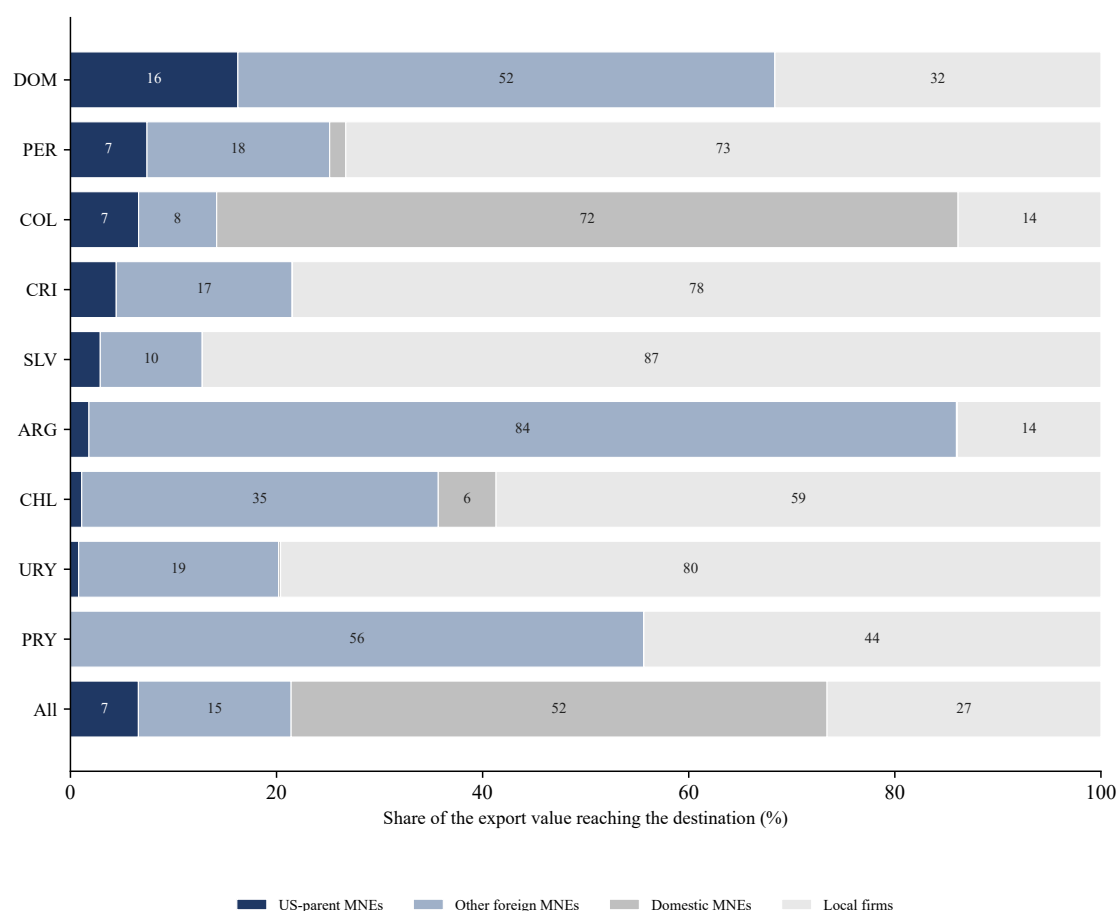
**Table 164: Mining and Fuels: Share of Foreign-Multinational Exports Shipped to the Parent Jurisdiction**

| Panel   | Parent | Home share (%) | Exports (US\$bn/yr) |
|---------|--------|----------------|---------------------|
| Panel A | GBR    | 0.6            | 11.9                |
| Panel A | USA    | 14.2           | 7.3                 |
| Panel A | CAN    | 26.1           | 5.7                 |
| Panel A | ESP    | 3.4            | 1.4                 |
| Panel A | AUS    | 0.0            | 0.9                 |
| Panel A | BRA    | 33.8           | 0.7                 |
| Panel A | BGR    | 0.0            | 0.5                 |
| Panel A | FRA    | 0.8            | 0.5                 |
| Panel A | CHL    | 11.5           | 0.4                 |
| Panel A | ZAF    | 0.0            | 0.3                 |
| Panel B | LIE    | 0.0            | 2.0                 |
| Panel B | PAN    | 0.4            | 1.0                 |
| Panel B | CYM    | 0.0            | 0.2                 |
| Panel B | CHE    | 7.2            | 0.1                 |
| Panel B | VGB    | 0.0            | 0.1                 |
| Panel B | MUS    | 0.0            | 0.1                 |
| Panel B | BMU    | 0.0            | 0.1                 |
| Panel B | LUX    | 0.0            | 0.0                 |
| Panel B | BHS    | 0.0            | 0.0                 |
| Panel B | IRL    | 0.0            | 0.0                 |

*Notes:* Home share: percent of the parent's exports shipped to its own jurisdiction. Panel A: largest parent countries, excluding tax havens. Panel B: largest tax-haven and conduit jurisdictions. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 20.5 Destination Markets



**Figure 119:** Mining and Fuels: Exports to the United States, by Type of Exporter and Exporting Country

*Notes:* Percent of each country's mining and fuels exports to the United States, by type of exporter.

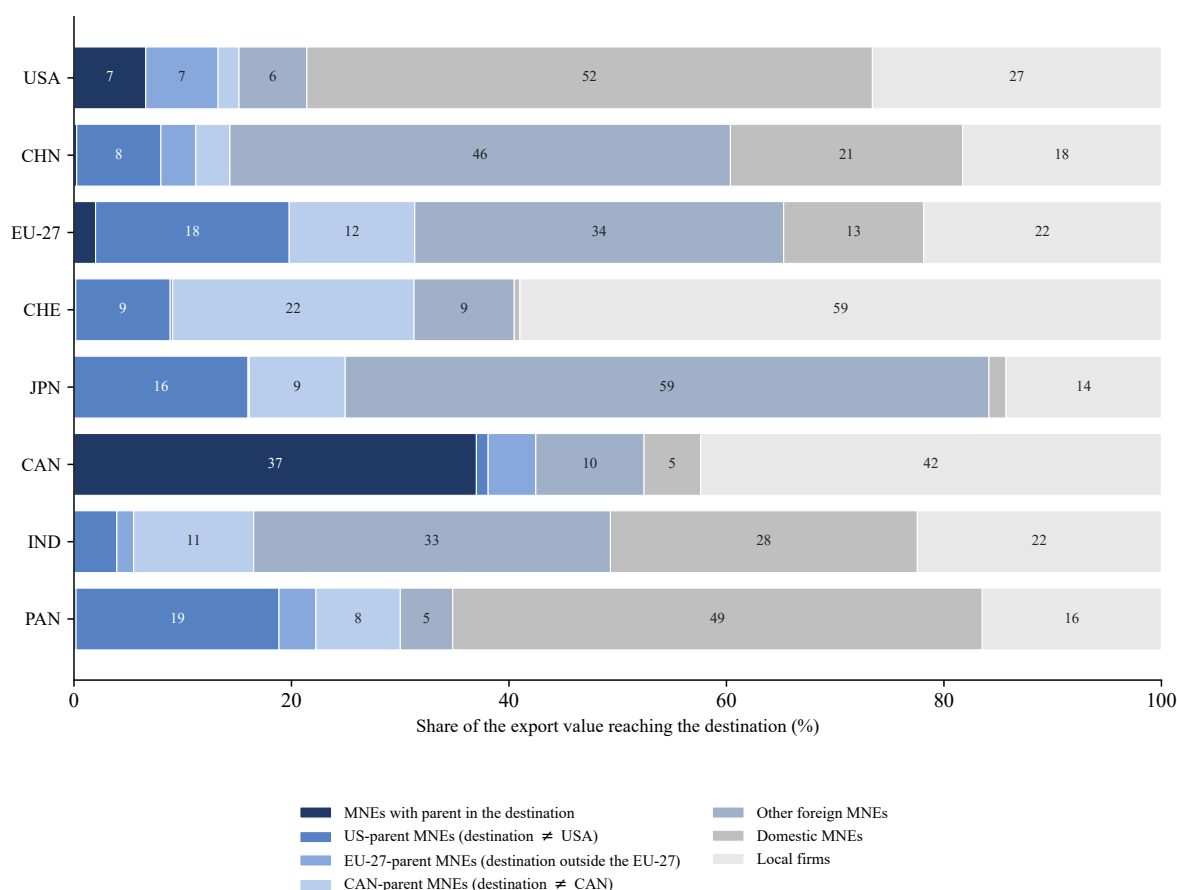
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 165:** Mining and Fuels: Exports to the United States, by Type of Exporter and Exporting Country

| Origin | Exports to the USA (US\$bn/yr) | US-parent MNEs (%) | Other foreign MNEs (%) | Domestic MNEs (%) | Local firms (%) |
|--------|--------------------------------|--------------------|------------------------|-------------------|-----------------|
| DOM    | 0.5                            | 16.2               | 52.1                   | 0.0               | 31.6            |
| PER    | 3.0                            | 7.4                | 17.7                   | 1.6               | 73.3            |
| COL    | 9.9                            | 6.6                | 7.6                    | 71.9              | 13.9            |
| CRI    | 0.1                            | 4.4                | 17.1                   | 0.0               | 78.4            |
| SLV    | 0.0                            | 2.9                | 9.9                    | 0.0               | 87.2            |
| ARG    | 0.9                            | 1.8                | 84.2                   | 0.1               | 13.9            |
| CHL    | 0.3                            | 1.1                | 34.6                   | 5.6               | 58.7            |
| URY    | 0.0                            | 0.8                | 19.4                   | 0.2               | 79.6            |
| PRY    | 0.0                            | 0.0                | 55.6                   | 0.0               | 44.3            |
| All    | 14.6                           | 6.6                | 14.8                   | 52.0              | 26.6            |

*Notes:* Percent of each country's mining and fuels exports to the United States, by type of exporter. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 120: Mining and Fuels: Ownership of Exports to the Main Destination Markets**

*Notes:* Percent of the export value reaching each destination, by type of exporter. The first segment is exports by MNEs whose parent is located in that destination. EU-27 member states are pooled.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 166: Mining and Fuels: Ownership of Exports to the Main Destination Markets**

| Destination | Exports (US\$bn/yr) | MNEs with parent in the destination (%) | US-parent MNEs (%) | EU-27-parent MNEs (%) | CAN-parent MNEs (%) | Other foreign MNEs (%) | Domestic MNEs (%) | Local firms (%) |
|-------------|---------------------|-----------------------------------------|--------------------|-----------------------|---------------------|------------------------|-------------------|-----------------|
| USA         | 14.6                | 6.6                                     | –                  | 6.7                   | 1.9                 | 6.2                    | 52.0              | 26.6            |
| CHN         | 13.8                | 0.3                                     | 7.7                | 3.2                   | 3.1                 | 46.0                   | 21.3              | 18.3            |
| EU-27       | 8.5                 | 2.0                                     | 17.8               | –                     | 11.5                | 33.9                   | 12.9              | 21.9            |
| CHE         | 4.9                 | 0.2                                     | 8.7                | 0.2                   | 22.2                | 9.2                    | 0.5               | 59.0            |
| JPN         | 3.8                 | 0.0                                     | 16.0               | 0.1                   | 8.8                 | 59.2                   | 1.6               | 14.3            |
| CAN         | 4.4                 | 37.0                                    | 1.1                | 4.4                   | –                   | 9.9                    | 5.2               | 42.4            |
| IND         | 3.2                 | 0.0                                     | 3.9                | 1.5                   | 11.0                | 32.8                   | 28.2              | 22.4            |
| PAN         | 2.5                 | 0.2                                     | 18.7               | 3.4                   | 7.8                 | 4.8                    | 48.7              | 16.5            |

*Notes:* Percent of the export value reaching each destination, by type of exporter. EU-27 member states are pooled. Values are annual averages in US\$ billion.

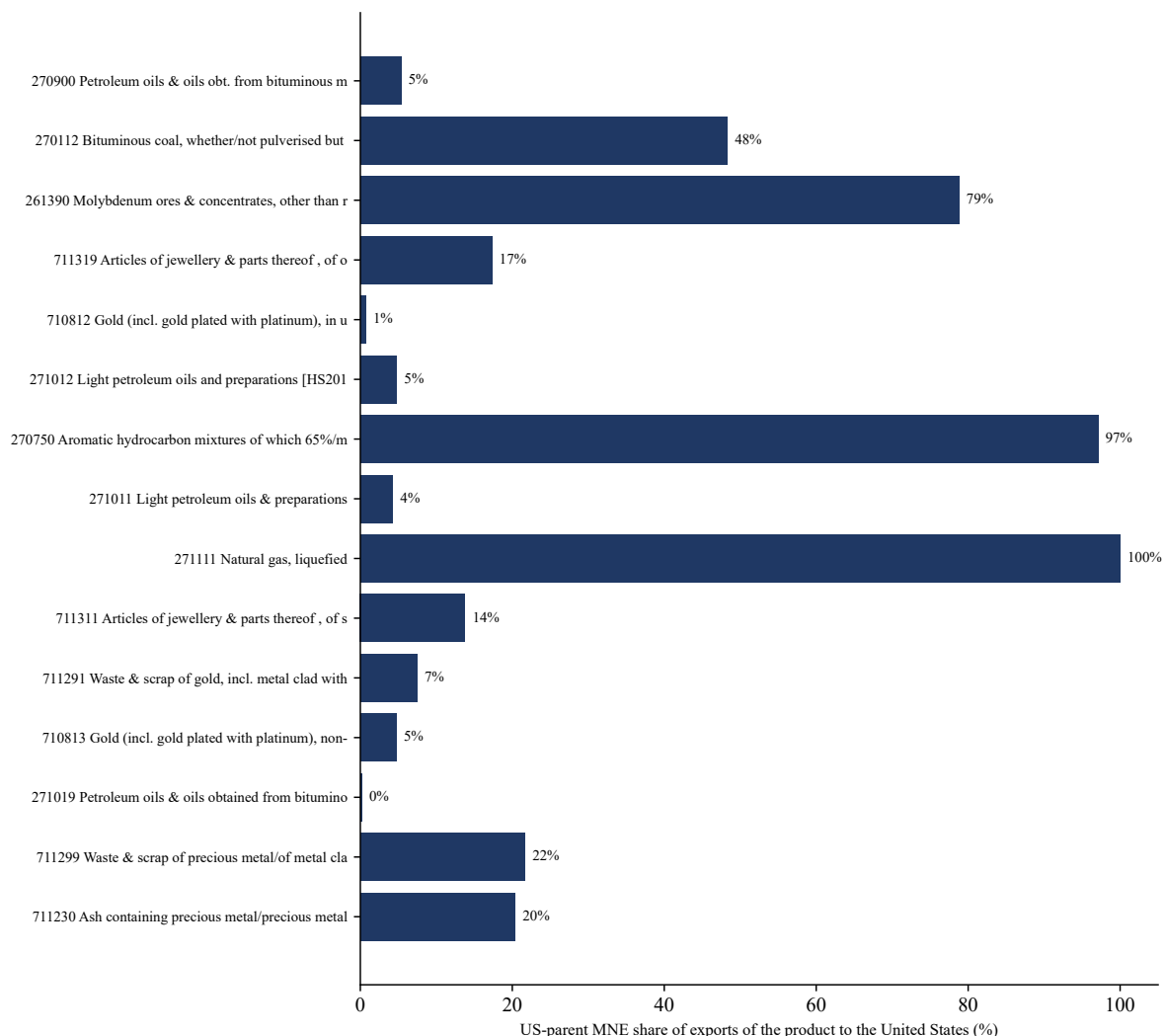
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 167: Mining and Fuels: Main Products Shipped by US-Parent Multinationals to the United States**

| HS6       | Description                                      | US-parent exports to the USA (US\$bn/yr) | % of all US-parent exports to the USA | % of US-parent exports of the product going to the USA | All exports of the product to the USA (US\$bn/yr) | US-parent share of exports of the product to the USA (%) | Main origins of the US-parent shipments (% of the line) |
|-----------|--------------------------------------------------|------------------------------------------|---------------------------------------|--------------------------------------------------------|---------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------|
| 270900    | Petroleum oils & oils obt. from bituminous mi... | 0.41                                     | 48.7                                  | 24                                                     | 7.84                                              | 5                                                        | COL 100%, ARG 0%                                        |
| 270112    | Bituminous coal, whether/not pulverised but n... | 0.19                                     | 22.6                                  | 9                                                      | 0.39                                              | 48                                                       | COL 100%                                                |
| 261390    | Molybdenum ores & concentrates, other than ro... | 0.15                                     | 14.8                                  | 99                                                     | 0.18                                              | 79                                                       | PER 100%                                                |
| 711319    | Articles of jewellery & parts thereof, of ot...  | 0.06                                     | 4.9                                   | 100                                                    | 0.35                                              | 17                                                       | DOM 100%                                                |
| 710812    | Gold (incl. gold plated with platinum), in un... | 0.02                                     | 1.7                                   | 4                                                      | 2.72                                              | 1                                                        | PER 56%, COL 23%, CHL 21%                               |
| 271012    | Light petroleum oils and preparations [HS2012... | 0.01                                     | 1.6                                   | 40                                                     | 0.27                                              | 5                                                        | COL 100%, CRI 0%, ARG 0%                                |
| 270750    | Aromatic hydrocarbon mixtures of which 65%/mo... | 0.02                                     | 1.4                                   | 98                                                     | 0.02                                              | 97                                                       | ARG 100%, COL 0%                                        |
| 271011    | Light petroleum oils & preparations              | 0.01                                     | 1.2                                   | 42                                                     | 0.69                                              | 4                                                        | COL 100%, SLV 0%, DOM 0%                                |
| 271111    | Natural gas, liquefied                           | 0.01                                     | 1.0                                   | 1                                                      | 0.01                                              | 100                                                      | PER 100%, DOM 0%                                        |
| 711311    | Articles of jewellery & parts thereof, of si...  | 0.01                                     | 0.7                                   | 100                                                    | 0.06                                              | 14                                                       | DOM 100%, ARG 0%, PER 0%                                |
| 711291    | Waste & scrap of gold, incl. metal clad with ... | 0.01                                     | 0.5                                   | 100                                                    | 0.06                                              | 7                                                        | DOM 72%, CRI 20%, COL 5%                                |
| 710813    | Gold (incl. gold plated with platinum), non-m... | 0.01                                     | 0.5                                   | 100                                                    | 0.12                                              | 5                                                        | DOM 99%, CRI 1%                                         |
| 271019    | Petroleum oils & oils obtained from bituminou... | 0.00                                     | 0.2                                   | 2                                                      | 1.14                                              | 0                                                        | SLV 91%, COL 5%, URY 3%                                 |
| 711299    | Waste & scrap of precious metal/of metal clad... | 0.00                                     | 0.1                                   | 83                                                     | 0.01                                              | 22                                                       | CRI 92%, COL 3%, DOM 3%                                 |
| 711230    | Ash containing precious metal/precious metal ... | 0.00                                     | 0.1                                   | 100                                                    | 0.00                                              | 20                                                       | DOM 100%                                                |
| All lines |                                                  | 0.9                                      | 100.0                                 | 12                                                     | 14.6                                              | 7                                                        |                                                         |

*Notes:* Fifteen HS6 products with the largest exports by US-parent MNEs to the United States. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Figure 121: Mining and Fuels: US-Parent Share of Exports to the United States, Main Products Shipped by US-Parent Multinationals**

*Notes:* Percent of the nine countries' exports of each product to the United States shipped by US-parent MNEs; the fifteen products with the largest US-parent exports to the United States.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 168:** Mining and Fuels: Largest Export Products to the United States, by Type of Exporter

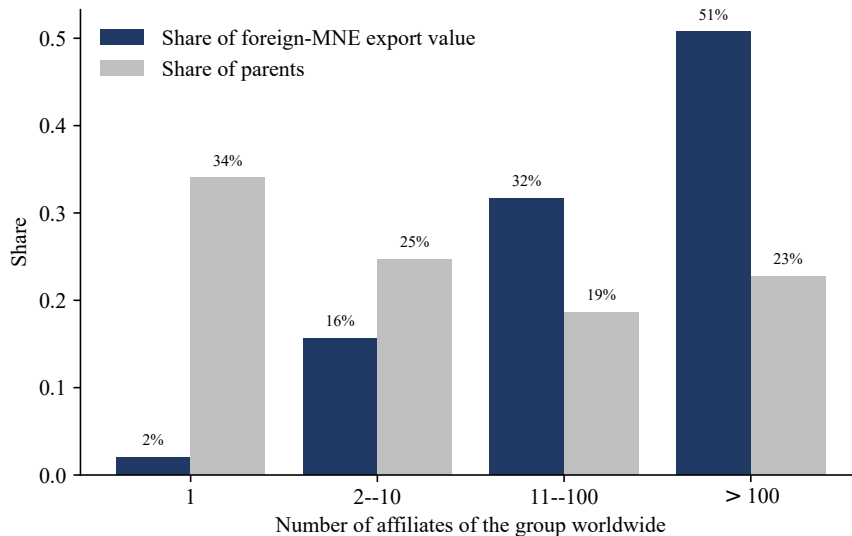
| HS6    | Description                                      | Exports to the<br>USA<br>(US\$bn/yr) | US-parent<br>MNEs (%) | Other foreign<br>MNEs (%) | Domestic<br>MNEs (%) | Local firms<br>(%) |
|--------|--------------------------------------------------|--------------------------------------|-----------------------|---------------------------|----------------------|--------------------|
| 270900 | Petroleum oils & oils obt. from bituminous mi... | 7.84                                 | 5                     | 10                        | 81                   | 4                  |
| 710812 | Gold (incl. gold plated with platinum), in un... | 2.72                                 | 1                     | 16                        | 9                    | 74                 |
| 271019 | Petroleum oils & oils obtained from bituminou... | 1.14                                 | 0                     | 14                        | 51                   | 35                 |
| 271011 | Light petroleum oils & preparations              | 0.69                                 | 4                     | 14                        | 5                    | 76                 |
| 270112 | Bituminous coal, whether/not pulverised but n... | 0.39                                 | 48                    | 32                        | 1                    | 19                 |
| 710691 | Silver (incl. silver plated with gold/platinu... | 0.34                                 | 0                     | 31                        | 0                    | 69                 |
| 271012 | Light petroleum oils and preparations [HS2012... | 0.27                                 | 5                     | 32                        | 43                   | 20                 |
| 711319 | Articles of jewellery & parts thereof, of ot...  | 0.35                                 | 17                    | 46                        | 0                    | 37                 |
| 261390 | Molybdenum ores & concentrates, other than ro... | 0.18                                 | 79                    | 18                        | 0                    | 3                  |
| 710813 | Gold (incl. gold plated with platinum), non-m... | 0.12                                 | 5                     | 51                        | 3                    | 41                 |
| 251010 | Natural calcium phosphates, natural aluminium... | 0.11                                 | 0                     | 100                       | 0                    | 0                  |
| 250100 | Salt (incl. table salt & denatured salt) & pu... | 0.06                                 | 0                     | 62                        | 9                    | 28                 |
| 710391 | Rubies, sapphires & emeralds, worked othw. th... | 0.06                                 | 0                     | 28                        | 0                    | 72                 |
| 711291 | Waste & scrap of gold, incl. metal clad with ... | 0.06                                 | 7                     | 5                         | 0                    | 87                 |
| 711311 | Articles of jewellery & parts thereof, of si...  | 0.06                                 | 14                    | 57                        | 0                    | 29                 |

*Notes:* Fifteen largest HS6 products exported to the United States; percent of each product's exports to the United States by type of exporter. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 21 Fact 4: A Small Set of Large Multinational Groups Accounts for Most Exports

### 21.1 Size of the Group's Global Network



**Figure 122:** Mining and Fuels: Foreign-Multinational Exports and Number of Parents, by Size of the Group's Global Network

*Notes:* Shares of foreign-MNE mining and fuels export value and of parents, by the number of affiliates the group has worldwide.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

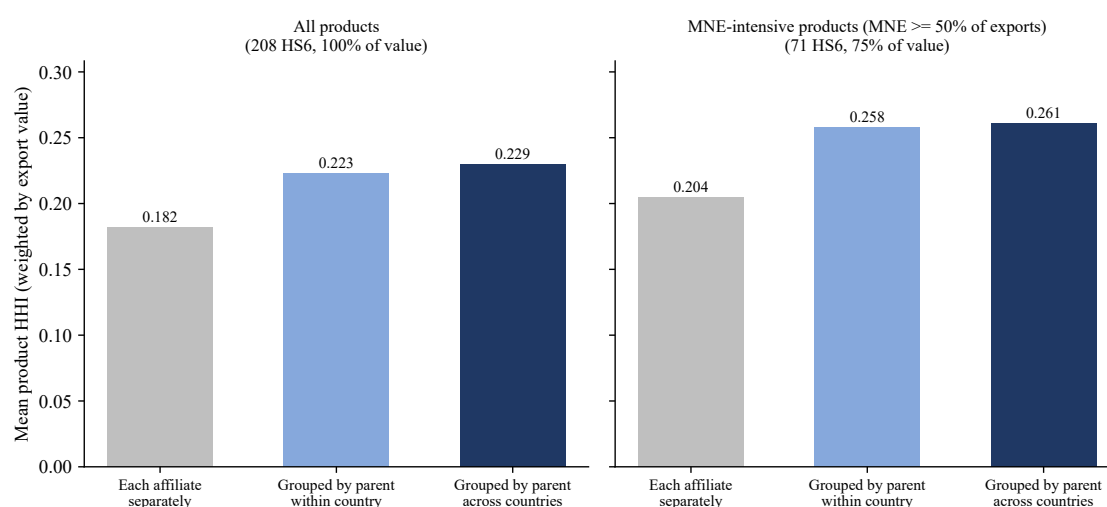
**Table 169:** Mining and Fuels: Foreign-Multinational Exports and Number of Parents, by Size of the Group's Global Network

| Affiliates worldwide | Parents | Share of parents (%) | Share of foreign-MNE export value (%) |
|----------------------|---------|----------------------|---------------------------------------|
| 1                    | 521     | 34.0                 | 2.0                                   |
| 2–10                 | 378     | 24.7                 | 15.6                                  |
| 11–100               | 285     | 18.6                 | 31.7                                  |
| >100                 | 348     | 22.7                 | 50.7                                  |
| All parents          | 1,532   | 100                  | 100                                   |

Notes: Number and share of parents, and share of foreign-MNE mining and fuels export value, by the number of affiliates the group has worldwide.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 21.2 Concentration within Products



**Figure 123:** Mining and Fuels: Export Concentration within Products: Firms and Groups

Notes: Mean Herfindahl index of exporters within each HS6 product, weighted by export value, counting each affiliate separately or grouping affiliates by parent within or across countries. Right panel: products where MNEs account for at least half of exports.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 170:** Mining and Fuels: Export Concentration within Products: Firms and Groups

| Sample                                                                     | Measure                               | Firm level | Groups within country | Groups across countries |
|----------------------------------------------------------------------------|---------------------------------------|------------|-----------------------|-------------------------|
| <i>All products: 208 HS6, 100% of value</i>                                |                                       |            |                       |                         |
|                                                                            | Mean HHI                              | 0.182      | 0.223                 | 0.229                   |
|                                                                            | Mean top-exporter share               | 31.1%      | 34.9%                 | 36.7%                   |
|                                                                            | Effective number of exporters (1/HHI) | 11.4       | 10.6                  | 7.9                     |
| <i>MNE-intensive products (MNE ≥ 50% of exports): 71 HS6, 75% of value</i> |                                       |            |                       |                         |
|                                                                            | Mean HHI                              | 0.204      | 0.258                 | 0.261                   |
|                                                                            | Mean top-exporter share               | 34.6%      | 39.7%                 | 39.7%                   |
|                                                                            | Effective number of exporters (1/HHI) | 7.7        | 6.7                   | 6.3                     |

Notes: Means across HS6 products, weighted by export value. Firm level: each affiliate counted separately. Groups: affiliates grouped by ultimate parent within each exporting country or across countries.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 21.3 Groups and Affiliates

**Table 171:** Mining and Fuels: Multinational Affiliates and Groups in Origin–Destination–Product–Year Cells

| Origin-destination-product-year cells with $\geq 1$ MNE   |        |
|-----------------------------------------------------------|--------|
| Cells                                                     | 17,619 |
| MNE affiliates per cell (mean)                            | 1.69   |
| MNE groups per cell (mean)                                | 1.67   |
| Affiliates per group, pooled                              | 1.01   |
| Affiliates per group, mean weighted by export value       | 1.09   |
| Cells where one group exports through $> 1$ affiliate (%) | 1.5    |

*Notes:* Cells with at least one MNE. Group: ultimate parent in Orbis or Dun & Bradstreet (parent name when the identifier is missing).

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 172:** Mining and Fuels: Multinational Presence and Exports: Affiliates and Groups

|                                                               | Origin $\times$ year, destination $\times$ year, product FE |                       |                       | Origin $\times$ dest. $\times$ product, origin $\times$ dest. $\times$ year FE |                       |                       |
|---------------------------------------------------------------|-------------------------------------------------------------|-----------------------|-----------------------|--------------------------------------------------------------------------------|-----------------------|-----------------------|
|                                                               | (1)                                                         | (2)                   | (3)                   | (4)                                                                            | (5)                   | (6)                   |
| <i>Panel A. Dependent variable: ln exports, all firms</i>     |                                                             |                       |                       |                                                                                |                       |                       |
| ln(# MNE firms)                                               | 2.5367***<br>(0.0797)                                       |                       |                       | 0.8789***<br>(0.0732)                                                          |                       |                       |
| ln(# MNE groups)                                              |                                                             | 2.5655***<br>(0.0800) | 2.5498***<br>(0.0794) |                                                                                | 0.8787***<br>(0.0735) | 0.8799***<br>(0.0734) |
| ln(affiliates per group)                                      |                                                             |                       | 1.7853***<br>(0.5380) |                                                                                |                       | 0.8051***<br>(0.2947) |
| <i>Panel B. Dependent variable: ln exports, non-MNE firms</i> |                                                             |                       |                       |                                                                                |                       |                       |
| ln(# MNE firms)                                               | 1.4518***<br>(0.0976)                                       |                       |                       | 0.1605*<br>(0.0952)                                                            |                       |                       |
| ln(# MNE groups)                                              |                                                             | 1.4559***<br>(0.0974) | 1.4511***<br>(0.0970) |                                                                                | 0.1684*<br>(0.0939)   | 0.1651*<br>(0.0947)   |
| ln(affiliates per group)                                      |                                                             |                       | 1.5168**<br>(0.7448)  |                                                                                |                       | -0.5378<br>(0.7197)   |
| Product FE                                                    | ✓                                                           | ✓                     | ✓                     |                                                                                |                       |                       |
| Origin $\times$ year FE                                       | ✓                                                           | ✓                     | ✓                     |                                                                                |                       |                       |
| Destination $\times$ year FE                                  | ✓                                                           | ✓                     | ✓                     |                                                                                |                       |                       |
| Origin $\times$ dest. $\times$ product FE                     |                                                             |                       |                       | ✓                                                                              | ✓                     | ✓                     |
| Origin $\times$ dest. $\times$ year FE                        |                                                             |                       |                       | ✓                                                                              | ✓                     | ✓                     |
| Observations (Panel A)                                        | 17,183                                                      | 17,183                | 17,183                | 13,668                                                                         | 13,668                | 13,668                |
| Observations (Panel B)                                        | 7,153                                                       | 7,153                 | 7,153                 | 5,898                                                                          | 5,898                 | 5,898                 |

*Notes:* Observations are origin–destination–HS6–year cells. Mining and Fuels products only. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Regressors: ln number of MNE affiliates or ln number of MNE groups (distinct ultimate parents). Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 22 Fact 5: Multinational Presence Is Associated with Higher Exports

**Table 173:** Mining and Fuels: Multinational Presence and Exports

|                                                               | Intensive: ln(# MNE firms) |                       |                       | Extensive: any MNE present |                       |                       |
|---------------------------------------------------------------|----------------------------|-----------------------|-----------------------|----------------------------|-----------------------|-----------------------|
|                                                               | (1)                        | (2)                   | (3)                   | (1)                        | (2)                   | (3)                   |
| <i>Panel A. Dependent variable: ln exports, all firms</i>     |                            |                       |                       |                            |                       |                       |
| ln(# MNE firms)                                               | 2.5346***<br>(0.0755)      | 2.5367***<br>(0.0797) | 0.8789***<br>(0.0732) |                            |                       |                       |
| Any MNE present                                               |                            |                       |                       | 1.3777***<br>(0.0890)      | 1.3943***<br>(0.0929) | 0.6626***<br>(0.0589) |
| <i>Panel B. Dependent variable: ln exports, non-MNE firms</i> |                            |                       |                       |                            |                       |                       |
| ln(# MNE firms)                                               | 1.4407***<br>(0.0898)      | 1.4518***<br>(0.0976) | 0.1605*<br>(0.0952)   |                            |                       |                       |
| Any MNE present                                               |                            |                       |                       | 1.6657***<br>(0.0962)      | 1.6631***<br>(0.1016) | 0.1846***<br>(0.0505) |
| Origin FE                                                     | ✓                          |                       |                       | ✓                          |                       |                       |
| Destination FE                                                | ✓                          |                       |                       | ✓                          |                       |                       |
| Product FE                                                    | ✓                          | ✓                     |                       | ✓                          | ✓                     |                       |
| Year FE                                                       | ✓                          |                       |                       | ✓                          |                       |                       |
| Origin × year FE                                              |                            | ✓                     |                       |                            | ✓                     |                       |
| Destination × year FE                                         |                            | ✓                     |                       |                            | ✓                     |                       |
| Origin × dest. × product FE                                   |                            |                       | ✓                     |                            |                       | ✓                     |
| Origin × dest. × year FE                                      |                            |                       | ✓                     |                            |                       | ✓                     |
| Observations (Panel A)                                        | 17,583                     | 17,183                | 13,668                | 33,290                     | 32,912                | 27,664                |
| Observations (Panel B)                                        | 7,425                      | 7,153                 | 5,898                 | 23,133                     | 22,836                | 18,666                |

*Notes:* Observations are origin–destination–HS6–year cells. Mining and Fuels products only. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Intensive margin: ln number of MNEs, cells with at least one MNE. Extensive margin: indicator for at least one MNE. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 174:** Mining and Fuels: Multinational Exports by Exporting Country and Destination Type

| Origin | MNE value (US\$bn/yr) | of which domestic (%) | Foreign-MNE value (US\$bn/yr) | % of foreign-MNE value, by destination |                                         |
|--------|-----------------------|-----------------------|-------------------------------|----------------------------------------|-----------------------------------------|
|        |                       |                       |                               | parent country                         | other (of which: group affiliate there) |
| ARG    | 5.0                   | 0.4                   | 5.0                           | 20.1                                   | 79.9 (38.6)                             |
| CHL    | 8.3                   | 9.4                   | 7.5                           | 1.8                                    | 98.2 (64.3)                             |
| COL    | 23.8                  | 64.8                  | 8.4                           | 8.6                                    | 91.4 (27.0)                             |
| CRI    | 0.1                   | 0.7                   | 0.1                           | 6.9                                    | 93.1 (60.5)                             |
| DOM    | 1.7                   | 0.0                   | 1.7                           | 43.7                                   | 56.3 (8.8)                              |
| PER    | 13.4                  | 2.0                   | 13.2                          | 3.6                                    | 96.4 (33.2)                             |
| PRY    | 0.0                   | 1.1                   | 0.0                           | 0.7                                    | 99.3 (23.0)                             |
| SLV    | 0.1                   | 0.0                   | 0.1                           | 4.9                                    | 95.1 (20.4)                             |
| URY    | 0.1                   | 0.1                   | 0.1                           | 2.4                                    | 97.6 (2.9)                              |
| All    | 52.4                  | 33.3                  | 36.0                          | 7.6                                    | 92.4 (39.6)                             |

*Notes:* MNE and foreign-MNE mining and fuels export value by country, and the split of foreign-MNE value by destination. Parent country: exports to the country of the ultimate parent. Other: all remaining destinations; in parentheses, destinations where the group has another affiliate. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 175:** Mining and Fuels: Multinational Presence and Exports: Intensive Margin, by Type of Multinational

|                                                               | All MNEs              |                       | Foreign / domestic    |                     | Foreign by destination, domestic |     |
|---------------------------------------------------------------|-----------------------|-----------------------|-----------------------|---------------------|----------------------------------|-----|
|                                                               | (1)                   | (2)                   | (3)                   | (4)                 | (5)                              | (6) |
| <i>Panel A. Dependent variable: ln exports, all firms</i>     |                       |                       |                       |                     |                                  |     |
| ln(# MNE firms)                                               | 2.5367***<br>(0.0797) | 0.8789***<br>(0.0732) |                       |                     |                                  |     |
| ln(# foreign MNEs)                                            |                       |                       | 1.1301***<br>(0.1799) | 0.2479<br>(0.1778)  |                                  |     |
| ln(# domestic MNEs)                                           |                       |                       | 0.3172<br>(0.2062)    | 0.3216*<br>(0.1639) |                                  |     |
| <i>Panel B. Dependent variable: ln exports, non-MNE firms</i> |                       |                       |                       |                     |                                  |     |
| ln(# MNE firms)                                               | 1.4518***<br>(0.0976) | 0.1605*<br>(0.0952)   |                       |                     |                                  |     |
| ln(# foreign MNEs)                                            |                       |                       | 0.9420***<br>(0.2641) | -0.1512<br>(0.3320) |                                  |     |
| ln(# domestic MNEs)                                           |                       |                       | -0.6594<br>(0.4830)   | -0.7273<br>(0.6536) |                                  |     |
| Product FE                                                    | ✓                     |                       | ✓                     |                     | ✓                                |     |
| Origin × year FE                                              | ✓                     |                       | ✓                     |                     | ✓                                |     |
| Destination × year FE                                         | ✓                     |                       | ✓                     |                     | ✓                                |     |
| Origin × dest. × product FE                                   |                       | ✓                     |                       | ✓                   |                                  | ✓   |
| Origin × dest. × year FE                                      |                       | ✓                     |                       | ✓                   |                                  | ✓   |
| Observations (Panel A)                                        | 17,183                | 13,668                | 1,417                 | 1,027               | 114                              | 114 |
| Observations (Panel B)                                        | 7,153                 | 5,898                 | 995                   | 726                 | 101                              | 101 |

*Notes:* Observations are origin–destination–HS6–year cells. Mining and Fuels products only. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Regressors: ln number of MNEs of each type. Columns (5)–(6) split foreign MNEs by whether the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 176: Mining and Fuels: Multinational Presence and Exports: Extensive Margin, by Type of Multinational**

|                                                               | All MNEs              |                       | Foreign / domestic    |                       | Foreign by destination, domestic |                       |
|---------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------------------------|-----------------------|
|                                                               | (1)                   | (2)                   | (3)                   | (4)                   | (5)                              | (6)                   |
| <i>Panel A. Dependent variable: ln exports, all firms</i>     |                       |                       |                       |                       |                                  |                       |
| Any MNE present                                               | 1.3943***<br>(0.0929) | 0.6626***<br>(0.0589) |                       |                       |                                  |                       |
| Any foreign MNE                                               |                       |                       | 1.4255***<br>(0.0892) | 0.7311***<br>(0.0560) |                                  |                       |
| Any foreign MNE, parent-country destination                   |                       |                       |                       |                       | 0.8592***<br>(0.2949)            | 0.5918***<br>(0.1276) |
| Any foreign MNE, other destination                            |                       |                       |                       |                       | 1.5219***<br>(0.0882)            | 0.7263***<br>(0.0563) |
| Any domestic MNE                                              |                       |                       | 1.7356***<br>(0.1413) | 0.5718***<br>(0.1044) | 1.7177***<br>(0.1388)            | 0.5692***<br>(0.1045) |
| <i>Panel B. Dependent variable: ln exports, non-MNE firms</i> |                       |                       |                       |                       |                                  |                       |
| Any MNE present                                               | 1.6631***<br>(0.1016) | 0.1846***<br>(0.0505) |                       |                       |                                  |                       |
| Any foreign MNE                                               |                       |                       | 1.5696***<br>(0.1099) | 0.1991***<br>(0.0547) |                                  |                       |
| Any foreign MNE, parent-country destination                   |                       |                       |                       |                       | 1.0557***<br>(0.3125)            | 0.2849<br>(0.1738)    |
| Any foreign MNE, other destination                            |                       |                       |                       |                       | 1.5363***<br>(0.1030)            | 0.1696***<br>(0.0568) |
| Any domestic MNE                                              |                       |                       | 1.1333***<br>(0.1497) | 0.2125**<br>(0.0899)  | 1.1058***<br>(0.1489)            | 0.2125**<br>(0.0895)  |
| Product FE                                                    | ✓                     |                       | ✓                     |                       | ✓                                |                       |
| Origin × year FE                                              | ✓                     |                       | ✓                     |                       | ✓                                |                       |
| Destination × year FE                                         | ✓                     |                       | ✓                     |                       | ✓                                |                       |
| Origin × dest. × product FE                                   |                       | ✓                     |                       | ✓                     |                                  | ✓                     |
| Origin × dest. × year FE                                      |                       | ✓                     |                       | ✓                     |                                  | ✓                     |
| Observations (Panel A)                                        | 32,912                | 27,664                | 32,912                | 27,664                | 32,912                           | 27,664                |
| Observations (Panel B)                                        | 22,836                | 18,666                | 22,836                | 18,666                | 22,836                           | 18,666                |

*Notes:* Observations are origin–destination–HS6–year cells. Mining and Fuels products only. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Regressors: indicators for at least one MNE of each type. Columns (5)–(6) split foreign MNEs by whether the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 23 Fact 6: Distance Is a Weaker Barrier for Multinationals

**Table 177: Mining and Fuels: Distance and Firm Exports: Multinationals by Presence at the Destination**

| Dep. var.: ln exports                         | (1)                   | (2)                   | (3)                   | (4)                   |
|-----------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| ln distance                                   | -0.0923<br>(0.0819)   | -0.0999<br>(0.0828)   | -0.0931<br>(0.0817)   | -0.1006<br>(0.0827)   |
| ln distance × MNE                             | 0.0584***<br>(0.0084) | 0.0571***<br>(0.0085) |                       |                       |
| ln distance × MNE, present at destination     |                       |                       | 0.0468***<br>(0.0158) | 0.0464***<br>(0.0160) |
| ln distance × MNE, not present at destination |                       |                       | 0.0622***<br>(0.0083) | 0.0607***<br>(0.0084) |
| Origin FE                                     | ✓                     |                       | ✓                     |                       |
| Destination FE                                | ✓                     |                       | ✓                     |                       |
| Year FE                                       | ✓                     |                       | ✓                     |                       |
| Origin × year FE                              |                       | ✓                     |                       | ✓                     |
| Destination × year FE                         |                       | ✓                     |                       | ✓                     |
| Product FE                                    | ✓                     | ✓                     | ✓                     | ✓                     |
| Observations                                  | 79,938                | 79,612                | 79,938                | 79,612                |

*Notes:* Observations are firm–destination–HS6–year cells. Mining and Fuels products only. Dependent variable: ln exports. Non-MNE firms are the omitted group. Present: the group has an affiliate in the destination or the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

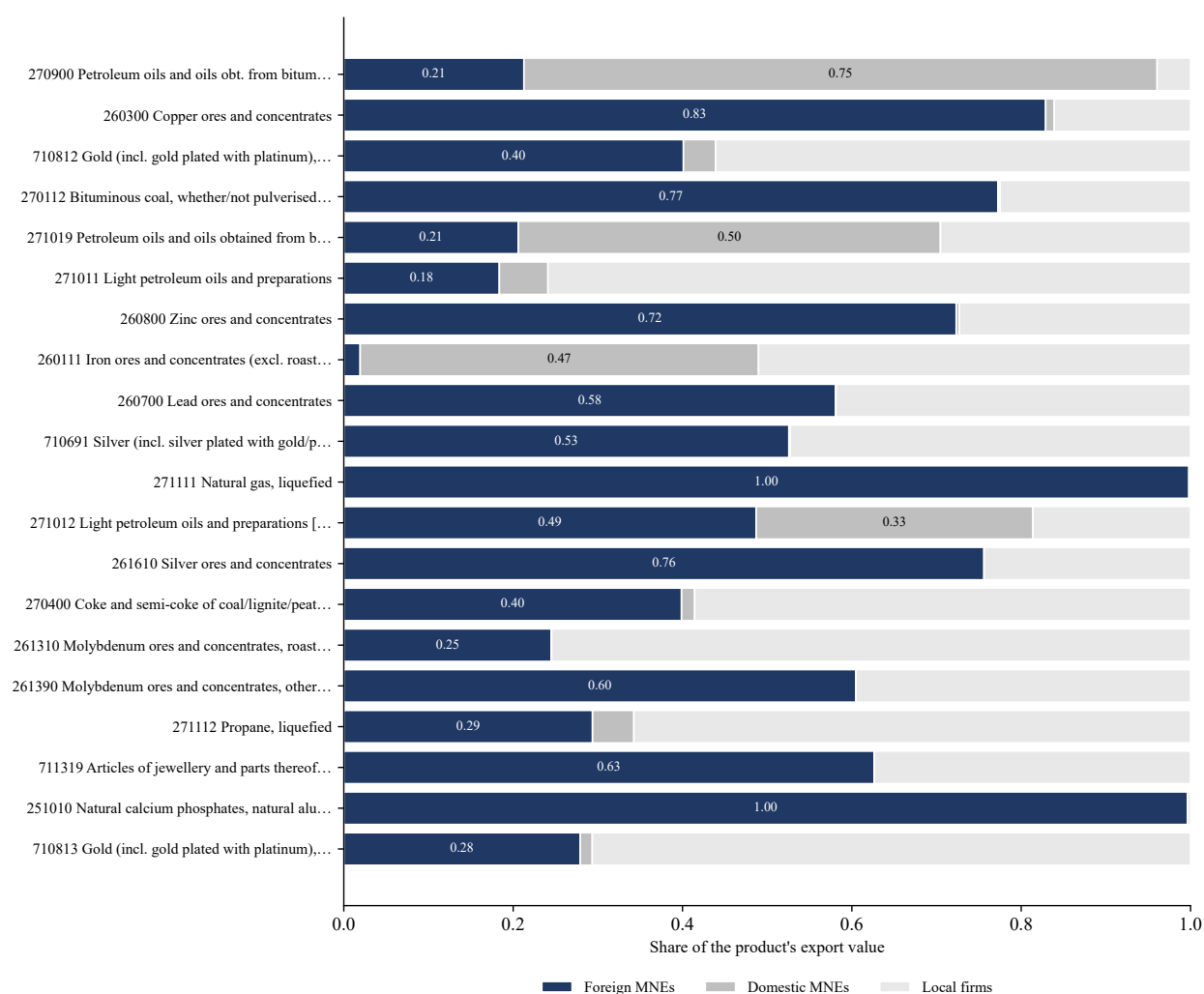
**Table 178:** Mining and Fuels: Distance and Firm Exports: Foreign and Domestic Multinationals

| Dep. var.: ln exports                                        | (1)                   | (2)                   | (3)                   | (4)                   |
|--------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| ln distance                                                  | -0.0926<br>(0.0819)   | -0.1001<br>(0.0828)   | -0.0895<br>(0.0816)   | -0.0975<br>(0.0827)   |
| ln distance $\times$ foreign MNE                             | 0.0566***<br>(0.0086) | 0.0557***<br>(0.0087) |                       |                       |
| ln distance $\times$ foreign MNE, parent-country destination |                       |                       | -0.0207<br>(0.0367)   | -0.0181<br>(0.0366)   |
| ln distance $\times$ foreign MNE, other destination          |                       |                       | 0.0615***<br>(0.0085) | 0.0604***<br>(0.0087) |
| ln distance $\times$ domestic MNE                            | 0.0706***<br>(0.0180) | 0.0673***<br>(0.0183) | 0.0712***<br>(0.0181) | 0.0678***<br>(0.0183) |
| Origin FE                                                    | ✓                     |                       | ✓                     |                       |
| Destination FE                                               | ✓                     |                       | ✓                     |                       |
| Year FE                                                      | ✓                     |                       | ✓                     |                       |
| Origin $\times$ year FE                                      |                       | ✓                     |                       | ✓                     |
| Destination $\times$ year FE                                 |                       | ✓                     |                       | ✓                     |
| Product FE                                                   | ✓                     | ✓                     | ✓                     | ✓                     |
| Observations                                                 | 79,938                | 79,612                | 79,938                | 79,612                |

*Notes:* Observations are firm–destination–HS6–year cells. Mining and Fuels products only. Dependent variable: ln exports. Non-MNE firms are the omitted group. Columns (3)–(4) split foreign MNEs by whether the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

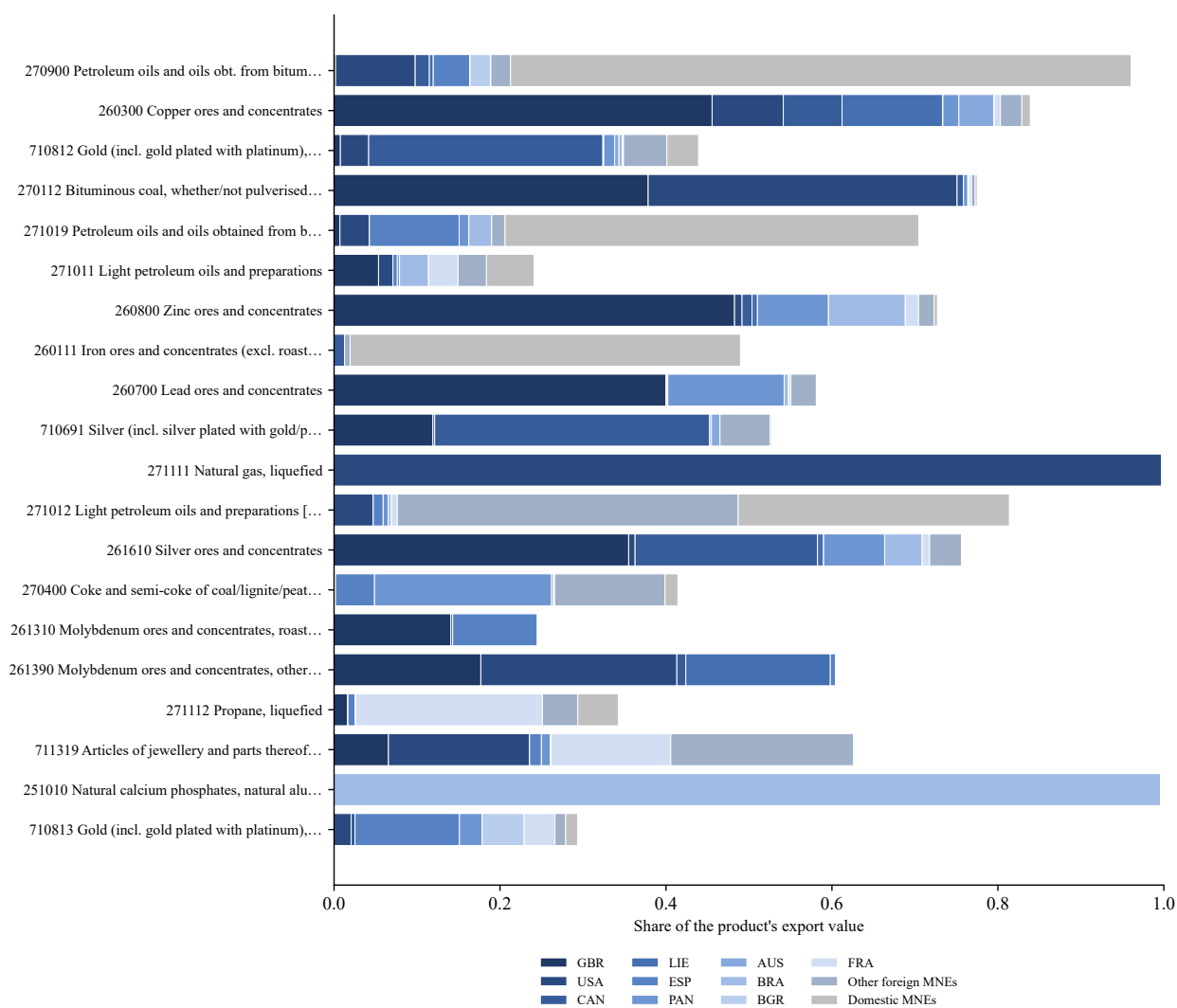
## 24 Products



**Figure 124: Mining and Fuels: Ownership of the Twenty Largest Export Products**

*Notes:* Shares of each product's export value. Twenty largest HS6 products by export value.

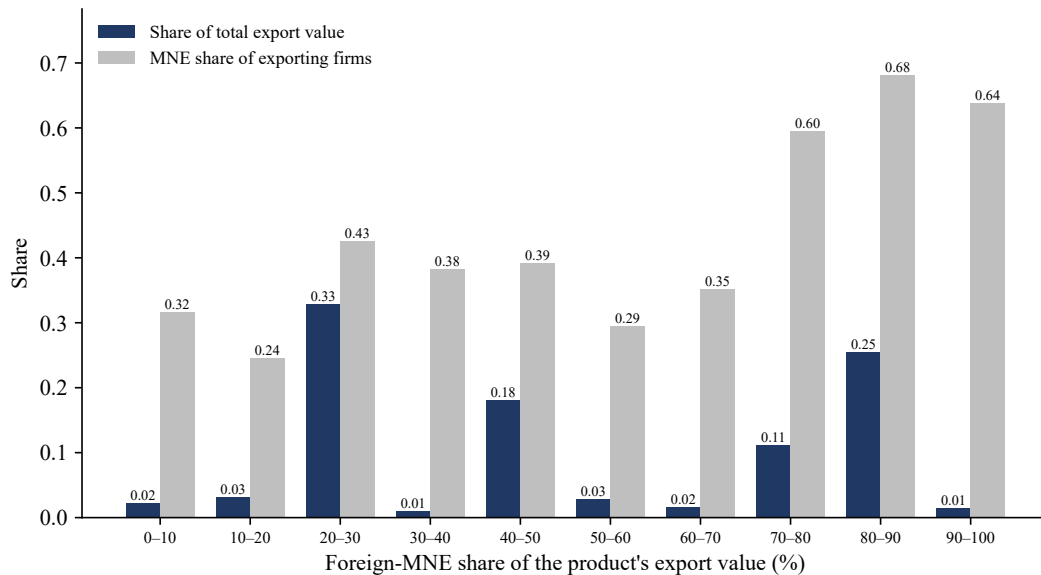
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 125:** Mining and Fuels: Ownership of the Twenty Largest Export Products, by Parent Country

*Notes:* Shares of each product's export value. Foreign-MNE share split by country of the ultimate parent; the unfilled part of each bar is local firms.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 126:** Mining and Fuels: Distribution of Export Value by Product-Level Foreign-Multinational Share

*Notes:* Share of mining and fuels export value and MNE share of exporting firms, by the foreign-MNE share of each HS6 product's export value.

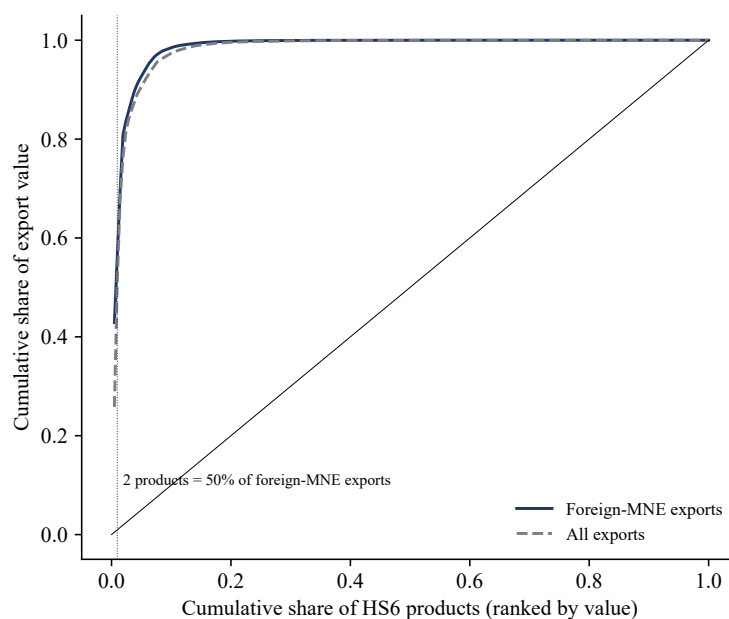
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 179:** Mining and Fuels: Distribution of Export Value by Product-Level Foreign-Multinational Share

| Foreign share bin (%) | Share of value | MNE share of exporting firms | HS6 products |
|-----------------------|----------------|------------------------------|--------------|
| 0-10                  | 0.022          | 0.317                        | 59           |
| 10-20                 | 0.031          | 0.245                        | 34           |
| 20-30                 | 0.329          | 0.425                        | 29           |
| 30-40                 | 0.011          | 0.382                        | 15           |
| 40-50                 | 0.181          | 0.391                        | 6            |
| 50-60                 | 0.029          | 0.294                        | 20           |
| 60-70                 | 0.016          | 0.351                        | 11           |
| 70-80                 | 0.112          | 0.595                        | 7            |
| 80-90                 | 0.254          | 0.681                        | 5            |
| 90-100                | 0.015          | 0.637                        | 22           |

*Notes:* Share of mining and fuels export value and MNE share of exporting firms, by the foreign-MNE share of each HS6 product's export value. MNE share of exporting firms: MNE firm-destination-product-year observations over all observations in the bin.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 127: Mining and Fuels: Concentration of Exports across Products**

*Notes:* Cumulative share of export value against the cumulative share of HS6 products, ranked by value, for foreign-MNE exports and for all exports.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 180: Mining and Fuels: Exports and Multinational Shares, by HS Section**

| HS section                     | US\$bn/yr | Foreign (%) | Domestic (%) | Local (%) | Leading parent |
|--------------------------------|-----------|-------------|--------------|-----------|----------------|
| V Mineral products             | 57.1      | 51          | 29           | 19        | GBR (0.42)     |
| XIV Precious metals and stones | 15.3      | 41          | 4            | 56        | CAN (0.65)     |
| All sections                   | 72.4      | 49          | 25           | 26        |                |

*Notes:* Foreign, domestic and local: percent of each section's export value. Leading parent: largest parent country and its share of the section's foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 181: Mining and Fuels: Thirty Largest Export Products**

| HS6    | Description                                                                                        | US\$bn/yr | Foreign (%) | Domestic (%) | Local (%) | Leading parent |
|--------|----------------------------------------------------------------------------------------------------|-----------|-------------|--------------|-----------|----------------|
| 270900 | Petroleum oils and oils obt. from bituminous mins., crude                                          | 17.9      | 21          | 75           | 4         | USA (0.45)     |
| 260300 | Copper ores and concentrates                                                                       | 17.6      | 83          | 1            | 16        | GBR (0.55)     |
| 710812 | Gold (incl. gold plated with platinum), in unwrought forms (excl. powder)                          | 13.4      | 40          | 4            | 56        | CAN (0.70)     |
| 270112 | Bituminous coal, whether/not pulverised but not agglomerated                                       | 5.7       | 77          | 0            | 23        | GBR (0.49)     |
| 271019 | Petroleum oils and oils obtained from bituminous minerals (other than crude) and preparations ...  | 3.6       | 21          | 50           | 30        | ESP (0.52)     |
| 271011 | Light petroleum oils and preparations                                                              | 2.2       | 18          | 6            | 76        | GBR (0.29)     |
| 260800 | Zinc ores and concentrates                                                                         | 1.5       | 72          | 0            | 27        | GBR (0.67)     |
| 260111 | Iron ores and concentrates (excl. roasted iron pyrites), non-agglomerated                          | 1.1       | 2           | 47           | 51        | CAN (0.65)     |
| 260700 | Lead ores and concentrates                                                                         | 1.3       | 58          | 0            | 42        | GBR (0.69)     |
| 710691 | Silver (incl. silver plated with gold/platinum), unwrought                                         | 0.8       | 53          | 0            | 47        | CAN (0.63)     |
| 271111 | Natural gas, liquefied                                                                             | 0.9       | 100         | 0            | 0         | USA (1.00)     |
| 271012 | Light petroleum oils and preparations [HS2012 code]                                                | 0.7       | 49          | 33           | 19        | CHL (0.80)     |
| 261610 | Silver ores and concentrates                                                                       | 0.8       | 76          | 0            | 24        | GBR (0.47)     |
| 270400 | Coke and semi-coke of coal/lignite/peat, whether/not agglomerated; retort carbon                   | 0.6       | 40          | 2            | 59        | PAN (0.53)     |
| 261310 | Molybdenum ores and concentrates, roasted                                                          | 0.5       | 25          | 0            | 75        | GBR (0.57)     |
| 261390 | Molybdenum ores and concentrates, other than roasted                                               | 0.6       | 60          | 0            | 40        | USA (0.39)     |
| 271112 | Propane, liquefied                                                                                 | 0.3       | 29          | 5            | 66        | FRA (0.77)     |
| 711319 | Articles of jewellery and parts thereof, of other precious metal (excl. silver), whether/not ...   | 0.4       | 63          | 0            | 37        | BEL (0.27)     |
| 251010 | Natural calcium phosphates, natural aluminium calcium phosphates and phosphatic chalk, underground | 0.3       | 100         | 0            | 0         | BRA (1.00)     |
| 710813 | Gold (incl. gold plated with platinum), non-monetary, in semi-manufactured forms                   | 0.3       | 28          | 1            | 71        | ESP (0.45)     |
| 271121 | Natural gas, in gaseous state                                                                      | 0.2       | 47          | 41           | 13        | USA (0.53)     |
| 271113 | Butanes, liquefied                                                                                 | 0.2       | 25          | 9            | 66        | FRA (0.83)     |
| 260112 | Iron ores and concentrates (excl. roasted iron pyrites), agglomerated                              | 0.1       | 7           | 88           | 4         | CHE (1.00)     |
| 710391 | Rubies, sapphires and emeralds, worked othw. than simply sawn/roughly shaped but not strung/mo...  | 0.1       | 22          | 1            | 77        | BGR (0.67)     |
| 271600 | Electrical energy (optional heading)                                                               | 0.1       | 20          | 0            | 80        | CAN (0.82)     |
| 262099 | Ash and residues (excl. from the manufacture of iron/steel), n.e.s. in Ch.26                       | 0.1       | 0           | 0            | 100       | CAN (0.89)     |
| 252329 | Portland cement (excl. white cement, whether/not artificially coloured), whether/not coloured      | 0.1       | 68          | 10           | 22        | MEX (0.34)     |
| 250100 | Salt (incl. table salt and denatured salt) and pure sodium chloride, whether/not in aqueous so...  | 0.1       | 62          | 8            | 29        | BEL (0.98)     |
| 271119 | Petroleum gases other than natural gas/propane/butanes/ethylene, propylene, butylene and butad...  | 0.1       | 52          | 0            | 48        | FRA (0.38)     |
| 711291 | Waste and scrap of gold, incl. metal clad with gold but excl. sweepings containing other preci...  | 0.1       | 18          | 0            | 81        | PAN (0.49)     |

*Notes:* Thirty largest HS6 products by export value. Foreign, domestic and local: percent of the product's export value. Leading parent: largest parent country and its share of the product's foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 182: Mining and Fuels: Thirty Largest Export Products of Foreign Multinationals**

| HS6    | Description                                                                                        | US\$bn/yr | Foreign (%) | Domestic (%) | Local (%) | Leading parent |
|--------|----------------------------------------------------------------------------------------------------|-----------|-------------|--------------|-----------|----------------|
| 260300 | Copper ores and concentrates                                                                       | 17.6      | 83          | 1            | 16        | GBR (0.55)     |
| 710812 | Gold (incl. gold plated with platinum), in unwrought forms (excl. powder)                          | 13.4      | 40          | 4            | 56        | CAN (0.70)     |
| 270112 | Bituminous coal, whether/not pulverised but not agglomerated                                       | 5.7       | 77          | 0            | 23        | GBR (0.49)     |
| 270900 | Petroleum oils and oils obt. from bituminous mins., crude                                          | 17.9      | 21          | 75           | 4         | USA (0.45)     |
| 260800 | Zinc ores and concentrates                                                                         | 1.5       | 72          | 0            | 27        | GBR (0.67)     |
| 271111 | Natural gas, liquefied                                                                             | 0.9       | 100         | 0            | 0         | USA (1.00)     |
| 271019 | Petroleum oils and oils obtained from bituminous minerals (other than crude) and preparations ...  | 3.6       | 21          | 50           | 30        | ESP (0.52)     |
| 260700 | Lead ores and concentrates                                                                         | 1.3       | 58          | 0            | 42        | GBR (0.69)     |
| 261610 | Silver ores and concentrates                                                                       | 0.8       | 76          | 0            | 24        | GBR (0.47)     |
| 710691 | Silver (incl. silver plated with gold/platinum), unwrought                                         | 0.8       | 53          | 0            | 47        | CAN (0.63)     |
| 271012 | Light petroleum oils and preparations [HS2012 code]                                                | 0.7       | 49          | 33           | 19        | CHL (0.80)     |
| 271011 | Light petroleum oils and preparations                                                              | 2.2       | 18          | 6            | 76        | GBR (0.29)     |
| 261390 | Molybdenum ores and concentrates, other than roasted                                               | 0.6       | 60          | 0            | 40        | USA (0.39)     |
| 270400 | Coke and semi-coke of coal/lignite/peat, whether/not agglomerated; retort carbon                   | 0.6       | 40          | 2            | 59        | PAN (0.53)     |
| 251010 | Natural calcium phosphates, natural aluminium calcium phosphates and phosphatic chalk, underground | 0.3       | 100         | 0            | 0         | BRA (1.00)     |
| 711319 | Articles of jewellery and parts thereof, of other precious metal (excl. silver), whether/not ...   | 0.4       | 63          | 0            | 37        | BEL (0.27)     |
| 261310 | Molybdenum ores and concentrates, roasted                                                          | 0.5       | 25          | 0            | 75        | GBR (0.57)     |
| 271121 | Natural gas, in gaseous state                                                                      | 0.2       | 47          | 41           | 13        | USA (0.53)     |
| 271112 | Propane, liquefied                                                                                 | 0.3       | 29          | 5            | 66        | FRA (0.77)     |
| 252329 | Portland cement (excl. white cement, whether/not artificially coloured), whether/not coloured      | 0.1       | 68          | 10           | 22        | MEX (0.34)     |
| 710813 | Gold (incl. gold plated with platinum), non-monetary, in semi-manufactured forms                   | 0.3       | 28          | 1            | 71        | ESP (0.45)     |
| 250100 | Salt (incl. table salt and denatured salt) and pure sodium chloride, whether/not in aqueous so...  | 0.1       | 62          | 8            | 29        | BEL (0.98)     |
| 271113 | Butanes, liquefied                                                                                 | 0.2       | 25          | 9            | 66        | FRA (0.83)     |
| 271119 | Petroleum gases other than natural gas/propane/butanes/ethylene, propylene, butylene and butad...  | 0.1       | 52          | 0            | 48        | FRA (0.38)     |
| 711311 | Articles of jewellery and parts thereof, of silver, whether/not plated/clad with other precio...   | 0.1       | 65          | 0            | 35        | FRA (0.40)     |
| 711719 | Imitation jewellery, of base metal, whether/not plated with precious metal, other than cuff-li...  | 0.0       | 63          | 3            | 34        | ALB (0.42)     |
| 710391 | Rubies, sapphires and emeralds, worked othw. than simply sawn/roughly shaped but not strung/mo...  | 0.1       | 22          | 1            | 77        | BGR (0.67)     |
| 252310 | Cement clinkers                                                                                    | 0.0       | 56          | 0            | 44        | MEX (0.46)     |
| 271600 | Electrical energy (optional heading)                                                               | 0.1       | 20          | 0            | 80        | CAN (0.82)     |
| 260111 | Iron ores and concentrates (excl. roasted iron pyrites), non-agglomerated                          | 1.1       | 2           | 47           | 51        | CAN (0.65)     |

*Notes:* Thirty largest HS6 products by foreign-MNE export value. Foreign, domestic and local: percent of the product's export value. Leading parent: largest parent country and its share of the product's foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 183: Mining and Fuels: Products with the Highest Foreign-Multinational Share**

| HS6    | Description                                                                                       | US\$bn/yr | Foreign (%) | Domestic (%) | Local (%) | Leading parent |
|--------|---------------------------------------------------------------------------------------------------|-----------|-------------|--------------|-----------|----------------|
| 271111 | Natural gas, liquefied                                                                            | 0.9       | 100         | 0            | 0         | USA (1.00)     |
| 251010 | Natural calcium phosphates, natural aluminium calcium phosphates and phosphatic chalk, un-ground  | 0.3       | 100         | 0            | 0         | BRA (1.00)     |
| 260300 | Copper ores and concentrates                                                                      | 17.6      | 83          | 1            | 16        | GBR (0.55)     |
| 270112 | Bituminous coal, whether/not pulverised but not agglomerated                                      | 5.7       | 77          | 0            | 23        | GBR (0.49)     |
| 261610 | Silver ores and concentrates                                                                      | 0.8       | 76          | 0            | 24        | GBR (0.47)     |
| 260800 | Zinc ores and concentrates                                                                        | 1.5       | 72          | 0            | 27        | GBR (0.67)     |
| 252329 | Portland cement (excl. white cement, whether/not artificially coloured), whether/not coloured     | 0.1       | 68          | 10           | 22        | MEX (0.34)     |
| 711311 | Articles of jewellery and parts thereof , of silver, whether/not plated/clad with other precio... | 0.1       | 65          | 0            | 35        | FRA (0.40)     |
| 711719 | Imitation jewellery, of base metal, whether/not plated with precious metal, other than cuff-li... | 0.0       | 63          | 3            | 34        | ALB (0.42)     |
| 711319 | Articles of jewellery and parts thereof , of other precious metal (excl. silver), whether/not ... | 0.4       | 63          | 0            | 37        | BEL (0.27)     |
| 250100 | Salt (incl. table salt and denatured salt) and pure sodium chloride, whether/not in aqueous so... | 0.1       | 62          | 8            | 29        | BEL (0.98)     |
| 261390 | Molybdenum ores and concentrates, other than roasted                                              | 0.6       | 60          | 0            | 40        | USA (0.39)     |
| 260700 | Lead ores and concentrates                                                                        | 1.3       | 58          | 0            | 42        | GBR (0.69)     |
| 710691 | Silver (incl. silver plated with gold/platinum), unwrought                                        | 0.8       | 53          | 0            | 47        | CAN (0.63)     |
| 271119 | Petroleum gases other than natural gas/propane/butanes/ethylene, propylene, butylene and butad... | 0.1       | 52          | 0            | 48        | FRA (0.38)     |
| 271012 | Light petroleum oils and preparations [HS2012 code]                                               | 0.7       | 49          | 33           | 19        | CHL (0.80)     |
| 271121 | Natural gas, in gaseous state                                                                     | 0.2       | 47          | 41           | 13        | USA (0.53)     |
| 710812 | Gold (incl. gold plated with platinum), in unwrought forms (excl. powder)                         | 13.4      | 40          | 4            | 56        | CAN (0.70)     |
| 270400 | Coke and semi-coke of coal/lignite/peat, whether/not agglomerated; retort carbon                  | 0.6       | 40          | 2            | 59        | PAN (0.53)     |
| 261690 | Precious metal ores and concentrates (excl. silver ores and concentrates)                         | 0.1       | 32          | 5            | 63        | PER (0.31)     |
| 271112 | Propane, liquefied                                                                                | 0.3       | 29          | 5            | 66        | FRA (0.77)     |
| 710813 | Gold (incl. gold plated with platinum), non-monetary, in semi-manufactured forms                  | 0.3       | 28          | 1            | 71        | ESP (0.45)     |
| 271113 | Butanes, liquefied                                                                                | 0.2       | 25          | 9            | 66        | FRA (0.83)     |
| 261310 | Molybdenum ores and concentrates, roasted                                                         | 0.5       | 25          | 0            | 75        | GBR (0.57)     |
| 710391 | Rubies, sapphires and emeralds, worked othw. than simply sawn/roughly shaped but not strung/mo... | 0.1       | 22          | 1            | 77        | BGR (0.67)     |
| 270900 | Petroleum oils and oils obt. from bituminous mins., crude                                         | 17.9      | 21          | 75           | 4         | USA (0.45)     |
| 271019 | Petroleum oils and oils obtained from bituminous minerals (other than crude) and preparations ... | 3.6       | 21          | 50           | 30        | ESP (0.52)     |
| 271600 | Electrical energy (optional heading)                                                              | 0.1       | 20          | 0            | 80        | CAN (0.82)     |
| 711291 | Waste and scrap of gold, incl. metal clad with gold but excl. sweepings containing other preci... | 0.1       | 18          | 0            | 81        | PAN (0.49)     |
| 271011 | Light petroleum oils and preparations                                                             | 2.2       | 18          | 6            | 76        | GBR (0.29)     |

*Notes:* HS6 products with exports of at least US\$500 million over the period, ranked by foreign-MNE share. Foreign, domestic and local: percent of the product's export value. Leading parent: largest parent country and its share of the product's foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 184: Mining and Fuels: Products with the Highest Domestic-Multinational Share**

| HS6    | Description                                                                                        | US\$bn/yr | Foreign (%) | Domestic (%) | Local (%) | Leading parent |
|--------|----------------------------------------------------------------------------------------------------|-----------|-------------|--------------|-----------|----------------|
| 260112 | Iron ores and concentrates (excl. roasted iron pyrites), agglomerated                              | 0.1       | 7           | 88           | 4         | CHE (1.00)     |
| 270900 | Petroleum oils and oils obt. from bituminous mins., crude                                          | 17.9      | 21          | 75           | 4         | USA (0.45)     |
| 271019 | Petroleum oils and oils obtained from bituminous minerals (other than crude) and preparations ...  | 3.6       | 21          | 50           | 30        | ESP (0.52)     |
| 260111 | Iron ores and concentrates (excl. roasted iron pyrites), non-agglomerated                          | 1.1       | 2           | 47           | 51        | CAN (0.65)     |
| 271121 | Natural gas, in gaseous state                                                                      | 0.2       | 47          | 41           | 13        | USA (0.53)     |
| 271012 | Light petroleum oils and preparations [HS2012 code]                                                | 0.7       | 49          | 33           | 19        | CHL (0.80)     |
| 271311 | Petroleum coke, not calcined                                                                       | 0.0       | 6           | 23           | 71        | GBR (0.80)     |
| 252329 | Portland cement (excl. white cement, whether/not artificially coloured), whether/not coloured      | 0.1       | 68          | 10           | 22        | MEX (0.34)     |
| 271113 | Butanes, liquefied                                                                                 | 0.2       | 25          | 9            | 66        | FRA (0.83)     |
| 250100 | Salt (incl. table salt and denatured salt) and pure sodium chloride, whether/not in aqueous so...  | 0.1       | 62          | 8            | 29        | BEL (0.98)     |
| 271011 | Light petroleum oils and preparations                                                              | 2.2       | 18          | 6            | 76        | GBR (0.29)     |
| 261690 | Precious metal ores and concentrates (excl. silver ores and concentrates)                          | 0.1       | 32          | 5            | 63        | PER (0.31)     |
| 271112 | Propane, liquefied                                                                                 | 0.3       | 29          | 5            | 66        | FRA (0.77)     |
| 710812 | Gold (incl. gold plated with platinum), in unwrought forms (excl. powder)                          | 13.4      | 40          | 4            | 56        | CAN (0.70)     |
| 711719 | Imitation jewellery, of base metal, whether/not plated with precious metal, other than cuff-li...  | 0.0       | 63          | 3            | 34        | ALB (0.42)     |
| 270400 | Coke and semi-coke of coal/lignite/peat, whether/not agglomerated; retort carbon                   | 0.6       | 40          | 2            | 59        | PAN (0.53)     |
| 710813 | Gold (incl. gold plated with platinum), non-monetary, in semi-manufactured forms                   | 0.3       | 28          | 1            | 71        | ESP (0.45)     |
| 710391 | Rubies, sapphires and emeralds, worked othw. than simply sawn/roughly shaped but not strung/mo...  | 0.1       | 22          | 1            | 77        | BGR (0.67)     |
| 260300 | Copper ores and concentrates                                                                       | 17.6      | 83          | 1            | 16        | GBR (0.55)     |
| 711291 | Waste and scrap of gold, incl. metal clad with gold but excl. sweepings containing other preci...  | 0.1       | 18          | 0            | 81        | PAN (0.49)     |
| 260800 | Zinc ores and concentrates                                                                         | 1.5       | 72          | 0            | 27        | GBR (0.67)     |
| 270112 | Bituminous coal, whether/not pulverised but not agglomerated                                       | 5.7       | 77          | 0            | 23        | GBR (0.49)     |
| 710691 | Silver (incl. silver plated with gold/platinum), unwrought                                         | 0.8       | 53          | 0            | 47        | CAN (0.63)     |
| 711311 | Articles of jewellery and parts thereof, of silver, whether/not plated/clad with other precio...   | 0.1       | 65          | 0            | 35        | FRA (0.40)     |
| 271119 | Petroleum gases other than natural gas/propane/butanes/ethylene, propylene, butylene and butad...  | 0.1       | 52          | 0            | 48        | FRA (0.38)     |
| 711319 | Articles of jewellery and parts thereof, of other precious metal (excl. silver), whether/not ...   | 0.4       | 63          | 0            | 37        | BEL (0.27)     |
| 261310 | Molybdenum ores and concentrates, roasted                                                          | 0.5       | 25          | 0            | 75        | GBR (0.57)     |
| 251010 | Natural calcium phosphates, natural aluminium calcium phosphates and phosphatic chalk, underground | 0.3       | 100         | 0            | 0         | BRA (1.00)     |
| 260700 | Lead ores and concentrates                                                                         | 1.3       | 58          | 0            | 42        | GBR (0.69)     |
| 262099 | Ash and residues (excl. from the manufacture of iron/steel), n.e.s. in Ch.26                       | 0.1       | 0           | 0            | 100       | CAN (0.89)     |

*Notes:* HS6 products with exports of at least US\$500 million over the period, ranked by domestic-MNE share. Foreign, domestic and local: percent of the product's export value. Leading parent: largest parent country and its share of the product's foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 25 Product Groups within the Sector

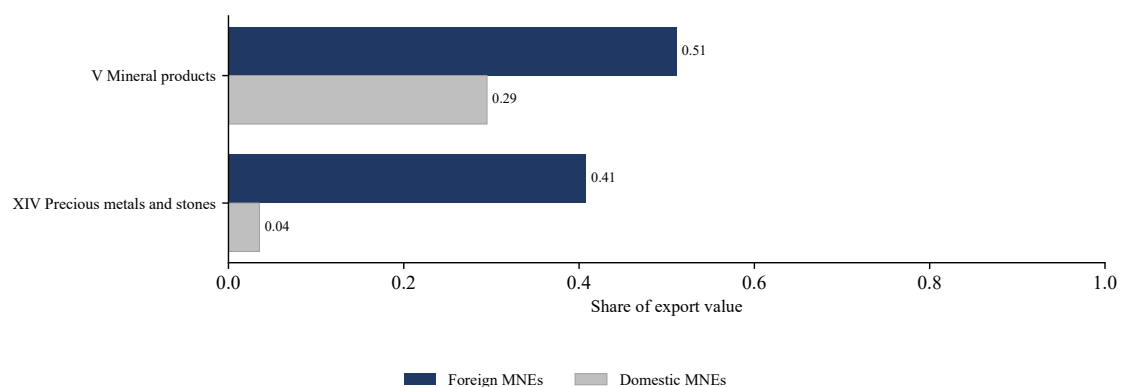
### 25.1 By HS Section

**Table 185:** Mining and Fuels: Exports and Multinational Shares, by HS Section

| HS section                     | Total exports (US\$bn/yr) | MNE exports (US\$bn/yr) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|--------------------------------|---------------------------|-------------------------|-------------|--------------|------------------------------------|
| V Mineral products             | 57.1                      | 45.4                    | 51.1        | 29.5         | GBR 0.42, USA 0.23, LIE 0.08       |
| XIV Precious metals and stones | 15.3                      | 7.0                     | 40.8        | 3.5          | CAN 0.65, USA 0.09, GBR 0.04       |
| All                            | 72.4                      | 52.4                    | 49.2        | 24.5         |                                    |

*Notes:* HS sections. Foreign and domestic: percent of the group's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

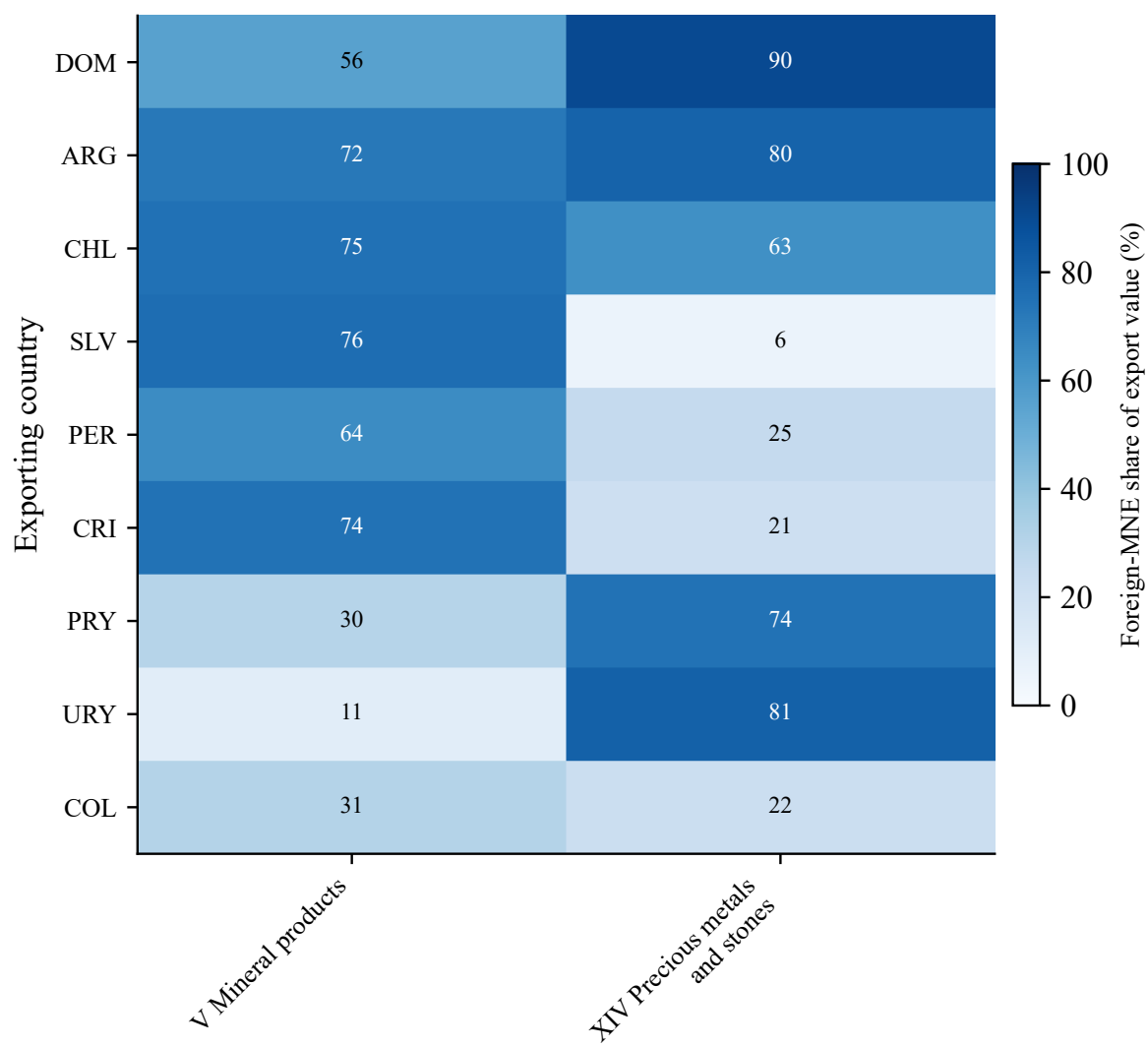
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 128:** Mining and Fuels: Multinational Share of Export Value, by HS Section

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. HS sections.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 129:** Mining and Fuels: Foreign-Multinational Share of Export Value, by Exporting Country and HS Section

*Notes:* Percent of each group's export value in each country shipped by foreign MNEs. HS sections.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 186:** Mining and Fuels: Exports by Exporting Country and HS Section

| Origin                                                                   | V Mineral products | XIV Precious metals and stones | All   |
|--------------------------------------------------------------------------|--------------------|--------------------------------|-------|
| <i>Panel A: composition of the origin's exports, % (rows sum to 100)</i> |                    |                                |       |
| ARG                                                                      | 63.5               | 36.5                           | 100.0 |
| CHL                                                                      | 92.8               | 7.2                            | 100.0 |
| COL                                                                      | 91.5               | 8.5                            | 100.0 |
| CRI                                                                      | 48.3               | 51.7                           | 100.0 |
| DOM                                                                      | 13.9               | 86.1                           | 100.0 |
| PER                                                                      | 68.8               | 31.2                           | 100.0 |
| PRY                                                                      | 67.9               | 32.1                           | 100.0 |
| SLV                                                                      | 74.8               | 25.2                           | 100.0 |
| URY                                                                      | 54.1               | 45.9                           | 100.0 |
| All origins                                                              | 80.9               | 19.1                           | 100.0 |
| <i>Panel B: foreign-MNE share of export value, %</i>                     |                    |                                |       |
| ARG                                                                      | 72.2               | 80.2                           | 75.1  |
| CHL                                                                      | 74.7               | 63.1                           | 73.9  |
| COL                                                                      | 30.8               | 22.4                           | 30.1  |
| CRI                                                                      | 74.4               | 21.1                           | 46.9  |
| DOM                                                                      | 55.7               | 90.1                           | 85.3  |
| PER                                                                      | 64.5               | 25.1                           | 52.2  |
| PRY                                                                      | 29.9               | 74.3                           | 44.1  |
| SLV                                                                      | 76.5               | 6.2                            | 58.8  |
| URY                                                                      | 10.6               | 81.0                           | 42.9  |
| All origins                                                              | 51.1               | 40.8                           | 49.2  |

*Notes:* Panel A: composition of each country's exports in the sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs. HS sections.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 187:** Mining and Fuels: Exports by HS Section and Parent Country

| HS section                     | GBR  | USA  | CAN  | LIE | ESP | PAN | AUS | BRA | BGR | FRA | Other | Domestic |
|--------------------------------|------|------|------|-----|-----|-----|-----|-----|-----|-----|-------|----------|
| V Mineral products             | 20.7 | 11.5 | 3.1  | 4.1 | 1.9 | 1.3 | 1.3 | 0.9 | 0.8 | 0.6 | 4.8   | 29.5     |
| XIV Precious metals and stones | 1.5  | 3.3  | 25.1 | 0.0 | 0.4 | 1.3 | 0.5 | 0.4 | 0.2 | 0.5 | 7.6   | 3.5      |

*Notes:* Percent of each group's export value, by country of the ultimate parent of foreign MNEs; domestic MNEs shown for comparison. HS sections.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

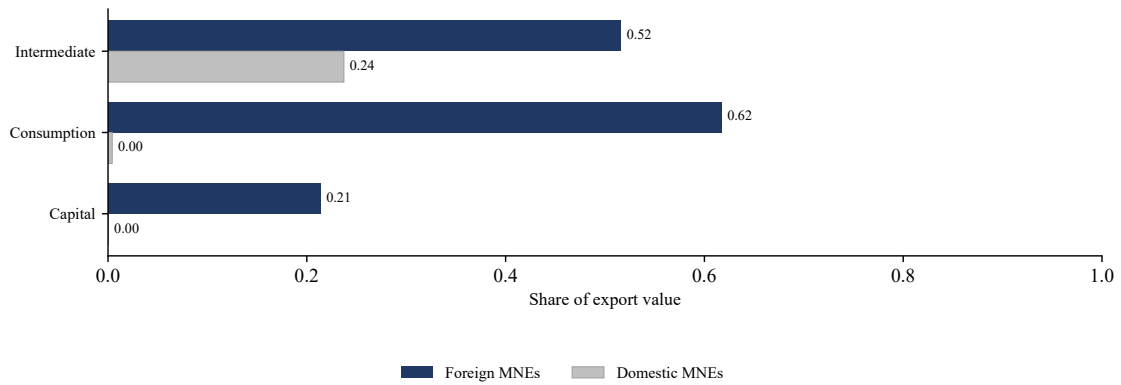
## 25.2 By End Use

**Table 188:** Mining and Fuels: Exports and Multinational Shares, by End Use

| BEC end use  | Total exports (US\$bn/yr) | MNE exports (US\$bn/yr) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|--------------|---------------------------|-------------------------|-------------|--------------|------------------------------------|
| Intermediate | 65.4                      | 48.6                    | 51.6        | 23.7         | GBR 0.37, USA 0.21, CAN 0.16       |
| Consumption  | 0.5                       | 0.3                     | 61.7        | 0.4          | USA 0.22, FRA 0.22, BEL 0.21       |
| Capital      | 0.0                       | 0.0                     | 21.4        | 0.0          | FRA 0.82, USA 0.08, GBR 0.03       |
| All          | 65.9                      | 48.9                    | 51.6        | 23.6         |                                    |

*Notes:* End use from BEC Rev. 4. Foreign and domestic: percent of the group's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

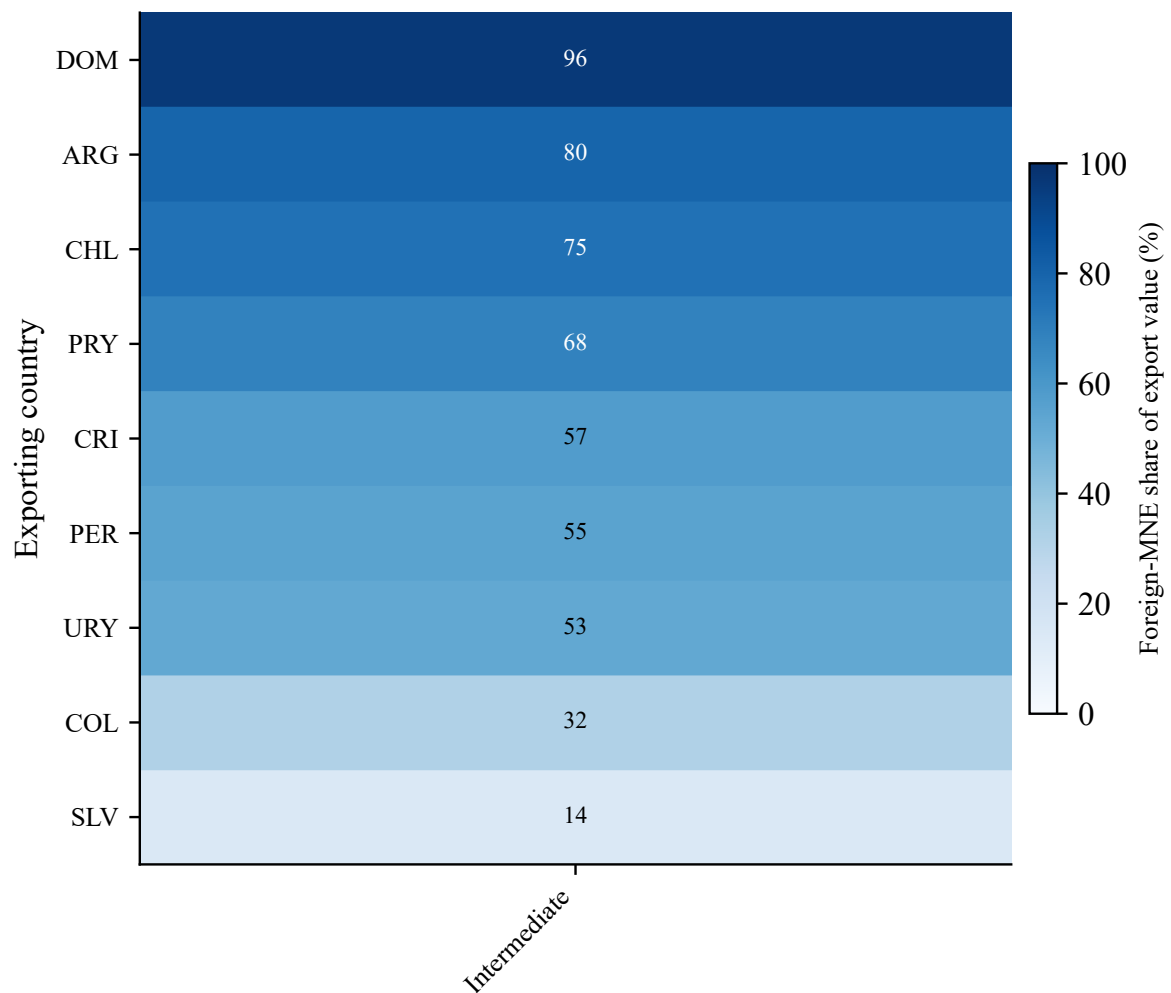
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 130:** Mining and Fuels: Multinational Share of Export Value, by End Use

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. End use from BEC Rev. 4.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 131:** Mining and Fuels: Foreign-Multinational Share of Export Value, by Exporting Country and End Use

*Notes:* Percent of each group's export value in each country shipped by foreign MNEs. End use from BEC Rev. 4.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 189: Mining and Fuels: Exports by Exporting Country and End Use**

| Origin                                                                   | Intermediate | Consumption | Capital | All   |
|--------------------------------------------------------------------------|--------------|-------------|---------|-------|
| <i>Panel A: composition of the origin's exports, % (rows sum to 100)</i> |              |             |         |       |
| ARG                                                                      | 100.0        | 0.0         | 0.0     | 100.0 |
| CHL                                                                      | 100.0        | 0.0         | 0.0     | 100.0 |
| COL                                                                      | 99.9         | 0.1         | 0.0     | 100.0 |
| CRI                                                                      | 82.5         | 17.5        | 0.1     | 100.0 |
| DOM                                                                      | 81.9         | 18.1        | 0.0     | 100.0 |
| PER                                                                      | 99.6         | 0.4         | 0.0     | 100.0 |
| PRY                                                                      | 98.1         | 0.1         | 1.8     | 100.0 |
| SLV                                                                      | 95.4         | 4.6         | 0.0     | 100.0 |
| URY                                                                      | 99.4         | 0.6         | 0.0     | 100.0 |
| All origins                                                              | 99.4         | 0.6         | 0.0     | 100.0 |
| <i>Panel B: foreign-MNE share of export value, %</i>                     |              |             |         |       |
| ARG                                                                      | 79.7         | 11.4        | 97.0    | 79.6  |
| CHL                                                                      | 74.8         | 24.4        | 28.6    | 74.8  |
| COL                                                                      | 31.7         | 68.0        | 22.0    | 31.8  |
| CRI                                                                      | 57.5         | 44.6        | 96.4    | 55.2  |
| DOM                                                                      | 96.1         | 57.9        | 0.0     | 89.2  |
| PER                                                                      | 54.7         | 77.2        | 0.2     | 54.8  |
| PRY                                                                      | 68.2         | 14.8        | 100.0   | 68.8  |
| SLV                                                                      | 14.4         | 0.4         | 0.0     | 13.7  |
| URY                                                                      | 52.7         | 69.3        | 58.2    | 52.8  |
| All origins                                                              | 51.6         | 61.7        | 21.4    | 51.6  |

Notes: Panel A: composition of each country's exports in the sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs. End use from BEC Rev. 4.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 190: Mining and Fuels: Exports by End Use and Parent Country**

| BEC end use  | GBR  | USA  | CAN | LIE | ESP | PAN | AUS | BRA | BGR | FRA  | Other | Domestic |
|--------------|------|------|-----|-----|-----|-----|-----|-----|-----|------|-------|----------|
| Intermediate | 18.6 | 10.6 | 8.1 | 3.6 | 1.4 | 1.4 | 1.3 | 0.7 | 0.8 | 0.5  | 4.5   | 23.7     |
| Consumption  | 6.2  | 13.4 | 0.0 | 0.0 | 2.2 | 2.6 | 0.0 | 0.1 | 0.0 | 13.3 | 23.9  | 0.4      |
| Capital      | 0.7  | 1.8  | 0.0 | 0.0 | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 17.5 | 1.2   | 0.0      |

Notes: Percent of each group's export value, by country of the ultimate parent of foreign MNEs; domestic MNEs shown for comparison. End use from BEC Rev. 4.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

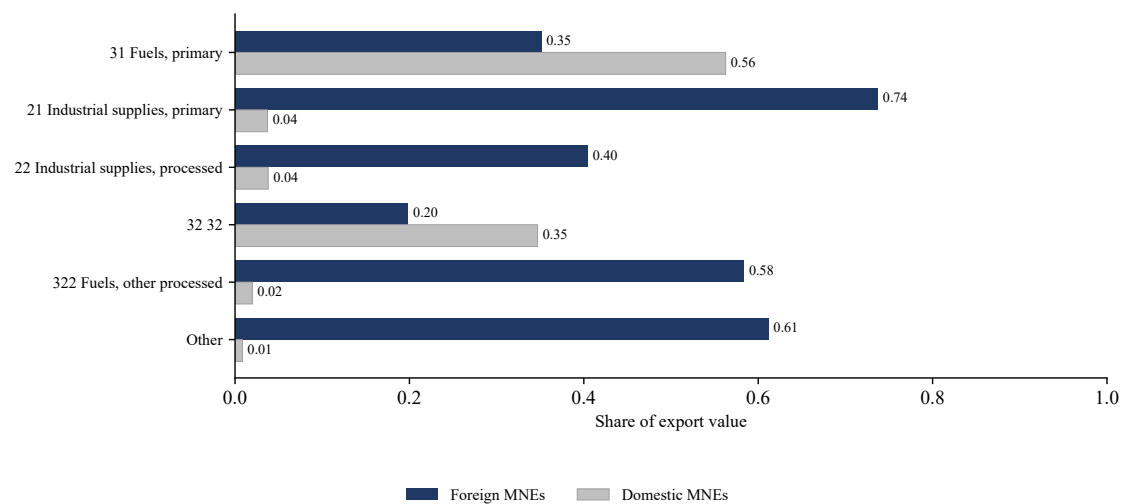
## 25.3 By BEC Category

**Table 191: Mining and Fuels: Exports and Multinational Shares, by BEC Category**

| BEC Rev. 4 category               | Total exports (US\$bn/yr) | MNE exports (US\$bn/yr) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|-----------------------------------|---------------------------|-------------------------|-------------|--------------|------------------------------------|
| 31 Fuels, primary                 | 23.8                      | 21.7                    | 35.1        | 56.3         | USA 0.47, GBR 0.27, ESP 0.10       |
| 21 Industrial supplies, primary   | 24.3                      | 18.6                    | 73.7        | 3.7          | GBR 0.54, LIE 0.13, USA 0.10       |
| 22 Industrial supplies, processed | 15.1                      | 6.9                     | 40.4        | 3.8          | CAN 0.67, USA 0.08, GBR 0.03       |
| 32 32                             | 5.8                       | 3.0                     | 19.8        | 34.7         | ESP 0.34, BRA 0.15, USA 0.14       |
| 322 Fuels, other processed        | 2.2                       | 1.4                     | 58.3        | 2.0          | USA 0.76, FRA 0.12, PAN 0.02       |
| Other                             | 0.5                       | 0.3                     | 61.2        | 0.8          | FRA 0.22, USA 0.22, BEL 0.21       |
| All                               | 71.7                      | 51.9                    | 49.2        | 24.5         |                                    |

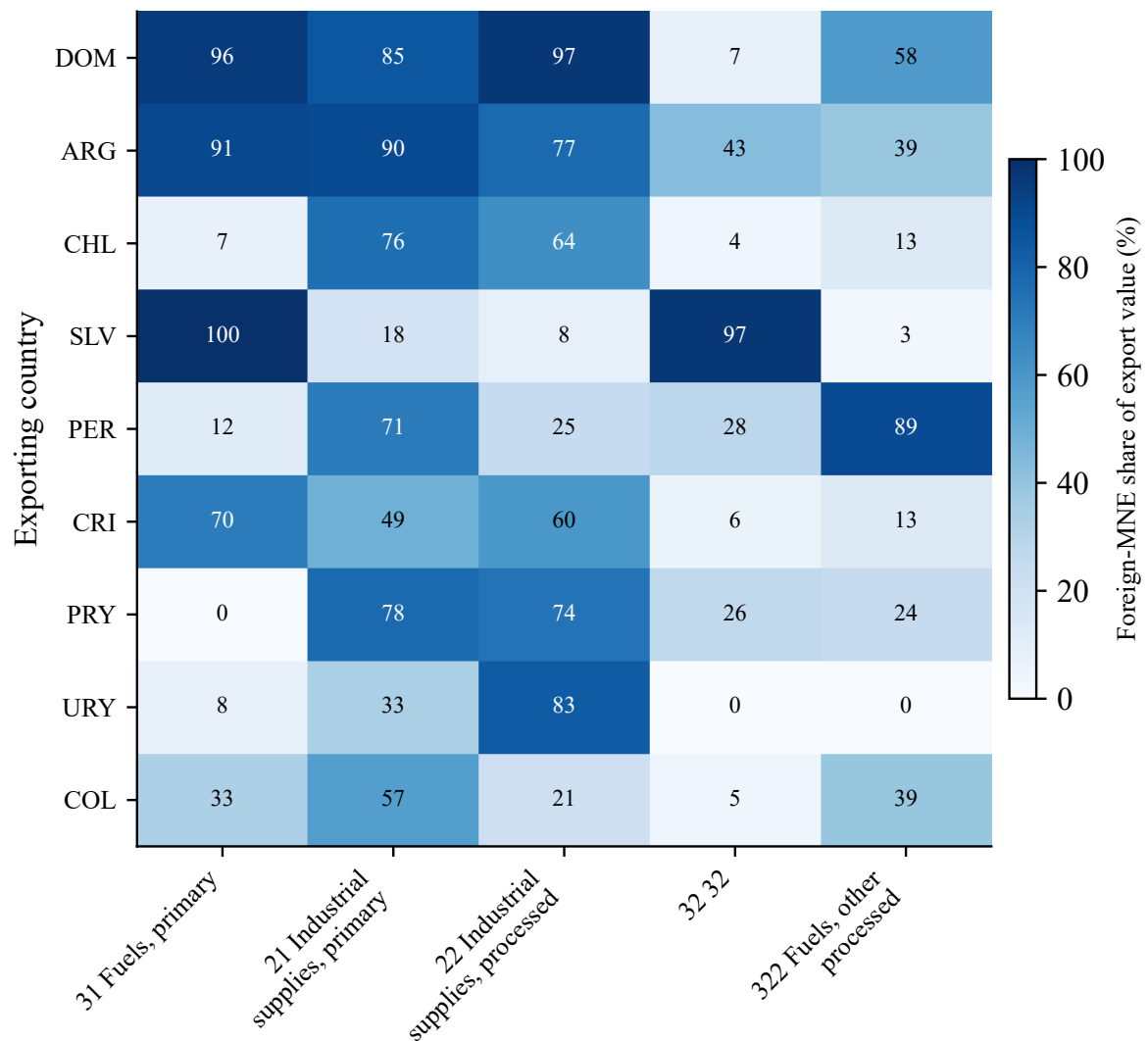
Notes: Categories of BEC Rev. 4. Foreign and domestic: percent of the group's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 132: Mining and Fuels: Multinational Share of Export Value, by BEC Category**

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. Categories of BEC Rev. 4.  
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 133: Mining and Fuels: Foreign-Multinational Share of Export Value, by Exporting Country and BEC Category**

*Notes:* Percent of each group's export value in each country shipped by foreign MNEs. Categories of BEC Rev. 4.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 192: Mining and Fuels: Exports by Exporting Country and BEC Category**

| Origin                                                                   | 31 Fuels, primary | 21 Industrial supplies, primary | 22 Industrial supplies, processed | 32 32 | 322 Fuels, other processed | Other | All   |
|--------------------------------------------------------------------------|-------------------|---------------------------------|-----------------------------------|-------|----------------------------|-------|-------|
| <i>Panel A: composition of the origin's exports, % (rows sum to 100)</i> |                   |                                 |                                   |       |                            |       |       |
| ARG                                                                      | 23.4              | 17.4                            | 38.8                              | 12.0  | 8.3                        | 0.0   | 100.0 |
| CHL                                                                      | 0.2               | 91.3                            | 7.2                               | 1.1   | 0.2                        | 0.0   | 100.0 |
| COL                                                                      | 80.4              | 0.4                             | 8.6                               | 8.1   | 2.4                        | 0.1   | 100.0 |
| CRI                                                                      | 0.0               | 15.8                            | 57.0                              | 11.5  | 0.2                        | 15.5  | 100.0 |
| DOM                                                                      | 0.0               | 5.1                             | 72.7                              | 4.8   | 0.2                        | 17.2  | 100.0 |
| PER                                                                      | 1.3               | 53.7                            | 31.0                              | 9.8   | 3.8                        | 0.4   | 100.0 |
| PRY                                                                      | 0.0               | 4.9                             | 31.3                              | 58.0  | 5.0                        | 0.8   | 100.0 |
| SLV                                                                      | 0.4               | 27.3                            | 9.0                               | 54.4  | 6.9                        | 2.1   | 100.0 |
| URY                                                                      | 0.0               | 10.1                            | 47.1                              | 18.8  | 23.4                       | 0.5   | 100.0 |
| All origins                                                              | 35.0              | 34.8                            | 19.1                              | 7.8   | 2.9                        | 0.5   | 100.0 |
| <i>Panel B: foreign-MNE share of export value, %</i>                     |                   |                                 |                                   |       |                            |       |       |
| ARG                                                                      | 90.5              | 89.8                            | 77.4                              | 43.2  | 38.7                       | 11.9  | 75.3  |
| CHL                                                                      | 7.4               | 76.0                            | 63.5                              | 4.1   | 13.0                       | 8.3   | 74.1  |
| COL                                                                      | 32.5              | 56.8                            | 21.1                              | 4.6   | 39.1                       | 67.8  | 29.6  |
| CRI                                                                      | 70.5              | 49.1                            | 59.8                              | 6.3   | 12.6                       | 44.9  | 49.6  |
| DOM                                                                      | 95.6              | 84.9                            | 97.0                              | 7.2   | 58.1                       | 57.9  | 85.3  |
| PER                                                                      | 12.0              | 70.7                            | 24.7                              | 27.8  | 89.4                       | 75.5  | 52.2  |
| PRY                                                                      | 0.0               | 77.7                            | 73.8                              | 26.3  | 24.4                       | 94.8  | 44.1  |
| SLV                                                                      | 99.8              | 18.3                            | 8.0                               | 96.9  | 2.5                        | 0.4   | 58.9  |
| URY                                                                      | 8.2               | 33.3                            | 82.9                              | 0.2   | 0.1                        | 68.3  | 42.9  |
| All origins                                                              | 35.1              | 73.7                            | 40.4                              | 19.8  | 58.3                       | 61.2  | 49.2  |

*Notes:* Panel A: composition of each country's exports in the sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs. Categories of BEC Rev. 4.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 193: Mining and Fuels: Exports by BEC Category and Parent Country**

| BEC Rev. 4 category               | GBR  | USA  | CAN  | LIE | ESP | PAN | AUS | BRA | BGR | FRA  | Other | Domestic |
|-----------------------------------|------|------|------|-----|-----|-----|-----|-----|-----|------|-------|----------|
| 31 Fuels, primary                 | 9.4  | 16.4 | 1.5  | 0.3 | 3.3 | 0.1 | 0.0 | 0.0 | 1.9 | 0.2  | 2.1   | 56.3     |
| 21 Industrial supplies, primary   | 38.9 | 6.9  | 5.8  | 9.3 | 0.2 | 2.7 | 3.1 | 1.6 | 0.0 | 0.7  | 4.5   | 3.7      |
| 22 Industrial supplies, processed | 1.3  | 3.0  | 25.4 | 0.0 | 0.3 | 1.2 | 0.5 | 0.5 | 0.2 | 0.2  | 7.8   | 3.8      |
| 32 32                             | 1.6  | 1.8  | 0.0  | 0.0 | 4.2 | 0.5 | 0.0 | 1.9 | 0.0 | 0.9  | 8.9   | 34.7     |
| 322 Fuels, other processed        | 0.7  | 36.6 | 1.0  | 0.0 | 0.4 | 1.0 | 0.0 | 0.0 | 0.0 | 5.7  | 12.9  | 2.0      |
| Other                             | 6.1  | 13.3 | 0.0  | 0.0 | 2.1 | 2.6 | 0.0 | 0.1 | 0.0 | 13.3 | 23.6  | 0.8      |

*Notes:* Percent of each group's export value, by country of the ultimate parent of foreign MNEs; domestic MNEs shown for comparison. Categories of BEC Rev. 4.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

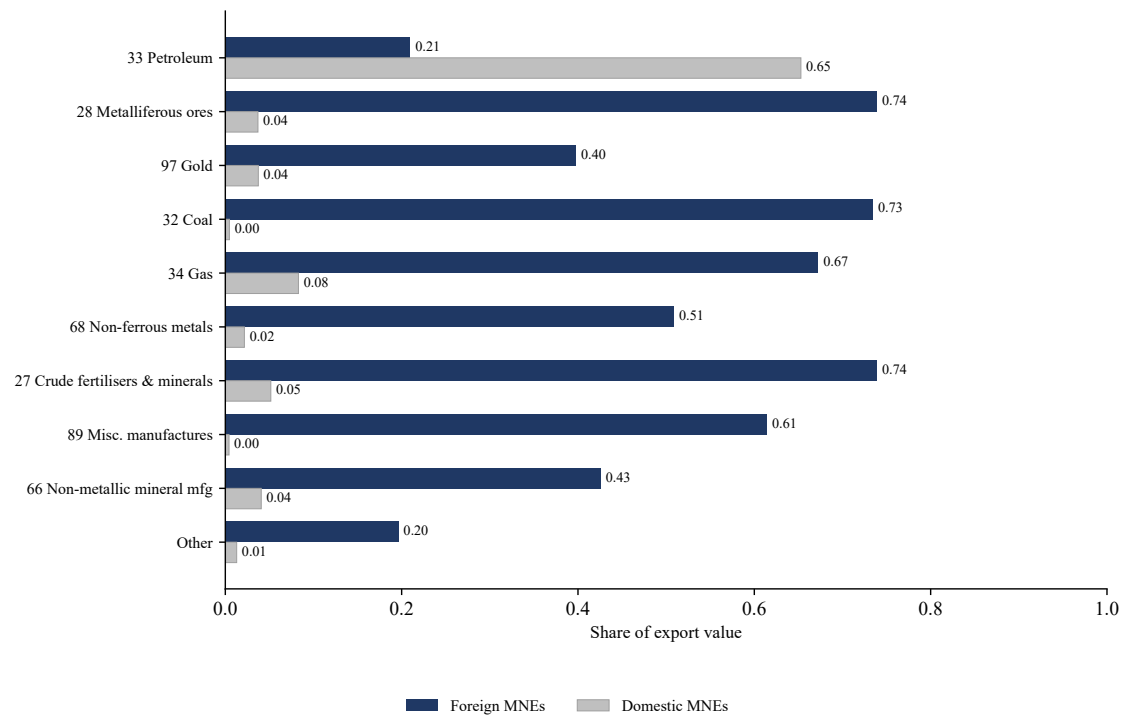
## 25.4 By SITC Division

**Table 194: Mining and Fuels: Exports and Multinational Shares, by SITC Division**

| SITC Rev. 3 division            | Total exports (US\$bn/yr) | MNE exports (US\$bn/yr) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|---------------------------------|---------------------------|-------------------------|-------------|--------------|------------------------------------|
| 33 Petroleum                    | 23.8                      | 20.2                    | 21.0        | 65.3         | USA 0.41, ESP 0.23, BGR 0.10       |
| 28 Metalliferous ores           | 23.7                      | 18.1                    | 73.9        | 3.7          | GBR 0.55, LIE 0.13, USA 0.10       |
| 97 Gold                         | 13.7                      | 6.2                     | 39.7        | 3.7          | CAN 0.69, USA 0.09, CYM 0.04       |
| 32 Coal                         | 6.3                       | 4.7                     | 73.4        | 0.5          | GBR 0.49, USA 0.48, PAN 0.01       |
| 34 Gas                          | 1.8                       | 1.3                     | 67.2        | 8.3          | USA 0.79, FRA 0.12, DEU 0.03       |
| 68 Non-ferrous metals           | 0.8                       | 0.5                     | 50.9        | 2.2          | CAN 0.61, GBR 0.22, CHE 0.06       |
| 27 Crude fertilisers & minerals | 0.5                       | 0.4                     | 73.9        | 5.2          | BRA 0.71, BEL 0.18, PAN 0.02       |
| 89 Misc. manufactures           | 0.5                       | 0.3                     | 61.4        | 0.4          | FRA 0.22, USA 0.22, BEL 0.21       |
| 66 Non-metallic mineral mfg     | 0.4                       | 0.2                     | 42.6        | 4.1          | MEX 0.30, UKR 0.13, ITA 0.12       |
| Other                           | 0.1                       | 0.0                     | 19.6        | 1.3          | CAN 0.82, CHL 0.08, ITA 0.07       |
| All                             | 71.7                      | 51.9                    | 49.2        | 24.5         |                                    |

*Notes:* SITC Rev. 3 divisions; HS6 products without an SITC correspondence are excluded. Foreign and domestic: percent of the group's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

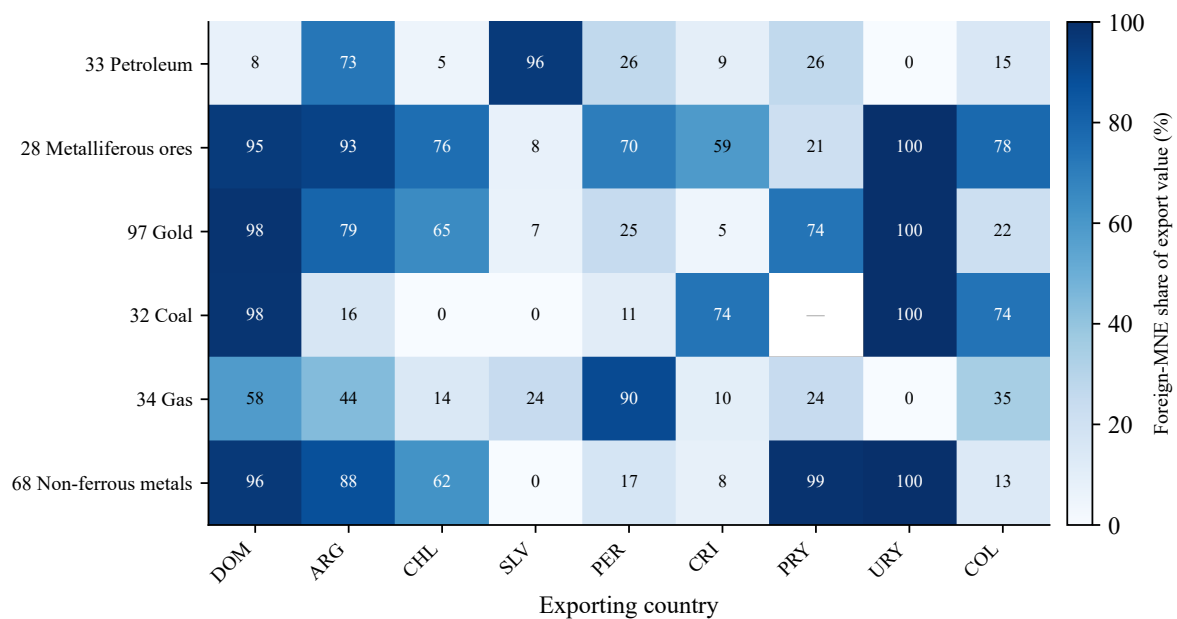
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 134: Mining and Fuels: Multinational Share of Export Value, by SITC Division**

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. SITC Rev. 3 divisions; HS6 products without an SITC correspondence are excluded.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 135: Mining and Fuels: Foreign-Multinational Share of Export Value, by Exporting Country and SITC Division**

*Notes:* Percent of each group's export value in each country shipped by foreign MNEs. SITC Rev. 3 divisions; HS6 products without an SITC correspondence are excluded.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 195: Mining and Fuels: Exports by Exporting Country and SITC Division**

| Sub-sector                                                                  | ARG   | CHL   | COL   | CRI   | DOM   | PER   | PRY   | SLV   | URY   | All   |
|-----------------------------------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| <i>Panel A: composition of the origin's exports, % (columns sum to 100)</i> |       |       |       |       |       |       |       |       |       |       |
| 33 Petroleum                                                                | 35.4  | 1.1   | 67.3  | 11.9  | 4.9   | 11.0  | 58.1  | 55.3  | 19.0  | 34.1  |
| 28 Metalliferous ores                                                       | 16.5  | 90.5  | 0.2   | 5.6   | 4.1   | 52.5  | 0.0   | 0.3   | 0.5   | 33.9  |
| 97 Gold                                                                     | 32.6  | 5.1   | 7.9   | 25.5  | 67.6  | 29.6  | 31.1  | 22.5  | 34.1  | 17.4  |
| 32 Coal                                                                     | 0.1   | 0.1   | 23.0  | 0.1   | 0.0   | 0.1   | 0.0   | 0.0   | 0.0   | 9.4   |
| 34 Gas                                                                      | 9.4   | 0.3   | 0.6   | 0.2   | 0.2   | 3.7   | 5.0   | 0.2   | 0.5   | 2.2   |
| 68 Non-ferrous metals                                                       | 4.0   | 2.0   | 0.2   | 1.5   | 1.0   | 1.2   | 0.1   | 0.2   | 0.0   | 1.1   |
| 27 Crude fertilisers & minerals                                             | 0.8   | 0.8   | 0.1   | 5.4   | 0.3   | 1.3   | 4.9   | 5.4   | 1.2   | 0.7   |
| 89 Misc. manufactures                                                       | 0.0   | 0.0   | 0.1   | 15.5  | 17.2  | 0.4   | 0.8   | 2.1   | 0.5   | 0.5   |
| 66 Non-metallic mineral mfg                                                 | 0.8   | 0.0   | 0.5   | 34.3  | 4.7   | 0.1   | 0.0   | 7.4   | 21.3  | 0.5   |
| Other                                                                       | 0.3   | 0.0   | 0.2   | 0.0   | 0.0   | 0.0   | 0.0   | 6.6   | 22.9  | 0.2   |
| All sub-sectors                                                             | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 |
| <i>Panel B: foreign-MNE share of export value, %</i>                        |       |       |       |       |       |       |       |       |       |       |
| 33 Petroleum                                                                | 73.0  | 5.0   | 15.2  | 9.4   | 7.8   | 26.2  | 26.3  | 95.9  | 0.2   | 21.0  |
| 28 Metalliferous ores                                                       | 92.9  | 76.1  | 78.1  | 58.6  | 95.4  | 70.2  | 21.5  | 7.6   | 100.0 | 73.9  |
| 97 Gold                                                                     | 79.4  | 64.9  | 21.7  | 4.6   | 98.4  | 24.8  | 73.8  | 6.9   | 100.0 | 39.7  |
| 32 Coal                                                                     | 16.4  | 0.0   | 73.8  | 73.7  | 98.0  | 11.3  | –     | 0.0   | 100.0 | 73.4  |
| 34 Gas                                                                      | 44.1  | 14.2  | 34.5  | 9.9   | 58.1  | 89.6  | 24.4  | 24.2  | 0.0   | 67.2  |
| 68 Non-ferrous metals                                                       | 87.7  | 61.8  | 13.3  | 8.1   | 96.1  | 17.1  | 98.6  | 0.1   | 100.0 | 50.9  |
| 27 Crude fertilisers & minerals                                             | 31.5  | 64.8  | 17.3  | 62.9  | 37.0  | 90.3  | 77.9  | 76.4  | 44.9  | 73.9  |
| 89 Misc. manufactures                                                       | 11.9  | 24.5  | 67.8  | 44.7  | 57.9  | 75.5  | 94.8  | 0.4   | 69.2  | 61.4  |
| 66 Non-metallic mineral mfg                                                 | 21.5  | 28.6  | 26.2  | 97.6  | 71.2  | 58.5  | 0.0   | 0.0   | 34.4  | 42.6  |
| Other                                                                       | 23.5  | 25.9  | 32.8  | 99.9  | 97.2  | 12.8  | 100.0 | 1.7   | 0.1   | 19.6  |
| All sub-sectors                                                             | 75.3  | 74.1  | 29.6  | 49.6  | 85.3  | 52.2  | 44.1  | 58.9  | 42.9  | 49.2  |

*Notes:* Panel A: composition of each country's exports in the sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs. SITC Rev. 3 divisions; HS6 products without an SITC correspondence are excluded.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 196: Mining and Fuels: Exports by SITC Division and Parent Country**

| SITC Rev. 3 division            | GBR  | USA  | CAN  | LIE | ESP | PAN | AUS | BRA  | BGR | FRA  | Other | Domestic |
|---------------------------------|------|------|------|-----|-----|-----|-----|------|-----|------|-------|----------|
| 33 Petroleum                    | 0.5  | 7.8  | 1.3  | 0.4 | 4.3 | 0.1 | 0.0 | 0.5  | 1.9 | 0.3  | 4.0   | 65.3     |
| 28 Metalliferous ores           | 39.8 | 7.0  | 5.9  | 9.5 | 0.2 | 2.7 | 3.2 | 0.6  | 0.0 | 0.7  | 4.2   | 3.7      |
| 97 Gold                         | 0.7  | 3.2  | 26.0 | 0.0 | 0.3 | 1.3 | 0.5 | 0.4  | 0.1 | 0.1  | 7.0   | 3.7      |
| 32 Coal                         | 34.0 | 33.4 | 0.7  | 0.0 | 0.1 | 0.8 | 0.0 | 0.0  | 0.2 | 0.3  | 4.0   | 0.5      |
| 34 Gas                          | 1.6  | 52.8 | 0.0  | 0.0 | 0.8 | 0.0 | 0.0 | 0.0  | 0.0 | 8.1  | 3.9   | 8.3      |
| 68 Non-ferrous metals           | 10.7 | 0.2  | 29.9 | 0.0 | 0.0 | 0.2 | 0.9 | 0.0  | 0.0 | 0.7  | 8.2   | 2.2      |
| 27 Crude fertilisers & minerals | 0.1  | 0.8  | 0.3  | 0.0 | 0.1 | 1.1 | 0.8 | 50.4 | 0.0 | 0.1  | 20.2  | 5.2      |
| 89 Misc. manufactures           | 6.1  | 13.3 | 0.0  | 0.0 | 2.1 | 2.6 | 0.0 | 0.1  | 0.0 | 13.3 | 23.7  | 0.4      |
| 66 Non-metallic mineral mfg     | 0.8  | 0.1  | 0.1  | 0.0 | 3.4 | 1.2 | 0.0 | 2.0  | 4.1 | 0.0  | 30.9  | 4.1      |
| Other                           | 0.0  | 0.6  | 16.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0  | 0.0 | 0.0  | 3.0   | 1.3      |

*Notes:* Percent of each group's export value, by country of the ultimate parent of foreign MNEs; domestic MNEs shown for comparison. SITC Rev. 3 divisions; HS6 products without an SITC correspondence are excluded.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

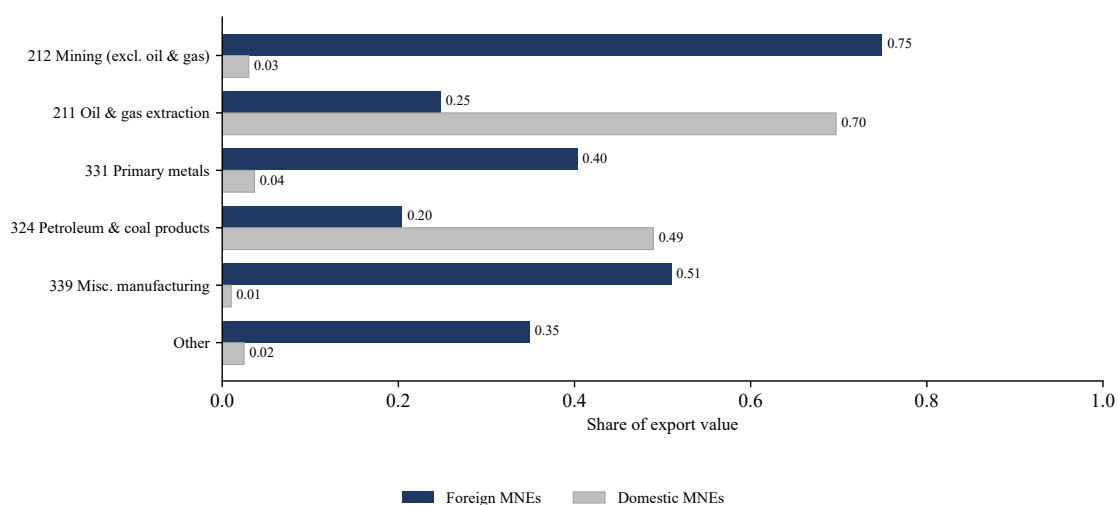
## 25.5 By NAICS Industry

**Table 197:** Mining and Fuels: Exports and Multinational Shares, by NAICS Industry

| NAICS 3-digit (product-based) | Total exports (US\$bn/yr) | MNE exports (US\$bn/yr) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|-------------------------------|---------------------------|-------------------------|-------------|--------------|------------------------------------|
| 212 Mining (excl. oil & gas)  | 29.7                      | 22.9                    | 74.9        | 3.0          | GBR 0.53, USA 0.18, LIE 0.10       |
| 211 Oil & gas extraction      | 19.6                      | 18.4                    | 24.8        | 69.7         | USA 0.52, ESP 0.17, BGR 0.09       |
| 331 Primary metals            | 15.0                      | 6.8                     | 40.3        | 3.7          | CAN 0.69, USA 0.08, PAN 0.03       |
| 324 Petroleum & coal products | 3.8                       | 2.6                     | 20.4        | 49.0         | ESP 0.52, USA 0.17, BRA 0.13       |
| 339 Misc. manufacturing       | 0.6                       | 0.3                     | 51.0        | 1.0          | USA 0.20, FRA 0.19, BEL 0.18       |
| Other                         | 0.8                       | 0.3                     | 34.9        | 2.5          | MEX 0.18, ESP 0.13, USA 0.10       |
| All                           | 69.5                      | 51.4                    | 50.0        | 25.0         |                                    |

*Notes:* NAICS 3-digit industries assigned to HS6 products. Foreign and domestic: percent of the group's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

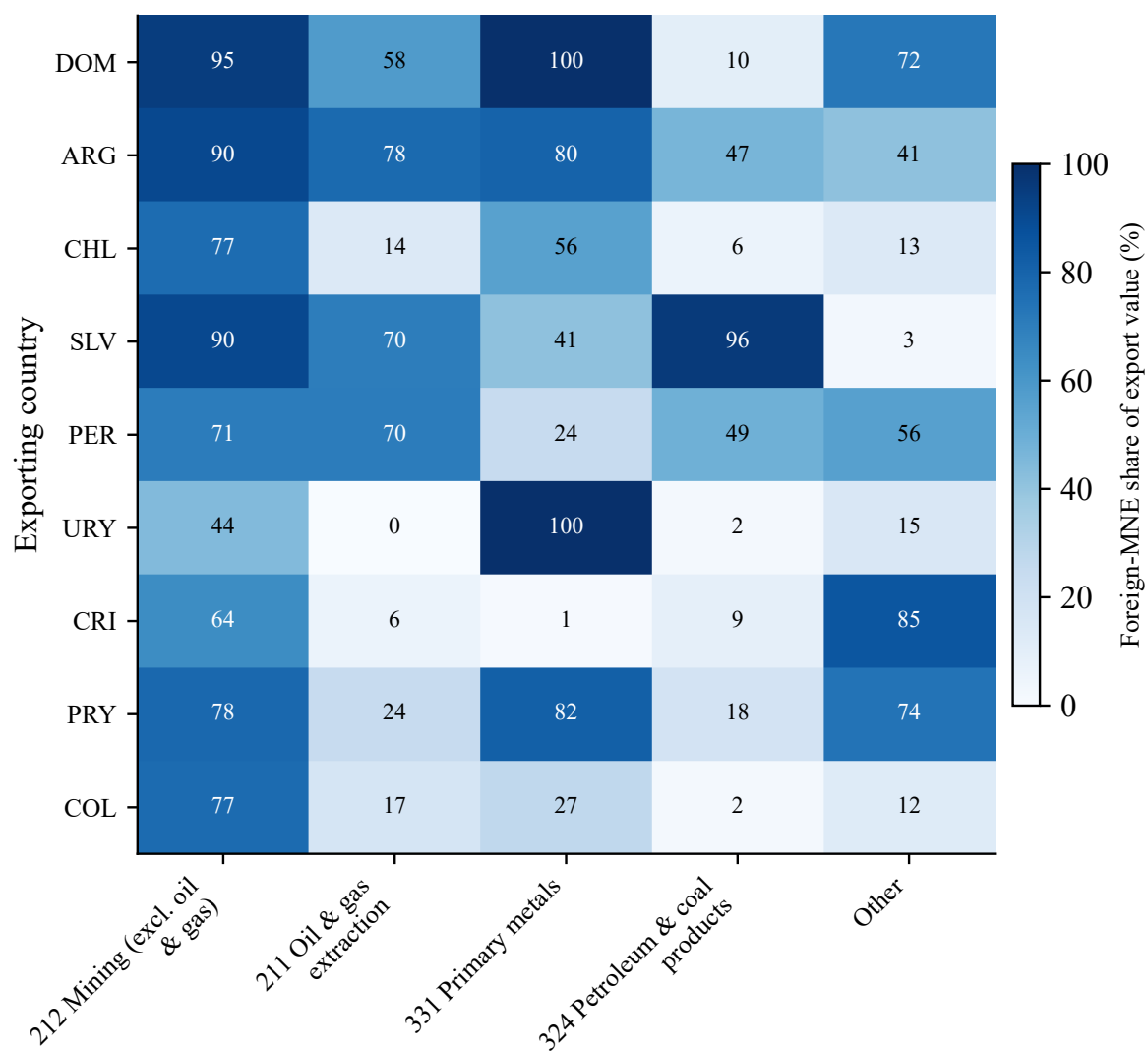
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 136:** Mining and Fuels: Multinational Share of Export Value, by NAICS Industry

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. NAICS 3-digit industries assigned to HS6 products.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 137:** Mining and Fuels: Foreign-Multinational Share of Export Value, by Exporting Country and NAICS Industry

*Notes:* Percent of each group's export value in each country shipped by foreign MNEs. NAICS 3-digit industries assigned to HS6 products.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 198: Mining and Fuels: Exports by Exporting Country and NAICS Industry**

| Origin                                                                   | 212 Mining (excl. oil & gas) | 211 Oil & gas extraction | 331 Primary metals | 324 Petroleum & coal products | 339 Misc. manufacturing | Other | All   |
|--------------------------------------------------------------------------|------------------------------|--------------------------|--------------------|-------------------------------|-------------------------|-------|-------|
| <i>Panel A: composition of the origin's exports, % (rows sum to 100)</i> |                              |                          |                    |                               |                         |       |       |
| ARG                                                                      | 18.8                         | 34.7                     | 40.5               | 4.1                           | 0.1                     | 1.9   | 100.0 |
| CHL                                                                      | 90.4                         | 0.3                      | 8.2                | 0.9                           | 0.0                     | 0.2   | 100.0 |
| COL                                                                      | 21.3                         | 60.0                     | 9.7                | 7.7                           | 0.6                     | 0.9   | 100.0 |
| CRI                                                                      | 5.2                          | 0.2                      | 21.9               | 11.8                          | 16.4                    | 44.5  | 100.0 |
| DOM                                                                      | 4.3                          | 0.3                      | 67.9               | 0.1                           | 18.1                    | 9.4   | 100.0 |
| PER                                                                      | 56.6                         | 5.2                      | 32.5               | 5.1                           | 0.4                     | 0.3   | 100.0 |
| PRY                                                                      | 5.3                          | 5.6                      | 1.9                | 53.7                          | 0.1                     | 33.4  | 100.0 |
| SLV                                                                      | 4.6                          | 0.6                      | 1.7                | 54.5                          | 2.4                     | 36.2  | 100.0 |
| URY                                                                      | 1.4                          | 0.6                      | 41.4               | 2.8                           | 13.9                    | 40.0  | 100.0 |
| All origins                                                              | 44.1                         | 29.0                     | 19.7               | 5.4                           | 0.8                     | 1.0   | 100.0 |
| <i>Panel B: foreign-MNE share of export value, %</i>                     |                              |                          |                    |                               |                         |       |       |
| ARG                                                                      | 90.4                         | 77.6                     | 80.2               | 46.6                          | 12.6                    | 41.4  | 79.0  |
| CHL                                                                      | 76.9                         | 13.5                     | 55.8               | 6.1                           | 6.8                     | 13.4  | 74.2  |
| COL                                                                      | 77.1                         | 16.9                     | 26.7               | 1.9                           | 35.0                    | 12.1  | 29.5  |
| CRI                                                                      | 64.1                         | 5.9                      | 1.0                | 8.8                           | 42.6                    | 85.4  | 49.6  |
| DOM                                                                      | 94.9                         | 58.1                     | 99.7               | 9.9                           | 57.9                    | 72.4  | 89.2  |
| PER                                                                      | 70.7                         | 70.4                     | 24.4               | 48.9                          | 76.1                    | 56.0  | 54.5  |
| PRY                                                                      | 78.5                         | 24.4                     | 81.6               | 18.0                          | 13.9                    | 73.9  | 41.4  |
| SLV                                                                      | 90.4                         | 70.0                     | 41.4               | 95.9                          | 0.3                     | 2.7   | 58.5  |
| URY                                                                      | 44.5                         | 0.0                      | 100.0              | 1.7                           | 24.4                    | 15.2  | 51.5  |
| All origins                                                              | 74.9                         | 24.8                     | 40.3               | 20.4                          | 51.0                    | 34.9  | 50.0  |

*Notes:* Panel A: composition of each country's exports in the sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs. NAICS 3-digit industries assigned to HS6 products.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 199: Mining and Fuels: Exports by NAICS Industry and Parent Country**

| NAICS 3-digit (product-based) | GBR  | USA  | CAN  | LIE | ESP | PAN | AUS | BRA | BGR | FRA | Other | Domestic |
|-------------------------------|------|------|------|-----|-----|-----|-----|-----|-----|-----|-------|----------|
| 212 Mining (excl. oil & gas)  | 39.0 | 12.9 | 4.8  | 7.5 | 0.1 | 2.3 | 2.5 | 1.3 | 0.0 | 0.6 | 3.8   | 3.0      |
| 211 Oil & gas extraction      | 0.3  | 12.8 | 1.6  | 0.4 | 4.1 | 0.0 | 0.0 | 0.0 | 2.2 | 0.7 | 2.6   | 69.7     |
| 331 Primary metals            | 1.3  | 2.9  | 25.3 | 0.0 | 0.1 | 1.3 | 0.5 | 0.4 | 0.0 | 0.1 | 8.5   | 3.7      |
| 324 Petroleum & coal products | 0.5  | 2.0  | 0.0  | 0.0 | 6.1 | 0.7 | 0.0 | 1.5 | 0.0 | 0.1 | 9.6   | 49.0     |
| 339 Misc. manufacturing       | 4.9  | 9.4  | 0.1  | 0.0 | 3.5 | 1.8 | 0.0 | 0.1 | 2.8 | 9.3 | 19.2  | 1.0      |
| Other                         | 0.2  | 3.5  | 3.1  | 0.0 | 4.5 | 2.4 | 0.0 | 2.0 | 1.8 | 1.3 | 16.0  | 2.5      |

*Notes:* Percent of each group's export value, by country of the ultimate parent of foreign MNEs; domestic MNEs shown for comparison. NAICS 3-digit industries assigned to HS6 products.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

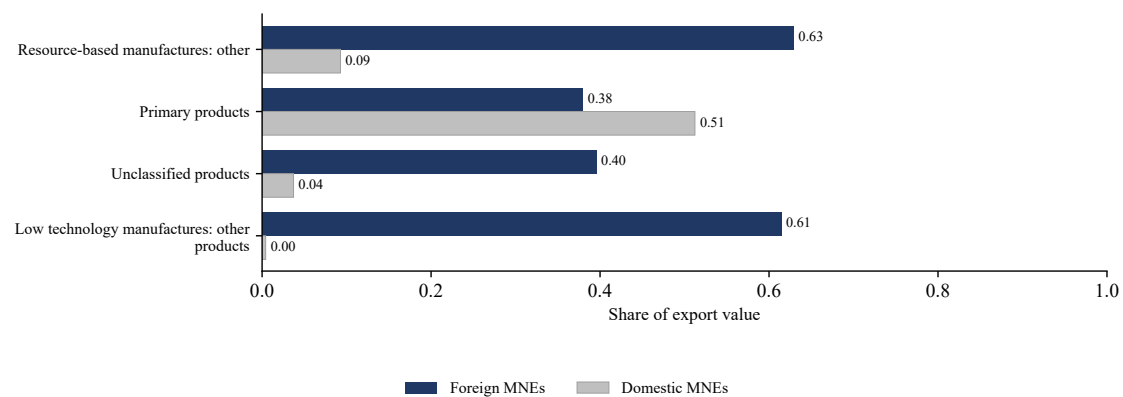
## 25.6 By Technology Class

**Table 200: Mining and Fuels: Exports and Multinational Shares, by Technology Class**

| Lall (2000) category                        | Total exports (US\$bn/yr) | MNE exports (US\$bn/yr) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|---------------------------------------------|---------------------------|-------------------------|-------------|--------------|------------------------------------|
| Resource-based manufactures: other          | 30.6                      | 21.7                    | 62.9        | 9.3          | GBR 0.53, LIE 0.13, USA 0.10       |
| Primary products                            | 26.7                      | 23.7                    | 38.0        | 51.2         | USA 0.47, GBR 0.23, ESP 0.08       |
| Unclassified products                       | 13.9                      | 6.2                     | 39.5        | 3.7          | CAN 0.69, USA 0.09, CYM 0.04       |
| Low technology manufactures: other products | 0.5                       | 0.3                     | 61.4        | 0.4          | FRA 0.22, USA 0.22, BEL 0.21       |
| All                                         | 71.7                      | 51.9                    | 49.2        | 24.5         |                                    |

*Notes:* Lall (2000) technology classes. Foreign and domestic: percent of the group's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

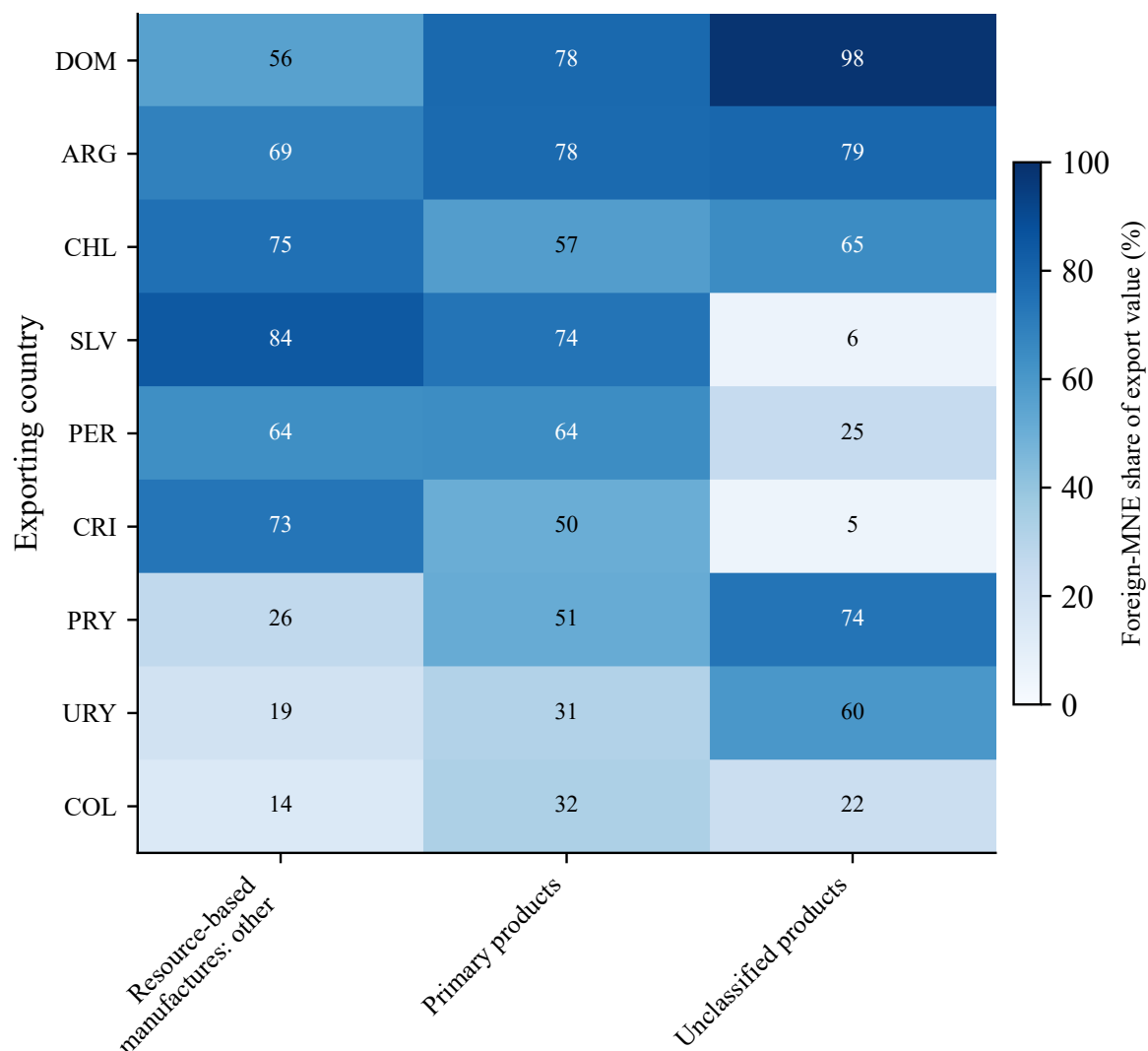
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 138:** Mining and Fuels: Multinational Share of Export Value, by Technology Class

*Notes:* Shares of mining and fuels export value (HS 25–27 and 71), all available years pooled. Lall (2000) technology classes.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 139:** Mining and Fuels: Foreign-Multinational Share of Export Value, by Exporting Country and Technology Class

*Notes:* Percent of each group's export value in each country shipped by foreign MNEs. Lall (2000) technology classes.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 201: Mining and Fuels: Exports by Exporting Country and Technology Class**

| Origin                                                                   | Resource-based manufactures: other | Primary products | Unclassified products | Low technology manufactures: other products | All   |
|--------------------------------------------------------------------------|------------------------------------|------------------|-----------------------|---------------------------------------------|-------|
| <i>Panel A: composition of the origin's exports, % (rows sum to 100)</i> |                                    |                  |                       |                                             |       |
| ARG                                                                      | 30.8                               | 36.3             | 32.8                  | 0.0                                         | 100.0 |
| CHL                                                                      | 91.7                               | 3.1              | 5.1                   | 0.0                                         | 100.0 |
| COL                                                                      | 11.1                               | 80.7             | 8.1                   | 0.1                                         | 100.0 |
| CRI                                                                      | 51.9                               | 7.1              | 25.6                  | 15.5                                        | 100.0 |
| DOM                                                                      | 13.6                               | 1.6              | 67.6                  | 17.2                                        | 100.0 |
| PER                                                                      | 62.4                               | 7.6              | 29.6                  | 0.4                                         | 100.0 |
| PRY                                                                      | 58.1                               | 10.0             | 31.1                  | 0.8                                         | 100.0 |
| SLV                                                                      | 62.6                               | 6.2              | 29.1                  | 2.1                                         | 100.0 |
| URY                                                                      | 40.8                               | 1.7              | 57.0                  | 0.5                                         | 100.0 |
| All origins                                                              | 43.3                               | 38.6             | 17.5                  | 0.5                                         | 100.0 |
| <i>Panel B: foreign-MNE share of export value, %</i>                     |                                    |                  |                       |                                             |       |
| ARG                                                                      | 68.7                               | 77.6             | 78.9                  | 11.9                                        | 75.3  |
| CHL                                                                      | 75.2                               | 57.1             | 64.7                  | 24.5                                        | 74.1  |
| COL                                                                      | 13.9                               | 32.5             | 22.0                  | 67.8                                        | 29.6  |
| CRI                                                                      | 73.1                               | 50.1             | 4.8                   | 44.7                                        | 49.6  |
| DOM                                                                      | 55.7                               | 78.0             | 98.4                  | 57.9                                        | 85.3  |
| PER                                                                      | 63.6                               | 64.3             | 24.8                  | 75.5                                        | 52.2  |
| PRY                                                                      | 26.3                               | 51.2             | 73.8                  | 94.8                                        | 44.1  |
| SLV                                                                      | 84.2                               | 73.7             | 5.7                   | 0.4                                         | 58.9  |
| URY                                                                      | 19.3                               | 31.2             | 59.9                  | 69.2                                        | 42.9  |
| All origins                                                              | 62.9                               | 38.0             | 39.5                  | 61.4                                        | 49.2  |

*Notes:* Panel A: composition of each country's exports in the sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs. Lall (2000) technology classes.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 202: Mining and Fuels: Exports by Technology Class and Parent Country**

| Lall (2000) category                        | GBR  | USA  | CAN  | LIE | ESP | PAN | AUS | BRA | BGR | FRA  | Other | Domestic |
|---------------------------------------------|------|------|------|-----|-----|-----|-----|-----|-----|------|-------|----------|
| Resource-based manufactures: other          | 31.5 | 5.9  | 4.6  | 7.4 | 1.0 | 2.3 | 2.5 | 0.9 | 0.0 | 0.7  | 6.0   | 9.3      |
| Primary products                            | 8.8  | 17.6 | 2.2  | 0.3 | 3.0 | 0.1 | 0.0 | 0.9 | 1.7 | 0.6  | 2.7   | 51.2     |
| Unclassified products                       | 0.7  | 3.2  | 25.9 | 0.0 | 0.3 | 1.3 | 0.5 | 0.4 | 0.1 | 0.1  | 7.0   | 3.7      |
| Low technology manufactures: other products | 6.1  | 13.3 | 0.0  | 0.0 | 2.1 | 2.6 | 0.0 | 0.1 | 0.0 | 13.3 | 23.7  | 0.4      |

*Notes:* Percent of each group's export value, by country of the ultimate parent of foreign MNEs; domestic MNEs shown for comparison. Lall (2000) technology classes.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 25.7 HS Section and End Use

**Table 203: Mining and Fuels: Exports by HS Section and End Use**

| HS section / BEC end use       | Intermediate | Consumption | Capital | Total |
|--------------------------------|--------------|-------------|---------|-------|
| V Mineral products             | 50.6         | 0.0         | 0.0     | 50.6  |
| XIV Precious metals and stones | 14.8         | 0.5         | 0.0     | 15.3  |
| Total                          | 65.4         | 0.5         | 0.0     | 65.9  |

*Notes:* Export value by HS section and BEC Rev. 4 end use. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 204: Mining and Fuels: Foreign-Multinational Share of Exports by HS Section and End Use**

| HS section / BEC end use       | Intermediate | Consumption | Capital |
|--------------------------------|--------------|-------------|---------|
| V Mineral products             | 54.5         | —           | —       |
| XIV Precious metals and stones | 40.2         | 61.7        | 21.4    |

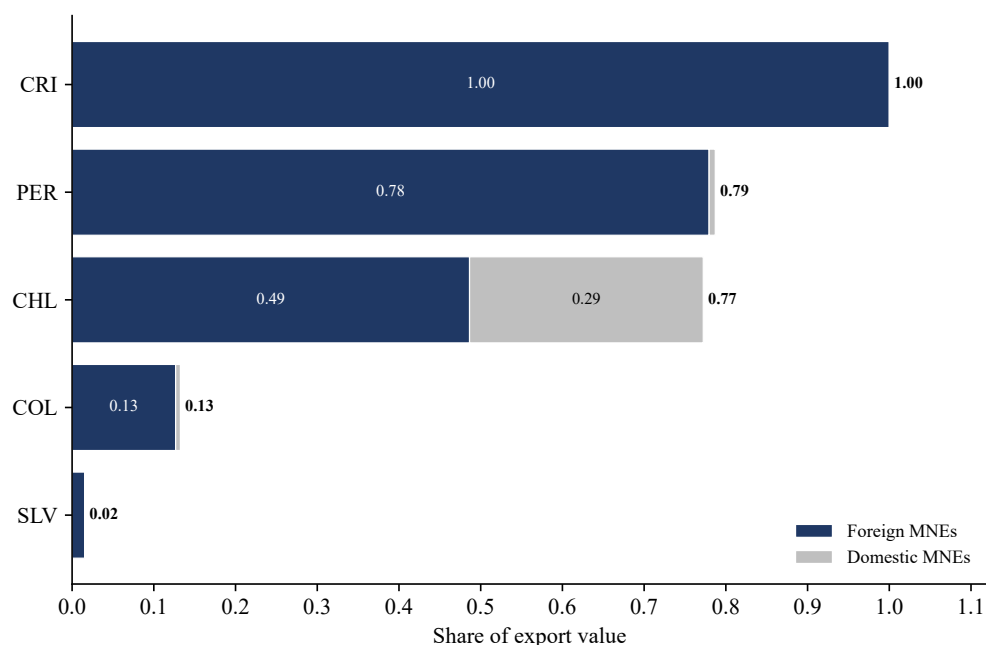
*Notes:* Percent of each cell's export value shipped by foreign MNEs.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## Part IV

# Other Goods

## 26 Fact 1: Multinationals Account for a Large Share of Exports



**Figure 140:** Other Goods: Multinational Share of Export Value, by Exporting Country

*Notes:* Shares of export value of other goods (HS 98–99 and unclassified codes), all available years pooled. The figure at the end of each bar is the total MNE share.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

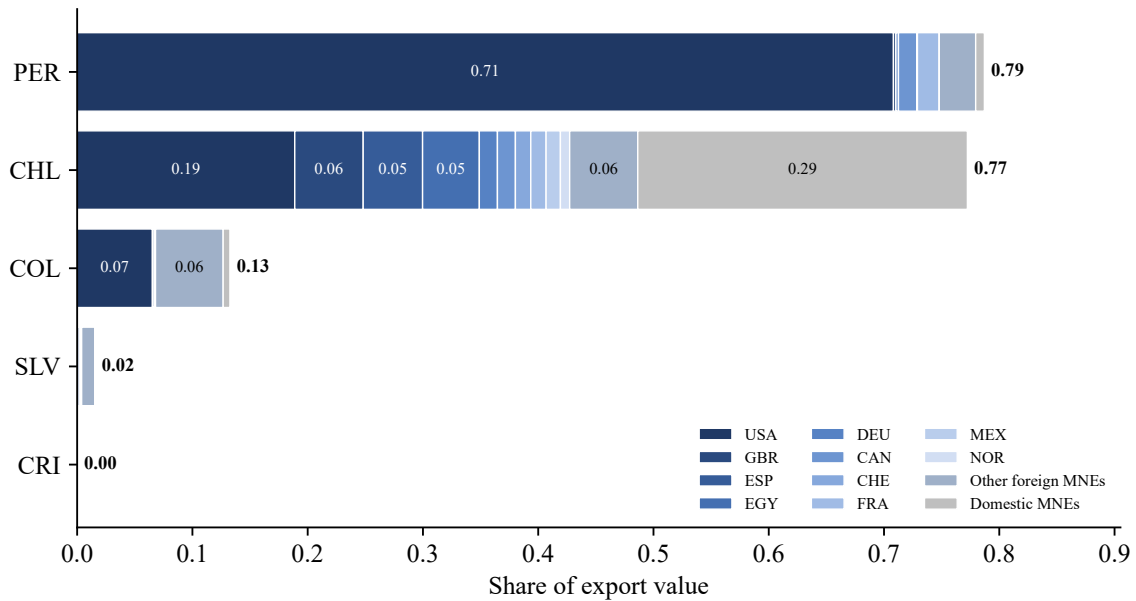
**Table 205:** Other Goods: Multinational Share of Export Value, by Exporting Country

| Origin | Foreign | Domestic | All MNEs | Value (US\$bn/yr) |
|--------|---------|----------|----------|-------------------|
| CRI    | 1.000   | 0.000    | 1.000    | 0.0               |
| PER    | 0.780   | 0.008    | 0.787    | 0.0               |
| CHL    | 0.486   | 0.286    | 0.772    | 0.5               |
| COL    | 0.127   | 0.006    | 0.133    | 0.0               |
| SLV    | 0.015   | 0.000    | 0.015    | 0.0               |
| All    | 0.479   | 0.280    | 0.759    | 0.5               |

*Notes:* Shares of export value of other goods (HS 98–99 and unclassified codes), all available years pooled. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 26.1 By Parent Country



**Figure 141:** Other Goods: Multinational Share of Export Value, by Exporting Country and Parent Country

*Notes:* Shares of export value of other goods (HS 98–99 and unclassified codes), all available years pooled. Foreign MNEs split by country of the ultimate parent; the ten largest parent countries are shown separately.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 206:** Other Goods: Multinational Share of Export Value, by Exporting Country and Parent Country

| Origin | USA   | GBR   | ESP   | EGY   | DEU   | CAN   | CHE   | FRA   | MEX   | NOR   | Other | Domestic | All MNEs | Value (US\$bn/yr) |
|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|----------|-------------------|
| PER    | 0.708 | 0.002 | 0.000 | 0.000 | 0.002 | 0.016 | 0.000 | 0.019 | 0.000 | 0.000 | 0.032 | 0.008    | 0.787    | 0.0               |
| CHL    | 0.189 | 0.059 | 0.052 | 0.049 | 0.016 | 0.016 | 0.013 | 0.013 | 0.012 | 0.008 | 0.059 | 0.286    | 0.772    | 0.5               |
| COL    | 0.065 | 0.001 | 0.001 | 0.000 | 0.000 | 0.000 | 0.000 | 0.001 | 0.000 | 0.000 | 0.059 | 0.006    | 0.133    | 0.0               |
| SLV    | 0.000 | 0.001 | 0.002 | 0.000 | 0.001 | 0.000 | 0.000 | 0.000 | 0.001 | 0.000 | 0.011 | 0.000    | 0.015    | 0.0               |
| CRI    | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000    | 0.000    | 0.0               |

*Notes:* Shares of export value of other goods (HS 98–99 and unclassified codes), all available years pooled. Other: all remaining foreign parents. Values are annual averages in US\$ billion.

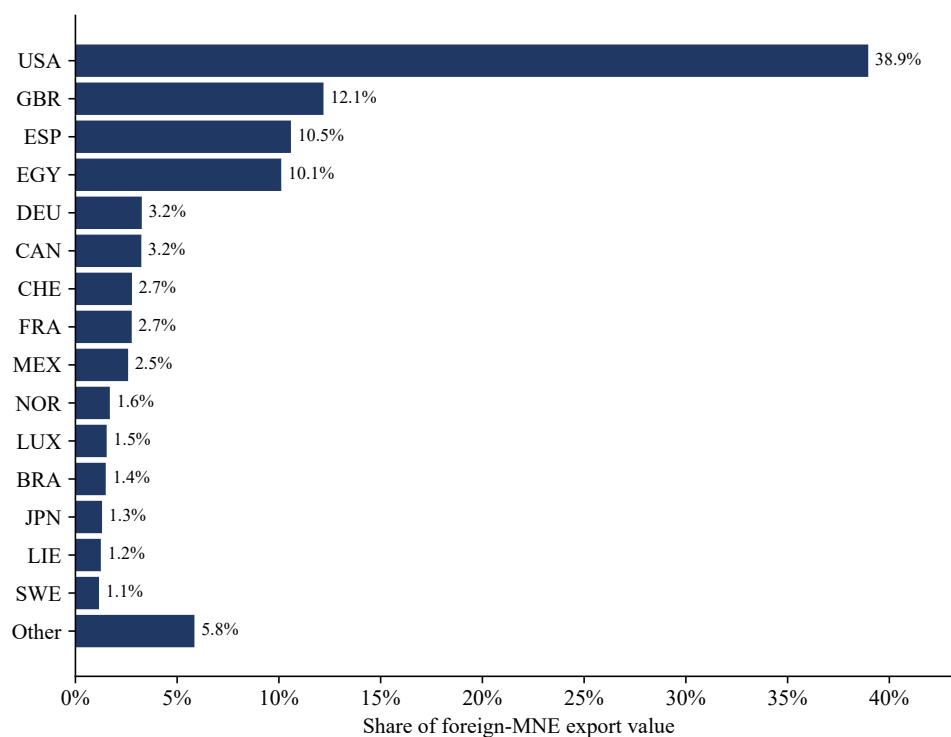
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 27 Fact 2: Foreign Multinationals Specialize in Complex Products, Domestic Multinationals in Primary Goods

Product characteristics (complexity, technology class, elasticities) are not available for these codes, so this fact cannot be computed for them.

## 28 Fact 3: Multinationals from a Small Set of Countries Dominate Exports

### 28.1 Parent Countries



**Figure 142:** Other Goods: Foreign-Multinational Exports, by Parent Country

*Notes:* Percent of foreign-MNE export value, by country of the ultimate parent.

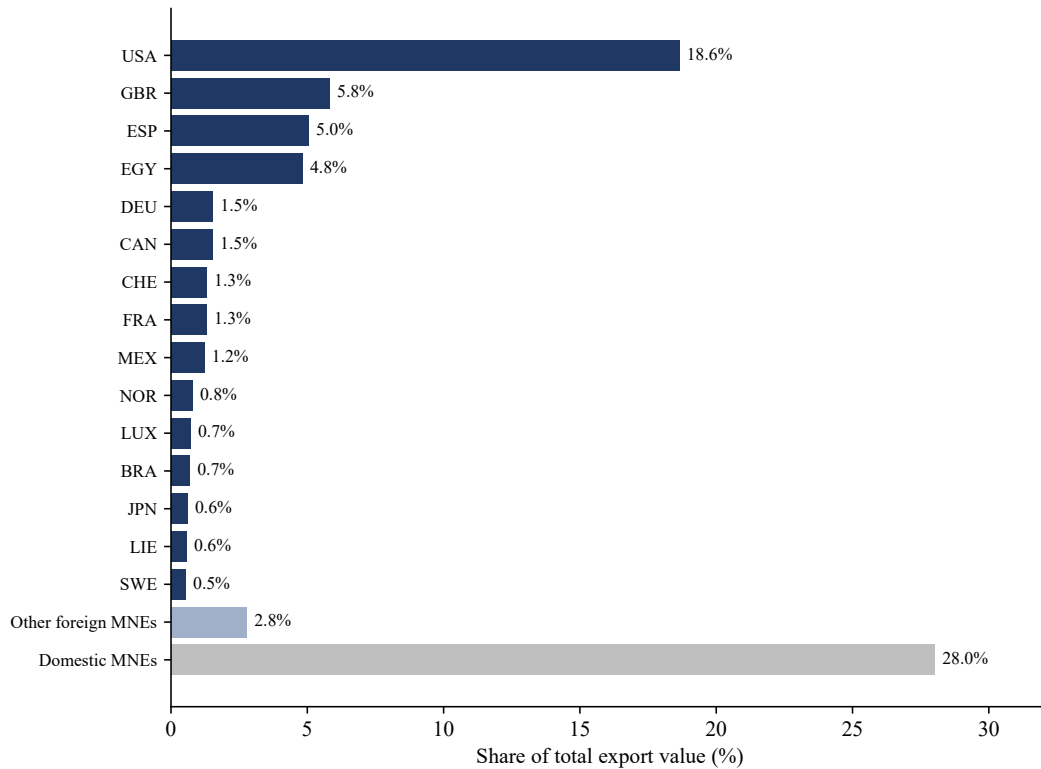
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 207:** Other Goods: Foreign-Multinational Exports, by Parent Country

| Parent country       | Share (%) | Value (US\$bn/yr) |
|----------------------|-----------|-------------------|
| USA                  | 38.9      | 0.1               |
| GBR                  | 12.1      | 0.0               |
| ESP                  | 10.5      | 0.0               |
| EGY                  | 10.1      | 0.0               |
| DEU                  | 3.2       | 0.0               |
| CAN                  | 3.2       | 0.0               |
| CHE                  | 2.7       | 0.0               |
| FRA                  | 2.7       | 0.0               |
| MEX                  | 2.5       | 0.0               |
| NOR                  | 1.6       | 0.0               |
| LUX                  | 1.5       | 0.0               |
| BRA                  | 1.4       | 0.0               |
| JPN                  | 1.3       | 0.0               |
| LIE                  | 1.2       | 0.0               |
| SWE                  | 1.1       | 0.0               |
| Other (64 countries) | 5.8       | 0.0               |
| All foreign MNEs     | 100.0     | 0.2               |

*Notes:* Percent of foreign-MNE export value, by country of the ultimate parent. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 143:** Other Goods: Share of Total Exports, by Parent Country

*Notes:* Percent of total export value shipped by foreign MNEs of each parent country, other foreign MNEs and domestic MNEs; the remainder is local firms.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 208:** Other Goods: Share of Total Exports, by Parent Country

| Exporter type / parent country | Share of total exports (%) |
|--------------------------------|----------------------------|
| USA                            | 18.6                       |
| GBR                            | 5.8                        |
| ESP                            | 5.0                        |
| EGY                            | 4.8                        |
| DEU                            | 1.5                        |
| CAN                            | 1.5                        |
| CHE                            | 1.3                        |
| FRA                            | 1.3                        |
| MEX                            | 1.2                        |
| NOR                            | 0.8                        |
| LUX                            | 0.7                        |
| BRA                            | 0.7                        |
| JPN                            | 0.6                        |
| LIE                            | 0.6                        |
| SWE                            | 0.5                        |
| Other foreign MNEs             | 2.8                        |
| Domestic MNEs                  | 28.0                       |
| All MNEs                       | 75.9                       |
| Local firms                    | 24.1                       |

*Notes:* Percent of total export value, by type of exporter and parent country.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 28.2 Parent Region and Destination Region

**Table 209:** Other Goods: Multinational Exports by Parent Region and Destination Region: Export Value

| Parent region / Destination region | Latin America | North America | Europe | Asia | Africa | Oceania | Rest of World | Total |
|------------------------------------|---------------|---------------|--------|------|--------|---------|---------------|-------|
| Latin America                      | 0.0           | 0.0           | 0.0    | 0.0  | 0.0    | 0.0     | 0.0           | 0.0   |
| North America                      | 0.0           | 0.0           | 0.0    | 0.0  | 0.0    | 0.0     | 0.0           | 0.1   |
| Europe                             | 0.0           | 0.0           | 0.0    | 0.0  | 0.0    | 0.0     | 0.0           | 0.1   |
| Asia                               | 0.0           | 0.0           | 0.0    | 0.0  | 0.0    | 0.0     | 0.0           | 0.0   |
| Africa                             | 0.0           | 0.0           | 0.0    | 0.0  | 0.0    | 0.0     | 0.0           | 0.0   |
| Oceania                            | 0.0           | 0.0           | 0.0    | 0.0  | 0.0    | 0.0     | 0.0           | 0.0   |
| Rest of World                      | 0.0           | 0.0           | 0.0    | 0.0  | 0.0    | 0.0     | 0.0           | 0.0   |
| Domestic MNEs                      | 0.1           | 0.0           | 0.0    | 0.0  | 0.0    | 0.0     | 0.0           | 0.1   |
| Local firms                        | 0.1           | 0.0           | 0.0    | 0.0  | 0.0    | 0.0     | 0.0           | 0.1   |
| Total                              | 0.3           | 0.1           | 0.1    | 0.0  | 0.0    | 0.0     | 0.0           | 0.4   |

*Notes:* Values are annual averages in US\$ billion. Rows: region of the ultimate parent; columns: destination region.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 210:** Other Goods: Multinational Exports by Parent Region and Destination Region: Row Percentages

| Parent region / Destination region | Latin America | North America | Europe | Asia | Africa | Oceania | Rest of World | Total |
|------------------------------------|---------------|---------------|--------|------|--------|---------|---------------|-------|
| Latin America                      | 33.1          | 22.2          | 39.9   | 3.8  | 0.1    | 0.2     | 0.7           | 100.0 |
| North America                      | 51.8          | 39.0          | 7.8    | 1.3  | 0.0    | 0.1     | 0.0           | 100.0 |
| Europe                             | 30.2          | 37.1          | 30.1   | 2.0  | 0.2    | 0.4     | 0.0           | 100.0 |
| Asia                               | 30.0          | 17.7          | 13.0   | 39.1 | 0.0    | 0.1     | 0.0           | 100.0 |
| Africa                             | 82.7          | 2.7           | 10.8   | 2.7  | 0.7    | 0.0     | 0.4           | 100.0 |
| Oceania                            | 78.6          | 7.2           | 0.3    | 12.3 | 0.0    | 1.6     | 0.0           | 100.0 |
| Rest of World                      | 5.0           | 0.0           | 0.0    | 95.0 | 0.0    | 0.0     | 0.0           | 100.0 |
| Domestic MNEs                      | 92.6          | 4.0           | 3.0    | 0.4  | 0.0    | 0.0     | 0.0           | 100.0 |
| Local firms                        | 50.6          | 18.3          | 26.5   | 4.0  | 0.3    | 0.2     | 0.0           | 100.0 |

*Notes:* Percent of each group's exports, by destination region (rows add to 100). Domestic MNEs and local firms are shown for comparison.

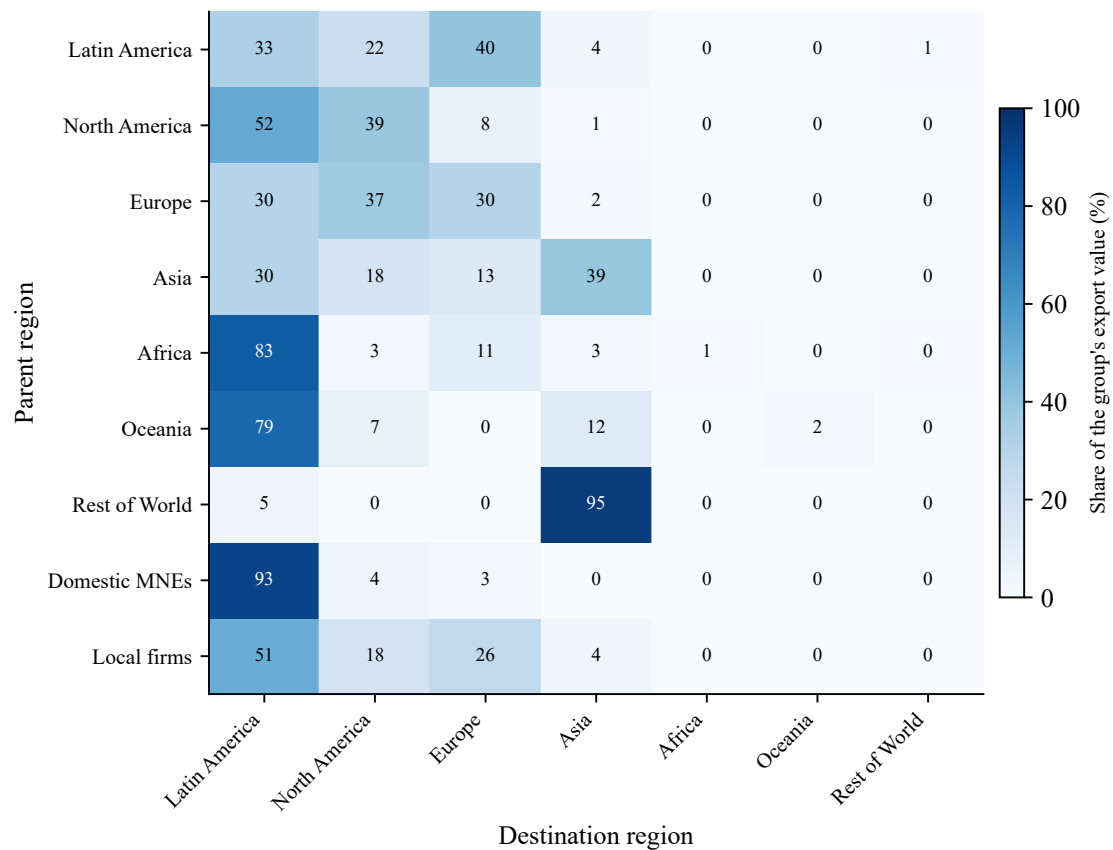
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 211:** Other Goods: Multinational Exports by Parent Region and Destination Region: Column Percentages

| Parent region / Destination region | Latin America | North America | Europe | Asia  | Africa | Oceania | Rest of World |
|------------------------------------|---------------|---------------|--------|-------|--------|---------|---------------|
| Latin America                      | 1.4           | 2.9           | 6.5    | 4.1   | 1.2    | 3.2     | 43.3          |
| North America                      | 15.2          | 34.8          | 8.8    | 9.9   | 4.6    | 6.0     | 0.0           |
| Europe                             | 8.3           | 30.8          | 31.6   | 14.1  | 16.9   | 41.5    | 2.7           |
| Asia                               | 0.4           | 0.7           | 0.7    | 13.2  | 0.0    | 0.7     | 0.0           |
| Africa                             | 5.9           | 0.6           | 2.9    | 4.8   | 20.0   | 0.5     | 42.9          |
| Oceania                            | 0.1           | 0.0           | 0.0    | 0.6   | 0.0    | 1.2     | 0.0           |
| Rest of World                      | 0.0           | 0.0           | 0.0    | 3.5   | 0.0    | 0.0     | 0.0           |
| Domestic MNEs                      | 46.7          | 6.2           | 5.7    | 5.0   | 0.6    | 7.4     | 0.6           |
| Local firms                        | 21.9          | 24.1          | 43.9   | 44.7  | 56.7   | 39.3    | 10.5          |
| Total                              | 100.0         | 100.0         | 100.0  | 100.0 | 100.0  | 100.0   | 100.0         |

*Notes:* Percent of the exports reaching each destination region, by parent region (columns add to 100).

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 144:** Other Goods: Multinational Exports by Parent Region and Destination Region

*Notes:* Percent of each group's exports, by destination region (rows add to 100).

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 212:** Other Goods: Foreign-Multinational Exports by Exporting Country and Destination Region

| Origin | Destination region | Latin America | North America | Europe | Asia | Africa | Oceania | Rest of World | Total |
|--------|--------------------|---------------|---------------|--------|------|--------|---------|---------------|-------|
| CHL    |                    | 37.3          | 42.1          | 17.3   | 2.9  | 0.1    | 0.2     | 0.1           | 0.2   |
| COL    |                    | 43.5          | 45.4          | 8.1    | 2.4  | 0.5    | 0.1     | 0.0           | 0.0   |
| CRI    |                    | 0.0           | 100.0         | 0.0    | 0.0  | 0.0    | 0.0     | 0.0           | 0.0   |
| PER    |                    | 12.9          | 85.6          | 0.5    | 1.0  | 0.0    | 0.0     | 0.0           | 0.0   |
| SLV    |                    | 65.5          | 13.1          | 14.6   | 2.9  | 3.8    | 0.0     | 0.0           | 0.0   |

*Notes:* Percent of each country's foreign-MNE exports, by destination region (rows add to 100). Last column: foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 213:** Other Goods: Local-Firm Exports by Exporting Country and Destination Region

| Origin | Destination region | Latin America | North America | Europe | Asia | Africa | Oceania | Rest of World | Total |
|--------|--------------------|---------------|---------------|--------|------|--------|---------|---------------|-------|
| CHL    |                    | 50.7          | 17.5          | 27.3   | 4.0  | 0.2    | 0.2     | 0.0           | 0.1   |
| COL    |                    | 48.1          | 28.0          | 16.8   | 4.9  | 1.4    | 0.7     | 0.0           | 0.0   |
| PER    |                    | 48.5          | 24.9          | 17.9   | 7.0  | 0.2    | 1.5     | 0.0           | 0.0   |
| SLV    |                    | 51.9          | 31.7          | 11.1   | 3.5  | 1.2    | 0.5     | 0.0           | 0.0   |

*Notes:* Percent of each country's exports by local firms, by destination region (rows add to 100). Last column: local-firm exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 28.3 Parent Country and Destination Country

**Table 214:** Other Goods: Foreign-Multinational Exports by Parent Country and Destination: Export Value

| Parent / Destination | USA | PER | ARG | ECU | GBR | ESP | CAN | MEX | DEU | COL | Other destinations | Total |
|----------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|--------------------|-------|
| USA                  | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0                | 0.1   |
| GBR                  | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0                | 0.0   |
| ESP                  | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0                | 0.0   |
| EGY                  | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0                | 0.0   |
| DEU                  | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0                | 0.0   |
| CAN                  | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0                | 0.0   |
| CHE                  | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0                | 0.0   |
| FRA                  | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0                | 0.0   |
| MEX                  | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0                | 0.0   |
| NOR                  | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0                | 0.0   |
| Other parents        | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0                | 0.0   |
| Total                | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0                | 0.2   |

*Notes:* Values are annual averages in US\$ billion. Ten largest parent countries and ten largest destinations by foreign-MNE export value.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 215:** Other Goods: Foreign-Multinational Exports by Parent Country and Destination: Row Percentages

| Parent / Destination | USA  | PER  | ARG  | ECU  | GBR  | ESP  | CAN  | MEX  | DEU  | COL  | Other destinations | Total |
|----------------------|------|------|------|------|------|------|------|------|------|------|--------------------|-------|
| USA                  | 39.3 | 32.8 | 10.0 | 0.5  | 1.5  | 1.5  | 2.3  | 1.0  | 0.6  | 1.0  | 9.5                | 100.0 |
| GBR                  | 66.4 | 9.6  | 1.6  | 0.5  | 10.7 | 0.0  | 1.0  | 1.0  | 0.2  | 3.0  | 6.2                | 100.0 |
| ESP                  | 14.1 | 3.9  | 8.2  | 0.7  | 5.0  | 24.6 | 3.1  | 9.4  | 1.4  | 3.4  | 26.3               | 100.0 |
| EGY                  | 2.6  | 4.8  | 4.9  | 66.1 | 2.5  | 0.3  | 0.0  | 0.2  | 2.1  | 0.0  | 16.6               | 100.0 |
| DEU                  | 33.2 | 2.8  | 1.4  | 0.3  | 0.0  | 0.0  | 5.9  | 0.5  | 51.2 | 0.8  | 4.0                | 100.0 |
| CAN                  | 0.8  | 27.8 | 26.6 | 0.9  | 0.0  | 18.2 | 7.2  | 0.2  | 1.2  | 11.8 | 5.3                | 100.0 |
| CHE                  | 9.8  | 16.6 | 4.7  | 2.9  | 2.0  | 0.3  | 24.6 | 6.0  | 0.3  | 3.6  | 29.1               | 100.0 |
| FRA                  | 17.9 | 2.5  | 14.7 | 1.0  | 0.0  | 0.6  | 1.4  | 1.5  | 0.3  | 2.3  | 57.9               | 100.0 |
| MEX                  | 13.1 | 2.1  | 7.0  | 0.3  | 60.4 | 0.7  | 7.8  | 3.8  | 0.4  | 0.1  | 4.3                | 100.0 |
| NOR                  | 1.3  | 9.2  | 30.9 | 3.3  | 0.0  | 0.4  | 0.0  | 19.3 | 3.3  | 13.8 | 18.5               | 100.0 |
| Other parents        | 21.8 | 10.8 | 5.2  | 2.0  | 8.1  | 9.4  | 2.7  | 3.9  | 1.7  | 5.4  | 28.9               | 100.0 |

*Notes:* Percent of each parent country's exports, by destination (rows add to 100). Ten largest parent countries and destinations.

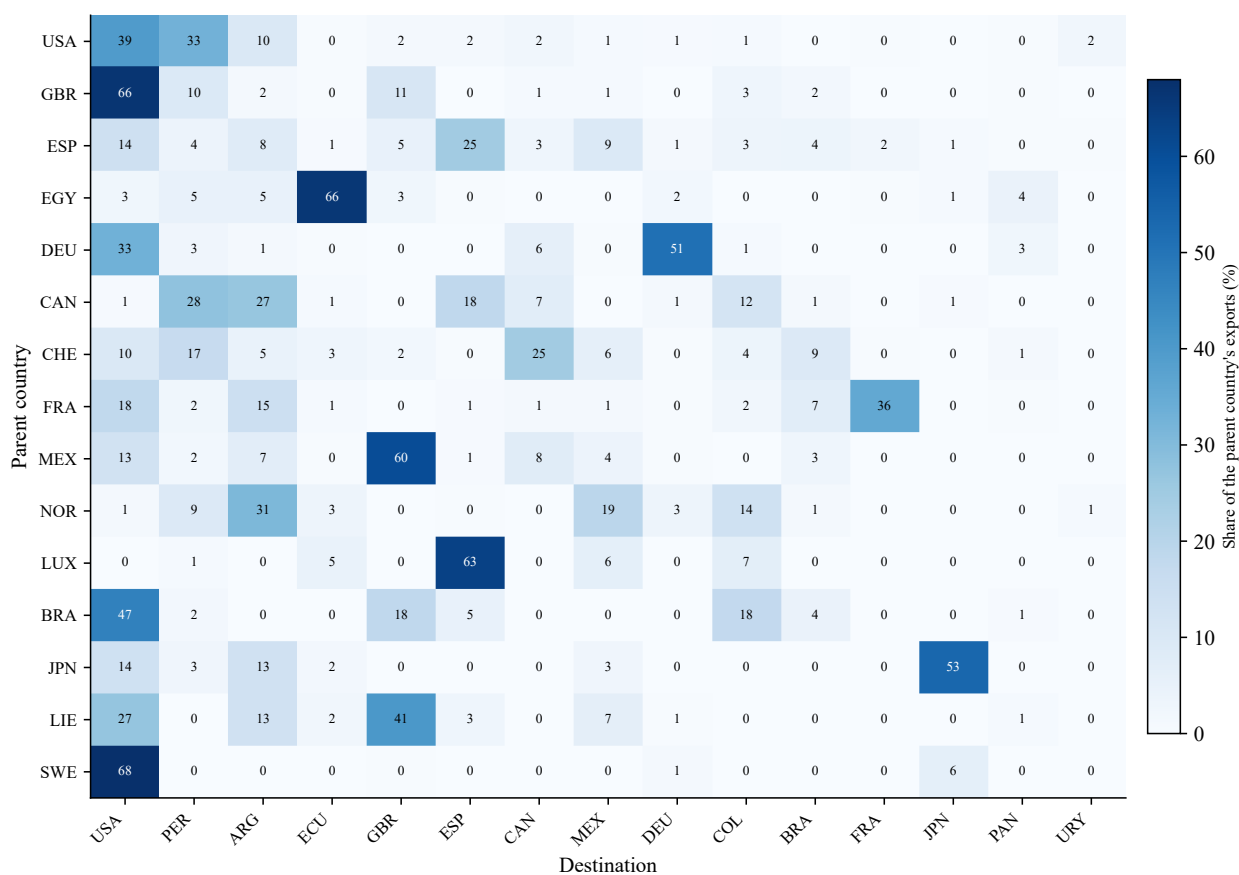
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 216:** Other Goods: Foreign-Multinational Exports by Parent Country and Destination: Column Percentages

| Parent / Destination | USA   | PER   | ARG   | ECU   | GBR   | ESP   | CAN   | MEX   | DEU   | COL   | Other destinations |
|----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------------------|
| USA                  | 51.0  | 71.5  | 47.6  | 2.5   | 11.3  | 11.8  | 29.6  | 14.3  | 9.7   | 15.7  | 23.8               |
| GBR                  | 26.9  | 6.5   | 2.3   | 0.8   | 24.6  | 0.0   | 3.9   | 4.5   | 0.8   | 14.2  | 4.8                |
| ESP                  | 4.9   | 2.3   | 10.5  | 1.0   | 10.1  | 51.8  | 10.8  | 37.7  | 5.8   | 13.8  | 17.8               |
| EGY                  | 0.9   | 2.7   | 6.0   | 89.5  | 4.8   | 0.6   | 0.0   | 0.8   | 8.0   | 0.0   | 10.8               |
| DEU                  | 3.6   | 0.5   | 0.5   | 0.1   | 0.0   | 0.0   | 6.3   | 0.6   | 63.2  | 1.0   | 0.8                |
| CAN                  | 0.1   | 5.0   | 10.4  | 0.4   | 0.0   | 11.6  | 7.8   | 0.3   | 1.5   | 14.5  | 1.1                |
| CHE                  | 0.9   | 2.5   | 1.6   | 1.1   | 1.0   | 0.2   | 22.5  | 6.2   | 0.3   | 3.8   | 5.1                |
| FRA                  | 1.6   | 0.4   | 4.9   | 0.4   | 0.0   | 0.3   | 1.2   | 1.5   | 0.3   | 2.5   | 10.1               |
| MEX                  | 1.1   | 0.3   | 2.2   | 0.1   | 29.3  | 0.4   | 6.7   | 3.7   | 0.3   | 0.1   | 0.7                |
| NOR                  | 0.1   | 0.8   | 6.2   | 0.7   | 0.0   | 0.1   | 0.0   | 12.1  | 2.1   | 8.7   | 2.0                |
| Other parents        | 9.0   | 7.5   | 7.9   | 3.4   | 18.9  | 23.1  | 11.1  | 18.4  | 7.9   | 25.7  | 22.9               |
| Total                | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0              |

*Notes:* Percent of the foreign-MNE exports reaching each destination, by parent country (columns add to 100). Ten largest parent countries and destinations.

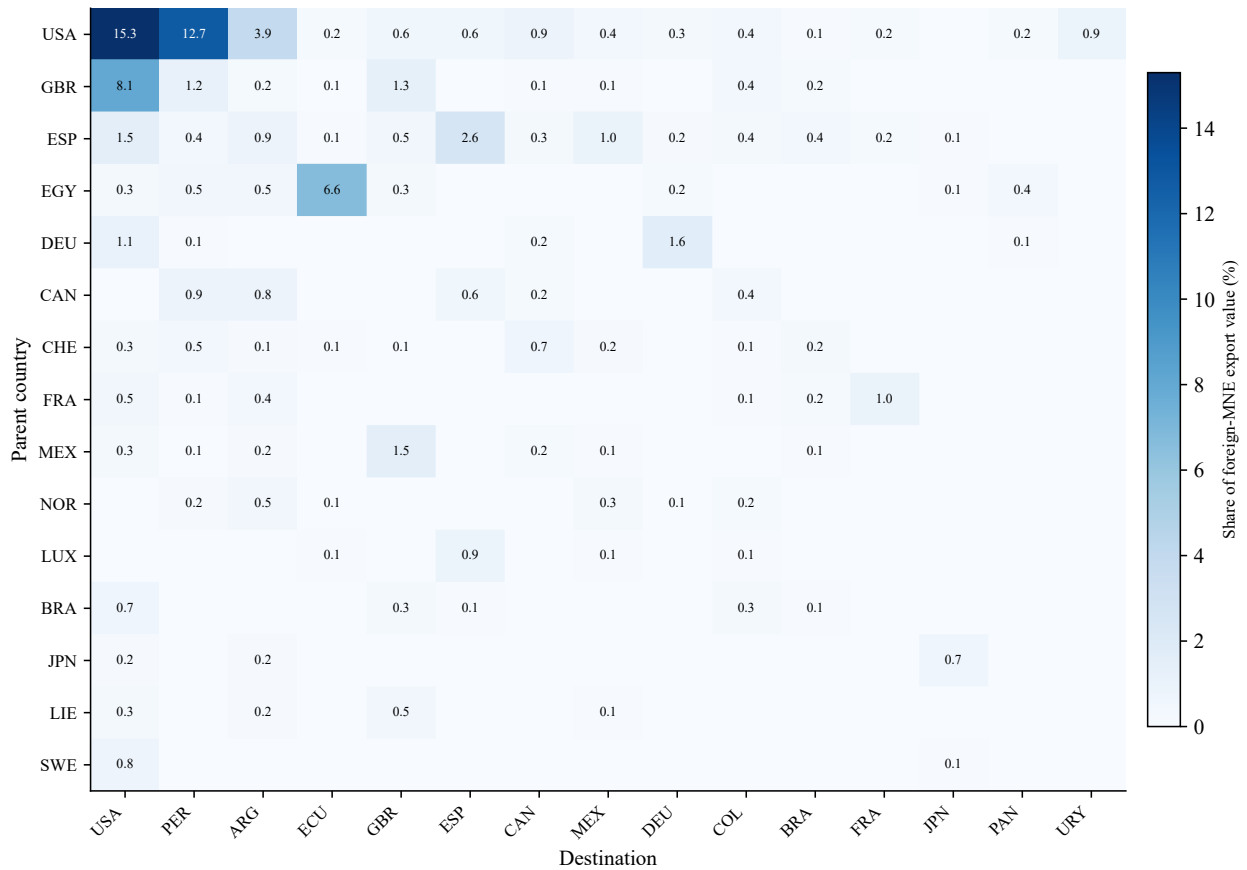
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 145:** Other Goods: Destination of Foreign-Multinational Exports: Fifteen Largest Parent Countries and Destinations

*Notes:* Percent of each parent country's exports going to each destination (rows add to 100 including other destinations).

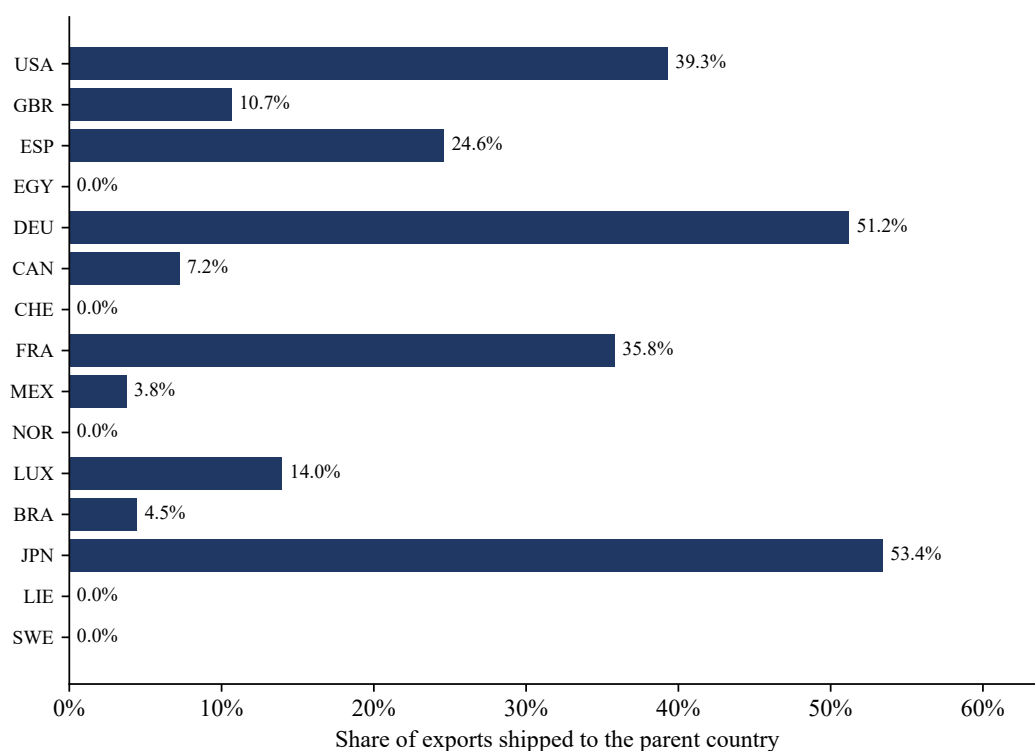
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 146: Other Goods: Foreign-Multinational Exports by Parent Country and Destination: Share of the Total**  
*Notes:* Percent of all foreign-MNE export value in each parent country and destination pair; fifteen largest parent countries and destinations.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 28.4 Exports to the Parent Country



**Figure 147:** Other Goods: Share of Foreign-Multinational Exports Shipped to the Parent Country

*Notes:* Percent of each parent country's exports shipped to that same country.

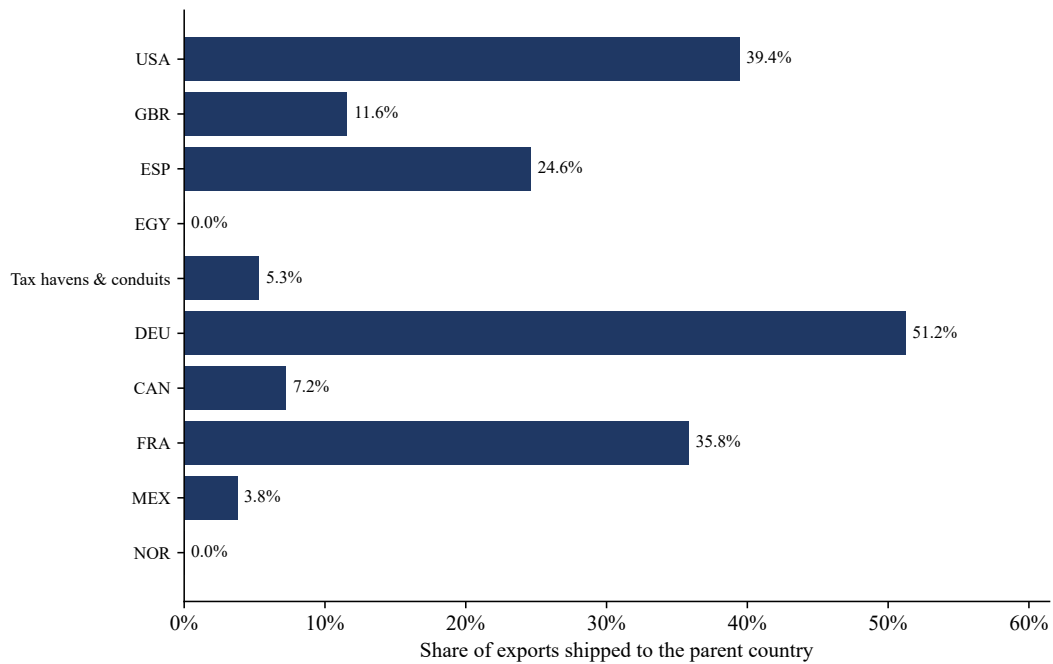
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 217:** Other Goods: Foreign-Multinational Exports to the Parent Countries, by Parent Country

| Parent / Destination | USA  | GBR  | ESP  | EGY | DEU  | CAN  | CHE | FRA  | MEX  | NOR | Other destinations | Total |
|----------------------|------|------|------|-----|------|------|-----|------|------|-----|--------------------|-------|
| USA                  | 39.3 | 1.5  | 1.5  | 0.0 | 0.6  | 2.3  | 0.0 | 0.5  | 1.0  | 0.1 | 53.1               | 100.0 |
| GBR                  | 66.4 | 10.7 | 0.0  | 0.0 | 0.2  | 1.0  | 0.0 | 0.0  | 1.0  | 0.0 | 20.8               | 100.0 |
| ESP                  | 14.1 | 5.0  | 24.6 | 0.0 | 1.4  | 3.1  | 0.6 | 2.1  | 9.4  | 0.1 | 39.7               | 100.0 |
| EGY                  | 2.6  | 2.5  | 0.3  | 0.0 | 2.1  | 0.0  | 0.1 | 0.4  | 0.2  | 3.5 | 88.3               | 100.0 |
| DEU                  | 33.2 | 0.0  | 0.0  | 0.0 | 51.2 | 5.9  | 0.0 | 0.2  | 0.5  | 0.0 | 9.1                | 100.0 |
| CAN                  | 0.8  | 0.0  | 18.2 | 0.0 | 1.2  | 7.2  | 0.0 | 0.1  | 0.2  | 0.0 | 72.2               | 100.0 |
| CHE                  | 9.8  | 2.0  | 0.3  | 0.0 | 0.3  | 24.6 | 0.0 | 0.3  | 6.0  | 0.0 | 56.7               | 100.0 |
| FRA                  | 17.9 | 0.0  | 0.6  | 0.0 | 0.3  | 1.4  | 0.0 | 35.8 | 1.5  | 0.0 | 42.5               | 100.0 |
| MEX                  | 13.1 | 60.4 | 0.7  | 0.0 | 0.4  | 7.8  | 0.2 | 0.0  | 3.8  | 0.0 | 13.6               | 100.0 |
| NOR                  | 1.3  | 0.0  | 0.4  | 0.0 | 3.3  | 0.0  | 0.0 | 0.0  | 19.3 | 0.0 | 75.7               | 100.0 |
| Other parents        | 21.8 | 8.1  | 9.4  | 0.0 | 1.7  | 2.7  | 2.1 | 0.5  | 3.9  | 0.0 | 49.8               | 100.0 |

*Notes:* Percent of each parent country's exports, by destination (rows add to 100). Destinations are the ten largest parent countries; the diagonal is the share shipped to the parent's own country.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 148:** Other Goods: Share of Foreign-Multinational Exports Shipped to the Parent Jurisdiction, Consolidated Parents

*Notes:* Percent of each parent's exports shipped to its own jurisdiction. Dependencies are assigned to their sovereign state and tax havens and conduit jurisdictions are pooled.

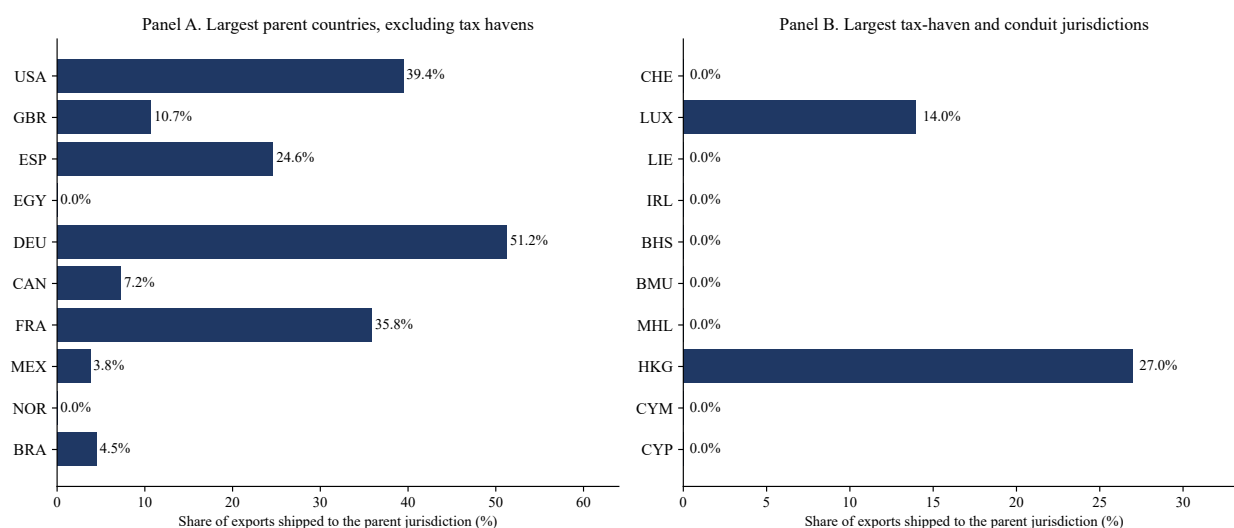
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 218:** Other Goods: Share of Foreign-Multinational Exports Shipped to the Parent Jurisdiction, Consolidated Parents

| Consolidated parent   | Home share (%) | Exports (US\$bn/yr) |
|-----------------------|----------------|---------------------|
| USA                   | 39.4           | 0.1                 |
| GBR                   | 11.6           | 0.0                 |
| ESP                   | 24.6           | 0.0                 |
| EGY                   | 0.0            | 0.0                 |
| Tax havens & conduits | 5.3            | 0.0                 |
| DEU                   | 51.2           | 0.0                 |
| CAN                   | 7.2            | 0.0                 |
| FRA                   | 35.8           | 0.0                 |
| MEX                   | 3.8            | 0.0                 |
| NOR                   | 0.0            | 0.0                 |

*Notes:* Home share: percent of the parent's exports shipped to its own jurisdiction. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 149: Other Goods: Share of Foreign-Multinational Exports Shipped to the Parent Jurisdiction**

*Notes:* Percent of each parent jurisdiction's exports shipped to that same jurisdiction. Panel A: largest parent countries, excluding tax havens; dependencies are assigned to their sovereign state. Panel B: largest tax-haven and conduit jurisdictions.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

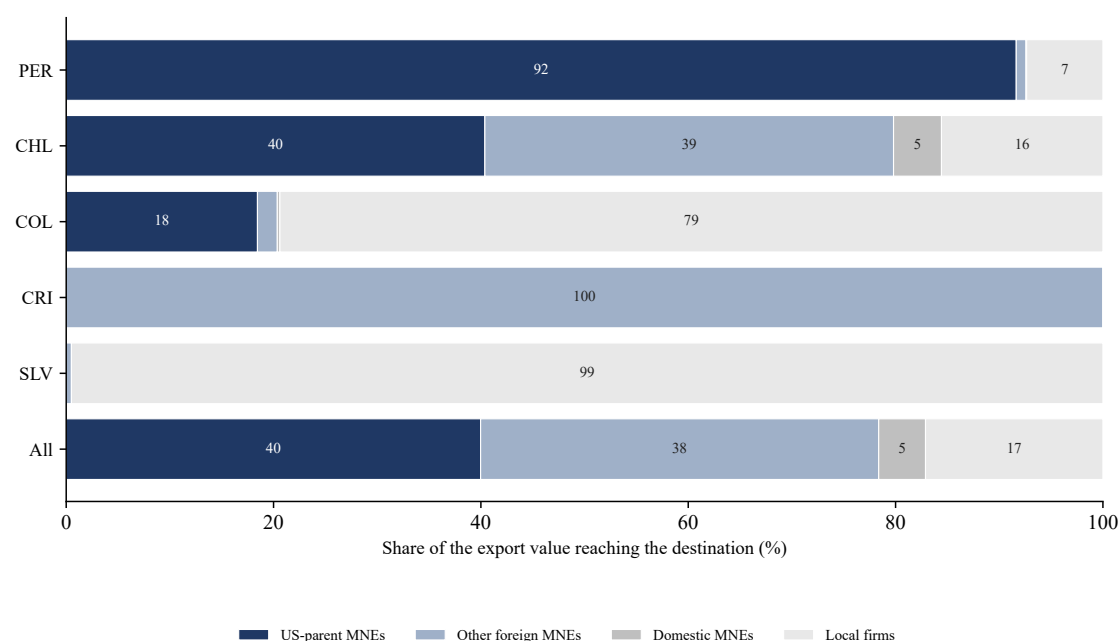
**Table 219: Other Goods: Share of Foreign-Multinational Exports Shipped to the Parent Jurisdiction**

| Panel   | Parent | Home share (%) | Exports (US\$bn/yr) |
|---------|--------|----------------|---------------------|
| Panel A | USA    | 39.4           | 0.1                 |
| Panel A | GBR    | 10.7           | 0.0                 |
| Panel A | ESP    | 24.6           | 0.0                 |
| Panel A | EGY    | 0.0            | 0.0                 |
| Panel A | DEU    | 51.2           | 0.0                 |
| Panel A | CAN    | 7.2            | 0.0                 |
| Panel A | FRA    | 35.8           | 0.0                 |
| Panel A | MEX    | 3.8            | 0.0                 |
| Panel A | NOR    | 0.0            | 0.0                 |
| Panel A | BRA    | 4.5            | 0.0                 |
| Panel B | CHE    | 0.0            | 0.0                 |
| Panel B | LUX    | 14.0           | 0.0                 |
| Panel B | LIE    | 0.0            | 0.0                 |
| Panel B | IRL    | 0.0            | 0.0                 |
| Panel B | BHS    | 0.0            | 0.0                 |
| Panel B | BMU    | 0.0            | 0.0                 |
| Panel B | MHL    | 0.0            | 0.0                 |
| Panel B | HKG    | 27.0           | 0.0                 |
| Panel B | CYM    | 0.0            | 0.0                 |
| Panel B | CYP    | 0.0            | 0.0                 |

*Notes:* Home share: percent of the parent's exports shipped to its own jurisdiction. Panel A: largest parent countries, excluding tax havens. Panel B: largest tax-haven and conduit jurisdictions. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 28.5 Destination Markets



**Figure 150:** Other Goods: Exports to the United States, by Type of Exporter and Exporting Country

*Notes:* Percent of each country's exports to the United States, by type of exporter.

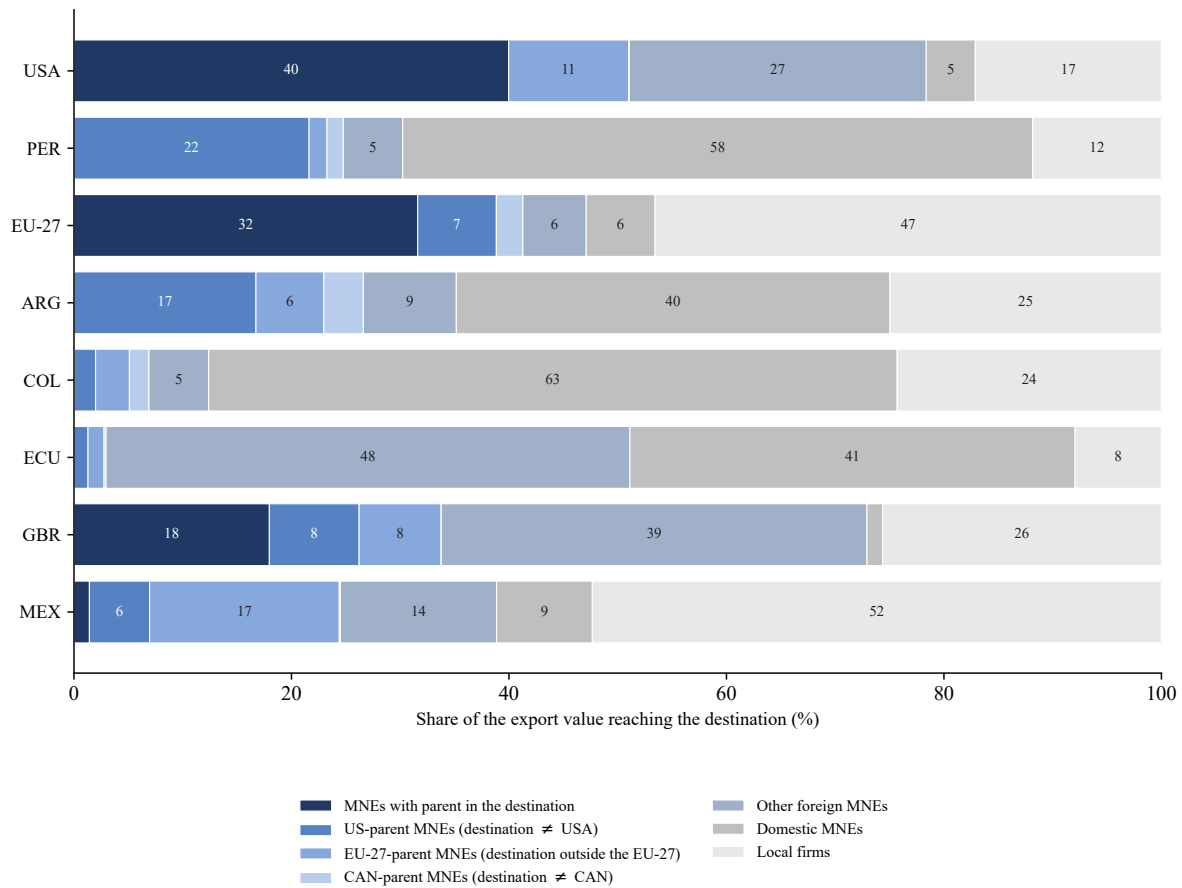
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 220:** Other Goods: Exports to the United States, by Type of Exporter and Exporting Country

| Origin | Exports to the USA (US\$bn/yr) | US-parent MNEs (%) | Other foreign MNEs (%) | Domestic MNEs (%) | Local firms (%) |
|--------|--------------------------------|--------------------|------------------------|-------------------|-----------------|
| PER    | 0.0                            | 91.6               | 0.9                    | 0.1               | 7.4             |
| CHL    | 0.1                            | 40.4               | 39.4                   | 4.6               | 15.6            |
| COL    | 0.0                            | 18.4               | 1.9                    | 0.2               | 79.4            |
| CRI    | 0.0                            | 0.0                | 100.0                  | 0.0               | 0.0             |
| SLV    | 0.0                            | 0.0                | 0.5                    | 0.0               | 99.5            |
| All    | 0.1                            | 40.0               | 38.4                   | 4.5               | 17.1            |

*Notes:* Percent of each country's exports to the United States, by type of exporter. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 151: Other Goods: Ownership of Exports to the Main Destination Markets**

*Notes:* Percent of the export value reaching each destination, by type of exporter. The first segment is exports by MNEs whose parent is located in that destination. EU-27 member states are pooled.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 221: Other Goods: Ownership of Exports to the Main Destination Markets**

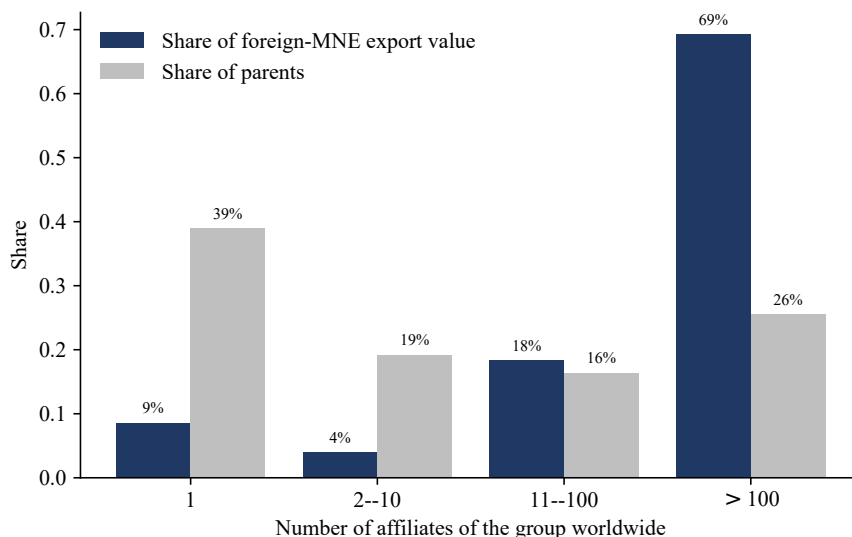
| Destination | Exports (US\$bn/yr) | MNEs with parent in the destination (%) | US-parent MNEs (%) | EU-27-parent MNEs (%) | CAN-parent MNEs (%) | Other foreign MNEs (%) | Domestic MNEs (%) | Local firms (%) |
|-------------|---------------------|-----------------------------------------|--------------------|-----------------------|---------------------|------------------------|-------------------|-----------------|
| USA         | 0.1                 | 40.0                                    | –                  | 11.0                  | 0.1                 | 27.3                   | 4.5               | 17.1            |
| PER         | 0.1                 | 0.0                                     | 21.6               | 1.6                   | 1.5                 | 5.5                    | 57.9              | 11.8            |
| EU-27       | 0.1                 | 31.6                                    | 7.2                | –                     | 2.4                 | 5.8                    | 6.3               | 46.6            |
| ARG         | 0.0                 | 0.0                                     | 16.7               | 6.2                   | 3.6                 | 8.5                    | 39.9              | 25.0            |
| COL         | 0.0                 | 0.0                                     | 2.0                | 3.1                   | 1.8                 | 5.5                    | 63.3              | 24.3            |
| ECU         | 0.0                 | 0.0                                     | 1.3                | 1.5                   | 0.2                 | 48.2                   | 40.9              | 7.9             |
| GBR         | 0.0                 | 18.0                                    | 8.3                | 7.6                   | 0.0                 | 39.1                   | 1.4               | 25.6            |
| MEX         | 0.0                 | 1.4                                     | 5.5                | 17.4                  | 0.1                 | 14.4                   | 8.8               | 52.3            |

*Notes:* Percent of the export value reaching each destination, by type of exporter. EU-27 member states are pooled. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 29 Fact 4: A Small Set of Large Multinational Groups Accounts for Most Exports

### 29.1 Size of the Group's Global Network



**Figure 152:** Other Goods: Foreign-Multinational Exports and Number of Parents, by Size of the Group's Global Network

*Notes:* Shares of foreign-MNE export value and of parents, by the number of affiliates the group has worldwide.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

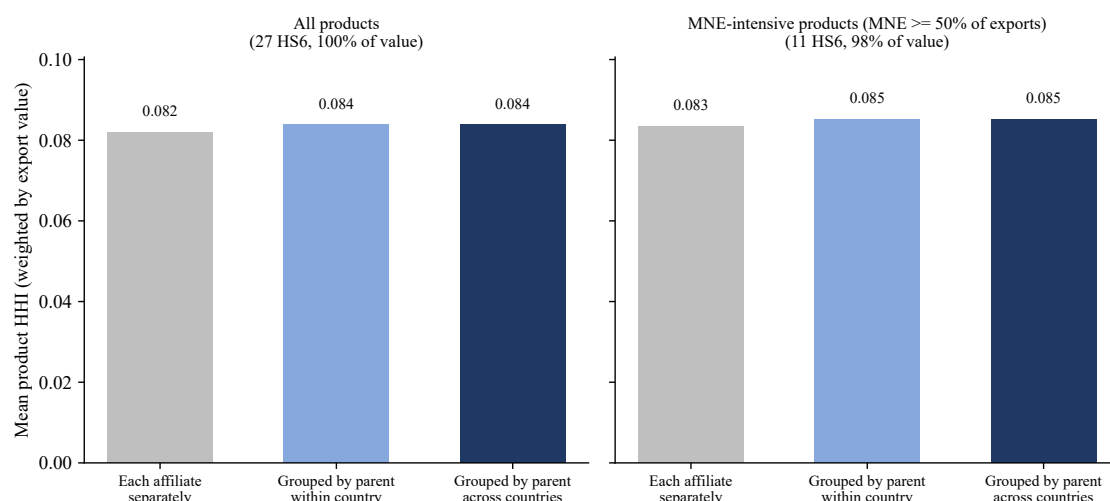
**Table 222:** Other Goods: Foreign-Multinational Exports and Number of Parents, by Size of the Group's Global Network

| Affiliates worldwide | Parents | Share of parents (%) | Share of foreign-MNE export value (%) |
|----------------------|---------|----------------------|---------------------------------------|
| 1                    | 221     | 38.9                 | 8.5                                   |
| 2–10                 | 109     | 19.2                 | 3.9                                   |
| 11–100               | 93      | 16.4                 | 18.3                                  |
| >100                 | 145     | 25.5                 | 69.2                                  |
| All parents          | 568     | 100                  | 100                                   |

*Notes:* Number and share of parents, and share of foreign-MNE export value, by the number of affiliates the group has worldwide.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 29.2 Concentration within Products



**Figure 153:** Other Goods: Export Concentration within Products: Firms and Groups

*Notes:* Mean Herfindahl index of exporters within each HS6 product, weighted by export value, counting each affiliate separately or grouping affiliates by parent within or across countries. Right panel: products where MNEs account for at least half of exports.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 223:** Other Goods: Export Concentration within Products: Firms and Groups

| Sample                                                                     | Measure                               | Firm level | Groups within country | Groups across countries |
|----------------------------------------------------------------------------|---------------------------------------|------------|-----------------------|-------------------------|
| <i>All products: 27 HS6, 100% of value</i>                                 |                                       |            |                       |                         |
|                                                                            | Mean HHI                              | 0.082      | 0.084                 | 0.084                   |
|                                                                            | Mean top-exporter share               | 24.9%      | 24.9%                 | 24.9%                   |
|                                                                            | Effective number of exporters (1/HHI) | 18.3       | 17.7                  | 17.7                    |
| <i>MNE-intensive products (MNE ≥ 50% of exports): 11 HS6, 98% of value</i> |                                       |            |                       |                         |
|                                                                            | Mean HHI                              | 0.083      | 0.085                 | 0.085                   |
|                                                                            | Mean top-exporter share               | 25.2%      | 25.2%                 | 25.2%                   |
|                                                                            | Effective number of exporters (1/HHI) | 13.2       | 12.6                  | 12.6                    |

*Notes:* Means across HS6 products, weighted by export value. Firm level: each affiliate counted separately. Groups: affiliates grouped by ultimate parent within each exporting country or across countries.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 29.3 Groups and Affiliates

**Table 224:** Other Goods: Multinational Affiliates and Groups in Origin–Destination–Product–Year Cells

| Origin-destination-product-year cells with ≥ 1 MNE      |       |
|---------------------------------------------------------|-------|
| Cells                                                   | 2,372 |
| MNE affiliates per cell (mean)                          | 5.05  |
| MNE groups per cell (mean)                              | 4.95  |
| Affiliates per group, pooled                            | 1.02  |
| Affiliates per group, mean weighted by export value     | 1.03  |
| Cells where one group exports through > 1 affiliate (%) | 8.4   |

*Notes:* Cells with at least one MNE. Group: ultimate parent in Orbis or Dun & Bradstreet (parent name when the identifier is missing).

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 225:** Other Goods: Multinational Presence and Exports: Affiliates and Groups

|                                                               | Origin × year, destination × year, product FE |                       |                       | Origin × dest. × product, origin × dest. × year FE |                       |                       |
|---------------------------------------------------------------|-----------------------------------------------|-----------------------|-----------------------|----------------------------------------------------|-----------------------|-----------------------|
|                                                               | (1)                                           | (2)                   | (3)                   | (4)                                                | (5)                   | (6)                   |
| <i>Panel A. Dependent variable: ln exports, all firms</i>     |                                               |                       |                       |                                                    |                       |                       |
| ln(# MNE firms)                                               | 1.8402***<br>(0.1590)                         |                       |                       | 1.1124***<br>(0.2255)                              |                       |                       |
| ln(# MNE groups)                                              |                                               | 1.8312***<br>(0.1604) | 1.8391***<br>(0.1588) |                                                    | 1.1119***<br>(0.2265) | 1.1077***<br>(0.2270) |
| ln(affiliates per group)                                      |                                               |                       | 2.3965**<br>(0.9466)  |                                                    |                       | 1.3037<br>(1.7782)    |
| <i>Panel B. Dependent variable: ln exports, non-MNE firms</i> |                                               |                       |                       |                                                    |                       |                       |
| ln(# MNE firms)                                               | 1.2338***<br>(0.2149)                         |                       |                       | 0.6712*<br>(0.3739)                                |                       |                       |
| ln(# MNE groups)                                              |                                               | 1.2217***<br>(0.2128) | 1.2370***<br>(0.2163) |                                                    | 0.6829*<br>(0.3667)   | 0.6460*<br>(0.3712)   |
| ln(affiliates per group)                                      |                                               |                       | 2.8459*<br>(1.5421)   |                                                    |                       | -4.9065<br>(4.0733)   |
| Product FE                                                    | ✓                                             | ✓                     | ✓                     |                                                    |                       |                       |
| Origin × year FE                                              | ✓                                             | ✓                     | ✓                     |                                                    |                       |                       |
| Destination × year FE                                         | ✓                                             | ✓                     | ✓                     |                                                    |                       |                       |
| Origin × dest. × product FE                                   |                                               |                       |                       | ✓                                                  | ✓                     | ✓                     |
| Origin × dest. × year FE                                      |                                               |                       |                       | ✓                                                  | ✓                     | ✓                     |
| Observations (Panel A)                                        | 1,333                                         | 1,333                 | 1,333                 | 641                                                | 641                   | 641                   |
| Observations (Panel B)                                        | 767                                           | 767                   | 767                   | 204                                                | 204                   | 204                   |

*Notes:* Observations are origin–destination–HS6–year cells. Other Goods products only. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Regressors: ln number of MNE affiliates or ln number of MNE groups (distinct ultimate parents). Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 30 Fact 5: Multinational Presence Is Associated with Higher Exports

**Table 226:** Other Goods: Multinational Presence and Exports

|                                                               | Intensive: ln(# MNE firms) |                       |                       | Extensive: any MNE present |                      |                       |
|---------------------------------------------------------------|----------------------------|-----------------------|-----------------------|----------------------------|----------------------|-----------------------|
|                                                               | (1)                        | (2)                   | (3)                   | (1)                        | (2)                  | (3)                   |
| <i>Panel A. Dependent variable: ln exports, all firms</i>     |                            |                       |                       |                            |                      |                       |
| ln(# MNE firms)                                               | 1.8059***<br>(0.1087)      | 1.8402***<br>(0.1590) | 1.1124***<br>(0.2255) |                            |                      |                       |
| Any MNE present                                               |                            |                       |                       | 0.1214<br>(0.1490)         | 0.2291<br>(0.2076)   | 1.0052***<br>(0.3256) |
| <i>Panel B. Dependent variable: ln exports, non-MNE firms</i> |                            |                       |                       |                            |                      |                       |
| ln(# MNE firms)                                               | 0.9826***<br>(0.1424)      | 1.2338***<br>(0.2149) | 0.6712*<br>(0.3739)   |                            |                      |                       |
| Any MNE present                                               |                            |                       |                       | 0.4614***<br>(0.1490)      | 0.5217**<br>(0.2381) | 0.6655<br>(0.5481)    |
| Origin FE                                                     | ✓                          |                       |                       | ✓                          |                      |                       |
| Destination FE                                                | ✓                          |                       |                       | ✓                          |                      |                       |
| Product FE                                                    | ✓                          | ✓                     |                       | ✓                          | ✓                    |                       |
| Year FE                                                       | ✓                          |                       |                       | ✓                          |                      |                       |
| Origin × year FE                                              |                            | ✓                     |                       |                            | ✓                    |                       |
| Destination × year FE                                         |                            | ✓                     |                       |                            | ✓                    |                       |
| Origin × dest. × product FE                                   |                            |                       | ✓                     |                            |                      | ✓                     |
| Origin × dest. × year FE                                      |                            |                       | ✓                     |                            |                      | ✓                     |
| Observations (Panel A)                                        | 2,350                      | 1,333                 | 641                   | 3,831                      | 2,928                | 903                   |
| Observations (Panel B)                                        | 1,330                      | 767                   | 204                   | 2,793                      | 2,125                | 383                   |

*Notes:* Observations are origin–destination–HS6–year cells. Other Goods products only. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Intensive margin: ln number of MNEs, cells with at least one MNE. Extensive margin: indicator for at least one MNE. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 227:** Other Goods: Multinational Exports by Exporting Country and Destination Type

| Origin | MNE value (US\$bn/yr) | of which domestic (%) | Foreign-MNE value (US\$bn/yr) | % of foreign-MNE value, by destination |                                         |
|--------|-----------------------|-----------------------|-------------------------------|----------------------------------------|-----------------------------------------|
|        |                       |                       |                               | parent country                         | other (of which: group affiliate there) |
| CHL    | 0.4                   | 37.0                  | 0.2                           | 18.7                                   | 81.3 (37.2)                             |
| COL    | 0.0                   | 4.4                   | 0.0                           | 34.5                                   | 65.5 (9.8)                              |
| CRI    | 0.0                   | 0.0                   | 0.0                           | 0.0                                    | 100.0 (0.0)                             |
| PER    | 0.0                   | 1.0                   | 0.0                           | 84.9                                   | 15.1 (0.2)                              |
| SLV    | 0.0                   | 0.0                   | 0.0                           | 4.6                                    | 95.4 (3.3)                              |
| All    | 0.4                   | 36.9                  | 0.2                           | 18.8                                   | 81.2 (37.0)                             |

*Notes:* MNE and foreign-MNE export value by country, and the split of foreign-MNE value by destination. Parent country: exports to the country of the ultimate parent. Other: all remaining destinations; in parentheses, destinations where the group has another affiliate. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 228:** Other Goods: Multinational Presence and Exports: Intensive Margin, by Type of Multinational

|                                                               | All MNEs  |           | Foreign / domestic |          | Foreign by destination, domestic |     |
|---------------------------------------------------------------|-----------|-----------|--------------------|----------|----------------------------------|-----|
|                                                               | (1)       | (2)       | (3)                | (4)      | (5)                              | (6) |
| <i>Panel A. Dependent variable: ln exports, all firms</i>     |           |           |                    |          |                                  |     |
| ln(# MNE firms)                                               | 1.8402*** | 1.1124*** |                    |          |                                  |     |
|                                                               | (0.1590)  | (0.2255)  |                    |          |                                  |     |
| ln(# foreign MNEs)                                            |           |           | 1.2219***          | 0.7610*  |                                  |     |
|                                                               |           |           | (0.2567)           | (0.3809) |                                  |     |
| ln(# domestic MNEs)                                           |           |           | 0.6260             | -0.0815  |                                  |     |
|                                                               |           |           | (0.3832)           | (0.1748) |                                  |     |
| <i>Panel B. Dependent variable: ln exports, non-MNE firms</i> |           |           |                    |          |                                  |     |
| ln(# MNE firms)                                               | 1.2338*** | 0.6712*   |                    |          |                                  |     |
|                                                               | (0.2149)  | (0.3739)  |                    |          |                                  |     |
| ln(# foreign MNEs)                                            |           |           | 0.2490             | 0.2668*  |                                  |     |
|                                                               |           |           | (0.3400)           | (0.1164) |                                  |     |
| ln(# domestic MNEs)                                           |           |           | 0.7441             | 0.0365   |                                  |     |
|                                                               |           |           | (0.5034)           | (0.5349) |                                  |     |
| Product FE                                                    | ✓         |           | ✓                  |          | ✓                                |     |
| Origin × year FE                                              | ✓         |           | ✓                  |          | ✓                                |     |
| Destination × year FE                                         | ✓         |           | ✓                  |          | ✓                                |     |
| Origin × dest. × product FE                                   |           | ✓         |                    | ✓        |                                  | ✓   |
| Origin × dest. × year FE                                      |           | ✓         |                    | ✓        |                                  | ✓   |
| Observations (Panel A)                                        | 1,333     | 641       | 145                | 70       | 141                              | 141 |
| Observations (Panel B)                                        | 767       | 204       | 125                | 54       | 139                              | 139 |

*Notes:* Observations are origin–destination–HS6–year cells. Other Goods products only. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Regressors: ln number of MNEs of each type. Columns (5)–(6) split foreign MNEs by whether the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 229:** Other Goods: Multinational Presence and Exports: Extensive Margin, by Type of Multinational

|                                                               | All MNEs             |                       | Foreign / domestic    |                       | Foreign by destination, domestic |                      |
|---------------------------------------------------------------|----------------------|-----------------------|-----------------------|-----------------------|----------------------------------|----------------------|
|                                                               | (1)                  | (2)                   | (3)                   | (4)                   | (5)                              | (6)                  |
| <i>Panel A. Dependent variable: ln exports, all firms</i>     |                      |                       |                       |                       |                                  |                      |
| Any MNE present                                               | 0.2291<br>(0.2076)   | 1.0052***<br>(0.3256) |                       |                       |                                  |                      |
| Any foreign MNE                                               |                      |                       | 0.5417***<br>(0.1593) | 0.7689**<br>(0.3199)  |                                  |                      |
| Any foreign MNE, parent-country destination                   |                      |                       |                       |                       | 1.6482***<br>(0.2403)            | 0.8616**<br>(0.3835) |
| Any foreign MNE, other destination                            |                      |                       |                       |                       | 0.4937***<br>(0.1689)            | 0.4724<br>(0.3114)   |
| Any domestic MNE                                              |                      |                       | 2.6655***<br>(0.2144) | 0.8535***<br>(0.2919) | 2.4719***<br>(0.2184)            | 0.8036**<br>(0.3075) |
| <i>Panel B. Dependent variable: ln exports, non-MNE firms</i> |                      |                       |                       |                       |                                  |                      |
| Any MNE present                                               | 0.5217**<br>(0.2381) | 0.6655<br>(0.5481)    |                       |                       |                                  |                      |
| Any foreign MNE                                               |                      |                       | 0.4939**<br>(0.2196)  | 0.3217<br>(0.5881)    |                                  |                      |
| Any foreign MNE, parent-country destination                   |                      |                       |                       |                       | 1.0152***<br>(0.2936)            | 0.4587<br>(0.2979)   |
| Any foreign MNE, other destination                            |                      |                       |                       |                       | 0.5331**<br>(0.2189)             | 0.2250<br>(0.5351)   |
| Any domestic MNE                                              |                      |                       | 1.4052***<br>(0.2128) | 0.3147<br>(0.5086)    | 1.2667***<br>(0.2017)            | 0.2834<br>(0.5337)   |
| Product FE                                                    | ✓                    |                       | ✓                     |                       | ✓                                |                      |
| Origin × year FE                                              | ✓                    |                       | ✓                     |                       | ✓                                |                      |
| Destination × year FE                                         | ✓                    |                       | ✓                     |                       | ✓                                |                      |
| Origin × dest. × product FE                                   |                      | ✓                     |                       | ✓                     |                                  | ✓                    |
| Origin × dest. × year FE                                      |                      | ✓                     |                       | ✓                     |                                  | ✓                    |
| Observations (Panel A)                                        | 2,928                | 903                   | 2,928                 | 903                   | 2,928                            | 903                  |
| Observations (Panel B)                                        | 2,125                | 383                   | 2,125                 | 383                   | 2,125                            | 383                  |

*Notes:* Observations are origin–destination–HS6–year cells. Other Goods products only. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Regressors: indicators for at least one MNE of each type. Columns (5)–(6) split foreign MNEs by whether the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 31 Fact 6: Distance Is a Weaker Barrier for Multinationals

**Table 230:** Other Goods: Distance and Firm Exports: Multinationals by Presence at the Destination

| Dep. var.: ln exports                         | (1)                 | (2)                 | (3)                    | (4)                  |
|-----------------------------------------------|---------------------|---------------------|------------------------|----------------------|
| ln distance                                   | -0.0720<br>(0.1077) | -0.0574<br>(0.1025) | -0.0698<br>(0.1080)    | -0.0545<br>(0.1023)  |
| ln distance × MNE                             | -0.0086<br>(0.0109) | 0.0029<br>(0.0103)  |                        |                      |
| ln distance × MNE, present at destination     |                     |                     | 0.0308<br>(0.0189)     | 0.0405**<br>(0.0185) |
| ln distance × MNE, not present at destination |                     |                     | -0.0270***<br>(0.0088) | -0.0151*<br>(0.0084) |
| Origin FE                                     | ✓                   |                     | ✓                      |                      |
| Destination FE                                | ✓                   |                     | ✓                      |                      |
| Year FE                                       | ✓                   |                     | ✓                      |                      |
| Origin × year FE                              |                     | ✓                   |                        | ✓                    |
| Destination × year FE                         |                     | ✓                   |                        | ✓                    |
| Product FE                                    | ✓                   | ✓                   | ✓                      | ✓                    |
| Observations                                  | 38,597              | 38,110              | 38,597                 | 38,110               |

*Notes:* Observations are firm–destination–HS6–year cells. Other Goods products only. Dependent variable: ln exports. Non-MNE firms are the omitted group. Present: the group has an affiliate in the destination or the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Source:* Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

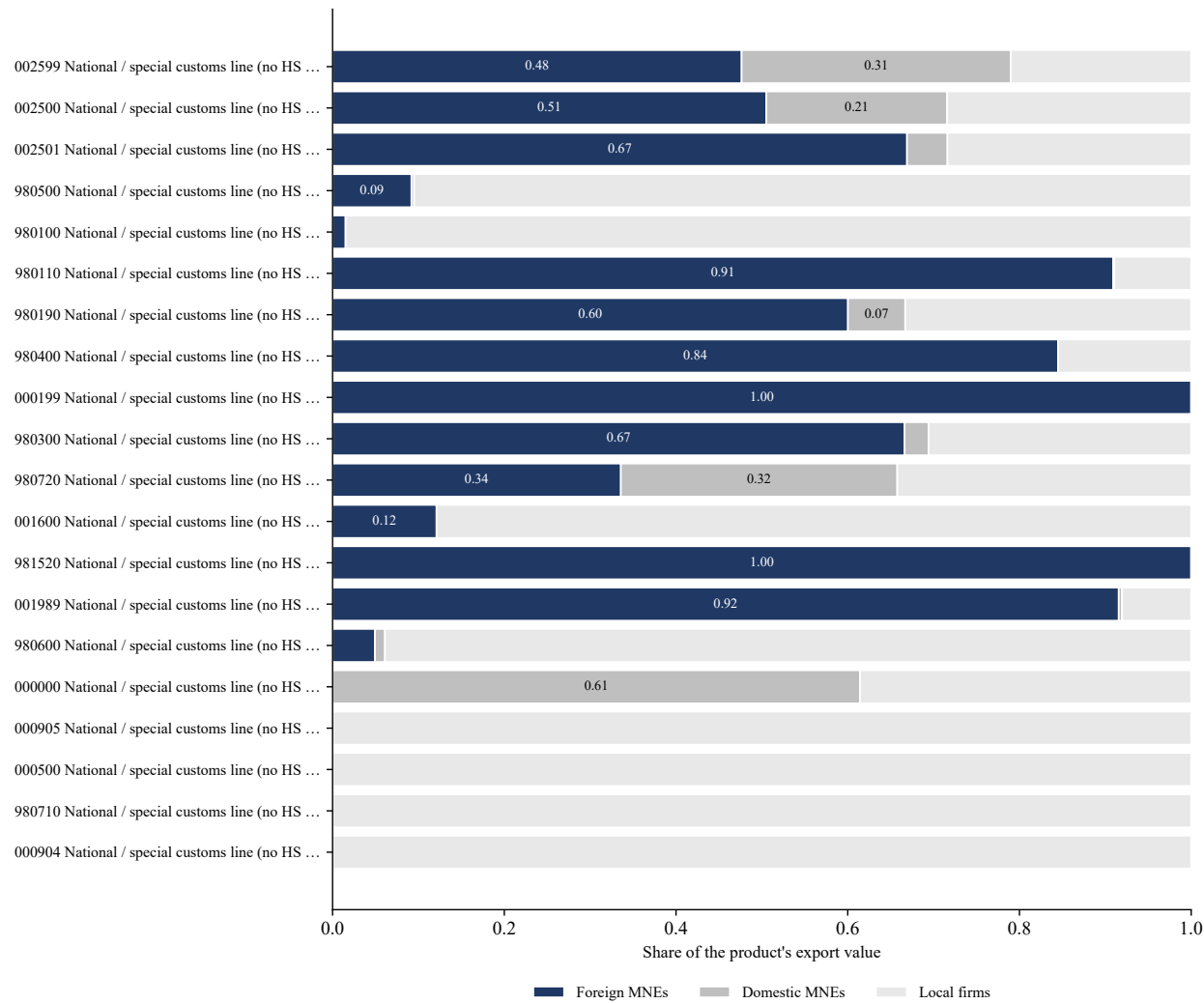
**Table 231:** Other Goods: Distance and Firm Exports: Foreign and Domestic Multinationals

| Dep. var.: ln exports                                        | (1)                 | (2)                 | (3)                   | (4)                   |
|--------------------------------------------------------------|---------------------|---------------------|-----------------------|-----------------------|
| ln distance                                                  | -0.0699<br>(0.1067) | -0.0568<br>(0.1018) | -0.0724<br>(0.1039)   | -0.0610<br>(0.0987)   |
| ln distance $\times$ foreign MNE                             | -0.0122<br>(0.0127) | 0.0017<br>(0.0120)  |                       |                       |
| ln distance $\times$ foreign MNE, parent-country destination |                     |                     | 0.1942***<br>(0.0205) | 0.1952***<br>(0.0210) |
| ln distance $\times$ foreign MNE, other destination          |                     |                     | -0.0253**<br>(0.0123) | -0.0110<br>(0.0117)   |
| ln distance $\times$ domestic MNE                            | 0.0070<br>(0.0101)  | 0.0078<br>(0.0106)  | 0.0060<br>(0.0100)    | 0.0068<br>(0.0105)    |
| Origin FE                                                    | ✓                   |                     | ✓                     |                       |
| Destination FE                                               | ✓                   |                     | ✓                     |                       |
| Year FE                                                      | ✓                   |                     | ✓                     |                       |
| Origin $\times$ year FE                                      |                     | ✓                   |                       | ✓                     |
| Destination $\times$ year FE                                 |                     | ✓                   |                       | ✓                     |
| Product FE                                                   | ✓                   | ✓                   | ✓                     | ✓                     |
| Observations                                                 | 38,597              | 38,110              | 38,597                | 38,110                |

*Notes:* Observations are firm–destination–HS6–year cells. Other Goods products only. Dependent variable: ln exports. Non-MNE firms are the omitted group. Columns (3)–(4) split foreign MNEs by whether the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

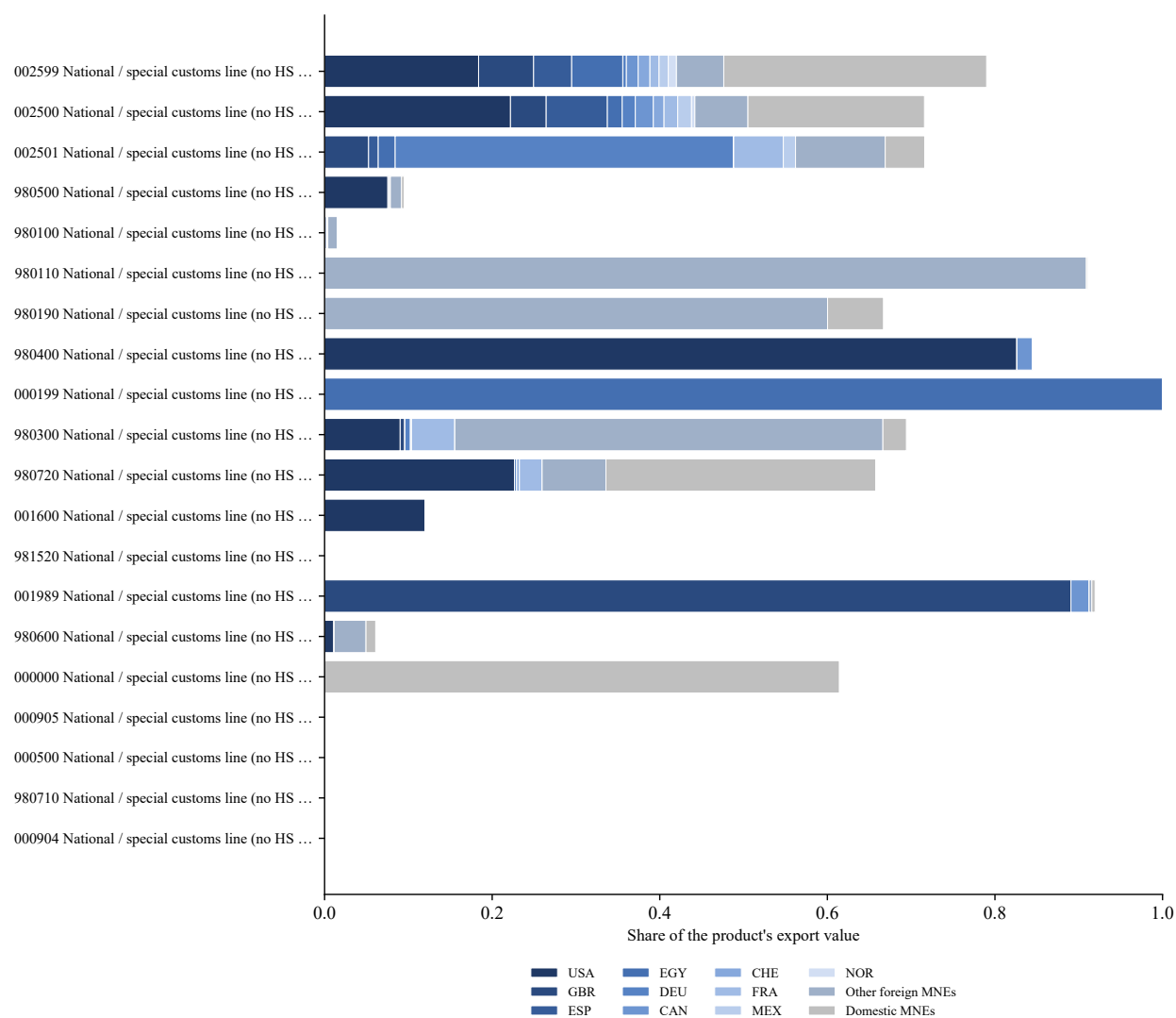
*Source:* Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

# 32 Products



**Figure 154:** Other Goods: Ownership of the Twenty Largest Export Products

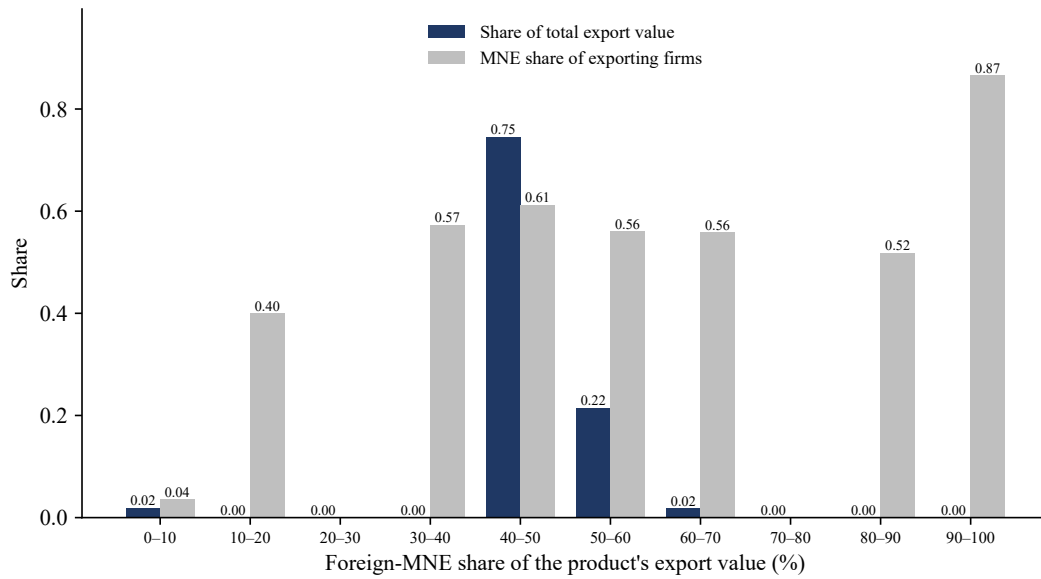
*Notes:* Shares of each product's export value. Twenty largest HS6 products by export value.  
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 155: Other Goods: Ownership of the Twenty Largest Export Products, by Parent Country**

*Notes:* Shares of each product's export value. Foreign-MNE share split by country of the ultimate parent; the unfilled part of each bar is local firms.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 156:** Other Goods: Distribution of Export Value by Product-Level Foreign-Multinational Share

*Notes:* Share of export value and MNE share of exporting firms, by the foreign-MNE share of each HS6 product's export value.

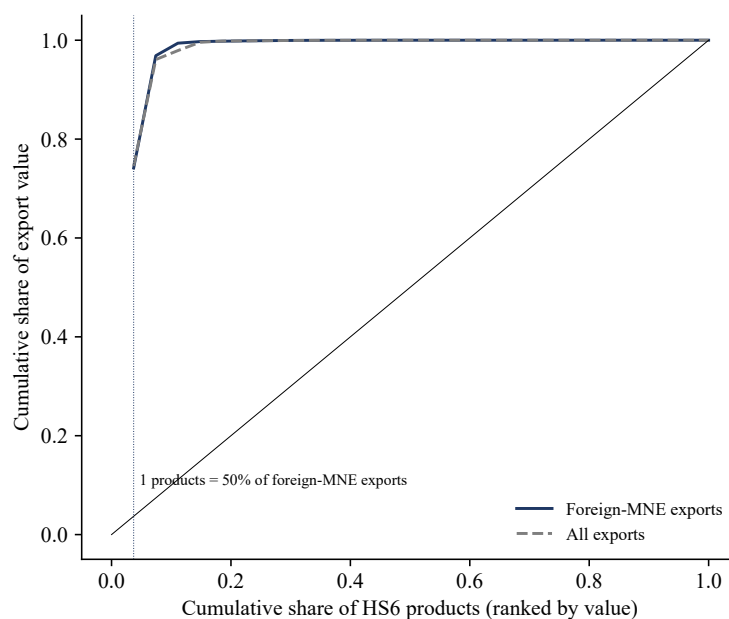
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 232:** Other Goods: Distribution of Export Value by Product-Level Foreign-Multinational Share

| Foreign share bin (%) | Share of value | MNE share of exporting firms | HS6 products |
|-----------------------|----------------|------------------------------|--------------|
| 0-10                  | 0.019          | 0.036                        | 13           |
| 10-20                 | 0.000          | 0.400                        | 1            |
| 20-30                 | 0.000          | –                            | 0            |
| 30-40                 | 0.000          | 0.573                        | 1            |
| 40-50                 | 0.746          | 0.613                        | 2            |
| 50-60                 | 0.215          | 0.560                        | 1            |
| 60-70                 | 0.019          | 0.559                        | 3            |
| 70-80                 | 0.000          | –                            | 0            |
| 80-90                 | 0.000          | 0.519                        | 1            |
| 90-100                | 0.001          | 0.866                        | 5            |

*Notes:* Share of export value and MNE share of exporting firms, by the foreign-MNE share of each HS6 product's export value. MNE share of exporting firms: MNE firm–destination–product–year observations over all observations in the bin.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 157:** Other Goods: Concentration of Exports across Products

*Notes:* Cumulative share of export value against the cumulative share of HS6 products, ranked by value, for foreign-MNE exports and for all exports.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 233:** Other Goods: Exports and Multinational Shares, by HS Section

| HS section                     | US\$bn/yr | Foreign (%) | Domestic (%) | Local (%) | Leading parent |
|--------------------------------|-----------|-------------|--------------|-----------|----------------|
| Unclassified                   | 0.5       | 49          | 29           | 23        | USA (0.39)     |
| Special / unclassified (98-99) | 0.0       | 13          | 1            | 87        | USA (0.57)     |
| All sections                   | 0.5       | 48          | 28           | 24        |                |

*Notes:* Foreign, domestic and local: percent of each section's export value. Leading parent: largest parent country and its share of the section's foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 234: Other Goods: Thirty Largest Export Products**

| HS6    | Description                                         | US\$bn/yr | Foreign (%) | Domestic (%) | Local (%) | Leading parent |
|--------|-----------------------------------------------------|-----------|-------------|--------------|-----------|----------------|
| 002599 | National / special customs line (no HS description) | 0.36      | 48          | 31           | 21        | USA (0.39)     |
| 002500 | National / special customs line (no HS description) | 0.11      | 51          | 21           | 28        | USA (0.44)     |
| 002501 | National / special customs line (no HS description) | 0.01      | 67          | 5            | 28        | DEU (0.60)     |
| 980500 | National / special customs line (no HS description) | 0.01      | 9           | 0            | 91        | USA (0.82)     |
| 980100 | National / special customs line (no HS description) | 0.00      | 2           | 0            | 98        | PRY (0.32)     |
| 980110 | National / special customs line (no HS description) | 0.00      | 91          | 0            | 9         | JPN (1.00)     |
| 980190 | National / special customs line (no HS description) | 0.00      | 60          | 7            | 33        | JPN (1.00)     |
| 980400 | National / special customs line (no HS description) | 0.00      | 84          | 0            | 16        | USA (0.98)     |
| 000199 | National / special customs line (no HS description) | 0.00      | 100         | 0            | 0         | EGY (1.00)     |
| 980300 | National / special customs line (no HS description) | 0.00      | 67          | 3            | 31        | ALB (0.55)     |
| 980720 | National / special customs line (no HS description) | 0.00      | 34          | 32           | 34        | USA (0.68)     |
| 001600 | National / special customs line (no HS description) | 0.00      | 12          | 0            | 88        | USA (0.99)     |
| 981520 | National / special customs line (no HS description) | 0.00      | 100         | 0            | 0         | –              |
| 001989 | National / special customs line (no HS description) | 0.00      | 92          | 0            | 8         | GBR (0.97)     |
| 980600 | National / special customs line (no HS description) | 0.00      | 5           | 1            | 94        | HUN (0.67)     |
| 000000 | National / special customs line (no HS description) | 0.00      | 0           | 61           | 39        | –              |
| 000905 | National / special customs line (no HS description) | 0.00      | 0           | 0            | 100       | –              |
| 000500 | National / special customs line (no HS description) | 0.00      | 0           | 0            | 100       | –              |
| 980710 | National / special customs line (no HS description) | 0.00      | 0           | 0            | 100       | –              |
| 000904 | National / special customs line (no HS description) | 0.00      | 0           | 0            | 100       | –              |
| 000901 | National / special customs line (no HS description) | 0.00      | 47          | 0            | 53        | GBR (1.00)     |
| 980900 | National / special customs line (no HS description) | 0.00      | 0           | 0            | 100       | –              |
| 001400 | National / special customs line (no HS description) | 0.00      | 100         | 0            | 0         | –              |
| 980200 | National / special customs line (no HS description) | 0.00      | 0           | 0            | 100       | –              |
| 002102 | National / special customs line (no HS description) | 0.00      | 0           | 0            | 100       | –              |
| 981000 | National / special customs line (no HS description) | 0.00      | 0           | 0            | 100       | –              |
| 000802 | National / special customs line (no HS description) | 0.00      | 0           | 0            | 100       | –              |

*Notes:* Thirty largest HS6 products by export value. Foreign, domestic and local: percent of the product's export value. Leading parent: largest parent country and its share of the product's foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 235: Other Goods: Thirty Largest Export Products of Foreign Multinationals**

| HS6    | Description                                         | US\$bn/yr | Foreign (%) | Domestic (%) | Local (%) | Leading parent |
|--------|-----------------------------------------------------|-----------|-------------|--------------|-----------|----------------|
| 002599 | National / special customs line (no HS description) | 0.36      | 48          | 31           | 21        | USA (0.39)     |
| 002500 | National / special customs line (no HS description) | 0.11      | 51          | 21           | 28        | USA (0.44)     |
| 002501 | National / special customs line (no HS description) | 0.01      | 67          | 5            | 28        | DEU (0.60)     |
| 980500 | National / special customs line (no HS description) | 0.01      | 9           | 0            | 91        | USA (0.82)     |
| 980110 | National / special customs line (no HS description) | 0.00      | 91          | 0            | 9         | JPN (1.00)     |
| 980400 | National / special customs line (no HS description) | 0.00      | 84          | 0            | 16        | USA (0.98)     |
| 980190 | National / special customs line (no HS description) | 0.00      | 60          | 7            | 33        | JPN (1.00)     |
| 000199 | National / special customs line (no HS description) | 0.00      | 100         | 0            | 0         | EGY (1.00)     |
| 980300 | National / special customs line (no HS description) | 0.00      | 67          | 3            | 31        | ALB (0.55)     |
| 980100 | National / special customs line (no HS description) | 0.00      | 2           | 0            | 98        | PRY (0.32)     |
| 981520 | National / special customs line (no HS description) | 0.00      | 100         | 0            | 0         | –              |
| 980720 | National / special customs line (no HS description) | 0.00      | 34          | 32           | 34        | USA (0.68)     |
| 001989 | National / special customs line (no HS description) | 0.00      | 92          | 0            | 8         | GBR (0.97)     |
| 001600 | National / special customs line (no HS description) | 0.00      | 12          | 0            | 88        | USA (0.99)     |
| 000901 | National / special customs line (no HS description) | 0.00      | 47          | 0            | 53        | GBR (1.00)     |
| 980600 | National / special customs line (no HS description) | 0.00      | 5           | 1            | 94        | HUN (0.67)     |
| 001400 | National / special customs line (no HS description) | 0.00      | 100         | 0            | 0         | –              |
| 002102 | National / special customs line (no HS description) | 0.00      | 0           | 0            | 100       | –              |
| 000905 | National / special customs line (no HS description) | 0.00      | 0           | 0            | 100       | –              |
| 000904 | National / special customs line (no HS description) | 0.00      | 0           | 0            | 100       | –              |
| 980200 | National / special customs line (no HS description) | 0.00      | 0           | 0            | 100       | –              |
| 000802 | National / special customs line (no HS description) | 0.00      | 0           | 0            | 100       | –              |
| 000500 | National / special customs line (no HS description) | 0.00      | 0           | 0            | 100       | –              |
| 980710 | National / special customs line (no HS description) | 0.00      | 0           | 0            | 100       | –              |
| 980900 | National / special customs line (no HS description) | 0.00      | 0           | 0            | 100       | –              |
| 981000 | National / special customs line (no HS description) | 0.00      | 0           | 0            | 100       | –              |
| 000000 | National / special customs line (no HS description) | 0.00      | 0           | 61           | 39        | –              |

*Notes:* Thirty largest HS6 products by foreign-MNE export value. Foreign, domestic and local: percent of the product's export value. Leading parent: largest parent country and its share of the product's foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 236: Other Goods: Products with the Highest Foreign-Multinational Share**

| HS6    | Description                                         | US\$bn/yr | Foreign (%) | Domestic (%) | Local (%) | Leading parent |
|--------|-----------------------------------------------------|-----------|-------------|--------------|-----------|----------------|
| 002500 | National / special customs line (no HS description) | 0.11      | 51          | 21           | 28        | USA (0.44)     |
| 002599 | National / special customs line (no HS description) | 0.36      | 48          | 31           | 21        | USA (0.39)     |

*Notes:* HS6 products with exports of at least US\$500 million over the period, ranked by foreign-MNE share. Foreign, domestic and local: percent of the product's export value. Leading parent: largest parent country and its share of the product's foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 237: Other Goods: Products with the Highest Domestic-Multinational Share**

| HS6    | Description                                         | US\$bn/yr | Foreign (%) | Domestic (%) | Local (%) | Leading parent |
|--------|-----------------------------------------------------|-----------|-------------|--------------|-----------|----------------|
| 002599 | National / special customs line (no HS description) | 0.36      | 48          | 31           | 21        | USA (0.39)     |
| 002500 | National / special customs line (no HS description) | 0.11      | 51          | 21           | 28        | USA (0.44)     |

*Notes:* HS6 products with exports of at least US\$500 million over the period, ranked by domestic-MNE share. Foreign, domestic and local: percent of the product's export value. Leading parent: largest parent country and its share of the product's foreign-MNE exports. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

## 33 Product Groups within the Sector

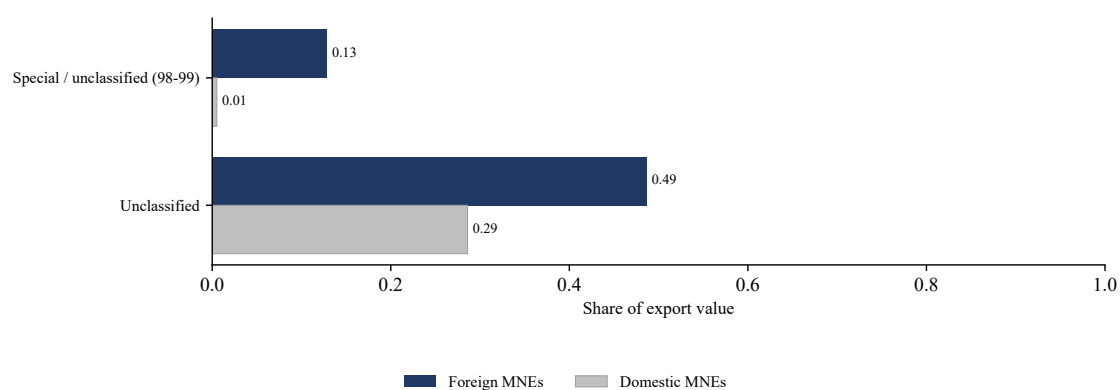
### 33.1 By HS Section

**Table 238:** Other Goods: Exports and Multinational Shares, by HS Section

| HS section                     | Total exports (US\$bn/yr) | MNE exports (US\$bn/yr) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|--------------------------------|---------------------------|-------------------------|-------------|--------------|------------------------------------|
| Special / unclassified (98-99) | 0.0                       | 0.0                     | 12.8        | 0.5          | USA 0.57, JPN 0.28, ALB 0.03       |
| Unclassified                   | 0.5                       | 0.4                     | 48.6        | 28.6         | USA 0.39, GBR 0.12, ESP 0.11       |
| All                            | 0.5                       | 0.4                     | 47.9        | 28.0         |                                    |

*Notes:* HS sections. Foreign and domestic: percent of the group's export value. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

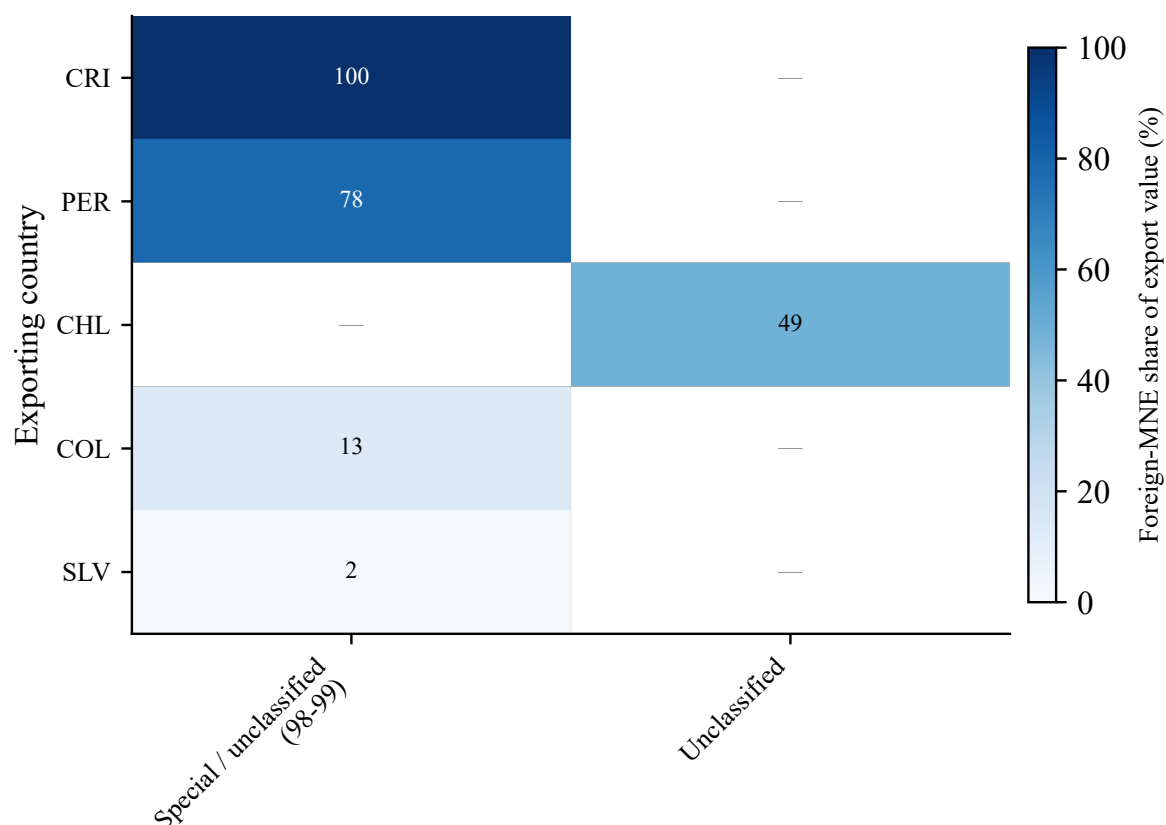
*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 158:** Other Goods: Multinational Share of Export Value, by HS Section

*Notes:* Shares of export value of other goods (HS 98–99 and unclassified codes), all available years pooled. HS sections.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.



**Figure 159:** Other Goods: Foreign-Multinational Share of Export Value, by Exporting Country and HS Section

*Notes:* Percent of each group's export value in each country shipped by foreign MNEs. HS sections.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 239:** Other Goods: Exports by Exporting Country and HS Section

| Origin                                                                   | Special / unclassified (98-99) | Unclassified | All   |
|--------------------------------------------------------------------------|--------------------------------|--------------|-------|
| <i>Panel A: composition of the origin's exports, % (rows sum to 100)</i> |                                |              |       |
| CHL                                                                      | 0.0                            | 100.0        | 100.0 |
| COL                                                                      | 100.0                          | 0.0          | 100.0 |
| CRI                                                                      | 100.0                          | 0.0          | 100.0 |
| PER                                                                      | 100.0                          | 0.0          | 100.0 |
| SLV                                                                      | 100.0                          | 0.0          | 100.0 |
| All origins                                                              | 2.1                            | 97.9         | 100.0 |
| <i>Panel B: foreign-MNE share of export value, %</i>                     |                                |              |       |
| CHL                                                                      | —                              | 48.6         | 48.6  |
| COL                                                                      | 12.7                           | —            | 12.7  |
| CRI                                                                      | 100.0                          | —            | 100.0 |
| PER                                                                      | 78.0                           | —            | 78.0  |
| SLV                                                                      | 1.5                            | —            | 1.5   |
| All origins                                                              | 12.8                           | 48.6         | 47.9  |

*Notes:* Panel A: composition of each country's exports in the sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs. HS sections.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 240:** Other Goods: Exports by HS Section and Parent Country

| HS section                     | USA  | GBR | ESP | EGY | DEU | CAN | CHE | FRA | MEX | NOR | Other | Domestic |
|--------------------------------|------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-------|----------|
| Special / unclassified (98-99) | 5.8  | 0.0 | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.1 | 0.0 | 0.0 | 6.7   | 0.5      |
| Unclassified                   | 15.3 | 4.8 | 4.2 | 4.0 | 1.3 | 1.3 | 1.1 | 1.1 | 1.0 | 0.7 | 14.0  | 28.6     |

*Notes:* Percent of each group's export value, by country of the ultimate parent of foreign MNEs; domestic MNEs shown for comparison. HS sections.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

### 33.2 HS Section and End Use

**Table 241:** Other Goods: Exports by HS Section and End Use

| HS section / BEC end use | Total |
|--------------------------|-------|
| Total                    | 0.0   |

*Notes:* Export value by HS section and BEC Rev. 4 end use. Values are annual averages in US\$ billion.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Table 242:** Other Goods: Foreign-Multinational Share of Exports by HS Section and End Use

| HS section / BEC end use |
|--------------------------|
|--------------------------|

*Notes:* Percent of each cell's export value shipped by foreign MNEs.

*Source:* Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.